Exact fluid constructions and the Navier–Stokes question

A cumulative proof and source notebook

8 September 2026

Abstract

This notebook reconstructs explicit mathematical maps suggested by earlier polynomial, arithmetic, and fluid calculations. It also develops selected mechanisms from the released smooth-forcing Euler and Boussinesq work, retaining the additional terms produced by positive viscosity. Every result proved here is stated at its actual scope. No counterexample to the classical three-dimensional Navier–Stokes problem has been established by this notebook. The reported separate classical Navier–Stokes proof has not been obtained. The notebook makes no prize claim. Attribution follows identifiable sources; it does not assign unverified contributions to a named person.

Contents

1 The exact target

We retain the dimensional viscosity $\nu > 0$. The unknowns are $u : \mathbb{R}^3 \times [0, \infty) \to \mathbb{R}^3$ and $p : \mathbb{R}^3 \times [0, \infty) \to \mathbb{R}$, with prescribed data $u^0 : \mathbb{R}^3 \to \mathbb{R}^3$ and $f : \mathbb{R}^3 \times [0, \infty) \to \mathbb{R}^3$. The sign convention is

$$\partial_t u_i + \sum_{j=1}^{3} u_j \partial_{x_j} u_i = \nu \sum_{j=1}^{3} \partial_{x_j}^2 u_i - \partial_{x_i} p + f_i, \qquad \sum_{i=1}^{3} \partial_{x_i} u_i = 0, \qquad u(x, 0) = u^0(x). \tag{1}$$

These are the equations and alternatives in Fefferman's official statement, including its appended pressure-periodicity correction[8]. No change of spatial metric is implicit in the Euclidean Laplacian above.

For the whole-space alternatives, the prescribed initial data must satisfy, for every multi-index $\alpha \in \mathbb{N}_0^3$ and every $K \in \mathbb{N}_0$,

$$|\partial_x^\alpha u^0(x)| \le C_{\alpha K}(1 + |x|)^{-K}. \tag{2}$$

The prescribed force must satisfy for every such α, K and $m \in \mathbb{N}_0$,

$$|\partial_x^\alpha \partial_t^m f(x, t)| \le C_{\alpha m K}(1 + |x| + t)^{-K} \quad (x \in \mathbb{R}^3, t \ge 0). \tag{3}$$

The required solution is smooth on the whole closed time half-space and has one finite bound C such that

$$\int_{\mathbb{R}^3} |u(x, t)|^2\, dx < C \quad \text{for every } t \ge 0. \tag{4}$$

For the periodic alternatives, the domain is exactly $\mathbb{R}^3/\mathbb{Z}^3$: u^0, f, u, and p are invariant under $x \mapsto x + e_j$, $j = 1, 2, 3$. Here e_j are the original unit coordinate vectors. The force condition is

$$|\partial_x^\alpha \partial_t^m f(x, t)| \le C_{\alpha m K}(1 + |t|)^{-K}. \tag{5}$$

In particular a spatially affine pressure cannot be inserted into a periodic counterexample while preserving the corrected target.

Alternative A asserts existence of a smooth solution with (4) for every divergence-free datum satisfying (2), with $f = 0$. Alternative B asserts smooth periodic existence for every smooth divergence-free periodic datum, with $f = 0$. Alternative C asks for data satisfying (2) and a force satisfying (3) for which no global smooth solution with (4) exists. Alternative D asks for smooth periodic data and a force satisfying (5) for which no global smooth periodic solution exists. All four alternatives retain $\nu > 0$ and dimension three. A velocity expression that is singular is therefore not by itself a counterexample: its prescribed data and force and the nonexistence conclusion must belong to one of these exact alternatives.

2 Released source material and provenance

The 8 September statement by Tristan Buckmaster identifies released results with smooth forcing for Euler, Boussinesq, and incompressible porous media, and reports a separate claimed classical Navier–Stokes proof that he had not seen[9]. The corresponding public papers and formalization repository are accessible[10, 11, 12]. This notebook does not independently certify their complete proofs merely by linking them. Its subsequent sections give complete derivations of the particular identities and constructions they use here.

The public repository snapshot inspected is commit `d0124689230b58b4f86e7b90ac59de0640 4b3b6b`. Its tree contains 3,777 files, including three comparator JSON configurations and three Lake dependency manifests. The inspected tree contains no JSONL file. This bounded inventory does not establish the absence of transcripts elsewhere. The retrieved JSON files configure proof verification or software dependencies; they are not a discovery transcript of classical Navier–Stokes blowup. Source hashes and retrieval scope are recorded in the machine-readable research files.

The original polynomial map used below is retained exactly from the earlier local proof corpus. Its discovery attribution and any claimed propagation to other conjectures are separate questions from the identities we explicitly verify. The requested personal credit must be based on identifiable contributions; the present fluid sources name Levent Alpöge and Tristan Buckmaster and credit the underlying program of Diego Córdoba and Luis Martínez-Zoroa. No identity between those names and another name is assumed.

3 The exact axisymmetric coordinate mechanism and its viscous extension

This section gives the complete coordinate and force reconstruction underlying the Euler system on pages 3–4, equations (2.1)–(2.6), of *Blowup for the Euler equations with smooth forcing*, the released manuscript examined in this repository. The coordinate identities are proved here independently. The compact force inverse is constructed and proved explicitly; no external inverse-divergence lemma is used. The same calculation also gives every additional term for classical viscosity $\nu > 0$. It proves an exact correspondence of smooth equations. It does not verify the manuscript's subsequent infinite construction or its terminal estimates.

3.1 Domains, signs, and the forward and inverse maps

Fix the original physical parameters

$$r_0 > 0, \qquad z_0 \in \mathbb{R}, \qquad 0 < R < r_0/2, \qquad r = (x_1^2 + x_2^2)^{1/2},$$
$$\mathcal{T}_R = \{(x_1, x_2, z) : (z - z_0)^2 + (r - r_0)^2 < R^2\}.$$

No dilation or translation of these parameters is performed. On $r > 0$, write

$$e_r = (\cos\phi, \sin\phi, 0), \qquad e_\phi = (-\sin\phi, \cos\phi, 0), \qquad e_z = (0, 0, 1).$$

Thus (e_r, e_ϕ, e_z) is positively oriented. An axisymmetric vector field has the form $u = u^r(r, z, t)e_r + u^\phi(r, z, t)e_\phi + u^z(r, z, t)e_z$, with components independent of ϕ.

The ordered meridional coordinates in this section are (z,r). Define the diffeomorphism and its inverse

$$\Phi:\mathbb{R}\times(0,\infty)\longrightarrow H:=\mathbb{R}\times(0,\infty),\qquad \Phi(z,r)=(y_1,y_2)=(z,r^2/2),\qquad \Phi^{-1}(y_1,y_2)=(y_1,\sqrt{2y_2}). \tag{6}$$

Their Jacobian determinants, in these orders, are r and $1/r$. In particular

$$dy_1\wedge dy_2=r\,dz\wedge dr,\qquad dy=r\,dz\,dr,$$
$$\partial_z=\partial_{y_1},\qquad \partial_r=r\partial_{y_2},\qquad \partial_{y_2}=r^{-1}\partial_r.$$

For a meridional set $K\subset H$, its physical lift is

$$\widetilde{K}=\{(r\cos\phi,r\sin\phi,z):(z,r^2/2)\in K,\ \phi\in\mathbb{R}/(2\pi\mathbb{Z})\}.$$

Let $D_0\Subset D$ be concentric open balls in H, centered at $y_*=(z_0,r_0^2/2)$, whose closures lie in the image under Φ of the physical meridional disk defining $\mathcal{T}_R$. Such balls exist: that image is open and contains y_*, so it contains a sufficiently small closed ball and a smaller concentric closed ball. The notation $D_0\Subset D$ means that $\overline{D_0}$ is a compact subset of D.

All fields below are smooth on a time interval I; at a closed endpoint, smoothness means compatible one-sided derivatives. Their spatial supports are contained in the indicated fixed compact subset of the stated ball, uniformly in $t\in I$. Fields in volume coordinates supported in H are extended by zero to $\mathbb{R}^2$.

Set

$$J(a_1,a_2)=(-a_2,a_1),\qquad A(y)=\begin{pmatrix}(2y_2)^{-1} & 0\\ 0 & 1\end{pmatrix},$$
$$L=\partial_{y_2}^2+(2y_2)^{-1}\partial_{y_1}^2,\qquad W=(2y_2)^{-2}. \tag{7}$$

The equality $L=\operatorname{div}_y(A\nabla_y)$ follows by expansion: the first diagonal coefficient depends on y_2, whereas the derivative in its divergence term is ∂_{y_1}; the other diagonal coefficient is identically one.

Proposition 3.1 (Velocity correspondence with a fixed streamfunction gauge)**.** *The linear map from pairs* (Γ,Ψ) *of smooth functions supported in* D_0 *to physical fields, defined by*

$$u^z=-\Psi_{y_2},\qquad u^r=\frac{\Psi_{y_1}}{r},\qquad u^\phi=\frac{\Gamma}{r},\qquad r=\sqrt{2y_2}, \tag{8}$$

is a bijection onto the smooth, divergence-free, axisymmetric vector fields supported in $\widetilde{D_0}$*. The inverse recovers*

$$\Gamma(y,t)=ru^\phi(r,z,t),\qquad v(y,t)=(u^z(r,z,t),ru^r(r,z,t)),$$
$$\Psi(y,t)=\int_{-\infty}^{y_1}v_2(s,y_2,t)\,ds. \tag{9}$$

In particular $v=J\nabla_y\Psi$.

Proof. For an axisymmetric field,

$$\operatorname{div}_x u=\partial_z u^z+r^{-1}\partial_r(ru^r)=\partial_{y_1}v_1+\partial_{y_2}v_2. \tag{10}$$

One direct derivation of the first identity uses $\partial_{x_1}r=\cos\phi$, $\partial_{x_2}r=\sin\phi$, $\partial_{x_1}\phi=-\sin\phi/r$, and $\partial_{x_2}\phi=\cos\phi/r$, substituted into the Cartesian divergence. The terms containing the azimuthal component cancel; those containing the radial component give $\partial_r u^r+u^r/r$. For (8), $v=(-\Psi_{y_2},\Psi_{y_1})$, so the right side of (10) is zero by equality of mixed derivatives. The factors $1/r$, the coordinate change, and the cylindrical basis are smooth on a neighborhood of the fixed support.

The field is identically zero on a fixed neighborhood of the axis. Consequently its extension to all of $\mathbb{R}^3$ is smooth, and its support is as stated.

Conversely, let u be a field in the specified target space. The functions Γ and v in (9) are smooth and compactly supported in D_0. Equation (10) gives $\partial_{y_1} v_1 + \partial_{y_2} v_2 = 0$. The integral in (9) is smooth and satisfies

$$\Psi_{y_1} = v_2, \qquad \Psi_{y_2} = \int_{-\infty}^{y_1} \partial_{y_2} v_2(s, y_2, t)\, ds = -\int_{-\infty}^{y_1} \partial_s v_1(s, y_2, t)\, ds = -v_1.$$

Here the lower boundary term vanishes by compact support. It remains to verify compact support of Ψ, which cannot simply be inferred from compact support of its gradient. Define $m(y_2, t) = \int_{\mathbb{R}} v_2(s, y_2, t)\, ds$. Its derivative is

$$\partial_{y_2} m = -\int_{\mathbb{R}} \partial_s v_1(s, y_2, t)\, ds = 0.$$

It vanishes for y_2 outside the projection of the support, hence $m = 0$ everywhere. Each horizontal slice of the ball D_0 is an interval. To the left of that interval the integral defining Ψ is zero; to its right it equals $m = 0$. Thus $\Psi = 0$ outside D_0. A smaller concentric ball still contains the fixed compact support of v, and the same argument places the support of Ψ in that smaller ball. Therefore Ψ has the required compact support. Any two streamfunctions giving this v have zero gradient, so their difference is spatially constant. Compact support makes that constant zero. The formulas recover all three components of u, proving the bijection, including its gauge and its time dependence. □

For every smooth axisymmetric scalar a, the material derivative is preserved exactly:

$$Da := (\partial_t + u^r \partial_r + u^z \partial_z) a = (\partial_t + v \cdot \nabla_y) a =: D_v a. \tag{11}$$

This identity applies to scalar components; the derivative of a vector also differentiates the cylindrical basis, as computed below. The measure identity also gives the exact energy formula

$$\begin{aligned} \int_{\mathbb{R}^3} |u(x,t)|^2\, dx &= 2\pi \int_H \left[\Psi_{y_2}^2 + \frac{\Psi_{y_1}^2 + \Gamma^2}{2y_2} \right] dy \\ &= 2\pi \int_H \left[v_1^2 + \frac{v_2^2 + \Gamma^2}{2y_2} \right] dy. \end{aligned} \tag{12}$$

Indeed the azimuthal integral of a scalar independent of ϕ is 2π, and $r\, dr\, dz = dy$; insertion of (8) proves both equalities.

3.2 Full vorticity and the exact physical gradient correspondence

The chosen orientation gives

$$\omega^r = -\partial_z u^\phi, \qquad \omega^\phi = \partial_z u^r - \partial_r u^z, \qquad \omega^z = r^{-1} \partial_r (r u^\phi), \qquad \omega = \nabla_x \times u. \tag{13}$$

These identities can be checked without a coordinate convention left implicit. At $\phi = 0$, $\partial_{x_1} = \partial_r$ and $\partial_{x_2} = r^{-1} \partial_\phi$. The basis derivatives are $\partial_\phi e_r = e_\phi$ and $\partial_\phi e_\phi = -e_r$. Substitution into the Cartesian curl therefore gives, in order, $-\partial_z u^\phi$, $\partial_z u^r - \partial_r u^z$, and $\partial_r u^\phi + u^\phi / r$. Rotational covariance gives the same components at every ϕ.

Define the reduced azimuthal vorticity and the squared circulation by

$$\xi = \frac{\omega^\phi}{r}, \qquad \Theta = \Gamma^2.$$

The velocity map gives every component, with its original radial factor:

$$\omega^r = -\frac{\Gamma_{y_1}}{r}, \qquad \omega^\phi = rL\Psi = r\xi, \qquad \omega^z = \Gamma_{y_2}, \qquad \xi = L\Psi. \tag{14}$$

For the middle component, direct differentiation gives

$$r^{-1}(\partial_z u^r - \partial_r u^z) = r^{-1}\left(\frac{\Psi_{y_1y_1}}{r} + r\Psi_{y_2y_2}\right) = (2y_2)^{-1}\Psi_{y_1y_1} + \Psi_{y_2y_2}.$$

The first and third components follow by differentiating Γ/r in (13). Consequently the full physical gradient, rather than just its coordinate representation, obeys

$$\nabla_x\Gamma = r\Gamma_{y_2}e_r + \Gamma_{y_1}e_z, \qquad |\omega|^2 = \frac{|\nabla_x\Gamma|^2}{r^2} + r^2\xi^2. \tag{15}$$

The two displayed vectors contributing to ω are orthogonal, which proves the sum of squares. For fields supported in $\mathcal{T}_R$, put $r_- = r_0 - R > 0$ and $r_+ = r_0 + R$. Pointwise on their support,

$$|\omega| \ge r_+^{-1}|\nabla_x\Gamma|, \qquad |\omega| \le r_-^{-1}|\nabla_x\Gamma| + r_+|\xi|.$$

Taking suprema preserves these inequalities. Thus growth of the physical circulation gradient forces full vorticity growth by an explicit bounded geometric factor. This statement alone gives no growth rate or time-integrability conclusion.

3.3 Derivation of the reduced Euler and Navier–Stokes equations

Keep $\nu \ge 0$ as an explicit constant and use the convention

$$\partial_t u + (u\cdot\nabla_x)u + \nabla_x p = \nu\Delta_x u + f, \qquad \operatorname{div}_x u = 0. \tag{16}$$

The value $\nu = 0$ is Euler. The force and pressure are axisymmetric in this calculation. Define the scalar cylindrical Laplacian $\Delta_0 = \partial_z^2 + \partial_r^2 + r^{-1}\partial_r$. The basis derivatives above, together with $u\cdot\nabla = u^r\partial_r + (u^\phi/r)\partial_\phi + u^z\partial_z$, give the acceleration components

$$Du^r - \frac{(u^\phi)^2}{r}, \qquad Du^\phi + \frac{u^r u^\phi}{r}, \qquad Du^z.$$

For completeness the scalar Laplacian follows by applying the divergence formula to the scalar gradient: $\Delta a = r^{-1}\partial_r(r\partial_r a) + r^{-2}\partial_\phi^2 a + \partial_z^2 a$. Apply this Cartesian-componentwise to a vector. Since $\partial_\phi^2 e_r = -e_r$, $\partial_\phi^2 e_\phi = -e_\phi$, and $\partial_\phi e_z = 0$, its axisymmetric components are

$$(\Delta_x u)^r = (\Delta_0 - r^{-2})u^r, \qquad (\Delta_x u)^\phi = (\Delta_0 - r^{-2})u^\phi, \qquad (\Delta_x u)^z = \Delta_0 u^z.$$

Thus (16) is precisely

$$Du^r - \frac{(u^\phi)^2}{r} + p_r = \nu(\Delta_0 - r^{-2})u^r + f^r, \tag{17}$$

$$Du^\phi + \frac{u^r u^\phi}{r} = \nu(\Delta_0 - r^{-2})u^\phi + f^\phi, \tag{18}$$

$$Du^z + p_z = \nu\Delta_0 u^z + f^z. \tag{19}$$

Multiplying (18) by r and using $Dr = u^r$ gives

$$D\Gamma = \nu\mathcal{D}_-\Gamma + rf^\phi, \qquad \mathcal{D}_- := \partial_z^2 + \partial_r^2 - r^{-1}\partial_r. \tag{20}$$

To verify the diffusion term explicitly,

$$\partial_r(\Gamma/r) = \Gamma_r/r - \Gamma/r^2, \qquad \partial_r^2(\Gamma/r) = \Gamma_{rr}/r - 2\Gamma_r/r^2 + 2\Gamma/r^3,$$

so $r(\Delta_0 - r^{-2})(\Gamma/r) = \Gamma_{zz} + \Gamma_{rr} - \Gamma_r/r$.

Next set $q = \partial_z u^r - \partial_r u^z = r\xi$. For arbitrary smooth u^r, u^z, expansion of the material derivative gives

$$\begin{aligned}\partial_z(Du^r) - \partial_r(Du^z) &= Dq + (\partial_z u^r)(\partial_r u^r) + (\partial_z u^z)(\partial_z u^r)\\ &\quad - (\partial_r u^r)(\partial_r u^z) - (\partial_r u^z)(\partial_z u^z)\\ &= Dq + (\partial_r u^r + \partial_z u^z)q = Dq - \frac{u^r}{r}q = rD\xi.\end{aligned}$$

The penultimate equality uses exactly the divergence equation, and the last uses $D(r\xi) = u^r\xi + rD\xi$. Apply ∂_z to (17) and subtract ∂_r of (19). The pressure cancels by equality of mixed derivatives. The centrifugal term becomes

$$-\partial_z((u^\phi)^2/r) = -r^{-3}\partial_z(\Gamma^2).$$

The viscous term is computed from the explicit commutator

$$\partial_r(\Delta_0 a) = \Delta_0(\partial_r a) - r^{-2}\partial_r a, \qquad \partial_z\Delta_0 a = \Delta_0\partial_z a.$$

These give

$$\begin{aligned}\partial_z[(\Delta_0 - r^{-2})u^r] - \partial_r\Delta_0 u^z &= (\Delta_0 - r^{-2})q\\ &= (\Delta_0 - r^{-2})(r\xi) = r\left(\xi_{zz} + \xi_{rr} + \frac{3}{r}\xi_r\right).\end{aligned}$$

In the last line, $(r\xi)_r = \xi + r\xi_r$ and $(r\xi)_{rr} = 2\xi_r + r\xi_{rr}$, so all zero-order terms cancel exactly. Division by the original radius now proves

$$D\xi = r^{-4}\partial_z(\Gamma^2) + \nu\mathcal{D}_+\xi + \frac{\partial_z f^r - \partial_r f^z}{r}, \qquad \mathcal{D}_+ := \partial_z^2 + \partial_r^2 + 3r^{-1}\partial_r. \tag{21}$$

For any scalar $a(y_1, y_2, t)$,

$$\partial_r a = r a_{y_2}, \qquad \partial_r^2 a = a_{y_2} + r^2 a_{y_2y_2}, \qquad \partial_z^2 a = a_{y_1y_1}.$$

Therefore the two distinct diffusion operators in volume coordinates are

$$Q := \partial_{y_1}^2 + 2y_2\partial_{y_2}^2, \qquad M := \partial_{y_1}^2 + 2y_2\partial_{y_2}^2 + 4\partial_{y_2}, \qquad \mathcal{D}_- = Q, \quad \mathcal{D}_+ = M. \tag{22}$$

In particular $Q = 2y_2L$. This is equality of operators applied to a function, with multiplication by $2y_2$ performed *after* applying L; it does not permit commuting derivatives through that factor.

For the meridional force use the covector components

$$b = (b_1, b_2) = (f^z, f^r/r), \qquad \operatorname{curl}_y b := \partial_{y_1}b_2 - \partial_{y_2}b_1. \tag{23}$$

They differ from the pushforward velocity components because

$$f^z\,dz + f^r\,dr = b_1\,dy_1 + b_2\,dy_2, \qquad \operatorname{curl}_y b = \frac{\partial_z f^r - \partial_r f^z}{r}.$$

The differential of this one-form is $(\operatorname{curl}_y b)dy_1 \wedge dy_2$, which agrees with $(\partial_z f^r - \partial_r f^z)dz \wedge dr$ by the Jacobian. This proves the exact map for the force, including its orientation.

The complete reduced system is consequently

$$\boxed{\begin{aligned} v &= J\nabla_y\Psi, \qquad \xi = L\Psi,\\ D_v\Gamma &= \nu Q\Gamma + rf^\phi,\\ D_v\xi &= W\partial_{y_1}(\Gamma^2) + \nu M\xi + \operatorname{curl}_y(f^z, f^r/r).\end{aligned}} \tag{24}$$

The equation for the original square $\Theta = \Gamma^2$, with its gradient-square term retained, is

$$D_v\Theta = \nu Q\Theta - 2\nu\big((\Gamma_{y_1})^2 + 2y_2(\Gamma_{y_2})^2\big) + 2\Gamma r f^{\phi}. \tag{25}$$

Indeed the product rule yields $Q(\Gamma^2) = 2\Gamma Q\Gamma + 2(\Gamma_{y_1})^2 + 4y_2(\Gamma_{y_2})^2$, and multiplication of the circulation equation by 2Γ gives (25). When $\nu = 0$, (24) and (25) are exactly the Euler formulas (2.6) in the source.

The momentum equations can also be displayed directly in volume coordinates, preserving the geometric quadratic terms:

$$D_v v_1 + p_{y_1} = f^z + \nu(Q + 2\partial_{y_2})v_1, \tag{26}$$

$$D_v v_2 - \frac{v_2^2 + \Gamma^2}{2y_2} + 2y_2 p_{y_2} = r f^r + \nu Q v_2. \tag{27}$$

The first follows from $v_1 = u^z$ and $\Delta_0 = Q + 2\partial_{y_2}$. For the second, $D(ru^r) = (u^r)^2 + rDu^r$; multiply (17) by r, use $rp_r = r^2 p_{y_2}$, and apply the already proved identity $r(\Delta_0 - r^{-2})(v_2/r) = Qv_2$.

3.4 An explicit compact curl inverse, with proof and bounds

We construct the force operator used below. Choose once and for all a smooth nonnegative function η supported in a compact subset of D, with $\int_{\mathbb{R}^2} \eta = 1$. Existence is explicit: choose a closed ball inside D, put $\eta_0(x) = \exp[-1/(1 - |x-a|^2/\rho^2)]$ for $|x - a| < \rho$ and zero otherwise, and divide it by its positive finite integral. Every derivative at the boundary vanishes because a polynomial in $(1 - |x-a|^2/\rho^2)^{-1}$ times the exponential tends to zero. This proves the required smoothness of the bump.

For a smooth g supported in D_0, define

$$(Bg)(x) = \int_{\mathbb{R}^2} g(y)(x-y)\left(\int_1^{\infty} \eta(y + t(x-y))t\,dt\right) dy, \qquad \mathcal{R}g = JBg. \tag{28}$$

The formula is interpreted by its absolutely locally integrable vector kernel, as justified next; equivalently it is given by the convergent smooth formulas in the proof.

Lemma 3.2 (Fixed compact inverse)**.** *The operator B is linear and smooth on smooth compactly supported inputs, and*

$$\operatorname{div} Bg = g - \eta\int_{\mathbb{R}^2} g, \qquad \operatorname{curl}_y \mathcal{R}g = g - \eta\int_{\mathbb{R}^2} g. \tag{29}$$

Its support is contained in the convex hull of $\operatorname{supp} g \cup \operatorname{supp}\eta$*, hence compactly contained in D. For every nonnegative integer k there is a finite constant C_k, depending only on the fixed balls, the bump, and k, such that*

$$\|Bg\|_{C^k(\mathbb{R}^2)} + \|\mathcal{R}g\|_{C^k(\mathbb{R}^2)} \le C_k\|g\|_{C^k(\mathbb{R}^2)}. \tag{30}$$

The same operator commutes with every time derivative.

Proof. Put $s = t - 1$ and $w = x - y$ in (28). Split the resulting s-integral at one. The portion from zero to one is

$$B_{\text{near}}g(x) = \int_0^1 (1+s)\int_{\mathbb{R}^2} wg(x-w)\eta(x+sw)\,dw\,ds. \tag{31}$$

In the portion from one to infinity put $a = sw$. It is

$$B_{\text{far}}g(x) = \int_1^{\infty} \frac{1+s}{s^3}\int_{\mathbb{R}^2} ag(x - a/s)\eta(x+a)\,da\,ds. \tag{32}$$

For x in any fixed bounded set, nonzero integrands in (31) have w in a fixed bounded set because $x-w \in \operatorname{supp} g \subset \overline{D_0}$. The integration domains in w, s then have finite measure and all x-derivatives are bounded by fixed finite linear combinations of derivatives of g and η. In (32), a lies in a fixed bounded set because $x + a \in \operatorname{supp} \eta$. The scalar weight is integrable: $\int_1^\infty (1+s)s^{-3} ds = 3/2$. Differentiating with respect to x differentiates only $g(x - a/s)$ and $\eta(x + a)$, with coefficients one. Dominated convergence therefore permits every derivative, proves smoothness, and bounds derivatives through order k by a constant times $\|g\|_{C^k}$. These arguments also prove absolute convergence of the original vector integral locally in x, by the indicated changes of variables.

If an integrand in (28) is nonzero, then $y \in \operatorname{supp} g$ and $a = y + t(x-y) \in \operatorname{supp} \eta$ for some $t \geq 1$. Consequently $x = (1 - 1/t)y + (1/t)a$ lies on the segment joining those two compact supports. Outside their closed convex hull the integral vanishes. Both supports lie in a smaller concentric closed ball inside the convex ball D, so their convex hull is a compact subset of D. Apply the preceding derivative bounds on one fixed bounded neighborhood of $\overline{D}$; outside that neighborhood the function and its derivatives vanish. This proves the global estimate (30), also for JB because J preserves Euclidean norm.

To prove the divergence identity, let φ be a smooth compactly supported test function. In the pairing with $\nabla\varphi$, change variables $a = y + t(x - y)$ in the x-integral. Then $dx = t^{-2} da$ and $x - y = (a - y)/t$. Thus

$$\begin{aligned} \int Bg(x) \cdot \nabla\varphi(x)\, dx &= \int g(y) \int \eta(a) \int_1^\infty \frac{a-y}{t^2} \cdot \nabla\varphi\left(y + \frac{a-y}{t}\right) dt\, da\, dy \\ &= \int g(y) \int \eta(a)\, [\varphi(a) - \varphi(y)]\, da\, dy. \end{aligned}$$

All integrations here are absolutely convergent: a, y range over fixed compact sets and the remaining bound is a constant times t^{-2}, whose integral is finite. The second equality uses

$$\frac{d}{dt}\varphi\left(y + \frac{a-y}{t}\right) = -\frac{a-y}{t^2} \cdot \nabla\varphi\left(y + \frac{a-y}{t}\right),$$

and the limit at infinity is $\varphi(y)$. Since $\int \eta = 1$, the definition of distributional divergence gives $\operatorname{div} Bg = g - \eta \int g$. Both sides are smooth, so equality also holds pointwise. Finally $\operatorname{curl}_y Jb = \partial_{y_1} b_1 + \partial_{y_2} b_2 = \operatorname{div} b$, proving the curl identity with the stated sign. The operator has no time-dependent coefficients, and the same dominated-convergence argument commutes any time derivative through it. This proves all claims of the lemma. $\square$

3.5 Force and pressure recovery in both directions

For an arbitrary pair (Γ, Ψ) as above define its exact reduced residuals at viscosity ν by

$$F_\nu^\Gamma = D_v\Gamma - \nu Q\Gamma, \qquad E_\nu^\xi = D_v(L\Psi) - W\partial_{y_1}(\Gamma^2) - \nu M(L\Psi). \tag{33}$$

No evolution equation has been assumed in this definition.

Proposition 3.3 (Complete compact reconstruction)**.** *For every such smooth pair and every fixed $\nu \geq 0$, the formulas*

$$b = \mathcal{R}E_\nu^\xi, \qquad f^z = b_1, \qquad f^r = rb_2, \qquad f^\phi = F_\nu^\Gamma / r \tag{34}$$

give a smooth axisymmetric force supported in $\widetilde{D}$. The velocity (8) then solves (16) with a smooth axisymmetric pressure on $\mathbb{R}^3 \times I$. All statements include mixed space-time derivatives. Conversely, every smooth axisymmetric solution of (16) in this compact velocity class satisfies (24) with its original force. Hence the reduced equations plus the proved pressure reconstruction are an exact correspondence of the stated systems.

Proof. First we verify the zero integral required by the compact inverse. All integrals are over $\mathbb{R}^2$, where the fields have compact support away from $y_2 = 0$. Because $L\Psi$ is a divergence, $\int L\Psi = 0$, and differentiation in time gives $\int \partial_t L\Psi = 0$. Also $v \cdot \nabla_y \xi = \operatorname{div}_y(v\xi)$, because $\operatorname{div}_y v = 0$, so its integral is zero. The function W depends only on y_2, and therefore $\int W\partial_{y_1}\Gamma^2 = 0$. Finally,

$$\int M\xi\, dy = \int (\xi_{y_1y_1} + 2y_2\xi_{y_2y_2} + 4\xi_{y_2})dy = 0:$$

the first and last terms integrate to zero, while integration by parts in the middle term gives $-2\int \xi_{y_2} = 0$. Thus $\int E^\xi_\nu = 0$, for Euler and for classical viscosity. Lemma 3.2 gives $\operatorname{curl}_y b = E^\xi_\nu$, so (34) satisfies both equations in (24). Its smoothness and support follow from that lemma and from the fixed positive lower bound on r on $\widetilde{D}$.

It remains to construct the pressure explicitly rather than leave it as a compatibility assertion. Define the smooth physical vector field

$$h = f + \nu\Delta_x u - \partial_t u - (u \cdot \nabla_x)u. \tag{35}$$

It is axisymmetric, compactly supported in $\widetilde{D}$, and identically zero in a fixed collar of the axis. The circulation equation gives $h^\phi = 0$. The previously proved meridional calculation gives the exact identity

$$\partial_z h^r - \partial_r h^z = r\left[\operatorname{curl}_y(f^z, f^r/r) + \nu M\xi - D_v\xi + W\partial_{y_1}\Gamma^2\right] = 0. \tag{36}$$

This also specifies the precise compatibility map from the momentum residual to the scalar residual, including the factor r.

For $r \geq 0$, extend the components of h by zero at the axis and put

$$p(z, r, t) = \int_0^r h^r(z, \rho, t)\, d\rho. \tag{37}$$

The integrand vanishes in a fixed neighborhood of $\rho = 0$, so there is no singular endpoint. Direct differentiation yields

$$p_r = h^r, \qquad p_z = \int_0^r \partial_z h^r(z, \rho, t)\, d\rho = \int_0^r \partial_\rho h^z(z, \rho, t)\, d\rho = h^z(z, r, t).$$

The last equality uses $h^z(z, 0, t) = 0$. There is no azimuthal derivative, so $\nabla_x p = h$ for $r > 0$. Moreover $p = 0$ in a fixed neighborhood of the axis. It therefore extends smoothly as an axisymmetric scalar on $\mathbb{R}^3$, with all time derivatives included. Substitution of (35) proves (16) everywhere, including the axis.

For the converse, Proposition 3.1 recovers the unique compact streamfunction and circulation from the original velocity. Applying the explicit component and curl calculations already proved gives (24). The original pressure is allowed: no step replaced it by a force or changed its derivatives. This proves the asserted correspondence. □

There are uniform finite derivative bounds in this reconstruction. For nonnegative integers k, b, the spatial chain and product rules, Lemma 3.2, and the fixed support give

$$\|\partial_t^b f\|_{C^k_x} \leq C_k \left(\|\partial_t^b E^\xi_\nu\|_{C^k_y} + \|\partial_t^b F^\Gamma_\nu\|_{C^k_y}\right), \tag{38}$$

where C_k depends on the fixed geometry and bump, not on the input pair or time. To see explicitly why the coordinate factors are bounded, on the compact support r has a positive minimum and a finite maximum, so every finite derivative of r, $1/r$, Φ, Φ^{-1}, e_r, and e_ϕ is bounded there. Each derivative in the product and chain rules is a finite sum of products of these fixed derivatives with derivatives of the indicated input. This proves (38) at every order. The pressure formula likewise preserves smoothness at each order by differentiation of its integral over a bounded radial interval on any compact spatial set.

Force recovery is unique up to an exactly characterized gradient. Suppose two smooth axisymmetric force fields supported away from the axis realize the same F_ν^Γ, E_ν^ξ. Their difference g has $g^\phi = 0$ and $\partial_z g^r - \partial_r g^z = 0$. The same formula $q(z,r,t) = \int_0^r g^r(z,\rho,t)\,d\rho$ then proves $g = \nabla_x q$. Conversely adding $\nabla_x q$ to the force and q to the pressure preserves (16) and both reduced residuals, by direct substitution and cancellation of mixed derivatives. Thus the missing scalar in a curl formulation is an explicit pressure gauge, with a proved reconstruction map.

3.6 What the viscosity calculation supplies for the next extension

At the level of arbitrary smooth fields, rather than just solutions, the same pair (Γ, Ψ) has residual differences

$$F_\nu^\Gamma - F_0^\Gamma = -\nu Q\Gamma, \qquad E_\nu^\xi - E_0^\xi = -\nu M\xi. \tag{39}$$

These identities exhibit the precise two additional terms that must be controlled in the circulation and reduced-vorticity equations. The coefficient $2y_2 = r^2$, the drift $4\partial_{y_2}$, the source coefficient $W = r^{-4}$, and the original r_0, z_0, R, ν have all been retained. The square equation (25) also retains its nonzero viscous gradient-square term.

There is an equally exact physical force identity. For the same smooth (u,p), define $f_0 = \partial_t u + (u\cdot\nabla)u + \nabla p$. Then the unique force that makes this same (u,p) satisfy the equation with viscosity ν is

$$f_\nu = f_0 - \nu\Delta_x u. \tag{40}$$

This follows by rearranging (16); it fixes the sign. The compact recovered force in (34) may use a different pressure gauge, but its difference from this expression has zero azimuthal component and zero meridional curl, by (39). The preceding explicit potential construction proves that the difference is a smooth gradient.

The coordinate mechanism therefore supplies a concrete translation between the equations. It does not estimate $\partial_t^b\partial_y^\alpha Q\Gamma$, $\partial_t^b\partial_y^\alpha M\xi$, or $\partial_t^b\partial_x^\alpha \Delta_x u$ at the terminal time of the released Euler construction. Smoothness on each closed interval strictly before that time gives no uniform terminal bound by itself: even the scalar function $(T-t)^{-1}$ is smooth on every such interval and is unbounded as $t \uparrow T$. A classical Navier–Stokes construction requires actual estimates or cancellations for the displayed viscous terms in its own complete evolution. Those estimates have not been established in this section.

3.7 Independent symbolic verification and its exact scope

The accompanying script `scripts/check_euler_coordinates.py` verifies fifteen exact differential identities with arbitrary smooth symbolic functions $\Psi, \Gamma, p, f^r, f^\phi, f^z$, retaining $r > 0$ and $\nu \geq 0$. All fifteen remainders are the exact integer zero; the recorded output is `checks/euler_coordinates_symbolic.json`. In particular, define the complete physical momentum residuals

$$\begin{aligned} R_r &= Du^r - (u^\phi)^2/r + p_r - \nu(\Delta_0 - r^{-2})u^r - f^r, \\ R_\phi &= Du^\phi + u^r u^\phi/r - \nu(\Delta_0 - r^{-2})u^\phi - f^\phi, \\ R_z &= Du^z + p_z - \nu\Delta_0 u^z - f^z. \end{aligned}$$

The independent expansion checks the identities

$$rR_\phi = D\Gamma - \nu\mathcal{D}_-\Gamma - rf^\phi, \tag{41}$$

$$\partial_z R_r - \partial_r R_z = r\left[D\xi - r^{-4}\partial_z\Gamma^2 - \nu\mathcal{D}_+\xi - \frac{\partial_z f^r - \partial_r f^z}{r}\right]. \tag{42}$$

Neither identity assumes that the residuals vanish. The remaining checks include the streamfunction signs, divergence, force-free pressure cancellation, vector-Laplacian curl, both diffusion coordinate changes, the material derivative, the vorticity operator, the swirl source, and the oriented Jacobian.

These algebraic checks corroborate the local differential calculation. The global support, smooth axis extension, inverse streamfunction, compact force inverse, and pressure recovery are established by the proofs above, not by the symbolic program.

4 The exact affine Boussinesq wave and its viscous counterpart

This section reconstructs the local calculation in Alpöge and Buckmaster's *Blowup for the Boussinesq equations with smooth forcing*, Sections 1.2 and 3.1, from its equations. The source is https://cims.nyu.edu/~tristanb/boussinesq.pdf. The pressure construction and the diffusion calculation below make explicit what the local wave does in the momentum equation. All coordinates, the upward buoyancy direction, the frequency parameter, and the signed amplitudes are retained. The fields in this section are defined on $\mathbb{R}^2$; their lack of spatial decay is part of their stated scope.

4.1 Coordinates, operators, and prescribed affine forces

Write $x = (x_1, x_2)$, $e_2 = (0, 1)$, and

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \nabla^\perp = J\nabla, \qquad \operatorname{curl}(w_1, w_2) = \partial_1 w_2 - \partial_2 w_1.$$

Thus $\operatorname{curl}(\nabla^\perp \psi) = \Delta\psi$. Let $I \subset \mathbb{R}$ be an open interval containing 0. Let $D : I \to \mathbb{R}^{2\times 2}$ and $G : I \to \mathbb{R}^2$ be smooth, with $\operatorname{tr} D = 0$. Put

$$u_b(x,t) = D(t)x, \qquad \theta_b(x,t) = G(t)\cdot x.$$

For any prescribed smooth pressure $p_b : I \times \mathbb{R}^2 \to \mathbb{R}$, define, without discarding any term,

$$f_{\theta,b}(x,t) = (\dot G + D^T G)\cdot x, \tag{43}$$
$$f_{u,b}(x,t) = (\dot D + D^2)x + \nabla p_b - (G\cdot x)e_2. \tag{44}$$

For constants $\kappa, \nu \geq 0$ these fields satisfy

$$\partial_t\theta_b + u_b\cdot\nabla\theta_b = \kappa\Delta\theta_b + f_{\theta,b}, \tag{45}$$
$$\partial_t u_b + (u_b\cdot\nabla)u_b + \nabla p_b = \nu\Delta u_b + \theta_b e_2 + f_{u,b}, \qquad \operatorname{div} u_b = 0. \tag{46}$$

Indeed, $\partial_t u_b = \dot D x$, $(u_b\cdot\nabla)u_b = D^2 x$, $\nabla\theta_b = G$, and both affine Laplacians vanish. These substitutions give exactly (43)–(44). In particular, an arbitrary time-dependent pair (D, G) is not being silently declared to be an unforced background.

The source uses the letter κ for $|\zeta|^2$ inside its local lemma. Here $q = |\zeta|^2$ denotes that exact quantity, while κ denotes the additional thermal diffusion coefficient. Neither quantity is set equal to one.

4.2 A general periodic profile and transported phase

Fix $\lambda > 0$ and $\zeta_0 \in \mathbb{R}^2 \setminus \{0\}$. The phase vector is the solution of

$$\dot\zeta = -D^T\zeta, \qquad \zeta(0) = \zeta_0. \tag{47}$$

It stays nonzero on I. Here is a direct justification of the linear ODE facts used throughout this section. On a compact time interval the matrix coefficient is bounded by some finite L. Successive integration of the equation $M' = DM$, $M(0) = \mathrm{Id}$ gives a matrix series whose nth term has norm at most $(L|t|)^n/n!$. It and its differentiated integral equation converge uniformly on that interval, producing a solution. The same construction solves $N' = -ND$, $N(0) = \mathrm{Id}$. Differentiation gives

$(NM)' = 0$, so $NM = \mathrm{Id}$. Smoothness follows by repeated differentiation of the equation. For uniqueness, the difference of two solutions is bounded, and iterating its homogeneous integral equation bounds its norm by a fixed constant times $(L|t|)^n/n!$ for every n; this tends to zero. The construction on overlapping compact intervals consequently agrees. Then $\zeta = M^{-T}\zeta_0$ gives (47) and cannot vanish. The same integral-series argument applies to the amplitude systems below.

Let $F : \mathbb{R} \to \mathbb{R}$ be smooth, odd, 2π-periodic, and not identically zero. Its integral over one period is zero: the integral on $[-\pi, \pi]$ vanishes by oddness, and periodicity makes all period integrals equal. Therefore

$$P_0(s) = \int_0^s F(a)\,da, \qquad P(s) = P_0(s) - \frac{1}{2\pi}\int_{-\pi}^{\pi} P_0(a)\,da$$

defines a smooth, even, mean-zero, periodic primitive with $P' = F$. For evenness, substitution $a = -b$ gives $P_0(-s) = P_0(s)$. For periodicity, $P_0(s+2\pi) - P_0(s) = \int_s^{s+2\pi} F(a)\,da = 0$. Subtracting the displayed constant supplies zero mean without changing the derivative.

Set

$$q(t) = |\zeta(t)|^2 > 0, \qquad s(x,t) = \lambda\zeta(t)\cdot x, \qquad C(t) = \frac{\Omega(t)}{\lambda q(t)}.$$

The signed amplitudes are $\Theta, \Omega : I \to \mathbb{R}$. Define the added fields

$$\vartheta = \Theta F(s), \qquad \psi = \frac{\Omega}{\lambda^2 q}P(s), \qquad v = CJ\zeta F(s), \qquad \varpi = \Omega F'(s). \tag{48}$$

Every identity below concerns these original fields.

Proposition 4.1 (Exact uncut inviscid wave, including pressure)**.** *For $\kappa = \nu = 0$, take the unique smooth amplitude solution with any real initial amplitudes and*

$$\dot\Theta = -\frac{J\zeta\cdot G}{\lambda q}\Omega, \qquad \dot\Omega = \lambda\zeta_1\Theta. \tag{49}$$

Define

$$h = \dot C J\zeta + 2CDJ\zeta - \Theta e_2, \qquad \pi(x,t) = -\frac{\zeta\cdot h}{\lambda q}P(s(x,t)). \tag{50}$$

Then

$$\theta = \theta_b + \vartheta, \qquad u = u_b + v, \qquad p = p_b + \pi$$

satisfy the full inviscid scalar and momentum equations (45)–(46) with exactly the same prescribed forces (43)–(44). Their vorticity is $\omega = D_{21} - D_{12} + \varpi$.

Proof. First $\nabla s = \lambda\zeta$. Since $P' = F$,

$$\nabla^\perp\psi = \frac{\Omega}{\lambda q}J\zeta F(s) = v, \qquad \Delta\psi = \frac{\Omega}{\lambda^2 q}\lambda^2 q F'(s) = \varpi.$$

Also

$$\operatorname{div} v = C\lambda(J\zeta\cdot\zeta)F'(s) = 0.$$

Write $L_b = \partial_t + (Dx)\cdot\nabla$. The precise phase calculation is

$$L_b s = \lambda\dot\zeta\cdot x + \lambda\zeta\cdot Dx = \lambda(\dot\zeta + D^T\zeta)\cdot x = 0.$$

The wave gradients are

$$\nabla\vartheta = \lambda\Theta\zeta F'(s), \qquad \nabla\varpi = \lambda\Omega\zeta F''(s).$$

Their scalar products with v vanish because $J\zeta\cdot\zeta = 0$. The background vorticity $D_{21} - D_{12}$ is spatially constant, so $v\cdot\nabla\omega_b = 0$ as well. Consequently the scalar residual increment is exactly

$$L_b\vartheta + v\cdot\nabla\theta_b + v\cdot\nabla\vartheta = \left(\dot\Theta + \frac{\Omega}{\lambda q}J\zeta\cdot G\right)F(s) = 0.$$

The vorticity residual increment is exactly

$$L_b\varpi + v\cdot\nabla\omega_b + v\cdot\nabla\varpi - \partial_1\vartheta = (\dot\Omega - \lambda\zeta_1\Theta)F'(s) = 0.$$

Thus far this is the scalar–vorticity calculation. We next prove its explicit lift to the momentum equation.

Write $D = \left(\begin{smallmatrix} a & b \\ c & -a\end{smallmatrix}\right)$. Multiplication gives $DJ + JD^T = 0$. Hence $J\dot\zeta = -JD^T\zeta = DJ\zeta$. The phase identity and $v\cdot\nabla v = 0$ give

$$\begin{aligned} L_b v + (v\cdot\nabla)u_b + (v\cdot\nabla)v - \vartheta e_2 &= (\dot C J\zeta + CJ\dot\zeta + CDJ\zeta - \Theta e_2)F(s) \\ &= hF(s). \end{aligned}$$

To show that the pressure in (50) cancels this exact vector, compute

$$\dot q = -2\zeta\cdot D^T\zeta = -2\zeta\cdot D\zeta, \qquad (J\zeta)\cdot DJ\zeta = -\zeta\cdot D\zeta = \frac{\dot q}{2}.$$

The second equality follows by multiplication with the displayed trace-free matrix, without symmetry assumptions on D. Therefore

$$(J\zeta)\cdot h = \dot C q + C\dot q - \Theta\zeta_1 = \frac{\dot\Omega}{\lambda} - \Theta\zeta_1 = 0.$$

The vectors $\zeta, J\zeta$ are orthogonal and have the same positive squared length q, so $h = q^{-1}(\zeta\cdot h)\zeta$. Finally

$$\nabla\pi = -\frac{\zeta\cdot h}{q}\zeta F(s) = -hF(s).$$

The momentum residual increment consequently vanishes component by component. The already computed divergence and curl establish the remaining claims. □

If $F(s) = \sin s$, the mean-zero primitive is $P(s) = -\cos s$. The signs in (48) then read

$$\psi = -\frac{\Omega}{\lambda^2 q}\cos s, \qquad v = \frac{\Omega}{\lambda q}J\zeta\sin s, \qquad \varpi = \Omega\cos s,$$

so they agree with the sine-wave presentation, including its negative streamfunction coefficient.

The profile can also be chosen exactly linear near zero. For completeness, fix $0 < a < b < \pi$, put $\rho(t) = e^{-1/t}$ for $t > 0$ and $\rho(t) = 0$ for $t \le 0$, and set

$$\chi(s) = \frac{\rho(b^2 - s^2)}{\rho(b^2 - s^2) + \rho(s^2 - a^2)}, \qquad F(s) = \sum_{m\in\mathbb{Z}} (s - 2\pi m)\chi(s - 2\pi m).$$

The denominator is strictly positive: its two arguments cannot both be nonpositive because $a < b$. The function ρ is smooth at zero since each one-sided derivative is a polynomial in $1/t$ times $e^{-1/t}$ and tends to zero as $t \downarrow 0$. Thus χ is smooth and even, equals one for $|s| \le a$, and vanishes for $|s| \ge b$. The sum is locally finite, periodic, odd, and equals s for $|s| \le a$. In that region,

$$\vartheta = \lambda\Theta\,\zeta\cdot x, \qquad v = \frac{\Omega}{q}J\zeta\,(\zeta\cdot x), \qquad \varpi = \Omega.$$

In particular

$$\nabla\vartheta(0,t) = \lambda\Theta\zeta, \qquad Dv(0,t) = \frac{\Omega}{q}J\zeta\otimes\zeta. \tag{51}$$

These two identities also hold for the sine profile since $F(0) = 0$ and $F'(0) = 1$.

4.3 Thermal diffusion and viscosity

For the same general profile, subtracting the affine background from the diffusive equations gives the exact scalar and vorticity residuals

$$R_\theta = \left(\dot\Theta + \frac{\Omega}{\lambda q} J\zeta \cdot G\right) F(s) - \kappa\lambda^2 q\,\Theta F''(s), \tag{52}$$

$$R_\omega = (\dot\Omega - \lambda\zeta_1\Theta)F'(s) - \nu\lambda^2 q\,\Omega F'''(s). \tag{53}$$

Indeed, $\Delta\vartheta = \lambda^2 q\,\Theta F''(s)$ and $\Delta\varpi = \lambda^2 q\,\Omega F'''(s)$, while all advective cancellations proved above still hold. Thus the general profile is not automatically closed under the diffusive equations.

For the sine profile, $F'' = -F$ and $F''' = -F'$, and the exact amplitude equations are instead

$$\dot\Theta = -\kappa\lambda^2 q\,\Theta - \frac{J\zeta \cdot G}{\lambda q}\Omega, \qquad \dot\Omega = \lambda\zeta_1\Theta - \nu\lambda^2 q\,\Omega. \tag{54}$$

These equations also lift to the full diffusive momentum equation without changing the prescribed background forces. In fact replace h in (50) by

$$h_\nu = \dot C J\zeta + 2CDJ\zeta + \nu\lambda^2 q\,CJ\zeta - \Theta e_2$$

and put $\pi_\nu = -(\zeta \cdot h_\nu)P(s)/(\lambda q)$. The additional momentum term is exactly $-\nu\Delta v = \nu\lambda^2 q\,v$. Moreover

$$J\zeta \cdot h_\nu = \frac{\dot\Omega + \nu\lambda^2 q\,\Omega}{\lambda} - \Theta\zeta_1 = 0$$

by (54). The same orthogonal decomposition and gradient calculation therefore give $\nabla\pi_\nu = -h_\nu F(s)$, proving the asserted full momentum identity.

4.4 The frozen matrix with every frequency factor retained

Fix

$$D = 0, \qquad G = -Ae_2, \qquad A > 0, \qquad \zeta = r(\sin\varphi, \cos\varphi), \qquad r > 0, \quad \varphi \in \mathbb{R}.$$

Here r, λ, A, φ are constants, and $k = \lambda r > 0$ is the magnitude of the physical wavevector. This is an exact stationary unforced background when

$$u_b = 0, \quad \theta_b = -Ax_2, \quad p_b = -\frac{A}{2}x_2^2 : \quad \nabla p_b = \theta_b e_2, \quad f_{\theta,b} = f_{u,b} = 0.$$

Its affine temperature does not decay in space.

Since $J\zeta = r(-\cos\varphi, \sin\varphi)$, we have $J\zeta \cdot G = -Ar\sin\varphi$ and $\zeta_1 = r\sin\varphi$. Equations (54) become

$$\frac{d}{dt}\begin{pmatrix}\Theta \\ \Omega\end{pmatrix} = B\begin{pmatrix}\Theta \\ \Omega\end{pmatrix}, \qquad B = \begin{pmatrix}-\kappa k^2 & A\sin\varphi/k \\ k\sin\varphi & -\nu k^2\end{pmatrix}. \tag{55}$$

No wavelength or coefficient has been absorbed into a new unit system.

Proposition 4.2 (Exact growth threshold for the frozen sine wave)**.** *The two eigenvalues of* (55) *are*

$$\mu_\pm = -\frac{(\kappa+\nu)k^2}{2} \pm \sqrt{\frac{(\kappa-\nu)^2k^4}{4} + A\sin^2\varphi}. \tag{56}$$

There is a strictly positive eigenvalue if and only if

$$A\sin^2\varphi > \kappa\nu\,\lambda^4 r^4. \tag{57}$$

Proof. Direct expansion yields

$$\det(\mu\mathrm{Id} - B) = (\mu + \kappa k^2)(\mu + \nu k^2) - A\sin^2\varphi = \mu^2 + (\kappa+\nu)k^2\mu + \kappa\nu k^4 - A\sin^2\varphi.$$

Completing the square gives (56). The expression under the square root is nonnegative. Since $\kappa, \nu \geq 0$, $\mu_- \leq 0$, and $\mu_+ > 0$ is equivalent to the square root exceeding $(\kappa+\nu)k^2/2$. Both quantities are nonnegative, so squaring is an equivalence. Subtracting $(\kappa+\nu)^2k^4/4$ from the radicand gives $A\sin^2\varphi - \kappa\nu k^4$. Finally $k^4 = \lambda^4 r^4$. This proves both directions, including the strict inequality. ☐

For $\kappa = \nu = 0$ and $0 < \varphi < \pi$, the positive eigenvalue is $\sqrt{A}\sin\varphi$ and its eigenline is

$$\Omega = \frac{\lambda r}{\sqrt{A}}\Theta.$$

Substitution into both rows of (55) proves this statement directly. Thus along that line

$$\Theta(t) = \Theta(0)e^{\sqrt{A}\sin\varphi\, t}, \qquad \Omega(t) = \frac{\lambda r}{\sqrt{A}}\Theta(0)e^{\sqrt{A}\sin\varphi\, t}.$$

Negative $\Theta(0)$ gives negative Ω and, by (51), a temperature-gradient increment pointing along $-\zeta$. Its Euclidean magnitude is $\lambda r|\Theta|$, whereas the sine temperature amplitude is $|\Theta|$. Also $Dv(0) = \Omega Je(\varphi) \otimes e(\varphi)$ for the original $e(\varphi) = (\sin\varphi, \cos\varphi)$.

If $\kappa, \nu > 0$ with A fixed, the threshold fails whenever $k^4 \geq A/(\kappa\nu)$, because $\sin^2\varphi \leq 1$. If $\kappa = 0$, $\nu > 0$, and $\sin\varphi \neq 0$, the positive rate is still strictly positive, but the exact rationalized expression is

$$\mu_+ = \frac{A\sin^2\varphi}{\sqrt{\nu^2k^4/4 + A\sin^2\varphi} + \nu k^2/2} \leq \frac{A\sin^2\varphi}{\nu k^2}.$$

The equality comes from multiplication by the conjugate, and the bound uses a denominator at least νk^2. Thus even in this one-diffusion case the inviscid growth rate cannot be carried over unchanged.

A fixed finite-dimensional matrix in (55) does not produce a finite-time singularity: its solution is $\sum_{n=0}^{\infty} t^n B^n/n!$ times the initial vector, bounded by $e^{|t|\|B\|}$ times that vector's norm on every bounded time interval. The spatial fields constructed from it are smooth at every finite time. A nonzero uncut wave also has no finite spatial L^2 norm: on a strip of positive width where its periodic profile has magnitude bounded below, the profile is constant in the unbounded direction perpendicular to ζ, so integrating its square in that direction diverges. The affine background separately fails the spatial decay requirements.

4.5 Exact common damping through growth, steering and holding

The released construction uses a growth interval followed by a rotation that returns the new vorticity amplitude to zero. Its holding interval then prepares the next layer. The following identity computes the effect of equal diffusion coefficients on the sine-wave amplitude system for the same prescribed affine background, including time-dependent phase length and steering. It does not assume that the background generated by a whole family of interacting layers is unchanged by viscosity.

Retain a compact interval $[t_0, t_1]$, the original D, G, ζ, λ, and $q = |\zeta|^2$. Define

$$A_0(t) = \begin{pmatrix} 0 & -(J\zeta\cdot G)/(\lambda q) \\ \lambda\zeta_1 & 0 \end{pmatrix}, \qquad d(t,t_0) = \exp\left(-\nu\lambda^2\int_{t_0}^{t} q(s)\,ds\right).$$

For $\kappa = \nu > 0$, the exact viscous amplitude matrix is $A_0(t) - \nu\lambda^2 q(t)\mathrm{Id}$. Let $Z_E = (\Theta_E, \Omega_E)$ be the uniquely defined solution of $Z_E' = A_0 Z_E$ with any real initial pair. Then the viscous solution with that same pair is

$$Z_\nu(t) = d(t,t_0)Z_E(t). \tag{58}$$

Indeed $d' = -\nu\lambda^2 q d$, and the ordinary product rule gives

$$(dZ_E)' = (A_0 - \nu\lambda^2 q\mathrm{Id})(dZ_E).$$

Also $d(t_0, t_0) = 1$. The existence and uniqueness argument for the linear system given earlier in this section proves the assertion and its converse after multiplication by d^{-1}. The scalar factor commutes with $A_0(t)$ even when the matrices $A_0(t)$ at different times do not commute with each other.

Since $d > 0$ at each finite time, zeros and signs of both amplitude components are preserved at exactly the same times in these two systems. In particular, a return $\Omega_E(t_*) = 0$ remains a return $\Omega_\nu(t_*) = 0$. During a subsequent interval where $\zeta_1 = 0$ and the initial vorticity is zero, uniqueness gives $\Omega_E = 0$, Θ_E constant, and

$$\Omega_\nu(t) = 0, \qquad \Theta_\nu(t) = \Theta_\nu(t_*) \exp\left(-\nu\lambda^2 \int_{t_*}^{t} q(s)\, ds\right).$$

Thus the held vorticity remains zero while the held temperature amplitude decays by this exact factor.

When $\Theta_E(t_0) \neq 0$ and $\Theta_E(t_1) \neq 0$, put $g = |\Theta_E(t_1)|/|\Theta_E(t_0)| > 0$. The viscous amplitude gain on the complete chosen interval is exactly

$$\frac{|\Theta_\nu(t_1)|}{|\Theta_\nu(t_0)|} = g \exp\left(-\nu\lambda^2 \int_{t_0}^{t_1} q(s)\, ds\right). \tag{59}$$

It exceeds one precisely when $\log g > \nu\lambda^2 \int_{t_0}^{t_1} q(s)\, ds$, by monotonicity of the logarithm. In the frozen growing eigenline, with $0 < \varphi < \pi$, this gives the retained rate $\sqrt{A}\sin\varphi - \nu\lambda^2 r^2$ and exactly the equal-diffusion case of (57).

For a general periodic phase profile, its Fourier mode m instead has the factor

$$d_m(t, t_0) = \exp\left(-\nu m^2\lambda^2 \int_{t_0}^{t} q(s)\, ds\right).$$

This follows by the same matrix product calculation after $\partial_s^2 e^{ims} = -m^2 e^{ims}$. The sine factor d_1 applies to modes $|m| = 1$. Nonzero modes with distinct absolute frequencies have different damping factors for $t > t_0$, so their sum generally has no common scalar damping factor; the full Fourier evolution is the one constructed in Section 13.

That section also gives exact finite localized corrections with their remaining residual. Section 15 gives the full three-dimensional swirl transfer and its computed force. Neither the frozen eigenvalue nor the common-damping identity establishes a singular limit of the coupled, localized sequence.

5 The report's Boussinesq coordinates, with the original scales retained

The incoming report, Section 6, writes the base gradient as $-e_0$, whereas Section 9.4 gives the original datum

$$\theta_{\mathrm{in}}(x) = -\frac{A_0}{\lambda_0}\sin(\lambda_0 x_2)\chi_0(x), \qquad A_0 \geq 1, \quad \lambda_0 = \left\lceil \frac{\sqrt{A_0}}{2} \right\rceil. \tag{60}$$

Here $\chi_0 : \mathbb{R}^2 \to [0, 1]$ is smooth, radial, compactly supported in B_{R_0}, equals one on B_2, and $R_0 > 2$. These formulas use different coordinates. The primary Boussinesq manuscript, Lemma 1.2 and equations (1.6)–(1.8), explicitly gives the dilation between them. We prove it here and retain both objects. The primary manuscript calls its time-dilation constant ν; in this section its exact value is denoted $\upsilon = \sqrt{A_0}$, so that it cannot be mistaken for physical viscosity. Set $\mu = \lambda_0$. Both constants are strictly positive.

5.1 An invertible map of fields, forces and both diffusion operators

On $\mathbb{R}^2_X \times [0, \widetilde{T})$ let $\widetilde{\theta}, \widetilde{u}, \widetilde{p}, \widetilde{f}_\theta, \widetilde{f}_u$ be smooth fields. Fix constants $\widetilde{\kappa}, \widetilde{\nu} \geq 0$, and define their residuals by

$$\widetilde{E}_\theta = \partial_\tau \widetilde{\theta} + \widetilde{u} \cdot \nabla_X \widetilde{\theta} - \widetilde{\kappa} \Delta_X \widetilde{\theta} - \widetilde{f}_\theta, \tag{61}$$

$$\widetilde{E}_u = \partial_\tau \widetilde{u} + (\widetilde{u} \cdot \nabla_X) \widetilde{u} + \nabla_X \widetilde{p} - \widetilde{\theta} e_2 - \widetilde{\nu} \Delta_X \widetilde{u} - \widetilde{f}_u. \tag{62}$$

Define $X = \mu x$, $\tau = \upsilon t$, $T = \widetilde{T}/\upsilon$, and evaluate every tilded expression below at $(\mu x, \upsilon t)$:

$$\theta = \frac{\upsilon^2}{\mu} \widetilde{\theta}, \qquad u = \frac{\upsilon}{\mu} \widetilde{u}, \qquad p = \frac{\upsilon^2}{\mu^2} \widetilde{p}, \tag{63}$$

$$f_\theta = \frac{\upsilon^3}{\mu} \widetilde{f}_\theta, \qquad f_u = \frac{\upsilon^2}{\mu} \widetilde{f}_u, \qquad \kappa = \frac{\upsilon}{\mu^2} \widetilde{\kappa}, \qquad \nu = \frac{\upsilon}{\mu^2} \widetilde{\nu}. \tag{64}$$

Let E_θ, E_u be the residuals in the original (x, t) coordinates, with the last two diffusion constants in (63). Then the following are identities of fields, without assuming that a residual vanishes:

$$E_\theta(x, t) = \frac{\upsilon^3}{\mu} \widetilde{E}_\theta(\mu x, \upsilon t), \quad E_u(x, t) = \frac{\upsilon^2}{\mu} \widetilde{E}_u(\mu x, \upsilon t), \quad \operatorname{div}_x u = \upsilon \operatorname{div}_X \widetilde{u}. \tag{65}$$

The map is a bijection of these field tuples and, in particular, of their zero-residual subsets. Its inverse is obtained by evaluating at $(x, t) = (X/\mu, \tau/\upsilon)$ and multiplying the five fields respectively by μ/υ^2, μ/υ, μ^2/υ^2, μ/υ^3, μ/υ^2; the inverse diffusion constants are $\widetilde{\kappa} = \mu^2 \kappa/\upsilon$, $\widetilde{\nu} = \mu^2 \nu/\upsilon$.

Proof. The chain rule gives

$$\partial_t \theta = \frac{\upsilon^3}{\mu} \partial_\tau \widetilde{\theta}, \qquad u \cdot \nabla_x \theta = \frac{\upsilon}{\mu} \widetilde{u} \cdot \upsilon^2 \nabla_X \widetilde{\theta}, \qquad \Delta_x \theta = \upsilon^2 \mu \Delta_X \widetilde{\theta}.$$

Multiplication of the last expression by $\kappa = \upsilon \widetilde{\kappa}/\mu^2$ gives the required factor υ^3/μ. For the vector equation,

$$\partial_t u = \frac{\upsilon^2}{\mu} \partial_\tau \widetilde{u}, \quad (u \cdot \nabla_x) u = \frac{\upsilon^2}{\mu} (\widetilde{u} \cdot \nabla_X) \widetilde{u}, \quad \nabla_x p = \frac{\upsilon^2}{\mu} \nabla_X \widetilde{p}, \quad \Delta_x u = \upsilon \mu \Delta_X \widetilde{u}.$$

Thus all its terms, including θe_2 with the unchanged laboratory vector $e_2 = (0, 1)$, acquire υ^2/μ. Taking the trace of the derivative of u proves the divergence identity. Every listed inverse multiplier is nonzero, and substitution in either order returns each original field and coefficient. This proves both directions, including the operators, domains and forces. $\square$

For the original datum, define $\widetilde{\chi}_0(X) = \chi_0(X/\mu)$ and $\widetilde{\theta}_{\text{in}}(X) = -\sin(X_2) \widetilde{\chi}_0(X)$. Then (63) gives exactly (60), since $\upsilon^2 = A_0$ and $\mu = \lambda_0$. The tilded cutoff equals one on $B_{2\mu}$ and is supported in $B_{\mu R_0}$. Support is therefore transported rather than discarded. For a compact streamfunction, the exact formulas are

$$\psi = \frac{\upsilon}{\mu^2} \widetilde{\psi}(\mu x, \upsilon t), \qquad u = J \nabla_x \psi, \qquad \omega = \Delta_x \psi = \upsilon \widetilde{\omega}(\mu x, \upsilon t), \qquad J(a, b) = (-b, a). \tag{66}$$

These follow by one and two spatial differentiations; no inverse Laplacian at zero frequency is needed.

For every spatial multi-index γ and integer $b \geq 0$,

$$\partial_x^\gamma \partial_t^b f_\theta = \upsilon^{3+b} \mu^{|\gamma|-1} (\partial_X^\gamma \partial_\tau^b \widetilde{f}_\theta)(\mu x, \upsilon t), \tag{67}$$

$$\partial_x^\gamma \partial_t^b f_u = \upsilon^{2+b} \mu^{|\gamma|-1} (\partial_X^\gamma \partial_\tau^b \widetilde{f}_u)(\mu x, \upsilon t). \tag{68}$$

The change of variables $dX = \mu^2 dx$ also proves, at corresponding times,

$$\|\theta\|_{L^2_x} = \frac{\upsilon^2}{\mu^2}\|\widetilde{\theta}\|_{L^2_X}, \qquad \|u\|_{L^2_x} = \frac{\upsilon}{\mu^2}\|\widetilde{u}\|_{L^2_X}, \tag{69}$$

$$\|\nabla_x\theta\|_\infty = \upsilon^2\|\nabla_X\widetilde{\theta}\|_\infty, \qquad \|\omega\|_\infty = \upsilon\|\widetilde{\omega}\|_\infty. \tag{70}$$

Positive finite factors preserve boundedness, full divergence to infinity, and divergent limsup with their different quantifiers. These identities transfer such properties of specified fields; they do not assert the global iteration estimates for unspecified fields.

5.2 The phase, shear and feedback map in the original variables

Retain a finite collection of layers at each time. For each layer j, let w_j, Ω_j, Θ_j be scalar functions, $\lambda_j > 0$, $\zeta_j \neq 0$, and $e_j = \zeta_j/|\zeta_j|$. Let $e_0 = R_\alpha e_2$, with R_α counterclockwise rotation. In the original variables the background arrays are

$$G_{<j} = -A_0 e_0 - \sum_{1\le i<j} w_i A_i e_i, \qquad A_i = -\lambda_i|\zeta_i|\Theta_i, \tag{71}$$

$$D_{<j} = \dot{\alpha}J + \sum_{1\le i<j} w_i\Omega_i J e_i \otimes e_i. \tag{72}$$

Here $(a\otimes b)c = a(b\cdot c)$. The unit-coordinate source formulas are related by the exact identities

$$\lambda_j = \mu\widetilde{\lambda}_j, \qquad \zeta_j(t) = \widetilde{\zeta}_j(\upsilon t), \qquad w_j(t) = \widetilde{w}_j(\upsilon t), \tag{73}$$

$$\Theta_j(t) = \frac{\upsilon^2}{\mu}\widetilde{\Theta}_j(\upsilon t), \qquad \Omega_j(t) = \upsilon\widetilde{\Omega}_j(\upsilon t), \qquad \alpha(t) = \widetilde{\alpha}(\upsilon t), \tag{74}$$

$$A_j(t) = \upsilon^2\widetilde{A}_j(\upsilon t), \qquad G_{<j}(t) = \upsilon^2\widetilde{G}_{<j}(\upsilon t), \qquad D_{<j}(t) = \upsilon\widetilde{D}_{<j}(\upsilon t). \tag{75}$$

In particular the phase is exactly $\lambda_j\zeta_j(t)\cdot x = \widetilde{\lambda}_j\widetilde{\zeta}_j(\tau)\cdot X$. Substitution into

$$\dot{\zeta}_j = -D^T_{<j}\zeta_j, \qquad \dot{\Theta}_j = -\frac{J\zeta_j\cdot G_{<j}}{\lambda_j|\zeta_j|^2}\Omega_j, \qquad \dot{\Omega}_j = \lambda_j\zeta_{j,1}\Theta_j$$

gives the source equations with respective common factors υ, υ^3/μ, and υ^2. This proves the dynamics in both coordinates. It also explains why the source's initial interpolation of duration one has duration $1/\sqrt{A_0}$ for (60).

We now prove the feedback identities without discarding any shear. Put $s_i = w_i\Omega_i$. Transposition of (71) and $J^T = -J$ give

$$\dot{\zeta}_j = \dot{\alpha}J\zeta_j - \sum_{i<j} s_i e_i(Je_i\cdot\zeta_j), \qquad \dot{\zeta}_{j,1} = -\dot{\alpha}\zeta_{j,2} - \sum_{i<j} s_i e_{i,1}(Je_i\cdot\zeta_j). \tag{76}$$

On the open set $\zeta_{j,2} \neq 0$ define

$$\mathcal{F}_j = -\frac{\sum_{i<j} s_i e_{i,1}(Je_i\cdot\zeta_j)}{\zeta_{j,2}}.$$

Then $\dot{\zeta}_{j,1} = (\mathcal{F}_j - \dot{\alpha})\zeta_{j,2}$. For a prescribed differentiable function Z, the identity $\dot{\alpha} = \mathcal{F}_j - \dot{Z}/\zeta_{j,2}$ implies $d(\zeta_{j,1} - Z)/dt = 0$ on that open set. Matching the values at one time therefore gives equality throughout the connected interval. This is an exact local feedback identity. It does not assert that its denominator stays nonzero in a constructed infinite cascade. Under (73), $\mathcal{F}_j(t) = \upsilon\widetilde{\mathcal{F}}_j(\upsilon t)$, so this identity too retains the original time scale.

On any time interval choose continuous angle lifts with $e_j = (\sin\phi_j, \cos\phi_j)$, and write $\rho_j = \log|\zeta_j|$, $d_{ij} = \phi_j - \phi_i$. Since $\partial_\phi e_j = -Je_j$, differentiation gives

$$\dot{\zeta}_j = |\zeta_j|(\dot{\rho}_j e_j - \dot{\phi}_j Je_j).$$

The exact inner products are $e_i \cdot e_j = \cos d_{ij}$, $Je_i \cdot e_j = -\sin d_{ij}$, and $e_i \cdot Je_j = \sin d_{ij}$. Taking inner products of (76) with e_j and Je_j, and dividing by the positive $|\zeta_j|$, proves

$$\dot{\rho}_j = \sum_{i<j} s_i \sin d_{ij} \cos d_{ij}, \tag{77}$$

$$\dot{\phi}_j = -\dot{\alpha} - \Sigma_j, \qquad \Sigma_j = \sum_{i<j} s_i \sin^2 d_{ij}, \qquad \dot{d}_{ij} = \Sigma_i - \Sigma_j. \tag{78}$$

Conversely these two equations for ρ_j, ϕ_j, inserted into the displayed derivative of $\zeta_j = e^{\rho_j} e_j$, recover both components of (76), because (e_j, Je_j) is an orthonormal basis. This proves the exact covector–angle correspondence, including its dynamics and all older shears. Common rotation cancels from the explicit relative-angle derivative, but the amplitudes and angles still satisfy a coupled system.

Finally, the affine-core derivative is $S_j = \Omega_j Je_j \otimes e_j$. Its trace is $\Omega_j e_j \cdot Je_j = 0$ and its square is $\Omega_j^2 Je_j (e_j \cdot Je_j) \otimes e_j = 0$. Its image equals $\operatorname{span}(Je_j)$ when $\Omega_j \neq 0$, whereas $S_j = 0$ when $\Omega_j = 0$. Thus the incoming report's unqualified "rank one" is corrected to rank one precisely for a nonzero amplitude and rank zero at its return endpoint.

5.3 All initial-angle cases in the report's illustrative subsystem

The report's equation (6.10) divides by $\sin \phi(0)$, so its endpoint cases must be restored. Keep exactly its equations

$$\frac{d\phi}{d\tau} = \sin \phi, \qquad \frac{d \log B}{d\tau} = \cos \phi, \qquad B(0) = B_0 > 0, \qquad \phi(0) = \phi_0 \in \mathbb{R}.$$

For $\phi_0 = n\pi$, uniqueness gives $\phi(\tau) = n\pi$ and direct integration gives $B(\tau) = B_0 e^{(-1)^n \tau}$. For $\phi_0 \notin \pi\mathbb{Z}$, set $z_0 = \tan(\phi_0/2)$. It is finite and nonzero. Choose the unique integer n for which $\phi_0 = 2 \arctan z_0 + 2n\pi$. Then on the whole real time line the exact formulas are

$$z(\tau) = e^{\tau} z_0, \qquad \phi(\tau) = 2 \arctan z(\tau) + 2n\pi, \qquad B(\tau) = B_0 \frac{e^{\tau}(1 + z_0^2)}{1 + e^{2\tau} z_0^2}.$$

Indeed $\phi' = 2z/(1 + z^2) = \sin \phi$ and $B'/B = 1 - 2z^2/(1 + z^2) = (1 - z^2)/(1 + z^2) = \cos \phi$. The positive denominator proves smoothness and $B > 0$ for finite time. The scalar ODE for ϕ has a globally Lipschitz right side; uniqueness follows by applying the integral inequality for its difference and iterating to the factorial bound. The linear equation $B' = B \cos \phi$ then has a unique solution by the same argument. Finally $\sin \phi = 2z/(1 + z^2)$ proves $B(\tau) = B_0 \sin \phi(\tau) / \sin \phi_0$ in exactly the nonstationary cases where that quotient is defined. Thus neither the constant-angle solutions nor their distinct exponential amplitudes have been omitted.

5.4 Source and verification scope

The primary formulas compared here are Lemma 1.2, (1.4), (1.6)–(1.8), and (3.3)–(3.6) of *Blowup for the Boussinesq Equations with Smooth Forcing*, `https://cims.nyu.edu/~tristanb/boussinesq.pdf`. The retrieved 76-page PDF is recorded by hash in `checks/boussinesq_retrieval.json`. The report's frequency schedule (9.1)–(9.5) matches the primary manuscript's (3.7)–(3.8), and its claimed force estimate (9.8) matches the statement of primary Theorem 7.3, (7.24). That comparison verifies attribution and transcription; it does not replace an independent proof of that theorem's quantitative induction. The identities proved in this section are independent of that induction. A direct symbolic replay is provided in `scripts/independent_boussinesq.py`.

6 Physical coordinates and exact finite viscous realization

We use the orientation

$$x = (x_1, x_2) \in \mathbb{R}^2, \quad e_2 = (0,1), \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \nabla^\perp = J\nabla, \quad \operatorname{curl} u = \partial_1 u_2 - \partial_2 u_1.$$

The physical equations throughout this note are

$$\theta_t + u \cdot \nabla\theta = \kappa\Delta\theta + f_\theta, \tag{79}$$

$$u_t + (u\cdot\nabla)u + \nabla p = \varepsilon\Delta u + \theta e_2 + f_u, \qquad \operatorname{div} u = 0. \tag{80}$$

Here $\kappa, \varepsilon \geq 0$ are respectively thermal diffusivity and physical viscosity. In particular ε is not the source's time-scaling parameter ν and is not its schedule error tolerance. The datum is retained exactly:

$$A_0 \geq 1, \quad \lambda_0 = \lceil \sqrt{A_0}/2 \rceil, \qquad \theta_{\text{in}}(x) = -\frac{A_0}{\lambda_0}\sin(\lambda_0 x_2)\chi_0(x), \qquad u_{\text{in}}(x) = 0.$$

The original radial cutoff $\chi_0 \in C_c^\infty(B_{R_0})$, $R_0 > 2$, equals one on B_2 and lies in $[0,1]$. Define

$$\mu = \lambda_0, \qquad \nu = \sqrt{A_0}, \qquad (X,T) = (\mu x, \nu t).$$

These are explicit maps between the source coordinates and the physical coordinates, rather than replacements of A_0 or λ_0 by one. For source fields decorated with a tilde their pullback is

$$\theta(x,t) = \frac{\nu^2}{\mu}\widetilde{\theta}(\mu x, \nu t), \qquad u(x,t) = \frac{\nu}{\mu}\widetilde{u}(\mu x, \nu t),$$

$$p(x,t) = \frac{\nu^2}{\mu^2}\widetilde{p}(\mu x, \nu t), \qquad \psi(x,t) = \frac{\nu}{\mu^2}\widetilde{\psi}(\mu x, \nu t). \tag{81}$$

Their forces pull back with factors ν^3/μ and ν^2/μ, respectively. Their source diffusion coefficients must be

$$\widetilde{\kappa} = \frac{\mu^2}{\nu}\kappa, \qquad \widetilde{\varepsilon} = \frac{\mu^2}{\nu}\varepsilon. \tag{82}$$

Indeed each term on the left of the scalar equation acquires ν^3/μ, whereas $\kappa\Delta_x\theta$ acquires $\kappa\nu^2\mu$. Each momentum and buoyancy term acquires ν^2/μ, whereas $\varepsilon\Delta_x u$ acquires $\varepsilon\nu\mu$. Division gives exactly (82). Also $\nabla_x^\perp\psi = u$, $\Delta_x\psi = \nu\widetilde{\omega}$, and $\operatorname{div}_x u = \nu \operatorname{div}_X \widetilde{u}$. The inverse map substitutes $(x,t) = (X/\mu, T/\nu)$ and reciprocates the displayed factors, proving a bijection for corresponding forces, coefficients, and initial values. The cutoff is $\widetilde{\chi}_0(X) = \chi_0(X/\mu)$, so the pulled-back initial temperature is exactly the displayed θ_{in}. These computations extend source Lemma 1.2, equations (1.6)–(1.7), pp. 6–7, by retaining both diffusion coefficients.

6.1 The released profile, the initial interpolation, and the base

Fix $0 < s_0 < 1$ and an even smooth cutoff χ supported in $(-\pi, \pi)$, equal to one on $[-s_0, s_0]$. On $[-\pi, \pi]$ put

$$F(s) = \sin s + \chi(s)(s - \sin s)$$

and extend periodically. Let

$$P(s) = \int_0^s F(r)\,dr - \frac{1}{2\pi}\int_{-\pi}^{\pi}\int_0^y F(r)\,dr\,dy.$$

Oddness and zero period integral of F show that P is even, periodic, has zero mean, and $P' = F$. Near zero, $F(s) = s$ and $P(s) = P(0) + s^2/2$. No sinusoidal identity is substituted for this profile.

Take a smooth nondecreasing step η that is zero for $t \le 0$ and one for $t \ge 1$. Every derivative of positive order vanishes at its endpoints: smoothness and its constant extension prove this.

Choose $R_H > R_0 + 1$ and radial $H \in C_c^\infty(B_{R_H})$ equal to $|x|^2/2$ on B_{R_0+1}. Let $\alpha = 0$ on $[0, 1/\nu]$ and $e_0 = R_\alpha e_2$. Define

$$\begin{aligned} \theta_0 &= -\frac{A_0}{\mu} F(\mu e_0 \cdot x)\chi_0(x), & \theta_{\mathrm{ref}} &= -\frac{A_0}{\mu} F(\mu x_2)\chi_0(x), \\ \theta_b &= \theta_0 + (1 - \eta(\nu t))(\theta_{\mathrm{in}} - \theta_{\mathrm{ref}}), & \psi_b &= \dot\alpha H, \qquad u_b = \dot\alpha J\nabla H. \end{aligned} \tag{83}$$

For $t \le 1/\nu$, $u_b = 0$ and $\partial_t \theta_b = \nu\eta'(\nu t)(\theta_{\mathrm{ref}} - \theta_{\mathrm{in}})$. For later times, $u_b = \dot\alpha J x$ on the support of θ_0. Since $\dot e_0 = \dot\alpha J e_0$,

$$(\partial_t + \dot\alpha J x \cdot \nabla)(\mu e_0 \cdot x) = \mu\dot\alpha(J e_0 \cdot x + e_0 \cdot J x) = 0.$$

Radiality gives $Jx \cdot \nabla\chi_0 = 0$. Therefore the exact zero-pressure base forces are

$$\begin{aligned} f^0_{\theta,b} &= \nu\eta'(\nu t)(\theta_{\mathrm{ref}} - \theta_{\mathrm{in}}), \\ f^0_{u,b} &= \ddot\alpha J\nabla H + \dot\alpha^2 (J\nabla H \cdot \nabla)(J\nabla H) - \theta_b e_2. \end{aligned} \tag{84}$$

On $|\mu e_0 \cdot x| < s_0$, $|x| < 2$, and $t \ge 1/\nu$, the base has the exact values

$$\theta_b = -A_0 e_0 \cdot x, \quad u_b = \dot\alpha J x, \quad G_b = -A_0 e_0, \quad D_b = \dot\alpha J.$$

This is source (4.9)–(4.10), pp. 38–39, in the original coordinates.

6.2 A compact realization of the complete first controlled stage

Use the first-stage coefficients constructed and proved below, with physical frequency $K_1 = \mu\lambda_1$, phase $\zeta = R_\alpha e(s_*)$, $|\zeta| = 1$, and signed amplitudes Θ, Ω satisfying

$$\dot\Theta = -\frac{J\zeta \cdot G_b}{K_1}\Omega, \qquad \dot\Omega = K_1 \zeta_1 \Theta. \tag{85}$$

Here $e(\phi) = (\sin\phi, \cos\phi)$ and $J\zeta \cdot G_b = -A_0 \sin s_*$, a rotationally invariant scalar. Choose $C_\ell \ge 4$, set $\ell = s_0/(C_\ell \mu)$, and take radial $f \in C_c^\infty(B_1)$ equal to one on $B_{1/2}$. Put

$$g(x) = f(x/\ell), \quad s = K_1 \zeta \cdot x, \quad w = \eta(\Gamma(t - 1/\nu)), \quad \Gamma = \nu \sin s_*, \quad T = w\Theta, \quad Q = \frac{w\Omega}{K_1^2}.$$

Although g is written in laboratory coordinates, it is exactly the paper's material envelope: $M = R_\alpha$, so $f(M^{-1}x/\ell) = f(x/\ell)$ by radiality. Set

$$\begin{aligned} V &= T F(s) g, & \Psi &= Q P(s) g, \\ v &= Q\{K_1 J\zeta F(s) g + P(s) J\nabla g\}, & \theta &= \theta_b + V, \quad u = u_b + v, \quad p = 0. \end{aligned} \tag{86}$$

The new fields are extended by zero before $t = 1/\nu$. They are smooth across that time because w is flat there. Their support is in $B_\ell \subset B_{s_0/\mu} \cap B_2$, where the base is affine for every angle. The phase and envelope satisfy

$$(\partial_t + D_b x \cdot \nabla)s = 0, \qquad (\partial_t + D_b x \cdot \nabla) g = 0.$$

For the phase this follows from $\dot\zeta = \dot\alpha J\zeta = -D_b^T \zeta$; for the envelope it follows from radiality.

Write $\partial_\perp g = J\zeta \cdot \nabla g$. Direct expansion gives the following complete added forces at zero diffusion:

$$E^0_\theta = \dot w \Theta F g + Q P (J\nabla g \cdot G_b) + K_1 Q T (F^2 - P F') g \partial_\perp g, \tag{87}$$

$$E^0_u = 2 D_b v + \dot Q\{K_1 J\zeta F g + P J\nabla g\} + (v \cdot \nabla) v - T F g e_2. \tag{88}$$

Every function of s in these equations is evaluated at the phase in (86). The vector derivative in the last formula can, equivalently, be evaluated by the explicitly specified matrix

$$\nabla v = QJ\left[K_1^2\zeta\otimes\zeta F'g + K_1F(\zeta\otimes\nabla g+\nabla g\otimes\zeta)+P\nabla^2 g\right], \qquad (v\cdot\nabla)v = (\nabla v)v.$$

Thus (88) is a direct component formula, without solving a pressure equation or an inverse-curl problem.

Here is a proof of both residual identities. Since $J\zeta\cdot\zeta = 0$,

$$v\cdot\nabla V = K_1QT(F^2 - PF')g(J\zeta\cdot\nabla g).$$

Also $v\cdot G_b = QK_1(J\zeta\cdot G_b)Fg + QP(J\nabla g\cdot G_b)$ and the material derivative of V is $\dot{T}Fg$. Equation (85) implies $\dot{T}+QK_1J\zeta\cdot G_b = \dot{w}\Theta$, proving (87). For any trace-free matrix D, direct multiplication proves $DJ + JD^T = 0$. Consequently

$$(\partial_t + Dx\cdot\nabla)J\nabla\Psi = DJ\nabla\Psi + J\nabla[(\partial_t + Dx\cdot\nabla)\Psi].$$

The material derivative of Ψ is $\dot{Q}Pg$, and $(v\cdot\nabla)u_b = D_bv$ on the new support. Adding these two terms, the self-advection, and the negative buoyancy increment proves (88). Away from the new support every increment vanishes, so this verification is global.

For every prescribed $\kappa,\varepsilon\geq 0$ the *same* controlled fields now solve (79)–(80) with the explicit forces

$$f_\theta^{\kappa,\varepsilon} = f_{\theta,b}^0 + E_\theta^0 - \kappa\Delta\theta_b - \kappa\Delta V, \tag{89}$$

$$f_u^{\kappa,\varepsilon} = f_{u,b}^0 + E_u^0 - \varepsilon\Delta u_b - \varepsilon\Delta v. \tag{90}$$

Indeed substituting these right-hand sides cancels the displayed diffusion terms exactly and leaves the two residual identities already proved. This proves a concrete finite controlled viscous stage, for arbitrary physical coefficients, retaining the paper's actual profile and motion. Its costs are calculated next; no claim that these are small or uniform over infinitely many stages is part of this construction.

All fields and forces are supported in the fixed spatial ball B_{R_H}. They are smooth on the complete closed finite interval, including interpolation, activation, growth, transition, steering, and any specified finite holding duration. The pressure $p=0$ satisfies $p(0,t)=0$. Since $\psi_b+\Psi$ is compactly supported, $u = J\nabla(\psi_b+\Psi)$ is divergence free and has finite spatial energy. It also has exactly the Biot–Savart relation $u = K\omega$: if $N = (2\pi)^{-1}\log|x|$, integration by parts against the compact streamfunction gives $N*\Delta(\psi_b+\Psi) = (\Delta N)*(\psi_b+\Psi) = \psi_b+\Psi$; applying $J\nabla$ proves the assertion in distributions and hence for these smooth fields. Odd F, even P, and radial cutoffs give odd θ, u, f_θ, f_u and even ω. The finite-slab forces can also be specified with compact temporal support. Extend the state by its stationary initial values to the left and its stationary held values to the right; flatness of the interpolation and the selected pulse proves smoothness of these extensions. For any fixed $\tau_{\mathrm{ext}}>0$, multiply the resulting force formulas by

$$\eta(1+t/\tau_{\mathrm{ext}})\eta(1+(T_f-t)/\tau_{\mathrm{ext}}).$$

Here T_f denotes the finite final time of the prescribed hold. This multiplier is one on $[0,T_f]$ and zero outside $(-\tau_{\mathrm{ext}}, T_f+\tau_{\mathrm{ext}})$. Thus the same solution on its specified slab has forces in $C_c^\infty(\mathbb{R}^2\times\mathbb{R})$, with explicitly specified spatial and temporal support.

On the smaller open set

$$|x|<\ell/2, \qquad |K_1\zeta\cdot x|<s_0,$$

the exact fields are

$$\theta = (-A_0e_0 + wK_1\Theta\zeta)\cdot x, \quad u = (\dot{\alpha}J + w\Omega J\zeta\otimes\zeta)x. \tag{91}$$

Their Laplacians vanish there, and so do the additional diffusive forces *and all their spatial derivatives* there. Therefore the evolving parent background for a next layer inside this open set is exactly the coupled gradient and shear in (91), with the actual time-dependent Θ,Ω,α,w. It has not been replaced by prescribed unchanging D,G. The finite-stage force supplies the compensating diffusion outside this affine region.

6.3 Every spatial diffusion cost, including the cutoff terms

Put $k = K_1\zeta$, so $|k| = K_1$. Direct differentiation gives

$$\Delta V = T\{K_1^2 F'' g + 2F' k \cdot \nabla g + F\Delta g\}, \tag{92}$$

$$\begin{aligned}\Delta v = QJ\{&K_1^2 k F'' g + K_1^2 F' \nabla g + 2F' k(k \cdot \nabla g) \\ &+ 2F\nabla(k \cdot \nabla g) + Fk\Delta g + P\nabla\Delta g\}.\end{aligned} \tag{93}$$

For the second formula first differentiate $\Delta\Psi = Q\{K_1^2 F' g + 2Fk \cdot \nabla g + P\Delta g\}$, and apply J. These are exact differential expressions on all of $\mathbb{R}^2$ and retain the phase, mixed, and envelope terms. For example, their zeroth-order norms satisfy

$$\begin{aligned}\|\Delta V\|_\infty &\le |T|\{K_1^2\|F''\|_\infty\|g\|_\infty + 2K_1\|F'\|_\infty\|\nabla g\|_\infty + \|F\|_\infty\|\Delta g\|_\infty\}, \\ \|\Delta v\|_\infty &\le |Q|\{K_1^3\|F''\|_\infty\|g\|_\infty + 3K_1^2\|F'\|_\infty\|\nabla g\|_\infty \\ &\quad + 2K_1\|F\|_\infty\|\nabla^2 g\|_\infty + K_1\|F\|_\infty\|\Delta g\|_\infty + \|P\|_\infty\|\nabla\Delta g\|_\infty\}.\end{aligned} \tag{94}$$

The Hessian norm is its operator norm, and vector norms are Euclidean. Thus no cutoff diffusion term has been absorbed into a frequency-only estimate. Here $\|\partial^\beta g\|_\infty = \ell^{-|\beta|}\|\partial^\beta f\|_\infty$, including every original s_0, C_ℓ, μ through $\ell = s_0/(C_\ell\mu)$.

For an arbitrary nonnegative integer m put

$$E_m = \max_{|\beta|=m} \|\partial^\beta g\|_\infty, \quad B_m(W) = \sum_{j=0}^{m} \binom{m}{j} K_1^j \|W^{(j)}\|_\infty E_{m-j}.$$

Leibniz' rule, $\partial^\beta W(k \cdot x) = k^\beta W^{(|\beta|)}(k \cdot x)$, and $\sum_{|\beta|=j,\, \beta\le\gamma} \binom{\gamma}{\beta} = \binom{m}{j}$ for $|\gamma| = m$ prove

$$\begin{aligned}\max_{|\gamma|=m} \|\partial^\gamma \Delta V\|_\infty &\le 2|T|B_{m+2}(F), \\ \max_{|\gamma|=m} \|\partial^\gamma \Delta v\|_\infty &\le 2\sqrt{2}|Q|B_{m+3}(P).\end{aligned} \tag{95}$$

The two in each line counts the two coordinate Laplacian terms; $\sqrt{2}$ counts the two components of $J\nabla$.

Let $S = |\Theta^{\mathrm{seed}}|$, let L be the prescribed growth gain, let Λ_1 be the paper's pulse compression, and let $H_t \ge 0$ be the specified physical holding duration. The proved selected stage below has durations L/Γ, $1/\Gamma$, $(1 + \Lambda_1^{-1})/\Gamma$, and H_t, respectively. On its growth and transition $|\Theta| = S\exp(\Gamma(t - t_{\mathrm{in}}))$ and $|\Omega| = (K_1/\nu)|\Theta|$. On its selected steering,

$$|\Theta| \le Se^{L+2+\Lambda_1^{-1}}, \qquad |\Omega| \le \frac{K_1}{\nu}|\Theta|.$$

On its hold $\Omega = 0$ and Θ is constant. Consequently the explicit costs

$$\begin{aligned}I_\Theta &= S\frac{e^{L+1} - 1}{\Gamma} + Se^{L+2+\Lambda_1^{-1}}\left(\frac{1 + \Lambda_1^{-1}}{\Gamma} + H_t\right), \\ I_\Omega &= \frac{K_1}{\nu}S\left\{\frac{e^{L+1} - 1}{\Gamma} + e^{L+2+\Lambda_1^{-1}}\frac{1 + \Lambda_1^{-1}}{\Gamma}\right\}\end{aligned} \tag{96}$$

satisfy $\int |T|\, dt \le I_\Theta$ and $\int |w\Omega|\, dt \le I_\Omega$ over the complete physical stage. This follows by integrating the exponential exactly, using $0 \le w \le 1$, and then using the stated bounds on the remaining intervals. Combining with (95) gives

$$\begin{aligned}\int \max_{|\gamma|=m} \|\partial^\gamma(-\kappa\Delta V)\|_\infty dt &\le 2\kappa B_{m+2}(F) I_\Theta, \\ \int \max_{|\gamma|=m} \|\partial^\gamma(-\varepsilon\Delta v)\|_\infty dt &\le \frac{2\sqrt{2}\varepsilon}{K_1^2} B_{m+3}(P) I_\Omega.\end{aligned} \tag{97}$$

In particular the holding scalar cost is proportional to H_t; there is no assertion that a forced hold has zero scalar diffusion cost.

The base costs are equally explicit. Write

$$C_m = \max_{|\beta|=m} \|\partial^\beta \chi_0\|_\infty, \qquad \mathcal{B}_m(W) = \sum_{j=0}^{m} \binom{m}{j} \mu^j \|W^{(j)}\|_\infty C_{m-j}.$$

On the initial interpolation $\theta_b = (1-\eta(\nu t))\theta_{\rm in} + \eta(\nu t)\theta_{\rm ref}$; afterwards it is a rotated F-profile with the same radial cutoff. Thus, writing $T_f = 1/\nu + (L+2+\Lambda_1^{-1})/\Gamma + H_t$,

$$\begin{aligned}
&\int_0^{T_f} \max_{|\gamma|=m} \|\partial^\gamma(\kappa\Delta\theta_b)\|_\infty dt \le \frac{2\kappa A_0}{\mu}\left\{\frac{\max(\mathcal{B}_{m+2}(\sin), \mathcal{B}_{m+2}(F))}{\nu} + (T_f - 1/\nu)\mathcal{B}_{m+2}(F)\right\},\\
&\int_0^{T_f} \max_{|\gamma|=m} \|\partial^\gamma(\varepsilon\Delta u_b)\|_\infty dt \le \varepsilon \max_{|\gamma|=m} \|\partial^\gamma J\nabla\Delta H\|_\infty \int_0^{T_f} |\dot\alpha|\, dt.
\end{aligned} \tag{98}$$

The last angle variation is exactly the finite integral of the explicit control and has the following fixed upper bound. For $z(\tau)$ in the stage proof and its selected pulse multiplier m_*, $|Z| \le 1/4$, and

$$\int |\dot\alpha| dt \le \frac{4}{\sqrt{15}} \int |\dot Z| dt \le \frac{4\sin s_*}{\sqrt{15}}\left(1 + m_*\Lambda_1 \int_0^1 |\eta_2'(y)| dy\right).$$

This uses $\alpha = s_* - \arcsin Z$, $\int_0^1 \eta' = 1$, and the change of variables $y = \Lambda_1(\tau - 1)$ on the pulse.

For completeness all mixed derivatives also have an exact finite replay formula; spatial bounds alone are not being called bounds for time derivatives. For $A = T, W = F$, or $A = Q, W = P$, and any multi-index γ, Leibniz gives

$$\partial_t^b \partial_x^\gamma [AW(k\cdot x)g] = \sum_{\beta\le\gamma} \binom{\gamma}{\beta} (\partial^\beta g) \sum_{a+c+d=b} \frac{b!}{a!c!d!} A^{(a)} (k^{\gamma-\beta})^{(c)} \partial_t^d W^{(|\gamma-\beta|)}(k\cdot x). \tag{99}$$

For $d > 0$ the remaining derivative is the finite sum

$$\partial_t^d W^{(n)}(k\cdot x) = \sum_{\substack{m_1,\dots,m_d\ge 0\\ \sum j m_j = d}} \frac{d!}{\prod_{j=1}^d m_j!(j!)^{m_j}} W^{(n+\sum m_j)}(k\cdot x) \prod_{j=1}^d (k^{(j)}\cdot x)^{m_j}.$$

For $d = 0$ it is $W^{(n)}(k\cdot x)$. This identity follows by induction: differentiation either increases the derivative order on W and creates $k'\cdot x$, or differentiates one of the displayed factors; grouping the resulting multiplicities gives exactly the factorial coefficients at order $d+1$. On the support use $|k^{(j)}\cdot x| \le \ell |k^{(j)}|$ and $|\partial^\beta g| \le E_{|\beta|}$ to obtain an explicit upper bound from each term. Sum the two coordinate second derivatives in (99) for ΔV and then differentiate once more spatially for Δv. The same formula with $k = \mu e_0$, $g = \chi_0$ handles θ_0; the initial interpolation additionally uses the exact derivatives $\partial_t^a \eta(\nu t) = \nu^a \eta^{(a)}(\nu t)$. For u_b, mixed diffusion derivatives are exactly $\varepsilon\alpha^{(b+1)}\partial^\gamma J\nabla\Delta H$. All coefficient derivatives are defined by the explicit smooth stage equations on a compact interval and the finitely many fixed profile derivatives, so these formulas quantify every specified mixed order without an asymptotic omission.

7 The actual first controlled stage in the original units

The source for this section is Alpöge–Buckmaster, *Blowup for the Boussinesq equations with smooth forcing*, https://cims.nyu.edu/~tristanb/boussinesq.pdf, released PDF, 76 pages. The source locations used here are Lemma 1.2 and (1.6), pp. 6–7; (3.1)–(3.12), pp. 8–10; the complete proofs of Lemmas 3.7–3.8 and (3.24)–(3.34), pp. 15–18; (3.58) and the following second-stage calculation, p. 27; and the first holding integral on p. 36. The proofs below are given in full. The source's unit-scale convention is not imposed on the fields in this section.

7.1 Parameters, coordinates, and the original coupled equations

Let $x = (x_1, x_2) \in \mathbb{R}^2$, $e_2 = (0, 1)$, and

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_\alpha = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}, \quad e(s) = (\sin s, \cos s).$$

In particular $R_\alpha e(s) = e(s-\alpha)$ and $\operatorname{curl} u = \partial_1 u_2 - \partial_2 u_1$. Retain $A_0 > 0$, $\lambda_0 \geq 1$, and the source's two scaling parameters

$$\mu = \lambda_0, \qquad \nu = \sqrt{A_0}, \qquad K_1 = \mu\lambda_1.$$

Here ν is the source's *time-scaling parameter*. Physical momentum viscosity below is denoted $\nu_{\text{phys}} \geq 0$; thermal diffusivity is $\kappa_{\text{phys}} \geq 0$. The first pulse compression is the source's Λ_1, which is different from the physical spatial frequency K_1.

Fix a smooth nondecreasing step $\eta : \mathbb{R} \to [0, 1]$ equal to zero on $(-\infty, 0]$ and one on $[1, \infty)$, and $h \in C_c^\infty((0, 1))$ with $h \geq 0$ and $\int_0^1 h = 1$. Write $H = \|h\|_\infty$. The source's choice is $h = \eta_2$, $c_p = 3$; the following calculation retains any

$$c_p > 1, \quad \Lambda_1 \geq 2c_p H, \quad 0 < s = s_* \leq \tfrac{1}{8}, \quad c_p\Lambda_1 H \sin s \leq \tfrac{1}{4}. \tag{100}$$

Since $\int h = 1$, $H \geq 1$ and $\Lambda_1 > 2$. Let $k_1 \geq 0$ be an integer, $\lambda_1 > 1$, and set

$$S = \frac{A_0}{\mu}\lambda_1^{-k_1-6}, \quad L = (k_1 + 5 + \tfrac{1}{8})\log\lambda_1 \geq 1, \quad P_* = Se^L = \frac{A_0}{\mu}\lambda_1^{-7/8}, \quad \Gamma = \nu\sin s, \quad t_0 = \nu^{-1}. \tag{101}$$

The condition $L \geq 1$ is ensured explicitly by $\lambda_1 \geq \exp((k_1 + 41/8)^{-1})$.

The first layer's complete preceding affine background, phase transport, and signed amplitude pair are

$$D_{<1} = \dot\alpha J, \qquad G_{<1} = -A_0 R_\alpha e_2, \quad \dot M_1 = D_{<1} M_1, \quad M_1(t_0) = I, \tag{102}$$

$$\zeta_1 = M_1^{-T} e(s), \quad \dot\Theta_1 = a_1\Omega_1, \qquad \dot\Omega_1 = b_1\Theta_1, \qquad a_1 = -\frac{J\zeta_1 \cdot G_{<1}}{K_1|\zeta_1|^2}, \quad b_1 = K_1\zeta_{1,1}. \tag{103}$$

The entry data are

$$\alpha(t_0) = 0, \quad \Theta_1(t_0) = -S, \quad \Omega_1(t_0) = -\frac{K_1}{\nu}S, \qquad w_1(t) = \eta(\Gamma(t - t_0)).$$

The activation factor multiplies the physical increment, not the homogeneous amplitude equation. It is flat at t_0 and equals one by $t_0 + \Gamma^{-1}$.

For this actual background, $M_1 = R_\alpha$: differentiation gives $\dot R_\alpha = \dot\alpha J R_\alpha$ and the initial values agree. Uniqueness follows, for example, by differentiating $R_\alpha^{-1} M_1$. Consequently

$$\zeta_1 = R_\alpha e(s), \quad |\zeta_1| = 1, \quad J\zeta_1 \cdot G_{<1} = -A_0\sin s, \quad a_1 = \frac{A_0 \sin s}{K_1}, \quad b_1 = K_1\sin(s - \alpha). \tag{104}$$

The determinant identity used here is explicit: $Je(s) = (-\cos s, \sin s)$ has scalar product $\sin s$ with e_2, and simultaneous rotation preserves that scalar product. In particular a_1 is constant because both the parent gradient and the new phase rotate; it has not been frozen independently of the evolving parent.

7.2 Exact growth and transition

For the first stage the feedback sums F_0 and F_1 in source (3.4) are empty and equal zero. Thus both the source's growth control $\dot\alpha = F_0$ and its transition control $(1 - \eta_b)F_0 + \eta_b F_1$ give $\alpha = 0$. Set

$$t_a = t_0 + \frac{L}{\Gamma}, \qquad t_1 = t_a + \Gamma^{-1}, \qquad t_b = t_1 + \Gamma^{-1}(1 + \Lambda_1^{-1}).$$

On $[t_0, t_1]$ the exact solution of the original pair is

$$P_1(t) := -\Theta_1(t) = Se^{\Gamma(t-t_0)}, \qquad \Omega_1(t) = -\frac{K_1}{\nu} P_1(t). \tag{105}$$

Indeed $a_1 K_1/\nu = \Gamma$ and $\Gamma K_1/\nu = K_1 \sin s = b_1$. This proves both differential equations and both initial data. Uniqueness follows from the linear integral equation: on a bounded interval with coefficient norm at most B, the n-fold iterated integral of the difference of two solutions is bounded by its supremum times $(B\ell)^n/n!$, which tends to zero. The same convergent integral series supplies existence on every finite interval. Since P_1 is strictly increasing, t_a is exactly the first crossing of P_*, and the transition contributes exactly one to $\log P_1$. The steering entry amplitude is therefore $P_{\text{ent}} = eP_*$.

7.3 The trial steering family and proof of its unique return

Use $m \in [0, c_p]$ for the pulse trial parameter, to distinguish it from the unchanged spatial parameter $\mu = \lambda_0$. With $\tau = \Gamma(t - t_1)$ and $\tau_b = 1 + \Lambda_1^{-1}$ set

$$z(\tau; m) = \begin{cases} 1 - \eta(\tau), & 0 \le \tau \le 1, \\ -m\Lambda_1 h(\Lambda_1(\tau - 1)), & 1 \le \tau \le \tau_b, \end{cases} \quad Z = \sin s\, z, \quad \alpha = s - \arcsin Z. \tag{106}$$

Extend $Z = \sin s$ before t_1 and $Z = 0$ after t_b. The two smooth pieces and extensions agree to every order, since the step is flat and h is supported in the interior. The source's selected pulse height in (3.12) is precisely $h_1 = m\Lambda_1 \sin s$; its allowed interval is $[0, c_p \Lambda_1 \sin s]$. The design gives

$$|Z| \le \tfrac{1}{4}, \quad \zeta_{1,1} = Z, \quad \zeta_{1,2} = \sqrt{1 - Z^2} \ge \frac{\sqrt{15}}{4}, \quad \dot\alpha = -\frac{\dot Z}{\zeta_{1,2}}, \quad 0 \le \alpha < \tfrac{1}{2}. \tag{107}$$

For the last bound, $Z \le \sin s$ implies $\alpha \ge 0$, and $\arcsin(1/4) \le (1/4)/\sqrt{1 - 1/16} < 1/3$ gives $\alpha \le 1/8 + 1/3 < 1/2$. These are the source's actual feedback and the original laboratory phase component; gravity remains e_2.

Define $\mathcal{P}$ first as the solution of the linear equation

$$\mathcal{P}_{\tau\tau} = z\mathcal{P}, \qquad \mathcal{P}(0) = \mathcal{P}_\tau(0) = 1. \tag{108}$$

The same linear integral series proves existence, uniqueness, and smoothness for every trial on the full finite interval. Moreover the exact inverse map to the original amplitude pair is

$$\Theta_1(t) = -P_{\text{ent}}\mathcal{P}(\tau), \qquad \Omega_1(t) = -\frac{K_1}{\nu} P_{\text{ent}} \mathcal{P}_\tau(\tau). \tag{109}$$

To check it, its component derivatives are exactly

$$-\Gamma P_{\text{ent}} \mathcal{P}_\tau = a_1 \Omega_1, \qquad -\frac{K_1}{\nu}\Gamma P_{\text{ent}} z\mathcal{P} = K_1 \sin s\, z\Theta_1 = b_1\Theta_1.$$

The data at t_1 match (105). Conversely the first amplitude equation with constant a_1 and $a_1 b_1 = \Gamma^2 z$ gives (108) for $-\Theta_1/P_{\text{ent}}$, with the stated initial derivative. Thus the auxiliary equation and original pair are explicitly inverse descriptions of the same dynamics, not a substituted oscillator.

On $0 \le \tau \le 1$, $z \ge 0$. Before any first zero of either $\mathcal{P}$ or $\mathcal{P}_\tau$, one has $\mathcal{P}_{\tau\tau} \ge 0$ and therefore $\mathcal{P}_\tau \ge 1$, $\mathcal{P} \ge 1 + \tau$. Continuity excludes that first zero and proves these inequalities through $\tau = 1$. Only now define $v = \mathcal{P}_\tau/\mathcal{P}$. It satisfies

$$v_\tau = 1 - \eta - v^2, \qquad v(0) = 1.$$

Since $(v-1)_\tau = -(v+1)(v-1) - \eta$, the integrating-factor formula gives $v \le 1$. Since $v > 0$ and $v_\tau \ge -v^2$, $(1/v)_\tau \le 1$, so

$$\frac{1}{1+\tau} \le v(\tau) \le 1, \quad \log 2 \le \int_0^1 v\, d\tau \le 1, \quad v_d := v(1) \in [1/2, 1], \quad 2 \le \mathcal{P}(1) \le e. \tag{110}$$

On the negative pulse set $y = \Lambda_1(\tau - 1)$ and $\chi(y) = \mathcal{P}(1 + y/\Lambda_1)/\mathcal{P}(1)$. There is no division by an unknown evolving amplitude in this definition. Its exact equation and data are

$$\chi_{yy} + \frac{mh(y)}{\Lambda_1}\chi = 0, \qquad \chi(0) = 1, \quad \chi_y(0) = v_d/\Lambda_1. \tag{111}$$

Before a possible first zero, χ is concave, so $\chi \le 1 + y/\Lambda_1 \le 2$. Since $mh/\Lambda_1 \le c_p H/\Lambda_1 \le 1/2$, one has $\chi_{yy} \ge -1$, hence

$$\chi(y) \ge 1 + \frac{v_d}{\Lambda_1} y - \frac{y^2}{2} \ge \frac{1}{2} \qquad (0 \le y \le 1).$$

The strict positive lower bound excludes a first zero. Thus every trial satisfies on the entire pulse

$$\tfrac{1}{2} \le \chi(y) \le 1 + y/\Lambda_1, \qquad \chi_y = \frac{v_d}{\Lambda_1} - \frac{m}{\Lambda_1}\int_0^y h(r)\chi(r)\, dr \ge -\frac{2c_p}{\Lambda_1}. \tag{112}$$

Consequently the original temperature is strictly negative for every trial, and the quotient is well defined. Writing $V(y;m) = v(1 + y/\Lambda_1; m) = \Lambda_1 \chi_y/\chi$ gives

$$V_y = -mh(y) - \frac{V^2}{\Lambda_1}, \quad V(0;m) = v_d, \qquad -4c_p \le V(y;m) \le v_d \le 1. \tag{113}$$

Here is the parameter dependence needed to select the return. In the first-order integral system for (χ, χ_y) the coefficient matrix depends affinely on m. On the compact parameter interval, its iterated integrals converge uniformly with factorial denominators. Differentiating r times in m introduces at most a degree-r polynomial in the iteration index, still dominated by that factorial. Thus this series and each parameter derivative converge uniformly. The lower bound $\chi \ge 1/2$ then gives smooth parameter dependence of V. Differentiating its exact equation and using an integrating factor gives

$$\partial_m V(1;m) = -\int_0^1 h(y) \exp\left(-\frac{2}{\Lambda_1}\int_y^1 V(r;m)\, dr\right) dy < 0. \tag{114}$$

The strict sign follows because $h \ge 0$ has mass one and the exponential is positive. Indeed the explicit bounds also give $e^{-2/\Lambda_1} \le -\partial_m V(1;m) \le e^{8c_p/\Lambda_1}$. At $m = 0$, direct solution of $V_y = -V^2/\Lambda_1$ gives $V(1;0) = v_d/(1 + v_d/\Lambda_1) > 0$. At $m = c_p$, integration gives $V(1;c_p) \le v_d - c_p < 0$. Continuity and strict monotonicity therefore give exactly one $m_* \in (0, c_p)$ with $V(1;m_*) = 0$. This is exactly the source's selected return: by (109) and positivity, it is equivalent to $\Omega_1(t_b) = 0$.

For this selected trial V is nonincreasing and ends at zero, so $0 \le V \le v_d$. Integration of (113) now gives the exact identity and bounds

$$m_* = v_d - \frac{1}{\Lambda_1}\int_0^1 V(y;m_*)^2\, dy, \qquad \frac{1}{2} - \frac{1}{4\Lambda_1} \le m_* \le 1. \tag{115}$$

For the lower bound, $v_d - v_d^2/\Lambda_1$ is increasing for $v_d \in [1/2, 1]$, because its derivative is $1 - 2v_d/\Lambda_1 > 0$.

For every trial and every steering time,

$$0 \le \log \mathcal{P}(\tau) \le 1 + \Lambda_1^{-1}.$$

On the ramp this follows from (110); on the pulse it follows from $\mathcal{P}(1) \geq 2$, $\chi \geq 1/2$, and $\log\chi \leq \log(1+\Lambda_1^{-1}) \leq \Lambda_1^{-1}$. For the selected trial the pulse logarithmic contribution is nonnegative, so the endpoint obeys

$$\log 2 \leq \log \mathcal{P}(\tau_b) \leq 1 + \Lambda_1^{-1}, \qquad 2eP_* \leq P_b := P_1(t_b) \leq e^{2+\Lambda_1^{-1}} P_*. \tag{116}$$

The total duration from entry to return is exactly

$$t_b - t_0 = \frac{L + 2 + \Lambda_1^{-1}}{\nu \sin s}. \tag{117}$$

Throughout the selected stage P_1 is nondecreasing, $S \leq P_1 \leq P_b$, and $-(K_1/\nu)P_1 \leq \Omega_1 \leq 0$. For every unselected trial the valid bound instead is $|\Omega_1| \leq 4c_p(K_1/\nu)e^{2+\Lambda_1^{-1}}P_*$; its sign need not stay negative.

Near t_b, $h = 0$, so $Z = 0$ and $\alpha = s$. The equation $\mathcal{P}_{\tau\tau} = 0$, together with $\mathcal{P}_\tau(\tau_b) = 0$, implies that $\mathcal{P}$ is constant on that whole neighborhood. Thus α, Θ_1, and Ω_1 match their stationary continuations to all orders. The endpoint state is

$$\alpha = s, \quad M_1 = R_s, \quad \zeta_1 = e_2, \quad w_1 = 1, \quad \Theta_1 = -P_b, \quad \Omega_1 = 0. \tag{118}$$

7.4 Control bounds and exact whole-stage integrals

For $j = 1, 2$ define the explicit finite profile quantities

$$B_j = \Gamma^j \sin s(\|\eta^{(j)}\|_\infty + c_p\Lambda_1^{j+1}\|h^{(j)}\|_\infty).$$

From (106), $|\partial_t^j Z| \leq B_j$. Differentiating the original control, including its denominator, gives

$$\begin{aligned} \dot\alpha &= -\dot Z(1 - Z^2)^{-1/2}, \qquad\qquad (119)\\ \ddot\alpha &= -\ddot Z(1 - Z^2)^{-1/2} - Z\dot Z^2(1 - Z^2)^{-3/2}, \end{aligned}$$

$$|\dot\alpha| \leq \frac{4}{\sqrt{15}}B_1, \qquad |\ddot\alpha| \leq \frac{4}{\sqrt{15}}B_2 + \frac{16}{15\sqrt{15}}B_1^2. \tag{120}$$

Higher derivatives exist because the displayed smooth composition has argument in $[-1/4, 1/4]$; no endpoint differentiability is omitted.

For exact coefficient costs use the invertible fixed map $Y = (\Theta_1, \nu\Omega_1/K_1)^T$. Its inverse is $(Y_1, K_1Y_2/\nu)^T$, and direct differentiation gives

$$\dot Y = \widehat{B}Y, \qquad \widehat{B} = \begin{pmatrix} 0 & \Gamma \\ \nu Z & 0 \end{pmatrix}.$$

Let $J_\eta = \int_0^1 (1 - \eta(r))\, dr \in [0, 1]$. On a hold of physical duration $T_h \geq 0$, the following are exact:

interval	duration	$\int \lvert\widehat{B}_{12}\rvert\, dt$	$\int \lvert\widehat{B}_{21}\rvert\, dt$
growth	L/Γ	L	L
transition	$1/\Gamma$	1	1
steering ramp	$1/\Gamma$	1	J_η
negative pulse	$1/(\Lambda_1\Gamma)$	$1/\Lambda_1$	m_*
holding	T_h	ΓT_h	0

For example, on the pulse substitution $y = \Lambda_1\Gamma(t - t_1 - \Gamma^{-1})$ gives $\int \nu|Z|\, dt = m_* \int_0^1 h = m_*$. Thus the two transition-and-steering costs are exactly $2 + \Lambda_1^{-1}$ and $1 + J_\eta + m_*$; their sum is at most $4 + c_p + \Lambda_1^{-1}$.

The activation and amplitude integrals needed by force estimates can also be retained without an asymptotic replacement. Set

$$J_a = \int_0^1 \eta'(r)e^r\,dr \in [1,e], \qquad J_g = \int_0^1 \eta(r)e^r\,dr = e - J_a.$$

For the selected stage including the hold, direct change of variable and integration by parts give

$$\int_{t_0}^{t_b} |\dot w_1\Theta_1|\,dt = SJ_a, \qquad \int_{t_0}^{t_b} |\dot w_1\Omega_1|\,dt = \frac{K_1}{\nu}SJ_a, \tag{121}$$

$$\begin{aligned} I_P := \int_{t_0}^{t_b+T_h} w_1P_1\,dt &= \frac{S}{\Gamma}(J_g + e^{L+1} - e) + P_bT_h \\ &\quad + \frac{eP_*}{\Gamma}\int_0^{1+\Lambda_1^{-1}} \mathcal{P}(\tau; m_*)\,d\tau, \end{aligned} \tag{122}$$

$$I_\Omega := \int_{t_0}^{t_b+T_h} w_1|\Omega_1|\,dt = \frac{P_b - SJ_a}{a_1}. \tag{123}$$

For the last identity, $P_1' = -a_1\Omega_1 = a_1|\Omega_1|$, $w_1(t_0) = 0$, and $w_1(t_b+T_h) = 1$, hence $\int w_1P_1' = P_b - \int \dot w_1P_1 = P_b - SJ_a$. In particular

$$I_P \le \frac{S}{\Gamma}(J_g + e^{L+1} - e) + \frac{eP_*}{\Gamma}(1+\Lambda_1^{-1})e^{1+\Lambda_1^{-1}} + P_bT_h.$$

These are finite specified integrals of the prescribed profiles and the uniquely selected exact linear solution.

7.5 The evolving background inherited by layer two

On the affine part of the first wave, the complete background for the next layer is

$$G_{<2} = -A_0R_\alpha e_2 + w_1K_1\Theta_1\zeta_1, \qquad D_{<2} = \dot\alpha J + w_1\Omega_1J\zeta_1\otimes\zeta_1. \tag{124}$$

These are source (3.3) with $A_1 = -K_1\Theta_1|\zeta_1|$ and the physical units retained. Their complete derivatives are

$$\begin{aligned} \dot G_{<2} &= \dot\alpha JG_{<2} + (K_1\dot w_1\Theta_1 + w_1A_0\sin s\,\Omega_1)\zeta_1, \end{aligned} \tag{125}$$

$$\begin{aligned} \dot D_{<2} &= \ddot\alpha J + (\dot w_1\Omega_1 + w_1K_1\zeta_{1,1}\Theta_1)J\zeta_1\otimes\zeta_1 \\ &\quad + w_1\Omega_1\dot\alpha(-\zeta_1\otimes\zeta_1 + J\zeta_1\otimes J\zeta_1). \end{aligned} \tag{126}$$

Indeed $\dot\zeta_1 = \dot\alpha J\zeta_1$, $J\dot\zeta_1 = -\dot\alpha\zeta_1$, and the product rule gives each displayed term. In particular, using $J\zeta_1\cdot G_{<2} = -A_0\sin s$ gives the exact identity

$$\dot G_{<2} + D_{<2}^TG_{<2} = K_1\dot w_1\Theta_1\zeta_1. \tag{127}$$

Thus the inherited gradient satisfies its true affine transport equation after activation, even while the first stage rotates it. The affine vorticity is $2\dot\alpha + w_1\Omega_1$ and its scalar source is $G_{<2,1} = A_0\sin\alpha + w_1K_1\Theta_1\zeta_{1,1}$; therefore its exact vorticity residual is

$$(2\dot\alpha + w_1\Omega_1)' - G_{<2,1} = 2\ddot\alpha - A_0\sin\alpha + \dot w_1\Omega_1. \tag{128}$$

At the selected endpoint define $A_1 = K_1P_b > 0$. Then

$$G_{<2}(t_b) = (A_0\sin s, -A_0\cos s - A_1), \quad D_{<2}(t_b) = 0, \quad \sigma_1^2 = |G_{<2}(t_b)| = \sqrt{A_0^2 + A_1^2 + 2A_0A_1\cos s}. \tag{129}$$

Consequently

$$2eA_0\lambda_1^{1/8} \le A_1 \le e^{2+\Lambda_1^{-1}} A_0\lambda_1^{1/8}, \qquad A_1 + A_0\cos s \le \sigma_1^2 \le A_1 + A_0.$$

The lower norm bound is its positive vertical component in absolute value; the upper bound is the triangle inequality, which also follows by squaring the displayed expression. No horizontal component of the parent gradient has been discarded.

On the first hold the actual feedback is $F_1 = 0$. The endpoint state (118) is stationary under the original equations: $b_1 = K_1\zeta_{1,1} = 0$ gives $\dot\Omega_1 = 0$, then $\dot\Theta_1 = a_1\Omega_1 = 0$, and $\dot\alpha = 0$ gives $\dot\zeta_1 = 0$. Thus (129) remains the actual inherited background throughout this hold. For an arbitrary perturbation of the weighted amplitude pair its propagator over length T_h is exactly

$$\exp\left(T_h\begin{pmatrix}0 & \Gamma\\ 0 & 0\end{pmatrix}\right) = \begin{pmatrix}1 & \Gamma T_h\\ 0 & 1\end{pmatrix}. \tag{130}$$

This follows by squaring the matrix and obtaining zero, and retains the nonzero first holding coefficient cost.

For clarity one can construct the next growth interval itself rather than prescribing an unspecified holding time. Retain $\lambda_2 = \lambda_1^{Q_2}$ with $Q_2 \ge 202$, an integer $k_2 \ge 0$, $L_2 = (k_2 + 41/8)\log\lambda_2$, and the source's

$$s_2 = L_2\nu/\sigma_1, \qquad K_2 = \mu\lambda_2, \qquad \Gamma_2 = \sigma_1\sin s_2.$$

An explicit sufficient finite choice ensuring $L_2 \ge 2$ and $0 < s_2 \le 1/8$ is $\log\lambda_1 \ge 4096(k_2 + 41/8)Q_2$. To prove this, put $B = (k_2 + 41/8)Q_2$ and $x = \log\lambda_1$. The gain implies $\sigma_1 \ge \sqrt{2e}\nu e^{x/16}$; the exponential series gives $e^{x/16} \ge x^2/512$. Hence $s_2 \le 512B/(\sqrt{2e}x) \le 1/8$. Also $L_2 = Bx \ge 4096B^2 > 2$.

During the second growth, $\alpha = s$, $\Omega_1 = 0$, and $D_{<2} = 0$, so $M_2 = I$, $\zeta_2 = e(s_2)$, and the original coefficient formulas yield exactly

$$a_2 = \frac{A_1\sin s_2 + A_0\sin(s+s_2)}{K_2}, \quad b_2 = K_2\sin s_2, \quad \gamma_2 = \sqrt{[A_1\sin s_2 + A_0\sin(s+s_2)]\sin s_2} > 0. \tag{131}$$

Both signs are positive because $0 < s + s_2 \le 1/4$. For its actual seed $S_2 = (A_0/\mu)\lambda_2^{-k_2-6}$ and entry vorticity $-\sqrt{b_2/a_2}\,S_2$, the exact solution is $P_2 = S_2e^{\gamma_2(t-t_b)}$, $\Omega_2 = -\sqrt{b_2/a_2}\,P_2$. Substitution verifies both equations; the first threshold time is $t_{a,2} = t_b + L_2/\gamma_2$. Its activation is the source's $\eta(\Gamma_2(t-t_b))$. To see that activation completes before threshold, note

$$R := \frac{\gamma_2^2}{\Gamma_2^2} = \frac{A_1 + A_0\cos s + A_0\sin s\cot s_2}{\sigma_1^2}, \qquad \cos s \le R \le 1 + \frac{\nu\sin s}{L_2\sigma_1}. \tag{132}$$

For the lower bound, discard the positive cotangent term, use $\sigma_1^2 \le A_1 + A_0$, and then $(A_1 + A_0\cos s)/(A_1 + A_0) \ge \cos s$. For the upper bound use $A_1 + A_0\cos s \le \sigma_1^2$ and $\cot s_2 \le 1/s_2$, the latter following from $(\sin r - r\cos r)' = r\sin r \ge 0$. Since $\sigma_1 \ge \nu$ and $L_2 \ge 2$, $R \le 1 + \sin s/2 < 2$, and $\Gamma_2L_2/\gamma_2 = L_2/\sqrt{R} > 1$. The exact first holding integral therefore is

$$H_1 = \Gamma\frac{L_2}{\gamma_2} = \sin s\,\frac{s_2}{\sin s_2}\frac{1}{\sqrt{R}}, \qquad \frac{\sin s}{\sqrt{1 + \nu\sin s/(L_2\sigma_1)}} \le H_1 \le \frac{\sin s}{\sqrt{\cos s}(1 - s_2^2/6)}. \tag{133}$$

Here $\sin r \le r$ follows by integrating $\cos r \le 1$; $\sin r \ge r - r^3/6$ follows by integrating $\cos r \ge 1 - r^2/2$, itself obtained from $1 - \cos r = \int_0^r \sin u\,du \le r^2/2$. Thus no small-angle replacement is used in the bounds. The second transition also has $F_2 = 0$, because its only shear coefficient is $\Omega_1 = 0$; it prolongs the first pair's stationarity by exactly Γ_2^{-1}. The source counts that transition in the first layer's tail, not in H_1.

7.6 Exact uncut forced fields and their retained diffusion cost

This subsection gives an explicit PDE realization of the just-proved finite stage on $\mathbb{R}^2$. Its fields have the spatial domain $\mathbb{R}^2$ and need not decay; compact spatial localization is a separate construction. Let F be any smooth odd 2π-periodic function. Define $Q_0(r) = \int_0^r F(v)\,dv$ and $\mathcal{Q}(r) = Q_0(r) - (2\pi)^{-1}\int_{-\pi}^{\pi} Q_0(v)\,dv$. Oddness gives zero period integral for F, so Q_0 is periodic; substitution $v = -u$ gives $Q_0(-r) = Q_0(r)$. Thus $\mathcal{Q}$ is even and mean-zero, and $\mathcal{Q}' = F$. Put $q(x,t) = K_1\zeta_1(t)\cdot x$ and

$$\theta = G_{<1}\cdot x + w_1\Theta_1 F(q),$$
$$u = \dot\alpha J x + \frac{w_1\Omega_1}{K_1} J\zeta_1 F(q), \tag{134}$$
$$p = \frac{\dot\alpha^2}{2}|x|^2 + \frac{1}{2}x_2(G_{<1}\cdot x) + \left(\frac{2w_1\Omega_1\dot\alpha}{K_1^2} + \frac{w_1\Theta_1\zeta_{1,2}}{K_1}\right)\mathcal{Q}(q). \tag{135}$$

An additive function of time can be subtracted if $p(0,t) = 0$ is desired. The exact inviscid forces are

$$f_\theta^0 = \dot w_1\Theta_1 F(q), \qquad f_u^0 = \left(\ddot\alpha - \frac{A_0}{2}\sin\alpha\right) Jx + \frac{\dot w_1\Omega_1}{K_1} J\zeta_1 F(q). \tag{136}$$

We verify every component. The background derivative satisfies $\dot G_{<1} = \dot\alpha J G_{<1}$, hence its scalar transport residual vanishes. The phase satisfies $(\partial_t + \dot\alpha Jx\cdot\nabla)q = 0$. The wave velocity has scalar product zero with ∇q, so it advects neither its own temperature nor its own velocity profile. Its advection of the background temperature equals $(w_1\Omega_1/K_1)(J\zeta_1\cdot G_{<1})F(q)$, which cancels $w_1\dot\Theta_1 F(q)$ by (103). The remaining scalar residual is exactly $\dot w_1\Theta_1 F(q)$.

For momentum, write $C = w_1\Omega_1/K_1$ and $v = CJ\zeta_1 F(q)$. Differentiating v and adding its transport of $\dot\alpha Jx$ gives, after subtracting the wave buoyancy,

$$\left[\frac{\dot w_1\Omega_1}{K_1}J\zeta_1 + \frac{w_1\dot\Omega_1}{K_1}J\zeta_1 - 2C\dot\alpha\zeta_1 - w_1\Theta_1 e_2\right]F(q).$$

The last three vector terms have zero scalar product with $J\zeta_1$: this is $w_1\dot\Omega_1/K_1 - w_1\Theta_1\zeta_{1,1} = 0$. Their scalar product with ζ_1 is $-2C\dot\alpha - w_1\Theta_1\zeta_{1,2}$. Since $(\zeta_1, J\zeta_1)$ is an orthonormal basis, the wave pressure in (135) cancels precisely these three terms, leaving the displayed activation force. The base momentum residual before pressure is $(\ddot\alpha J - \dot\alpha^2 I - e_2\otimes G_{<1})x$. The gradient of the first two pressure terms removes its symmetric part. Its antisymmetric part is $(\ddot\alpha - G_{<1,1}/2)Jx = (\ddot\alpha - A_0\sin\alpha/2)Jx$, as claimed. Finally $\operatorname{div} u = 0$ since $\operatorname{tr} J = 0$ and $J\zeta_1\cdot\zeta_1 = 0$.

For the retained-viscosity equations

$$\partial_t\theta + u\cdot\nabla\theta = \kappa_{\mathrm{phys}}\Delta\theta + f_\theta, \qquad \partial_t u + (u\cdot\nabla)u + \nabla p = \nu_{\mathrm{phys}}\Delta u + \theta e_2 + f_u, \qquad \nabla\cdot u = 0,$$

the very same state and controls solve them with the explicitly specified compensating forces

$$f_\theta = f_\theta^0 - \kappa_{\mathrm{phys}} w_1\Theta_1 K_1^2 F''(q),$$
$$f_u = f_u^0 - \nu_{\mathrm{phys}} w_1\Omega_1 K_1 J\zeta_1 F''(q). \tag{137}$$

Indeed the affine Laplacians vanish, while two spatial derivatives of the profile multiply by $K_1^2|\zeta_1|^2 = K_1^2$. This proves the asserted equations by substitution, without changing either ν or ν_{phys} and without assuming damping leaves the parent coefficients unchanged. The exact cumulative L_x^∞ norms of the two added forces through the entire stage and hold are

$$\kappa_{\mathrm{phys}} K_1^2\|F''\|_\infty I_P, \qquad \nu_{\mathrm{phys}} K_1\|F''\|_\infty I_\Omega. \tag{138}$$

Equality holds because $x \mapsto K_1\zeta_1\cdot x$ is surjective onto $\mathbb{R}$ and $|J\zeta_1| = 1$; this verifies the norm statement as well as an upper bound. The scalar holding contribution is exactly $\kappa_{\mathrm{phys}} K_1^2\|F''\|_\infty P_b T_h$;

the velocity holding contribution is zero. These finite costs are not asserted to be summable over the source's infinite frequency sequence.

The construction proves a finite controlled stage, its actual evolving affine background, its return, and its next growth holding interval. It does not establish an infinite viscous cascade or a smooth terminal-time viscous force. In particular, simply replacing the first temperature by an exponentially damped amplitude during holding would replace A_1 in (129) by a time-dependent quantity and would change both coefficients and the rate in (131). The compensating forces in (137) give an exact specified relation that retains the original coupled finite trajectory; their costs have been retained explicitly rather than claimed negligible.

8 Solving the next growth interval with a diffusing parent

This section continues an uncompensated equal-diffusion variant of the first stage through the next growth interval. It keeps the laboratory direction and the original coefficients. The exact amplitude system is realized below both on a forced affine parent and on an actual two-wave state with an explicitly computed interaction force. The latter force specifies the difference between the affine-gradient calculation and the full spatial equations.

8.1 The inherited gradient and the original second pair

Retain the preceding section's $A_0, \mu, \nu, K_1, s, \lambda_1$, and let $\nu_{\mathrm{phys}} = \kappa_{\mathrm{phys}} = d > 0$. For a sine first wave on its rotating affine base, the exact amplitude equations with retained diffusion are

$$\dot{\Theta}_{1,d} = a_1\Omega_{1,d} - dK_1^2\Theta_{1,d}, \qquad \dot{\Omega}_{1,d} = b_1\Theta_{1,d} - dK_1^2\Omega_{1,d}.$$

If (Θ_1, Ω_1) denotes the first inviscid pair of the preceding section on its specified control and times, direct differentiation proves

$$(\Theta_{1,d}, \Omega_{1,d})(t) = e^{-dK_1^2(t-t_0)}(\Theta_1, \Omega_1)(t).$$

Both the forward map and its inverse multiplication by the positive exponential satisfy the indicated systems, so they are a bijection of solution spaces. In particular the selected vorticity still returns to zero at the specified t_b. This uses the inviscid times and does not assert the damped temperature crosses the same growth threshold. Its exact log-gain from the original seed is

$$\log(P_b/S) - \frac{dK_1^2}{\nu \sin s}(L + 2 + \Lambda_1^{-1}).$$

Set $P_{d,b} = e^{-dK_1^2(t_b-t_0)}P_b > 0$, $A_{1,d} = K_1 P_{d,b}$, and $r = t - t_b \geq 0$. During the fixed first holding control $\alpha = s$ the exact first pair is

$$\Theta_{1,d}(r) = -P_{d,b}e^{-qr}, \quad \Omega_{1,d} = 0, \quad q = dK_1^2, \quad G_d(r) = (A_0 \sin s, -A_0 \cos s - A_{1,d}e^{-qr}).$$

The gradient therefore changes at the rate $G_d'(r) = (0, qA_{1,d}e^{-qr})$. Let the new laboratory phase be $\zeta_2 = e(s_2)$ with $0 < s_2 \leq 1/8$, and let $K_2 = \mu\lambda_2 > 0$. The actual first hold has zero velocity gradient, so $\zeta_2' = 0$. The second affine-wave coefficients are exactly

$$a_{2,d}(r) = -\frac{J\zeta_2 \cdot G_d(r)}{K_2} = \frac{A_{1,d}e^{-qr}\sin s_2 + A_0 \sin(s + s_2)}{K_2}, \quad b_2 = K_2 \sin s_2, \quad \delta_2 = dK_2^2. \tag{139}$$

Their signs are positive because $0 < s + s_2 \leq 1/4$. The exact second sine amplitude system is

$$\Theta_2' = a_{2,d}(r)\Omega_2 - \delta_2\Theta_2, \qquad \Omega_2' = b_2\Theta_2 - \delta_2\Omega_2. \tag{140}$$

In particular the changing inherited gradient has not been replaced by its endpoint value. An affine realization alone would require the scalar force $G_d'(r) \cdot x$: the affine Laplacian is zero, so that force cannot be inferred to vanish from the old sine-wave diffusion.

8.2 Exact operator change and convergent fundamental solutions

Define

$$c_0 = A_0 \sin(s+s_2)\sin s_2 > 0, \qquad c_1 = A_{1,d}\sin^2 s_2 > 0, \qquad W(r) = e^{\delta_2 r}\Omega_2(r).$$

Because b_2 is constant, differentiating the second original equation gives the exact system equivalence

$$W'' = (c_0 + c_1 e^{-qr})W, \qquad \Omega_2 = e^{-\delta_2 r}W, \qquad \Theta_2 = \frac{e^{-\delta_2 r}}{b_2}W'. \tag{141}$$

Conversely these last two formulas give $\Omega_2' = b_2\Theta_2 - \delta_2\Omega_2$ and $\Theta_2' = e^{-\delta_2 r}W''/b_2 - \delta_2\Theta_2 = a_{2,d}\Omega_2 - \delta_2\Theta_2$, since $b_2 a_{2,d} = c_0 + c_1 e^{-qr}$. Thus the initial-data map is the invertible map $(\Theta_2(0), \Omega_2(0)) \mapsto (W(0), W'(0)) = (\Omega_2(0), b_2\Theta_2(0))$.

Put

$$x(r) = x_0 e^{-qr/2}, \quad x_0 = \frac{2\sqrt{c_1}}{q}, \quad \vartheta = \frac{2\sqrt{c_0}}{q} > 0, \qquad W(r) = \mathcal{W}(x(r)).$$

Here $x(r)$ is a scalar change of independent variable and is unrelated to either physical spatial coordinate x_1, x_2, which remain fixed in the PDE below. Since $x' = -qx/2$ and $x'' = q^2x/4$, one has

$$W'' - (c_0 + c_1 e^{-qr})W = \frac{q^2}{4}\left[x^2\mathcal{W}_{xx} + x\mathcal{W}_x - (x^2+\vartheta^2)\mathcal{W}\right].$$

This proves the operator transformation including its nonzero factor. Rather than invoking a special-function name, define on $x > 0$

$$f(x) = x^{\vartheta}\sum_{n=0}^{\infty} a_n x^{2n}, \quad a_0 = 1, \quad a_n = \frac{a_{n-1}}{4n(n+\vartheta)}, \qquad g(x) = f(x)\int_{x_0}^{x}\frac{d\xi}{\xi f(\xi)^2}. \tag{142}$$

For every compact interval in $x > 0$, the consecutive-term absolute ratio $x^2/[4n(n+\vartheta)]$ tends uniformly to zero. Every fixed derivative multiplies these terms by a polynomial in n and a bounded power of x^{-1} on that compact interval; its series therefore converges uniformly as well. Termwise substitution is valid. The coefficient of $x^{\vartheta+2n}$ in $x^2f'' + xf' - (x^2+\vartheta^2)f$ is $4n(n+\vartheta)a_n - a_{n-1} = 0$ for $n \geq 1$; the $n = 0$ coefficient is also zero. All coefficients are positive, so $f > 0$. The defining integral for g consequently exists on every compact subinterval of $x > 0$. If $I' = 1/(xf^2)$, differentiation yields

$$f(fI)' - f'(fI) = f^2 I' = 1/x.$$

Also $I''/I' = -1/x - 2f'/f$; substituting $g = fI$ into $g'' + g'/x - (1+\vartheta^2/x^2)g$ leaves $fI'' + (2f' + f/x)I' = 0$. Thus g solves the same equation and its Wronskian with f is the nonvanishing function $1/x$. For completeness this proves completeness of the two solutions: at any $x > 0$ their value-and-derivative matrix is invertible, so one linear combination matches arbitrary Cauchy data; linear uniqueness then identifies that combination with any solution on overlapping compact subintervals of $(0, \infty)$.

For the original data let $W_0 = \Omega_2(0)$ and $W_1 = b_2\Theta_2(0)$. The exact solution is

$$W(r) = c_f f(x(r)) + c_g g(x(r)), \quad c_f = \frac{W_0}{f(x_0)}, \quad c_g = -\frac{2f(x_0)}{q}\left(W_1 + \frac{qx_0}{2}c_f f'(x_0)\right). \tag{143}$$

Indeed $g(x_0) = 0$, $g'(x_0) = 1/[x_0 f(x_0)]$, and $x'(0) = -qx_0/2$ verify both initial conditions directly. For every finite r, $x(r) > 0$, so the displayed functions and their derivatives are finite there. If $d = 0$, the preceding substitutions dividing by q are not used: the original equation is $W'' = (c_0 + c_1)W$, with the exact solution $W = W_0\cosh(\sqrt{c_0+c_1}\,r) + W_1\sinh(\sqrt{c_0+c_1}\,r)/\sqrt{c_0+c_1}$.

8.3 Signs and quantitative physical gain with this feedback

Let $\gamma_0 = \sqrt{c_0}$ and $\gamma_i = \sqrt{c_0 + c_1}$. Use the original instantaneous growing eigenline at entry: $\Theta_2(0) < 0$ and $\Omega_2(0) = \sqrt{b_2/a_{2,d}(0)}\Theta_2(0) < 0$. Then $Y = -W$ has $Y(0) = Y_0 > 0$ and $Y'(0) = \gamma_i Y_0 > 0$. As long as $Y > 0$, its equation gives $Y'' > 0$ and $Y' \geq Y'(0) > 0$; this excludes a first zero by continuity. Define

$$Y_- = Y_0\left(\cosh(\gamma_0 r) + \frac{\gamma_i}{\gamma_0}\sinh(\gamma_0 r)\right), \qquad Y_+ = Y_0 e^{\gamma_i r}.$$

The differences have zero initial value and derivative and satisfy

$$(Y - Y_-)'' - \gamma_0^2(Y - Y_-) = c_1 e^{-qr} Y,$$
$$(Y_+ - Y)'' - \gamma_i^2(Y_+ - Y) = c_1(1 - e^{-qr})Y.$$

For $a > 0$, the formula $Z(r) = \int_0^r \sinh(a(r-u))S(u)/a\,du$ has $Z(0) = Z'(0) = 0$ and $Z'' - a^2 Z = S$, by two differentiations. Uniqueness proves it is the solution of that initial-value problem. Both sources above are nonnegative; both this kernel and its first r derivative are nonnegative for $0 \leq u \leq r$. Therefore $Y_- \leq Y \leq Y_+$ and $Y_-' \leq Y' \leq Y_+'$ for every finite $r \geq 0$. The inverse physical map now gives both amplitudes negative and the precise temperature-gain bounds

$$e^{-\delta_2 r}\left(\cosh(\gamma_0 r) + \frac{\gamma_0}{\gamma_i}\sinh(\gamma_0 r)\right) \leq \frac{-\Theta_2(r)}{-\Theta_2(0)} \leq e^{(\gamma_i - \delta_2)r}. \tag{144}$$

In particular, if the actual numerical coefficients satisfy $\delta_2 \geq \gamma_i$, this particular growth control never raises the second temperature above its entry amplitude. This is a proved obstruction to that specified uncompensated control, not a statement about other controls or the existence of an infinite viscous sequence.

8.4 The exact map to spatial fields and its interaction force

The affine parent $\theta_b = G_d(r)\cdot x$, $u_b = 0$ is a solution with pressure $p_b = x_2 G_d(r)\cdot x/2$ and forces $f_{\theta,b} = G_d'(r)\cdot x$ and $f_{u,b} = -(A_0 \sin s/2)Jx$. The scalar statement follows from $\Delta\theta_b = 0$. For momentum, direct differentiation gives $\nabla p_b - \theta_b e_2 = (G_d x_2 - e_2(G_d \cdot x))/2 = -G_{d,1}Jx/2$. An added sine wave whose amplitudes solve (140) has zero additional PDE residual except for its activation, by the same explicit phase, transverse-vector, and pressure calculations below. Thus the solved system has an exact forced affine-background realization.

There is also a full two-wave realization specifying the original spatial discrepancy. Let F_1, F_2 be smooth odd periodic profiles, each with mean-zero primitive $Q_j' = F_j$, and assume $F_1'(0) = 1$. This includes both sine and the released profile which is linear near zero. Set

$$G_0 = (A_0 \sin s, -A_0 \cos s), \quad T(r) = -P_{d,b}e^{-qr}, \quad q_1 = K_1 x_2, \quad q_2 = K_2 \zeta_2 \cdot x,$$

and let $w_2(r)$ be any specified smooth activation, for example the source's $\eta(\Gamma_2 r)$. Define on $\mathbb{R}^2$

$$\theta = G_0\cdot x + TF_1(q_1) + w_2\Theta_2 F_2(q_2),$$
$$u = \frac{w_2\Omega_2}{K_2}J\zeta_2 F_2(q_2),$$
$$p = \frac{1}{2}x_2(G_0\cdot x) + \frac{T}{K_1}Q_1(q_1) + \frac{w_2\Theta_2\zeta_{2,2}}{K_2}Q_2(q_2). \tag{145}$$

Their complete retained-diffusion forces are

$$\begin{aligned} f_\theta = &-dK_1^2 T\,[F_1(q_1) + F_1''(q_1)] + w_2'\Theta_2 F_2(q_2) \\ &- dK_2^2 w_2\Theta_2\,[F_2(q_2) + F_2''(q_2)] + f_{\text{cross}}, \end{aligned} \tag{146}$$

$$f_{\text{cross}} = \frac{w_2\Omega_2}{K_2}\sin s_2\, K_1 T\,[F_1'(q_1) - 1]F_2(q_2), \tag{147}$$

$$f_u = -\frac{A_0 \sin s}{2}Jx + \frac{w_2'\Omega_2}{K_2}J\zeta_2 F_2(q_2) - dK_2 w_2\Omega_2 J\zeta_2[F_2(q_2) + F_2''(q_2)]. \tag{148}$$

Here is a direct verification. The old wave has derivative $T' = -dK_1^2 T$ and Laplacian $K_1^2 T F_1''$, giving the first scalar force. The old temperature gradient differs from the coefficient gradient $G_d = G_0 + K_1 T e_2$ by precisely $K_1 T[F_1'(q_1) - 1]e_2$. Its advection by u is the displayed f_{cross} because $J\zeta_2 \cdot e_2 = \sin s_2$. The remainder of that old-temperature advection is $(w_2\Omega_2/K_2)J\zeta_2 \cdot G_d\, F_2 = -w_2 a_{2,d}\Omega_2 F_2$; it cancels the coupling part of $w_2\Theta_2' F_2$. The new wave advects its own temperature and velocity by zero since $J\zeta_2 \cdot \zeta_2 = 0$. Its amplitude damping and its Laplacian give the two new-profile terms in the forces. The old wave pressure has gradient $TF_1(q_1)e_2$, so it cancels old wave buoyancy exactly. The new pressure has gradient $w_2\Theta_2\zeta_{2,2}\zeta_2 F_2$. The new momentum coupling vector before that pressure is $w_2\Theta_2(\zeta_{2,1}J\zeta_2 - e_2)F_2 = -w_2\Theta_2\zeta_{2,2}\zeta_2 F_2$, by resolving e_2 in the orthonormal basis $(\zeta_2, J\zeta_2)$. Hence it cancels as well. The base pressure supplies exactly the first vector-force term, as proved above, and divergence is zero by transversality. This verifies both full equations including all interactions.

For the sine profiles, $F_j + F_j'' = 0$ identically. The remaining scalar interaction force is then exactly

$$f_{\mathrm{cross}} = -\frac{A_{1,d}e^{-qr}}{K_2} w_2\Omega_2 \sin s_2\, [\cos(K_1 x_2) - 1] \sin(q_2), \qquad \|f_{\mathrm{cross}}(r)\|_\infty \le \frac{2A_{1,d}e^{-qr}}{K_2} |w_2\Omega_2| \sin s_2.$$

The bound follows from $|\cos u - 1| \le 2$ and $|\sin v| \le 1$. For the released linear-core profile the same explicit formulas hold, and $F_1' - 1 = 0$ on its original linear interval, so the interaction force vanishes there. Its old-profile diffusion defect is instead $-dK_1^2 T q_1$ on that interval, since $F_1(q_1) = q_1$ and $F_1''(q_1) = 0$. Its gradient is exactly $G_d'(r)$. Thus exponential damping of that released linear core requires the stated core force, and is not an unforced consequence of its zero Laplacian. The explicit forces, inverse solution maps, and convergent formulas above carry the feedback calculation to an actual spatial solution without conflating the sine state, its affine gradient, and the released profile. No compact-support or infinite-iteration claim is made in this section.

9 Exact coupled feedback and the scope of common damping

Retain the source layer frequencies λ_j, and write $K_j = \mu\lambda_j$ for physical frequencies. Let Θ_j, Ω_j denote physical amplitudes. For each activated layer j, let $\zeta_j \neq 0$, $r_j = |\zeta_j|$, $e_j = \zeta_j/r_j$, $S_j = Je_j \otimes e_j$, and $A_j = -K_j r_j \Theta_j$. The exact physical counterpart of source (3.3)–(3.5) is

$$\begin{aligned} G_{<q} &= -A_0 e_0 + \sum_{j<q} w_j K_j \Theta_j \zeta_j, & D_{<q} &= \dot\alpha J + \sum_{j<q} w_j \Omega_j S_j, \\ \dot M_q &= D_{<q} M_q, \quad M_q(t_q^{\mathrm{in}}) = I, & \zeta_q &= M_q^{-T} p_q, \\ \dot\zeta_q &= -D_{<q}^T \zeta_q, & \dot\Theta_q &= -\frac{J\zeta_q \cdot G_{<q}}{K_q r_q^2}\Omega_q, \quad \dot\Omega_q = K_q \zeta_{q,1}\Theta_q. \end{aligned} \tag{149}$$

Each sum begins at $j = 1$, and contains finitely many terms at every finite preterminal time. The rotating base coefficient is the original A_0, not a reset unit value. The physical growth-scale definition is $\sigma_q^2 = |G_{<q+1}(t_q^b)|$, so $\sigma_q = \nu\widetilde\sigma_q$.

For $\zeta_{j,2} \neq 0$ define

$$\mathcal{F}_j = -\frac{\sum_{i<j} w_i \Omega_i e_{i,1}(Je_i \cdot \zeta_j)}{\zeta_{j,2}}, \qquad \mathcal{F}_0 = 0. \tag{150}$$

Transposing S_i gives $S_i^T \zeta_j = e_i(Je_i \cdot \zeta_j)$, and the first component of $-\dot\alpha J^T \zeta_j$ is $-\dot\alpha\zeta_{j,2}$. Hence

$$\dot\zeta_{j,1} = (\mathcal{F}_j - \dot\alpha)\zeta_{j,2}. \tag{151}$$

It follows by direct differentiation that $\dot\alpha = \mathcal{F}_j - \dot Z/\zeta_{j,2}$ gives $\dot\zeta_{j,1} = \dot Z$; equality of initial values gives $\zeta_{j,1} = Z$ on its entire interval of definition. This proves the control map exactly, including its denominator and the laboratory component that drives buoyancy.

There is no omitted parent evolution in (149). For example, differentiation of the definitions gives

$$\dot{G}_{<q} = -A_0\dot{\alpha}Je_0 + \sum_{j<q} K_j\{(\dot{w}_j\Theta_j + w_j\dot{\Theta}_j)\zeta_j - w_j\Theta_j D_{<j}^T\zeta_j\}, \tag{152}$$

$$\dot{D}_{<q} = \ddot{\alpha}J + \sum_{j<q}\{(\dot{w}_j\Omega_j + w_j\dot{\Omega}_j)S_j + w_j\Omega_j\dot{S}_j\}, \tag{153}$$

$$\dot{S}_j = -\frac{JD_{<j}^T\zeta_j\otimes\zeta_j + J\zeta_j\otimes D_{<j}^T\zeta_j}{r_j^2} + \frac{2\zeta_j\cdot D_{<j}^T\zeta_j}{r_j^2}S_j. \tag{154}$$

To verify the last line, use $S_j = J\zeta_j\otimes\zeta_j/r_j^2$, $\dot{\zeta}_j = -D_{<j}^T\zeta_j$, and $(r_j^2)' = -2\zeta_j\cdot D_{<j}^T\zeta_j$. These three identities give exactly the three displayed terms by the quotient rule. Equations (152)–(154), together with the amplitude rows, are exact equations for both inherited backgrounds, including activation and the common angular control.

9.1 Damped amplitude equations as an exact controlled realization

For a sine profile on an affine background, retaining physical diffusion gives the amplitude rows

$$\begin{aligned}\dot{\Theta}_j &= -\kappa K_j^2 r_j^2\Theta_j - \frac{J\zeta_j\cdot G_{<j}}{K_j r_j^2}\Omega_j + c_{\theta,j},\\ \dot{\Omega}_j &= -\varepsilon K_j^2 r_j^2\Omega_j + K_j\zeta_{j,1}\Theta_j + c_{\omega,j}.\end{aligned} \tag{155}$$

For an actual finite family these equations are coupled to the phase rows, D, G, and the control; the symbols $c_{\theta,j}$ and $c_{\omega,j}$ here explicitly record added amplitude forcing. Inserting them in (152)–(153) produces the exact diffusion contributions

$$\begin{aligned}(\dot{G}_{<q})_{\text{amplitude diffusion}} &= -\kappa\sum_{j<q} w_j K_j^3 r_j^2\Theta_j\zeta_j,\\ (\dot{D}_{<q})_{\text{amplitude diffusion}} &= -\varepsilon\sum_{j<q} w_j K_j^2 r_j^2\Omega_j S_j,\end{aligned} \tag{156}$$

and additional terms $\sum w_j K_j c_{\theta,j}\zeta_j$ and $\sum w_j c_{\omega,j} S_j$, respectively. The phase geometry and feedback then use those changing backgrounds, rather than their inviscid values.

For the released general F, (155) has a different precise meaning: after substitution into the exact uncut scalar and vorticity residuals, the wave forces are

$$\begin{aligned}R_{\theta,j} &= c_{\theta,j}F - \kappa K_j^2 r_j^2\Theta_j(F + F''),\\ R_{\omega,j} &= c_{\omega,j}F' - \varepsilon K_j^2 r_j^2\Omega_j(F' + F''').\end{aligned} \tag{157}$$

These formulas follow because $\Delta[\Theta_j F(K_j\zeta_j\cdot x)] = K_j^2 r_j^2\Theta_j F''$ and $\Delta[\Omega_j F'(K_j\zeta_j\cdot x)] = K_j^2 r_j^2\Omega_j F'''$. On an affine core they are $c_{\theta,j}s - \kappa K_j^2 r_j^2\Theta_j s$ and $c_{\omega,j} - \varepsilon K_j^2 r_j^2\Omega_j$. Thus damping these amplitudes for this profile requires force *inside* the affine core. Taking $c_{\theta,j} = \kappa K_j^2 r_j^2\Theta_j$ and $c_{\omega,j} = \varepsilon K_j^2 r_j^2\Omega_j$ restores exactly the original coupled amplitude rows and leaves instead $-\kappa K_j^2 r_j^2\Theta_j F''$ and $-\varepsilon K_j^2 r_j^2\Omega_j F'''$. These vanish in the core and are precisely the compensation used in (89)–(90), with all cutoff terms additionally retained there.

9.2 The exact defect of multiplying all old amplitudes by damping

To quantify a proposed map, let a finite inviscid coupled trajectory be decorated with E, and suppose for this calculation that $\kappa = \varepsilon = \delta > 0$. Keep its phases and angle and put

$$d_j(t) = \exp\left[-\delta K_j^2\int_{t_j^{\text{in}}}^t r_j(s)^2\,ds\right], \qquad \Theta_j^d = d_j\Theta_j^E, \quad \Omega_j^d = d_j\Omega_j^E.$$

The backgrounds computed from these candidate fields differ by

$$\Delta G_{<q} = \sum_{j<q} w_j K_j (d_j - 1)\Theta_j^E \zeta_j, \qquad \Delta D_{<q} = \sum_{j<q} w_j (d_j - 1)\Omega_j^E S_j. \tag{158}$$

Differentiating the candidate fields and using the inviscid equations gives the exact defects

$$\dot\zeta_q + (D^d_{<q})^T \zeta_q = \Delta D^T_{<q}\zeta_q, \tag{159}$$

$$\dot\Theta^d_q + \delta K_q^2 r_q^2 \Theta^d_q + \frac{J\zeta_q \cdot G^d_{<q}}{K_q r_q^2}\Omega^d_q = \frac{J\zeta_q \cdot \Delta G_{<q}}{K_q r_q^2}\Omega^d_q, \tag{160}$$

$$\dot\Omega^d_q + \delta K_q^2 r_q^2 \Omega^d_q - K_q \zeta_{q,1}\Theta^d_q = 0. \tag{161}$$

For instance the derivative of d_q cancels the diagonal diffusion term, leaving the difference of the two temperature couplings; this is precisely (160), with its positive sign. Also the feedback computed at these same phases has the exact difference

$$\mathcal{F}^d_j - \mathcal{F}^E_j = -\frac{\sum_{i<j} w_i (d_i - 1)\Omega_i^E e_{i,1}(Je_i \cdot \zeta_j)}{\zeta_{j,2}}. \tag{162}$$

Changing $\dot\alpha$ by this expression cancels the first component of (159) for that controlled index, as follows from (151); it does not in general cancel its second component or the equations for other phases. Equations (159)–(162) are an exact relation between the two candidate systems, rather than an assertion that the systems are unrelated.

The defects are nonzero even in a particularly transparent actual holding geometry. After the first stage, $\zeta_1 = e_2$, $\Omega_1 = 0$, and $\Theta_1 < 0$. At the start of the next growth interval $\zeta_2 = e(s_2)$, $s_2 > 0$. For the damping candidate with $d_1 < 1$,

$$J\zeta_2 \cdot \Delta G_{<2} = (d_1 - 1)K_1\Theta_1 \sin s_2 \neq 0.$$

All factors on the right are nonzero with the indicated signs. Thus even when the old shear vanishes exactly on a hold, its decaying temperature changes the next layer's amplification coefficient. The common scalar damping identity for a fixed prescribed D, G has not proved a conjugacy of this coupled system.

For the compact finite stage of Section 6, the stronger exact forced relation is available: retain the original coupled fields and add the explicitly calculated Laplacian forces. It preserves both backgrounds and all controls, and quantifies the entire cost. The next section records precisely why this particular relation has no smooth-force extension along the unchanged infinite source trajectory.

10 The exact cost along the released infinite trajectory

The finite construction above proves no infinite viscous singularity. There is a precise calculation for the particular proposed extension that retains the released infinite fields and changes only the forces by their Laplacians. We record both that calculation and its scope.

The source trajectory in this section means the specific construction of Alpöge–Buckmaster, including their compact corrections. Its existence and infinite inviscid conclusions are the source's Theorems 3.2, 7.3, 8.2, and 1.1, not new conclusions of this note. We use the following exact properties established in those proofs: at the growth crossing t_q^a, activation is complete and

$$\Theta_q(t_q^a) = -\frac{\nu^2}{\mu}\lambda_q^{-7/8}, \quad r_q(t_q^a) \geq r_* > \frac{1}{2}, \quad \left|\frac{\Omega_q(t_q^a)}{\Theta_q(t_q^a)}\right| \geq \frac{K_q}{2\sigma_{q-1}}. \tag{163}$$

The first is the definition of the stopping time, source (3.8) and item (i) after (3.11), p. 10. Here $r_* := 1 - C_r\epsilon_{\rm sch} > 1/2$ is the improved phase-length bound, with the fixed constant C_r from the source estimate; it is not the separate bootstrap constant called ζ_- in source (3.14). The second is source (3.18), p. 13 and the tolerance choice on p. 12. The last is the tracking estimate in the complete growth proof, pp. 29–30, between (3.63) and (3.64), with the low-order tolerance decreased within its permitted fixed range. For clarity the extraction of a numerical factor does not presume equality with the reference growing eigenline. With $u = \Omega/\Theta$, $u_* = (b/a)^{1/2}$, and $z = u/u_*$, direct differentiation gives

$$z' = \sqrt{ab}(1 - z^2) - z(\log u_*)'.$$

The source seed is chosen on the growing eigenline, so $z(t_q^{\mathrm{in}}) = 1$. For $q = 1$ the eigenline and the ratio are exact throughout growth, as proved in the finite-stage section; the following tracking computation applies to later stages. If the source error bound is at most $\delta \le 1/4$, its Riccati barrier gives $z \ge 1-\delta$: at $z = 1-\delta$, $z' \ge \sqrt{ab}\{1-(1-\delta)^2-\delta(1-\delta)\} = \delta\sqrt{ab} > 0$. At $z = 1+\delta$ the corresponding upper derivative is $-\delta\sqrt{ab} < 0$. A first exit is therefore impossible. To extract the coefficient ratio, use the exact source (3.55)–(3.56): during growth the preceding direction is vertical, $a_q = \mathcal{G}_q/(K_q r_q)$ and $b_q = K_q d_q$, where $d_q = r_q \sin(\phi_q - \phi_{q-1})$ and $\mathcal{G}_q = \sum_{i<q} w_i A_i \sin(\phi_q - \phi_i)$ includes the base term with A_0 and $w_0 = 1$. Hence $u_* = K_q\sqrt{d_q r_q/\mathcal{G}_q}$. The three source estimates are $r_q \ge 1 - \delta$, $d_q/\sin s_q \ge 1 - \delta$, and $\mathcal{G}_q/(\sigma_{q-1}^2 \sin s_q) \le 1 + \delta$. Together they give $u_* \ge (K_q/\sigma_{q-1})(1-\delta)/\sqrt{1+\delta}$. Their product is at least $K_q/(2\sigma_{q-1})$, because $(3/4)^2/\sqrt{5/4} > 1/2$. This proves the displayed extraction from the source's explicit tracking estimates without assigning a sign to an uncontrolled error.

The source also gives

$$\sigma_{q-1} \le \nu\sqrt{C_\sigma}\lambda_{q-1}^{1/16}, \quad \lambda_q = \lambda_{q-1}^{Q_q}, \quad Q_q = Q_* + q, \quad Q_* \ge 200, \tag{164}$$

by (3.7),(3.18). The material radius is $\ell_q^{\mathrm{phys}} = \lambda_{q-1}^{-3}/\mu$ for $q \ge 2$. Each older layer and the base are affine on the full new support (Lemma 4.3, pp. 37–38), while all corrections to the newest layer vanish on its entire material plateau (Proposition 6.1, pp. 54–55). These spatial statements are essential to the following pointwise computation: a bound on a central affine ball alone would not provide the points used here.

Choose one fixed $s_\dagger \in (-\pi, \pi)$ with $F''(s_\dagger) \ne 0$. Such a point exists for the released profile. Otherwise F'' would vanish on the full period and by periodicity everywhere, so F would be an affine periodic function, hence constant; oddness would make it zero, contradicting $F'(0) = 1$. Fix this point once for every layer. At the growth crossing put

$$a_q = \frac{s_\dagger}{K_q} p_q, \qquad x_q = M_q(t_q^a) a_q. \tag{165}$$

The activation vectors have $|p_q| = 1$ and $\zeta_q = M_q^{-T} p_q$, so $K_q \zeta_q \cdot x_q = s_\dagger$ exactly. The point belongs to the plateau as soon as

$$\frac{|s_\dagger|}{K_q} < \frac{\ell_q^{\mathrm{phys}}}{2}, \quad \text{equivalently} \quad 2|s_\dagger| < \lambda_{q-1}^{Q_q - 3}.$$

This holds for all sufficiently large q by (164). The entire neighborhood of this point in the plateau has constant envelope, zero correction fields, and affine older fields. No later layer is yet activated at t_q^a. Consequently at this point the Laplacians of the *full released fields*, not merely those of an isolated formal wave, are exactly

$$\begin{aligned} \Delta\theta_E(x_q, t_q^a) &= \Theta_q K_q^2 r_q^2 F''(s_\dagger), \\ \Delta u_E(x_q, t_q^a) &= \Omega_q K_q J\zeta_q F''(s_\dagger). \end{aligned} \tag{166}$$

For the second line use $u_q = \Omega_q J\zeta_q F(s)/(K_q r_q^2)$ on the plateau; applying the Laplacian contributes $K_q^2 r_q^2$ and cancels the displayed denominator. This verifies both the vector direction and every frequency and phase-length factor.

Inserting (163) gives

$$|\kappa\Delta\theta_E(x_q,t_q^a)| \ge \kappa\nu^2\mu r_*^2|F''(s_\dagger)|\lambda_q^{9/8}, \tag{167}$$
$$|\varepsilon\Delta u_E(x_q,t_q^a)| \ge \frac{\varepsilon\nu\mu r_*}{2\sqrt{C_\sigma}}|F''(s_\dagger)|\lambda_q^{9/8}\lambda_{q-1}^{-1/16}. \tag{168}$$

For the second line first use $|\Omega_q| \ge |\Theta_q|K_q/(2\sigma_{q-1})$ and $|J\zeta_q| = r_q$, and then (164). For fixed positive κ the first right-hand side tends to infinity. For fixed positive ε the second does also, since

$$\lambda_q^{9/8}\lambda_{q-1}^{-1/16} = \lambda_{q-1}^{9Q_q/8-1/16} \longrightarrow \infty.$$

In particular even the case $\kappa = 0$, $\varepsilon > 0$ has an explicit diverging vector diffusion cost along these unchanged fields.

The original forces extend smoothly through the source terminal time and have fixed compact support, and hence are bounded; this is Proposition 9.4(b),(c), pp. 72–73, proved using Lemma 9.3, pp. 71–72. For the specific exact extension

$$f_\theta^{\kappa,\varepsilon} = f_\theta^E - \kappa\Delta\theta_E, \qquad f_u^{\kappa,\varepsilon} = f_u^E - \varepsilon\Delta u_E, \qquad p^{\kappa,\varepsilon} = p_E,$$

the reverse triangle inequality and (167)–(168) therefore prove unbounded forces at the corresponding positive coefficient, before the terminal time. They cannot have a continuous, and thus cannot have a smooth, terminal extension. This proves failure of the specified unchanged-pressure map. There is a stronger, pressure-independent conclusion for these same fields. Choose $s_\ddagger$ with $F'''(s_\ddagger) \neq 0$. It exists: otherwise F'' would be constant, and periodicity would successively force $F'' = F' = 0$, contradicting $F'(0) = 1$. Use the plateau point in (165) with this phase. The vorticity there equals a spatially constant older vorticity plus $\Omega_q F'(s)$, so

$$\Delta\omega_E = \Omega_q K_q^2 r_q^2 F'''(s_\ddagger), \qquad |\varepsilon\Delta\omega_E| \ge \frac{\varepsilon\nu\mu^2 r_*^2}{2\sqrt{C_\sigma}}|F'''(s_\ddagger)|\lambda_q^{17/8}\lambda_{q-1}^{-1/16} \longrightarrow \infty.$$

For any smooth pressure p_{new} and any vector force producing the same u_E, θ_E in the viscous equation, subtraction of the inviscid momentum equation and application of curl give the exact identity

$$\operatorname{curl} f_{u,\text{new}} = \operatorname{curl} f_u^E - \varepsilon\Delta\omega_E,$$

because mixed derivatives of $p_{\text{new}} - p_E$ commute. The source curl force is bounded. The displayed divergence therefore excludes a terminal extension with bounded first spatial derivatives for *every* pressure and force that retain these exact state fields. In fact the failure is local at the origin and terminal time. Source Theorem 3.2(iv) gives $\|M_q\| \le \sqrt{C_\sigma}\lambda_{q-1}^{1/16}$, so the test points satisfy

$$|x_q| \le \frac{|s_\ddagger|\sqrt{C_\sigma}}{\mu}\lambda_{q-1}^{1/16-Q_q} \longrightarrow 0, \qquad t_q^a \longrightarrow T_*.$$

A continuous derivative on a neighborhood of $(0, T_*)$ is bounded on a smaller compact neighborhood, contradicting the computed curl divergence there. If a vector field has bounded first spatial derivatives then its curl is bounded by their sum, which proves this implication explicitly. This still does not exclude a modified coupled trajectory, a different state construction, or viscous blowup; none of those claims follows from evaluating the unchanged source state.

The completed result of this note is the actual finite forced stage proved above and below, including its original datum, control, background feedback, return, hold, support and every diffusion term. The released infinite inviscid sequence is reconstructed as source mathematics. A new infinite viscous sequence with forces smooth through its terminal time is not established by these calculations.

11 Full source controls and aggregate background identities

Source and scope. The source is Levent Alpöge and Tristan Buckmaster, *Blowup for the Boussinesq equations with smooth forcing*, the supplied 76-page released PDF. Page numbers below are the printed PDF page numbers. The companion source receipt identifies the exact local PDF by SHA-256. The equations and proofs here reconstruct the finite-dimensional system in Section 3, especially pp. 8–10 and 26–33, and its exact affine realization in Section 4, pp. 37–39. The source's local comparison bounds are identified as source results where mentioned; the new claims proved below are the coordinate, operator, feedback, and aggregate-background identities. No infinite retained-viscosity construction is asserted.

Original data and the distinct parameters. Keep the original data of source (1.4), p. 2:

$$A_0 \geq 1, \qquad \lambda_0 = \left\lceil \frac{\sqrt{A_0}}{2} \right\rceil, \qquad \theta_{\text{in}}(x) = -\frac{A_0}{\lambda_0}\sin(\lambda_0 x_2)\chi_0(x), \qquad u_{\text{in}} = 0.$$

Here $\chi_0 \in C_c^\infty(B_{R_0})$ is radial, $R_0 > 2$, $0 \leq \chi_0 \leq 1$, and $\chi_0 = 1$ on B_2. Write

$$\mu = \lambda_0, \qquad \nu_{\text{AB}} = \sqrt{A_0}, \qquad X = \mu x, \quad T = \nu_{\text{AB}} t.$$

The symbol ν_{AB} is precisely the time-scaling parameter called ν in source Lemma 1.2, pp. 6–7. It is not physical viscosity. Physical momentum viscosity is $\varepsilon_{\text{v}} \geq 0$ and thermal diffusivity is $\kappa_{\text{th}} \geq 0$. The source uses μ again locally for its shooting amplitude; we display that independent parameter as μ_{p}, and never identify it with $\mu = \lambda_0$.

Use a hat for a source Section 3 quantity, which is expressed there in (X, T). The exact dictionary, keeping both presentations, is

$$\begin{aligned}
\theta(x,t) &= \frac{\nu_{\text{AB}}^2}{\mu}\widehat{\theta}(X,T), & u(x,t) &= \frac{\nu_{\text{AB}}}{\mu}\widehat{u}(X,T), & p(x,t) &= \frac{\nu_{\text{AB}}^2}{\mu^2}\widehat{p}(X,T), \\
\omega(x,t) &= \nu_{\text{AB}}\widehat{\omega}(X,T), & \psi(x,t) &= \frac{\nu_{\text{AB}}}{\mu^2}\widehat{\psi}(X,T), \\
f_\theta(x,t) &= \frac{\nu_{\text{AB}}^3}{\mu}\widehat{f}_\theta(X,T), & f_u(x,t) &= \frac{\nu_{\text{AB}}^2}{\mu}\widehat{f}_u(X,T).
\end{aligned} \tag{169}$$

The transformed diffusivities are

$$\widehat{\kappa_{\text{th}}} = \kappa_{\text{th}}\frac{\mu^2}{\nu_{\text{AB}}}, \qquad \widehat{\varepsilon_{\text{v}}} = \varepsilon_{\text{v}}\frac{\mu^2}{\nu_{\text{AB}}}. \tag{170}$$

These statements follow directly from $\partial_t = \nu_{\text{AB}}\partial_T$, $\nabla_x = \mu\nabla_X$, and $\Delta_x = \mu^2\Delta_X$: both scalar transport terms have factor ν_{AB}^3/μ, both momentum transport terms and gravity have factor ν_{AB}^2/μ, and the respective Laplacian terms have factors $\kappa_{\text{th}}\nu_{\text{AB}}^2\mu$ and $\varepsilon_{\text{v}}\nu_{\text{AB}}\mu$. The divergence scales by ν_{AB}, curl by ν_{AB}, and $J\nabla_x\psi = (\nu_{\text{AB}}/\mu)J\nabla_X\widehat{\psi}$. Thus the laboratory gravity vector $e_2 = (0,1)$ is unchanged. The inverse transformations divide by the displayed nonzero factors and evaluate at $(x,t) = (X/\mu, T/\nu_{\text{AB}})$, so these are bijections of the corresponding smooth fields, with support radii divided by μ and time intervals divided by ν_{AB}.

Layer dictionary. For $j \geq 1$ retain the original source $\widehat{\lambda}_j$, and define its actual spatial frequency $\lambda_j = \mu\widehat{\lambda}_j$. The remaining dictionary is

$$\begin{aligned}
M_j(t) &= \widehat{M}_j(\nu_{\text{AB}}t), & \zeta_j(t) &= \widehat{\zeta}_j(\nu_{\text{AB}}t), & p_j &= \widehat{p}_j, \\
\Theta_j(t) &= \frac{\nu_{\text{AB}}^2}{\mu}\widehat{\Theta}_j(\nu_{\text{AB}}t), & \Omega_j(t) &= \nu_{\text{AB}}\widehat{\Omega}_j(\nu_{\text{AB}}t), & \alpha(t) &= \widehat{\alpha}(\nu_{\text{AB}}t), \\
A_j(t) &= \nu_{\text{AB}}^2\widehat{A}_j(\nu_{\text{AB}}t), & \sigma_j &= \nu_{\text{AB}}\widehat{\sigma}_j, & \Gamma_j &= \nu_{\text{AB}}\widehat{\Gamma}_j, \\
D_{<j}(t) &= \nu_{\text{AB}}\widehat{D}_{<j}(\nu_{\text{AB}}t), & G_{<j}(t) &= \nu_{\text{AB}}^2\widehat{G}_{<j}(\nu_{\text{AB}}t), & \ell_j &= \widehat{\ell}_j/\mu.
\end{aligned} \tag{171}$$

In particular $\sigma_0 = \nu_{\rm AB}$, $A_0 = \nu_{\rm AB}^2$, and the original $\lambda_0 = \mu$ has not been reset. All unadorned derivatives below are in original physical time t.

The exact coupled variables. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad e(\phi) = (\sin\phi, \cos\phi), \quad e_0 = R_\alpha e_2 = e(-\alpha), \quad w_0 = 1.$$

For $j \geq 1$ set $r_j = |\zeta_j|$, $e_j = \zeta_j/r_j = e(\phi_j)$, $P_j = -\Theta_j$, $A_j = \lambda_j r_j P_j$, and $B_j = \lambda_j \Theta_j \zeta_j = -A_j e_j$. The positive-amplitude variables are used only on intervals where their positivity has been established. Put $B_0 = -A_0 e_0$, and

$$\begin{aligned} &G_{<j} = B_0 + \sum_{1\leq i<j} w_i B_i, && D_{<j} = \dot\alpha J + \sum_{1\leq i<j} w_i S_i, && S_i = \Omega_i J e_i \otimes e_i, \\ &\dot M_j = D_{<j} M_j, && M_j(t_j^{\rm in}) = I, && \zeta_j = M_j^{-T} p_j, \\ &\dot\Theta_j = a_j \Omega_j, && a_j = -\frac{J\zeta_j \cdot G_{<j}}{\lambda_j r_j^2}, && \dot\Omega_j = b_j \Theta_j, \quad b_j = \lambda_j \zeta_{j,1}. \end{aligned} \tag{172}$$

These are source (3.2)–(3.3), p. 8, in original units. To check their scaling, $a_j = (\nu_{\rm AB}^2/\mu)\widehat a_j$ and $b_j = \mu \widehat b_j$; consequently both amplitude equations have their correct factors $\nu_{\rm AB}^3/\mu$ and $\nu_{\rm AB}^2$. Since $\operatorname{tr} J = \operatorname{tr}(Je_i \otimes e_i) = 0$, Jacobi's determinant formula gives $\det M_j = 1$ throughout every finite existence interval.

Derivation of the wave equations. Let F be the exact odd smooth periodic profile of source (4.1), and let $\mathcal{P}$ be its even mean-zero primitive. Put $s = \lambda_j \zeta_j \cdot x$ and

$$\vartheta_j = \Theta_j F(s), \quad \psi_j = \frac{\Omega_j}{\lambda_j^2 r_j^2}\mathcal{P}(s), \quad v_j = \frac{\Omega_j}{\lambda_j r_j^2} J\zeta_j F(s), \quad \varpi_j = \Omega_j F'(s).$$

Differentiating $M_j^{-T} p_j$ gives $\dot\zeta_j = -D_{<j}^T \zeta_j$, hence $(\partial_t + D_{<j} x \cdot \nabla)s = 0$. Moreover $J\zeta_j \cdot \zeta_j = 0$, so $v_j \cdot \nabla\vartheta_j = v_j \cdot \nabla\varpi_j = 0$. On an affine earlier state $G_{<j}\cdot x, D_{<j}x$, the added scalar residual is $(\dot\Theta_j + \Omega_j J\zeta_j \cdot G_{<j}/(\lambda_j r_j^2))F$, and the added vorticity residual is $(\dot\Omega_j - \lambda_j \zeta_{j,1}\Theta_j)F'$. This proves the exact map from (172) to the unactivated local wave equations, including the fixed sign and gravity component.

The control maps and every stage. Source (3.4), p. 9, defines

$$F_j^{\rm ctl} = -\frac{\sum_{1\leq i<j} w_i \Omega_i e_{i,1}(Je_i \cdot \zeta_j)}{\zeta_{j,2}}, \qquad F_0^{\rm ctl} = 0. \tag{173}$$

The superscript distinguishes the paper's feedback F_j from its periodic profile F. Transposing $D_{<j}$ in (172) and taking component one gives

$$\dot\zeta_{j,1} = (F_j^{\rm ctl} - \dot\alpha)\zeta_{j,2}.$$

Thus $\dot\alpha = F_j^{\rm ctl} - \dot Z/\zeta_{j,2}$ implies $\zeta_{j,1}(t) = Z(t)$ whenever equality holds initially and the second component is nonzero. This proves the feedback map; the second component is not replaced by a constant.

Here is the complete source schedule (3.7)–(3.12), pp. 9–10, preserving its units. Fix $\beta = 1/8$, $Q_* \geq 200$, $Q_j = Q_* + j$, and

$$\begin{aligned} &\widehat\lambda_1 \gg 1, \qquad \widehat\lambda_j = \widehat\lambda_{j-1}^{Q_j} \quad (j \geq 2), \\ &\widehat\ell_1 = s_0/C_\ell, \quad C_\ell \geq 4, \qquad \widehat\ell_j = \widehat\lambda_{j-1}^{-3} \quad (j\geq 2), \\ &\widehat\Pi_1 = (\log\widehat\lambda_1)^{10}, \qquad \widehat\Pi_j = \widehat\lambda_{j-1}^4 (\log\widehat\lambda_j)^{10} \quad (j \geq 2), \\ &k_j = \max\{k \in \mathbb{N}_0 : 120k \leq Q_j,\ K(k) \leq \log\widehat\lambda_j\}, \\ &J_j = 2k_j + 8, \qquad L_j = (k_j + 5 + \beta)\log\widehat\lambda_j. \end{aligned} \tag{174}$$

The prescribed seed is $\Theta_j^{\text{seed}} = -(\nu_{\text{AB}}^2/\mu)\widehat{\lambda}_j^{-k_j-6}$, and its gain target is

$$P_j(t_j^a) = \frac{\nu_{\text{AB}}^2}{\mu}\widehat{\lambda}_j^{-7/8}, \qquad \frac{P_j(t_j^a)}{|\Theta_j^{\text{seed}}|} = e^{L_j}.$$

In particular the physical frequency recursion is $\lambda_j = \mu(\lambda_{j-1}/\mu)^{Q_j}$, not $\lambda_{j-1}^{Q_j}$. Let η be smooth nondecreasing, zero on $(-\infty, 0]$ and one on $[1, \infty)$; let $\eta_2 \in C_c^\infty((0,1))$, $\eta_2 \geq 0$, $\int_0^1 \eta_2 = 1$, and $H_2 = \|\eta_2\|_\infty$. The fixed design is

$$c_p = 3, \quad \bar{v} = 2, \quad \Lambda_1 \geq 2c_p H_2, \quad 0 < s_* \leq 1/8, \quad c_p \Lambda_1 H_2 \sin s_* \leq 1/4.$$

Set $t_1^{\text{in}} = 1/\nu_{\text{AB}}$, $s_1 = s_*$, $\Gamma_1 = \nu_{\text{AB}} \sin s_*$, and for $j \geq 2$

$$t_j^{\text{in}} = t_{j-1}^b, \quad s_j = L_j \frac{\sigma_{j-2}}{\sigma_{j-1}}, \quad \Gamma_j = \sigma_{j-1} \sin s_j, \quad \Lambda_j = L_j.$$

At activation $p_j = e(s_j)$, $\Theta_j = \Theta_j^{\text{seed}}$, $\Omega_j = \sqrt{b_j/a_j}\Theta_j$, where source Theorem 3.2 proves $a_j, b_j > 0$ on the constructed growth interval. The activation multiplier is $w_j(t) = \eta(\Gamma_j(t - t_j^{\text{in}}))$.

1. Growth: $\dot{\alpha} = F_{j-1}^{\text{ctl}}$ until the first amplitude threshold t_j^a displayed above.
2. Transition: $t_j^1 = t_j^a + \Gamma_j^{-1}$, $\eta_b = \eta(\Gamma_j(t - t_j^a))$, and $\dot{\alpha} = (1-\eta_b)F_{j-1}^{\text{ctl}} + \eta_b F_j^{\text{ctl}}$.
3. Steering: write $z_1 = \zeta_{j,1}(t_j^1)$ and $\chi_1 = r_j(t_j^1)$. For $0 \leq h \leq h_j^{\max} = c_p \Lambda_j \sin s_j$ define

   $$Z_j(t) = z_1[1 - \eta(\Gamma_j(t - t_j^1))] - h\chi_1 \eta_2(\Lambda_j \Gamma_j(t - t_j^1 - \Gamma_j^{-1})),$$

   and $\dot{\alpha} = F_j^{\text{ctl}} - \dot{Z}_j/\zeta_{j,2}$. The fixed endpoint is $t_j^b = t_j^1 + \Gamma_j^{-1}(1 + \Lambda_j^{-1})$. Select the least h in the compact zero set $\Omega_j(t_j^b; h) = 0$ proved nonempty in the source. No uniqueness is asserted for $j \geq 2$.
4. Holding: use $\dot{\alpha} = F_j^{\text{ctl}}$ through t_{j+1}^a, concurrently with next-layer growth; evaluate $\sigma_j^2 = |G_{<j+1}(t_j^b)|$ only after selecting the stage. Since $\zeta_{j,1} = \Omega_j = 0$, both $\dot{\Omega}_j = 0$ and $\dot{\Theta}_j = 0$. This proves stationarity of this amplitude pair, while the older state and M_j may continue to evolve.

All these control maps are in physical time; $F_j^{\text{ctl}} = \nu_{\text{AB}} \widehat{F}_j^{\text{ctl}}$, and Z_j is dimensionless. The source's smooth time joins follow from flatness of η, interior support of η_2, and uniqueness for the stationary endpoint pair.

Exact aggregate parent evolution. The following identities explicitly give the background feedback rather than prescribing D, G independently.

Proposition 1. For every finite activated system (172), put $G = G_{<q}$, $D = D_{<q}$, and $C = D_{21} - D_{12} = 2\dot{\alpha} + \sum_{i<q} w_i \Omega_i$. Then

$$\dot{G} + D^T G = \sum_{1 \leq i < q} \dot{w}_i B_i, \tag{175}$$

$$\dot{C} - G_1 = 2\ddot{\alpha} - A_0 \sin\alpha + \sum_{1 \leq i < q} \dot{w}_i \Omega_i, \tag{176}$$

$$\begin{aligned} \dot{D} = \ddot{\alpha} J &+ \sum_{1 \leq i < q} (\dot{w}_i \Omega_i + w_i b_i \Theta_i) J e_i \otimes e_i \\ &+ \sum_{1 \leq i < q} w_i \Omega_i (\dot{\alpha} + \Sigma_i)(J e_i \otimes J e_i - e_i \otimes e_i), \end{aligned} \tag{177}$$

$$\Sigma_i = \sum_{1 \leq k < i} w_k \Omega_k \sin^2(\phi_i - \phi_k).$$

Proof. Differentiate $B_i = \lambda_i \Theta_i \zeta_i$ and use (172):

$$\dot{B}_i + D_{<i}^T B_i = -\Omega_i e_i (Je_i \cdot G_{<i}) = -S_i^T G_{<i}.$$

Also $\dot{B}_0 + (\dot{\alpha}J)^T B_0 = 0$. Expand $\dot{G} + D^T G$, insert these equalities, and keep all cross terms. Its nonactivation terms are

$$\sum_i w_i S_i^T B_0 - \sum_i w_i S_i^T \Big(B_0 + \sum_{k<i} w_k B_k\Big) + \sum_i w_i \sum_{k \geq i} w_k S_k^T B_i.$$

The B_0 terms cancel. The terms with $k > i$ in the last sum cancel the preceding double sum after exchanging its indices. Its diagonal terms are $w_i^2 S_i^T B_i = 0$, because $Je_i \cdot e_i = 0$. This proves (175) with every activation derivative retained. Next $\dot{C} = 2\ddot{\alpha} + \sum_i \dot{w}_i \Omega_i + \sum_i w_i \lambda_i \zeta_{i,1} \Theta_i$, while $G_1 = B_{0,1} + \sum_i w_i \lambda_i \Theta_i \zeta_{i,1}$ and $B_{0,1} = A_0 \sin\alpha$. Subtraction proves (176). Finally direct projection of $\dot{\zeta}_i = -D_{<i}^T \zeta_i$ onto e_i and Je_i gives $\dot{e}_i = (\dot{\alpha} + \Sigma_i) Je_i$. Differentiate $S_i = \Omega_i Je_i \otimes e_i$, use $J^2 = -I$, and sum with the activation derivatives. This proves (177). □

For completeness these affine dynamics have an exact momentum realization. On an affine region set $u = Dx$, $\theta = G \cdot x$, $H = D' + D^2 - e_2 \otimes G$, and

$$p = -\tfrac{1}{2} x^T (\operatorname{sym} H) x, \qquad f_\theta = (\dot{G} + D^T G) \cdot x, \qquad f_u = \tfrac{1}{2}(\dot{C} - G_1) Jx.$$

Indeed $H - \operatorname{sym} H$ is the antisymmetric matrix with entry difference $H_{21} - H_{12} = \dot{C} - G_1$; here $D^2 = -\det(D) I$ for a trace-free 2×2 matrix, verified by direct matrix multiplication. Thus $u_t + u \cdot \nabla u + \nabla p - \theta e_2 = f_u$. Both Laplacians $\Delta\theta$ and Δu are identically zero, so these same fields and forces solve the retained-viscosity equations on that region for every $\kappa_{\text{th}}, \varepsilon_{\text{v}} \geq 0$.

Relative geometry and its proof. Set $\rho_j = \log r_j$, $d_{ij} = \phi_j - \phi_i$, $\phi_0 = -\alpha$, and

$$\mathcal{G}_j = \sum_{0 \leq i < j} w_i A_i \sin d_{ij}.$$

Projection of the phase equation, using $Je_i \cdot e_j = -\sin d_{ij}$ and $e_i \cdot e_j = \cos d_{ij}$, gives

$$\begin{aligned} &\dot{\rho}_j = \sum_{1 \leq i < j} w_i \Omega_i \sin d_{ij} \cos d_{ij}, && \dot{\phi}_j = -\dot{\alpha} - \Sigma_j, \qquad \dot{d}_{ij} = \Sigma_i - \Sigma_j, \\ &a_j = \mathcal{G}_j / (\lambda_j r_j), && \dot{\Omega}_j = -A_j \sin\phi_j, \quad (\log P_j)' = -\mathcal{G}_j \Omega_j / A_j, \\ &F_j^{\text{ctl}} = -\Sigma_j + \dot{\rho}_j \tan\phi_j = \frac{\sum_{1 \leq i < j} w_i \Omega_i \sin\phi_i \sin d_{ij}}{\cos\phi_j}. \end{aligned} \tag{178}$$

The last equality follows from $\sin\phi_i = \sin\phi_j \cos d_{ij} - \cos\phi_j \sin d_{ij}$. These are source (3.6), p. 9, and (3.55), p. 26, with A_0 retained.

For adjacent phases $\vartheta_j = \phi_j - \phi_{j-1}$, the matrix difference is $D_{<j} - D_{<j-1} = w_{j-1} \Omega_{j-1} Je_{j-1} \otimes e_{j-1}$. Its transpose annihilates ζ_{j-1}, so both ζ_{j-1} and ζ_j solve the same transport matrix equation. Its trace is zero, hence the derivative of their determinant is zero. At activation $\zeta_{j-1} = r_{j-1}(t_j^{\text{in}}) e_2$ and $\zeta_j = e(s_j)$, and the determinant in that order is $-r_{j-1}(t_j^{\text{in}}) \sin s_j$. Therefore, with the sign of this orientation kept,

$$d_j^{\text{adj}} := r_j \sin\vartheta_j = \sin s_j \, e^{-\rho_{j-1}(t) + \rho_{j-1}(t_j^{\text{in}})}. \tag{179}$$

This is source (3.56), p. 26; its symbol d_j^{adj} denotes the source's separate $\mathfrak{d}_j$ rather than an angle d_{ij}.

For $q \geq 2$, all older activation weights are one on stage q. Define

$$E_2 = \sum_{i=1}^{q-2} \Omega_i \sin d_{iq} \cos d_{iq}, \qquad E_3 = \dot{\rho}_{q-1}, \qquad E_1 = \sin\vartheta_q \sum_{i=1}^{q-2} \Omega_i \sin(d_{iq} + d_{i,q-1}),$$

and $R_q = \sum_{i=0}^{q-2} A_i \sin d_{iq}$. Then

$$\dot\vartheta_q = -E_1 - \Omega_{q-1}\sin^2\vartheta_q, \quad \dot\rho_q = E_2 + \Omega_{q-1}\sin\vartheta_q\cos\vartheta_q, \quad \mathcal{G}_q = A_{q-1}\sin\vartheta_q + R_q. \tag{180}$$

The first equality follows by subtracting the two angle equations and using $\sin^2 a - \sin^2 b = \sin(a-b)\sin(a+b)$; the other two are the corresponding split sums. At $q = 2$ the shear sums E_1, E_2, E_3 vanish, but $R_2 = A_0 \sin d_{02}$ is retained. The source makes this exceptional base term explicit on p. 28.

Exact later-stage model, including parent response. Use $\sigma = \sigma_{q-1}$, $s = s_q$, $\Gamma = \sigma\sin s$, $L = L_q$, $\tau = \Gamma(t - t_q^1)$, $\chi = r_q$, $\chi_1 = r_q(t_q^1)$, $W = \Omega_{q-1}/\sigma$, and $\mu_{\mathrm{p}} = h/(L\sin s)$. Write $d = d_q^{\mathrm{adj}}$, $d_1^* = d(t_q^1)$, and $e_4 = z_1/d_1^* - 1$. To avoid collision with the error coefficient d_1, the asterisk here marks the fixed adjacent determinant factor at steering entry. Then source (3.66), p. 31, is exactly

$$\begin{aligned}
c_1 &= (d/\sin s)\cos\vartheta_q, & d_1 &= \chi E_2/\Gamma,\\
c_2 = c_5 &= (A_{q-1}/\sigma^2)(d/\sin s), & c_3 &= \sin s/d,\\
d_3 &= 1 - (d_1^*/d)(1+e_4), & c_4 &= (d_1^*/\sin s)(1+e_4) = z_1/\sin s,\\
d_4 &= \frac{\chi_1 R_q}{\chi\sigma^2\sin s}, & d_2 &= \frac{A_{q-1}}{\sigma^2\sin s}\,\frac{d(\cos\phi_q - 1) - Z(\cos\vartheta_q - 1)}{\chi}.
\end{aligned} \tag{181}$$

Define $\widetilde{k} = c_5\chi_1/\chi^2 + d_4$ and

$$g(\tau) = \begin{cases} \eta(\tau) + d_3(1-\eta(\tau)), & 0 \le \tau \le 1,\\ 1 + \mu_{\mathrm{p}} c_3 L\chi_1\eta_2(L(\tau-1)), & 1 \le \tau \le 1 + L^{-1},\end{cases}$$

$$f(\tau) = \begin{cases} (c_4/\chi_1)(1-\eta(\tau)), & 0 \le \tau \le 1,\\ -\mu_{\mathrm{p}} L\eta_2(L(\tau-1)), & 1 \le \tau \le 1 + L^{-1}.\end{cases}$$

On every interval where $\Theta_q \neq 0$, put $v = \sigma\Omega_q/(\lambda_q\chi_1\Theta_q)$. The exact system is

$$\chi' = c_1 W + d_1, \qquad W' = c_2 g/\chi + d_2, \qquad v' = f - \widetilde{k}v^2, \qquad (\log P_q)' = \widetilde{k}v, \tag{182}$$

where these primes mean τ derivatives.

Proof. From (180), $\dot r_q = r_q E_2 + \Omega_{q-1} d\cos\vartheta_q$, proving the first equation. Next

$$\dot\Omega_{q-1} = A_{q-1}\sin(\vartheta_q - \phi_q) = \frac{A_{q-1}}{\chi}(d\cos\phi_q - Z\cos\vartheta_q).$$

Split its last parentheses as $d - Z + d(\cos\phi_q - 1) - Z(\cos\vartheta_q - 1)$. The profile gives $1 - Z/d = g$ on each of its two intervals, proving the second equation including the full d_2. Also

$$\frac{a_q\lambda_q\chi_1}{\sigma\Gamma} = \frac{\chi_1}{\sigma^2\sin s}\left(\frac{A_{q-1}d}{\chi^2} + \frac{R_q}{\chi}\right) = \widetilde{k}, \qquad \frac{\sigma b_q}{\Gamma\lambda_q\chi_1} = \frac{Z}{\chi_1\sin s} = f.$$

Differentiate the quotient and $\log P_q$ to obtain the last equations. This proves the exact substitution without freezing $A_{q-1}, \Omega_{q-1}, \rho_{q-1}$, or any older layer. $\square$

Where source selection is proved. The source first proves interval-uniform bounds for (182) in Lemma 3.9, pp. 18–21, then excludes zeros of Θ_q in Proposition 3.10, pp. 21–22. For the actual stage it first constructs the closed older/geometry subsystem on a common compact subset of

$$\det M_j > 1/2, \quad |M_j^{-T} p_j| > 1/4, \quad \zeta_{q-1,2} > z_*/2, \quad \zeta_{q,2} > z_*/2,$$

by the simultaneous bounds in Sections 3.7.2–3.7.6, pp. 26–32. In original units the pair data and coefficient sizes change by the dictionary above, while this geometric domain is unchanged.

The newest pair is absent from that subsystem: F_q^{ctl} uses only older vorticities and phases, and M_q uses only $D_{<q}$. One then solves the newest linear pair over the constructed subsystem. This order matters for the exact map (182). The source's $c_i = 1 + O(\epsilon_{\mathrm{sch}})$, $d_i = O(\epsilon_{\mathrm{sch}})$ estimates are consequences of its coupled induction, not independent prescriptions available after changing those dynamics.

For reference, source (3.67), p. 32, gives on the selected actual stage

$$\frac{\Omega_{q-1}(t_q^b)}{\sigma} \in [1/3 - C\epsilon_{\mathrm{sch}},\, 1 + \bar{v} + C\epsilon_{\mathrm{sch}}], \qquad \log \frac{P_q(t_q^b)}{(\nu_{\mathrm{AB}}^2/\mu)\widehat{\lambda}_q^{-7/8}} \in [1 + I_0 - C\epsilon_{\mathrm{sch}},\, 1 + \bar{v} + C\epsilon_{\mathrm{sch}}],$$

where $I_0 = \int_0^1 [(1+\tau^2/2)^2(1+\tau)]^{-1}\mathrm{d}\tau \in [2/9, \log 2]$. Source (3.58), p. 27, gives the separate first-stage gain interval $[1 + \log 2, 2 + \Lambda_1^{-1}]$ and exact duration $(L_1 + 2 + \Lambda_1^{-1})/(\nu_{\mathrm{AB}} \sin s_*)$. These are source bounds, not claims that the diffusive evolution below satisfies them without further estimates.

An explicit growth-end vorticity lower bound. The source growth proof is Section 3.7.5, pp. 29–30, including (3.63)–(3.64), not the later steering quotient alone. Its identities admit the following explicit extraction from the source's choice of low-order tolerance. Put $u = \Omega_q/\Theta_q$, $u_* = (b_q/a_q)^{1/2}$, $\gamma = (a_q b_q)^{1/2}$, and $z = u/u_*$. The source chooses its fixed error tolerance before its frequency threshold; require the following finite additional strict comparison when making that same choice: all the growth errors used here have absolute value at most $\delta \le 1/4$. Thus its proved geometric and coefficient estimates give

$$r_q \ge 1 - \delta, \qquad \frac{d_q^{\mathrm{adj}}}{\sin s_q} \ge 1 - \delta, \qquad \frac{\mathcal{G}_q}{\sigma_{q-1}^2 \sin s_q} \le 1 + \delta, \qquad |(\log u_*)'| \le \delta\gamma.$$

The original source's error rates in (3.63) are $C\epsilon_{\mathrm{sch}}\Gamma_q/L_q$, while $\gamma = (1 + O(\epsilon_{\mathrm{sch}}))\Gamma_q$, so this comparison is obtained by the same fixed-data choice of ϵ_{sch}; it imposes no new order-dependent frequency condition.

Here is the complete quotient proof used for the extraction. Direct differentiation before any possible zero of Θ_q gives

$$u' = b_q - a_q u^2, \qquad z' = \gamma(1 - z^2) - z(\log u_*)', \qquad z(t_q^{\mathrm{in}}) = 1.$$

At $z = 1 + \delta$, the right side is at most $\gamma[-2\delta - \delta^2 + \delta(1+\delta)] = -\delta\gamma < 0$; at $z = 1 - \delta$ it is at least $\gamma[2\delta - \delta^2 - \delta(1-\delta)] = \delta\gamma > 0$. The first-exit argument proves $1 - \delta \le z \le 1 + \delta$. This bound also excludes a first zero of Θ_q: at such a zero it would force $\Omega_q = 0$, contradicting backward uniqueness of the continuous-coefficient linear system from its nonzero entry datum. Consequently the bound holds through t_q^a. On growth the preceding phase is vertical, so $b_q = \lambda_q d_q^{\mathrm{adj}}$ and $a_q = \mathcal{G}_q/(\lambda_q r_q)$. It follows exactly that

$$u_* = \lambda_q \sqrt{\frac{d_q^{\mathrm{adj}} r_q}{\mathcal{G}_q}}, \qquad \frac{|\Omega_q|}{|\Theta_q|} = z u_* \ge \frac{\lambda_q}{\sigma_{q-1}} \frac{(1-\delta)^2}{\sqrt{1+\delta}} \ge \frac{\lambda_q}{2\sigma_{q-1}}.$$

For the last inequality the function $(1-\delta)^2/\sqrt{1+\delta}$ is decreasing on $[0, 1/4]$ by direct logarithmic differentiation, and its value at $1/4$ is $9/(8\sqrt{5}) > 1/2$ because $81 > 80$. Thus the exact physical growth threshold satisfies

$$|\Omega_q(t_q^a)| \ge \frac{\lambda_q}{2\sigma_{q-1}} \frac{\nu_{\mathrm{AB}}^2}{\mu} \widehat{\lambda}_q^{-7/8} = \frac{\nu_{\mathrm{AB}}^2}{2\sigma_{q-1}} \widehat{\lambda}_q^{1/8}, \qquad r_q(t_q^a) \ge 3/4. \tag{183}$$

For $q = 1$ the exact growing eigenline instead gives equality $|\Omega_1/\Theta_1| = \lambda_1/\nu_{\mathrm{AB}}$ and $r_1 = 1$, which is stronger than the displayed lower bound. No unproved viscous tracking assertion is used in this extraction.

The actual profile and retained viscosity. The source profile, (4.1), p. 37, is

$$F(s) = \sin s + \chi(s)(s - \sin s), \qquad |s| \le \pi,$$

periodically continued, where χ is even, smooth, compactly supported in $(-\pi, \pi)$ and one on $[-s_0, s_0]$. Thus $F = s$ on that interval. Direct differentiation of the local waves gives the retained-diffusion residual increments

$$\begin{aligned}\mathcal{R}_\theta &= (\dot{\Theta}_j - a_j\Omega_j)F(s) - \kappa_{\mathrm{th}}\lambda_j^2 r_j^2 \Theta_j F''(s), \\ \mathcal{R}_\omega &= (\dot{\Omega}_j - b_j\Theta_j)F'(s) - \varepsilon_{\mathrm{v}}\lambda_j^2 r_j^2 \Omega_j F'''(s).\end{aligned} \tag{184}$$

Consequently the actual fixed-profile, finite-stage retained-viscosity realization uses the original coupled system and the explicit additional forces $-\kappa_{\mathrm{th}}\Delta\theta$ and $-\varepsilon_{\mathrm{v}}\Delta u$. The signs in (184) are the signs of these forces when diffusion is placed on the right of the PDE. They are not positive damping terms in the forcing.

This repair preserves the exact actual D, G in Proposition 1: on every older affine region, $F'' = F''' = 0$, all envelope derivatives vanish, and the source's finite corrections vanish on that plateau. Hence the repaired force increments vanish in each affine core, and both Laplacians of the full older affine fields are zero. In physical coordinates the affine core of layer j contains

$$B_{r_j^{\mathrm{aff}}(t)}, \qquad r_j^{\mathrm{aff}}(t) = \min\left\{\frac{\ell_j}{2\|M_j^{-1}\|}, \frac{s_0}{\lambda_j r_j}\right\}.$$

The exact source nesting criterion is $K_j K_q \ell_q < \ell_j/2$ and $\lambda_j r_j K_q \ell_q < s_0$ for every older layer $j < q$; scaling by μ preserves both inequalities. The full-vector repair and the unchanged support follow because differentiation of a smooth compactly supported field does not enlarge its support. Quantitative force costs are provided separately in the parent calculation; the algebraic identity alone gives no uniform terminal force bound.

The exact feedback terms in a scalar-damped comparison. For transparency, compute also the altered finite-dimensional equations obtained from Fourier eigenprofiles. They are not the released fixed profile. Let $d_j^\theta = \kappa_{\mathrm{th}}\lambda_j^2 r_j^2$ and $d_j^\omega = \varepsilon_{\mathrm{v}}\lambda_j^2 r_j^2$, and replace only the two amplitude equations by

$$\dot{\Theta}_j = a_j\Omega_j - d_j^\theta\Theta_j, \qquad \dot{\Omega}_j = b_j\Theta_j - d_j^\omega\Omega_j, \tag{185}$$

while computing all D, G, F_j^{ctl} from the altered amplitudes and phases. For a sine profile these are precisely the homogeneous local wave equations, since $F'' = -F$ and $F''' = -F'$. Repeating the proof of Proposition 1 gives exactly

$$\begin{aligned}\dot{G} + D^T G &= \sum_{i<q} \dot{w}_i B_i - \sum_{i<q} w_i d_i^\theta B_i, \\ \dot{C} - G_1 &= 2\ddot{\alpha} - A_0 \sin\alpha + \sum_{i<q} \dot{w}_i \Omega_i - \sum_{i<q} w_i d_i^\omega \Omega_i.\end{aligned} \tag{186}$$

In (177), replace $b_i\Theta_i$ by $b_i\Theta_i - d_i^\omega\Omega_i$. The geometric equations (178), (179), and (180) retain their formulas but now use altered amplitudes. The exact gradient-amplitude drift becomes

$$\frac{\dot{A}_i}{A_i} = \dot{\rho}_i - \frac{\mathcal{G}_i\Omega_i}{A_i} - d_i^\theta. \tag{187}$$

This follows by differentiating $A_i = \lambda_i r_i P_i$ and (185). In particular

$$\begin{aligned}\dot{\mathcal{G}}_j = {}& A_0 \cos d_{0j}(\Sigma_0 - \Sigma_j) + \sum_{1 \le i < j} \dot{w}_i A_i \sin d_{ij} \\ &+ \sum_{1 \le i < j} w_i A_i \left[\left(\dot{\rho}_i - \frac{\mathcal{G}_i\Omega_i}{A_i} - d_i^\theta\right)\sin d_{ij} + \cos d_{ij}(\Sigma_i - \Sigma_j)\right], \qquad \Sigma_0 = 0.\end{aligned}$$

No base term or activation derivative has been discarded.

The feedback derivative is also explicit. With $N_{ij} = w_i\Omega_i \sin\phi_i \sin d_{ij}$,

$$\dot{F}_j^{\mathrm{ctl}} = \frac{1}{\cos\phi_j}\sum_{i<j}\big[(\dot{w}_i\Omega_i + w_i(b_i\Theta_i - d_i^{\omega}\Omega_i))\sin\phi_i\sin d_{ij} \\ + w_i\Omega_i\cos\phi_i(-\dot{\alpha}-\Sigma_i)\sin d_{ij} + w_i\Omega_i\sin\phi_i\cos d_{ij}(\Sigma_i-\Sigma_j)\big] + F_j^{\mathrm{ctl}}\tan\phi_j(-\dot{\alpha}-\Sigma_j).$$

The direct damping term is therefore $-\sum_{i<j} w_i d_i^{\omega}\Omega_i \sin\phi_i\sin d_{ij}/\cos\phi_j$, in addition to all changes in the evolving state and the other terms displayed above.

For the later-stage variables and coefficients (181), direct differentiation gives the exact altered model

$$\chi' = c_1 W + d_1, \qquad W' = c_2 g/\chi + d_2 - \frac{d_{q-1}^{\omega}}{\Gamma}W, \tag{188}$$
$$v' = f - \widetilde{k}v^2 + \frac{d_q^{\theta}-d_q^{\omega}}{\Gamma}v, \qquad (\log P_q)' = \widetilde{k}v - \frac{d_q^{\theta}}{\Gamma}.$$

To prove the extra terms, $W' = \dot{\Omega}_{q-1}/(\sigma\Gamma)$ supplies its damping term; differentiating Ω_q/Θ_q supplies $(d_q^{\theta}-d_q^{\omega})\Omega_q/\Theta_q$; differentiating $\log(-\Theta_q)$ supplies $-d_q^{\theta}$. The previous derivation supplies all remaining terms. These are identities on every interval where their displayed denominators are nonzero, and do not assert a new existence theorem from assumed coefficient bounds.

Even when $\kappa_{\mathrm{th}} = \varepsilon_{\mathrm{v}}$, only the explicit damping term in the quotient equation cancels. The preceding-layer term $-d_{q-1}^{\omega}W/\Gamma$ remains, A_{q-1} and R_q obey the altered gradient drifts, and hence c_2, c_5, d_4 evolve differently. On holding, $\zeta_{q,1} = \Omega_q = 0$ still solves those two equations, but

$$P_q(t) = P_q(t_q^b)\exp\left(-\kappa_{\mathrm{th}}\lambda_q^2\int_{t_q^b}^{t} r_q(s)^2\mathrm{d}s\right)$$

is no longer stationary if $\kappa_{\mathrm{th}} > 0$. This follows by solving $\dot{P}_q = -d_q^{\theta}P_q$; r_q retains its evolving value inside the integral.

Finite-stage gain and exact diffusion integrals. For the damped comparison, (188) proves the exact identity

$$\log\frac{P_q(t_b)}{P_q(t_a)} = \int_{t_a}^{t_b}\left(-\frac{\mathcal{G}_q\Omega_q}{A_q}\right)\mathrm{d}t - \kappa_{\mathrm{th}}\lambda_q^2\int_{t_a}^{t_b} r_q^2\mathrm{d}t.$$

For a stage partitioned into growth, transition, positive steering, negative steering, and holding, the last integral is the sum over all five actual intervals. Its original-unit coefficient is

$$\kappa_{\mathrm{th}}\lambda_q^2\int r_q^2\mathrm{d}t = \frac{\kappa_{\mathrm{th}}\mu^2}{\nu_{\mathrm{AB}}}\widehat{\lambda}_q^2\int \widehat{r}_q^2\mathrm{d}T.$$

On any finite interval of length H with proved bounds $m \le r_q \le M$, its exact value is bracketed by $\kappa_{\mathrm{th}}\lambda_q^2 m^2 H$ and $\kappa_{\mathrm{th}}\lambda_q^2 M^2 H$. The inequality is simply the integrated pointwise inequality $m^2 \le r_q^2 \le M^2$. It does not permit retaining the original source gain integral when the coupled parent trajectory has changed. Conversely, in the actual fixed-profile force repair the trajectory and the gain integral remain exactly those of the source, and the diffusion cost is paid by the additional explicit force, not by a damping loss in that trajectory.

Status of the infinite sequence. The source constructs its inviscid infinite sequence after fixing low-order design bounds, then all coefficient/correction constants, then the nondecreasing majorant K of (7.21), p. 63, and finally one initial frequency threshold in (8.1)–(8.6), pp. 66–67. It proves the coupled stages before evaluating each new σ_q, and it proves full-lifetime bounds before applying the finite corrections. The retained-viscosity finite-stage force repair proved here commutes exactly with every finite activated sum. Extending its force series smoothly to the limiting time requires estimates for the additional Laplacian terms; no such terminal estimates have been established in this document. The exact Fourier comparison (188) does not supply them, and its altered parent terms prevent importing the source's inviscid coefficient bounds without a new calculation. Neither statement is a disproof of another retained-viscosity construction.

12 Exact periodic profile evolution and the affine region

This calculation concerns the precise periodic profile in Alpöge–Buckmaster, *Blowup for the Boussinesq equations with smooth forcing*, released 76-page source, Sections 3.1 and 4.1, equations (3.1), (3.2), (4.1)–(4.4). It proves exact identities and a profile evolution for a prescribed affine background, including the exact force that keeps the original profile. It also identifies the change to the affine region when that force is omitted. No conclusion about an infinite viscous controlled sequence follows from these profile identities alone.

12.1 The original profile and all coefficients

Let $0 < s_0 < 1$ and let $\chi \in C_c^\infty((-\pi,\pi))$ be even and equal to one on $[-s_0, s_0]$. The actual source profile is the 2π-periodic extension of

$$F(s) = \sin s + \chi(s)(s - \sin s), \qquad -\pi < s < \pi. \tag{189}$$

It is smooth because it equals $\sin s$ near the endpoints of this period; it is odd because χ is even. Hence its period mean is zero. It equals s on $[-s_0, s_0]$, in particular $F'(0) = 1$. Let P be its unique mean-zero periodic primitive. Indeed $\widetilde{P}(s) = \int_0^s F(r)\,\mathrm{d}r$ is even and periodic, and $P = \widetilde{P} - \langle \widetilde{P} \rangle$ has the claimed properties. Here $\langle h \rangle = (2\pi)^{-1} \int_{-\pi}^{\pi} h(s)\,\mathrm{d}s$.

Fix a closed finite time interval $I = [t_0, t_1]$, smooth $D : I \to \mathbb{R}^{2\times 2}$ with $\operatorname{tr} D = 0$, smooth $G : I \to \mathbb{R}^2$, a frequency $\lambda > 0$, and a nonzero initial vector $\zeta(t_0)$. Define $J(x_1, x_2) = (-x_2, x_1)$ and

$$\dot{\zeta} = -D^T \zeta, \qquad s(x,t) = \lambda \zeta(t)\cdot x, \qquad \kappa(t) = |\zeta(t)|^2, \qquad K(t) = \lambda^2 \kappa(t). \tag{190}$$

The notation $\kappa = |\zeta|^2$ is the source's geometric notation. The physical temperature diffusivity is $\kappa_{\mathrm{th}} \ge 0$, and the physical momentum viscosity is $\nu_{\mathrm{phys}} \ge 0$; neither is the paper's time-scaling parameter named ν. These constants are not set to one. This calculation does not rescale A_0, λ_0, μ, the physical coordinates, time, or laboratory gravity $e_2 = (0, 1)$.

The nonvanishing of ζ follows directly from the invertible fundamental matrix of $\dot{M} = DM$, $M(t_0) = \mathrm{Id}$: differentiation gives $\zeta = M^{-T}\zeta(t_0)$, and differentiation of the determinant gives $\det M = \exp \int_{t_0}^t \operatorname{tr} D = 1$. Write

$$a(t) = -\frac{J\zeta \cdot G}{\lambda \kappa}, \qquad b(t) = \lambda \zeta_1. \tag{191}$$

In particular the buoyancy coefficient uses the first laboratory component ζ_1.

For any smooth scalar θ and divergence-free velocity u, with $\omega = \partial_1 u_2 - \partial_2 u_1$, define the full retained-diffusion residuals by

$$\mathcal{R}_\theta(\theta, u) = \partial_t \theta + u \cdot \nabla \theta - \kappa_{\mathrm{th}} \Delta \theta, \qquad \mathcal{R}_\omega(\theta, u) = \partial_t \omega + u \cdot \nabla \omega - \partial_1 \theta - \nu_{\mathrm{phys}} \Delta \omega. \tag{192}$$

Thus these residuals are the required scalar force and curl of the required momentum force. On the region under consideration the older fields are $\theta_{\mathrm{old}} = G \cdot x$, $u_{\mathrm{old}} = Dx$; they need not have zero residual. All statements below subtract their actual residuals.

12.2 Exact residual and failure of constant-profile damping

Proposition 1. For arbitrary smooth real amplitudes Θ, Ω, define the source wave

$$\vartheta = \Theta F(s), \quad \psi = \frac{\Omega}{\lambda^2 \kappa} P(s), \quad v = \nabla^\perp \psi = \frac{\Omega}{\lambda \kappa} J\zeta F(s), \quad \varpi = \Delta \psi = \Omega F'(s). \tag{193}$$

The scalar and curl residual increments are exactly

$$\mathcal{R}_\theta(\theta_{\text{old}} + \vartheta, u_{\text{old}} + v) - \mathcal{R}_\theta(\theta_{\text{old}}, u_{\text{old}}) = (\dot{\Theta} - a\Omega)F - \kappa_{\text{th}} K \Theta F'', \tag{194}$$
$$\mathcal{R}_\omega(\theta_{\text{old}} + \vartheta, u_{\text{old}} + v) - \mathcal{R}_\omega(\theta_{\text{old}}, u_{\text{old}}) = (\dot{\Omega} - b\Theta)F' - \nu_{\text{phys}} K \Omega F'''. \tag{195}$$

Every profile in these formulas is evaluated at $s(x, t)$.

Proof. Equation (190) gives $(\partial_t + Dx \cdot \nabla)s = \lambda(\dot{\zeta} + D^T \zeta) \cdot x = 0$. The orthogonality $J\zeta \cdot \zeta = 0$ gives both $v \cdot \nabla \vartheta = 0$ and $v \cdot \nabla \varpi = 0$. The older vorticity $D_{21} - D_{12}$ is spatially constant, so $v \cdot \nabla \omega_{\text{old}} = 0$. Moreover $v \cdot G = -a\Omega F$, $\partial_1 \vartheta = b\Theta F'$, $\Delta \vartheta = K\Theta F''$, and $\Delta \varpi = K\Omega F'''$. Expanding (192), canceling the older residual, and retaining all these terms proves (194)–(195). The displayed velocity and vorticity in (193) follow by differentiating $P' = F$ twice; their signs follow from the specified J. $\square$

The profile derivatives, including their transition-region terms, are

$$F' = \cos s + \chi'(s)(s - \sin s) + \chi(s)(1 - \cos s), \tag{196}$$
$$F'' = (\chi - 1)\sin s + 2\chi'(1 - \cos s) + \chi''(s - \sin s), \tag{197}$$
$$F''' = (\chi - 1)\cos s + 3\chi' \sin s + 3\chi''(1 - \cos s) + \chi'''(s - \sin s). \tag{198}$$

These follow by the product rule on $(-\pi, \pi)$ and extend smoothly across the period endpoints.

Proposition 2. There is no constant $c \in \mathbb{R}$ such that $F'' = cF$ on the circle, and no constant $c \in \mathbb{R}$ such that $F''' = cF'$ on the circle. Consequently, at a time with $\kappa_{\text{th}} > 0$ and $\Theta \neq 0$, no choice of scalar $\dot{\Theta} - a\Omega$ makes (194) vanish for every phase. At a time with $\nu_{\text{phys}} > 0$ and $\Omega \neq 0$, no choice of scalar $\dot{\Omega} - b\Theta$ makes (195) vanish for every phase.

Proof. On $(-s_0, s_0)$, $F(s) = s$ and $F'' = 0$. An identity $F'' = cF$ therefore implies $cs = 0$ for every such s, so $c = 0$. Then $F'' = 0$ everywhere would make F globally affine. Its periodicity would force its slope to be zero, contradicting $F'(0) = 1$. If $F''' = cF'$, evaluation inside that same interval gives $0 = c$, since $F' = 1$ there. Then $F''' = 0$ everywhere makes F a quadratic polynomial on $\mathbb{R}$. Periodicity forces both its quadratic and linear coefficients to vanish, again contradicting $F'(0) = 1$. Since $K > 0$, each asserted residual cancellation with a nonzero amplitude would imply precisely one of these excluded constant-coefficient identities, by division by $\kappa_{\text{th}} K\Theta$ or $\nu_{\text{phys}} K\Omega$. $\square$

For example a proposed common damping equation $\dot{\Theta} = a\Omega - d\Theta$, $\dot{\Omega} = b\Theta - d\Omega$ gives $-d\Theta F - \kappa_{\text{th}} K\Theta F''$ and $-d\Omega F' - \nu_{\text{phys}} K\Omega F'''$ as its two residuals. On the exact affine core they are $-d\Theta s$ and $-d\Omega$. This proves failure of that fixed-profile amplitude prescription, not failure of a viscous controlled construction.

12.3 An exact evolving-profile map for all diffusivities

We now construct the profile evolution instead of replacing the missing profile dynamics by scalar damping. Let $T(s, t), V(s, t)$ be smooth, mean-zero periodic functions, and let $Q_s = V$, $\langle Q \rangle = 0$. Set

$$\vartheta = T(s, t), \quad \psi = \frac{Q(s, t)}{\lambda^2 \kappa}, \quad v = \frac{J\zeta}{\lambda \kappa} V(s, t), \quad \varpi = V_s(s, t). \tag{199}$$

The time derivative T_t here and below holds phase s fixed.

Proposition 3. The residual increments for (199) are

$$T_t - aV - \kappa_{\mathrm{th}} K T_{ss}, \qquad \partial_s (V_t - bT - \nu_{\mathrm{phys}} K V_{ss}). \tag{200}$$

Their vanishing is equivalent in the mean-zero class to

$$T_t = aV + \kappa_{\mathrm{th}} K T_{ss}, \qquad V_t = bT + \nu_{\mathrm{phys}} K V_{ss}. \tag{201}$$

For the initial data $T(s,t_0) = \Theta_0 F(s)$, $V(s,t_0) = \Omega_0 F(s)$, this system has a unique smooth periodic solution on I given by the following exact Fourier evolution. Write

$$F(s) = \sum_{n\ge 1} f_n \sin(ns), \qquad f_n = \frac{1}{\pi}\int_{-\pi}^{\pi} F(s)\sin(ns)\,\mathrm{d}s. \tag{202}$$

For each $n \ge 1$, let $U_n(t,t_0)$ solve

$$\partial_t U_n = \begin{pmatrix} -\kappa_{\mathrm{th}} K n^2 & a \\ b & -\nu_{\mathrm{phys}} K n^2 \end{pmatrix} U_n, \qquad U_n(t_0,t_0) = \mathrm{Id}. \tag{203}$$

Then

$$\begin{pmatrix} T(s,t) \\ V(s,t) \end{pmatrix} = \sum_{n\ge 1} f_n U_n(t,t_0) \begin{pmatrix} \Theta_0 \\ \Omega_0 \end{pmatrix} \sin(ns). \tag{204}$$

Proof. The phase and tangency computations in Proposition 1 remain valid, including $v\cdot\nabla T = v\cdot\nabla V_s = 0$. Now $v \cdot G = -aV$ and $\partial_1 T = bT_s$, which proves (200). The expression inside its second derivative has zero period mean. If its phase derivative vanishes, it is constant in s, and its zero mean forces this constant to be zero. This proves the exact equivalence to (201).

For completeness, each matrix solution in (203) can be defined by the convergent iterated-integral series $U_n = \mathrm{Id} + \sum_{j\ge 1}\int_{t_0<r_j<\cdots<r_1<t} A_n(r_1)\cdots A_n(r_j)\,\mathrm{d}r_j\cdots\mathrm{d}r_1$, where A_n is the displayed matrix. For a fixed n, its jth term is bounded by $(|I|\sup_I \|A_n\|)^j/j!$, so termwise differentiation gives the ODE. Uniqueness follows by integrating the difference and applying the iterated integral estimate, which tends to zero as $j \to \infty$. For its column solution $y = (y_1, y_2)$, direct differentiation gives

$$\tfrac{1}{2}\partial_t |y|^2 = -Kn^2(\kappa_{\mathrm{th}} y_1^2 + \nu_{\mathrm{phys}} y_2^2) + (a+b) y_1 y_2 \le \tfrac{1}{2}(|a|+|b|)|y|^2.$$

Multiplication by $\exp(-\int_{t_0}^t (|a|+|b|))$ and differentiation therefore proves the bound, uniform in n,

$$|y(t)| \le e^{\frac{1}{2}\int_{t_0}^t (|a|+|b|)} |y(t_0)|.$$

Repeated differentiation of (203) gives, for each fixed j, $\sup_I |\partial_t^j y| \le C_j (1+n)^{2j} |y(t_0)|$, by induction: the coefficient derivatives of A_n are bounded by constants times $(1+n)^2$, and the differentiated product has finitely many terms.

Integration by parts any prescribed number of times in (202) shows $|f_n| \le C_N n^{-N}$ for every N, without endpoint terms by periodicity. Thus (204) and every mixed derivative converge uniformly on $\mathbb{T} \times I$, and termwise differentiation proves the equations and the prescribed initial values. The Fourier expansion indeed equals F: its absolutely convergent series has the same Fourier coefficients as F, and a continuous function with all Fourier coefficients zero is zero. To see this last assertion directly, convolve with the Fejér kernel $\mathfrak{F}_N(r) = N^{-1}(\sin(Nr/2)/\sin(r/2))^2$. It has integral 2π, is nonnegative, and its integral outside every fixed neighborhood of 0 tends to zero because its numerator is at most one and the denominator is bounded below there. Uniform continuity therefore gives uniform convergence of its convolution to the continuous function. When all Fourier coefficients are zero each such convolution is zero, proving the assertion. The same Fourier coefficient ODE and its uniqueness show uniqueness among smooth solutions of (201). ☐

Proposition 4. The map in Proposition 3 also realizes the full momentum equation. Let $S_s = T$, $\langle S \rangle = 0$, and retain $Q_s = V$, $\langle Q \rangle = 0$. For a solution of (201), the exact pressure increment is

$$p^{\mathrm{wave}}(x,t) = -\frac{2\zeta \cdot DJ\zeta}{\lambda^2\kappa^2} Q(s,t) + \frac{\zeta_2}{\lambda\kappa} S(s,t). \tag{205}$$

With this pressure increment, adding (T, v) to any affine older state preserves its actual vector momentum residual, including retained viscosity ν_{phys} and the unchanged laboratory buoyancy θe_2.

Proof. Put $c = J\zeta/(\lambda\kappa)$, a vector in this proof. The vector momentum residual increment before adding pressure is

$$(\dot{c} + Dc)V + c(V_t - \nu_{\mathrm{phys}} K V_{ss}) - Te_2.$$

Indeed the transported phase has zero material derivative under Dx, $v\cdot\nabla v = 0$, $v\cdot\nabla(Dx) = Dv$, and $\Delta v = KcV_{ss}$. For every trace-free 2×2 matrix, $DJ = -JD^T$; multiplying its four entries verifies the identity. It gives $\dot{c} = Dc - c\dot{\kappa}/\kappa$ and $\dot{\kappa} = -2\zeta \cdot D\zeta$. The identity $(J\zeta)\cdot D(J\zeta) + \zeta \cdot D\zeta = \kappa \operatorname{tr} D = 0$ shows that $\dot{c} + Dc$ is orthogonal to $J\zeta$. Its component along ζ is therefore

$$\dot{c} + Dc = \frac{2\zeta \cdot DJ\zeta}{\lambda\kappa^2}\zeta.$$

Also $bc - e_2 = -\zeta_2\zeta/\kappa$, by the two explicit components of $J\zeta = (-\zeta_2, \zeta_1)$. Substituting the second profile equation reduces the residual to

$$\zeta \left[\frac{2\zeta \cdot DJ\zeta}{\lambda\kappa^2} V - \frac{\zeta_2}{\kappa} T \right].$$

The spatial gradient of (205) is its negative, because $\nabla s = \lambda\zeta$, $Q_s = V$ and $S_s = T$. This proves the full momentum statement. Mean-zero periodic primitives exist because the profiles have zero mean, by the same integration argument used for P above. □

Proposition 5. If $\kappa_{\mathrm{th}} = \nu_{\mathrm{phys}} = \delta \geq 0$, put

$$\tau(t) = \delta\lambda^2 \int_{t_0}^{t} |\zeta(r)|^2 \, \mathrm{d}r, \qquad F_\tau(s) = \mathcal{H}_\tau F(s) = \sum_{n\geq 1} f_n e^{-n^2\tau} \sin(ns). \tag{206}$$

Let $\dot{\Theta} = a\Omega$, $\dot{\Omega} = b\Theta$ with the given initial amplitudes. The exact solution in Proposition 3 is

$$T = \Theta F_\tau, \quad V = \Omega F_\tau, \quad Q = \Omega \mathcal{H}_\tau P, \quad \varpi = \Omega \partial_s F_\tau. \tag{207}$$

In particular this is an exact retained-diffusion map with the same prescribed D, G, ζ, all original frequency factors, and laboratory gravity. Each nonzero temperature mode has exact gain

$$\frac{T_n(t)}{T_n(t_0)} = \frac{\Theta(t)}{\Theta_0} e^{-n^2\tau(t)} \quad (f_n\Theta_0 \neq 0). \tag{208}$$

For vorticity the corresponding ratio is $(\Omega(t)/\Omega_0)e^{-n^2\tau(t)}$ when $nf_n\Omega_0 \neq 0$.

Proof. The matrix in (203) is $B(t) - \delta K(t) n^2 \mathrm{Id}$, where $B = \left(\begin{smallmatrix} 0 & a \\ b & 0 \end{smallmatrix}\right)$. If U solves $\dot{U} = BU$, differentiation of $e^{-n^2\tau(t)}U(t,t_0)$ gives exactly (203), including its initial identity. Hence uniqueness proves this factorization and (207). Since derivatives commute termwise with the Fourier series, $(\mathcal{H}_\tau P)_s = \mathcal{H}_\tau F$ and its mean remains zero. This verifies the streamfunction as well as the scalar and vorticity profiles. Formula (208) is division of the actual mode coefficients, retaining its stated nonzero denominator. □

The exact temperature phase norm is

$$\|T(\cdot,t)\|^2_{L^2(-\pi,\pi)} = \pi\Theta(t)^2 \sum_{n\ge 1} f_n^2 e^{-2n^2\tau(t)}. \tag{209}$$

This follows by integrating the uniformly convergent squared series and using the exact orthogonality identity

$$\int_{-\pi}^{\pi} \sin(ns)\sin(ms)\,\mathrm{d}s = \pi\mathbf{1}_{n=m}.$$

Thus $\|F_\tau\|_2 \le e^{-\tau}\|F\|_2$, and for any index with $f_n \neq 0$ one has $\|F_\tau\|_2 \ge \sqrt{\pi}|f_n|e^{-n^2\tau}$. These are inequalities about phase profiles; an unlocalized affine wave need not be integrable over $\mathbb{R}^2$.

12.4 Positive heat time and the affine core

Proposition 6. For every $\tau > 0$, $\mathcal{H}_\tau F$ has no affine restriction to any nonempty open interval of $\mathbb{R}$. In particular it cannot retain the source's nontrivial exact linear phase interval. Also F has infinitely many nonzero Fourier modes.

Proof. For any $R > 0$ and any nonnegative integer m, the series of mth complex derivatives of $\sum_{n\ge1} f_n e^{-n^2\tau}\sin(nz)$ is bounded on $|\operatorname{Im} z| \le R$ by $\sum_{n\ge1}|f_n|n^m e^{-n^2\tau+nR} < \infty$. For instance boundedness of f_n and completion of the square make the tail bounded by a polynomial times $e^{-\tau n^2/2}$. The sum therefore defines an entire function $h(z)$ and has its termwise derivatives. If $h(s) = cs + d$ on a nonempty real open interval, the entire function $h(z) - cz - d$ vanishes identically. The identity principle used here follows from Taylor expansion: a nonzero holomorphic function has a first nonzero Taylor coefficient at each zero where its Taylor series is not identically zero, hence that zero is isolated; an accumulating set of zeros forces the Taylor series to vanish locally, and this local vanishing propagates through the connected domain. Periodicity now gives $2\pi c = h(z+2\pi) - h(z) = 0$. Oddness gives $d = 0$. Thus every Fourier coefficient $f_n e^{-n^2\tau}$ would vanish, implying $f_n = 0$ for all n. The Fourier uniqueness argument in Proposition 3 would imply $F = 0$, contradicting $F'(0) = 1$. If F had only finitely many nonzero modes, the same entire-function argument, applied to its finite trigonometric sum at heat time zero, would contradict its interval $F(s) = s$. □

There is nevertheless an exact estimate of the defect deep inside the original interval. Define, for $\tau > 0$ and $d > 0$,

$$\mathcal{E}(d,\tau) = \frac{2}{\sqrt{\pi}}\int_{d/(2\sqrt{\tau})}^{\infty} e^{-r^2}\,\mathrm{d}r \le \frac{2\sqrt{\tau}}{d\sqrt{\pi}}e^{-d^2/(4\tau)}. \tag{210}$$

The inequality follows from $r \ge d/(2\sqrt{\tau})$ and $\int_a^\infty re^{-r^2}\mathrm{d}r = e^{-a^2}/2$.

Proposition 7. For $0 < r_0 < s_0$, $d = s_0 - r_0$, and $|s| \le r_0$,

$$|\partial_s F_\tau(s) - 1| \le \|F' - 1\|_\infty \mathcal{E}(d,\tau), \tag{211}$$

$$|F_\tau(s) - s| \le |s|\|F' - 1\|_\infty \mathcal{E}(d,\tau), \tag{212}$$

$$|\partial_s^m F_\tau(s)| \le \|F^{(m)}\|_\infty \mathcal{E}(d,\tau) \quad (m \ge 2). \tag{213}$$

Proof. The periodized Gaussian kernel of $\mathcal{H}_\tau$ is $\sum_{k\in\mathbb{Z}}(4\pi\tau)^{-1/2}e^{-(r+2\pi k)^2/(4\tau)}$. To verify this representation, its nth Fourier multiplier is the Gaussian integral $\int_{\mathbb{R}}(4\pi\tau)^{-1/2}e^{-r^2/(4\tau)}e^{-inr}\mathrm{d}r = e^{-n^2\tau}$. One direct verification of this integral differentiates it with respect to n, integrates the Gaussian derivative by parts, and obtains $I'(n) = -2\tau nI(n)$ with $I(0) = 1$; the last normalization is the usual Gaussian integral, obtained by squaring it and integrating in polar coordinates. Fourier uniqueness then proves the representation. For any periodic bounded function h vanishing on

$[-s_0, s_0]$, its periodic lift satisfies $h(s-r) = 0$ whenever $|s| \le r_0$ and $|r| < d$. The Gaussian representation on the line therefore gives

$$|\mathcal{H}_\tau h(s)| \le \|h\|_\infty \int_{|r|\ge d} (4\pi\tau)^{-1/2} e^{-r^2/(4\tau)}\,\mathrm{d}r = \|h\|_\infty \mathcal{E}(d,\tau).$$

Apply this to $h = F' - 1$ and $h = F^{(m)}$ for $m \ge 2$, using commutation of derivatives with $\mathcal{H}_\tau$ and $\mathcal{H}_\tau 1 = 1$. Finally $F_\tau(0) = 0$ by oddness; integrating the first bound between 0 and s proves the second bound with the same d. □

The source background in equation (3.3) is obtained because each older layer is exactly affine on the newer support, by equations (4.3)–(4.6). For a heat-evolved older layer with nonzero scalar or velocity amplitude, Proposition 6 removes this exact affine premise. Its gradient at the origin can still be computed, but its values away from the origin are not that gradient times x. The tail estimates give explicit defects rather than an identification of those two fields. Consequently Proposition 5 alone supplies no coupled feedback law for the heat-modified older background.

12.5 Exact preservation by force and the localized diffusion costs

For the original inviscid amplitude equations $\dot\Theta = a\Omega$, $\dot\Omega = b\Theta$, Proposition 1 shows that the exact additional scalar and curl forces are

$$f_{\theta,\mathrm{diff}} = -\kappa_{\mathrm{th}} K\Theta F'', \qquad E_{\mathrm{diff}} = -\nu_{\mathrm{phys}} K\Omega F'''. \tag{214}$$

They vanish on the entire affine interval $|s| < s_0$. This retains exactly the original core, amplitudes, phase and hence its local affine feedback; it pays a nonzero diffusion force in the rest of the profile. In the unlocalized wave this force is periodic in phase and is not a compactly supported force on $\mathbb{R}^2$.

Here is the exact compact counterpart for the leading fields in source equation (4.2). Let M solve $\dot M = DM$, $M(t_0) = \mathrm{Id}$, let $p = \zeta(t_0)$, and let $\ell > 0$. Fix the source's radial $f \in C_c^\infty(B_1)$, $f = 1$ on $B_{1/2}$, and put

$$\begin{aligned} a_{\mathrm{mat}} &= M^{-1}x, \qquad g(a_{\mathrm{mat}}) = f(a_{\mathrm{mat}}/\ell), \\ \theta^L &= w\Theta F(s) g(a_{\mathrm{mat}}), \qquad \psi^L = \frac{w\Omega}{\lambda^2\kappa} P(s) g(a_{\mathrm{mat}}), \quad u^L = \nabla^\perp \psi^L. \end{aligned} \tag{215}$$

The activation $w(t)$ remains the original smooth scalar function; no time differentiation of it is omitted from the existing inviscid residual. The following formulas give exactly the additional diffusion terms, with w retained. At a fixed time put

$$B = M^{-1}, \quad H = BB^T, \quad q = B\zeta, \quad \mathscr{L} = \lambda^2\kappa\partial_s^2 + 2\lambda q\cdot\nabla_a\partial_s + H : \nabla_a^2. \tag{216}$$

Although $s = \lambda p \cdot a_{\mathrm{mat}}$ upon evaluation, s, a are independent variables in this differential operator.

Proposition 8. For every smooth profile $A(s,a)$, $\Delta_x[A(\lambda\zeta\cdot x, Bx)] = (\mathscr{L}A)(\lambda\zeta\cdot x, Bx)$. Consequently the exact added forces for (215) are

$$f^L_{\theta,\mathrm{diff}} = -\kappa_{\mathrm{th}} w\Theta\big[\lambda^2\kappa F'' g + 2\lambda F' q\cdot\nabla g + F(H : \nabla^2 g)\big], \tag{217}$$

$$f^L_{u,\mathrm{diff}} = -\nu_{\mathrm{phys}}\Delta u^L = -\nu_{\mathrm{phys}}\nabla^\perp\Delta\psi^L, \qquad E^L_{\mathrm{diff}} = -\nu_{\mathrm{phys}}\frac{w\Omega}{\lambda^2\kappa}\mathscr{L}^2(Pg). \tag{218}$$

Writing $\mathscr{H} = H : \nabla_a^2$, the last operator is explicitly

$$\begin{aligned} \mathscr{L}^2(Pg) = {}& \lambda^4\kappa^2 F''' g + 4\lambda^3\kappa F'' q\cdot\nabla g + 2\lambda^2\kappa F'\mathscr{H} g + 4\lambda^2 F'(q\cdot\nabla)^2 g \\ & + 4\lambda F(q\cdot\nabla)\mathscr{H} g + P\mathscr{H}^2 g. \end{aligned} \tag{219}$$

All these forces are supported in $M\overline{B}_\ell$. They vanish in the exact affine region $MB_{\ell/2} \cap \{|s| < s_0\}$.

Proof. The chain rule gives $\partial_{x_i} A = \lambda\zeta_i A_s + \sum_j B_{ji} A_{a_j}$. Applying it a second time and summing in i gives the three terms in (216), since $\sum_i B_{ji}\zeta_i = (B\zeta)_j$ and $\sum_i B_{ji}B_{ki} = (BB^T)_{jk}$. Applying the same identity twice to Pg proves (218). Expanding the square of the commuting constant-in-(s,a) operators in (216), and using $P' = F$, $P'' = F'$, $P''' = F''$, $P'''' = F'''$, proves all six terms of (219). Spatial differentiation does not enlarge the support of a smooth compactly supported function, proving the support statement. On the plateau $g = 1$ all positive derivatives of g vanish. On the linear phase interval $F'' = F''' = 0$. Thus the scalar and curl formulas vanish on their intersection. There $\psi^L = (w\Omega/(\lambda^2\kappa))(P(0) + s^2/2)$ is quadratic in x, so u^L is linear and $\Delta u^L = 0$, proving the vector-force statement as well. Finally $\operatorname{curl}(-\nu_{\text{phys}}\Delta u^L) = -\nu_{\text{phys}}\Delta^2\psi^L$, so the specified vector force realizes the curl force without any inverse-curl ambiguity. $\square$

For explicit quantitative costs let $C_j = \sup_a \|\nabla^j f(a)\|_{\text{HS}}$, with the Euclidean Hilbert–Schmidt tensor norm, and $C_0 = \|f\|_\infty$. Then $\|\nabla^j g\|_\infty = C_j \ell^{-j}$ by the chain rule. Write $h = \|H\|_{\text{HS}}$ and $q_* = |q|$. Cauchy–Schwarz on the tensor contractions in (217) and (219) proves

$$\|f^L_{\theta,\text{diff}}\|_\infty \le \kappa_{\text{th}}|w\Theta|\big[\lambda^2\kappa\|F''\|_\infty C_0 + 2\lambda q_*\|F'\|_\infty C_1\ell^{-1} + h\|F\|_\infty C_2\ell^{-2}\big], \tag{220}$$

$$\begin{aligned}\|E^L_{\text{diff}}\|_\infty \le \frac{\nu_{\text{phys}}|w\Omega|}{\lambda^2\kappa}\big[&\lambda^4\kappa^2\|F'''\|_\infty C_0 + 4\lambda^3\kappa q_*\|F''\|_\infty C_1\ell^{-1} \\ &+ (2\lambda^2\kappa h + 4\lambda^2 q_*^2)\|F'\|_\infty C_2\ell^{-2} + 4\lambda q_* h\|F\|_\infty C_3\ell^{-3} + h^2\|P\|_\infty C_4\ell^{-4}\big].\end{aligned} \tag{221}$$

The same reasoning applies to every compact correction: for a specified smooth correction (θ^c, ψ^c) the extra scalar and vector forces are exactly $-\kappa_{\text{th}}\Delta\theta^c$ and $-\nu_{\text{phys}}\nabla^\perp\Delta\psi^c$, with curl $-\nu_{\text{phys}}\Delta^2\psi^c$. This is the exact linear map from an already constructed inviscid forced field to a retained-diffusion forced field:

$$(\theta, u, p, f_\theta, f_u) \longmapsto (\theta, u, p, f_\theta - \kappa_{\text{th}}\Delta\theta, f_u - \nu_{\text{phys}}\Delta u). \tag{222}$$

Substitution in the scalar and momentum equations proves the map: adding $\kappa_{\text{th}}\Delta\theta$ or $\nu_{\text{phys}}\Delta u$ to the new right-hand side cancels its respective force decrement. Thus pressure, divergence, initial data and the entire trajectory are retained exactly. On each closed finite smooth stage these are smooth finite forces; where the original fields and their derivatives are compactly supported, the new terms have the same support. This statement supplies no uniform bound as the layer index tends to infinity. The explicit factors λ^2, ℓ^{-2}, and ℓ^{-4} above must be retained in such an analysis.

Source and scope receipt

The primary source read here is the 76-page PDF *Blowup for the Boussinesq equations with smooth forcing*, Levent Alpöge and Tristan Buckmaster; source equations (3.1)–(3.6) are on pages 8–9 and (4.1)–(4.6) on pages 37–38. The exact file SHA-256 is

`895a628d1783bcb039374686f50b895b5f450f53b8ef8aa173523487a7a4a21b.`

The companion replay script checks the profile differentiation, full two-dimensional residual algebra, localized Laplacian square, common diffusivity matrix factorization, and eigenfunction obstruction identities. Its checks supplement the proofs above; they do not certify an infinite-stage existence or force-smoothness theorem.

13 Exact localization and finite corrections with both diffusion coefficients

We now reconstruct the localization identities and finite correction mechanism of Alpöge and Buckmaster, *Blowup for the Boussinesq equations with smooth forcing*, Sections 4 and 6, keeping the coordinates and frequency parameters of Section 4. We also derive the corresponding finite correction equations when the thermal diffusion coefficient κ and the viscosity ν are nonnegative. This is an exact finite-layer calculation. The limiting many-layer construction and uniform force bounds up to a singular time are not asserted here.

13.1 Material coordinates and the full two residuals

Let $[t_0, t_1]$ be a compact time interval. Retain the smooth trace-free matrix $D(t)$ and vector $G(t)$, the backgrounds $u_b = Dx$ and $\theta_b = G \cdot x$, and their prescribed forces from (43)–(44). Solve

$$M' = DM, \qquad M(t_0) = \mathrm{Id}, \qquad p = \zeta(t_0) \neq 0, \qquad \zeta = M^{-T} p.$$

The existence and invertibility proof in Section 4 applies with initial time t_0. Differentiating the determinant by its polynomial cofactor formula gives $(\det M)' = (\operatorname{tr} D) \det M = 0$, so $\det M = 1$. Define, without changing units,

$$a = M^{-1}x, \qquad s = \lambda\zeta \cdot x, \qquad C = M^{-1}M^{-T},$$
$$q = p \cdot Cp = |\zeta|^2, \qquad \widetilde{G} = M^T G, \qquad c = M^{-1}e_1.$$

The phase variable s and material variable a are independent variables of a profile, even though its evaluation on physical space ties them together. For a smooth 2π-periodic profile $A(s, a, t)$ put

$$A^\natural(x, t) = A(\lambda\zeta(t) \cdot x, M(t)^{-1}x, t).$$

All profile time derivatives below fix (s, a). Introduce the operators

$$\begin{aligned}
&\partial_\perp = Jp \cdot \nabla_a, && B = (Cp) \cdot \nabla_a, && \Delta_C = \operatorname{div}_a(C\nabla_a),\\
&D_1 = 2B\partial_s, && L_\lambda = \lambda^2 q \partial_s^2 + \lambda D_1 + \Delta_C, && \mathcal{P} = \lambda\zeta\partial_s + M^{-T}\nabla_a,\\
&J_s(A, B_0) = A_s \partial_\perp B_0 - (B_0)_s \partial_\perp A, && \{A, B_0\}_a = J\nabla_a A \cdot \nabla_a B_0. &&
\end{aligned}$$

Here B_0 in the last line is an arbitrary profile, not the operator B. The original source denotes q by κ; we retain q so that κ can continue to mean thermal diffusivity.

The exact chain rules are

$$\nabla_x A^\natural = (\mathcal{P}A)^\natural, \qquad (\partial_t + Dx \cdot \nabla_x)A^\natural = (A_t)^\natural, \tag{223}$$
$$\Delta_x A^\natural = (L_\lambda A)^\natural, \qquad \nabla_x^\perp \Psi^\natural \cdot \nabla_x A^\natural = (\lambda J_s(\Psi, A) + \{\Psi, A\}_a)^\natural. \tag{224}$$

For the first line, differentiate $a = M^{-1}x$ and $s = \lambda\zeta \cdot x$ and use $\zeta' = -D^T\zeta$. Squaring the spatial gradient gives the three terms in L_λ. For the last identity, expand both gradients. In dimension two a matrix of determinant one satisfies $M^{-1}JM^{-T} = J$, as direct multiplication by the four entries shows. Thus the slow-slow term is $\{\Psi, A\}_a$, the fast-fast term is zero, and the two cross terms are respectively $\lambda\Psi_s\partial_\perp A$ and $-\lambda A_s \partial_\perp \Psi$. This proves every sign in (224).

For arbitrary smooth profiles V, Ψ, add

$$\theta = \theta_b + V^\natural, \qquad u = u_b + v, \qquad v = \nabla_x^\perp \Psi^\natural, \qquad \omega = \omega_b + (L_\lambda\Psi)^\natural, \quad \omega_b = D_{21} - D_{12}.$$

The exact increments to scalar force and curl of vector force are

$$E_\theta = V_t + \lambda(J\zeta \cdot G)\Psi_s + J\nabla_a\Psi \cdot \widetilde{G} + \lambda J_s(\Psi, V) + \{\Psi, V\}_a - \kappa L_\lambda V, \tag{225}$$
$$E_\omega = \partial_t(L_\lambda\Psi) + \lambda J_s(\Psi, L_\lambda\Psi) + \{\Psi, L_\lambda\Psi\}_a - \lambda\zeta_1 V_s - c \cdot \nabla_a V - \nu L_\lambda^2 \Psi. \tag{226}$$

Indeed $v \cdot \nabla\theta_b = v \cdot G$ gives the two displayed background-gradient terms in (225) because $JM^{-T} = MJ$. The background vorticity is spatially constant, so $v \cdot \nabla\omega_b = 0$. Applying (223)–(224) to every other term of the scalar and vorticity equations gives the formulas exactly. In particular, $\partial_t(L_\lambda\Psi)$ differentiates q and C.

These curl residuals have a direct full momentum realization. Put $\mathsf{v} = J\mathcal{P}\Psi$ and define the profile vector

$$\mathsf{E}_u = 2D\mathsf{v} + J\mathcal{P}\Psi_t + (\mathsf{v} \cdot \mathcal{P})\mathsf{v} - \nu J\mathcal{P}L_\lambda\Psi - Ve_2. \tag{227}$$

Then with the unchanged pressure p_b the full momentum equation holds for $f_u = f_{u,b} + \mathsf{E}_u^\natural$, and $\mathrm{curl}_x\, \mathsf{E}_u^\natural = E_\omega^\natural$. To verify the vector formula, differentiating the coefficients of $J\mathcal{P}$ gives $DJ\mathcal{P}$: for example $J\zeta' = DJ\zeta$, and $J(M^{-T})' = DJM^{-T}$. It follows that

$$(\partial_t + Dx \cdot \nabla_x)v = Dv + (J\mathcal{P}\Psi_t)^\natural.$$

The additional term $(v \cdot \nabla)u_b = Dv$ accounts for the second copy of Dv. The self-advection, diffusion, and buoyancy increments are exactly the remaining terms of (227). For the curl assertion, expansion in coordinates gives, for any smooth divergence-free two-dimensional u,

$$\mathrm{curl}((u \cdot \nabla)u) = u \cdot \nabla(\partial_1 u_2 - \partial_2 u_1) + (\partial_1 u_1 + \partial_2 u_2)(\partial_1 u_2 - \partial_2 u_1) = u \cdot \nabla\omega.$$

Taking the curl of the already proved vector equation therefore gives (226). For compact material support, both force increments and the added fields have compact spatial support on this finite interval. The prescribed affine background separately retains the spatial behavior stated in Section 4.

13.2 The complete leading localized residual

Retain the odd periodic F, its even mean-zero primitive P, and $P' = F$. Allow arbitrary smooth signed amplitudes $T(t), Q(t)$ and an arbitrary smooth material envelope $g(a,t)$. Set

$$V_0 = TFg, \qquad \Psi_0 = QPg, \qquad \mathcal{O} = \lambda^2 qQ, \qquad h = Bg, \qquad d = \Delta_C g.$$

Products here and below are evaluated at (s, a, t) only at the end. Direct application of L_λ gives

$$L_\lambda V_0 = T(\lambda^2 qF''g + 2\lambda F'h + Fd), \tag{228}$$
$$L_\lambda \Psi_0 = \mathcal{O}F'g + 2\lambda QFh + QPd, \tag{229}$$
$$L_\lambda^2 \Psi_0 = Q[\lambda^4 q^2 F'''g + 4\lambda^3 qF''h + \lambda^2 F'(2qd + 4Bh) + 4\lambda FBd + P\Delta_C d]. \tag{230}$$

The last line follows by applying the three terms of L_λ to each of the three terms in (229). Its two mixed second-order contributions are $2\lambda^3 qF''h$ each, its two $F'd$ contributions are $\lambda^2 qF'd$ each, and $\Delta_C h = B\Delta_C g = Bd$ because C and Cp are constant in a. These observations give exactly the coefficients $4, 2, 4, 4$ in (230), with no term omitted.

Substitution in (225) gives

$$\begin{aligned} E_\theta(V_0, \Psi_0) = (T' + \lambda(J\zeta \cdot G)Q)Fg + TFg_t + QP(J\nabla_a g \cdot \widetilde{G}) \\ + \lambda QT(F^2 - PF')g\partial_\perp g - \kappa T(\lambda^2 qF''g + 2\lambda F'h + Fd). \end{aligned} \tag{231}$$

The complete vorticity formula is

$$\begin{aligned} E_\omega(V_0, \Psi_0) = {} & (\mathcal{O}' - \lambda\zeta_1 T)F'g + \mathcal{O}F'g_t + \partial_t(2\lambda QFh + QPd) - TF(c \cdot \nabla_a g) \\ & + \lambda^3 qQ^2(FF' - PF'')g\partial_\perp g \\ & + 2\lambda^2 Q^2(F^2 g\partial_\perp h - PF'h\partial_\perp g) + 2\lambda Q^2 PF\{g, h\}_a \\ & + \lambda Q^2 PF(g\partial_\perp d - d\partial_\perp g) + Q^2P^2\{g, d\}_a \\ & - \nu Q[\lambda^4 q^2 F'''g + 4\lambda^3 qF''h + \lambda^2 F'(2qd + 4Bh) + 4\lambda FBd + P\Delta_C d]. \end{aligned} \tag{232}$$

For a verification of all nonlinear terms, the slow-slow bracket of QPg with TFg vanishes, and their fast bracket is $QT(F^2 - PF')g\partial_\perp g$. In the vorticity equation, bracket QPg successively with the three summands of (229). The first gives $\lambda^3 qQ^2(FF' - PF'')g\partial_\perp g$ and zero slow bracket. The second gives

$$2\lambda^2 Q^2(F^2 g\partial_\perp h - PF'h\partial_\perp g) \quad \text{and} \quad 2\lambda Q^2 PF\{g, h\}_a.$$

The third gives $\lambda Q^2 PF(g\partial_\perp d - d\partial_\perp g)$ and $Q^2P^2\{g,d\}_a$. This exhausts the two brackets and proves (232). Time differentiation in that formula keeps the derivatives of C, q, g, Q, h, and d.

For the exact inviscid amplitudes of (49), a smooth activation $w(t)$, and a time-independent material envelope, put

$$T = w\Theta, \qquad Q = \frac{w\Omega}{\lambda^2 q}.$$

Then the first terms in (231) and (232) are respectively $w'\Theta F g$ and $w'\Omega F' g$. Setting $\kappa = \nu = 0$ therefore reproduces every term of the source's equation (6.5). The compact support is generated by the material envelope; no spatial coordinate or frequency has been set equal to one in this calculation.

Define phase average at fixed (a,t) by

$$\langle A\rangle(a,t) = \frac{1}{2\pi}\int_0^{2\pi} A(s,a,t)\,ds, \qquad \mu_2 = \langle F^2\rangle, \qquad \mu_P = \langle P^2\rangle.$$

The exact means, including arbitrary g_t, κ, and ν, are

$$\langle E_\theta\rangle = 2\mu_2\lambda Q T g\partial_\perp g, \qquad \langle E_\omega\rangle = 2\mu_2\lambda^2 Q^2\partial_\perp(gh) + \mu_P Q^2\{g,d\}_a. \tag{233}$$

Indeed $\langle F\rangle = \langle P\rangle = 0$ and all derivatives have zero mean. Integration by parts gives $\langle PF'\rangle = -\mu_2$, $\langle FF'\rangle = \langle PF''\rangle = 0$, and $\langle PF\rangle = 0$. Substitution of these identities in the two expanded residuals proves (233) term by term. These averages are functions of material position. They are not physical spatial integrals and need not vanish.

13.3 A finite phase correction that includes viscosity

We prove an exact finite cancellation identity. It extends the source's inviscid phase recursion; the construction here treats diffusion in the evolution equation itself. It does not replace viscosity by an uncontrolled forcing term.

Fix $\lambda > 0$, $\kappa,\nu \geq 0$ and the smooth coefficients above on $[t_0,t_1]$. Since $q > 0$, compactness gives a positive lower bound for q on this interval. Let $g \in C_c^\infty(\mathbb{R}^2)$ be time-independent and equal to one on an open material set U. For the parity assertion below, additionally let $g(-a) = g(a)$. Let w be smooth and flat at t_0, meaning $w^{(j)}(t_0) = 0$ for every nonnegative integer j. Set

$$K = \lambda^2 q, \qquad b = J\zeta\cdot G, \qquad \alpha = \frac{b}{\lambda q}, \qquad \beta = \lambda\zeta_1.$$

Here K is an actual coefficient, not a change of spatial or temporal units. Start with any smooth mean-zero odd initial phase pair and solve

$$\partial_t\begin{pmatrix}\mathcal{T}\\ \mathcal{W}\end{pmatrix} = \begin{pmatrix}\kappa K\partial_s^2 & -\alpha\\ \beta & \nu K\partial_s^2\end{pmatrix}\begin{pmatrix}\mathcal{T}\\ \mathcal{W}\end{pmatrix}. \tag{234}$$

The existence and smoothness proof is given below. Define the leading profiles and streamfunctions by

$$V_0 = wg\mathcal{T}, \qquad W_0 = wg\mathcal{W}, \qquad \Psi_n = K^{-1}\partial_s^{-1}W_n \quad (n \geq 0). \tag{235}$$

The inverse ∂_s^{-1} is the unique mean-zero periodic primitive of a mean-zero profile. Explicitly, for such a profile H,

$$\partial_s^{-1}H(s,a,t) = \int_0^s H(r,a,t)\,dr - \frac{1}{2\pi}\int_0^{2\pi}\int_0^v H(r,a,t)\,dr\,dv.$$

The zero average makes the first integral periodic. Differentiation and averaging verify respectively that its derivative is H and its mean is zero. Two such primitives differ by a constant with mean zero, so uniqueness follows. All these operations fix a and preserve its support and vanishing on U.

Write $\mathsf{P}_0 H = H - \langle H\rangle$. All negatively indexed profiles are zero. For $j \geq 0$ define

$$Z_j = \partial_s W_j + \lambda D_1 \Psi_{j-1} + \Delta_C \Psi_{j-2}, \qquad Z_j^\flat = Z_j - \partial_s W_j. \tag{236}$$

For $n \geq 1$, the following sources involve only profiles of index less than n:

$$\begin{aligned} S_n^\theta &= J\nabla_a \Psi_{n-1} \cdot \widetilde{G} + \lambda \sum_{i+j=n-1} J_s(\Psi_i, V_j) + \sum_{i+j=n-2} \{\Psi_i, V_j\}_a \\ &\quad - \kappa(\lambda D_1 V_{n-1} + \Delta_C V_{n-2}), \end{aligned} \tag{237}$$

$$\begin{aligned} S_n^\omega &= \partial_t Z_n^\flat - c \cdot \nabla_a V_{n-1} + \lambda \sum_{i+j=n-1} J_s(\Psi_i, Z_j) + \sum_{i+j=n-2} \{\Psi_i, Z_j\}_a \\ &\quad - \nu\big(K\partial_s^2 Z_n^\flat + \lambda D_1 Z_{n-1} + \Delta_C Z_{n-2}\big). \end{aligned} \tag{238}$$

The sums have nonnegative indices; an empty sum is zero. In particular, $Z_n^\flat$ contains only Ψ_{n-1} and Ψ_{n-2}, so it does not introduce an unknown of level n into its own source.

For each $n \geq 1$ solve the zero-data equation

$$\partial_t \begin{pmatrix} V_n \\ W_n \end{pmatrix} = \begin{pmatrix} \kappa K \partial_s^2 & -\alpha \\ \beta & \nu K \partial_s^2 \end{pmatrix} \begin{pmatrix} V_n \\ W_n \end{pmatrix} + \begin{pmatrix} -\mathsf{P}_0 S_n^\theta \\ -\partial_s^{-1} \mathsf{P}_0 S_n^\omega \end{pmatrix}, \qquad (V_n, W_n)|_{t=t_0} = 0. \tag{239}$$

This is a fully specified recursion at every finite depth; no unsolved equation is substituted for its solution. The explicit solution follows from the two-dimensional propagators described next.

For $m \in \mathbb{Z}$ let $\Phi_m(t,\tau)$ solve

$$\partial_t \Phi_m(t,\tau) = A_m(t)\Phi_m(t,\tau), \qquad \Phi_m(\tau,\tau) = \mathrm{Id}, \qquad A_m(t) = \begin{pmatrix} -\kappa K(t) m^2 & -\alpha(t) \\ \beta(t) & -\nu K(t) m^2 \end{pmatrix}.$$

An explicit convergent definition for $t \geq \tau$ is the ordered integral series

$$\Phi_m(t,\tau) = \mathrm{Id} + \sum_{r=1}^\infty \int_{\tau \leq t_r \leq \cdots \leq t_1 \leq t} A_m(t_1) \cdots A_m(t_r)\, dt_r \cdots dt_1. \tag{240}$$

For fixed m, if $L_m = \sup \|A_m\|$, the rth summand has norm at most $L_m^r (t-\tau)^r / r!$. Uniform convergence and the fundamental theorem of calculus give the matrix differential equation. Uniqueness follows by iterating the integral equation for a difference, as in Section 4.

For a profile H write $\widehat{H}_m = (2\pi)^{-1} \int_0^{2\pi} H(s) e^{-ims}\, ds$. Put $\widehat{R}_{n,0} = 0$ and, for $m \neq 0$,

$$\widehat{R}_{n,m}(a,t) = \begin{pmatrix} -\widehat{S}_{n,m}^\theta(a,t) \\ -(im)^{-1} \widehat{S}_{n,m}^\omega(a,t) \end{pmatrix}.$$

Then the required solution is exactly

$$\begin{pmatrix} V_n \\ W_n \end{pmatrix}(s,a,t) = \sum_{m \neq 0} e^{ims} \int_{t_0}^t \Phi_m(t,\tau) \widehat{R}_{n,m}(a,\tau)\, d\tau. \tag{241}$$

The initial pair in (234) has the analogous series with coefficient $\Phi_m(t,t_0)$ times its initial Fourier coefficient. For $\kappa = \nu = 0$, A_m is independent of m, so (241) becomes the pointwise phase Duhamel integral of the same 2×2 amplitude system as in the source.

Here are convergence details needed to justify this formula as a smooth solution, including the degenerate cases where a diffusion coefficient is zero. For a complex solution $z' = A_m z$, its squared Euclidean norm satisfies

$$\frac{d}{dt}|z|^2 = -2\kappa K m^2 |z_1|^2 - 2\nu K m^2 |z_2|^2 + 2\,\mathrm{Re}\big((\beta - \alpha)\bar{z}_2 z_1\big) \leq |\beta - \alpha||z|^2.$$

Multiplication by $\exp(-\int_\tau^t |\beta-\alpha|)$ and differentiation therefore give

$$\|\Phi_m(t,\tau)\| \le \exp\left(\frac{1}{2}\int_\tau^t |\beta(r)-\alpha(r)|\,dr\right), \tag{242}$$

uniformly in m. Smooth periodic sources have Fourier coefficients decaying faster than every inverse power of $|m|$: integration by parts N times bounds each coefficient by $|m|^{-N}$ times the supremum of its Nth phase derivative. The same argument applies after any fixed material or time derivative, uniformly on the compact time interval and compact material support. Bound (242) therefore proves uniform convergence of (241) after arbitrarily many phase or material derivatives. Its first time derivative is obtained by the fundamental theorem of calculus and the matrix equation; multiplication by A_m loses at most two powers of m. Repetition proves convergence of every mixed time derivative, since coefficient derivatives are bounded on the compact interval and every finite power loss is available from source smoothness. Induction on n now constructs all levels as smooth profiles. Complex conjugation of the Fourier coefficients for m and $-m$ proves that the resulting profiles are real. Their zero phase means follow either from the displayed sum or from the homogeneous zero-mode equation with zero initial data. This proves existence and the explicit formula. Fourier expansion and uniqueness of the matrix ODE at each frequency prove uniqueness.

Proposition 13.1 (Exact finite residual with two diffusion coefficients)**.** *For any integer $N \ge 0$, construct levels $0,\dots,N$ as above and put*

$$V = \sum_{j=0}^{N} V_j, \qquad \Psi = \sum_{j=0}^{N} \Psi_j.$$

All levels $j \ge 1$ vanish on U and outside $\operatorname{supp} g$ *for every phase. They have zero phase means and are flat at t_0. If g is even, V_j and W_j are odd, and Ψ_j is even, under $(s,a)\mapsto(-s,-a)$.*

Set the primary profiles V_j, W_j, Ψ_j equal to zero for $j > N$, while retaining the definition (236); thus

$$Z_{N+1} = \lambda D_1 \Psi_N + \Delta_C \Psi_{N-1}, \qquad Z_{N+2} = \Delta_C \Psi_N, \qquad Z_j = 0 \quad (j > N+2).$$

Use these truncated profiles in the source formulas for all positive indices. The exact residuals are

$$E_\theta = w' g \mathcal{T} + \sum_{n=1}^{N} \langle S_n^\theta \rangle + \sum_{n=N+1}^{2N+2} S_n^\theta, \tag{243}$$

$$E_\omega = w' g \partial_s \mathcal{W} + \sum_{n=1}^{N} \langle S_n^\omega \rangle + \sum_{n=N+1}^{2N+4} S_n^\omega. \tag{244}$$

Every summand is explicitly determined by the finite Duhamel formula; no infinite-layer convergence statement is included in these identities.

Proof. At level zero the profiles are independent of a on U and vanish outside $\operatorname{supp} g$. Every term in each positive-level source contains a material derivative: this is explicit in the linear terms, in $Z_n^\flat$, and in both brackets. If every earlier positive level vanishes on U, its derivatives vanish there. The only possible zeroth-level factors are independent of a there, so the entire source vanishes. The same argument holds outside the support. Phase averaging, phase inversion, and the propagator integral do not change a. Induction proves both support assertions.

The factor w is flat initially, so the leading levels and all their mixed derivatives are zero at t_0. The source formulas preserve this property through sums, products, spatial derivatives, and time derivatives. In (239), differentiating successively in time at t_0 shows that each higher profile and all its mixed derivatives are zero there. This proves flatness.

For parity, a derivative in either s or a material coordinate reverses simultaneous parity. Thus L_λ preserves parity, both brackets have the product parity of their two arguments, Z_j is even when W_j is odd and Ψ_j is even, S_n^θ is odd, and S_n^ω is even. The phase average has the same simultaneous parity as its original profile, and phase inversion reverses it. Consequently both source components in (239) are odd. The equation and its unique zero-data solution preserve this parity. Since $\Psi_j = K^{-1}\partial_s^{-1}W_j$, it is even. Induction begins with the odd initial phase pair and even g.

It remains to prove the residual equalities, retaining the time dependence of K. The definition of Ψ_n gives the identity $K\partial_s^2\Psi_n = \partial_s W_n$ for every t. Therefore the principal scalar expression at level n is

$$\partial_t V_n + \alpha W_n - \kappa K \partial_s^2 V_n,$$

and the principal vorticity expression is

$$\partial_s\big(\partial_t W_n - \beta V_n - \nu K \partial_s^2 W_n\big).$$

For $n \geq 1$, equation (239) makes these $-\mathsf{P}_0 S_n^\theta$ and $-\mathsf{P}_0 S_n^\omega$, respectively. At $n = 0$, (234) leaves precisely $w' g \mathcal{T}$ and $w' g \partial_s \mathcal{W}$.

To account for all remaining terms, the finite index shift in (236) gives

$$L_\lambda \Psi = \sum_{j=0}^{N+2} Z_j.$$

The two scalar brackets give grades $i + j + 1$ and $i + j + 2$, the background-gradient and mixed-diffusion terms give grades one and two higher than their primary profile, and hence all are exactly the terms in (237). Their largest possible grade is $2N + 2$. In the vorticity equation, after removing the principal expression, the linear time derivative is $\partial_t Z_n^\flat$, and the remaining viscosity terms are exactly

$$-\nu\big(K\partial_s^2 Z_n^\flat + \lambda D_1 Z_{n-1} + \Delta_C Z_{n-2}\big).$$

The two vorticity brackets give grades $i + j + 1$ and $i + j + 2$ with $i \leq N$ and $j \leq N + 2$. Thus all remaining terms are exactly (238), with maximal grade $2N + 4$. Adding each source to its canceled mean-zero part leaves its phase mean through grade N and the whole source at the higher grades. This is precisely (243)–(244). No differentiation of q has been omitted: it was incorporated in the exact identity $K\partial_s^2\Psi_n = \partial_s W_n$ before differentiating. □

The affine core in the source can now be stated exactly. In the inviscid case, take $\mathcal{T} = \Theta(t)F(s)$ and $\mathcal{W} = \Omega(t)F(s)$, where (49) holds. These solve (234). On $a \in U$, every finite corrected field therefore agrees with the uncut leading field. If also $F(s) = s$ for $|s| < s_0$, then on

$$\{x : M^{-1}x \in U,\ |\lambda\zeta \cdot x| < s_0\}$$

one has, with the original signs and coefficients,

$$\vartheta = w\Theta\lambda\zeta \cdot x, \qquad v = \frac{w\Omega}{q}(J\zeta \otimes \zeta)x, \qquad \varpi = w\Omega.$$

The activation terms $w' g \mathcal{T}$ and $w' g \partial_s \mathcal{W}$ are still present there; the corrections to the state vanish, not necessarily the force. For a sine phase pair and nonzero diffusion, the damped amplitude system (54) gives an explicit solution of (234), and the same material plateau is preserved. For a general profile initially linear near zero, the diffusive phase evolution is given instead by its Fourier propagator; the preceding proof does not replace that evolution by the inviscid linear-core identities. In every case the finite residual is the explicit quantity (243)–(244), including its phase means and final uncanceled grades.

14 An explicit three-dimensional viscous strain layer

The preceding phase-amplitude construction has a directly computable three-dimensional counterpart. This section derives that counterpart in the Cartesian Navier–Stokes equation, with its pressure and viscosity intact, and places any finite interval of its central dynamics inside a smooth, compactly supported forced solution. This is a local building block. The constructed compact solution is global and therefore establishes no nonexistence alternative from Section 1.

14.1 The full velocity and its exact evolution

Fix $\sigma > 0$, $\nu > 0$, $a_0 \in \mathbb{R}$, and $(K_1, K_3) \in \mathbb{R}^2 \setminus \{(0,0)\}$. Coordinates remain $x = (x_1, x_2, x_3) \in \mathbb{R}^3$ and time is $t \geq 0$. Define

$$k_1(t) = K_1 e^{-\sigma t}, \qquad k_3(t) = K_3, \qquad k(t) = (k_1(t), 0, K_3), \tag{245}$$

$$q(t) = K_1^2 e^{-2\sigma t} + K_3^2, \qquad \varphi(x,t) = k_1(t)x_1 + K_3 x_3. \tag{246}$$

$$a(t) = a_0 \exp\left\{\sigma t - \nu\left[\frac{K_1^2}{2\sigma}(1 - e^{-2\sigma t}) + K_3^2 t\right]\right\}. \tag{247}$$

In particular $q(t) > 0$ at each finite time. Put

$$u(x,t) = (\sigma x_1, -\sigma x_2 + a(t)\cos\varphi(x,t), 0), \qquad p_0(x,t) = -\frac{\sigma^2}{2}(x_1^2 + x_2^2). \tag{248}$$

Proposition 14.1 (Exact viscous wave over a strain). *The fields* (248) *solve*

$$u_t + (u \cdot \nabla)u + \nabla p_0 = \nu \Delta u, \qquad \operatorname{div} u = 0$$

at every $(x,t) \in \mathbb{R}^3 \times [0,\infty)$. They are smooth on this entire time interval. Their vorticity and its supremum norm are exactly

$$\omega = a(t)\sin\varphi\,(K_3, 0, -k_1(t)), \qquad \|\omega(t)\|_\infty = |a(t)|\sqrt{q(t)}. \tag{249}$$

The full velocity has infinite spatial energy at every time.

Proof. Write $D = \operatorname{diag}(\sigma, -\sigma, 0)$, $u_b = Dx$, and $v = a\cos\varphi\, e_2$. Direct differentiation gives

$$k' = -D^T k, \quad (\partial_t + u_b \cdot \nabla)\varphi = 0, \quad a' = (\sigma - \nu q)a.$$

The second component v is independent of x_2, so $(v \cdot \nabla)v = 0$ and $\operatorname{div} v = 0$. Moreover $(v \cdot \nabla)u_b = Dv = -\sigma v$ and $\Delta v = -qv$. Thus

$$v_t + (u_b \cdot \nabla)v + (v \cdot \nabla)u_b = (a' - \sigma a)\cos\varphi\, e_2 = -\nu q v.$$

The base equations hold because $(u_b \cdot \nabla)u_b = D^2 x = (\sigma^2 x_1, \sigma^2 x_2, 0)$, $\nabla p_0 = -D^2 x$, $\Delta u_b = 0$, and $\operatorname{tr} D = 0$. Adding these identities proves every momentum component and divergence. Taking the Cartesian curl gives (249). Since $k(t) \neq 0$, the linear map $x \mapsto k(t) \cdot x$ takes every real value, so $|\sin\varphi|$ attains one; this proves the supremum formula. Every displayed coefficient is smooth for finite t, proving global smoothness. Finally $|u(x,t)|^2 \geq \sigma^2 x_1^2$; its integral over $[1,2] \times \mathbb{R} \times [0,1]$ is infinite, proving the energy assertion. □

For $a_0 \neq 0$ the exact logarithmic vorticity rate is

$$\frac{d}{dt}\log\|\omega(t)\|_\infty = \sigma - \nu q - \frac{\sigma k_1(t)^2}{q} = \frac{\sigma K_3^2}{q} - \nu q. \tag{250}$$

Indeed $q' = -2\sigma k_1^2$ and differentiation of $\log|a| + \frac{1}{2}\log q$ gives the displayed equality. Consequently vorticity increases at a particular time exactly when $\sigma K_3^2 > \nu q(t)^2$. In the retained subfamily $K_1 = 0$, $K_3 \neq 0$, the full formula becomes

$$\|\omega(t)\|_\infty = |a_0 K_3| e^{(\sigma - \nu K_3^2)t}.$$

This grows when $\sigma > \nu K_3^2$, but has no finite-time singularity. In the different subfamily $K_3 = 0$, $K_1 \neq 0$,

$$\|\omega(t)\|_\infty = |a_0 K_1| \exp\left[-\frac{\nu K_1^2}{2\sigma}(1 - e^{-2\sigma t})\right].$$

Here the changing wavevector cancels the amplitude's strain growth. These two exact expressions keep the distinction between amplitude, spatial frequency, and vorticity visible in a common family.

14.2 A compact realization with an unchanged interior

Fix $0 < R_0 < R_1$ and $0 < T_0 < T_1$. The following explicit cutoffs specify the entire constructed data and force. Define

$$b(s) = \begin{cases} e^{-1/s}, & s > 0, \\ 0, & s \leq 0, \end{cases} \qquad H(s) = \frac{b(s)}{b(s) + b(1-s)},$$

and

$$\chi(x) = H\left(\frac{R_1^2 - |x|^2}{R_1^2 - R_0^2}\right), \qquad \eta(t) = H\left(\frac{T_1 - t}{T_1 - T_0}\right). \tag{251}$$

The denominator in H is positive for all real s: if $s \leq 0$, then $1 - s > 0$; if $s \geq 1$, then $s > 0$; and both are positive when $0 < s < 1$. Each positive-side derivative of b is $e^{-1/s}$ times a polynomial in $1/s$, as induction under differentiation shows. For each integer $m \geq 0$, $z^m e^{-z} \to 0$ as $z \to \infty$: the series for $e^{z/2}$ bounds $z^m e^{-z/2}$ by a constant, leaving an additional factor $e^{-z/2} \to 0$. Thus all these derivatives vanish at zero and b, H, χ, η are smooth. In particular $0 \leq \chi, \eta \leq 1$, $\chi = 1$ on $|x| \leq R_0$, $\chi = 0$ on $|x| \geq R_1$, $\eta = 1$ for $0 \leq t \leq T_0$, and $\eta = 0$ for $t \geq T_1$.

An explicit vector potential for (248) is

$$A(x,t) = -\frac{1}{3}x \times (Dx) + \frac{a(t)}{q(t)}(e_2 \times k(t)) \sin\varphi(x,t). \tag{252}$$

Here the curl orientation is the original right-handed Cartesian one. To check the base term, the product identity

$$\operatorname{curl}(X \times Y) = X \operatorname{div} Y - Y \operatorname{div} X + (Y \cdot \nabla)X - (X \cdot \nabla)Y$$

follows by expanding its three components. For $X = x, Y = Dx$, its right side is $0 - 3Dx + Dx - Dx = -3Dx$. For the wave term, $k \times (e_2 \times k) = q e_2$ since $k \cdot e_2 = 0$. Taking its curl therefore gives $a \cos\varphi\, e_2$. Together these calculations prove $\operatorname{curl} A = u$.

Set

$$w = \nabla\chi \times A, \qquad V = \chi u + w = \operatorname{curl}(\chi A), \qquad U = \eta V, \qquad P = \eta^2 \chi p_0. \tag{253}$$

Prescribe the force by the following expanded, finite expression, where $\nabla\chi \cdot \nabla u = \sum_{j=1}^3 (\partial_j \chi)\partial_j u$:

$$\begin{aligned} F = {} & \eta' V + \eta \partial_t w - \nu\eta\left[2\nabla\chi \cdot \nabla u + (\Delta\chi)u + \Delta w\right] \\ & + \eta\chi(\eta\chi - 1)(u \cdot \nabla)u + \eta\chi(\eta - 1)\nabla p_0 \\ & + \eta^2\big[\chi(u \cdot \nabla\chi)u + \chi(u \cdot \nabla)w + \chi(w \cdot \nabla)u \\ & \qquad + (w \cdot \nabla\chi)u + (w \cdot \nabla)w + p_0\nabla\chi\big]. \end{aligned} \tag{254}$$

All factors in this formula have been explicitly defined, including their initial values and cutoffs. In particular no solution-dependent unknown force is left to choose.

Proposition 14.2 (Exact localization and admissibility)**.** *The fields (U, P, F) in (253)–(254) satisfy*

$$U_t + (U \cdot \nabla)U + \nabla P = \nu \Delta U + F, \qquad \operatorname{div} U = 0.$$

They are smooth for all $t \geq 0$. Their spatial supports lie in $\{|x| \leq R_1\}$, and all three vanish for $t \geq T_1$. The initial datum $U(x, 0) = \operatorname{curl}(\chi A(x, 0))$ satisfies (2), the force satisfies (3), and U satisfies (4). On $|x| < R_0$, $0 \leq t \leq T_0$, one has exactly $(U, P, F) = (u, p_0, 0)$.

Proof. The divergence of a curl is zero because each mixed second derivative appears twice with opposite signs. Since η is spatially constant, this proves $\operatorname{div} U = 0$. Direct products give

$$\begin{aligned}
\Delta V &= \chi \Delta u + 2\nabla\chi \cdot \nabla u + (\Delta\chi)u + \Delta w, \\
(V \cdot \nabla)V &= \chi^2 (u \cdot \nabla)u + \chi(u \cdot \nabla\chi)u + \chi(u \cdot \nabla)w + \chi(w \cdot \nabla)u \\
&\quad + (w \cdot \nabla\chi)u + (w \cdot \nabla)w, \\
\nabla(\chi p_0) &= \chi \nabla p_0 + p_0 \nabla\chi.
\end{aligned}$$

Insert these three identities into $U_t + (U \cdot \nabla)U + \nabla P - \nu\Delta U$. The terms $\eta\chi(u_t - \nu\Delta u)$ equal $-\eta\chi[(u \cdot \nabla)u + \nabla p_0]$ by the already proved uncut equation. Combining their two coefficients with the displayed convection and pressure expansions produces $\eta\chi(\eta\chi - 1)$ and $\eta\chi(\eta - 1)$ respectively. Every remaining term is exactly (254), proving the momentum equation with the prescribed force.

Smoothness follows from the explicit formulas, $q > 0$ at finite time, and flatness of the cutoffs at their endpoints. Outside the spatial cutoff, χ and all its derivatives vanish, hence w, V, U, P, F vanish. For $t \geq T_1$, η and every time derivative vanish, so U, P, F vanish as well. Each mixed derivative of F is consequently bounded on the compact set $\{|x| \leq R_1, 0 \leq t \leq T_1\}$, say by $M_{\alpha m}$. For every $K \in \mathbb{N}_0$, the choice

$$C_{\alpha m K} = M_{\alpha m}(1 + R_1 + T_1)^K$$

proves (3), including the exterior where the derivative is zero. The same argument at $t = 0$ proves the data condition.

For a quantitative energy bound define

$$a_* = |a_0| e^{\sigma T_1}, \qquad k_* = \sqrt{K_1^2 e^{-2\sigma T_1} + K_3^2} > 0, \qquad c_\chi = \|\nabla\chi\|_\infty.$$

The viscosity exponent in (247) is nonpositive for $0 \leq t \leq T_1$, so $|a| \leq a_*$ there. Also $\sqrt{q} \geq k_*$, $|Dx| \leq \sigma R_1$ on the spatial support, and (252) gives $|A| \leq \sigma R_1^2/3 + a_*/k_*$. Consequently

$$\int_{\mathbb{R}^3} |U(x, t)|^2 \, dx \leq \frac{4\pi R_1^3}{3} \left[\sigma R_1 + a_* + c_\chi \left(\frac{\sigma R_1^2}{3} + \frac{a_*}{k_*} \right) \right]^2 \qquad (t \geq 0).$$

Adding one to this finite right side provides the strict bound in (4). Finally, on the stated interior plateau $\chi = \eta = 1$, $w = 0$ and all cutoff derivatives vanish; direct substitution proves $(U, P, F) = (u, p_0, 0)$ there. ☐

The original strain, both initial wavevector components, amplitude, viscosity, inner and outer spatial radii, and both time endpoints have all been retained. Localization supplies an actual map from this nondecaying exact solution to admissible compact data and forcing with an identical finite interior evolution. Increasing an amplification parameter creates a different smooth solution; the displayed family alone does not produce a finite-time singular solution for fixed prescribed data.

15 The exact swirl-rate and positive-carrier transfer

We retain the original axisymmetric coordinates of Section 3: $y_1 = z$, $y_2 = r^2/2 > 0$, $v = (u^z, ru^r) = J\nabla\Psi$, $\Gamma = ru^\phi$, and $\xi = L\Psi = \omega^\phi/r$. The viscosity remains the fixed physical constant $\nu > 0$. The following change of dependent variable makes the meridional source exactly a derivative of a square, while retaining the geometric terms in its transport equation. It then gives a complete compact transfer from a signed scalar layer into actual three-dimensional data and force.

15.1 A common viscous operator for swirl rate and vorticity

Define the signed angular velocity, or swirl rate, by

$$h = \frac{\Gamma}{2y_2} = \frac{u^\phi}{r}, \qquad Q = \partial_1^2 + 2y_2\partial_2^2, \quad M = Q + 4\partial_2, \quad D_v = \partial_t + v\cdot\nabla_y.$$

This is an invertible change of variables on the retained domain $y_2 > 0$: its inverse is $\Gamma = 2y_2 h$. No time or spatial coordinate has been rescaled. Product differentiation gives

$$Q(2y_2h) = 2y_2h_{11} + 2y_2(4h_2 + 2y_2h_{22}) = 2y_2Mh, \tag{255}$$

$$D_v(2y_2h) = 2y_2D_vh + 2v_2h, \qquad W\partial_1(\Gamma^2) = \partial_1(h^2), \quad W = (2y_2)^{-2}. \tag{256}$$

Thus the complete reduced system (24) is equivalent to

$$\boxed{\begin{aligned} &v = J\nabla\Psi, \qquad \xi = L\Psi,\\ &D_vh = \nu Mh - \frac{v_2}{y_2}h + \frac{f^\phi}{r},\\ &D_v\xi = \nu M\xi + \partial_1(h^2) + \mathrm{curl}_y(f^z, f^r/r).\end{aligned}} \tag{257}$$

Both scalar equations now use the same operator M. The multiplication term $-v_2h/y_2$ is part of the exact swirl equation. In original physical coordinates it is $-2u^rh/r$, since $v_2 = ru^r$ and $y_2 = r^2/2$.

For scalar functions a, b write

$$\langle\nabla a, \nabla b\rangle_Q = a_1b_1 + 2y_2a_2b_2, \qquad |\nabla a|_Q^2 = \langle\nabla a, \nabla a\rangle_Q.$$

The product rule, including the drift $4\partial_2$, is

$$M(ab) = aMb + bMa + 2\langle\nabla a, \nabla b\rangle_Q.$$

It follows either by differentiating twice in both coordinates, or by noting that the drift obeys the first-order product rule and contributes no additional quadratic term. Set $B = h^2$, a new square distinct from the original $\Theta = \Gamma^2$ in (25). Multiplying the swirl equation by $2h$ gives, everywhere including $h = 0$,

$$D_vB = \nu MB - 2\nu|\nabla h|_Q^2 - \frac{2v_2}{y_2}B + \frac{2h}{r}f^\phi. \tag{258}$$

On a component where $h \neq 0$, $B > 0$ and $|\nabla B|_Q^2 = 4B|\nabla h|_Q^2$. Hence there the same equation is

$$D_vB = \nu MB - \frac{\nu}{2B}|\nabla B|_Q^2 - \frac{2v_2}{y_2}B + \frac{2h}{r}f^\phi. \tag{259}$$

This last formula is only used on $B > 0$. The globally smooth equation (258) is used at zeros, so no division by a vanishing swirl has been inserted.

15.2 A signed layer, positive carrier and explicit compact forces

Retain a bounded convex ball $D \Subset \{y_2 > 0\}$ and the compact curl inverse $\mathcal{R}$ constructed in Section 3. Fix $0 < T_0 < T_1$ and an open time interval I containing $[0, T_1]$. Take real smooth fields Ψ, ϑ on $\mathbb{R}^2 \times I$ whose spatial supports lie in fixed compact subsets of D. Choose a smooth cutoff $0 \le \chi \le 1$, compactly supported in D, equal to one on a neighborhood of the spatial support of ϑ at every time. To give an explicit choice, write $D = B_R(y_*)$ and let K be the union of the fixed compact support set with $\{y_*\}$. It is a nonempty compact subset of D, including when $\vartheta = 0$. Choose $\max_{y \in K} |y - y_*| < R_0 < R_1 < R$ and use the spatial cutoff in (251) with $|x|$ replaced by $|y - y_*|$. Compactness of $K \Subset D$ guarantees these strict inequalities.

Retain an amplitude conversion factor $\mu > 0$. Define

$$A_\vartheta = \sup_{\mathbb{R}^2 \times [0,T_1]} |\vartheta| < \infty, \qquad c > \mu A_\vartheta, \qquad B_0 = c + \mu\vartheta, \qquad H = \sqrt{B_0} > 0.$$

The finite supremum exists by smoothness on the fixed compact support and compact time interval. Put $\delta = c - \mu A_\vartheta > 0$. The bound $B_0 \ge \delta$ holds on $[0, T_1]$, and uniform continuity on a slightly larger compact cylinder gives $B_0 \ge \delta/2 > 0$ on some time neighborhood of $[0, T_1]$. The positive root H and all derivatives are consequently smooth there. The coefficient μ is part of the scalar amplitude dictionary: $\mu\vartheta$ has the units of the angular velocity squared, as does c. No claim identifying unrelated scalar units is implicit in this definition.

Let $\eta(t)$ be the explicit smooth time cutoff in (251), with these same T_0, T_1. Thus $\eta = 1$ on $[0, T_0]$, $\eta = 0$ on $[T_1, \infty)$, and every derivative vanishes at T_1. Put

$$\widetilde{\Psi} = \eta\Psi, \quad v_0 = J\nabla\Psi, \quad \xi_0 = L\Psi, \quad v = \eta v_0, \quad \xi = \eta\xi_0, \quad h = \eta\chi H, \quad \Gamma = 2y_2 h. \tag{260}$$

The tilde marks the actual target streamfunction; Ψ is the original input and is not silently replaced in a derivative. Define two original-input residuals

$$E_\vartheta = \partial_t\vartheta + v \cdot \nabla\vartheta - \nu M\vartheta, \tag{261}$$

$$E_0 = \partial_t\xi_0 + v_0 \cdot \nabla\xi_0 - \nu M\xi_0 - \mu\partial_1\vartheta. \tag{262}$$

Their different velocities, v and v_0, are deliberate and retained in the following full formulas. Prescribe

$$\begin{aligned} R_h = \eta'\chi H &+ \eta\chi\left[\frac{\mu E_\vartheta}{2H} + \frac{\nu\mu^2}{4H^3}|\nabla\vartheta|_Q^2 + \frac{v_2}{y_2}H\right] \\ &+ \eta H(v \cdot \nabla\chi - \nu M\chi) - \frac{\eta\nu\mu}{H}\langle\nabla\chi, \nabla\vartheta\rangle_Q, \end{aligned} \tag{263}$$

$$E_\xi = \eta E_0 + \eta'\xi_0 + \eta(\eta - 1)(v_0 \cdot \nabla\xi_0 - \mu\partial_1\vartheta) - \eta^2 c\partial_1(\chi^2). \tag{264}$$

In the actual fixed-support construction the last inner product in (263) is zero, because χ is identically one near the support of ϑ. It is retained to expose the complete product rule and the additional term for a nonflat envelope.

Proposition 15.1 (Complete finite-interval transfer)**.** *For the explicitly defined fields above, put*

$$b = \mathcal{R}E_\xi, \qquad f^z = b_1, \quad f^r = rb_2, \quad f^\phi = rR_h. \tag{265}$$

The physical velocity with streamfunction $\widetilde{\Psi}$ and circulation Γ solves the original three-dimensional viscous equation for a smooth pressure. Its data and force satisfy (2)–(3), and its energy has a uniform bound for all $t \ge 0$. On $0 \le t \le T_0$ and the plateau $\chi = 1$, the exact identities are

$$h^2 = c + \mu\vartheta, \qquad \partial_1(h^2) = \mu\partial_1\vartheta, \qquad E_\xi = E_0,$$

and

$$\frac{f^\phi}{r} = \frac{\mu E_\vartheta}{2\sqrt{c + \mu\vartheta}} + \frac{\nu\mu^2|\nabla\vartheta|_Q^2}{4(c + \mu\vartheta)^{3/2}} + \frac{v_2}{y_2}\sqrt{c + \mu\vartheta}. \tag{266}$$

Proof. The chain rule with $H=\sqrt{c+\mu\vartheta}$ gives

$$D_vH-\nu MH=\frac{\mu E_\vartheta}{2H}+\frac{\nu\mu^2}{4H^3}|\nabla\vartheta|_Q^2,\qquad \nabla H=\frac{\mu}{2H}\nabla\vartheta.$$

For verification, the first two derivatives of the function $s\mapsto\sqrt{c+\mu s}$ are $\mu/(2\sqrt{c+\mu s})$ and $-\mu^2/(4(c+\mu s)^{3/2})$. Inserting them into the second-order part of M supplies its displayed positive residual term; the drift obeys the ordinary first derivative rule. Apply the product rule to $h=\eta\chi H$, remembering that $D_v\eta=\eta'$, $D_v\chi=v\cdot\nabla\chi$, and η is constant in space. The result is exactly

$$D_vh-\nu Mh+(v_2/y_2)h=R_h$$

with every summand in (263).

The plateau condition gives the global identity $\chi^2\vartheta=\vartheta$, and therefore

$$h^2=\eta^2(c\chi^2+\mu\vartheta),\qquad \partial_1(h^2)=\eta^2[c\partial_1(\chi^2)+\mu\partial_1\vartheta].$$

Next $D_v\xi=\eta'\xi_0+\eta\partial_t\xi_0+\eta^2v_0\cdot\nabla\xi_0$ and $M\xi=\eta M\xi_0$. Subtracting the preceding source and $\nu M\xi$ gives (264) exactly, so $D_v\xi-\nu M\xi-\partial_1(h^2)=E_\xi$.

This residual has zero spatial integral. Indeed $\int\xi=\int L\widetilde{\Psi}=0$, since integration of each second derivative in its own coordinate gives zero and the coefficient of ∂_1^2 is independent of y_1. Divergence-free v gives $\int v\cdot\nabla\xi=0$; $\int\partial_1(h^2)=0$; and $\int M\xi=0$ by the explicit integration by parts already proved after (33). Thus the compact inverse applies and $\operatorname{curl}_y b=E_\xi$. Equations (265) satisfy both equations in (257). Equivalently $F_\nu^\Gamma=2y_2R_h$ and the force is precisely the force in Proposition 3.3. That proposition supplies the full velocity and the explicitly integrated smooth pressure, so no pressure compatibility or missing vector component is assumed here.

All input supports and the cutoff support are fixed compact subsets of D. Division by H is uniformly harmless because of its positive lower bound, and r has a fixed positive lower bound there. The compact inverse has the smoothness and support proved in that proposition. Every displayed field is therefore smooth on a time neighborhood of $[0,T_1]$ within its support. At T_1, η and all its derivatives vanish; the formulas for E_ξ and R_h show the same flatness for the force. Extending the velocity, force and the reconstructed pressure by zero after T_1 is smooth. The velocity and force have spatial support in the fixed solid torus corresponding to D. Hence all initial-data derivatives are rapidly decaying in space. Every force derivative is bounded on its compact spacetime support, which proves (3) for every stated multi-index, time derivative and polynomial weight.

For an explicit energy bound let $y_-=\inf_D y_2>0$, $y_+=\sup_D y_2<\infty$, and let $S_j=\sup_{D\times[0,T_1]}|\partial_j\Psi|$. The energy identity proved in the coordinate section becomes exactly

$$\int_{\mathbb{R}^3}|u|^2\,dx=2\pi\eta^2\int_D\left[\Psi_2^2+\frac{\Psi_1^2}{2y_2}+2y_2\chi^2(c+\mu\vartheta)\right]dy.$$

It is at most $2\pi|D|[S_2^2+S_1^2/(2y_-)+2y_+(c+\mu A_\vartheta)]$ on $[0,T_1]$, and is zero after T_1. This finite number plus one supplies (4). The plateau formulas follow by setting $\eta=\chi=1$ and all their derivatives to zero in the exact residuals. $\square$

The carrier changes the meridional source only by the explicit fixed edge term $c\partial_1(\chi^2)$ during the uncut time interval. Its swirl force retains both the geometric term and the positive viscosity gradient-square term in (266). These formulas are an exact reconstruction map with all forces computed. They construct global smooth solutions for fixed smooth inputs and finite cutoffs; they do not establish a singular limit.

16 An axis-crossing transfer to five dimensions with exact force and energy

The coordinates in this section are the original Euclidean Cartesian coordinates (x_1, x_2, z), with $r = (x_1^2 + x_2^2)^{1/2}$ and $s = r^2$. Physical time is t, and viscosity is the prescribed $\nu > 0$. In particular $s = 2y_2$ in the preceding meridional coordinate calculation. The formulas below extend through the physical axis. They give a complete map of smooth equations, their forcing and their energy; the finite smooth constructions proved here do not assert a terminal singularity.

16.1 The spaces and the velocity map, including its inverse

Let $I \subset \mathbb{R}$ be an open time interval. Take real functions $q(s, z, t), h(s, z, t)$ that are smooth on an actual open neighborhood of $[0, \infty) \times \mathbb{R} \times I$. Only their restrictions to $s \geq 0$ enter any field below. Assume that there are fixed $S, Z > 0$ such that these restrictions vanish when $s > S$ or $|z| > Z$, for every $t \in I$. They need not vanish at $s = 0$. This hypothesis fixes the support and guarantees bounded derivatives on each compact time subinterval. Define

$$\begin{aligned} a = -q_z, \qquad c = 2q + 2sq_s, \\ u = (x_1 a - x_2 h,\ x_2 a + x_1 h,\ c), \qquad s = x_1^2 + x_2^2. \end{aligned} \tag{267}$$

Composition with this polynomial expression for s proves that u is smooth on all of $\mathbb{R}^3 \times I$, with fixed compact spatial support. In the positively oriented cylindrical frame $e_r = (x_1/r, x_2/r, 0)$, $e_\phi = (-x_2/r, x_1/r, 0)$, $e_z = (0, 0, 1)$, valid for $r > 0$, its components are

$$u^r = -rq_z, \qquad u^\phi = rh, \qquad u^z = 2q + 2sq_s. \tag{268}$$

The streamfunction and circulation in the earlier convention are exactly $\Psi = -sq$ and $\Gamma = sh$: the formulas $u^r = \Psi_z/r$ and $u^z = -\Psi_r/r$ give (268). No cylindrical unit vector is assigned a value at $r = 0$; the Cartesian formula defines the field there.

Direct differentiation, before using any equation of motion, gives

$$\begin{aligned} \operatorname{div}_3 u = 2a + 2sa_s + c_z = 0, \\ \xi = a_z - 2c_s = -(q_{zz} + 4sq_{ss} + 8q_s), \\ \omega = \operatorname{curl}_3 u = (-x_1 h_z - x_2 \xi,\ -x_2 h_z + x_1 \xi,\ 2h + 2sh_s). \end{aligned} \tag{269}$$

For example, $\omega_1 = \partial_{x_2} c - \partial_z(x_2 a + x_1 h) = -x_2(a_z - 2c_s) - x_1 h_z$; the other two components follow by the same differentiation. Thus $\omega^\phi = r\xi$ off the axis, and ξ is the smooth extension of that quotient.

There is also an inverse, with its domain specified. Consider fields of the form $(x_1 a - x_2 h, x_2 a + x_1 h, c)$, where a, c, h have the smoothness and fixed support just imposed, and $2a + 2sa_s + c_z = 0$. Put

$$q(s, z, t) = \frac{1}{2} \int_0^1 c(\tau s, z, t)\, d\tau. \tag{270}$$

This integral proves smoothness through $s = 0$. For $s > 0$, it is equivalent to $2sq = \int_0^s c(\sigma, z, t)\, d\sigma$, whence $c = 2q + 2sq_s$. Integration of the divergence identity gives $\partial_z \int_0^s c\, d\sigma = -2sa$, so that $q_z = -a$. To check support, let $J(z, t) = \int_0^\infty c(\sigma, z, t)\, d\sigma$. Then $J_z = -2[sa]_0^\infty = 0$; since $J = 0$ for $|z| > Z$, it vanishes for every z. Therefore $q = 0$ for $s > S$, as well as for $|z| > Z$. Finally, a difference of two inverse functions satisfies $\partial_s(sq) = 0$; smoothness at zero forces the resulting constant sq to vanish. This proves the bijection onto the stated coefficient class, with no unmentioned axis gauge or lost flux.

16.2 The complete material and diffusion operators

For smooth functions of (s, z, t), set

$$\begin{aligned} Lg &= 4sg_{ss} + 4g_s + g_{zz}, \\ Mg &= 4sg_{ss} + 8g_s + g_{zz}, \\ Dg &= g_t + 2sag_s + cg_z. \end{aligned} \tag{271}$$

The scalar three-dimensional Laplacian is L, whereas $\Delta_3(x_j g) = x_j Mg$ for $j = 1, 2$. For example $\partial_{x_j} g = 2x_j g_s$, and summing the differentiated products gives these two coefficients 4 and 8. The full vector advection is

$$\begin{aligned} ((u \cdot \nabla)u)_1 &= x_1(a^2 - h^2 + 2saa_s + ca_z) - x_2(2ah + 2sah_s + ch_z), \\ ((u \cdot \nabla)u)_2 &= x_2(a^2 - h^2 + 2saa_s + ca_z) + x_1(2ah + 2sah_s + ch_z), \\ ((u \cdot \nabla)u)_3 &= 2sac_s + cc_z. \end{aligned} \tag{272}$$

The terms $-h^2$ and $2ah$ have both been retained.

Define the exact residuals

$$R_h = Dh - \nu Mh + 2ah, \qquad E_\xi = D\xi - \nu M\xi - \partial_z(h^2). \tag{273}$$

Equivalently, the two scalar equations are

$$Dh = \nu Mh + 2q_z h + R_h, \qquad D\xi = \nu M\xi + \partial_z(h^2) + E_\xi. \tag{274}$$

These are exact equations for the original functions, including all time, convection, stretching and diffusion terms.

16.3 A force inverse through the axis and the full pressure

Set

$$F(s, z, t) = \frac{1}{2}\int_s^\infty E_\xi(\sigma, z, t)\, d\sigma, \qquad f = (-x_2 R_h,\ x_1 R_h,\ F). \tag{275}$$

All derivatives of the integrand have fixed compact support. Differentiation under the integral is consequently justified on every compact time interval, and $F_s = -E_\xi/2$. More explicitly, near $s = 0$ one can write $F = \frac{1}{2}\int_0^\infty E_\xi\, d\sigma - \frac{1}{2}\int_0^s E_\xi\, d\sigma$ using the given smooth extension; this proves smoothness on an actual neighborhood of the axis. The force is zero for $s > S$ and for $|z| > Z$. Its support can fill the interval between the support of E_ξ and $s = 0$, which is still compact in physical space. In particular, no zero-integral or mean-zero condition on E_ξ has been imposed. Its value at the axis is $f^z(0, z, t) = \frac{1}{2}\int_0^\infty E_\xi(\sigma, z, t)\, d\sigma$. The off-axis cylindrical components and their exact curl relation are

$$f^r = 0, \qquad f^\phi = rR_h, \qquad f^z = F, \qquad \frac{(\operatorname{curl} f)^\phi}{r} = \frac{\partial_z f^r - \partial_r f^z}{r} = -2F_s = E_\xi. \tag{276}$$

In Cartesian form the entire curl is

$$\operatorname{curl} f = (-x_1(R_h)_z - x_2 E_\xi,\ -x_2(R_h)_z + x_1 E_\xi, 2R_h + 2s(R_h)_s),$$

so its axis extension is explicit as well.

For completeness, put $\mathcal{F} = f + \nu\Delta_3 u - u_t - (u \cdot \nabla)u$. Equations (271)–(273) give

$$\begin{aligned} \mathcal{F} &= (x_1 A,\ x_2 A,\ B), \\ A &= \nu Ma - a_t - a^2 - 2saa_s - ca_z + h^2, \\ B &= F + \nu Lc - c_t - 2sac_s - cc_z. \end{aligned} \tag{277}$$

The azimuthal coefficient vanished because it was exactly $R_h + \nu M h - h_t - 2sah_s - ch_z - 2ah = 0$. In particular $\mathcal{F}^r/r = A$ is a specified smooth function, even though the quotient notation itself applies only off the axis.

The closure identity, including its sign, is

$$A_z - 2B_s = -2F_s - E_\xi = 0. \tag{278}$$

Here is a full algebraic verification. The time terms are $-\xi_t$. The viscosity terms are $\nu M\xi$, since $(Lc)_s = M(c_s)$ and $(Ma)_z = M(a_z)$. The force and swirl terms are $-2F_s + \partial_z(h^2)$. The remaining advection terms are

$$N = -\partial_z(a^2 + 2saa_s + ca_z) + 2\partial_s(2sac_s + cc_z).$$

Expanding the products groups them as

$$N = -2sa\,\partial_s(a_z - 2c_s) - c\,\partial_z(a_z - 2c_s) - (a_z - 2c_s)(2a + 2sa_s + c_z).$$

The final factor is the divergence already proved to be zero, leaving $-2sa\xi_s - c\xi_z$ and proving (278).

Define the pressure, with a fixed time-dependent additive gauge, by

$$p(s,z,t) = \int_0^z B(0,\zeta,t)\,d\zeta + \frac{1}{2}\int_0^s A(\sigma,z,t)\,d\sigma. \tag{279}$$

Both integrals have smooth integrands through the axis. They give $p_s = A/2$ and, using $A_z = 2B_s$,

$$p_z = B(0,z,t) + \frac{1}{2}\int_0^s A_z(\sigma,z,t)\,d\sigma = B(s,z,t).$$

Thus the Cartesian gradient is $\nabla_3 p = (x_1 A, x_2 A, B) = \mathcal{F}$ everywhere, including the axis. We have proved, with the full force and pressure displayed,

$$u_t + (u\cdot\nabla_3)u + \nabla_3 p = \nu\Delta_3 u + f, \qquad \operatorname{div}_3 u = 0. \tag{280}$$

The pressure in this gauge has a constant value on the exterior of the physical support cylinder, and that constant can be removed explicitly. Indeed, $A_z = 2B_s$ implies $\partial_s \int_{\mathbb{R}} B(s,z,t)\,dz = 0$. This integral vanishes for $s > S$, hence also at $s = 0$; no unequal upper and lower axial pressure constants arise. The Cartesian gradient vanishes outside the closed cylinder $s \le S, |z| \le Z$, whose exterior is connected. At the exterior point $(s,z) = (S+1,0)$ the value of the pressure is

$$C(t) = \frac{1}{2}\int_0^{S+1} A(\sigma,0,t)\,d\sigma.$$

Thus $p - C(t)$ is a smooth pressure supported in that closed cylinder and gives precisely the same momentum equation. This conclusion follows from the constructed force and the closure identity; a pressure-decay hypothesis was not imposed separately.

16.4 The five-dimensional operator morphism and its Newton inverse

Use distinct coordinates $Y = (X,z) \in \mathbb{R}^4 \times \mathbb{R}$. For each scalar coefficient define its lift by $\mathcal{J}g(X,z,t) = g(|X|^2,z,t)$, and write $Q = \mathcal{J}q$, $H = \mathcal{J}h$, $\Xi = \mathcal{J}\xi$. This map is injective: every $s \ge 0$ is $|X|^2$ for some X. Its codomain here is exactly the displayed lifts of smooth coefficient functions, so no regularity claim about a larger invariant function space is being assumed. Define

$$V(X,z,t) = (-q_z(|X|^2,z,t)X,\ 2q + 2|X|^2 q_s). \tag{281}$$

Differentiating in the four original X coordinates proves

$$\begin{gathered}\Delta_5\mathcal{J}g = \mathcal{J}(Mg), \qquad (\partial_t + V\cdot\nabla_5)\mathcal{J}g = \mathcal{J}(Dg),\\ \Xi = -\Delta_5 Q, \qquad \operatorname{div}_5 V = 4a + 2sa_s + c_z = -2q_z = -2Q_z,\\ V\cdot\nabla_5 Q = -2sq_zq_s + cq_z = 2qq_z, \qquad \operatorname{div}_5(QV) = 0.\end{gathered} \tag{282}$$

Consequently the full transported scalar equations are

$$\begin{aligned} H_t + V \cdot \nabla_5 H &= \nu \Delta_5 H + 2Q_z H + \mathcal{J} R_h, \\ \Xi_t + V \cdot \nabla_5 \Xi &= \nu \Delta_5 \Xi + \partial_z (H^2) + \mathcal{J} E_\xi. \end{aligned} \tag{283}$$

This calculation retains the nonzero five-dimensional divergence. It establishes an exact map of material operators and diffusion, rather than identifying V with an incompressible five-dimensional velocity.

The elliptic inverse for these compactly supported coefficients has the exact constant

$$Q(Y,t) = \frac{1}{8\pi^2} \int_{\mathbb{R}^5} \frac{\Xi(Y',t)}{|Y-Y'|^3} \, dY'. \tag{284}$$

To prove it, the Gaussian integral in five variables is $\pi^{5/2}$. In polar coordinates it is $|\mathbb{S}^4| \int_0^\infty e^{-\rho^2} \rho^4 \, d\rho = |\mathbb{S}^4| \, 3\sqrt{\pi}/8$, where two integrations by parts give the last integral. Therefore $|\mathbb{S}^4| = 8\pi^2/3$. For $\Phi(Y) = |Y|^{-3}/(8\pi^2)$, differentiation shows $\Delta_5 \Phi = 0$ away from zero, and the inward flux for $-\Delta_5$ across a sphere is $3|\mathbb{S}^4|/(8\pi^2) = 1$. Green's identity on a punctured ball, followed by shrinking its inner sphere, gives $-\Delta_5 \Phi = \delta_0$: the term involving $\Phi \partial_r \varphi$ has magnitude $O(\rho)$, whereas the normal-derivative term converges to $\varphi(0)$ for each smooth compactly supported test function φ. The kernel is locally integrable because its radial integral at zero is $\int_0^\epsilon \rho \, d\rho$. Hence, using compact support of Q, distributional integration by parts gives $\Phi * (-\Delta_5 Q) = (-\Delta_5 \Phi) * Q = Q$, which is exactly (284). These equalities can also be tested against a compact test function and justified by Fubini with the locally integrable kernel; they introduce no boundary term at infinity.

The compact support in this assertion belongs to q, and therefore to the particular $\Xi = -\Delta_5 Q$. It is not a conclusion for an arbitrary compactly supported right-hand side. Indeed, if a nonnegative smooth right-hand side G, supported in $|Y| \le R$, has mass $m > 0$, its Newton potential satisfies for $|Y| > 2R$

$$(\Phi * G)(Y) \ge \frac{m}{8\pi^2(|Y|+R)^3}.$$

It is consequently neither compactly supported nor rapidly decreasing. For the actual image of our map, integration by parts additionally gives $\int \Xi P \, dY = 0$ for every harmonic polynomial P, including $\int \Xi \, dY = 0$. These facts describe real restrictions on the elliptic image; no arbitrary compact Ξ is silently substituted.

16.5 The exact measures, energy, dissipation and force work

The two measures differ. Angular integration in physical space gives $2\pi r \, dr \, dz = \pi \, ds \, dz$. In $\mathbb{R}^5$, integration over the four-dimensional radial variable gives $|\mathbb{S}^3| r^3 dr \, dz = \pi^2 s \, ds \, dz$, since $|\mathbb{S}^3| = 2\pi^2$. The latter constant follows, for example, from $\pi^2 = |\mathbb{S}^3| \int_0^\infty e^{-r^2} r^3 dr = |\mathbb{S}^3|/2$. All integrals in (s,z) below have domain $[0,\infty) \times \mathbb{R}$.

The complete physical energy, without a factor $1/2$ in its definition, satisfies

$$\begin{aligned} E_3(t) := \int_{\mathbb{R}^3} |u|^2 \, dx &= \pi \int \left[s(q_z^2 + h^2) + 4(q + s q_s)^2 \right] ds \, dz \\ &= \frac{1}{\pi} \int_{\mathbb{R}^5} (|\nabla_5 Q|^2 + H^2) \, dY. \end{aligned} \tag{285}$$

To see the last equality with all boundary terms, note that $|\nabla_5 Q|^2 = 4 s q_s^2 + q_z^2$, and the difference of the two meridional integrands is exactly

$$4(q + s q_s)^2 - 4 s^2 q_s^2 = 4 \partial_s (s q^2).$$

Its integral is zero at infinity by compact support and at zero by smoothness and the factor s. Neither measure was replaced by the other in this computation.

The complete enstrophy identity is

$$\int_{\mathbb{R}^3} |\operatorname{curl} u|^2\, dx = \pi \int \left[s(\xi^2 + h_z^2) + 4(h + sh_s)^2 \right] ds\, dz = \frac{1}{\pi} \int_{\mathbb{R}^5} (\Xi^2 + |\nabla_5 H|^2)\, dY. \tag{286}$$

The cross terms in the first two Cartesian vorticity components in (269) cancel, and the remaining difference is $4\partial_s(sh^2)$, whose two boundary values vanish. Also $\int |\nabla_3 u|^2 = \int |\operatorname{curl} u|^2$: expansion of the squared curl gives $\int |\nabla u|^2 - \int \partial_i u_j \partial_j u_i$, and two integrations by parts identify the latter integral with $\int (\operatorname{div} u)^2 = 0$. Compact support justifies each integration.

The force inverse is compatible with these exact measures as follows:

$$\int_{\mathbb{R}^3} u \cdot f\, dx = \pi \int s(qE_\xi + hR_h)\, ds\, dz = \frac{1}{\pi} \int_{\mathbb{R}^5} (Q\mathcal{J}E_\xi + H\mathcal{J}R_h)\, dY. \tag{287}$$

Indeed $u \cdot f = shR_h + cF$, and

$$\int_0^\infty cF\, ds = [2sqF]_0^\infty - \int_0^\infty 2sqF_s\, ds = \int_0^\infty sqE_\xi\, ds.$$

The boundary at zero vanishes even when $F(0, z, t) \neq 0$, because q, F are smooth and the factor is s.

These identities also prove the full energy balance directly from the five-dimensional equations. Differentiate (285) and use $\Xi = -\Delta_5 Q$ and its time derivative to obtain

$$\tfrac{\pi}{2} E_3' = \int_{\mathbb{R}^5} (Q\Xi_t + HH_t)\, dY.$$

Substitution of (283) gives the dissipation $-\nu \int (\Xi^2 + |\nabla H|^2)$, and the following individual transport and coupling terms:

$$\begin{aligned}
-\int QV \cdot \nabla \Xi &= \int \Xi \operatorname{div}(QV) = 0, \\
\int Q\partial_z(H^2) &= -\int Q_z H^2, \\
-\int HV \cdot \nabla H &= \tfrac{1}{2} \int \operatorname{div} V\, H^2 = -\int Q_z H^2, \\
\int H(2Q_z H) &= 2 \int Q_z H^2.
\end{aligned}$$

Their sum is exactly zero. The remaining terms are the force work in (287). Division by π, followed by (286), proves

$$\frac{1}{2} \frac{d}{dt} \int_{\mathbb{R}^3} |u|^2\, dx + \nu \int_{\mathbb{R}^3} |\nabla_3 u|^2\, dx = \int_{\mathbb{R}^3} u \cdot f\, dx. \tag{288}$$

All differentiated integrals are justified by fixed compact spatial support and bounded derivatives on compact time intervals.

The corresponding local five-dimensional energy flux is also exact. With $e_z = (0, 0, 0, 0, 1)$, set

$$\begin{aligned}
\mathcal{B} = {} & Q\nabla_5 Q_t - Q\Xi V + \nu(Q\nabla_5 \Xi - \Xi \nabla_5 Q) \\
& + QH^2 e_z - \tfrac{1}{2} H^2 V + \nu H \nabla_5 H.
\end{aligned} \tag{289}$$

Then, pointwise on all of $\mathbb{R}^5 \times I$,

$$\partial_t \frac{|\nabla_5 Q|^2 + H^2}{2} + \nu(\Xi^2 + |\nabla_5 H|^2) = Q\mathcal{J}E_\xi + H\mathcal{J}R_h + \operatorname{div}_5 \mathcal{B}. \tag{290}$$

The product identities proving this formula are

$$\begin{aligned}
\nabla Q \cdot \nabla Q_t &= Q\Xi_t + \operatorname{div}(Q\nabla Q_t),\\
-QV \cdot \nabla \Xi &= -\operatorname{div}(Q\Xi V),\\
Q\Delta\Xi &= -\Xi^2 + \operatorname{div}(Q\nabla\Xi - \Xi\nabla Q),\\
Q\partial_z(H^2) &= \operatorname{div}(QH^2 e_z) - Q_z H^2,\\
-HV \cdot \nabla H &= -\operatorname{div}(H^2 V/2) - Q_z H^2,\\
H\Delta H &= -|\nabla H|^2 + \operatorname{div}(H\nabla H).
\end{aligned}$$

Here all differential operators in these six lines are five-dimensional. The first uses $\Xi_t = -\Delta Q_t$, the second uses $\operatorname{div}(QV) = 0$, the third uses $\Delta Q = -\Xi$, and the fifth uses $\operatorname{div} V = -2Q_z$. Adding them with the two scalar equations proves (290), including the cancellation with $2Q_z H^2$. Every transverse gradient of a smooth lifted coefficient and every transverse component of V is $O(|X|)$ on compact time intervals. The flux through an inner tube $|X| = \epsilon$, over its bounded axial support, is therefore $O(\epsilon^4)$ and tends to zero. This proves explicitly that the local balance has no extra axis flux.

16.6 Global smooth finite fields and every time-cutoff term

For explicit global-in-time instances retain arbitrary $0 < T_0 < T_1$ and initial coefficients q_0, h_0 satisfying the hypotheses on an open interval containing $[0, T_1]$. Define

$$b(\tau) = \begin{cases} e^{-1/\tau}, & \tau > 0,\\ 0, & \tau \le 0, \end{cases} \qquad \eta(t) = \frac{b(T_1 - t)}{b(T_1 - t) + b(t - T_0)}.$$

Every derivative of b for positive τ is an exponential times a polynomial in $1/\tau$ and tends to zero as $\tau \downarrow 0$. Thus b, and then η, are smooth; the denominator never vanishes because $T_0 < T_1$. The values are $\eta = 1$ on $t \le T_0$ and $\eta = 0$ on $t \ge T_1$. Take $q = \eta q_0$, $h = \eta h_0$ through T_1 and extend both by zero thereafter. Flatness at T_1 proves that this extension is smooth. Write $\xi_0 = -Mq_0$, let $v_0 \cdot \nabla_{s,z} = -2s(q_0)_z\partial_s + (2q_0 + 2s(q_0)_s)\partial_z$, and define

$$R_0 = (h_0)_t + v_0 \cdot \nabla h_0 - \nu M h_0 - 2(q_0)_z h_0, \quad E_0 = (\xi_0)_t + v_0 \cdot \nabla \xi_0 - \nu M \xi_0 - \partial_z(h_0^2).$$

The complete residuals for the actual cutoff fields are

$$\begin{aligned}
R_h &= \eta R_0 + \eta' h_0 + \eta(\eta - 1)\big(v_0 \cdot \nabla h_0 - 2(q_0)_z h_0\big),\\
E_\xi &= \eta E_0 + \eta' \xi_0 + \eta(\eta - 1)\big(v_0 \cdot \nabla \xi_0 - \partial_z(h_0^2)\big).
\end{aligned} \tag{291}$$

These formulas follow by the product rule: time and diffusion terms have one factor η, and the displayed quadratic terms have two. In particular the η' and $\eta(\eta - 1)$ terms are part of the force (275), and have not been discarded. That integral, together with (279), now gives a solution of (280) on the entire time half-line.

All force components and their mixed space-time derivatives are supported in the fixed cylinder $s \le S, |z| \le Z, 0 \le t \le T_1$. Smoothness makes each derivative bounded there. Consequently, for every multi-index α, integer $m \ge 0$, and integer $K \ge 0$,

$$\sup_{x \in \mathbb{R}^3, t \ge 0} (1 + |x| + t)^K |\partial_x^\alpha \partial_t^m f(x,t)| < \infty.$$

The initial velocity has the corresponding Schwartz bounds and is divergence-free. To give a completely explicit finite energy estimate, let Q_0, Q_s, Q_z, H_0 bound respectively $|q|, |q_s|, |q_z|, |h|$ on the fixed space-time cylinder. Then

$$\sup_{t \ge 0} E_3(t) \le 2\pi Z \left\{ \frac{S^2}{2}(Q_z^2 + H_0^2) + 4S(Q_0 + SQ_s)^2 \right\} < \infty. \tag{292}$$

This follows by bounding the two integrands in (285) and integrating s and 1 over $[0, S]$. The solution itself is global and smooth, so these instances do not disprove global regularity. The established transfer supplies exact axis, material, elliptic, force, pressure and energy formulas for subsequent constructions without importing an unproved singular limit.

17 The complete axisymmetric Gaussian family and its terminal forcing

17.1 Original parameters, fields and domains

Fix the original constants

$$\nu > 0, \quad T > 0, \quad \tau_* > 0, \quad L_* > 0, \qquad U_*, H_*, \beta \in \mathbb{R}.$$

For physical time $0 \le t < T$, put

$$\begin{gathered} \tau = T - t, \quad \sigma = \tau/\tau_*, \quad L = L_*\sigma^\beta, \quad k = L^{-2}, \\ A = U_*\sigma^{\beta-1}, \quad B = H_*\sigma^{-1}, \quad s = x_1^2 + x_2^2 = r^2, \quad \rho = s + z^2, \quad v = k\rho, \quad G = e^{-k\rho}. \end{gathered} \tag{293}$$

Here v is an explicit polynomial argument, not a replacement of the original spatial variables. The centre is exactly $z = 0$. In particular $\sigma(0) = T/\tau_*$, which has not been set to one. The given coefficients and their time derivatives are

$$\begin{gathered} q = AG, \qquad h = BG, \\ A_t = (1-\beta)A/\tau, \quad B_t = B/\tau, \quad k_t = 2\beta k/\tau, \\ q_t = q(1-\beta-2\beta v)/\tau, \qquad h_t = h(1-2\beta v)/\tau. \end{gathered} \tag{294}$$

Every formula is smooth for all real s, z and for $t < T$; this provides an actual smooth extension through $s = 0$. The physical field, its streamfunction and its circulation are

$$\begin{gathered} u = (2kzq\, x_1 - hx_2,\ 2kzq\, x_2 + hx_1,\ 2q(1-ks)), \\ u^r = 2kzrq, \quad u^\phi = rh, \quad u^z = 2q(1-ks), \qquad \Psi = -sq, \quad \Gamma = sh. \end{gathered} \tag{295}$$

The cylindrical equalities use the positive frame $e_r = (x_1/r, x_2/r, 0)$, $e_\phi = (-x_2/r, x_1/r, 0)$, $e_z = (0, 0, 1)$ only when $r > 0$. The Cartesian formula defines the axis values.

With $a = 2kzq = -q_z$ and $c = 2q(1-ks)$, direct differentiation gives $2a + 2sa_s + c_z = 0$; hence $\operatorname{div}_3 u = 0$. The complete vorticity is

$$\begin{gathered} \xi = -(q_{zz} + 4sq_{ss} + 8q_s) = AkG(10 - 4v), \\ \omega = (2kzh\, x_1 - x_2\xi,\ 2kzh\, x_2 + x_1\xi,\ 2h(1-ks)). \end{gathered} \tag{296}$$

For instance $\omega_1 = \partial_{x_2}c - \partial_z(x_2 a + x_1 h) = -x_2(a_z - 2c_s) - x_1 h_z$; the other two derivatives give the displayed expression. Thus $\omega^\phi = r\xi$ off the axis.

All fields in this section have Gaussian tails, rather than compact support. On every closed subinterval of $[0, T)$, k has a strictly positive lower bound and finite upper bound, and all its time derivatives, as well as those of A, B, are bounded. Every mixed derivative of a displayed field is a finite sum of polynomials in (x_1, x_2, z) times $e^{-k\rho}$ or $e^{-2k\rho}$, with coefficients bounded on that subinterval. Multiplication by any fixed power of $1 + |x|$ leaves a bounded function. This proves smoothness, Schwartz decay at each time, and locally uniform differentiated Gaussian tails before the endpoint.

17.2 Every residual, the inward force, and the full pressure

Retain the operators

$$M = 4s\partial_s^2 + 8\partial_s + \partial_z^2, \qquad L_3 = 4s\partial_s^2 + 4\partial_s + \partial_z^2, \qquad D = \partial_t + 2sa\partial_s + c\partial_z.$$

Thus $\Delta_3 g = L_3 g$ for scalar coefficients, while $\Delta_3(x_j g) = x_j M g$ for $j = 1, 2$. In this family

$$(2sa\partial_s + c\partial_z)\rho = 4zq.$$

The following four polynomials retain all their coefficients:

$$\begin{aligned} P_\beta(v) &= 10(1+\beta) - (4+32\beta)v + 8\beta v^2, \\ Q(v) &= 140 - 112v + 16v^2, \\ R_\beta(v) &= 6(1-\beta) - (4+16\beta)v + 8\beta v^2, \\ S(v) &= 60 - 80v + 16v^2. \end{aligned} \tag{297}$$

Their specific roles follow by differentiating the preceding expressions:

$$\begin{gathered} \xi_t = AkGP_\beta(v)/\tau, \qquad M\xi = -Ak^2GQ(v), \\ (2sa\partial_s + c\partial_z)\xi = zA^2k^2G^2(16v - 56), \\ (2sa\partial_s + c\partial_z)h = -4kzqh = 2q_z h. \end{gathered} \tag{298}$$

For example, the third line is $4zq$ times the ρ-derivative $AG(-14k^2 + 4k^3\rho)$. The last identity is the exact cancellation of convection with stretching for this particular common Gaussian shape. It does not discard those terms in other coefficient families.

Define the complete scalar residuals by $R_h = Dh - \nu Mh - 2q_z h$ and $E_\xi = D\xi - \nu M\xi - \partial_z(h^2)$. Their fully evaluated values are

$$\begin{aligned} R_h &= BG\left[\frac{1-2\beta v}{\tau} + \nu k(10 - 4v)\right], \\ E_\xi &= AkG\left[\frac{P_\beta(v)}{\tau} + \nu kQ(v)\right] + zG^2\left[A^2k^2(16v - 56) + 4kB^2\right]. \end{aligned} \tag{299}$$

In particular the swirl-square contribution in E_ξ has positive coefficient $4kB^2zG^2$.

The required axial integral converges with every differentiated tail, by the bounds proved above. It evaluates exactly to

$$\begin{aligned} F(s,z,t) &= \frac{1}{2}\int_s^\infty E_\xi(\zeta, z, t)\, d\zeta \\ &= \frac{AG}{2}\left[\frac{R_\beta(v)}{\tau} + \nu kS(v)\right] + zG^2\left[4A^2k(v-3) + B^2\right], \\ f &= (-x_2R_h,\ x_1R_h,\ F). \end{aligned} \tag{300}$$

One verification of the integral retains the change of integration variable $w = k(\zeta + z^2)$, whose differential is $k\,d\zeta$. The identities $R_\beta - R_\beta' = P_\beta$ and $S - S' = Q$ integrate the linear terms. For the quadratic terms one uses

$$\int_\rho^\infty e^{-2k\rho'}\, d\rho' = \frac{G^2}{2k}, \qquad \int_\rho^\infty \rho' e^{-2k\rho'}\, d\rho' = G^2\left(\frac{\rho}{2k} + \frac{1}{4k^2}\right).$$

They give precisely the last term of (300). Alternatively, differentiating it gives $-2F_s = E_\xi$, and both its value and every differentiated tail tend to zero as $s \to \infty$. In either proof the factor $1/2$, the sign and the infinite endpoint are retained. The integral also extends smoothly through the axis: write it as $F(0,z,t) - \frac{1}{2}\int_0^s E_\xi\, d\zeta$. No zero-integral condition is needed.

The complete force curl is

$$\operatorname{curl} f = (-x_1(R_h)_z - x_2E_\xi,\ -x_2(R_h)_z + x_1E_\xi, 2R_h + 2s(R_h)_s). \tag{301}$$

It follows by three Cartesian derivatives and $-2F_s = E_\xi$, and holds on the axis as well as off it.

A pressure with spatially decaying gauge is

$$p = zq\left[\frac{1-\beta-2\beta v}{\tau} + \nu k(10-4v)\right] + \frac{q^2}{2}(1-2v) - \frac{h^2}{4k}. \tag{302}$$

Here is a direct full momentum verification. With $\mathcal{F} = f + \nu\Delta_3 u - u_t - (u\cdot\nabla_3)u$, the radial, azimuthal and axial coefficients are respectively

$$\mathcal{F} = (x_1\mathcal{A}, x_2\mathcal{A}, \mathcal{B}), \quad \mathcal{A} = \nu Ma - a_t - a^2 - 2saa_s - ca_z + h^2, \quad \mathcal{B} = F + \nu L_3c - c_t - 2sac_s - cc_z.$$

The azimuthal coefficient is exactly $R_h + \nu Mh - h_t - 2sah_s - ch_z - 2ah = 0$. For the radial coefficient, substitution gives

$$\mathcal{A} = AzG\left[\nu k^2(-28+8v) - \frac{2k}{\tau}(1+\beta-2\beta v)\right] - 4kA^2G^2(1-v) + B^2G^2. \tag{303}$$

The advection terms here follow from $a^2 + (2sa\partial_s + c\partial_z)a = 4kq^2(1-v)$, so their centrifugal contribution has not been removed. Integrating $-\mathcal{A}/2$ from s to infinity yields (302). Thus $2p_s = \mathcal{A}$. The exact curl closure is $\mathcal{A}_z - 2\mathcal{B}_s = 0$: its time, diffusion, advection and swirl contributions are respectively $-\xi_t$, $\nu M\xi$, $-2sa\xi_s - c\xi_z$ and $\partial_z(h^2)$, together with $-2F_s$. These sum to $-E_\xi - 2F_s = 0$. Since $\mathcal{B}$ and its differentiated tails decay at infinity, differentiation of the pressure integral gives $p_z = -\frac{1}{2}\int_s^\infty \mathcal{A}_z\,d\zeta = \mathcal{B}$. We have therefore proved all three Cartesian equations

$$u_t + (u\cdot\nabla_3)u + \nabla_3 p = \nu\Delta_3 u + f, \qquad \operatorname{div}_3 u = 0. \tag{304}$$

The axis-path pressure $\int_0^z \mathcal{B}(0,\zeta,t)d\zeta + \frac{1}{2}\int_0^s \mathcal{A}(\zeta,z,t)d\zeta$ is precisely $p - p(0,0,t)$, where $p(0,0,t) = A^2/2 - B^2/(4k)$. This proves the pressure gauge relation, including its time-dependent constant. The decaying pressure and its gradient are Schwartz at every preterminal time.

17.3 The five-dimensional map and the complete energy integrals

For $Y = (X,z) \in \mathbb{R}^4\times\mathbb{R}$, put $Q_5 = Ae^{-k|Y|^2}$, $H_5 = Be^{-k|Y|^2}$, and $\Xi_5 = Ake^{-k|Y|^2}(10-4k|Y|^2)$. The lift substitutes $s = |X|^2$ and retains the original time and coefficients. Differentiation in all four X coordinates shows

$$\Delta_5 Q_5 = -\Xi_5, \quad V = (-q_zX, 2q + 2|X|^2q_s), \quad \operatorname{div}_5 V = -2(Q_5)_z, \quad \operatorname{div}_5(Q_5V) = 0.$$

For every smooth coefficient g, the lift carries Mg into $\Delta_5(g(|X|^2,z,t))$ and Dg into its material derivative $(\partial_t + V\cdot\nabla_5)g(|X|^2,z,t)$. In particular it carries both full equations with residuals (299). The Newton formula also holds for these Gaussian tails:

$$Q_5(Y,t) = \frac{1}{8\pi^2}\int_{\mathbb{R}^5}\frac{\Xi_5(Y',t)}{|Y-Y'|^3}\,dY'.$$

Indeed $|\mathbb{S}^4| = 8\pi^2/3$, obtained by integrating $e^{-|Y|^2}$ both as $\pi^{5/2}$ and in polar coordinates. The kernel $|Y|^{-3}/(8\pi^2)$ is harmonic away from zero and has unit $-\Delta_5$ flux. Green's identity on punctured expanding balls gives the formula: the inner boundary converges to $Q_5(Y,t)$, and the outer boundary tends to zero because Q_5 and its gradient decay faster than every power. The convolution is absolutely convergent at infinity and locally integrable at the kernel singularity.

The measures after transverse angular integration are, respectively, $\pi\,ds\,dz$ in physical space and $\pi^2 s\,ds\,dz$ in five dimensions. The identity $4(q+sq_s)^2 - 4s^2q_s^2 = 4\partial_s(sq^2)$ has zero boundary integral at both ends, by axis smoothness and Gaussian decay. The same argument with h and the explicitly computed vorticity gives

$$\begin{aligned} E(t) &:= \int_{\mathbb{R}^3} |u|^2\, dx = \frac{1}{\pi}\int_{\mathbb{R}^5} (|\nabla Q_5|^2 + H_5^2)\, dY, \\ D_2(t) &:= \int_{\mathbb{R}^3} |\nabla u|^2\, dx = \int_{\mathbb{R}^3} |\omega|^2\, dx = \frac{1}{\pi}\int_{\mathbb{R}^5} (\Xi_5^2 + |\nabla H_5|^2)\, dY. \end{aligned} \tag{305}$$

The equality between gradient and curl integrals follows by integrating $\partial_i u_j \partial_j u_i$ twice and using $\operatorname{div} u = 0$; all boundary terms vanish by the proved tails.

For an explicit evaluation, in dimension n the three Gaussian moments are

$$\int e^{-2k|Y|^2} dY = (\pi/(2k))^{n/2}, \qquad \frac{\int |Y|^2 e^{-2k|Y|^2} dY}{\int e^{-2k|Y|^2} dY} = \frac{n}{4k}, \qquad \frac{\int |Y|^4 e^{-2k|Y|^2} dY}{\int e^{-2k|Y|^2} dY} = \frac{n(n+2)}{16k^2}.$$

They follow by differentiating the first Gaussian integral in k. Using $n = 5$, all terms in (305) evaluate to

$$\begin{aligned} E(t) &= C_E \sigma^{5\beta-2}, \qquad C_E = \frac{\pi^{3/2}}{2^{5/2}}(5U_*^2 L_*^3 + H_*^2 L_*^5), \\ D_2(t) &= C_D \sigma^{3\beta-2}, \qquad C_D = \frac{\pi^{3/2}}{2^{5/2}}(35U_*^2 L_* + 5H_*^2 L_*^3). \end{aligned} \tag{306}$$

For example the square $(10 - 4k|Y|^2)^2$ has Gaussian mean $100 - 100 + 35 = 35$, which fixes the poloidal dissipation coefficient.

The force-work identity follows directly from its radial integral:

$$\int_0^\infty cF\, ds = [2sqF]_0^\infty - \int_0^\infty 2sqF_s\, ds = \int_0^\infty sqE_\xi\, ds.$$

Both boundary terms vanish, including the one at the actual axis. Thus $\int u \cdot f = \pi \int s(qE_\xi + hR_h) ds dz$, and Gaussian integration, or the full material cancellation $\operatorname{div}(Q_5 V) = 0$, yields

$$\int_{\mathbb{R}^3} u \cdot f\, dx = \frac{2 - 5\beta}{2\tau_*} C_E \sigma^{5\beta-3} + \nu C_D \sigma^{3\beta-2} = \tfrac{1}{2}E'(t) + \nu D_2(t). \tag{307}$$

This also follows by multiplying (304) by u; the pressure and transport boundary terms vanish by the established Gaussian tails. Every coefficient is retained, including the sign of $2 - 5\beta$.

For nonzero amplitude pair (U_*, H_*), both C_E, C_D are strictly positive. Consequently energy remains bounded at T exactly when $\beta \geq 2/5$, is constant when $\beta = 2/5$, and tends to zero when $\beta > 2/5$. Writing $\sigma_0 = T/\tau_*$, the accumulated dissipation is exactly

$$\int_0^t D_2(t')dt' = \begin{cases} \dfrac{C_D \tau_*}{3\beta - 1}(\sigma_0^{3\beta-1} - \sigma^{3\beta-1}), & \beta \neq 1/3, \\ C_D \tau_* \log(\sigma_0/\sigma), & \beta = 1/3. \end{cases}$$

It is finite up to T exactly when $\beta > 1/3$.

17.4 Exact force and curl norms, and bounds for every mixed derivative

The following evaluated L^2 norms retain their cross terms. For short coefficients set $d = 1/\tau$, $\delta = \nu k$, and $I_m = (\pi/(mk))^{3/2}$, for $m = 2, 4$. Define

$$
\begin{aligned}
\mathsf{F}_s &= \frac{B^2}{8k}[d^2(35\beta^2 - 20\beta + 4) + d\delta(40 - 60\beta) + 140\delta^2],\\
\mathsf{F}_e &= \frac{15A^2}{16}[d^2(39\beta^2 - 20\beta + 4) + d\delta(56 - 84\beta) + 252\delta^2],\\
\mathsf{F}_o &= \frac{371A^4k^2 - 76A^2B^2k + 4B^4}{32k},\\
\mathsf{C}_s &= \frac{5B^2}{8}[d^2(23\beta^2 - 12\beta + 4) + d\delta(56 - 28\beta) + 252\delta^2],\\
\mathsf{C}_e &= \frac{35A^2k}{8}[d^2(27\beta^2 - 12\beta + 4) + d\delta(72 - 36\beta) + 396\delta^2],\\
\mathsf{C}_o &= \frac{455A^4k^2 - 84A^2B^2k + 4B^4}{8}.
\end{aligned}
$$

Then

$$
\|f\|_2^2 = I_2(\mathsf{F}_s + \mathsf{F}_e) + I_4\mathsf{F}_o, \qquad \|\operatorname{curl} f\|_2^2 = I_2(\mathsf{C}_s + \mathsf{C}_e) + I_4\mathsf{C}_o. \tag{308}
$$

Here is a finite integral derivation of every coefficient. For a polynomial $W(v)$, define the explicitly evaluated functional

$$
\mathcal{G}_m[W] = \sum_{j\ge 0}[v^j]W\,\frac{(3/2)(5/2)\cdots(3/2+j-1)}{m^j},
$$

where the product is one at $j = 0$. Polar integration proves $\int_{\mathbb{R}^3} W(k|x|^2)e^{-mk|x|^2}dx = I_m\mathcal{G}_m[W]$. Set, without discarding any term,

$$
\begin{aligned}
&C_R(v) = d(1 - 2\beta v) + \delta(10 - 4v),\\
&C_F(v) = dR_\beta(v) + \delta S(v), \qquad C_\xi(v) = dP_\beta(v) + \delta Q(v),\\
&C_N(v) = 4A^2k(v-3) + B^2, \qquad D_N(v) = A^2k^2(16v - 56) + 4kB^2.
\end{aligned}
$$

The even and odd axial force terms are orthogonal by reflection in z. For a radial integrand the angular means of s, z^2 and sz^2 are respectively $2\rho/3$, $\rho/3$ and $2\rho^2/15$. The six coefficients before expansion are therefore exactly

$$
\begin{aligned}
\mathsf{F}_s &= \frac{2B^2}{3k}\mathcal{G}_2[vC_R^2], & \mathsf{F}_e &= \frac{A^2}{4}\mathcal{G}_2[C_F^2], & \mathsf{F}_o &= \frac{1}{3k}\mathcal{G}_4[vC_N^2],\\
\mathsf{C}_s &= \frac{8B^2}{3}\mathcal{G}_2[v^2(C_R' - C_R)^2], & \mathsf{C}_e &= \frac{2A^2k}{3}\mathcal{G}_2[vC_\xi^2], & \mathsf{C}_o &= \frac{2}{15k^2}\mathcal{G}_4[v^2D_N^2].
\end{aligned}
$$

For the first curl coefficient, the curl of the swirl force is obtained from (296) by replacing h with R_h and setting $q = 0$. Its integral equals the five-dimensional weighted gradient integral of the lift of R_h; equivalently, integration of $\partial_v(v^{3/2}e^{-2v}C_R^2)$ removes the identical boundary term in the Cartesian expansion. This gives the displayed coefficient. The other two curl coefficients follow directly from (301) and the even/odd reflection. Evaluating the finite products gives exactly the six expanded polynomials above.

There are also explicit bounds for every L^p norm, including $p = \infty$, and every mixed derivative. For $a > 0$ and integer $j \ge 0$, put

$$
\mathcal{N}_{j,p}(a) = \begin{cases} [2\pi\,\Gamma((jp+3)/2)(pa)^{-(jp+3)/2}]^{1/p}, & 1 \le p < \infty,\\ 1, & p = \infty,\ j = 0,\\ (j/(2ae))^{j/2}, & p = \infty,\ j > 0. \end{cases} \tag{309}
$$

This is exactly $\||x|^j e^{-a|x|^2}\|_{L^p(\mathbb{R}^3)}$: polar integration proves the first case, and differentiating $r^j e^{-ar^2}$ proves the two supremum cases. Here $\Gamma(b) = \int_0^\infty t^{b-1}e^{-t}dt$ for $b > 0$, so the displayed constants are defined by a convergent integral. For example, with $r_j = [v^j]C_R$, $f_j = [v^j]C_F$ and $n_j = [v^j]C_N$, the full vector force satisfies

$$\begin{aligned}\|f\|_p \le |B| \sum_{j=0}^{1} |r_j| k^j \mathcal{N}_{2j+1,p}(k) \\ + \frac{|A|}{2} \sum_{j=0}^{2} |f_j| k^j \mathcal{N}_{2j,p}(k) + \sum_{j=0}^{1} |n_j| k^j \mathcal{N}_{2j+1,p}(2k).\end{aligned} \tag{310}$$

This follows from $r \le |x|$, $|z| \le |x|$, the exact three force terms and the triangle inequality; no signed term was removed from the field itself.

For a completely specified bound at every derivative order, expand any displayed field into its finite terms

$$c\sigma^a x^\gamma \exp\{-m|x|^2/(L_*^2\sigma^{2\beta})\}, \qquad m \in \{1,2\}.$$

This expansion is obtained just by substituting (293) and expanding the explicitly given polynomials. Its constants still contain all of $U_*, H_*, \nu, \tau_*, L_*, \beta$. Represent a term by (c, a, γ, m). Its exact spatial derivative in coordinate i is the sum of the following two terms, with a zero coefficient term omitted only when its coefficient is actually zero:

$$(c\gamma_i, a, \gamma - e_i, m), \qquad (-2mc/L_*^2, a - 2\beta, \gamma + e_i, m). \tag{311}$$

Its exact time derivative is the following four-term sum:

$$(-ac/\tau_*, a-1, \gamma, m), \qquad \left(-\frac{2\beta mc}{\tau_* L_*^2}, a - 2\beta - 1, \gamma + 2e_i, m\right), \quad i = 1, 2, 3. \tag{312}$$

These formulas are the full product and chain rules, including $\sigma_t = -1/\tau_*$. Iterating them gives every requested mixed derivative in finitely many exact terms. For the resulting finite list $\mathcal{L}$, a rigorous numerical or symbolic bound is

$$\|\partial_x^\alpha \partial_t^j f\|_p \le \sum_{(c,a,\gamma,m)\in\mathcal{L}} |c|\sigma^a \mathcal{N}_{|\gamma|,p}(mk). \tag{313}$$

For a vector field the lists for its three components are all included. The same recurrence supplies bounds for u, p, ω, R_h, E_ξ, or for the force curl using its displayed Cartesian components. For every nonnegative integer weight K, the corresponding weighted bound is obtained explicitly by replacing each $\mathcal{N}_{|\gamma|,p}(mk)$ in (313) by $\sum_{j=0}^K \binom{K}{j} \mathcal{N}_{|\gamma|+j,p}(mk)$. Indeed this is the binomial expansion of $(1+|x|)^K$ followed by the same monomial bound and triangle inequality. This gives explicit bounds at all orders, rather than assuming that undifferentiated Gaussian decay controls time derivatives at T.

17.5 Every force L^p endpoint, retaining the spatial dilation map

The endpoint sizes of the force can be determined for all $1 \le p \le \infty$. Use the specified auxiliary map $x = L_*\sigma^\beta y$, whose Jacobian is $L_*^3\sigma^{3\beta}$; write $w = |y|^2$. The original fields remain those in (300). Define the following two fixed functions with every original constant present:

$$\begin{aligned}\mathscr{F}_0(y) &= \frac{H_* L_*}{\tau_*} e^{-w}(1 - 2\beta w)(-y_2, y_1, 0) + \frac{U_*}{2\tau_*} e^{-w} R_\beta(w) e_z \\ &\quad + L_* y_3 e^{-2w} \left[\frac{4U_*^2}{L_*^2}(w-3) + H_*^2\right] e_z,\end{aligned} \tag{314}$$

$$\mathscr{F}_1(y) = \frac{H_*}{L_*} e^{-w}(10 - 4w)(-y_2, y_1, 0) + \frac{U_*}{2L_*^2} e^{-w} S(w) e_z. \tag{315}$$

Substitution into the full field proves the exact equality

$$f(L_*\sigma^\beta y,t) = \sigma^{\beta-2}\mathscr{F}_0(y) + \nu\sigma^{-\beta-1}\mathscr{F}_1(y). \tag{316}$$

This identity includes the complete nonlinear axial term in $\mathscr{F}_0$. No amplitude, length, viscosity or reference time has been fixed to a unit value.

For any nonzero amplitude pair, both functions are nonzero. If $H_* \neq 0$, their horizontal polynomials are nonzero. If $H_* = 0$ and $U_* \neq 0$, the even axial part of $\mathscr{F}_0$ contains R_β, which is not the zero polynomial for any β: its quadratic coefficient would require $\beta = 0$, and its constant coefficient would then be six. Its odd part cannot cancel that even part. Also $\mathscr{F}_1(0) = 30U_*e_z/L_*^2 \neq 0$. At $\beta = 1/2$, $\mathscr{F}_0 + \nu\mathscr{F}_1$ is nonzero as well: for $U_* \neq 0$ its value at zero is $U_*[3/(2\tau_*) + 30\nu/L_*^2]e_z$; otherwise its horizontal linear coefficient is $H_*[L_*/\tau_* + 10\nu/L_*] \neq 0$.

The exact Jacobian and continuity of the norm give, with $3/\infty = 0$,

$$\begin{aligned}
\|f\|_p &\sim L_*^{3/p}\|\mathscr{F}_0\|_p\sigma^{\beta(1+3/p)-2}, && \beta < 1/2,\\
\|f\|_p &= L_*^{3/p}\|\mathscr{F}_0 + \nu\mathscr{F}_1\|_p\sigma^{-3/2+3/(2p)}, && \beta = 1/2,\\
\|f\|_p &\sim \nu L_*^{3/p}\|\mathscr{F}_1\|_p\sigma^{\beta(3/p-1)-1}, && \beta > 1/2.
\end{aligned} \tag{317}$$

The notation $a \sim b$ here means their quotient tends to one, with the displayed strictly positive coefficient. For example, in the first case divide (316) by $\sigma^{\beta-2}$; the other term tends to zero in every indicated norm because $\nu\sigma^{1-2\beta} \to 0$. The reverse triangle inequality proves convergence of the norms, including the supremum norm. The last case follows in the same way with the other term retained.

It follows that for nonzero amplitudes

$$\sup_{t \text{ near } T}\|f(t)\|_p < \infty \quad \Longleftrightarrow \quad 1 \le p < 3 \ \text{ and } \ \beta \ge \frac{p}{3-p}. \tag{318}$$

Indeed, for $\beta < 1/2$ the first exponent is strictly negative for every $p \ge 1$. At equality $\beta = 1/2$, it is zero only for $p = 1$. For $\beta > 1/2$, the last exponent is nonnegative exactly in the range stated. At its boundary the norm tends to a positive constant, and above it the norm tends to zero. In particular the force L^2 norm is bounded exactly when $\beta \ge 2$, while $\int_0^T \|f(t)\|_2^2 dt$ is finite exactly when $\beta > 1$. The last statement follows from the squared exponents $5\beta - 4$ for $\beta < 1/2$, $-3/2$ at $1/2$, and $\beta - 2$ for $\beta > 1/2$; the equality $\beta = 1$ gives a logarithmic divergence. These integral norm facts do not imply smooth terminal forcing.

17.6 Exact terminal tests, all amplitudes and all exponents

The terminal failure of a nonzero velocity in this family is a physical field failure, with explicit original-coordinate tests. If $H_* \neq 0$, then $\omega_3(0,t) = 2H_*\sigma^{-1}$. If $H_* = 0$ and $U_* \neq 0$, then

$$\omega_2(L,0,0,t) = \frac{6U_*}{eL_*}\sigma^{-1}.$$

For $\beta \ge 0$ these points stay bounded, precluding a C^1 velocity extension. For $\beta < 0$, the fixed-origin value $u_3(0,t) = 2U_*\sigma^{\beta-1}$ already diverges. For clarity, large positive exponents need not produce an unbounded velocity supremum. The exact dilation formula is

$$u(Ly,t) = \sigma^{\beta-1}e^{-|y|^2}\big(2U_*y_3y_1 - H_*L_*y_2, 2U_*y_3y_2 + H_*L_*y_1, 2U_*(1 - y_1^2 - y_2^2)\big).$$

The displayed fixed function is nonzero for every nonzero amplitude pair, so $\|u\|_\infty$ is exactly its positive supremum times $\sigma^{\beta-1}$. For $\beta > 1$ the velocity tends uniformly to zero while the preceding vorticity tests still forbid a smooth extension. No auxiliary-coordinate singularity is substituted for these physical statements.

First, for nonzero swirl, the fixed-origin curl coefficient is

$$(\operatorname{curl} f)_3(0,t) = \frac{2H_*}{\tau_*}\sigma^{-2} + \frac{20\nu H_*}{L_*^2}\sigma^{-1-2\beta}. \tag{319}$$

Both terms have the sign of H_*, so its absolute value is at least $2|H_*|\sigma^{-2}/\tau_*$ for every real β. A force with $H_* \neq 0$ cannot be C^1 through T.

For any nonzero poloidal amplitude, the canonical axial force at the same original point is

$$F(0,0,t) = \frac{3(1-\beta)U_*}{\tau_*}\sigma^{\beta-2} + \frac{30\nu U_*}{L_*^2}\sigma^{-\beta-1}. \tag{320}$$

For $\beta < 1$ both coefficients have the sign of U_*, and the first term alone diverges. At $\beta = 1$ the first term is exactly zero and the second diverges. For $\beta > 1$, the ratio of the first term to the second is $(1-\beta)L_*^2\sigma^{2\beta-1}/(10\nu\tau_*)$, which tends to zero. Hence the magnitude is eventually at least $15\nu|U_*|\sigma^{-\beta-1}/L_*^2$, and again diverges. This proves that the specified force fails continuity through T for every $U_* \neq 0$, whether or not swirl is present.

To test other pressure and force choices for the same velocity, a pressure-independent calculation is needed in addition to (320). For $U_* \neq 0$, let

$$c_* = (7-\sqrt{14})/2 > 0, \qquad x(t) = (L\sqrt{c_*}, 0, 0).$$

The exact identities $Q(c_*) = 0$ and $P_\beta(c_*) = (1-6\beta)(10-4c_*)$ follow from (297). At $z = 0$, the nonlinear part of E_ξ is zero and $(R_h)_z = 0$, even when $H_* \neq 0$. Therefore, if $\beta \neq 1/6$,

$$(\operatorname{curl} f)_2(x(t),t) = \frac{U_*}{L_*\tau_*}\sigma^{-2}\sqrt{c_*}e^{-c_*}(1-6\beta)(10-4c_*). \tag{321}$$

Every constant multiplying U_* is nonzero: in particular $10-4c_* = -4+2\sqrt{14} > 0$. The exceptional exponent is handled explicitly. At $\beta = 1/6$, $P_{1/6} = Q/12$; use $x(t) = (L,0,0)$ instead. Since $P_{1/6}(1) = 11/3$ and $Q(1) = 44$, one obtains

$$(\operatorname{curl} f)_2(x(t),t) = \frac{U_*}{eL_*\tau_*}\sigma^{-2}\left[\frac{11}{3} + \frac{44\nu\tau_*}{L_*^2}\sigma^{2/3}\right]. \tag{322}$$

Its terms have the same sign. Thus, for every real exponent and every nonzero poloidal amplitude, the spatial supremum of the force curl diverges at least as a positive constant times σ^{-2}. For $\beta \geq 0$ the test points remain in a fixed bounded set, so this is also a local obstruction to a C^1 extension. For $\beta < 0$, the width expands and the test points escape to infinity; in that case this test proves failure of a uniform force-derivative bound. It is not being presented as a fixed-point local obstruction. The canonical force nevertheless fails continuity at the fixed origin by (320).

Suppose now that a force $\widetilde{f}$ and pressure $\widetilde{p}$ give (304) with exactly this same velocity. Subtraction of the two full equations gives

$$\widetilde{f} - f = \nabla(\widetilde{p} - p), \qquad \operatorname{curl}\widetilde{f} = \operatorname{curl} f.$$

The latter equality holds in distributions as well if the pressure is initially specified distributionally. Thus the tests above exclude every alternative force whose first spatial derivatives are uniformly bounded up to T, including the forcing class in the stated Navier–Stokes problem. A gradient or pressure choice cannot remove this obstruction.

There is a second complete check that uses the original energy requirement and only fixed-origin coefficients. For $H_* = 0$,

$$\partial_{x_1}(\operatorname{curl} f)_2(0,t) = E_\xi(0,0,t) = \frac{10(1+\beta)U_*}{\tau_* L_*^2}\sigma^{-\beta-2} + \frac{140\nu U_*}{L_*^4}\sigma^{-3\beta-1}. \tag{323}$$

When $\beta \geq 2/5$ and $U_* \neq 0$, both terms have the same sign and diverge. This excludes any alternative force locally smooth through T, independently of its pressure. At negative exponents one cannot infer the same conclusion just from this particular coefficient: for example, its first coefficient vanishes at $\beta = -1$. The uniform tests already cover these cases without making that inference.

For completeness, if $\beta < 2/5$, any smooth-through-T Schwartz force for the same preterminal branch would contradict the energy identity. Here the pressure integration needs justification, since it is no longer the explicit pressure (302). The alternative equation itself gives $\nabla\widetilde{p} = \widetilde{f} + \nu\Delta u - u_t - (u\cdot\nabla)u$. At each preterminal time its right side is Schwartz. A smooth scalar with such a gradient has a bounded gauge and a single limiting constant at infinity: integrate its radial derivative along rays, and compare two directions along a sphere of radius R, whose path length is at most πR. A bound $|\nabla\widetilde{p}| \leq C_N(1+R)^{-N}$, with $N > 2$, makes the directional difference tend to zero and the radial integral converge. Subtracting the resulting time-dependent constant makes the pressure decay. Consequently its product with the Gaussian velocity has zero boundary flux, and full energy integration gives

$$\tfrac{1}{2}E'(t) + \nu D_2(t) = \int u\cdot\widetilde{f}\,dx, \qquad \sqrt{E(t)} \leq \sqrt{E(0)} + \int_0^t \|\widetilde{f}(t')\|_2 dt'.$$

The last inequality follows from Cauchy–Schwarz and $E(t) > 0$. The admissible forcing bounds imply a uniform L^2 bound on $[0,T]$: for example $|\widetilde{f}(x,t)| \leq M(1+|x|)^{-2}$ gives $\|\widetilde{f}(t)\|_2 \leq M\sqrt{4\pi/3}$ by polar integration. The right side stays finite, whereas (306) diverges when $\beta < 2/5$. This proves that contradiction with the full pressure term justified.

All amplitude and exponent cases are now accounted for. If $U_* = H_* = 0$, the velocity, vorticity, residuals, canonical force and decaying pressure are identically zero for every β; they extend smoothly through T. If at least one amplitude is nonzero, the canonical force cannot extend smoothly through T, and no pressure modification can turn this same family into a solution with the required smooth-through-T Schwartz force. The exponents 0 (fixed width), 1/6 (the exceptional polynomial test), 1/3 (logarithmic accumulated dissipation), 2/5 (constant energy), 1/2 (equal force scales), and 1 (vanishing canonical time coefficient at the origin) are included in the computations above; the formulas and inequalities cover all other real exponents as well. This conclusion concerns the fully specified Gaussian family and its exact force morphism. It does not exclude different profiles, dynamics or singular constructions.

18 Every smooth moving centre of the original Gaussian family

Keep all original parameters and fields of Section 17: $\nu, T, \tau_*, L_* > 0$, real U_*, H_*, β, $\tau = T - t$, $\sigma = \tau/\tau_*$, $L = L_*\sigma^\beta$, $k = L^{-2}$, $A = U_*\sigma^{\beta-1}$ and $B = H_*\sigma^{-1}$. The spatial coordinates in this section are denoted $y = (y_1, y_2, y_3)$ before translation; thus $s = y_1^2 + y_2^2$, $z = y_3$, $\rho = |y|^2$, and $G = e^{-k|y|^2}$ in the unchanged Gaussian formulas. This is only a naming of the input of the translation, with its inverse given below.

Let $d : [0,T) \to \mathbb{R}^3$ be any smooth curve, with a smooth one-sided extension at 0. No bound on d, d', or their preterminal growth is imposed. In the original physical coordinates $x \in \mathbb{R}^3$ define

$$u_d(x,t) = u(x - d(t), t), \qquad p_d(x,t) = p(x - d(t), t),$$
$$f_d(x,t) = f(x - d(t), t) - \sum_{j=1}^{3} d_j'(t)\partial_{y_j} u(x - d(t), t). \tag{324}$$

The differential of $y = x - d(t)$ with respect to x is I_3, and its time derivative is $-d'(t)$. Hence

$$\partial_t u_d = (u_t - d'\cdot\nabla_y u)(x - d, t), \quad (u_d\cdot\nabla_x)u_d = ((u\cdot\nabla_y)u)(x - d, t), \quad \Delta_x u_d = (\Delta_y u)(x - d, t).$$

The pressure gradient and divergence translate in the same way. Substitution of the already proved full Gaussian momentum equation therefore proves

$$\partial_t u_d + (u_d \cdot \nabla_x) u_d + \nabla_x p_d = \nu \Delta_x u_d + f_d, \qquad \operatorname{div}_x u_d = 0.$$

The inverse on this triple is

$$u(y,t) = u_d(y+d,t), \quad p(y,t) = p_d(y+d,t), \quad f(y,t) = f_d(y+d,t) + d' \cdot \nabla_x u_d(y+d,t).$$

Substitution in both orders gives the identity on all three components. Thus the full force correction, and not only the velocity coordinates, is part of this exact map. Its spatial determinant is one, so the original energy and dissipation formulas are unchanged:

$$\int_{\mathbb{R}^3} |u_d|^2 \, dx = C_E \sigma^{5\beta-2}, \qquad \int_{\mathbb{R}^3} |\nabla_x u_d|^2 \, dx = C_D \sigma^{3\beta-2}.$$

The constants C_E, C_D are exactly those in (306). Each translated field is Schwartz at every preterminal time. Indeed translation and differentiation preserve that class, and $d'(t)$ is a finite vector at each such time. This statement does not infer uniform endpoint bounds.

18.1 Exact vorticity and force-curl translation

Write $\omega = \operatorname{curl}_y u$ and $C = \operatorname{curl}_y f$ for the full Gaussian expressions (296) and (301). Spatial derivatives commute with translation, and $d'(t)$ is independent of space. The full identities are consequently

$$\operatorname{curl}_x u_d(x,t) = \omega(x-d,t), \qquad \operatorname{curl}_x f_d(x,t) = (C - d' \cdot \nabla_y \omega)(x-d,t). \tag{325}$$

For example, the ith component of the second equality is $\sum_{j,\ell} \epsilon_{ij\ell} \partial_j f_\ell - \sum_{m,j,\ell} d'_m \epsilon_{ij\ell} \partial_j \partial_m u_\ell$ at $y = x - d$. Commuting the two smooth spatial derivatives gives $C_i - \sum_m d'_m \partial_m \omega_i$, which proves each sign and every component of (325).

Theorem 18.1 (Translation cannot regularize the force of this family)**.** *For every nonzero amplitude pair (U_*, H_*) and every real β, the translated velocity u_d cannot satisfy the same Navier–Stokes equation with a force whose first spatial derivatives are uniformly bounded on $\mathbb{R}^3 \times [0,T)$, for any choice of pressure. This conclusion holds for every smooth centre curve d specified above.*

Proof. First suppose $H_* \neq 0$. The third vorticity component is

$$\omega_3(y,t) = 2B e^{-k|y|^2} (1 - k(y_1^2 + y_2^2)).$$

It is even in each Cartesian coordinate, so $\nabla_y \omega_3(0,t) = 0$. The full poloidal vorticity has zero third component, including when $U_* \neq 0$. Evaluating (325) at the actual moving centre gives

$$(\operatorname{curl}_x f_d)_3(d(t),t) = \frac{2H_*}{\tau_*} \sigma^{-2} + \frac{20\nu H_*}{L_*^2} \sigma^{-1-2\beta}. \tag{326}$$

Both terms have the sign of H_*. Its absolute value is at least $2|H_*|\sigma^{-2}/\tau_*$ and tends to infinity, irrespective of the centre velocity.

It remains to consider $H_* = 0$ and $U_* \neq 0$. The complete vorticity then has the form

$$\omega(y,t) = (-y_2 \xi, y_1 \xi, 0), \qquad \xi = AkG(10 - 4k|y|^2).$$

The scalar ξ is even under $y \mapsto -y$, and hence $\omega(-y,t) = -\omega(y,t)$. Differentiating this identity with respect to y_j gives $\partial_j \omega(-y,t) = \partial_j \omega(y,t)$. Thus the full term $d' \cdot \nabla_y \omega$ in (325) has identical values at opposite points, for each actual vector $d'(t)$.

For every $c > 0$ let $y_c(t) = (L\sqrt{c}, 0, 0)$. At both y_c and $-y_c$ the nonlinear part

$$zA^2k^2G^2(16k|y|^2 - 56)$$

of E_ξ vanishes exactly because $z = 0$. The remaining scalar value is $Ake^{-c}[P_\beta(c)/\tau + \nu k Q(c)]$. Since $R_h = 0$ when $H_* = 0$, the force-curl components at these two points are $(0, \pm L\sqrt{c}\,E_\xi, 0)$. Subtracting the two translated values in (325) therefore gives the exact identity

$$\begin{aligned}\frac{1}{2}\Big[(\operatorname{curl}_x f_d)_2(d + y_c, t) - (\operatorname{curl}_x f_d)_2(d - y_c, t)\Big]& \\ = \frac{U_*\sqrt{c}\,e^{-c}}{L_*\tau_*}\sigma^{-2}\left[P_\beta(c) + \frac{\nu\tau_*}{L_*^2}\sigma^{1-2\beta}Q(c)\right]&. \qquad (327)\end{aligned}$$

The source polynomials P_β, Q are unchanged from (297); neither the nonlinear term nor the translation term was discarded before its exact evaluation or cancellation.

For $\beta \neq 1/6$ choose $c = c_* = (7 - \sqrt{14})/2 > 0$. The already verified identities $Q(c_*) = 0$ and $P_\beta(c_*) = (1 - 6\beta)(10 - 4c_*)$ give the absolute value of the right side of (327) as

$$\frac{|U_*|\sqrt{c_*}\,e^{-c_*}}{L_*\tau_*}|1 - 6\beta|(10 - 4c_*)\,\sigma^{-2}.$$

Its coefficient is strictly positive, since $10 - 4c_* = -4 + 2\sqrt{14} > 0$. For $\beta = 1/6$, choose $c = 1$ instead. The right side is

$$\frac{U_*}{eL_*\tau_*}\sigma^{-2}\left[\frac{11}{3} + \frac{44\nu\tau_*}{L_*^2}\sigma^{2/3}\right].$$

Its absolute value is at least $11|U_*|\sigma^{-2}/(3eL_*\tau_*)$. For any real numbers a, b, $\max(|a|, |b|) \geq |a-b|/2$ by the triangle inequality. Applying this to the two force-curl values proves that their spatial supremum diverges in every case.

Finally, let any other pressure $\widetilde{p}_d$ and force $\widetilde{f}_d$ solve the full equation with exactly the same velocity u_d. Subtracting their equation from the proved one gives

$$\widetilde{f}_d - f_d = \nabla_x(\widetilde{p}_d - p_d), \qquad \operatorname{curl}_x \widetilde{f}_d = \operatorname{curl}_x f_d.$$

The second equality is also valid distributionally. Every curl component is a difference of first spatial derivatives of force components; uniform bounds on all those derivatives would bound the curl. Equations (326)–(327) contradict that bound. This proves the assertion for every pressure. □

The target whole-space forcing condition (3), already at $K = 0$ and first spatial derivative order, requires the uniform bound just excluded. The test locations may escape to infinity when d or L does. The proof then concerns this global condition; it does not claim a singularity at a fixed finite spatial point. If the relevant test locations stay in a fixed compact set, their divergent curl values also preclude a smooth local extension there. When $U_* = H_* = 0$, every field in (324) is zero for every such d and β, so this degenerate case has no obstruction.

There is a different moving-frame map which adds a constant background velocity; its constant must also be retained. For $\widehat{u}(x,t) = u(x - d(t), t) + d'(t)$ and $\widehat{p}(x,t) = p(x - d(t), t)$, the time derivative's $-d' \cdot \nabla u$ term cancels the identical contribution with positive sign in convection. The full resulting force is $f(x - d, t) + d''(t)$. Equivalently the affine pressure $\widehat{p} - d''(t) \cdot x$ removes this acceleration from the force. Whenever $d'(t) \neq 0$, however, $\widehat{u}(t)$ has infinite energy: if it and the translated Gaussian were both in L^2, their difference, the nonzero constant vector $d'(t)$, would be in L^2, whereas its squared integral over a radius-R ball is $|d'(t)|^2(4\pi R^3/3)$. Thus this complete alternative map does not evade the energy requirement. These results concern the specified Gaussian fields and their exact moving-centre maps; they do not exclude different profiles or dynamics.

19 The positive-radius circle argument

This section audits the incoming report's Section 13 (PDF pages 26–27, printed pages 21–22). The geometric implication is valid. The proof below specifies its solution class and proves the passage from a classical solution before a terminal time to an interior singular set. It uses the Caffarelli–Kohn–Nirenberg partial-regularity and suitable-existence theorems as external theorems; it does not claim to reprove those theorems or to certify the incoming report as a whole.

19.1 The equation, suitable solutions, and the external theorem

Let $\nu > 0$ be the original viscosity. On an open set $D \subset \mathbb{R}^3 \times \mathbb{R}$, retain the equation

$$\partial_t u_i + \sum_{j=1}^{3} u_j \partial_{x_j} u_i - \nu \sum_{j=1}^{3} \partial_{x_j}^2 u_i + \partial_{x_i} p = f_i, \qquad \sum_{j=1}^{3} \partial_{x_j} u_j = 0. \tag{328}$$

Here $u : D \to \mathbb{R}^3$, $p : D \to \mathbb{R}$, and $f : D \to \mathbb{R}^3$. Write $e = |u|^2/2$ and, for $\phi \in C_c^\infty(D)$, define

$$\mathcal{L}_\nu(u, p, f; \phi) = \int_D \Big[e(\partial_t \phi + \nu \Delta \phi) + (e + p) u \cdot \nabla \phi + (f \cdot u)\phi - \nu |\nabla u|^2 \phi \Big] \, dx \, dt. \tag{329}$$

The local energy inequality is $\mathcal{L}_\nu(u, p, f; \phi) \geq 0$ for every nonnegative such ϕ. In this section the sufficient suitable class has, on each compactly contained cylinder $B \times I \Subset D$,

$$u \in L^\infty(I; L^2(B)) \cap L^2(I; H^1(B)), \qquad p \in L^{3/2}(B \times I), \qquad f \in L^q(B \times I), \quad q > 5/2, \tag{330}$$

and satisfies (328) distributionally and (329). This is a sufficient class for the cited CKN result; no assertion that these are minimal hypotheses is needed. The products in the local energy inequality are integrable: the local energy class embeds into $L^{10/3}$ in spacetime, hence into L^3 on a bounded cylinder; $pu \in L^1$ follows from $p \in L^{3/2}$ and $u \in L^3$, and $fu \in L^1$ follows from $q > 5/2$ and the same finite-measure embeddings. For clarity, the energy embedding follows from the spatial inequality $\|u\|_{10/3} \leq C_B \|u\|_2^{2/5} \|u\|_{H^1}^{3/5}$: raising to $10/3$ and integrating gives $\int_I \|u\|_{10/3}^{10/3} \leq C_B^{10/3} (\operatorname{ess\,sup}_I \|u\|_2^{4/3}) \int_I \|u\|_{H^1}^2$.

A regular point is a point with an open spacetime neighborhood on which u is essentially bounded. Denote the complementary set by $S(u)$. Define

$$d_{\mathrm{par}}((x,t),(y,s)) = \max\{|x - y|, |t - s|^{1/2}\}. \tag{331}$$

For a metric d, our Hausdorff measure convention is

$$\mathcal{H}_d^1(E) = \lim_{\delta \downarrow 0} \inf \left\{ \sum_j \operatorname*{diam}_d U_j : E \subset \bigcup_j U_j, \quad \operatorname*{diam}_d U_j \leq \delta \right\}. \tag{332}$$

Theorem 19.1 (External CKN inputs)**.** *For a suitable solution with divergence-free force locally in L^q, $q > 5/2$, the singular set has zero one-dimensional parabolic Hausdorff measure. In the whole space, divergence-free L^2 initial data and a divergence-free force in $L^2(\mathbb{R}^3 \times (0, S)) \cap L^q_{\mathrm{loc}}$, $q > 5/2$, admit a suitable energy weak solution on $(0, S)$ with the initial trace, global energy inequality, and pressure in $L^{5/3}$ locally. These are the inputs of CKN Theorem B, Theorem A', and Appendix A; the original paper uses unit viscosity. At regular points, its cited interior regularity result gives bounded spatial derivatives on smaller neighborhoods when the force has bounded smooth spatial derivatives.*

The exact source and page checks are recorded in the accompanying source inventory; the original publication is [1]. The circle consequence is also stated in the introduction of Jiu–Xin [2], for their unforced equation. The forced application here is obtained from the CKN hypotheses and the explicit maps below, not by changing the force in Jiu–Xin's statement. The readable research paper [3], pp.2–3, independently corroborates the force threshold, sufficient pressure class, and spatial regularity assertion. The original CKN page images were unavailable; that source-access limitation is retained in the inventory. CKN's parabolic covering measure and (332) have the same null sets. For the direction used here it suffices that a parabolic cylinder of radius r has d_{par}-diameter at most $2r$ (a temporal interval of length $2r^2$ also has this bound). Every CKN cylinder cover therefore yields a Hausdorff cover with diameter sum at most twice the radius sum. Taking infima and then the small-radius limit proves that CKN zero measure implies $\mathcal{H}^1_{d_{\text{par}}} = 0$.

Exact transfer of the external theorem for the original ν. No viscosity is discarded. Introduce a second spacetime variable $s = \nu t$ and the invertible map

$$\begin{aligned} D_\nu = \{(x, \nu t) : (x,t) \in D\}, \quad & v(x,s) = \nu^{-1} u(x, s/\nu), \\ \pi(x,s) = \nu^{-2} p(x, s/\nu), \quad & g(x,s) = \nu^{-2} f(x, s/\nu). \end{aligned} \tag{333}$$

The inverse is $u(x,t) = \nu v(x, \nu t)$, $p(x,t) = \nu^2 \pi(x, \nu t)$, $f(x,t) = \nu^2 g(x, \nu t)$. Each of $\partial_s v$, $(v \cdot \nabla) v$, $-\Delta v$, $\nabla \pi$, and g is respectively ν^{-2} times $\partial_t u$, $(u \cdot \nabla) u$, $-\nu \Delta u$, ∇p, and f, evaluated at $(x, s/\nu)$. Also $\nabla \cdot v = \nu^{-1} \nabla \cdot u$. Thus the equations, including their distributional versions, are mapped exactly to the unit-viscosity equations on D_ν. For $\psi(x,s) = \phi(x, s/\nu)$ and $ds = \nu\, dt$, direct substitution in every term gives

$$\mathcal{L}_1(v, \pi, g; \psi) = \nu^{-2} \mathcal{L}_\nu(u, p, f; \phi). \tag{334}$$

For example, the time term contributes ν^{-2}, the diffusion-test term contributes ν^{-1} before comparison with its original factor ν, and the cubic, pressure, and force terms contribute $\nu^{-3}\nu = \nu^{-2}$. The dissipation term contributes $\nu^{-2}\nu = \nu^{-1}$ before comparison with its original factor ν. All coefficients therefore agree with (334). The spacetime L^q force integral is multiplied by ν^{1-2q}, the pressure $L^{3/2}$ integral by ν^{-2}, the essential supremum of the spatial velocity L^2 norm squared by ν^{-2}, and the spacetime gradient L^2 integral by ν^{-1}. These positive finite factors preserve the stated classes. The same substitution transfers the global energy inequality and initial trace. Finally, for $F_\nu(x,t) = (x, \nu t)$,

$$\min\{1, \sqrt{\nu}\}\, d_{\text{par}}(a,b) \le d_{\text{par}}(F_\nu a, F_\nu b) \le \max\{1, \sqrt{\nu}\}\, d_{\text{par}}(a,b). \tag{335}$$

Applying these inequalities to every covering set proves preservation of Hausdorff null sets. Boundedness neighborhoods correspond under F_ν and multiplication of velocity by ν^{-1}. Consequently the external theorem gives its claimed zero-measure and existence conclusions for the original equation with its original positive ν.

19.2 Rotation as an exact action on the equation and energy inequality

For $\theta \in \mathbb{R}$, let

$$\begin{gathered} R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ \mathcal{R}_R u(x,t) = R u(R^T x, t), \quad \mathcal{R}_R p(x,t) = p(R^T x, t), \quad \mathcal{R}_R f(x,t) = R f(R^T x, t). \end{gathered} \tag{336}$$

These maps have domain the fields on D and codomain the corresponding fields on $RD = \{(Rx, t) : (x,t) \in D\}$, and inverse $\mathcal{R}_{R^T}$. For two rotations R, Q,

$$\mathcal{R}_R \mathcal{R}_Q u(x,t) = RQ\, u(Q^T R^T x, t) = \mathcal{R}_{RQ} u(x,t).$$

The scalar formula is identical without the prefactor. This proves the group law, including the domains and inverses.

Write $y = R^T x$. With every index ranging from 1 to 3, the derivative identities are

$$\begin{aligned}
\partial_t(\mathcal{R}_R u)_i &= \sum_a R_{ia}\partial_t u_a(y,t),\\
\partial_{x_j}(\mathcal{R}_R u)_i &= \sum_{a,b} R_{ia}R_{jb}\partial_{y_b}u_a(y,t),\\
\sum_i \partial_{x_i}(\mathcal{R}_R u)_i &= \sum_a \partial_{y_a}u_a(y,t),\\
\Delta_x(\mathcal{R}_R u)_i &= \sum_a R_{ia}\Delta_y u_a(y,t),\\
\sum_j (\mathcal{R}_R u)_j\partial_{x_j}(\mathcal{R}_R u)_i &= \sum_{a,b} R_{ia}u_b\partial_{y_b}u_a(y,t),\\
\partial_{x_i}(\mathcal{R}_R p) &= \sum_b R_{ib}\partial_{y_b}p(y,t).
\end{aligned} \tag{337}$$

For divergence and Laplacian, use $\sum_i R_{ia}R_{ib} = \delta_{ab}$; for the nonlinear term, insert $(\mathcal{R}_R u)_j = \sum_c R_{jc}u_c$ and contract $\sum_j R_{jc}R_{jb} = \delta_{cb}$. Thus the full momentum residual transforms to R times the original residual at $(R^T x, t)$, with the same ν, and the divergence constraint is preserved. These calculations remain valid in distributions: in the defining integrals for distributional derivatives make the linear substitution $x = Ry$, whose Jacobian is $\det R = 1$, and use the displayed chain rule on each compactly supported test function. Tensor products transform by $u^R \otimes u^R = R(u\otimes u)R^T$, so the distributional divergence form of convection has the same identity.

Moreover

$$|u^R(x,t)| = |u(y,t)|, \qquad \nabla u^R(x,t) = R\nabla u(y,t)R^T, \qquad |\nabla u^R(x,t)|^2 = |\nabla u(y,t)|^2. \tag{338}$$

The last equality follows by expanding the Frobenius norm and contracting both orthogonal factors. For $\phi \in C_c^\infty(RD)$ put $\psi(y,t) = \phi(Ry,t)$. Then $\nabla_y\psi = R^T\nabla_x\phi$, $\Delta_y\psi = \Delta_x\phi$, and $\partial_t\psi = \partial_t\phi$ at corresponding points. The force contraction is $(Rf)\cdot(Ru) = f\cdot u$ and the transport contraction is $(Ru)\cdot\nabla_x\phi = u\cdot\nabla_y\psi$. Consequently every integral in (329) satisfies

$$\mathcal{L}_\nu(u^R, p^R, f^R; \phi) = \mathcal{L}_\nu(u,p,f;\psi). \tag{339}$$

Nonnegative tests correspond bijectively, proving exact preservation of suitability, the force exponent, and all local energy norms. For completeness, the vorticity action has the same orientation:

$$\nabla_x \times u^R(x,t) = R(\nabla_y \times u)(y,t). \tag{340}$$

Indeed the alternating trilinear form obeys $\det(Ra,Rb,Rc) = \det(R)\det(a,b,c) = \det(a,b,c)$; expanding this identity in coordinate vectors gives the alternating-tensor contraction needed when (337) is inserted in $(\nabla\times u^R)_i = \sum_{j,k}\epsilon_{ijk}\partial_j u^R_k$. The sign in (340) is therefore positive.

Axisymmetry means precisely

$$u(R_\theta x, t) = R_\theta u(x,t) \quad \text{for every } \theta\text{, as an equality of measurable fields.} \tag{341}$$

It is vector equivariance, not invariance of each Cartesian component. If the full triple is axisymmetric, it is a fixed point of the action. For the singular-set argument only (341) is needed. For each fixed θ, the equality holds outside a null set; no common null set for uncountably many angles is required, because essential boundedness is compared separately for each fixed rotation.

19.3 The fixed-time circle has positive parabolic measure

Fix $r_0 > 0$, $z_0 \in \mathbb{R}$, and a time T. Define the actual orbit

$$\mathcal{C}_{r_0,z_0,T} = \{(r_0\cos\theta, r_0\sin\theta, z_0, T) : 0 \le \theta < 2\pi\}. \tag{342}$$

The embedding $\iota_T : x \mapsto (x,T)$ is an isometry from Euclidean space onto the time slice for (331). This also gives equality of the corresponding Hausdorff measures for subsets of that slice: intersecting an ambient cover with the slice does not increase diameter, while lifting a spatial cover preserves diameter. These operations prove the two inequalities between the covering infima in (332) for every δ, hence their equality after the limit.

Here is a direct positive lower bound that needs no curve-measure theorem. The projection $(x,t) \mapsto x_1$ has Lipschitz constant one for d_{par}, and maps (342) onto $[-r_0, r_0]$. If (U_j) is any countable cover of that circle, the projections of $U_j \cap \mathcal{C}_{r_0,z_0,T}$ can be enclosed in intervals of length at most $\operatorname{diam}_{d_{\mathrm{par}}} U_j + \varepsilon 2^{-j}$. Those intervals cover $[-r_0, r_0]$. Countable subadditivity of interval length therefore yields $2r_0 \le \sum_j \operatorname{diam}_{d_{\mathrm{par}}} U_j + \varepsilon$. Letting $\varepsilon \downarrow 0$, taking the covering infimum and then $\delta \downarrow 0$ proves

$$\mathcal{H}^1_{d_{\mathrm{par}}}(\mathcal{C}_{r_0,z_0,T}) \ge 2r_0 > 0. \tag{343}$$

The measure is finite as well: partition $[0, 2\pi]$ into intervals of length at most δ/r_0. The corresponding arcs cover the circle, each arc has diameter no greater than r_0 times the interval length by integration of the derivative of the circle parametrization, and the total diameter sum is at most $2\pi r_0$. Thus the precise bounds $2r_0 \le \mathcal{H}^1_{d_{\mathrm{par}}}(\mathcal{C}) \le 2\pi r_0$ are sufficient for the contradiction; no circumference normalization of Hausdorff measure is silently imposed.

Proposition 19.2 (Interior suitable circle exclusion)**.** *Let D be rotationally invariant and let (u, p, f) belong to the suitable class to which Theorem 19.1 applies. If u is axisymmetric, every point of $S(u)$ has $x_1 = x_2 = 0$.*

Proof. An orthogonal rotation maps any neighborhood of (x,t) bijectively onto a neighborhood of (Rx, t) and preserves $|u|$ and spacetime null sets. Therefore (341) says that either both points are regular or both are singular. If a singular point has radius $r_0 > 0$, every point of its orbit (342) is singular. Monotonicity of the covering definition and (343) imply $\mathcal{H}^1_{d_{\mathrm{par}}}(S(u)) \ge 2r_0 > 0$, contrary to Theorem 19.1. If the domain is not the whole space, the complete orbit must be contained in the open domain, as ensured here by rotational invariance; this is an interior argument. $\square$

19.4 The prescribed force need not be divergence-free

For the whole-space application, take the original smooth force f on $\mathbb{R}^3 \times [0, S]$, with its original spatial decay. Define

$$\chi = \Delta^{-1}\nabla \cdot f = -\frac{1}{4\pi}\int_{\mathbb{R}^3} \frac{\nabla_y \cdot f(y,t)}{|x-y|}\,dy, \qquad g = f - \nabla\chi, \qquad \pi = p - \chi. \tag{344}$$

The sign follows from the fundamental solution $\Delta(-1/(4\pi|x|)) = \delta_0$. To check that identity, the function is harmonic away from zero and its outward radial flux through a sphere of radius ϵ is $4\pi\epsilon^2(1/(4\pi\epsilon^2)) = 1$; integration by parts against a test function and passage to $\epsilon \downarrow 0$ gives δ_0. Consequently $\Delta\chi = \nabla \cdot f$ and $\nabla \cdot g = 0$. Substituting (344) gives

$$\partial_t u + (u \cdot \nabla)u - \nu\Delta u + \nabla\pi = g. \tag{345}$$

This is an exact equivalence: the inverse restores $p = \pi + \chi$ and $f = g + \nabla\chi$. It changes neither u nor the original force being prescribed in (328).

The local energy functional is unchanged. The difference of the projected and original integrands in the pressure and force terms is $-\chi u \cdot \nabla\phi - (\nabla\chi \cdot u)\phi = -u \cdot \nabla(\chi\phi)$, whose integral vanishes by $\nabla \cdot u = 0$. Hence

$$\mathcal{L}_\nu(u, \pi, g; \phi) = \mathcal{L}_\nu(u, p, f; \phi). \tag{346}$$

All distributional products here are legitimate because χ is smooth on each compact spacetime set. To see this smoothness without estimating a singular differentiated kernel, differentiate the smooth rapidly decaying function $\nabla \cdot f$ under convolution with $|x-y|^{-1}$, which is locally integrable; uniform decay bounds dominate the tail.

There is also a direct global bound needed for suitable existence. For spatial Fourier frequency $\xi \neq 0$, the multiplier of $f \mapsto g$ is $P(\xi) = I - \xi\xi^T/|\xi|^2$. Its square equals itself, its transpose equals itself, and $|P(\xi)a|^2 = |a|^2 - |\xi \cdot a|^2/|\xi|^2 \leq |a|^2$. Define its value at zero arbitrarily, since that set has zero measure. Plancherel's identity proves, for each integer $m \geq 0$,

$$\|g(t)\|_{H^m(\mathbb{R}^3)} \leq \|f(t)\|_{H^m(\mathbb{R}^3)}, \qquad \|g\|_{L^2(\mathbb{R}^3 \times (0,S))} \leq \|f\|_{L^2(\mathbb{R}^3 \times (0,S))}. \tag{347}$$

Smooth force decay uniformly on this finite time interval makes every right-hand side finite, and the local smoothness just proved gives $g \in L^q_{\mathrm{loc}}$ for every finite q. Thus the original force, even when not divergence-free, leads exactly to the force class of Theorem 19.1. Adding χ to the resulting pressure preserves local $L^{3/2}$ and restores the original equation and local energy inequality. In the global energy inequality the work is also unchanged: $g = Pf$, $Pv = v$ for divergence-free L^2 velocity v, and self-adjointness of the matrix multiplier gives $\langle g, v\rangle = \langle Pf, v\rangle = \langle f, Pv\rangle = \langle f, v\rangle$.

19.5 Existence beyond the terminal time and agreement before it

Fix $0 < T < S < \infty$, divergence-free smooth decaying data u^0, and an original force f smooth on $\mathbb{R}^3 \times [0, S]$ with spatial decay and derivative bounds there, for example the force class in the official whole-space problem statement. Suppose u is its classical strong solution on $[0, T)$, in $C([0, t_0]; H^m)$ for every $t_0 < T$ and some integer $m \geq 3$. This is the usual smooth strong-solution branch from these data. These hypotheses specify the branch; mere pointwise smoothness with $\|u(t)\|_2 < \infty$ at each time is not being silently substituted for them.

For every $t < T$, the classical global energy equality is

$$\frac{1}{2}\|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 \, ds = \frac{1}{2}\|u^0\|_2^2 + \int_0^t \langle f(s), u(s)\rangle \, ds. \tag{348}$$

To derive it, pair (328) with u. The convection term is $\int \nabla \cdot (|u|^2 u/2) = 0$, the pressure term is $\int \nabla \cdot (pu) = 0$ after the standard whole-space cutoff limit, and integration by parts makes $-\nu \int u \cdot \Delta u = \nu \int |\nabla u|^2$. The strong Sobolev regularity on every shorter interval justifies these limits and time integration; equivalently one may pair the projected equation, whose pressure term vanishes by the orthogonal projection. Set $F(t) = \int_0^t \|f(s)\|_2 \, ds$ and $M(t) = \|u^0\|_2 + F(t)$. Regularizing the square root of $\|u\|_2^2$ by $(\|u\|_2^2 + \epsilon^2)^{1/2}$ in its differential inequality gives derivative at most $\|f\|_2$. Integration and $\epsilon \downarrow 0$ yield

$$\|u(t)\|_2 \leq M(t) \leq M(T), \qquad \nu \int_0^T \|\nabla u(s)\|_2^2 \, ds \leq \frac{1}{2} M(T)^2. \tag{349}$$

For the second inequality insert $\|u(s)\|_2 \leq M(s)$ into (348), use $\int_0^t F'(s)M(s)\, ds = (M(t)^2 - M(0)^2)/2$, discard the nonnegative terminal energy, and pass monotonically to T. This proves the uniform energy and dissipation bounds from the specified strong branch and force control; they are not inferred from finiteness of individual time-slice energies alone.

Theorem 19.1, the force map (344), and the viscosity map (333) give a suitable energy weak solution (v, P) for the original data and original force on $(0, S)$. The suspected time T is now an interior time. We next prove the needed agreement; uniqueness after T is neither used nor asserted.

Fix $t_0 < T$. Put $w = v - u$ and let $\kappa(t) = \|\nabla u(t)\|_{L^\infty}$, where the matrix norm is the Frobenius norm. The assumed strong branch gives $\int_0^{t_0} \kappa(t)\, dt < \infty$ by the H^m embedding for $m \geq 3$. Testing the weak equation of v with the smooth divergence-free u and using the strong equation of u gives the cross identity

$$\langle v(t), u(t)\rangle - \|u^0\|_2^2 = \int_0^t \int_{\mathbb{R}^3} \Big[\sum_{i,j} v_i(v_j - u_j)\partial_j u_i - 2\nu \nabla v : \nabla u + f \cdot (u + v)\Big] \, dx\, ds. \tag{350}$$

Here the term $\sum v_i v_j \partial_j u_i$ is the weak convection term; the strong convection in $\partial_t u$ gives $-\sum v_i u_j \partial_j u_i$; and the two diffusion pairings each give $-\nu \nabla v : \nabla u$. This identifies every sign and factor. The test is justified by time smoothing and divergence-free spatial approximation of u: the convection pairing is bounded by $\kappa \|v\|_2^2$, the diffusion pairing by $\|\nabla v\|_2 \|\nabla u\|_2$, and the force pairings by their L^2 norms, so the approximations converge. The weak initial trace and strong initial trace give the displayed initial cross term.

Add the energy inequality for v to (348), then subtract (350). Expanding the norm squares yields, for almost every $t \le t_0$,

$$\frac{1}{2}\|w(t)\|_2^2 + \nu \int_0^t \|\nabla w(s)\|_2^2 \, ds \le -\int_0^t \int_{\mathbb{R}^3} \sum_{i,j} w_i w_j \partial_j u_i \, dx \, ds \le \int_0^t \kappa(s) \|w(s)\|_2^2 \, ds. \tag{351}$$

In the first step the extra term with $v_i = u_i + w_i$ vanishes because $\sum_{i,j} u_i w_j \partial_j u_i = w \cdot \nabla(|u|^2/2)$ and $\nabla \cdot w = 0$. The original forcing cancels exactly. For an explicit zero-data Gronwall argument, put $Y(t) = \|w(t)\|_2^2$ and $Z(t) = 2\int_0^t \kappa(s) Y(s) \, ds$. Then $0 \le Y \le Z$, $Z(0) = 0$, and $Z' \le 2\kappa Z$ almost everywhere. The derivative of $Z(t) \exp(-2\int_0^t \kappa)$ is nonpositive, so this nonnegative function is identically zero. Thus $v = u$ almost everywhere on $(0, t_0)$. Taking a countable increasing sequence $t_0 \uparrow T$ proves

$$v = u \quad \text{almost everywhere on } \mathbb{R}^3 \times (0, T). \tag{352}$$

No terminal strong trace at T and no uniqueness beyond T entered this argument.

Theorem 19.3 (First-time off-axis exclusion for the original forced equation)**.** *For the smooth decaying data, force defined through $S > T$, and classical strong branch just specified, suppose that u is axisymmetric on $[0, T)$. Then every x_0 with $r_0 = (x_{01}^2 + x_{02}^2)^{1/2} > 0$ has $\rho > 0$ such that*

$$\operatorname*{ess\,sup}_{B_\rho(x_0) \times (T - \rho^2, T)} |u| < \infty, \qquad T - \rho^2 > 0. \tag{353}$$

In particular a first singularity in this backward-local velocity sense cannot lie at positive radius. The continuation used in the proof need not be axisymmetric at or after T.

Proof. Use (v, P) on $(0, S)$ and (352). If no ρ satisfies (353), then u is essentially unbounded in every sufficiently small backward cylinder at (x_0, T). Preterminal axisymmetry and rotation invariance of cylinders imply the same statement at every $(R_\theta x_0, T)$. If any of those points were regular for v, an open neighborhood where v is bounded would contain a sufficiently small backward cylinder, contradicting (352). Thus every point of the positive-radius circle is in $S(v)$. By (343) this has positive parabolic one-dimensional measure, whereas Theorem 19.1 gives zero. This contradiction proves (353). □

For smooth force, the spatial interior regularity included in the external input turns a bounded neighborhood for v into bounds for its spatial derivatives on smaller neighborhoods. Thus the same argument also excludes a positive-radius orbit of preterminal spatial-derivative singularities: an unbounded derivative along a sequence converging to an orbit point would contradict those smaller-neighborhood bounds and (352). This specifies the extra regularity fact used when the incoming report describes vorticity blow-up, rather than velocity blow-up. It is not legitimate to apply a velocity singular-set theorem directly to an arbitrary Euler vorticity singularity.

The conclusion is local in space. For a compact set K disjoint from the axis, the neighborhoods in (353) have a finite subcover of K. The minimum of their positive time lengths and the maximum of their finite bounds give one velocity bound near T on K. An assertion about a supremum on an unbounded region additionally requires control at spatial infinity. A compact toroidal concentration region bounded away from the axis has no such escape-to-infinity issue.

19.6 Why preterminal smoothness alone is an insufficient replacement

The following exact examples describe the missing force hypothesis; they are not candidates in the prescribed smooth-through-T force class. Let $s = x_1^2 + x_2^2 + x_3^2$, let $W(x) = (-x_2, x_1, 0)$, and choose $\eta \in C_c^\infty((0,\infty))$ equal to one on an interval around r_0^2, where $r_0 > 0$. Put $V(x) = \eta(s)W(x)$. Direct differentiation gives

$$\begin{aligned}\nabla \cdot V &= -2x_1x_2\eta'(s) + 2x_1x_2\eta'(s) = 0,\\ (V\cdot\nabla)V &= -\eta(s)^2(x_1, x_2, 0),\\ \Delta V &= (4s\eta''(s) + 10\eta'(s))W(x).\end{aligned} \tag{354}$$

For the second identity, $W \cdot \nabla s = 0$ and $(W\cdot\nabla)W = -(x_1, x_2, 0)$. For the third, $\Delta\eta(s) = 4s\eta''(s) + 6\eta'(s)$ and $2\nabla\eta(s)\cdot\nabla W_i = 4\eta'(s)W_i$. The field is smooth, compactly supported, and axisymmetric in exactly the sense (341). For any smooth scalar $a : [0,T) \to \mathbb{R}$, define

$$u(x,t) = a(t)V(x), \quad p(x,t) = 0, \quad f(x,t) = a'(t)V(x) + a(t)^2(V\cdot\nabla)V(x) - \nu a(t)\Delta V(x). \tag{355}$$

Substitution of $u_t = a'V$, $(u\cdot\nabla)u = a^2(V\cdot\nabla)V$, and $\Delta u = a\Delta V$ proves the original forced NS equation exactly. Every time slice has finite energy $a(t)^2\|V\|_2^2$.

Take first $a(t) = (T-t)^{-1}$. On the plateau of η,

$$f(x,t) = (T-t)^{-2}[(-x_2, x_1, 0) - (x_1, x_2, 0)], \qquad |f(x,t)| = \sqrt{2}\,(x_1^2 + x_2^2)^{1/2}(T-t)^{-2}. \tag{356}$$

In a small spatial neighborhood of $(r_0, 0, 0)$ the radius is bounded below by $r_0/2$, so $\int |f|^q$ up to T diverges for every $q \geq 1/2$. The velocity blows up at the entire circle, with finite energy at each $t < T$ but without a uniform energy bound. This exact example proves that pointwise preterminal smoothness and finite time-slice energies do not imply the force or uniform-energy hypotheses.

Take instead $a(t) = \sin((T-t)^{-1})$. Then $\sup_{t<T}\|u(t)\|_2 \leq \|V\|_2$ and $\int_0^T \|\nabla u\|_2^2 \leq T\|\nabla V\|_2^2$. Nevertheless u has no limit at a point of the plateau where $V \neq 0$, and the tangential part of its force there has magnitude $r(T-t)^{-2}|\cos((T-t)^{-1})|$. Its time integral diverges, since the substitution $z = (T-t)^{-1}$ gives $\int^\infty |\cos z|\,dz = \infty$. Thus even uniform energy and integrated dissipation do not manufacture an admissible continuation of a force that is only smooth before T. The terminal theorem above avoids both defects by using the actual prescribed force through $S > T$ and a suitable solution for that force.

Precise integration conclusion. The checked addition is the circle exclusion, its explicit force and viscosity maps, and the preterminal weak–strong argument. The incoming sentence about obtaining an axisymmetric weak continuation can be replaced by the stronger proved statement that any suitable energy continuation suffices. The result excludes retention of a fixed positive-radius axisymmetric singular orbit for the original classical viscous equation in the stated class. It does not assert a disproof of another geometry, nonaxisymmetric mechanisms, concentration approaching the axis, or an altered exact bridge between the Euler and viscous equations. No existence theorem for any such alternative is inferred.

20 The dissipative vortex-layer programme: exact scales and residuals

The four-page statement points to the programme of Diego Córdoba and Luis Martínez-Zoroa. A primary dissipative predecessor is their joint paper with Fan Zheng, *Finite time blow-up for the hypodissipative Navier–Stokes equations with a force in $L_t^1 C_x^{1,\epsilon} \cap L_t^\infty L_x^2$*, arXiv:2407.06776v2, and its published version, Archive for Rational Mechanics and Analysis 250 (2026), article 38. Their operator is

$$\widehat{\Lambda^\alpha v}(\xi) = |\xi|^\alpha \hat{v}(\xi), \qquad \Lambda^\alpha = (-\Delta)^{\alpha/2}.$$

In this convention the classical viscosity is $\nu\Lambda^2 = -\nu\Delta$. The theorem stated in those sources concerns $0 \le \alpha < (22 - 8\sqrt{7})/9$ and a force in the indicated time-integrated Hölder class. It does not state smoothness of the force through the terminal time. The following calculations reproduce the actual scale mechanism and repair specified transcription and residual identities. They do not reproduce the source's analytic flow-map and singular-integral estimates in full. Whenever an estimate from that paper is used below, it is identified as a source estimate; the algebra applied to its explicit right-hand side is proved here. In particular this section asserts no new blow-up theorem.

20.1 All scale variables and the source's initial four constraints

Retain $N > 1$, $R > 1$, and integer $n \ge 1$. Define

$$M_n = N^{R^n}, \quad A = N^a, \quad A_n = A^{R^n} = M_n^a, \quad L = N^\ell, \quad L_n = L^{R^n} = M_n^\ell.$$

The index n is never substituted for R^n in these definitions. The parameters a, ℓ are exponents, while A_n, L_n, M_n are positive magnitudes. Let K_n denote the stretching factor of layer n. The source's introductory scale calculation associates the four quantities

$$\frac{M_n^\alpha}{A_{n-1}}, \qquad \frac{A_n M_n^s}{K_n}, \qquad \frac{K_n M_{n-1}}{L_n}, \qquad \frac{A_n^2 L_n M_n^{s-1}}{A_{n-1}} \tag{FS1}$$

with dissipation versus stretching, force needed to start a layer, spatial support within the outer wavelength, and integrated quadratic interaction, respectively. Their being at most one is a model of those four costs, not an identity for the full velocity field.

Here is the exact algebra of that model. The second and third inequalities give $K_n \ge A_n M_n^s$ and $L_n \ge A_n M_n^s M_{n-1}$. Substituting this lower bound into the fourth gives

$$A_n^3 M_n^{2s-1} M_{n-1} \le A_{n-1}.$$

On setting $A_{n-1} = M_n^\alpha$ and therefore $A_n = M_{n+1}^\alpha$, the left side divided by the right side is

$$M_{n+1}^{3\alpha} M_n^{2s-1-\alpha} M_{n-1} = N^{R^{n-1}[3\alpha R^2 + (2s-1-\alpha)R + 1]}.$$

Since $N > 1$, its being at most one is exactly

$$3\alpha R^2 + (2s - 1 - \alpha)R + 1 \le 0. \tag{FS2}$$

For $\alpha > 0$, this quadratic can be nonpositive at positive R only if $2s-1-\alpha < 0$ and $(2s-1-\alpha)^2 \ge 12\alpha$. The latter two requirements together are exactly $s \le (1+\alpha)/2 - \sqrt{3\alpha}$. Indeed, the product of the two roots is $1/(3\alpha) > 0$; their positive sum is equivalent to the stated sign condition. This is the introductory calculation. The full source chooses a smaller regularity margin to control additional interactions, as follows.

20.2 The precise additional exponent that produces the cutoff

For the rest of this section set

$$\begin{aligned}
\alpha_0 &= \frac{22 - 8\sqrt{7}}{9}, & 0 &< \alpha < \alpha_0, \\
R &= \sqrt{\frac{2}{7\alpha}}, & a &= \sqrt{\frac{2\alpha}{7}}, \\
s &= \frac{2 + 3\alpha - 2\sqrt{14\alpha}}{4}, & \ell &= 1 + \alpha - s - 2a.
\end{aligned} \tag{357}$$

These are the parameters in Section 4 of the source. In particular

$$a = \alpha R, \qquad \frac{1}{R} = \frac{7}{2}a, \qquad \sqrt{14\alpha} = 7a, \qquad A_{n-1} = M_n^{\alpha}, \qquad \ell = \frac{2+\alpha}{4} + \frac{3}{2}a. \tag{FS3}$$

Each identity follows by squaring the positive square roots, or substituting $4s = 2 + 3\alpha - 14a$; no coordinate or physical coefficient has been changed.

The source's outer-velocity error, after integrating its duration, has the explicit scale

$$A_n M_n^{r+1} L_n^{-(3-\delta)} N^{2R^{n-1}} \log M_n. \tag{FS4}$$

For arbitrary $a = \alpha R$ and $\ell = 1 + \alpha - s - 2a$, its exponent at $\delta = 0$ is exactly

$$a + r + 1 - 3\ell + \frac{2}{R} = 7\alpha R + \frac{2}{R} - 2 - 3\alpha + r + 3s.$$

At $r = s$, the R-dependent part satisfies the identity

$$7\alpha R + \frac{2}{R} - 2\sqrt{14\alpha} = \left(\sqrt{7\alpha R} - \sqrt{\frac{2}{R}}\right)^2 \geq 0.$$

Equality holds exactly at the value of R in (357). Thus s in that equation is exactly the value at which the optimized outer-error exponent at $r = s$ is zero. For the actual $r < s$ it becomes $r - s$. This explains the numerical cutoff directly from the extra interaction cost.

We prove the admissible range without decimal approximations. The inequality $s > 0$ is $2 + 3\alpha > 2\sqrt{14\alpha}$. Both sides are positive, so squaring is equivalent and gives

$$9\alpha^2 - 44\alpha + 4 > 0, \qquad 9\alpha^2 - 44\alpha + 4 = 9(\alpha - \alpha_0)\left(\alpha - \frac{22 + 8\sqrt{7}}{9}\right).$$

The scale requirement $R > 1$ implies $\alpha < 2/7$. In this interval the second root exceeds $2/7$, so $s > 0$ is precisely $\alpha < \alpha_0$. Furthermore $8\sqrt{7} > 21$ because $448 > 441$, giving $\alpha_0 < 1/9$. The following inequalities and identities will be used without losing their strict margins:

$$0 < s < \frac{1}{2}, \quad 0 < a < \frac{1}{5}, \quad a > \alpha, \quad 0 < \ell < \frac{5}{6}, \quad s + a < 1, \tag{358}$$

$$\ell - a - \frac{1}{R} - s = \frac{a - \alpha}{2} > 0, \qquad 3\ell - 1 - \frac{2}{R} = s + a > 0. \tag{359}$$

For $s < 1/2$, square $3\alpha < 2\sqrt{14\alpha}$, obtaining $9\alpha < 56$, valid for $\alpha < 1/9$. Next $a^2 = 2\alpha/7 < 2/63 < 1/25$, and $a > \alpha$ follows from $\alpha < 2/7$. The displayed alternative expression for ℓ makes it positive and gives

$$\ell < \frac{19}{36} + \frac{3}{10} = \frac{149}{180} < \frac{5}{6}.$$

Also $s + a = (2 + 3\alpha - 10a)/4 < 7/12 < 1$. Finally direct substitution of $4s = 2 + 3\alpha - 14a$ and $1/R = 7a/2$ proves both identities in (359). Consequently $L > AN^{1/R+s}$ and $L^3 > N^{1+2/R}$, with the exact positive margins displayed.

20.3 Diffusion inside the evolving layer and its backward factor

For clarity, this paragraph proves a finite-layer identity independently of all singular limits. Let $U(x,t)$ be a smooth velocity on a region and time interval whose flow $\Phi_{t,1}(x)$ exists. Write $J(x,t) = D_x\Phi_{t,1}(x)$ and take a smooth terminal vector field w_1 and a scalar function $g(t)$. Set

$$G(t) = \exp\left(\int_t^1 g(\tau)\,d\tau\right), \qquad w(x,t) = G(t)J(x,t)^{-1}w_1(\Phi_{t,1}(x)). \tag{FS5}$$

The inverse exists because a smooth flow has a smooth inverse. Along a trajectory, $\Phi_{t,1}$ is constant, and differentiation of its spatial derivative gives $(\partial_t + U\cdot\nabla)J = -JDU$. Differentiate $J^{-1}J = I$ to obtain $(\partial_t + U\cdot\nabla)J^{-1} = DUJ^{-1}$. Finally $G' = -gG$, so the product rule proves

$$\partial_t w + U\cdot\nabla w = w\cdot\nabla U - gw. \tag{FS6}$$

Thus $G \geq 1$ for $g \geq 0$ is the correct *backward* amplification factor for a forward damping term. Removing it would alter the terminal-value problem.

The actual source uses the cyclic axes $e_{n+1}, e_{n+2}, e_{n+3}$, with indices read modulo three, and the transverse deformation factors g_{n2}, g_{n3}. Its scalar damping is

$$g_n(t) = \nu M_n^\alpha \Big((\partial_x g_{n2})(0,t)^2 + g_{n3}(0,t)^2\Big)^{\alpha/2}. \tag{FS7}$$

The two phases are those in the actual transported layer, not a replacement Cartesian Fourier mode. The scalar approximates the principal diffusion on that layer. The force receives the residual $\nu\Lambda^\alpha w_n - g_n w_n$.

For each use of Λ^α below, take a globally defined smooth compactly supported vorticity on $\mathbb{R}^3$, as in the source's layers; the fractional operator then has its usual Fourier-multiplier meaning. The transport formula above applies on every region where that field has the stated representation. Define the bilinear vector-field expression $\mathcal{B}(v,w) = v\cdot\nabla w - w\cdot\nabla v$. Equation (FS6) gives, for every smooth u,

$$\partial_t w + \mathcal{B}(u,w) + \nu\Lambda^\alpha w = \mathcal{B}(u-U,w) + \nu\Lambda^\alpha w - gw. \tag{FS8}$$

This follows by adding and subtracting $\mathcal{B}(U,w)$ and using its equation; it fixes both signs. The dissipative term has been incorporated in the evolution, and only its approximation error appears in this formula.

The published Section 4.3.1 chooses the *physical viscosity*

$$d = s - r > 0, \quad 0 < r < s, \qquad \nu = \frac{d}{200}. \tag{FS9}$$

It retains the factor G_n omitted from the corresponding initial-layer bound in the 2024 arXiv text. Here is the exact arithmetic of the correction. The source's deformation and duration bounds are

$$|\partial_x g_{n2}(0,t)| \leq 1, \quad |g_{n3}(0,t)| < 2, \qquad 1 - T_n < 34R^n A^{-R^{n-1}} \log(AN^s). \tag{FS10}$$

Since $5^{1/18} < 2$ (raise both positive sides to power 18), these imply

$$0 \leq g_n(t) < \frac{d}{100} M_n^\alpha.$$

Using $t \geq T_n$, $A^{R^{n-1}} = M_n^\alpha$, and $R^n \log(AN^s) = (a+s)\log M_n$ gives

$$\begin{aligned} \log G_n(t) &= \int_t^1 g_n(\tau)\,d\tau \\ &< \frac{d}{100} M_n^\alpha (1 - T_n) < \frac{17}{50} d(a+s)\log M_n < \frac{2}{5} d \log M_n. \end{aligned}$$

Thus the source's coarser bound $G_n < M_n^{2d/5}$ follows while the factor 34 and the time duration $1 - T_n$ are retained. Its additional estimate $G_n \leq K_n$ is used in quadratic products: $G_n^2/K_n \leq G_n$. The switching-time estimate in that section has right-hand side

$$G_n(T_n)K_n(T_n)^{-1}A_n M_n^r, \qquad K_n(T_n) = A_n M_n^s.$$

Its exponent is therefore at most $-3d/5$, and in particular at most $-d/2$.

20.4 Every remaining exponent and its positive summation margin

The five error classes in published Sections 4.3.1–4.3.5 use the following right-hand sides. We retain all logarithms when multiplying the pointwise bound by the duration; $M = M_n$ in this subsection.

The switching term is bounded by a source constant times $M^{-3d/5}$ as just computed. The quadratic self-interaction, including the one remaining factor of G_n, has integrated scale

$$A_n^2 L_n M^{r+2d/5-1+2\delta-\alpha}(\log M)^2 = M^{-3d/5+2\delta}(\log M)^2. \tag{FS11}$$

The equality uses $2a+\ell-1-\alpha = -s$. The inner-velocity/outer-vorticity scale is

$$M^{r+2d/5+2\delta}(A/L)^{R^n} N^{R^{n-1}} \log M = M^{-3d/5-(a-\alpha)/2+2\delta} \log M. \tag{FS12}$$

Here (359) gives $a - \ell + 1/R = -s - (a-\alpha)/2$. The outer-velocity/inner-vorticity scale (FS4) is

$$M^{a+r+1+2/R-(3-\delta)\ell} \log M = M^{-d+\ell\delta} \log M. \tag{FS13}$$

This is the optimized exponent already proved above.

The diffusion estimate in published Section 4.3.5 has two terms before time integration. Retaining the factor $G_n \le M^{2d/5}$, their exponents are

$$\begin{aligned}
E_1 &= \frac{2}{5}d + a + \ell + \alpha + (r-1)(1-\delta) \\
&= 2\alpha - a - \frac{3}{5}d + (1-r)\delta, \\
E_2 &= \frac{2}{5}d + 2a + \alpha + r + s + \frac{2}{R} - 2\ell \\
&= 2\alpha - a - \frac{3}{5}d.
\end{aligned}$$

For the second equality use $2/R = 7a$, $\ell = 1 + \alpha - s - 2a$, and $4s = 2 + 3\alpha - 14a$; the terms combine without discarding the r dependence. Multiplication by the duration $M^{-\alpha}\log M$ leaves the exponents

$$\alpha - a - \frac{3}{5}d + (1-r)\delta, \qquad \alpha - a - \frac{3}{5}d, \tag{FS14}$$

with $(\log M)^2$ allowed for both terms. The source displays one logarithm in its final integrated bound; the two logarithms directly obtained by multiplication are kept here and have the summability proved next.

One explicit simultaneous selection is

$$0 < \delta \le \delta_* := \min\left\{\frac{d}{12}, \frac{a-\alpha}{4}, \frac{1-1/R}{2}, \frac{1}{10}\right\}. \tag{FS15}$$

All four entries are positive by (358) and $R > 1$. The self exponent is at most $-13d/30$, and the inner exponent is no larger. The outer exponent is smaller than $-d + (5/6)(d/12) = -67d/72$. The first diffusion exponent is at most $-3(a-\alpha)/4 - 3d/5$, since $0 < r < 1$ gives $(1-r)\delta \le \delta$; the second is negative with larger margin. In particular every displayed exponent is strictly negative. This also ensures $M^{1/R} \le M^{1-\delta}$, an inequality used in the source's inner-interaction estimate.

For completeness, no logarithmic summation step is being suppressed. Given $\epsilon > 0$ and integer $m \ge 0$, the function $y^m e^{-\epsilon y/2}$ on $y \ge 0$ is bounded: for $m > 0$ its maximum is $(2m/(e\epsilon))^m$, found by differentiating; for $m = 0$ its bound is one. Therefore

$$M_n^{-\epsilon}(\log M_n)^m \le C_{m,\epsilon} N^{-\epsilon R^n/2}.$$

Bernoulli's inequality $R^n \ge 1 + n(R-1)$ follows by induction from $(1+h)^{n+1} \ge (1+nh)(1+h) \ge 1 + (n+1)h$ for $h = R - 1 > 0$. Consequently the last sequence is bounded by the summable geometric sequence $N^{-\epsilon/2}[N^{-\epsilon(R-1)/2}]^n$. This proves summability for every explicit scale above, including both logarithms in the diffusion term.

As an exact rational example, take

$$\alpha=\frac{1}{14},\quad R=2,\quad a=\frac{1}{7},\quad s=\frac{3}{56},\quad \ell=\frac{41}{56},$$
$$r=\frac{3}{112},\quad d=\frac{3}{112},\quad \nu=\frac{3}{22400},\quad \delta=\frac{1}{448}.$$

Substitution in (357) proves all equalities; this δ is $d/12$ and satisfies the other three bounds in (FS15). The script `scripts/check_forced_scaling.py` computes each exponent as an exact rational number and verifies the identities symbolically.

20.5 Keeping the physical viscosity and testing the classical endpoint

For $\alpha>0$, a solution on $\mathbb{R}^3$ with coefficient $\nu_*>0$ has an exact spatial transfer to any $\nu>0$. Retain

$$b=(\nu_*/\nu)^{1/\alpha},\quad \widetilde{u}(x,t)=b^{-1}u(bx,t),\quad \widetilde{p}(x,t)=b^{-2}p(bx,t),\quad \widetilde{f}(x,t)=b^{-1}f(bx,t). \tag{FS16}$$

The chain rule gives $\widetilde{u}_t=b^{-1}u_t$, $\widetilde{u}\cdot\nabla\widetilde{u}=b^{-1}(u\cdot\nabla u)$, and $\nabla\widetilde{p}=b^{-1}\nabla p$, all evaluated at (bx,t). The Fourier transform of $u(b\,\cdot)$ is $b^{-3}\widehat{u}(\xi/b)$. Multiplication by $|\xi|^\alpha$ and inverse transformation prove $\Lambda^\alpha\widetilde{u}=b^{\alpha-1}(\Lambda^\alpha u)(bx,t)$. Since $\nu b^\alpha=\nu_*$, every term in the new equation equals b^{-1} times the old equation. Divergence vanishes by the same chain rule. The exact L^2 factor for u and f is $b^{-5/2}$, obtained by the change of variables $y=bx$. The supremum norm of ∇u is unchanged. A spatial r-Hölder seminorm of ∇f is multiplied by b^r, because its numerator is unchanged and its denominator rescales by b^{-r}. The time interval is unchanged. This proves the claimed coefficient transfer and explicitly records its data, forcing, and norm transformations.

At the classical value $\alpha=2$, the very same expressions instead give

$$R=\frac{1}{\sqrt{7}}<1,\qquad a=\frac{2}{\sqrt{7}},\qquad s=2-\sqrt{7}<0,\qquad \alpha-a=2-\frac{2}{\sqrt{7}}>0. \tag{FS17}$$

All three statements follow from $\sqrt{7}>2>1$. Thus this particular layer choice loses increasing frequencies, positive force regularity margin, and the negative diffusion-error exponent at once. These are exact failures of this parametrization at $\alpha=2$. They establish no nonexistence theorem for a different classical construction. The equations above identify the terms whose new cancellations or estimates a reconstruction must actually compute.

20.6 Literal source discrepancies retained in the audit

The saved published JATS XML has stable equation identifiers. Its `Equ148` uses the damping sign in (FS6). In `Equ192`–`Equ193`, however, the displayed force decomposition uses $-\nu\Lambda^\alpha$ and labels a bracket ending in $-g_nw_n$ as zero. Substitution in (FS6) makes that bracket $-2g_nw_n$, whereas the corrected bracket ends in $+g_nw_n$ and is zero. Identity (FS8) proves the corrected residual, without a sign convention change. The definition of K_n in `Equ154` omits an exponential; the exponential is present in `Equ17` and `Equ199`. As a stretching factor with terminal value one it must be

$$K_n(t)=\exp\left(\int_t^1-\partial_{n+2}(U_n)_{n+2}(0,0,0,\tau)\,d\tau\right).$$

At $t=1$ the integral alone is zero, immediately contradicting that terminal value. In `Equ197`, the duration in the first displayed exponent is T_n; the integral over $[t,1]$ gives $1-t\le 1-T_n$, as calculated above. Some powers in that XML are literally A^n or $(AN^s)^n$, while the layer definition and the arXiv TeX use A^{R^n} and $(AN^s)^{R^n}$. The calculation here explicitly fixes its scale dictionary in (FS3); the XML and TeX snapshots remain unchanged for inspection. These discrepancies are not silently treated as independently validated formulas.

There is also an exact correction when time cutoffs are inserted. For a finite family of smooth compactly supported vorticities w_j on $\mathbb{R}^3$, set $v_j = \mathcal{K} * w_j$, $w = \sum_j \rho_j w_j$, and $u = \sum_j \rho_j v_j$, where ρ_j depends only on time. Linearity of the time derivative and of Λ^α, and bilinearity of $\mathcal{B}$, give

$$\begin{aligned}\partial_t w + \mathcal{B}(u, w) + \nu\Lambda^\alpha w = \sum_j \rho_j' w_j + \sum_j \rho_j(\partial_t w_j + \nu\Lambda^\alpha w_j) \\ + \sum_{j,k} \rho_j \rho_k \mathcal{B}(v_j, w_k).\end{aligned}$$

In particular the self term has coefficient ρ_j^2, not ρ_j. Their difference is $(\rho_j^2 - \rho_j)\mathcal{B}(v_j, w_j)$. For $0 \le \rho_j \le 1$, its coefficient has absolute value at most $1/4$, since $\rho_j(1 - \rho_j) = 1/4 - (\rho_j - 1/2)^2$. The already displayed self-interaction bound therefore controls this extra term with the factor $1/4$. Ordered disjoint switching intervals make every older cutoff one while a newer cutoff changes; this eliminates the corresponding mixed-coefficient discrepancy. The full finite identity above remains valid without that ordering. A locally finite family on $t < 1$ satisfies the same identity on each compact time interval because only finitely many summands are present.

The source snapshots, equation locators, exact arithmetic results, and the independent sign derivation are retained in `research/forced_program_evidence.json` and `research/forced_program_sign_audit.md`. They record reconstruction evidence and calculations, not an original discovery-session transcript.

21 Exact bridges from the retained Fable polynomial and the S6 cusp

21.1 Source and scope

The source corpus is *S6 Proof Audit, Reconstruction & Overleaf Release*; its audited artifact names and hashes are retained with this notebook. Its current source root is `s6_consolidated_canonical_2026-08-26`. All source locators in this section are relative to that root. The dated status in `supporting_materials/workbench/research/status.tex`, lines 145–154 and 527–587, says that all six topology bridges are closed and records a compact complex threefold X with $X \cong_{\text{diff}} S^6$, $a(X) = 1$, and $\langle c_3(T_X), [X]\rangle = 2$. It explicitly defines "Verified" to mean that a complete proof is written in the workbench. The older unconfirmed audit verdict at `main.tex`, lines 5989 and 6241, is not the current workbench verdict. The present bounded audit checks the concrete polynomial, metric, differential-operator and Fourier calculations below; it does not independently re-certify every analytic gluing and integral geometric map of the complete S6 workbench.

For example, the displayed terminal group presentation at status lines 578–580 is trivial by an exact calculation: $xy = 1$ gives $y = x^{-1}$; $x^3 = c$ and $y^4 = c^{-1}$ give $x^{-4} = x^{-3}$, hence $x = 1$, then $y = c = 1$. This verifies the algebra of that presentation. Whether it is the geometric fundamental group is a different claim, whose geometric proof is in the source; this short algebra calculation does not replace that proof.

The polynomial source is `supporting_materials/workbench/comparative_context/prior_fable_context.tex`, lines 8–28. That file describes a pinned JGPTech certificate for a map attributed there to Fable; lines 31–38 expressly say that an original model-session transcript was not recovered there. The JSON supplied with this audit is a new exact replay record, not that missing discovery transcript.

21.2 The original polynomial, its determinant, and its collision

Retain the original ordered polynomial coordinates (x, y, t) and define $F : \mathbb{C}^3 \longrightarrow \mathbb{C}^3$ by

$$F_1 = (1 + xy)^3 t + y^2(1 + xy)(4 + 3xy), \tag{360}$$

$$F_2 = y + 3x(1 + xy)^2 t + 3xy^2(4 + 3xy), \tag{361}$$

$$F_3 = 2x - 3x^2 y - x^3 t. \tag{362}$$

The symbol t in these three equations is a spatial coordinate, never fluid time. Fluid time below is s. To compare with a convention using spatial coordinates (z_1, z_2, z_3), use the explicitly defined linear isometry $R(z_1, z_2, z_3) = (x, y, t) = (z_1, z_2, z_3)$. Its derivative is I_3, determinant is 1, and its pullback sends $(\partial_x, \partial_y, \partial_t)$ to $(\partial_{z_1}, \partial_{z_2}, \partial_{z_3})$ in the same order. Thus $F \circ R$ is an exact renaming preserving every coefficient, derivative, volume form and Euclidean operator. We continue to use (x, y, t).

Here is a determinant proof retaining also the locus $x = 0$. On $x \neq 0$, put $v = 1 + xy$ and $h = 2 - 3xy - x^2 t$. Direct substitution gives

$$F_1 = \frac{v^2 + v - v^3 h}{x^2}, \qquad F_2 = \frac{4v + 2 - 3v^2 h}{x}, \qquad F_3 = xh.$$

The coordinate-change determinant is

$$\det \frac{\partial(x, v, h)}{\partial(x, y, t)} = \det \begin{pmatrix} 1 & 0 & 0 \\ y & x & 0 \\ -3y - 2xt & -3x & -x^2 \end{pmatrix} = -x^3.$$

Writing $A = v^2 + v - v^3 h$ and $B = 4v + 2 - 3v^2 h$, the other determinant is

$$\det \frac{\partial(F_1, F_2, F_3)}{\partial(x, v, h)} = x^{-3} \det \begin{pmatrix} -2A & 2v + 1 - 3v^2 h & -v^3 \\ -B & 4 - 6vh & -3v^2 \\ h & 0 & 1 \end{pmatrix} = \frac{2}{x^3}.$$

Indeed the displayed numerator expands to

$$(4 - 6vh)(-2v^2 - 2v + 3v^3 h) + (2v + 1 - 3v^2 h)(4v + 2 - 6v^2 h) = 2.$$

Consequently $\det DF = -2$ on $x \neq 0$. Both sides are polynomials in the original coordinates, so equality on this open set is a polynomial identity and includes $x = 0$. Independently, substitution at $x = 0$ into the original Jacobian gives determinant -2 there as well.

The three retained points

$$p_0 = (0, 0, -1/4), \quad p_+ = (1, -3/2, 13/2), \quad p_- = (-1, 3/2, 13/2)$$

are distinct and have common image $(-1/4, 0, 0)$. At p_0 this follows directly from the original equations. At $p_\pm$, one has $xy = -3/2$, $1 + xy = -1/2$, $4 + 3xy = -1/2$, $y^2 = 9/4$ and $t = 13/2$. Thus $F_1 = -13/16 + 9/16 = -1/4$. At p_+, $F_2 = -12/8 + 39/8 - 27/8 = 0$ and $F_3 = 2 + 9/2 - 13/2 = 0$. At p_- each summand in F_2, F_3 changes sign, so both remain zero. This is an exact algebraic certificate about the displayed map, independent of authorship or discovery claims. In fact these are the entire fibre, including over $\mathbb{C}$. For a point with image $(-1/4, 0, 0)$ and $x = 0$, the original second equation gives $y = 0$ and the first then gives $t = -1/4$. For $x \neq 0$, $F_3 = xh = 0$ gives $h = 0$; the retained equations give $xF_2 = 4v + 2 = 0$, hence $v = -1/2$, and $x^2 F_1 = v^2 + v = -1/4$. Since $F_1 = -1/4$, this forces $x^2 = 1$. Then $y = (v - 1)/x = -3/(2x)$ and $t = (5 - 3v)/x^2 = 13/2$. Both possibilities are exactly p_+ and p_-.

21.3 A polynomial incompressible velocity and an escaping trajectory

Restrict F to $\mathbb{R}^3$ and let $J = DF$ in the original coordinate order. Since $\det J = -2$, the field $W = J^{-1}e_1$ is globally polynomial. Its full components are

$$W_x = 3x^6yt + 9x^5y^2 + 3x^5t + 3x^4y - 4x^3, \tag{363}$$

$$W_y = 3x^4yt + 9x^3y^2 + 3x^3t + 3x^2y - 3x, \tag{364}$$

$$W_t = -9x^5yt^2 - 45x^4y^2t - 9x^4t^2 - 54x^3y^3 - 30x^3yt - 27x^2y^2 + 9x^2t + 21xy + 1. \tag{365}$$

These components can be verified without an inverse-matrix convention: cofactor expansion gives

$$W = \frac{\nabla F_2 \times \nabla F_3}{\det J} = -\frac{1}{2}\nabla F_2 \times \nabla F_3 = \operatorname{curl}\Big(-\frac{1}{2}F_2\nabla F_3\Big). \tag{366}$$

The dot product of ∇F_1 with the numerator is $\det J$; the dot products with $\nabla F_2, \nabla F_3$ are zero. Hence $JW = e_1$ exactly, proving the inverse claim. Expanding the three cross-product components gives precisely (365). The identity $\operatorname{curl}(a\nabla b) = \nabla a \times \nabla b$ follows by the product rule and cancellation of the symmetric second derivatives of b. Mixed derivatives likewise give $\operatorname{div}\operatorname{curl} = 0$. In particular $\operatorname{div} W = 0$. An independent component check is

$$\begin{aligned}
\partial_x W_x &= 18x^5yt + 45x^4y^2 + 15x^4t + 12x^3y - 12x^2,\\
\partial_y W_y &= 3x^4t + 18x^3y + 3x^2,\\
\partial_t W_t &= -18x^5yt - 45x^4y^2 - 18x^4t - 30x^3y + 9x^2,
\end{aligned}$$

whose sum is zero coefficient by coefficient.

For $s < 1/4$ use the positive real square root and define

$$\gamma(s) = \big((1-4s)^{-1/2}, -\tfrac{3}{2}(1-4s)^{1/2}, \tfrac{13}{2}(1-4s)\big). \tag{367}$$

The entire displayed interval is part of its definition; in particular $\gamma(0) = p_+$. Along the curve, write $X = (1-4s)^{-1/2}$. Then $y = -3/(2X)$ and $t = 13/(2X^2)$. Substituting into (365) gives $W(\gamma(s)) = (2X^3, 3X, -26)$, which is exactly $\gamma'(s)$. The restricted original map gives

$$F(\gamma(s)) = (-1/(4X^2), 0, 0) = (s - 1/4, 0, 0).$$

Equivalently, this last equality and $J\gamma' = e_1$ give the same trajectory identity because J is invertible everywhere. As $s \uparrow 1/4$, the first coordinate tends to $+\infty$. Therefore this integral curve has no continuous extension to time $1/4$ in $\mathbb{R}^3$: any such extension would have a finite first coordinate. This is finite-time escape of a particle trajectory in a stationary smooth polynomial vector field.

21.4 Exact Euclidean Navier–Stokes residual

Use the Euclidean Laplacian $\Delta = \partial_x^2 + \partial_y^2 + \partial_t^2$ and viscosity $\nu > 0$. A stationary unforced Navier–Stokes solution with velocity W would satisfy

$$(W\cdot\nabla)W - \nu\Delta W = -\nabla p. \tag{368}$$

The right-hand side has zero curl, even for a distributional pressure, because distributional mixed derivatives commute. Direct differentiation of the retained polynomial gives

$$\operatorname{curl}\big((W\cdot\nabla)W - \nu\Delta W\big)\big|_{(x,y,t)=(0,0,t)} = (0, 0, -18\nu t). \tag{369}$$

Here is a local calculation of that identity in addition to the exact replay. Along this axis $W = (0, 0, 1)$ and the only possibly nonzero entry of DW is $\partial_x W_y = -3$; $\partial_x W_t = 21y$ vanishes there. Thus $(W \cdot \nabla)W = \partial_t W = 0$ there. Differentiating this advection expression before restriction gives zero first derivatives there: the product-rule term from $\partial_x W_y = -3$ multiplies $\partial_y W = 0$, and the remaining term $\partial_{xt} W$ also vanishes; derivatives in y, t vanish by the same displayed polynomial. Consequently its curl is zero there. The viscous curl is read from

$$\begin{aligned} W_x &= -4x^3 + 3x^4 y + 3x^5 t + 9x^5 y^2 + 3x^6 yt, \\ W_y &= -3x + 3x^2 y + 3x^3(t + 3y^2) + 3x^4 yt, \\ W_t &= 1 + 21xy + x^2(9t - 27y^2) + x^3(-30yt - 54y^3) \\ &\qquad + x^4(-9t^2 - 45y^2 t) - 9x^5 yt^2. \end{aligned}$$

At $x = y = 0$, $\partial_y \Delta W_t - \partial_t \Delta W_y = 0$, $\partial_t \Delta W_x - \partial_x \Delta W_t = 0$, and $\partial_x \Delta W_y - \partial_y \Delta W_x = 18t$. These are respectively the three components of $\operatorname{curl} \Delta W$. They prove (369), including its sign. At the single point $(0, 0, 1)$ the residual curl is $(0, 0, -18\nu) \neq 0$. Thus no pressure makes W an unforced stationary Euclidean Navier–Stokes solution for any positive viscosity.

There is also an independent initial-data obstruction. W has infinite Euclidean kinetic energy. For completeness, a nonzero polynomial P of degree d cannot lie in $L^2(\mathbb{R}^3)$: its top homogeneous part is nonzero on some open spherical patch, and is bounded in absolute value below by a positive constant on a smaller compact patch. On the associated cone, for all sufficiently large radii, lower-degree terms are bounded by half the leading term, so $|P(r\theta)| \geq cr^d$. The polar-coordinate integral then dominates $c^2 \int_R^\infty r^{2d+2}\, dr = \infty$ times a positive solid angle. Apply this to a nonzero component of W; its first component has degree 8. Moreover W is not unit-periodic: on $y = t = 0$ its first component is $-4x^3$, whose values at $x = 0, 1$ differ. None of these facts establishes anything about every possible velocity constructed from F; they establish the precise limitations of the displayed W.

21.5 Exact local-isometry correspondence and incomplete metric

There is a stronger geometric bridge. Equip the same underlying $\mathbb{R}^3$ with

$$g = J^\top J, \qquad g_{ab} = \sum_{i=1}^{3} (\partial_a F_i)(\partial_b F_i). \tag{370}$$

This is positive definite because J is invertible. Its determinant is 4, so its positive volume density is $2\, dx\, dy\, dt$; the signed pullback of the target orientation is instead $F^*(dX_1 \wedge dX_2 \wedge dX_3) = -2\, dx \wedge dy \wedge dt$. The real inverse function theorem gives local coordinates $X_i = F_i$ around every point. In exactly these coordinates the metric is $dX_1^2 + dX_2^2 + dX_3^2$. Thus F is a flat local isometry, with the specified orientation reversal; it is not asserted to be a global coordinate chart. The metric change already appears at the original origin:

$$J(0) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 2 & 0 & 0 \end{pmatrix}, \qquad g(0) = \operatorname{diag}(4, 1, 1).$$

For a direct scalar diffusion comparison, the target coordinate X_1 is Euclidean harmonic, while $F^* X_1 = F_1$ has $\Delta F_1(0) = 8$, from its exact quadratic term $4y^2$. The metric correspondence therefore does not intertwine the two ordinary Euclidean Laplacians.

Let $\Omega = F(\mathbb{R}^3)$, an open set, and let a smooth target velocity $v(s, X)$ and pressure $p(s, X)$ be defined on an open time interval times Ω. Define

$$u(s, x, y, t) = J(x, y, t)^{-1} v(s, F(x, y, t)), \qquad p_g(s, x, y, t) = p(s, F(x, y, t)). \tag{371}$$

This is a globally defined pullback map on those fields. In each of the local coordinates $X = F(x, y, t)$ its components and metric are precisely v and the Euclidean metric. Consequently the following operator identities hold on every local chart, and hence globally:

$$\begin{aligned} \operatorname{div}_g u &= (\operatorname{div} v) \circ F, \\ \nabla^g_u u &= J^{-1}((v \cdot \nabla_X)v) \circ F, \\ \operatorname{grad}_g p_g &= J^{-1}(\nabla_X p) \circ F, \\ (\Delta_{H,g} u^\flat)^\sharp &= J^{-1}(-\Delta_X v) \circ F. \end{aligned}$$

Here $\Delta_{H,g} = d\delta_g + \delta_g d$ is the nonnegative Hodge Laplacian; on Euclidean one-forms it is the negative componentwise Laplacian. One may verify the identities directly in X coordinates: the metric components are constant, Christoffel symbols vanish, the divergence is $\sum_i \partial_{X_i} v_i$, and the Hodge Laplacian is $-\sum_i \partial^2_{X_i}$ on each component. Coordinate tensor transformation gives exactly the displayed J^{-1} factors. The g-divergence also equals ordinary coordinate divergence because its volume density is the constant 2: $\operatorname{div}_g u = \frac{1}{2} \sum_a \partial_a(2u^a) = \operatorname{div} u$.

It follows by substitution, retaining the same s and ν, that pullback carries the Euclidean equation $\partial_s v + (v \cdot \nabla_X)v = -\nabla_X p + \nu \Delta_X v$ to the exact intrinsic equation

$$\partial_s u + \nabla^g_u u = -\operatorname{grad}_g p_g - \nu(\Delta_{H,g} u^\flat)^\sharp, \qquad \operatorname{div}_g u = 0.$$

This is a proved morphism between specified PDE solution spaces. Conversely a smooth field u descends under this morphism precisely when $J(x)u(x)$ is constant on each fibre of F; a pressure descends precisely when it too is constant on each fibre. Necessity follows from (371). For sufficiency these equalities define a field and pressure on Ω; local inverses of F prove smoothness, and fibre constancy makes the local definitions agree. The local operator identities then prove the equations for the descended field. There is therefore a full local correspondence and a stated exact global descent condition, not an assertion of a nonexistent global inverse.

In particular the target solution $v = e_1$, $p = 0$ pulls back to $u = W$. It is parallel for g, has $|W|_g = 1$, and solves this metric equation for every s and every $\nu > 0$. Its metric energy is also infinite: $\int_{\mathbb{R}^3} |W|_g^2 \, d\operatorname{vol}_g = 2 \int_{\mathbb{R}^3} 1 \, dx \, dy \, dt = \infty$. Nevertheless g is incomplete. The g-speed of (367) is $|J\gamma'| = |e_1| = 1$. For any sequence $s_n \uparrow 1/4$, the tail of the curve has length $|s_n - s_m|$, so the points $\gamma(s_n)$ are Cauchy in the g distance. They cannot converge to a point of $\mathbb{R}^3$: the first coordinate is unbounded, and a smooth positive-definite metric induces the ordinary local topology. To check the latter assertion at a putative limit, take a small closed Euclidean ball around it. The minimum eigenvalue of g on this compact ball is positive; every curve reaching the ball's boundary has g length at least that lower length factor times the Euclidean radius. Thus sufficiently small metric balls lie in any chosen Euclidean neighborhood of the limit. This contradicts the unbounded coordinate of the sequence. The metric field remains smooth at every finite spatial point and at every time; particle escape in this incomplete metric is not finite-time loss of smoothness in the complete Euclidean spatial geometry.

21.6 A compactly supported incompressible initial-data map

An explicit localization retains a genuine map to smooth admissible spatial data. Define $b(r) = e^{-1/r}$ for $r > 0$ and $b(r) = 0$ for $r \le 0$, and

$$\eta(r) = \frac{b(4-r)}{b(4-r) + b(r-1)}, \qquad \chi(z) = \eta(|z - p_+|^2/R^2), \quad R > 0.$$

The denominator is positive for every real r. Each derivative of b on the positive side is an exponential times a polynomial in $1/r$, which tends to zero as $r \downarrow 0$; hence b and η are smooth. The function χ equals 1 on the closed radius-R ball about p_+ and 0 outside the radius-$2R$ ball. Set

$$U_\chi = -\frac{1}{2} \operatorname{curl}(\chi F_2 \nabla F_3) = \chi W - \frac{1}{2} F_2 (\nabla\chi \times \nabla F_3). \tag{372}$$

The product rule proves the second equality. All factors are smooth, and the potential has compact support, so U_χ is smooth and compactly supported. Its divergence is zero by commuting mixed derivatives. On the plateau $\chi \equiv 1$ its derivative is zero, giving $U_\chi = W$ exactly. Every derivative of U_χ is bounded and compactly supported; in particular the field is in every Sobolev space of nonnegative integral order, has finite energy, and decays faster than every inverse power. It is thus an explicit map to smooth finite-energy divergence-free initial data for the Euclidean problem.

The autonomous particle equation for this compact field is complete. Indeed $|U_\chi| \le M$ globally and its first derivatives are bounded, so it is globally Lipschitz. On every finite time interval a trajectory stays within distance M times the length of that interval from its initial point; standard local ODE existence and uniqueness extend it past any finite endpoint in that compact ball. This completeness statement concerns its autonomous ODE and makes no assertion about the subsequent nonlinear Navier–Stokes evolution of U_χ. The escaping curve γ leaves the plateau in finite time before $1/4$ because its first coordinate diverges. Thus the exact local agreement does not preserve the escaping final segment.

21.7 Retained cusp Fourier data mapped to exact periodic solutions

The source `supporting_materials/workbench/research/cusp_spectral_geometry.tex`, lines 83–185, specifies an additional Euclidean metric on a regular four-dimensional real torus. Retain all original coefficients:

$$P_T = \begin{pmatrix} 6\alpha_T & \eta_T & 1 & 0 \\ 6m_T & L_T & 0 & 0 \\ c_T & \alpha_T & 0 & 1 \\ -q_T & m_T & 0 & 0 \end{pmatrix}, \qquad D_T = L_T q_T + 6m_T^2 > 0, \quad F_T = \mathbb{R}^4 / P_T \mathbb{Z}^4.$$

Its oriented determinant is $-D_T$. For the original dual label $k = (r, s_0, n_1, n_2) \in \mathbb{Z}^4$ define

$$R_T = r - 6\alpha_T n_1 - c_T n_2, \qquad S_T = s_0 - \eta_T n_1 - \alpha_T n_2.$$

Here s_0 denotes the source's second integer Fourier label, solely to avoid collision with fluid time s; the map $s_{\text{source}} \mapsto s_0$ is the identity on that integer, with no change to the label or lattice. Solving $P_T^{\mathsf{T}} \ell = k$ gives exactly

$$\ell = \left(n_1, \frac{m_T R_T + q_T S_T}{D_T}, n_2, \frac{-L_T R_T + 6m_T S_T}{D_T} \right).$$

The first and third components come from the last two equations; the other two solve $\begin{pmatrix} 6m_T & -q_T \\ L_T & m_T \end{pmatrix} (\ell_2, \ell_4)^{\mathsf{T}} = (R_T, S_T)^{\mathsf{T}}$, whose determinant is D_T. This proves the inverse-transpose formula, including every sign. The character $\exp(2\pi i \langle \ell, x \rangle)$ is well-defined on F_T since $\langle \ell, P_T n \rangle = k \cdot n \in \mathbb{Z}$. The nonnegative scalar Laplacian Δ_T therefore has the eigenvalue $4\pi^2 |\ell|^2$ on this character by two differentiations.

Fix any nonzero original label k and put $a = |\ell| > 0$. Consider the exact closed sector of functions $h([x]) = f(\langle \ell, x \rangle \bmod 1)$ with smooth real 1-periodic f. The phase map is surjective onto the circle: its lift is a nonzero real linear functional, so takes every real value. Hence this representation determines f uniquely. On this sector define

$$\mathcal{I}_k h(z_1, z_2, z_3) = e_2 f(z_1), \qquad (z_1, z_2, z_3) \in \mathbb{R}^3 / \mathbb{Z}^3.$$

This is an injective linear map with fully specified source and target. If $\Delta_3 = -\sum_{j=1}^{3} \partial_{z_j}^2$ is the nonnegative componentwise operator on the target, the chain rule proves

$$\Delta_T h = -a^2 f''(\langle \ell, x \rangle), \qquad \Delta_3 \mathcal{I}_k h = e_2(-f''(z_1)) = a^{-2} \mathcal{I}_k \Delta_T h.$$

Thus the source heat solution $h(\tau) = e^{-\nu\tau\Delta_T} h_0$ gives the target velocity

$$u(s,z) = \mathcal{I}_k h(s/a^2)(z), \qquad p(s,z) = 0, \quad s \geq 0.$$

It satisfies $\partial_s u = -\nu\Delta_3 u$. Also $\operatorname{div} u = \partial_{z_2} f(z_1) = 0$ and $(u\cdot\nabla)u = f(z_1)\partial_{z_2}(e_2 f(z_1)) = 0$. Therefore it solves the full unforced periodic three-dimensional Navier–Stokes equation with viscosity ν, exactly.

For an explicit global regularity proof in this sector, write $f_0(\theta) = \sum_{n\in\mathbb{Z}} c_n e^{2\pi i n\theta}$ with $c_{-n} = \overline{c_n}$. Integration by parts N times in the Fourier coefficient integral bounds $|c_n| \leq C_N |n|^{-N}$ for every N and $n \neq 0$. The target solution is

$$u(s,z) = e_2 \sum_{n\in\mathbb{Z}} c_n e^{-4\pi^2\nu n^2 s} e^{2\pi i n z_1}.$$

After any prescribed finite number of time and space derivatives, choose N larger than the resulting power of n plus 2. The differentiated series then converges absolutely and uniformly for every $s \geq 0$. It follows that this solution is smooth for all nonnegative time; its energy on the unit torus is finite, and its mean is zero when $c_0 = 0$. No singularity is obtained by this exact Fourier-sector morphism.

The source gauge operator at `supporting_materials/workbench/research/cusp_gauge_operators.tex`, lines 64–95, is the Hessian $g_{\mathrm{YM}}^{-2} d^*d$ at a product connection. On a co-closed one-form, d^*d agrees with the Hodge Laplacian because dd^* vanishes. This identifies its differential expression with the positive linear diffusion operator; it supplies no nonzero convection term. In the exact periodic sector above that term was computed explicitly and is zero.

21.8 Reproducibility and precise conclusion

The exact rational-polynomial script `scripts/s6_bridge_check.py` retains the original F, checks $\det DF = -2$, all three collision images, $JW = e_1$, the vector potential, divergence, trajectory, residual curl, the full retained cusp matrix determinant and inverse transpose. It writes `research/s6_bridge_audit.json`, including SHA-256 hashes of the seven source files used. All assertions passed with SymPy 1.13.1 on 8 September 2026; a separate agent independently replayed the velocity, trajectory, potential and residual-curl calculations and confirmed their signs. The fixed source hashes are packaged in `sources/s6_source_hashes.json`. Default replay uses that recorded provenance and requires no sibling source repository; it does not claim to reread the seven source files. The optional `-source-root` argument additionally rereads those original files and fails on a hash mismatch. The result records which of these two provenance modes was used.

The exact relations established here are a polynomial developing map, its divergence-free inverse-Jacobian field, its flat local-isometry PDE pullback, its compactly supported divergence-free localization, and its cusp Fourier-sector map to classical periodic Navier–Stokes solutions. The escaping polynomial field fails the unforced Euclidean PDE by the nonzero curl in (369); the metric pullback is smooth but has incomplete spatial geometry; the localized fields are admissible initial data without an established singular evolution; and the Fourier-sector solutions are globally smooth. Accordingly these calculations provide no verified disproof of the classical three-dimensional Navier–Stokes existence-and-smoothness assertion. They also do not assert absence of other mathematical relations or rule out other constructions from the retained objects.

22 The entire inverse fibre and its incompressible coefficient heat flow

The source for this calculation is the current zeta task's `tex/satellites/23_source_mechanism_transfer.tex`, with the companion `certificates/rh_counterexample_20260908/verify_retained_mechanism.py`. Both files were read without alteration. Their hashes are retained in `sources/inverse_fibre_heat_source_hashes.json`. The calculation below verifies their complete inverse-fibre map and then constructs its exact dynamical lift as a polynomial incompressible field. The fluid coordinates, metric, orientation, and diffusion operator are stated throughout.

22.1 Every finite inverse point, including the excluded roots

Retain the original map, with the identity renaming of its third spatial coordinate from t to w:

$$
\begin{aligned}
F_1 &= (1+xy)^3 w + y^2(1+xy)(4+3xy),\\
F_2 &= y + 3x(1+xy)^2 w + 3xy^2(4+3xy),\\
F_3 &= 2x - 3x^2 y - x^3 w.
\end{aligned}
\tag{373}
$$

The target coordinates are (a,b,c). All algebraic statements first hold over $\mathbb{C}$. Real parameters and real roots give precisely the real points. For the determinant, on $x \neq 0$ set

$$
v = 1+xy, \qquad h = 2 - 3xy - x^2 w, \qquad A = v^2 + v - v^3 h, \qquad B = 4v + 2 - 3v^2 h.
$$

Direct substitution gives $F = (A/x^2, B/x, xh)$ and

$$
\det \frac{\partial(x,v,h)}{\partial(x,y,w)} = -x^3.
$$

The other determinant is x^{-3} times

$$
\det \begin{pmatrix} -2A & 2v+1-3v^2h & -v^3 \\ -B & 4-6vh & -3v^2 \\ h & 0 & 1 \end{pmatrix} = 2.
$$

Indeed expansion gives

$$
(4-6vh)(-2v^2-2v+3v^3h) + (2v+1-3v^2h)(4v+2-6v^2h) = 2.
$$

Their product is -2. The original Jacobian determinant is a polynomial; agreement on $x \neq 0$ proves

$$
\det DF = -2 \quad \text{everywhere.} \tag{374}
$$

For fixed (a,b,c) define

$$
P_{a,b,c}(r) = cr^3 - 2r^2 + br - 2a, \qquad \alpha = \tfrac{1}{2} P_r(r).
$$

For each simple root, the entire inverse point is

$$
H(r,\alpha,c) = \left(\alpha^{-1},\ r-\alpha,\ 5\alpha^2 - 3r\alpha - c\alpha^3\right). \tag{375}
$$

Here and below $\alpha \neq 0$ is part of the domain. To prove both directions, start with $x \neq 0$, set $\alpha = 1/x$, $r = y + 1/x$, and solve $F_3 = c$. This gives exactly (375). Before either other target equation is imposed, direct substitution gives

$$
F \circ H = \left(r^2 + r\alpha - cr^3,\ 4r + 2\alpha - 3cr^2,\ c\right). \tag{376}
$$

The second equation equals b precisely when $2\alpha = 3cr^2 - 4r + b = P_r(r)$. After imposing that identity, $2(F_1 - a) = P(r)$. Thus the formulas give every point with $x \neq 0$, and recovering $r = y + 1/x$ proves injectivity of the parametrization. A multiple root has $\alpha = 0$ and gives no finite point in this chart.

The excluded hyperplane is handled without division:

$$F(0,y,w) = (w+4y^2, y, 0).$$

It contributes exactly $(0, b, a-4b^2)$ when $c=0$, and contributes no point when $c \neq 0$. These statements exhaust the whole fibre.

For completeness the multiplicities are explicit. If $c \neq 0$, put

$$\mathscr{D} = 4(b^2 - cb^3 + 18abc - 16a - 27a^2c^2).$$

Substitution of $(c, -2, b, -2a)$ in the cubic discriminant gives this formula. If $\mathscr{D} \neq 0$, there are three simple complex roots and three finite complex inverse points. If $\mathscr{D} = 0$ and there is a double root and a simple root, the entire fibre is the single point corresponding to the simple root. The remaining case is a triple root. Comparing coefficients of $c(r-r_*)^3$ gives exactly

$$r_* = \frac{2}{3c}, \qquad b = \frac{4}{3c}, \qquad a = \frac{4}{27c^2}. \tag{377}$$

All three root multiplicities then belong to the excluded $\alpha = 0$ locus, and the original finite fibre is empty. These are conclusions about this displayed polynomial and its inverse formulas, not about any other coordinate model.

When $c = 0$, the degree is exactly two, since its leading coefficient remains -2. Its discriminant is $\delta = b^2 - 16a$. If $\delta \neq 0$, its two simple roots and the separate $x = 0$ point give three complex points. If $\delta = 0$, its double root supplies no chart point and the separate point is the whole fibre. For real parameters, a real cubic with $\mathscr{D} > 0$ has three real simple roots, while $\mathscr{D} < 0$ has one real simple root and a nonreal conjugate pair. This follows, for example, from $c^4 \prod_{i<j}(r_i - r_j)^2$: it is positive for three distinct real roots and negative for one real root and a conjugate pair. In the double-root cubic case both the double and simple root are real, because a nonreal multiple root would require its conjugate with the same multiplicity. The real quadratic case has two real chart points when $\delta > 0$ and none when $\delta < 0$, always retaining the additional real boundary point.

22.2 The coefficient heat flow and every time constant

Let τ denote forward heat time. The complete target flow is

$$\Phi_\tau(a,b,c) = (a + 2\tau,\ b + 6c\tau,\ c). \tag{378}$$

It is a group of affine maps with determinant one:

$$D\Phi_\tau = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 6\tau \\ 0 & 0 & 1 \end{pmatrix}, \qquad \Phi_{\tau+s} = \Phi_\tau \circ \Phi_s.$$

For $P(\tau, r) = P_{\Phi_\tau(a,b,c)}(r)$,

$$P_\tau = 6cr - 4 = P_{rr}. \tag{379}$$

On a simple root the implicit function theorem applies because $P_r \neq 0$; differentiating $P(\tau, r(\tau)) = 0$ gives $r' = -P_{rr}/P_r$. No derivative formula is assigned to a multiple root.

Two further identities determine this coefficient motion exactly:

$$\frac{d}{d\tau}(b - 3ca) = 0,$$
$$\frac{d}{d\tau}\mathscr{D} = (2\partial_a + 6c\partial_b)\mathscr{D} = -8(3bc-4)^2. \tag{380}$$

Indeed differentiating the displayed polynomial $\mathscr{D}$ gives $4(48bc - 18c^2b^2 - 32)$, which equals the last expression. Writing $A_0 = 3bc - 4$ at $\tau = 0$ and integrating the exact square gives

$$\mathscr{D}(\tau) = \mathscr{D}(0) - 8A_0^2\tau - 144A_0c^2\tau^2 - 864c^4\tau^3.$$

For real $c \neq 0$ this is strictly decreasing: the integral of the nonnegative square over every interval of positive length is positive, since $A_0 + 18c^2\tau$ cannot vanish throughout such an interval. Its cubic leading term gives opposite infinite limits at the two ends of the real line. Thus each such real heat orbit meets the discriminant locus exactly once. For $c = 0$, the derivative is instead the constant -128, with the degree-two interpretation above still required.

For $c \neq 0$, the exact coordinate substitution $r = z + 2/(3c)$ gives

$$P = cz^3 + \left(b - \frac{4}{3c}\right) z + K, \qquad K = \frac{2b}{3c} - \frac{16}{27c^2} - 2a.$$

The original coefficient c is retained, and K is constant under (378). If the orbit passes through (377) at τ_c, its polynomial is exactly

$$P = cz\big(z^2 + 6(\tau - \tau_c)\big).$$

For $\tau \neq \tau_c$, the central root has $\alpha = 3c(\tau - \tau_c)$; either outer root has $\alpha = -6c(\tau - \tau_c)$. These explicit denominators record the escape of every chart inverse point at the triple-root time. At that time the finite fibre is empty, as proved above.

Newman time is the specific reversal $t = -\tau$. Consequently

$$\mathcal{P}(t, r) = P_{a_0-2t, b_0-6ct, c}(r), \qquad \mathcal{P}_t = -\mathcal{P}_{rr}.$$

For $b_0 = c = 0$,

$$\mathcal{P}(t, r) = -2\big(r^2 - 2(t - a_0/2)\big).$$

Thus $a_0 = -1/4$ gives the model collision time $-1/8$ in this Newman convention, while its forward heat time is $+1/8$. Neither number specifies the Newman constant of the zeta family: the polynomial here has its own explicitly fixed initial datum.

22.3 A global polynomial incompressible lift

Write $J = DF$ and define all three inverse columns, with cyclic indices,

$$W_i = J^{-1}e_i = -\tfrac{1}{2}\nabla F_{i+1} \times \nabla F_{i+2}.$$

The scalar triple products and (374) prove $JW_i = e_i$. Commutation of mixed derivatives proves $\operatorname{div} W_i = 0$. Each column is polynomial. Moreover $W_iF_j = \delta_{ij}$, and $[W_i, W_j]F_k = 0$ for every k; invertibility of J proves $[W_i, W_j] = 0$.

The exact heat lift is

$$U = 2W_1 + 6F_3W_2, \qquad JU = (2, 6F_3, 0)^{\mathsf{T}}. \tag{381}$$

It is a polynomial vector field on the whole original $\mathbb{R}^3$. Its ordinary Euclidean divergence is zero, since

$$\operatorname{div} U = 2\operatorname{div} W_1 + 6F_3 \operatorname{div} W_2 + 6W_2F_3 = 0.$$

A global vector potential is also explicit:

$$U = \operatorname{curl}\left(-F_2\nabla F_3 - \tfrac{3}{2}F_3^2\nabla F_1\right). \tag{382}$$

Expanding the curl gives $-\nabla F_2 \times \nabla F_3 - 3F_3\nabla F_3 \times \nabla F_1$, which is exactly (381).

The complete component formula can be kept polynomial without an inverse denominator. With the same original $h = 2 - 3xy - x^2w$, put $Q = 3h(1 + xy) - 2$. Then

$$\begin{aligned}
U_x &= x^3(Q^2 - 3h),\\
U_y &= x(Q^2 - Q - 3h),\\
U_w &= 3Q + (3h + 3xy - 7)Q^2 + (21 - 9xy)h - 9h^2.
\end{aligned} \tag{383}$$

These identities follow either by expanding (381), or by substituting the inverse chart into the differential calculation below and using $F_3 = xh$. They hold on $x \neq 0$ and then everywhere by polynomial identity. At $x = 0$ they give $U = (0, 0, 2)$, so the separate inverse point has exactly the required boundary evolution.

Let φ_τ be the maximal local flow of U. Smooth polynomial coefficients give local existence and uniqueness. Differentiating $F(\varphi_\tau(q))$ using (381), and using uniqueness for the affine target ODE, proves

$$F \circ \varphi_\tau = \Phi_\tau \circ F \tag{384}$$

on every common interval of definition. This identity includes the time variable, not only a matching of fibres. Differentiating it in the original initial point gives

$$D\varphi_\tau(q) = J(\varphi_\tau(q))^{-1} D\Phi_\tau(F(q))J(q), \qquad \det D\varphi_\tau = 1. \tag{385}$$

Both determinants of J are -2, so their ratio and $\det D\Phi_\tau = 1$ give the last equality without a limit argument. For a retained initial covector ξ_0, its transported covector is

$$(D\varphi_\tau(q))^{-\mathsf{T}}\xi_0 = J(\varphi_\tau(q))^{\mathsf{T}}(D\Phi_\tau)^{-\mathsf{T}}J(q)^{-\mathsf{T}}\xi_0.$$

This is the full inverse-transpose material dictionary in these coordinates.

There is an exact scalar operator identity as well. Define

$$\Pi(q, r) = F_3(q)r^3 - 2r^2 + F_2(q)r - 2F_1(q).$$

Using $UF_1 = 2$, $UF_2 = 6F_3$, $UF_3 = 0$ gives, on the entire product of the original space with the root coordinate,

$$U\Pi = 6F_3r - 4 = \partial_r^2\Pi. \tag{386}$$

Thus $\Pi(\varphi_\tau(q), r)$ obeys the exact forward heat equation in the displayed auxiliary variable r. This variable remains the inverse-root coordinate; it is not relabelled as a Euclidean spatial diffusion coordinate of the three-dimensional fluid.

22.4 The full inverse-chart dynamics and its preserved area form

In (375), (r, α, c) are actual coordinates on $x \neq 0$, with inverse $(y + 1/x, 1/x, F_3)$. Direct determinants of (375) and (376) give

$$\det DH = -\alpha, \qquad \det D(F \circ H) = 2\alpha.$$

They multiply consistently with $\det J = -2$. Put $q_r = 3cr - 2$, where the subscript is a name for this polynomial, not a derivative. The complete lifted ODE in this chart is

$$r' = -\frac{q_r}{\alpha}, \qquad \alpha' = 3c - \frac{q_r^2}{\alpha}, \qquad c' = 0. \tag{387}$$

Indeed $P_{rr} = 2q_r$, $P_r = 2\alpha$ give the first equation. Differentiating $\alpha = (3cr^2 - 4r + b)/2$ along $b' = 6c$ gives $\alpha' = q_r r' + 3c$, which is the second equation. Differentiating each component of H then yields exactly $U \circ H$. For example $x' = -\alpha'/\alpha^2 = -3cx^2 + q_r^2x^3$ and $y' = r' - \alpha' = (q_r^2 - q_r)x - 3c$. For the third component the unaltered expression is

$$w' = -3\alpha r' + (10\alpha - 3r - 3c\alpha^2)\alpha',$$

which gives (383) after $c = xh$, $q_r = 3h(1 + xy) - 2$ are substituted.

The heat invariant $I = b - 3ca$ becomes the specific polynomial

$$I(r, \alpha, c) = 4r + 2\alpha - 6cr^2 - 3cr\alpha + 3c^2r^3.$$

Its two derivatives are
$$I_\alpha = -q_r, \qquad I_r = q_r^2 - 3c\alpha.$$
Consequently, for each fixed real c, the nondegenerate two-form
$$\omega_c = \alpha\, dr \wedge d\alpha \quad (\alpha \neq 0)$$
satisfies $\iota_{(r',\alpha')}\omega_c = dI$. Indeed its contraction is $\alpha r'\, d\alpha - \alpha\alpha'\, dr = I_\alpha\, d\alpha + I_r\, dr$. This states the exact Hamiltonian convention and proves the claimed Hamiltonian correspondence.

The preserved signed volume can also be checked directly:
$$\partial_r(\alpha r') + \partial_\alpha(\alpha\alpha') = \partial_r(-q_r) + \partial_\alpha(3c\alpha - q_r^2) = 0.$$
The Euclidean volume of the original source is $|\alpha|\, dr\, d\alpha\, dc$ in this chart. On either component of $\alpha \neq 0$, the absolute value has a fixed sign relative to α, so this same calculation proves its preservation. The density tending to zero at $\alpha = 0$ describes a boundary of this inverse chart; $x = 1/\alpha$ shows that boundary is not a finite original coordinate point.

22.5 The exact positive-viscosity fluid equation and escape

On the target $\mathbb{R}^3_{a,b,c}$ with Euclidean metric, standard orientation, and any retained viscosity $\nu > 0$, set
$$v(a,b,c) = (2, 6c, 0), \qquad p = 0, \qquad f = 0.$$
Every term of its stationary Navier–Stokes equation can be computed: $\operatorname{div} v = 0$, $(v \cdot \nabla)v = (2\partial_a + 6c\partial_b)v = 0$, and $\Delta v = 0$. Its oriented curl is $(-6, 0, 0)$. Its material flow is exactly (378).

On the original source set $g = J^{\mathsf{T}} J$, using the standard source orientation. This positive metric has $\det g = 4$ and volume $2\, dx\, dy\, dw$. In a local F chart it is precisely the Euclidean target metric. The Levi–Civita connection pulls back under this local isometry: this follows from its metric compatibility and zero torsion, which characterize that connection uniquely. The nonpositive vector diffusion operator is
$$\mathcal{L}_g Y = \left[-(d\delta_g + \delta_g d) Y_g^\flat\right]_g^\sharp.$$
The Hodge Laplacian commutes with local metric pullback, including an orientation-reversing pullback, because the two orientation signs in the codifferential cancel. In Euclidean coordinates this operator is the componentwise Laplacian. Therefore
$$\operatorname{div}_g U = 0, \qquad \nabla^g_U U = 0, \qquad \mathcal{L}_g U = 0.$$
These prove the complete intrinsic stationary unforced positive-viscosity Navier–Stokes equation with constant pressure. Its vorticity retains the orientation sign:
$$\operatorname{curl}_g U = 6W_1.$$
In fact $\det J = -2$ makes $*_g F^*\eta = -F^*(*_{\text{target}}\eta)$. Applying this to $d(v^\flat)$ negates the pullback of $\operatorname{curl} v = -6e_1$, giving the displayed result.

There are explicit real finite-time escaping material trajectories. On the initial fibre $(-1/4, 0, 0)$, for $\tau < 1/8$ take
$$\gamma_\pm(\tau) = \left(\pm(1-8\tau)^{-1/2}, \mp\tfrac{3}{2}(1-8\tau)^{1/2}, \tfrac{13}{2}(1-8\tau)\right).$$
Substitution in the full map gives $F(\gamma_\pm(\tau)) = (-1/4 + 2\tau, 0, 0)$. Differentiation and J^{-1} prove $\gamma_\pm' = U(\gamma_\pm)$. Their first coordinate escapes to infinity at $\tau = 1/8$; their g-speed is $|v| = 2$ throughout. Hence their length from $\tau = 0$ to the endpoint is $1/4$. Their times approaching $1/8$ form a metric Cauchy sequence because the remaining curve length tends to zero. It has no point limit: a smooth positive Riemannian metric induces the ordinary coordinate topology locally, while x diverges. Thus the metric is incomplete. The separate inverse point $(0, 0, -1/4 + 2\tau)$ remains a finite trajectory throughout. The coefficient heat collision and its escaping branches have all been retained by these formulas.

22.6 Its exact force in the original Euclidean metric

The same polynomial U, now using the original Euclidean metric and its component Laplacian, has stationary Navier–Stokes triples precisely of the form

$$p = q, \qquad f = (U \cdot \nabla)U - \nu \Delta U + \nabla q.$$

Substitution proves sufficiency; rearrangement of the momentum equation proves necessity. Thus every pressure adjustment has been retained.

The force has the following exact curl on the original coordinate axis:

$$\operatorname{curl} f(0,0,w) = (0,0,108\nu w). \tag{388}$$

Here is a direct derivative certificate from the full component formula. Its transverse polynomial terms of degree at most three are

$$U_x = 10x^3 + R_x, \quad U_y = 6x - 12x^2 y - 18wx^3 + R_y, \quad U_w = 2 - 54xy - 18wx^2 + R_w,$$

where the remainders are defined by these exact differences and

$$R_x, R_y \in (x,y)^5 \mathbb{R}[x,y,w], \qquad R_w \in (x,y)^4 \mathbb{R}[x,y,w].$$

These ideal memberships follow by expanding (383); thus no unspecified coefficient enters a derivative of the required order. On the axis,

$$U = 2e_3, \qquad DU = \begin{pmatrix} 0 & 0 & 0 \\ 6 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} =: N.$$

The derivative of the convective acceleration there is $N^2 + 2\partial_w DU = 0$, so its curl vanishes. For the diffusive part,

$$\Delta \operatorname{curl} U(0,0,w) = (0,0,-108w):$$

the third component is $\partial_x \Delta U_y - \partial_y \Delta U_x = -108w$, from $-18wx^3$, and the first two components vanish from the displayed terms and remainder ideals. Commuting derivatives and using $\operatorname{curl} \nabla q = 0$ proves (388), also distributionally wherever a pressure is interpreted as a distribution.

Thus this exact same Euclidean velocity cannot have a rapidly decaying smooth force for any pressure choice: its force curl is unbounded on the axis for $\nu > 0$. The source field is also a nonzero polynomial, hence has infinite Euclidean L^2 norm. To see the latter fact directly, a nonzero leading homogeneous component of a polynomial is bounded away from zero on some open patch of the unit sphere. The full polynomial then has a positive multiple of its leading radial power on the corresponding cone for all sufficiently large radii. Radial integration diverges. A nonzero constant component already gives the same conclusion at degree zero. Here the field is nonconstant, as $U_y = 6x$ to first order at the origin proves, and a polynomial periodic in every unit coordinate direction must be constant by the one-variable polynomial difference identity. It therefore does not define the required periodic velocity either.

These calculations identify both fluid realizations completely: the target shear and its intrinsic lift are exact unforced viscous solutions, while the original Euclidean lift requires the precisely computed nondecaying force. The finite-time material escape belongs to the incomplete metric realization. No classical Euclidean Navier–Stokes counterexample or off-critical zeta zero is asserted.

22.7 Standalone exact replay

The script `scripts/check_inverse_fibre_heat.py` contains the original polynomials, inverse chart, determinants, source boundary, multiplicity formulas, heat constants, full polynomial fluid lift, Hamiltonian identities, and Euclidean force-curl calculation. Its standalone receipt is `checks/inverse_fibre_heat_checks.json`. The analytic local-flow, metric, and fibre-classification arguments are the proofs written above. The script makes no network request, runs no Lean process, and needs no mutable sibling repository.

23 The full spatial material generator of the inverse-fibre flow

Keep the original real coordinates $q = (x, y, w)$, polynomial F, Jacobian $J = DF$, and vector field $U = 2W_1 + 6F_3W_2$ from Section 22. That section proves $U\Pi = \Pi_{rr}$ on the product with the inverse-root variable r. Here we calculate the actual three-dimensional spatial diffusion, covectors, vorticity, pressure, function spaces and time changes of this same flow.

23.1 The maximal flow and its complete differential

Use $\xi = (\xi_1, \xi_2, \xi_3)$ for the original initial point and $X_\tau(\xi)$ for the maximal solution of

$$\partial_\tau X_\tau(\xi) = U(X_\tau(\xi)), \qquad X_0(\xi) = \xi.$$

Local existence at every finite ξ follows directly from the integral equation: on a closed ball, U and DU are bounded, so for sufficiently short times $Y \mapsto \xi + \int_0^\tau U(Y(s))\,ds$ preserves the ball and is a contraction in the supremum norm. Its fixed point solves the ODE; the same Lipschitz estimate gives uniqueness. Differentiating the integral equation gives the variational equation, and successive differentiation gives smooth dependence on time and initial point. Continuation over overlapping such neighborhoods defines the maximal flow.

Let $\mathcal{D}_\tau$ be the open set of initial points whose trajectory exists throughout the interval joining 0 to τ. Openness follows by covering that finite trajectory segment with the local neighborhoods just constructed. Uniqueness gives the inverse $X_{-\tau}$, so $X_\tau : \mathcal{D}_\tau \to \mathcal{D}_{-\tau}$ is a diffeomorphism. We do not replace either domain by all of $\mathbb{R}^3$. Define

$$A_\tau = D_\xi X_\tau, \quad C_\tau = A_\tau^{-1}, \quad B_\tau = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 6\tau \\ 0 & 0 & 1 \end{pmatrix}.$$

Differentiating $F \circ X_\tau = \Phi_\tau \circ F$ gives the exact formulas

$$A_\tau = J(X_\tau)^{-1} B_\tau J(\xi), \qquad C_\tau = J(\xi)^{-1} B_\tau^{-1} J(X_\tau), \qquad \det A_\tau = 1. \tag{389}$$

Both determinants of J are the retained value -2, and $\det B_\tau = 1$, which proves the determinant assertion. Direct differentiation of the ODE also gives $\partial_\tau A_\tau = DU(X_\tau)A_\tau$.

The material metric and its inverse are therefore

$$g_\tau = A_\tau^\top A_\tau, \quad K_\tau = g_\tau^{-1} = C_\tau C_\tau^\top, \quad \mathrm{dvol}_{g_\tau} = d\xi_1\, d\xi_2\, d\xi_3. \tag{390}$$

In particular the full tensor is

$$K_\tau = J(\xi)^{-1} B_\tau^{-1} J(X_\tau) J(X_\tau)^\top B_\tau^{-\top} J(\xi)^{-\top}.$$

Every matrix retains its evaluation point; a constant metric or a scalar multiple of the identity has not been substituted for this tensor.

23.2 Scalar diffusion, function spaces and energy

For a scalar $f(\tau, q)$ put $\widehat{f} = f \circ X_\tau$. The time and spatial chain rules are

$$\partial_\tau \widehat{f} = ((\partial_\tau + U \cdot \nabla_q) f) \circ X_\tau, \qquad (\nabla_q f) \circ X_\tau = C_\tau^\top \nabla_\xi \widehat{f}. \tag{391}$$

With repeated indices summed from 1 to 3 and $\epsilon_{123} = 1$, the cofactor formula in this index convention gives

$$(C_\tau)_{ik} = \tfrac{1}{2} \epsilon_{iab} \epsilon_{k\ell m} (A_\tau)_{\ell a} (A_\tau)_{mb}, \qquad \partial_{\xi_i} (C_\tau)_{ik} = 0.$$

The first equality uses $\det A_\tau = 1$. For the second, differentiate the first equality. The derivative of the first factor contains $\partial_i \partial_a X^\ell$, whose symmetry in i, a makes its contraction with ϵ_{iab} zero. The derivative of the second factor similarly vanishes by symmetry in i, b. Consequently the complete spatial diffusion operator is

$$(\Delta_q f) \circ X_\tau = \mathcal{L}_\tau \widehat{f}, \qquad \mathcal{L}_\tau h = \partial_{\xi_i}((K_\tau)_{ij} \partial_{\xi_j} h). \tag{392}$$

Indeed after composition $\partial_{q_k} = C_{ik} \partial_{\xi_i}$; applying it twice and using the cofactor divergence formula proves (392). The first-order coefficient $\partial_{\xi_i}(K_\tau)_{ij}$ is part of this formula. For every retained $\nu > 0$ we have equality of full operators,

$$((\partial_\tau + U \cdot \nabla_q - \nu \Delta_q) f) \circ X_\tau = (\partial_\tau - \nu \mathcal{L}_\tau) \widehat{f}.$$

For an open $D \Subset \mathcal{D}_\tau$, change of variables gives

$$\begin{aligned} \int_{X_\tau(D)} |f|^2 \, dq &= \int_D |\widehat{f}|^2 \, d\xi, \\ \int_{X_\tau(D)} |\nabla_q f|^2 \, dq &= \int_D (\nabla_\xi \overline{\widehat{f}})^\mathsf{T} K_\tau \nabla_\xi \widehat{f} \, d\xi. \end{aligned} \tag{393}$$

For compact smooth h, integration by parts also gives $-\int_D \overline{h} \, \mathcal{L}_\tau h = \int_D (\nabla \overline{h})^\mathsf{T} K_\tau \nabla h$. The continuous positive-definite matrix K_τ on $\overline{D}$ has a strictly positive minimum eigenvalue and a finite maximum eigenvalue. These bounds depend on D, τ and are not asserted uniform at escape.

Define $H_0^1(D)$ by completion of $C_c^\infty(D)$ in the ordinary H^1 norm, and use the analogous definition on $X_\tau(D)$. The two eigenvalue bounds and (393) show that composition by X_τ extends to a bounded invertible map between these spaces, an isometry for the displayed material Dirichlet norm, and a unitary map of their L^2 spaces. The exact Dirichlet operator domains are carried bijectively as follows:

$$\{f \in H_0^1(X_\tau(D)) : \Delta_q f \in L^2\} \xrightarrow{f \mapsto f \circ X_\tau} \{h \in H_0^1(D) : \mathcal{L}_\tau h \in L^2\}.$$

Both derivative conditions are distributional. To prove the assertion, test a distributional derivative against a compact smooth function, use (393) in bilinear form, and change variables. The inverse change of variables proves the converse. Completion of compact smooth functions supplies the specified boundary conditions.

For a real phase φ, a material covector $\ell = d_\xi \widehat{\varphi}$ corresponds to the physical covector $C_\tau^\mathsf{T} \ell$. Its exact principal diffusive quadratic form is

$$\nu \, \ell^\mathsf{T} K_\tau \ell = \nu \, |C_\tau^\mathsf{T} \ell|^2. \tag{394}$$

Thus stretching of this actual covector and its diffusion cost are related by equality, including the original viscosity.

23.3 Vectors, covectors, pressure and vorticity

For a physical vector V use material vector coefficients $v = C_\tau(V \circ X_\tau)$. Their metric dual is $v^{\flat_{g_\tau}} = g_\tau v = A_\tau^\mathsf{T}(V \circ X_\tau)$. The exact divergence and connection formulas are

$$(\operatorname{div}_q V) \circ X_\tau = \partial_{\xi_i} v^i, \qquad (\Gamma_\tau)^k_{ij} = (C_\tau)_{k\ell} \partial_{\xi_i} \partial_{\xi_j} X^\ell_\tau.$$

The first is the proved cofactor product rule. To prove the connection formula, differentiate $A_\tau v$ in a material direction and multiply by C_τ. Its lower indices are symmetric; differentiating $g = A^\mathsf{T} A$ proves metric compatibility, identifying precisely the Levi–Civita connection. Consequently, if $z = C_\tau(Z \circ X_\tau)$,

$$C_\tau(((Z \cdot \nabla_q) V) \circ X_\tau) = \nabla^{g_\tau}_z v, \qquad C_\tau((\nabla_q p) \circ X_\tau) = K_\tau \nabla_\xi \widehat{p}.$$

The full vector diffusion is

$$\begin{aligned}\mathscr{L}_\tau v &:= C_\tau \mathcal{L}_\tau (A_\tau v) = C_\tau ((\Delta_q V) \circ X_\tau) \\ &= \mathcal{L}_\tau v + 2(K_\tau)_{ij} C_\tau (\partial_i A_\tau)\partial_j v + C_\tau (\mathcal{L}_\tau A_\tau) v. \end{aligned} \tag{395}$$

The last equality is the full product rule, using the symmetry of K; $\mathcal{L}_\tau$ acts componentwise on each displayed matrix or vector. This also equals the nonpositive Hodge vector Laplacian of g_τ: the orientation-preserving local isometry X_τ transports d and the metric codifferential to their Euclidean versions, whose negative Hodge Laplacian is the Cartesian component Laplacian.

Put $E_\tau = C_\tau \partial_\tau A_\tau$ and $u = C_\tau (U \circ X_\tau)$. Differentiating $C_\tau A_\tau = I$ gives

$$\partial_\tau v + E_\tau v = C_\tau (((\partial_\tau + U \cdot \nabla_q)V) \circ X_\tau).$$

Therefore the entire Navier–Stokes residual of any smooth V, p, f is carried to

$$\partial_\tau v + E_\tau v + \nabla^{g_\tau}_{v-u} v + K_\tau \nabla_\xi \widehat{p} - \nu \mathscr{L}_\tau v - \mathfrak{f}, \qquad \mathfrak{f} = C_\tau (f \circ X_\tau). \tag{396}$$

This expression is C_τ times the composed original residual, by adding the proved time and connection identities. Its vanishing is equivalent to vanishing of that residual because C_τ is invertible.

For the actual stationary U we have exactly $u(\tau, \xi) = U(\xi)$: $J(\xi)U(\xi) = (2, 6F_3(\xi), 0)$, $F_3(X_\tau) = F_3(\xi)$, and B_τ fixes this vector; (389) therefore gives $A_\tau U(\xi) = U(X_\tau)$. The material vector coefficients are constant in time, while g_τ, E_τ and both diffusion operators vary. Physical velocity and vorticity norms keep these changes:

$$|U(X_\tau)|^2 = u^\mathsf{T} g_\tau u, \qquad |\omega(X_\tau)|^2 = W^\mathsf{T} g_\tau W, \quad W = C_\tau (\omega \circ X_\tau), \quad \omega = \operatorname{curl}_q U.$$

Choose any smooth pressure p and the actual full force $f = (U \cdot \nabla)U - \nu \Delta U + \nabla p$ already computed in Section 22. Put $\beta = X_\tau^*(U^\flat) = g_\tau u$ and $\Omega = d_\xi \beta$. In the original Euclidean coordinates,

$$(\partial_\tau + \mathcal{L}_U)U^\flat = \nu \Delta_q U^\flat + f^\flat - d_q(p - |U|^2/2).$$

Here $\mathcal{L}_U$ means Lie derivative, distinct from the scalar operator $\mathcal{L}_\tau$. Its extra covector term is $(DU)^\mathsf{T} U = \nabla_q(|U|^2/2)$, proving the displayed identity. Differentiating $A_\tau^\mathsf{T}(U \circ X_\tau)$ shows that pullback of this Lie derivative is the ordinary material time derivative. Hence, with $\Delta_H = d\delta + \delta d$ nonnegative by convention,

$$\begin{aligned}\partial_\tau \beta &= -\nu \Delta_{H,g_\tau} \beta + g_\tau \mathfrak{f} - d_\xi (\widehat{p} - u^\mathsf{T} g_\tau u/2), \\ \partial_\tau \Omega &= -\nu \Delta_{H,g_\tau} \Omega + d_\xi (g_\tau \mathfrak{f}). \end{aligned} \tag{397}$$

The second line follows by applying d_ξ: both $d^2 = 0$ and $d\Delta_H = \Delta_H d$ follow from the displayed definition of Δ_H. This commutation is at each fixed metric time and does not differentiate the time-dependent metric codifferential.

Equivalently, the contravariant vorticity obeys

$$\partial_\tau W = \nu \mathscr{L}_\tau W + \operatorname{curl}_{g_\tau} \mathfrak{f}. \tag{398}$$

Indeed taking Cartesian curl and using $\operatorname{div} U = 0$ gives $(\partial_\tau + U \cdot \nabla)\omega = DU\,\omega + \nu \Delta \omega + \operatorname{curl} f$. The time product rule for C_τ cancels precisely the stretching term, since $E_\tau W = C_\tau DU(X_\tau) A_\tau W$. The orientation-preserving isometry transports the force curl as $C_\tau((\operatorname{curl} f) \circ X_\tau) = \operatorname{curl}_{g_\tau} \mathfrak{f}$. In material coordinates this last vector is $\epsilon_{ijk} \partial_{\xi_j} ((g_\tau)_{k\ell} \mathfrak{f}^\ell)$; its volume denominator is one because $\det g_\tau = 1$. Pressure disappears from this curl identity by $d^2 = 0$, with the full pressure still present in (396).

23.4 Accumulated time keeps its acceleration and diffusion costs

Define an explicit time change, with its full inverse,

$$\rho(s) = \frac{1-e^{-8s}}{8}, \quad \sigma(s) = \rho'(s) = e^{-8s}, \quad s \geq 0, \qquad s = -\tfrac{1}{8}\log(1-8\tau).$$

It maps $[0,\infty)$ bijectively to $[0,1/8)$. The exact spatial map is $Y_s = X_{\rho(s)} : \mathcal{D}_{\rho(s)} \to \mathcal{D}_{-\rho(s)}$, and it satisfies $\partial_s Y_s = \sigma(s)U(Y_s)$. For a scalar material function h, on $\{(s,\xi) : s \geq 0,\ \xi \in \mathcal{D}_{\rho(s)}\}$ the complete chain rule is

$$[(\partial_\tau - \nu\mathcal{L}_\tau)h]_{\tau=\rho(s)} = \sigma(s)^{-1}(\partial_s - \nu\sigma(s)\mathcal{L}_{\rho(s)})h(\rho(s),\xi).$$

Thus this time change transports the diffusion coefficient as displayed.

For the actual stationary fields set $\widetilde{U}(s,q) = \sigma(s)U(q)$ and $\widetilde{p}(s,q) = \sigma(s)^2 p(q)$. The velocity remains divergence-free. At the same viscosity $\nu > 0$ its exact force is

$$\widetilde{f} = \sigma' U + \sigma^2 f + \nu(\sigma^2 - \sigma)\Delta_q U. \tag{399}$$

The time derivative gives $\sigma' U$, convection and pressure give their old values multiplied by σ^2, and diffusion gives $-\nu\sigma\Delta U$; inserting the original complete f proves the formula. The previously computed axis values $\operatorname{curl} U = (0,0,6)$, $\Delta \operatorname{curl} U = (0,0,-108w)$ and $\operatorname{curl} f = (0,0,108\nu w)$ give the exact resulting curl

$$\operatorname{curl} \widetilde{f}(s,0,0,w) = (0,0,108\nu e^{-8s} w - 48e^{-8s}).$$

It remains unbounded in w at every finite s for every pressure choice and $\nu > 0$. The corresponding actual trajectory is

$$Y_s(1,-3/2,13/2) = (e^{4s}, -\tfrac{3}{2}e^{-4s}, \tfrac{13}{2}e^{-8s}).$$

It escapes as $s \to \infty$ in this coordinate. These maps preserve the original escape at $\tau = 1/8$, its spatial location at infinity, the full spatial diffusion and the exact force. They do not establish an admissible finite-energy Euclidean Navier–Stokes singularity.

The exact replay `scripts/check_material_generator.py` checks the initial scalar and vector generator derivatives on arbitrary formal test functions, including all first-order and matrix terms, and the full vorticity and retained-time force identities. Its thirteen checks are recorded in `checks/material_generator_checks.json`. The finite-time operator and domain proofs are the written arguments above; the replay does not replace them.

23.5 The full material derivative at the retained escaping orbit

All coordinates and constants below belong to the original polynomial

$$\begin{aligned}
F_1 &= (1+xy)^3 w + y^2(1+xy)(4+3xy),\\
F_2 &= y + 3x(1+xy)^2 w + 3xy^2(4+3xy),\\
F_3 &= 2x - 3x^2 y - x^3 w.
\end{aligned}$$

Here w is the identity renaming of the original third spatial coordinate t. Evolution time is τ. Put $J = DF$, $W_i = J^{-1}e_i$, and

$$U = 2W_1 + 6F_3 W_2, \qquad q_0 = (1,-3/2,13/2).$$

The original determinant is $\det J = -2$; hence these inverse columns are defined everywhere. The identities

$$W_i = -\frac{1}{2}\nabla F_{i+1} \times \nabla F_{i+2}, \qquad JU = (2, 6F_3, 0)^{\mathsf{T}}$$

use cyclic indices and the original negative determinant.

23.6 Original time and the positive shorthand

On the full interval $-\infty < \tau < 1/8$, introduce

$$z = \sqrt{1-8\tau} > 0.$$

This is a bijection onto $z > 0$, with the exact inverse and derivative

$$\tau = \frac{1-z^2}{8}, \qquad \frac{dz}{d\tau} = -\frac{4}{z}, \qquad \frac{d}{d\tau} = -\frac{4}{z}\frac{d}{dz}.$$

No evolution equation or time coefficient is changed. The initial time $\tau = 0$ is $z = 1$, and the endpoint $\tau \uparrow 1/8$ is $z \downarrow 0$.

The trajectory and target material matrix are

$$\gamma(\tau) = \left(z^{-1}, -\frac{3}{2}z, \frac{13}{2}z^2\right),$$
$$B_\tau = I + 6\tau E_{23} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{3}{4}(1-z^2) \\ 0 & 0 & 1 \end{pmatrix}.$$

Here E_{23} has its only nonzero entry 1 in row 2, column 3. Direct substitution gives

$$F(\gamma(\tau)) = (-1/4 + 2\tau, 0, 0).$$

Differentiation and $JU = (2, 6F_3, 0)^\mathsf{T}$ give $\gamma' = U(\gamma)$, with every factor of $2, 6, 8$ retained.

23.7 The complete material derivative and its inverse

Differentiating the original polynomials and substituting the retained points gives

$$J_0 := J(q_0) = \begin{pmatrix} -\frac{9}{16} & -\frac{3}{8} & -\frac{1}{8} \\ \frac{3}{8} & \frac{25}{4} & \frac{3}{4} \\ -\frac{17}{2} & -3 & -1 \end{pmatrix},$$

$$J_\gamma = \begin{pmatrix} -\frac{9}{16}z^3 & -\frac{3}{8}z & -\frac{1}{8} \\ \frac{3}{8}z^2 & \frac{25}{4} & \frac{3}{4z} \\ -\frac{17}{2} & -\frac{3}{z^2} & -\frac{1}{z^3} \end{pmatrix}.$$

Both determinants are exactly -2. The full material derivative, with no coordinate restriction on its input vector, is

$$A_\tau = J_\gamma^{-1} B_\tau J_0.$$

Its entire coefficient matrix is

$$A_\tau = \begin{pmatrix} \frac{17z^3-9}{8z^3} & \frac{3z^3-3}{4z^3} & \frac{z^3-1}{4z^3} \\ \frac{51z^3-24z-27}{16z} & \frac{9z^3+8z-9}{8z} & \frac{3z^3-3}{8z} \\ -\frac{153}{8}z^3 + \frac{9}{2}z + \frac{117}{8} & -\frac{27}{4}z^3 - 3z + \frac{39}{4} & \frac{13}{4} - \frac{9}{4}z^3 \end{pmatrix}.$$

Its full inverse is

$$C_\tau := A_\tau^{-1} = J_0^{-1} B_\tau^{-1} J_\gamma,$$

with the entire matrix

$$C_\tau = \begin{pmatrix} \frac{17-9z^3}{8} & \frac{3-3z^3}{4z^2} & \frac{1-z^3}{4z^3} \\ \frac{51-24z^2-27z^3}{16} & \frac{9+8z^2-9z^3}{8z^2} & \frac{3-3z^3}{8z^3} \\ \frac{-153+36z^2+117z^3}{8} & \frac{-27-12z^2+39z^3}{4z^2} & \frac{-9+13z^3}{4z^3} \end{pmatrix}.$$

The factored formulas prove both inverse products:

$$\begin{aligned} C_\tau A_\tau &= J_0^{-1} B_\tau^{-1} J_\gamma J_\gamma^{-1} B_\tau J_0 = I, \\ A_\tau C_\tau &= I. \end{aligned}$$

The displayed rational matrices are independently checked entry by entry against those products. At $z = 1$, they both equal I. Their determinants satisfy

$$\begin{aligned} \det A_\tau &= (\det J_\gamma)^{-1} \det B_\tau \det J_0 \\ &= (-2)^{-1} \cdot 1 \cdot (-2) = 1. \end{aligned}$$

23.8 Full variational equation in the retained time

Directly differentiating the original polynomial U, before substituting γ, gives

$$DU(\gamma) = \begin{pmatrix} -\frac{27}{2z^2} & -\frac{9}{z^4} & -\frac{3}{z^5} \\ -\frac{129}{4} & -\frac{27}{2z^2} & -\frac{9}{2z^3} \\ \frac{423}{2}z & \frac{93}{z} & \frac{27}{z^2} \end{pmatrix}.$$

Its trace is zero. Applying $d/d\tau = (-4/z)d/dz$ to the complete A_τ yields

$$\frac{dA_\tau}{d\tau} = \begin{pmatrix} -\frac{27}{2z^5} & -\frac{9}{z^5} & -\frac{3}{z^5} \\ -\frac{51}{2} - \frac{27}{4z^3} & -9 - \frac{9}{2z^3} & -3 - \frac{3}{2z^3} \\ \frac{459}{2}z - \frac{18}{z} & 81z + \frac{12}{z} & 27z \end{pmatrix}.$$

Multiplying the preceding displayed $DU(\gamma)$ by the displayed A_τ gives this same matrix in all nine entries. Therefore

$$\frac{dA_\tau}{d\tau} = DU(\gamma(\tau))A_\tau, \qquad A_0 = I, \qquad \det A_\tau = 1.$$

This also confirms that the inverse-Jacobian formula is the actual material derivative for the original polynomial vector field, not merely a matrix with the same determinant.

23.9 The entire transported covector

Let e_3 mean the third standard initial-coordinate covector, using the original (x, y, w) basis at q_0. Its transport is

$$\zeta_\tau = A_\tau^{-\mathsf{T}} e_3 = C_\tau^{\mathsf{T}} e_3 = \begin{pmatrix} \frac{-153+36z^2+117z^3}{8} \\ \frac{-27-12z^2+39z^3}{4z^2} \\ \frac{-9+13z^3}{4z^3} \end{pmatrix}.$$

Thus its exact Euclidean squared norm is the sum of three squares

$$\begin{aligned} |\zeta_\tau|^2 = &\frac{(-153 + 36z^2 + 117z^3)^2}{64} \\ &+ \frac{(-27 - 12z^2 + 39z^3)^2}{16z^4} + \frac{(-9 + 13z^3)^2}{16z^6}. \end{aligned}$$

In particular

$$\begin{aligned} \lim_{z\downarrow 0} z^3 \zeta_\tau &= (0, 0, -9/4)^{\mathsf{T}}, \\ \lim_{z\downarrow 0} z^6 |\zeta_\tau|^2 &= \frac{81}{16}, \end{aligned}$$

so

$$|\zeta_\tau| \sim \frac{9}{4} z^{-3} = \frac{9}{4}(1 - 8\tau)^{-3/2}.$$

Here $f \sim g$ means the ratio tends to 1 at the specified endpoint.

There is also an explicit bound on an exact time interval. For $0 < z \le 1/2$, equivalently $3/32 \le \tau < 1/8$,

$$|\zeta_\tau| \ge \left|\frac{-9 + 13z^3}{4z^3}\right| = \frac{9 - 13z^3}{4z^3}$$
$$\ge \frac{59}{32} z^{-3}.$$

The last constant follows from $13z^3 \le 13/8$; it is a rational inequality, not a numerical estimate.

23.10 Every entry of the diffusion tensor

The full tensor in the initial-coordinate basis is

$$K_\tau = A_\tau^{-1} A_\tau^{-\mathsf{T}} = C_\tau C_\tau^{\mathsf{T}}.$$

To state all six independent entries without suppressing any term, introduce the following seven polynomial abbreviations, local to this endpoint calculation:

$$\begin{aligned}
&\pi_1(z) = 17 - 9z^3, && \pi_2(z) = 1 - z^3,\\
&\pi_3(z) = 51 - 24z^2 - 27z^3, && \pi_4(z) = 9 + 8z^2 - 9z^3,\\
&\pi_5(z) = -153 + 36z^2 + 117z^3, && \pi_6(z) = -27 - 12z^2 + 39z^3,\\
&\pi_7(z) = -9 + 13z^3.
\end{aligned}$$

They denote the original numerators. They introduce no identification with any physical coordinate, density, time-conversion parameter, or previous scalar field. In the next formulas each π_j means the displayed $\pi_j(z)$. The complete symmetric tensor is specified by

$$\begin{aligned}
K_{11} &= \frac{\pi_1^2}{64} + \frac{9\pi_2^2}{16z^4} + \frac{\pi_2^2}{16z^6},\\
K_{12} = K_{21} &= \frac{\pi_1\pi_3}{128} + \frac{3\pi_2\pi_4}{32z^4} + \frac{3\pi_2^2}{32z^6},\\
K_{13} = K_{31} &= \frac{\pi_1\pi_5}{64} + \frac{3\pi_2\pi_6}{16z^4} + \frac{\pi_2\pi_7}{16z^6},\\
K_{22} &= \frac{\pi_3^2}{256} + \frac{\pi_4^2}{64z^4} + \frac{9\pi_2^2}{64z^6},\\
K_{23} = K_{32} &= \frac{\pi_3\pi_5}{128} + \frac{\pi_4\pi_6}{32z^4} + \frac{3\pi_2\pi_7}{32z^6},\\
K_{33} &= \frac{\pi_5^2}{64} + \frac{\pi_6^2}{16z^4} + \frac{\pi_7^2}{16z^6}.
\end{aligned}$$

Each equality follows from the corresponding row dot product of the explicit C_τ, and the replay also compares all nine entries with the fully expanded rational tensor. In particular $K_{33} = |\zeta_\tau|^2$.

For every finite $\tau < 1/8$, $z > 0$, and A_τ, C_τ are invertible. Therefore, for every nonzero real column ξ,

$$\xi^{\mathsf{T}} K_\tau \xi = |C_\tau^{\mathsf{T}} \xi|^2 > 0.$$

This proves strict positive definiteness at every such time. Also

$$\det K_\tau = (\det C_\tau)^2 = 1, \qquad K_0 = I.$$

On each compact interval $0 \le \tau \le T' < 1/8$, continuity of A_τ, C_τ gives finite upper bounds on both operator norms. The exact inequalities

$$\|A_\tau\|_{\mathrm{op}}^{-2} |\xi|^2 \le \xi^{\mathsf{T}} K_\tau \xi \le \|C_\tau\|_{\mathrm{op}}^2 |\xi|^2$$

then give uniform ellipticity on that compact interval.

23.11 Exact entry powers and all three eigenvalue powers

Define the following fixed row and column, local to the endpoint calculation:

$$\ell = \left(-\frac{9}{8}, -\frac{3}{4}, -\frac{1}{4}\right), \qquad a_* = \left(\frac{1}{4}, \frac{3}{8}, -\frac{9}{4}\right)^{\mathsf{T}}.$$

The symbol a_* denotes this fixed column and is not a coefficient coordinate of the fibre cubic. Their squared Euclidean norms are

$$|\ell|^2 = \frac{121}{64}, \qquad |a_*|^2 = \frac{337}{64}.$$

The displayed rational expressions give the leading term of every entry of A_τ by rows:

$$\begin{aligned} z^3(A_\tau)_{1,*} &\longrightarrow \ell, \\ z(A_\tau)_{2,*} &\longrightarrow \frac{3}{2}\ell, \\ (A_\tau)_{3,*} &\longrightarrow -13\ell. \end{aligned}$$

All entries of these limiting rows are nonzero, so the powers are exact. The leading terms of every entry of C_τ, by columns, are

$$\begin{aligned} (C_\tau)_{*,1} &\longrightarrow \frac{17}{2}a_*, \\ z^2(C_\tau)_{*,2} &\longrightarrow 3a_*, \\ z^3(C_\tau)_{*,3} &\longrightarrow a_*. \end{aligned}$$

Again all coefficients are nonzero. Thus the two full matrix limits are

$$z^3 A_\tau \longrightarrow e_1\ell, \qquad z^3 C_\tau \longrightarrow a_* e_3^{\mathsf{T}}.$$

Every entry of K_τ has exact pole order z^{-6}, with the entire leading matrix

$$z^6 K_\tau \longrightarrow a_* a_*^{\mathsf{T}} = \frac{1}{64}\begin{pmatrix} 4 & 6 & -36 \\ 6 & 9 & -54 \\ -36 & -54 & 324 \end{pmatrix}.$$

The scaled limiting matrix has rank one. This does not change the previously proved positive definiteness and determinant one of the unscaled K_τ at every finite time.

Let

$$0 < \lambda_1(\tau) \le \lambda_2(\tau) \le \lambda_3(\tau)$$

be the three eigenvalues of K_τ. The rank-one limits determine their full leading behavior without numerical eigenvalue calculations.

Indeed, entrywise convergence of a finite matrix implies convergence in Frobenius norm, which bounds operator norm. The triangle inequality

$$\big|\|M\|_{\mathrm{op}} - \|N\|_{\mathrm{op}}\big| \le \|M - N\|_{\mathrm{op}} \le \|M - N\|_{\mathrm{F}}$$

therefore proves convergence of the operator norms in the two matrix limits. For any rank-one matrix uv^{T}, Cauchy–Schwarz gives $\|uv^{\mathsf{T}}\|_{\mathrm{op}} \le |u||v|$; equality holds on $v/|v|$ when $v \neq 0$. Here u, v in this one elementary norm identity are arbitrary finite-dimensional columns, not fluid fields. Applying the identity to the two displayed limits gives

$$\begin{aligned} z^3\|A_\tau\|_{\mathrm{op}} &\longrightarrow \frac{11}{8}, \\ z^3\|C_\tau\|_{\mathrm{op}} &\longrightarrow \frac{\sqrt{337}}{8}. \end{aligned}$$

Because $K_\tau = C_\tau C_\tau^{\mathsf{T}}$ and $K_\tau^{-1} = A_\tau^{\mathsf{T}} A_\tau$, the largest eigenvalue and the reciprocal of the smallest are respectively these squared norms. Consequently

$$\lambda_1(\tau) \sim \frac{64}{121} z^6 = \frac{64}{121}(1-8\tau)^3,$$
$$\lambda_3(\tau) \sim \frac{337}{64} z^{-6} = \frac{337}{64}(1-8\tau)^{-3}.$$

Finally $\det K_\tau = \lambda_1\lambda_2\lambda_3 = 1$ proves

$$\lambda_2(\tau) = \frac{1}{\lambda_1(\tau)\lambda_3(\tau)} \longrightarrow \frac{121}{337}.$$

In particular the middle eigenvalue stays of order one even though every coordinate entry has a z^{-6} leading term. The exact condition-number behavior is

$$\frac{\lambda_3(\tau)}{\lambda_1(\tau)} \sim \frac{40777}{4096} z^{-12} = \frac{40777}{4096}(1-8\tau)^{-6}.$$

23.12 Direct exact failure of the two endpoint uniform bounds

The fixed initial unit covector e_3 has

$$e_3^{\mathsf{T}} K_\tau e_3 = |\zeta_\tau|^2 \sim \frac{81}{16} z^{-6}.$$

Hence there is no finite uniform upper ellipticity bound over $0 \le \tau < 1/8$.

For the lower bound, retain the exact time-dependent unit covector

$$\xi_\tau = \frac{A_\tau^{\mathsf{T}} e_1}{|A_\tau^{\mathsf{T}} e_1|}.$$

It is defined for every finite time because A_τ is invertible. The first row of the full A_τ gives

$$|A_\tau^{\mathsf{T}} e_1|^2 = \frac{(17z^3-9)^2 + 40(z^3-1)^2}{64z^6} = \frac{329z^6 - 386z^3 + 121}{64z^6}.$$

The polynomial $329z^6 - 386z^3 + 121$ is strictly positive: the two squared factors cannot vanish simultaneously. Since $C_\tau^{\mathsf{T}} A_\tau^{\mathsf{T}} e_1 = e_1$, its exact Rayleigh quotient is

$$\xi_\tau^{\mathsf{T}} K_\tau \xi_\tau = \frac{64z^6}{329z^6 - 386z^3 + 121} \sim \frac{64}{121} z^6 \longrightarrow 0.$$

Thus no positive uniform lower ellipticity constant holds on the full interval. This is loss of uniform endpoint bounds, with strict positive definiteness preserved at every finite $\tau < 1/8$.

23.13 Endpoint replay and scope

The standalone script `scripts/check_material_endpoint.py` retains the original map, original initial point, original τ, full matrices, and all coefficients. Its receipt `checks/material_endpoint_checks.json` contains the expanded entries as rational functions. All 34 exact checks pass, including the complete variational equation and all scaled matrix limits. The receipt binds the present `tex/material_endpoint.tex` by its SHA-256 hash.

The result describes the material derivative and tensor along the specified trajectory. No Lean or numerical inference is used. The calculation does not assert a classical Navier–Stokes singularity at a finite original spatial point.

24 The current S6 dynamics and an exact viscous periodic image

The current source task is *S6 Proof Audit, Reconstruction & Overleaf Release*. Its dynamical source is `supporting_materials/workbench/research/quaternionic_4441_reconciliation.tex`, in the cumulative `s6_consolidated_canonical_2026-08-26` workbench. The three-real-field equations occur at lines 3396–3460 and the periodic continuation at lines 3485–3583 in the audited snapshot. The source hashes are recorded in `sources/s6_dynamics_source_hashes.json`. The construction below uses these actual later field equations, rather than repeating the earlier polynomial map. It proves an explicit map to classical positive viscosity Navier–Stokes on the unit flat torus. The image consists of global smooth solutions. No conclusion about every possible map from the workbench, or about its full geometric reconstruction, follows from this calculation.

24.1 The retained full gauge–scalar equations

Use the source real state space and metric

$$\mathcal{E} = \mathrm{Herm}_3(\mathbb{C})_0 \oplus \mathbb{C}^3, \qquad \langle(\phi, c), (\psi, d)\rangle = \mathrm{tr}(\phi\psi) + 2\,\mathrm{Re}(c^* d).$$

Here the subscript means trace zero. Retain signature $(+\,-\,-\,-)$, source time τ, the gauge action $(\Phi, b) \mapsto (g\Phi g^*, gb)$ for $g \in SU(3)$, and the potential

$$V = \frac{m^2}{2}s + \frac{\Gamma}{4}s^2 - \kappa(\det\Phi - b^*\Phi b), \qquad s = \mathrm{tr}\,\Phi^2 + 2b^*b. \tag{400}$$

The scalar kinetic density is $\frac{1}{2}\,\mathrm{tr}(D_\mu\Phi D^\mu\Phi) + (D_\mu b)^* D^\mu b$. We use the identically zero connection $\mathcal{A}_\mu = 0$ and retain $Z_{\mathrm{tf}} = Z - \mathrm{tr}(Z)I_3/3$. Set $a = m^2 + \Gamma s$. Direct variation gives

$$\begin{aligned} dV(\delta\Phi, \delta b) = {} & \mathrm{tr}([a\Phi - \kappa(\mathrm{adj}\,\Phi - bb^*)_{\mathrm{tf}}]\delta\Phi) \\ & + 2\,\mathrm{Re}[((aI_3 + \kappa\Phi)b)^*\delta b]\,. \end{aligned} \tag{401}$$

The determinant derivative used here is the polynomial identity $d(\det\Phi)[H] = \mathrm{tr}((\mathrm{adj}\,\Phi)H)$, which holds also for singular matrices by polynomial continuation from the invertible ones. Integration by parts with the displayed kinetic metric therefore gives

$$\begin{aligned} \Box\Phi &= -a\Phi + \kappa(\mathrm{adj}\,\Phi - bb^*)_{\mathrm{tf}}, \\ \Box b &= -(aI_3 + \kappa\Phi)b, \qquad \Box = \partial_\tau^2 - \sum_{j=1}^{3}\partial_{x_j}^2. \end{aligned} \tag{402}$$

The matter current paired with an anti-Hermitian connection variation is

$$J_\mu = [\Phi, \partial_\mu\Phi] + b(\partial_\mu b)^* - (\partial_\mu b)b^*.$$

For the fields used below, this matrix is zero, so its trace-free projection is zero as well. The Yang–Mills field strength of the zero connection is zero. Thus all connection equations, including Gauss, hold for the full variational system, not only for variations restricted to the scalar ansatz.

Put

$$\Phi = \mathrm{diag}(h + k, h - k, -2h), \qquad b = \beta e_3, \qquad h, k, \beta \in \mathbb{R}. \tag{403}$$

All commutators in J_μ vanish because the matrices are diagonal; the remaining two products are equal because β and its derivatives are real. Further,

$$\begin{aligned} s &= 6h^2 + 2k^2 + 2\beta^2, \\ \det\Phi - b^*\Phi b &= 2h(k^2 + \beta^2 - h^2), \\ \mathrm{adj}\,\Phi - bb^* &= \mathrm{diag}\big(-2h(h-k), -2h(h+k), h^2 - k^2 - \beta^2\big). \end{aligned}$$

The last trace is $-3h^2 - k^2 - \beta^2$. Removing exactly one third of it in (402), taking half the sum and difference of the first two diagonal entries, and using the third component of b, gives

$$\begin{aligned}
\Box h &= -ah - \kappa h^2 + \frac{\kappa}{3}(k^2 + \beta^2),\\
\Box k &= -(a - 2\kappa h)k, \qquad \Box\beta = -(a - 2\kappa h)\beta,\\
a &= m^2 + \Gamma(6h^2 + 2k^2 + 2\beta^2).
\end{aligned} \tag{404}$$

Conversely these equations give every component of (402); all omitted entries are zero. This proves the exact invariant sector. For homogeneous fields the kinetic density is $3\dot{h}^2 + \dot{k}^2 + \dot{\beta}^2$, with Euler–Lagrange weights $6, 2, 2$.

24.2 The source oscillator, including its parameter interval

Retain the source hypotheses

$$0 < m^2 < \frac{\kappa^2}{27\Gamma}, \qquad \Gamma > 0, \quad \kappa > 0, \qquad d_0 = \sqrt{\kappa^2 - 24\Gamma m^2}, \qquad h_0 = \frac{\kappa + d_0}{24\Gamma}.$$

Choose $h_0 < h < \kappa/(6\Gamma)$ and define

$$\begin{aligned}
\Delta(h) &= 1 - \frac{6\Gamma h}{\kappa}, & a(h) &= \frac{m^2 + 12\Gamma h^2}{\Delta(h)},\\
R^2(h) &= \frac{3a(h)h}{\kappa} + 3h^2, & \omega^2(h) &= a(h) - 2\kappa h = \frac{m^2 - 2\kappa h + 24\Gamma h^2}{\Delta(h)}.
\end{aligned} \tag{405}$$

Take the positive roots R, ω. The square root d_0 is real and $0 < d_0 < \kappa$; in particular $0 < h_0 < \kappa/(12\Gamma)$, so the interval for h is nonempty. The denominator is positive there, as are a and R^2. The numerator of ω^2 factors as

$$24\Gamma\left(h - \frac{\kappa - d_0}{24\Gamma}\right)(h - h_0),$$

so $\omega^2 > 0$. Direct substitution gives the three exact identities

$$a = m^2 + \Gamma(6h^2 + 2R^2), \qquad -ah - \kappa h^2 + \frac{\kappa R^2}{3} = 0, \qquad \omega^2 = a - 2\kappa h. \tag{406}$$

Consequently

$$h(\tau) = h, \qquad k(\tau) = R\cos(\omega\tau), \qquad \beta(\tau) = R\sin(\omega\tau), \qquad \mathcal{A}_\mu = 0 \tag{407}$$

solves the full equations for every real τ. Indeed its squared radius is R^2, its last two second derivatives are $-\omega^2$ times the original coordinates, and its first second derivative is zero; these are precisely (404). The observable $\|b\|^2 = R^2\sin^2(\omega\tau)$ is gauge invariant and nonconstant. In the source polar convention $k = r\cos(2\theta)$, $\beta = -r\sin(2\theta)$, this same trajectory is $r = R$, $\theta = -\omega\tau/2$, preserving the original minus sign.

Its conserved energy density, obtained by adding its kinetic and potential densities without rescaling, is

$$E(h) = R^2\omega^2 + m^2(3h^2 + R^2) + \Gamma(3h^2 + R^2)^2 - 2\kappa h(R^2 - h^2). \tag{408}$$

On the source flat torus of volume $\mathcal{V}_3$, the total energy is $\mathcal{V}_3 E(h)$. This volume and this scalar energy are not identified with the fluid volume or kinetic energy in the map below.

24.3 A coefficient map with two nonparallel wave directions

The target is exactly $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$, with coordinates (x_1, x_2, x_3), Euclidean metric, standard orientation, and volume one. Choose and retain

$$n \in \mathbb{N}_{>0}, \qquad \lambda = 2\pi n, \qquad \nu > 0, \quad \eta > 0, \quad \sigma > 0.$$

Here $\tau = \sigma t$ is the stated source-to-fluid time dictionary and η converts scalar amplitude to fluid velocity; these are independent parameters. Define

$$B_1 = (\sin(\lambda x_3), \cos(\lambda x_3), 0), \qquad B_2 = (0, \sin(\lambda x_1), \cos(\lambda x_1)). \tag{409}$$

Both fields are unit-periodic. Their wave directions are e_3 and e_1. For example $\partial_3(B_1)_2 = -\lambda\sin(\lambda x_3)$ and $\partial_3(B_1)_1 = \lambda\cos(\lambda x_3)$; inserting these in the oriented curl gives $\operatorname{curl} B_1 = \lambda B_1$. The same calculation with $\partial_1(B_2)_3 = -\lambda\sin(\lambda x_1)$ and $\partial_1(B_2)_2 = \lambda\cos(\lambda x_1)$ gives $\operatorname{curl} B_2 = \lambda B_2$. Componentwise differentiation further proves

$$\operatorname{div} B_j = 0, \quad \Delta B_j = -\lambda^2 B_j, \quad |B_j|^2 = 1, \quad B_1 \cdot B_2 = c(x) := \cos(\lambda x_3)\sin(\lambda x_1). \tag{410}$$

The mean of c is zero, by integration of either sine or cosine over the unit interval. Therefore (B_1, B_2) is an orthonormal pair in real $L^2(\mathbb{T}^3; \mathbb{R}^3)$, despite its nonzero pointwise inner product.

Put $d(t) = e^{-\nu\lambda^2 t}$. For each finite t, the map

$$L_t : \mathbb{R}^2 \longrightarrow C^\infty(\mathbb{T}^3; \mathbb{R}^3), \qquad L_t(K, B) = \eta d(t)(KB_1 + BB_2) \tag{411}$$

has kernel zero and image $\operatorname{span}_{\mathbb{R}}\{B_1, B_2\}$. Its inverse on that image and its inner product are exactly

$$\begin{aligned} K &= \frac{\int_{\mathbb{T}^3} L_t(K,B)\cdot B_1\,dx}{\eta d(t)}, \qquad\qquad B = \frac{\int_{\mathbb{T}^3} L_t(K,B)\cdot B_2\,dx}{\eta d(t)}, \\ \langle L_t(K,B), L_t(K',B')\rangle_{L^2} &= \eta^2 d(t)^2 (KK' + BB'). \end{aligned} \tag{412}$$

These statements follow by the two orthogonality integrals, not by a dimension count. On the source oscillator circle $K^2 + B^2 = R^2$, its image is the circle of L^2 radius $\eta d(t) R$ in that two-dimensional plane. The original h is retained as a fixed parameter defining this circle; the construction does not identify the full three-coordinate state space or the full gauge configuration space with this plane.

There is also an exact flow relation on this image. Define $J_E B_1 = B_2$ and $J_E B_2 = -B_1$ on the two-dimensional image. The orthogonality already proved gives $J_E^* = -J_E$. For any trajectory $q(\tau) = (k(\tau), \beta(\tau))$ with $q' = \omega(-\beta, k)$, direct differentiation of (411) gives

$$\frac{d}{dt} L_t(q(\sigma t)) = (-\nu\lambda^2 I + \sigma\omega J_E) L_t(q(\sigma t)).$$

Thus the full coefficient evolution, time conversion, and damping are included in the map; it is not only a correspondence of time slices.

24.4 Exact pressure, force, energy, and regularity

For readability write $K(t) = k(\sigma t)$ and $B(t) = \beta(\sigma t)$. Then $K' = -\sigma\omega B$, $B' = \sigma\omega K$, and $K^2 + B^2 = R^2$. Define the following complete periodic data and solution:

$$\begin{aligned} u(x,t) &= \eta d(t)(K(t)B_1(x) + B(t)B_2(x)), \\ p(x,t) &= -\eta^2 d(t)^2 K(t)B(t)c(x), \\ f(x,t) &= \eta\sigma\omega d(t)(-B(t)B_1(x) + K(t)B_2(x)), \\ u^0(x) &= \eta R B_1(x). \end{aligned} \tag{413}$$

These satisfy the full three-dimensional classical system (1), with the stated positive viscosity and its original sign. To prove this, (410) first gives $\operatorname{div} u = 0$, $\operatorname{curl} u = \lambda u$, and $\Delta u = -\lambda^2 u$. The component identity

$$((u \cdot \nabla)u)_i = \partial_i \frac{|u|^2}{2} - (u \times \operatorname{curl} u)_i$$

follows by expanding the two Levi–Civita symbols in the cross product. Here its cross product is zero. Since

$$\frac{|u|^2}{2} = \frac{\eta^2 d(t)^2 R^2}{2} + \eta^2 d(t)^2 K(t)B(t)c(x),$$

the first term has zero spatial gradient and $(u \cdot \nabla)u + \nabla p = 0$ exactly. This pressure is spatially nonconstant whenever $KB \neq 0$: $\partial_1 c = \lambda \cos(\lambda x_3)\cos(\lambda x_1)$ is not the zero function. Thus the nonlinear interaction has been calculated, including its nonzero pressure. Finally $d' = -\nu\lambda^2 d$ and the two oscillator derivatives give $\partial_t u = \nu \Delta u + f$, completing every component of the equation. At $t = 0$ the displayed initial condition follows from $(K, B) = (R, 0)$.

The fluid kinetic energy and its exact balance are

$$\mathcal{E}_{\text{fluid}}(t) = \frac{1}{2}\int_{\mathbb{T}^3} |u|^2 dx = \frac{\eta^2 R^2}{2} e^{-2\nu\lambda^2 t}, \qquad \mathcal{E}'_{\text{fluid}}(t) + \nu \int_{\mathbb{T}^3} |\nabla u|^2 dx = 0. \tag{414}$$

Indeed integration of $u \cdot f$ gives $\eta^2 \sigma\omega d^2(-KB + BK) = 0$. Integration by parts on the periodic cube gives $\int |\nabla u|^2 = -\int u \cdot \Delta u = \lambda^2 \eta^2 d^2 R^2$, which proves the balance and agrees with differentiating the energy. The force does zero *integrated* work; no pointwise zero-work claim is made, since $u \cdot f = \eta^2 \sigma\omega d^2 (K^2 - B^2)c(x)$.

All fields in (413) are smooth for $t \geq 0$. For each spatial multi-index α and each nonnegative integer r, differentiation of the finite trigonometric expressions gives

$$\sup_{x \in \mathbb{T}^3} |\partial_x^\alpha \partial_t^r f(x,t)| \leq C_{\alpha r} e^{-\nu\lambda^2 t},$$

with a finite constant depending on the retained parameters. For every integer $N \geq 0$, $(1+t)^N e^{-\nu\lambda^2 t}$ is bounded on $[0, \infty)$: it is continuous on $[0, 1]$, and for $t \geq 1$ the exponential series $e^{\nu\lambda^2 t} \geq (\nu\lambda^2 t)^{N+1}/(N+1)!$ gives decay of its product with $(1+t)^N$. Thus the force satisfies precisely the periodic decay condition (5), and the initial data are smooth, divergence-free, and periodic. The construction gives an explicit global solution for these admissible data. Its decaying fluid energy (414) and the constant source energy (408) show exactly that this map preserves the displayed differential equations with the displayed force dictionary; it is not an energy-preserving identification of the two systems.

24.5 Gauge equivalence and the exact translation quotient

The coefficient map above uses the marked diagonal sector. It does not silently identify gauge-equivalent source matrices with a single fixed coordinate velocity. This issue has an exact resolution for the fixed h circle used here. For two states (403) with this same $h > 0$, full $SU(3)$ gauge equivalence implies

$$\beta'^2 = \beta^2, \qquad 2h(k'^2 - h^2) = \det \Phi' = \det \Phi = 2h(k^2 - h^2),$$

so $k'^2 = k^2$. Conversely every independent change of signs of k, β is a gauge transformation. In fact

$$g_b = \operatorname{diag}(-1, 1, -1), \qquad g_s = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

are unitary matrices of determinant one. The first sends (k, β) to $(k, -\beta)$, and the second sends it to $(-k, -\beta)$; their product realizes $(-k, \beta)$. These statements remain valid when one coordinate is zero. This proves both directions of the orbit classification within the fixed-h sector, with no generic-orbit exception.

For a torus translation $T_a v(x) = v(x+a)$, define

$$a_1 = (0, 0, 1/(2n)), \qquad a_2 = (1/(2n), 0, 0).$$

Then

$$T_{a_1} B_1 = -B_1, \quad T_{a_1} B_2 = B_2, \qquad T_{a_2} B_1 = B_1, \quad T_{a_2} B_2 = -B_2.$$

Consequently the $SU(3)$ orbit relation just calculated corresponds exactly under L_t to the orbit relation generated by T_{a_1}, T_{a_2} on its image. Their action on this image factors through $(\mathbb{Z}/2\mathbb{Z})^2$; the translations themselves need not have order two on the whole torus. The map therefore descends to a bijection between the fixed-h source circle modulo the stated gauge equivalence and the image circle modulo these translations. The inverse is supplied by (412) followed by the sign orbit.

For entire trajectories, a constant sign transformation also transforms the derivatives (K', B') with the same signs. Applying (413) using these transformed derivatives in $f = \eta d(K'B_1 + B'B_2)$ intertwines all three fields (u, p, f) with the corresponding fixed torus translation. A single sign reversal changes the orientation of the source oscillator; it is not to be inserted into $K' = -\sigma\omega B$, $B' = \sigma\omega K$ while retaining those first-order equations. The second-order source equations are unchanged. This proves equivariance of the full construction without discarding its orientation data.

24.6 What the fixed curl sector actually evolves

Let E_λ be the real smooth periodic fields satisfying $\operatorname{div} v = 0$ and $\operatorname{curl} v = \lambda v$, for the same $\lambda = 2\pi n > 0$. Let P be the flat torus orthogonal projection onto divergence-free fields, including the constant fields. In Fourier coordinates its symbol is $P_j = I - jj^{\mathsf{T}}/|j|^2$ for $j \in \mathbb{Z}^3 \setminus \{0\}$ and $P_0 = I$. This definition proves $P\nabla q = 0$ for every smooth periodic scalar q. For any time-dependent field that remains in E_λ, the component identity used above gives

$$P((u \cdot \nabla)u) = 0, \qquad \Delta u = -\lambda^2 u.$$

Projection of the full Navier–Stokes equation therefore yields the *exact* evolution

$$\partial_t u + \nu\lambda^2 u = Pf, \qquad u(t) = e^{-\nu\lambda^2 t} u(0) + \int_0^t e^{-\nu\lambda^2 (t-s)} Pf(s)\, ds. \tag{415}$$

The second identity follows by differentiating $e^{\nu\lambda^2 t} u(t)$ and integrating; no nonlinear estimate is substituted for an equality. Necessarily $Pf(s) \in E_\lambda$ for a solution that stays in that space. Conversely a smooth force with this property and an initial field in E_λ gives a velocity in E_λ by this formula. The complete pressure reconstruction is

$$\widehat{q}(0,t) = 0, \qquad \widehat{q}(j,t) = \frac{j \cdot \widehat{f}(j,t)}{2\pi i |j|^2} \quad (j \neq 0), \qquad p = q - \frac{|u|^2}{2} + c_0(t).$$

Here c_0 is any smooth real function. Smoothness of f gives rapid decay of its Fourier coefficients, including every time derivative on finite intervals, so this series defines a smooth real q. Its reality follows from the conjugate symmetry of $\widehat{f}$. Multiplication of its coefficient by $2\pi i j$ proves $\nabla q = (I - P)f$. Substitution then gives the full equation, since $Pf + \nabla(|u|^2/2) + \nabla p - f = 0$.

There are only finitely many Fourier frequencies in E_λ. Indeed $\Delta u = -\lambda^2 u$ gives $(4\pi^2|j|^2 - \lambda^2)\widehat{u}(j) = 0$; hence $|j| = n$, a finite set of integer triples, and $\widehat{u}(0) = 0$. Every spatial derivative norm is consequently bounded by a fixed finite sum of absolute values of these coefficients. Smooth prescribed forcing on a finite closed time interval has bounded coefficients there. Formula (415) bounds every coefficient and all its time derivatives on that interval, by differentiation of the finite scalar integrals. Thus this entire exact invariant sector has no finite-time singularity under smooth prescribed forcing. This establishes the behavior of this particular image, not an obstruction to maps whose images contain interacting curl eigenvalues or an unbounded frequency sequence.

24.7 The September 8 Wilson-box vacuum gives a second exact map

The later source extension `ym_bulk_plaquette_extension.tex` retains an open cubic spatial graph with vertices $\{-L, \dots, L\}^3$, $L \in \mathbb{Z}$, $L \geq 2$, all $N = 3(2L)(2L+1)^2$ positive links and all $M = 3(2L)^2(2L+1)$ elementary faces. The exact configuration space is the compact connected manifold

$$\mathcal{C}_L = SU(2)^N, \qquad \dim_{\mathbb{R}} \mathcal{C}_L = 3N.$$

The following symbols are the *Wilson source* parameters: its $\kappa = 2g_{\mathrm{YM}}^2/a$ is a kinetic coefficient, with $g_{\mathrm{YM}} > 0$, and its $a > 0$ is a lattice spacing. They are not the scalar cubic coefficient or $a(h)$ of (405). Retain

$$b = \frac{1}{2g_{\mathrm{YM}}^2 a}, \qquad H = -\kappa\Delta + V, \qquad V = b\sum_p (2 - W_p), \qquad W_p = \operatorname{tr}(U_{e_1}U_{e_2}U_{e_3}^{-1}U_{e_4}^{-1}),$$

with the oriented boundary of each square as in the source. The metric on each factor is bi-invariant with orthonormal $T_j = -i\sigma_j/2$ and inner product $-2\operatorname{tr}(T_jT_k) = \delta_{jk}$. Haar probability measure is its volume measure up to the fixed total volume factor; this factor does not affect integration by parts.

The lowest eigenfunction ψ may be chosen smooth, strictly positive, and L^2 normalized. To record the analytical ingredients explicitly, the bounded smooth real potential and compact manifold give a Rayleigh minimizer: a bounded-energy minimizing sequence has bounded H^1 norm, its subsequence converges strongly in L^2 by compact Sobolev embedding, and weak lower semicontinuity gives the minimum. Replacing it by its absolute value preserves its L^2 norm and does not increase the Dirichlet integral. The weak Euler–Lagrange equation and elliptic regularity give a smooth nonnegative eigenfunction. The strong maximum principle applied after a sufficiently large constant shift makes it strictly positive. Its eigenvalue E is the minimum. Uniqueness up to scale also follows directly: if another real ground eigenfunction is ψF, subtracting F times the equation for ψ and integrating against $F\psi$ gives $\int \psi^2|\nabla F|^2 = 0$; thus F is constant. The real and imaginary parts give the corresponding complex uniqueness. This argument retains the actual interacting potential, with every link and face present.

Set $\phi = \log\psi$ and $\rho = \psi^2$. Expanding the Laplacian of ψF proves the ground-state transform

$$\psi^{-1}(H - E)(\psi F) = -\kappa(\Delta F + \nabla\log\rho \cdot \nabla F).$$

The forward heat drift is $\beta_{\mathrm{W}} = \kappa\nabla\log\rho$. Define its negative as the potential velocity $u_{\mathrm{W}} = -2\kappa\nabla\phi = -\beta_{\mathrm{W}}$. To fix the geometric diffusion sign, define

$$\Delta_{\text{rough}} Y = \sum_j \left(\nabla_{E_j}\nabla_{E_j}Y - \nabla_{\nabla_{E_j}E_j}Y\right), \qquad \mathcal{D}Y = \Delta_{\text{rough}}Y - \operatorname{Ric}^\sharp Y.$$

The commutation of covariant derivatives on the gradient gives

$$\Delta_{\text{rough}}\nabla q = \nabla\Delta q + \operatorname{Ric}^\sharp \nabla q, \qquad \mathcal{D}\nabla q = \nabla\Delta q.$$

For completeness, the bi-invariant connection on each factor is $\nabla_X Y = [X,Y]/2$ on invariant fields. The Koszul formula gives this identity from invariance of the metric. Substitution in the definition of curvature gives sectional curvature $|[X,Y]|^2/4$ for orthonormal X, Y. Since $[T_j, T_k] = \epsilon_{jk\ell}T_\ell$, this curvature is $1/4$, and summing its two transverse directions gives $\operatorname{Ric} = g/2$. Mixed factors have zero curvature, so on the product $\mathcal{D} = \Delta_{\text{rough}} - I/2$.

The original ground-state equation, divided by its positive ψ, now yields

$$\kappa(\Delta\phi + |\nabla\phi|^2) = V - E.$$

The symmetry of the Hessian gives $\nabla_{\nabla\phi}\nabla\phi = \nabla|\nabla\phi|^2/2$. Taking a gradient of the preceding identity therefore proves the exact stationary viscous equation

$$\begin{aligned}\nabla_{u_{\mathrm{W}}} u_{\mathrm{W}} - \kappa\mathcal{D}u_{\mathrm{W}} &= 2\kappa^2\nabla(|\nabla\phi|^2 + \Delta\phi)\\ &= 2\kappa\nabla V = -\frac{2}{a^2}\nabla\sum_p W_p.\end{aligned} \tag{416}$$

The last coefficient uses $2\kappa b = 2/a^2$. This is an exact forced potential Burgers equation on the retained configuration manifold. The velocity sign is essential: the forward drift itself satisfies $\nabla_{\beta_{\mathrm{W}}}\beta_{\mathrm{W}} + \kappa\mathcal{D}\beta_{\mathrm{W}} = 2\kappa\nabla V$.

Its incompressible component can be computed exactly on this same manifold. If $\operatorname{div} u_{\mathrm{W}} = -2\kappa\Delta\phi$ vanished, then integration against ϕ would give $0 = \int \phi\Delta\phi = -\int|\nabla\phi|^2$, so ϕ and ψ would be constant by connectedness. The ground-state equation would then force V to be constant. It is not: all identity links give $V = 0$, whereas replacing one strictly interior link by $-I$ leaves all other links at the identity and changes precisely its four incident face traces from 2 to -2, giving $V = 16b > 0$. Therefore this actual vacuum velocity is not divergence-free. Moreover for every smooth divergence-free vector field Y,

$$\langle u_{\mathrm{W}}, Y\rangle_{L^2} = 2\kappa\int\phi\operatorname{div} Y = 0.$$

By continuity this holds for the L^2 closure of those fields, so the orthogonal incompressible projection of u_{W} is exactly zero. The force in (416) is also a gradient and has zero such projection. This proves both the differential-equation correspondence and what its specific incompressible projection retains. It does not erase the interacting vacuum, replace the graph by three coordinates, or exclude different maps from excited states or currents.

24.8 Exact replay record

The standalone script `scripts/check_s6_dynamics.py` checks the matrix sector variation, all three oscillator parameter identities, the oriented curls, divergences and Laplacians of both modes, their orthogonality, the full viscous momentum residual, the nonlinear pressure, the gauge generators and translation signs, the energy identity, and the Wilson parameter coefficient. Its deterministic record is `checks/s6_dynamics_checks.json`. These finite symbolic identities accompany the analytic arguments above; they do not certify the full source workbench or establish a Navier–Stokes counterexample.

25 Arithmetic heat observables and exact three-dimensional fluid maps

The formulas audited in this section come from the maintained zeta source, `satellites/22_rh_counterexample_routes.tex`: fluid metric and time coordinates at lines 14–108; BC state and modular time at 112–179; heat and Cole–Hopf at 928–980; completed arithmetic heat at 982–1218. The finite identities below were independently replayed by `scripts/verify_arithmetic_fluid_bridge.py` (29 exact checks). The hydrodynamic vacuum reconstruction is a sourced construction; the coordinate, scalar PDE, incompressibility, pressure and shear calculations are proved below. The original polynomial and its metric pullback are proved in the separate polynomial section of this notebook.

25.1 The actual fluid/vacuum map and its preserved constants

The local source cites Bredberg–Keeler–Lysov–Strominger, §5.1, equations (14)–(20). Its coordinates are $(\tau, r, x^1, \ldots, x^p)$, cutoff $r = r_c > 0$, and future Rindler horizon $r = 0$. It retains the metric

$$
\begin{aligned}
ds^2 = & -r\,d\tau^2 + 2\,d\tau\,dr + \delta_{ij}dx^i dx^j \\
& -2(1-r/r_c)v_i\,dx^i d\tau - 2r_c^{-1}v_i\,dx^i dr \\
& +(1-r/r_c)[(v^2+2P)d\tau^2 + r_c^{-1}v_iv_jdx^idx^j] + r_c^{-1}(v^2+2P)d\tau\,dr \\
& -r_c^{-1}(r^2-r_c^2)\Delta v_i\,dx^i d\tau + \cdots .
\end{aligned}
$$

The expansion assigns $v = O(\epsilon)$, $P = O(\epsilon^2)$, $\partial_i = O(\epsilon)$, $\partial_\tau = O(\epsilon^2)$. It imposes

$$
\partial_i v^i = 0, \qquad \partial_\tau v_i + v^j\partial_j v_i + \partial_i P - r_c\Delta v_i = 0.
$$

The source gives a hydrodynamic expansion with flat cutoff metric, horizon regularity and specified incoming data. It does not exhibit an independently constructed vacuum solution whose boundary velocity becomes singular while meeting the three-dimensional problem's data requirements. For $p = 3$, the bulk has five coordinates (τ, r, x^1, x^2, x^3), a point relevant to any attempted transfer from four-dimensional relativity.

The temperature calculation can be checked exactly without solving the perturbative Einstein equations. On $r > 0$, define

$$
\eta = \tau - \log r, \quad \xi = 2\sqrt{r}, \qquad X^0 = \xi\sinh(\eta/2), \quad X^1 = \xi\cosh(\eta/2).
$$

Since $d\tau = d\eta + dr/r$, direct multiplication gives

$$
-r(d\eta + dr/r)^2 + 2(d\eta + dr/r)dr = -r\,d\eta^2 + dr^2/r = -(\xi^2/4)d\eta^2 + d\xi^2.
$$

Direct differentiation of the hyperbolic coordinates gives $-(dX^0)^2 + (dX^1)^2$ for the final expression. Their image is the right Minkowski wedge, with inverse

$$
\xi = \sqrt{(X^1)^2 - (X^0)^2}, \qquad \eta = 2\,\mathrm{arctanh}(X^0/X^1).
$$

Thus the inverse-chart singularity at the horizon is not a curvature singularity of this zeroth-order metric: its components become constant Minkowski components under the displayed map.

On the cutoff orbit, set

$$
t = \sqrt{r_c}\tau, \quad V_i(t,x) = r_c^{-1/2}v_i(t/\sqrt{r_c}, x), \quad \Pi(t,x) = r_c^{-1}P(t/\sqrt{r_c}, x).
$$

The inverse is $v_i(\tau,x) = \sqrt{r_c}V_i(\sqrt{r_c}\tau, x)$ and $P(\tau,x) = r_c\Pi(\sqrt{r_c}\tau, x)$. The four terms in the equation become, in order,

$$
r_c\partial_t V_i, \qquad r_cV^j\partial_jV_i, \qquad r_c\partial_i\Pi, \qquad r_c^{3/2}\Delta V_i.
$$

Consequently the proper-time equation has viscosity $\nu_{\mathrm{loc}} = \sqrt{r_c}$. In the source's units $\hbar = c = k_B = 1$, its acceleration temperature is $T_{\mathrm{loc}} = 1/(4\pi\sqrt{r_c})$, so $\nu_{\mathrm{loc}} = 1/(4\pi T_{\mathrm{loc}})$. Neither viscosity is set to one in this calculation. Sending r_c to zero changes the viscosity and degenerates the time map; it is not a singularity at a fixed positive viscosity and fixed physical time.

25.2 BC state, modular flow, and heat observable

Let H on $\ell^2(\mathbb{N}^\times)$ be the self-adjoint diagonal operator

$$
H\varepsilon_n = (\log n)\varepsilon_n, \quad \mathrm{Dom}\,H = \{a : \sum_n (\log n)^2|a_n|^2 < \infty\}.
$$

For real $\beta > 1$, define $Z(\beta) = \zeta(\beta)$, $p_n = n^{-\beta}/Z(\beta)$, $D_\beta\varepsilon_n = p_n\varepsilon_n$, and $\varphi_\beta(A) = \mathrm{Tr}(D_\beta A)$ on bounded operators. All p_n are positive and sum to one. In the convention used by the source,

$$
\alpha_u(A) = D_\beta^{-iu}AD_\beta^{iu} = e^{i\beta uH}Ae^{-i\beta uH}.
$$

Indeed $D_\beta^{-iu} = Z(\beta)^{iu} e^{i\beta u H}$, and the scalar factors cancel. On matrix units the factor is $(m/n)^{i\beta u}$. The opposite modular sign changes βu to $-\beta u$. The positive-temperature partition function supplies no pole of this real-time unitary conjugation.

The actual heat observable is

$$F_\beta(\tau, x) = \varphi_\beta(e^{-\tau H^2} e^{ixH}) = \frac{1}{\zeta(\beta)} \sum_{n \geq 1} n^{-\beta} e^{-\tau(\log n)^2 + ix \log n}.$$

For real $\tau \geq 0, x$, every derivative of prescribed finite order is uniformly dominated on $\tau \geq 0$ by a constant times $\sum n^{-\beta}(\log n)^k$, which converges by the integral test after $n = e^s$. Differentiating in τ or twice in x introduces the same factor $-(\log n)^2$. Therefore

$$\partial_\tau F_\beta = \partial_x^2 F_\beta, \qquad F_\beta(0, x) = \zeta(\beta - ix)/\zeta(\beta).$$

This is a scalar complex heat equation on a real spatial line. τ varies the observable, while the Gibbs state and modular automorphism remain fixed. Every displayed observable commutes with H; its modular evolution is constant. This is an exact dynamical test, not an argument from different terminology.

On any open set where a heat solution F with viscosity $\nu > 0$ is nonzero, define $u = -2\nu F_x/F$. A global logarithm is unnecessary; local logarithms $L = \log F$ suffice and their derivatives agree. Since

$$L_\tau = \nu(L_{xx} + L_x^2),$$

one gets

$$u_\tau = -2\nu^2 L_{xxx} - 4\nu^2 L_x L_{xx}, \quad u u_x = 4\nu^2 L_x L_{xx}, \quad \nu u_{xx} = -2\nu^2 L_{xxx}.$$

Their sum proves $u_\tau + u u_x = \nu u_{xx}$. If $F = (x - x_0)^m a(x)$ with $a(x_0) \neq 0$, the exact local result is

$$u = -\frac{2\nu m}{x - x_0} - 2\nu \frac{a'}{a}.$$

These poles are real mathematical singularities of the logarithmic-derivative map; the following tests determine what happens to them under candidate three-dimensional maps.

For the specific BC solution at $\tau = x = 0$, write

$$\mu = \sum_n p_n \log n, \qquad \sigma^2 = \sum_n p_n (\log n)^2 - \mu^2.$$

Then $F = 1$, $F_x = i\mu$, $F_{xx} = -\sum p_n (\log n)^2$, so the viscosity-one Burgers field satisfies

$$u(0,0) = -2i\mu, \qquad u_x(0,0) = 2\sigma^2.$$

Here $\mu > 0$, since the $n = 2$ summand is positive and all summands are nonnegative. Also $\sigma^2 = \sum p_n (\log n - \mu)^2 > 0$: equality would force every $\log n$ with $p_n > 0$ to equal μ, contradicted by $\log 1 = 0$ and $\log 2 > 0$. Thus this particular source-derived velocity is nonreal and has strictly positive divergence under the direct longitudinal lift.

The exact map to two real fields is also available. Write $u = a + ib$, with real a, b. Multiplication gives $u u_x = a a_x - b b_x + i(a b_x + b a_x)$, hence

$$a_t + a a_x - b b_x = \nu a_{xx}, \qquad b_t + a b_x + b a_x = \nu b_{xx}.$$

Thus the real-part map retains a coupled forcing term $b b_x$. Taking the real part of the heat observable before constructing a shear in the arithmetic shear construction below has the exactly different, explicitly proved linear heat evolution.

25.3 Longitudinal and shear lifts: exact scalar-pressure test

Use the three-dimensional unforced equation with real coordinates (x, y, z) and unchanged positive viscosity ν:

$$\partial_t U + (U \cdot \nabla)U + \nabla p - \nu \Delta U = 0, \qquad \nabla \cdot U = 0.$$

First let $U = (u(t, x), 0, 0)$. Its divergence is u_x. A Burgers solution solves its first momentum equation with spatially constant pressure, but it is incompressible only when $u_x = 0$. Burgers then gives $u_t = 0$. This map therefore retains only constant Burgers fields as incompressible solutions.

Next let $U = (0, u(t, x), 0)$. Its divergence vanishes, and $(U \cdot \nabla)U = 0$. Its momentum residual before pressure is

$$R = (0, u_t - \nu u_{xx}, 0) = (0, -uu_x, 0).$$

A twice differentiable scalar pressure can cancel this residual only if

$$p_x = 0, \qquad p_y = uu_x, \qquad p_z = 0.$$

Commutation of mixed derivatives yields the necessary condition $(uu_x)_x = 0$. It is also sufficient on a rectangular coordinate patch: if $uu_x = c(t)$, choose $p = c(t)y + p_0(t)$. On a periodic domain, integrating $uu_x = (u^2/2)_x = c(t)$ over one period gives $c(t) = 0$; a smooth real function with u^2 spatially constant is spatially constant on the connected circle.

The pressure obstruction is not merely generic rhetoric. For $0 < \epsilon < 1$,

$$F(t, x) = 1 + \epsilon e^{-\nu t} \cos x, \qquad u(t, x) = \frac{2\nu a(t) \sin x}{1 + a(t) \cos x}, \qquad a(t) = \epsilon e^{-\nu t},$$

is a globally positive real heat solution for $t \geq 0$, together with its smooth real Burgers image. Indeed $0 < a(t) \leq \epsilon < 1$ on that time half-line, so $F \geq 1-\epsilon > 0$. At $x = 0$, $u = 0$, $u_x = 2\nu a/(1+a)$, and

$$\partial_x(uu_x)(t, 0) = \frac{4\nu^2 a(t)^2}{(1 + a(t))^2} > 0.$$

Consequently no scalar pressure makes its shear lift an unforced incompressible Navier–Stokes solution even locally near $x = 0$. Conversely, the explicit force $f = (0, -uu_x, 0)$ does make that shear lift satisfy the forced equation with constant pressure. If a logarithmic-derivative pole develops, that force carries products of its singular derivatives; smoothness of an admissible forcing cannot be inferred from this formula.

25.4 A stronger divergence-free completion and all its surviving real Burgers fields

The map

$$U(t, x, y, z) = (u(t, x), -yu_x(t, x), 0)$$

preserves the longitudinal component and cancels its divergence exactly: $u_x - u_x = 0$. Suppose u is a real smooth Burgers solution. Direct differentiation gives

$$R_1 = u_t + uu_x - \nu u_{xx} = 0,$$

$$R_2 = y[-u_{xt} - uu_{xx} + u_x^2 + \nu u_{xxx}].$$

Differentiating Burgers gives $u_{xt} + u_x^2 + uu_{xx} = \nu u_{xxx}$, so

$$R = (0, 2yu_x^2, 0).$$

The scalar-pressure equations are $p_x = 0$, $p_y = -2yu_x^2$, $p_z = 0$. Their xy compatibility is $(u_x^2)_x = 0$. Conversely, when this expression vanishes, $p = -y^2u_x^2 + p_0(t)$ has the required derivatives. On a connected real spatial interval, continuity implies that a real u_x with constant square is constant: if the square is zero it vanishes; if positive, it cannot change sign without crossing zero. Hence $u = a(t)x + b(t)$.

Substitution into Burgers yields exactly

$$a' + a^2 = 0, \qquad b' + ab = 0.$$

For data $a(0) = a_0$, $b(0) = b_0$, the solutions on $1 + a_0t \neq 0$ are

$$a(t) = \frac{a_0}{1+a_0t}, \quad b(t) = \frac{b_0}{1+a_0t}, \quad U = \left(\frac{a_0x + b_0}{1+a_0t}, -\frac{a_0y}{1+a_0t}, 0\right), \quad p = -\frac{a_0^2y^2}{(1+a_0t)^2} + p_0(t).$$

These are exact incompressible Navier–Stokes solutions for every $\nu > 0$; all velocity Laplacians vanish. If $a_0 < 0$, their coefficients diverge at $t = -1/a_0 > 0$. At $t = 0$, however, their velocity grows linearly in space and is nonperiodic. Its squared integral on $\mathbb{R}^3$ is infinite. If $a_0 = 0$ and $b_0 \neq 0$, the constant field also has infinite $\mathbb{R}^3$ energy, although it is a globally smooth periodic solution. Thus this explicitly constructed blowup family does not have the decay, finite energy or periodic initial data of the target problem. No admissibility issue is hidden in the pressure.

25.5 Exact arithmetic maps into genuine three-dimensional Navier–Stokes

The arithmetic heat observable does supply solutions directly, without passing through its logarithmic derivative. On $\mathbb{R}^3$, set

$$q(t,x) = \operatorname{Re} F_\beta(\nu t, x), \quad U(t,x,y,z) = (0, q(t,x), 0), \quad p = p_0(t).$$

The chain rule gives $q_t = \nu q_{xx}$. Hence the divergence, convection, pressure gradient and heat residual all vanish in the Navier–Stokes equation. This is a genuine real-valued, viscosity-preserving morphism. The derivative bounds in the BC heat calculation above prove global smoothness for $t \geq 0$. For every such time, all summands at $x = 0$ are nonnegative and the $n = 1$ summand gives $q(t, 0) \geq 1/\zeta(\beta) > 0$. Continuity therefore gives a nonempty spatial interval on which its magnitude is bounded below at that time. Since the field is independent of (y, z), its $\mathbb{R}^3$ energy is infinite at every $t \geq 0$.

A periodic finite-energy version follows from an explicitly specified bounded arithmetic projection. Fix an integer $b \geq 2$, a length $\ell > 0$, and let P_b be the orthogonal projection onto the closed span of $\{\varepsilon_{b^k} : k \geq 0\}$. Define

$$c = \frac{2\pi}{\ell \log b}, \qquad G(t,x) = \operatorname{Re} \varphi_\beta \left(P_b e^{-\nu c^2 t H^2} e^{icxH}\right).$$

The state is unchanged. Direct basis evaluation gives every coefficient and frequency:

$$G(t,x) = \frac{1}{\zeta(\beta)} \sum_{k=0}^{\infty} b^{-k\beta} e^{-\nu(2\pi k/\ell)^2 t} \cos(2\pi k x/\ell).$$

For each finite derivative order, geometric decay in k dominates its polynomial factor, uniformly for $t \geq 0$. Thus this is a globally smooth periodic heat solution satisfying $G_t = \nu G_{xx}$. On the cubic torus $(\mathbb{R}/\ell\mathbb{Z})^3$,

$$U = (0, G(t,x), 0), \qquad p = p_0(t)$$

is an exact incompressible Navier–Stokes solution. Taking $\ell = 1$ gives precisely the specified unit torus $\mathbb{R}^3/\mathbb{Z}^3$, with the same viscosity and pressure. Orthogonality of the cosine modes, established by integrating their product over $[0, \ell]$, gives the finite energy

$$\int_{[0,\ell]^3} |U|^2 = \frac{\ell^3}{\zeta(\beta)^2} \left[1 + \frac{1}{2} \sum_{k \geq 1} b^{-2k\beta} e^{-2\nu(2\pi k/\ell)^2 t} \right].$$

Each nonconstant summand decreases in time. Termwise differentiation is justified by the same geometric bound and gives $\frac{d}{dt} \int |U|^2 = -2\nu \int |\nabla U|^2$. This supplies the exact state, observable, time, spatial period, real velocity, pressure and viscosity dictionary for a completed fluid bridge. It produces globally smooth shears, not a singularity.

25.6 Multidimensional Cole–Hopf and finite-energy incompressibility

Let a positive real F solve $F_t = \nu \Delta F$ on $\mathbb{R}^3$, and let $L = \log F$, $U = -2\nu \nabla L$. Direct differentiation gives

$$L_t = \nu(\Delta L + |\nabla L|^2), \quad U_t = -2\nu^2 \nabla \Delta L - 2\nu^2 \nabla |\nabla L|^2.$$

The Hessian of L is symmetric, so $(U \cdot \nabla)U = 4\nu^2 \operatorname{Hess}(L) \nabla L = 2\nu^2 \nabla |\nabla L|^2$, while $\nu \Delta U = -2\nu^2 \nabla \Delta L$. Thus U solves the vector Burgers equation. Its incompressibility is the exact additional equation $\Delta L = 0$. In that case every component of U is harmonic. If U is square integrable on $\mathbb{R}^3$, the harmonic mean-value property and Cauchy–Schwarz give, for each component and any radius R,

$$|U_i(x)| \leq |B_R|^{-1/2} \|U_i\|_{L^2(\mathbb{R}^3)}.$$

Letting R tend to infinity proves $U_i(x) = 0$. Therefore the real positive scalar Cole–Hopf construction cannot yield a nonzero finite-energy incompressible velocity on all of $\mathbb{R}^3$.

On a torus, a smooth periodic gradient velocity with both zero curl and zero divergence is spatially constant: integrate by parts to obtain

$$\int |\nabla U|^2 = \int (|\nabla \times U|^2 + |\nabla \cdot U|^2) = 0.$$

If L itself is periodic, integration of each derivative over its period also forces that constant to be zero. These statements concern this exact map; they exclude no other nonlinear arithmetic-to-fluid correspondence.

25.7 Completed zeta and Newman time: coordinate and singularity checks

The source retains the Hermite seed

$$h(x) = \frac{\pi}{2} x^2 (2\pi x^2 - 3) e^{-\pi x^2}, \quad S(\lambda) = \sum_{n \geq 1} h(n\lambda), \quad k(v) = e^{v/2} S(e^v).$$

The Mellin constants can be checked completely. For $\operatorname{Re} s > -2$, integration with $q = \pi x^2$ gives

$$\begin{aligned} \mathcal{M}h(s) &= \pi^{-s/2} \left[\tfrac{1}{2}\Gamma(s/2 + 2) - \tfrac{3}{4}\Gamma(s/2 + 1) \right] \\ &= \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2). \end{aligned}$$

The expression is interpreted at removable singularities by its convergent integral. For the real Fourier convention $\widehat{f}(q) = \int_{\mathbb{R}} f(x) e^{-2\pi i x q}\, dx$, differentiation of the Gaussian transform gives

$$\widehat{x^2 e^{-\pi x^2}} = (1/(2\pi) - q^2) e^{-\pi q^2}, \qquad \widehat{x^4 e^{-\pi x^2}} = (q^4 - 3q^2/\pi + 3/(4\pi^2)) e^{-\pi q^2}.$$

Their linear combination with coefficients π^2 and $-3\pi/2$ is $h(q)$. Thus $\widehat{h} = h$ and $\int h = h(0) = 0$. Poisson summation for this Schwartz function yields $2S(\lambda) = 2\lambda^{-1}S(1/\lambda)$; no constant term remains. This proves $k(-v) = k(v)$. For $\lambda \geq 1$, the absolute series is bounded by $C\lambda^4 e^{-\pi\lambda^2}$, because $\sum_{n\geq 1} n^4 e^{-\pi(n^2-1)}$ converges. Applying symmetry to the signed sum gives $|S(\lambda)| \leq C\lambda^{-5}e^{-\pi/\lambda^2}$ for $0 < \lambda \leq 1$. This latter estimate uses cancellation; it does not bound the series of absolute values of the original small-λ summands. The two bounds for S, including the extra logarithmic powers introduced by differentiation in its Mellin variable, make its Mellin integral entire.

For $\operatorname{Re} s > 1$, the sum of absolute termwise integrals is $\sum n^{-\operatorname{Re} s} \int_0^\infty |h(x)| x^{\operatorname{Re} s - 1} dx < \infty$. Fubini and the substitution $x = n\lambda$ consequently prove

$$\mathcal{M}S(s) = \zeta(s)\mathcal{M}h(s) = \xi(s)/4, \quad \xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The entire continuation follows from the just-proved bounds and the identity theorem. The Newman function is

$$H_t(z) = \frac{1}{2}\int_{\mathbb{R}} e^{tv^2/4} k(v) e^{izv/2}\, dv, \qquad H_0(z) = \frac{1}{8}\xi(1/2 + iz/2).$$

The value at zero time follows by substituting $\lambda = e^v$ in the preceding Mellin identity and retaining its extra factor $1/2$. On any compact complex (t, z) set, the factors introduced by a fixed number of derivatives are bounded by a polynomial in $|v|$ times $e^{C(v^2+|v|)}$. The two displayed estimates for S dominate that expression at both ends, proving the differentiation under the integral used next.

Differentiating its integrand in t multiplies it by $v^2/4$; twice differentiating in z multiplies it by $-v^2/4$. Hence H_t obeys backward heat in its displayed real Newman time. With $F(\tau, z) = H_{-\tau}(z)$, forward heat is $F_\tau = F_{zz}$, and $u = -2F_z/F$ has positive viscosity one. The completed coordinate $s = 1/2 + iz/2$ is invertible with $z = -2i(s - 1/2)$; for $X_t(s) = 8H_t(-2i(s - 1/2))$, the chain rule gives the unambiguous equation $\partial_t X_t(s) = \frac{1}{4}\partial_s^2 X_t(s)$. The logarithmic derivative $w = -(1/2)X_s/X$ has viscosity $1/4$ in that complex coordinate.

For a zero $\rho = \sigma + i\gamma$ of ξ, its Newman coordinate is $z = 2\gamma + i(1 - 2\sigma)$. A critical-line zero becomes a real spatial pole of the Cole–Hopf field already at $\tau = 0$. The source's interval certificate at $t = 0$ is a real simple zero in $[28.269450279469, 28.269450287469]$. If that certificate is used as input, the resulting real-axis Burgers velocity is initially singular there; it is not a solution emerging from smooth data at time zero. A nonreal zero is a pole at a complex spatial coordinate, which does not by itself make a velocity on real $\mathbb{R}^3$ singular. Time reversal also reverses the sign in the viscosity equation as displayed above; it cannot be suppressed when transporting a collision.

The arithmetic spectral fibre coordinates in source lines 328–335 are different and explicit: $z = i\rho = -\gamma + i\sigma$, with inverse $\rho = -iz$. The map between these spectral and Newman coordinates is $z_{\text{Newman}} = -2z_{\text{spectral}} + i$. Substitution gives $-2(-\gamma + i\sigma) + i = 2\gamma + i(1 - 2\sigma)$. This is an exact affine relation between the two zero-coordinate presentations, but it assigns no three real spatial coordinates, divergence-free real velocity, pressure or initial-time regularity to a fibre.

25.8 Primary-source and verification boundary

The fluid/gravity source is Bredberg, Keeler, Lysov and Strominger, *From Navier–Stokes To Einstein*, https://arxiv.org/abs/1101.2451, §5.1, equations (14)–(20). The horizon temperature convention is from Eling, Fouxon and Oz, *The Incompressible Navier–Stokes Equations From Black Hole Membrane Dynamics*, https://arxiv.org/abs/0905.3638. The real-zero interval quoted above is a source certificate, not a new interval replay in this section. The source audit and its exact file hashes are retained in `research/arithmetic_fluid_bridge.md`. The actual periodic arithmetic shears meet the velocity, pressure and positive-viscosity equations and remain globally smooth. The tested logarithmic-derivative lifts supply the exact defects computed above; none gives an admissible classical counterexample.

26 Original coordinates and the retained cover

The original source coordinate triple is (x, y, w) and its target is (a, b, c). Its polynomial is

$$\begin{aligned} F_1 &= (1+xy)^3 w + y^2(1+xy)(4+3xy), \\ F_2 &= y + 3x(1+xy)^2 w + 3xy^2(4+3xy), \\ F_3 &= 2x - 3x^2 y - x^3 w. \end{aligned} \tag{417}$$

All coordinate spaces here are complex unless a real locus is stated. The original $x \neq 0$ inverse chart is

$$\begin{aligned} P_{a,b,c}(r) = cr^3 - 2r^2 + br - 2a, \qquad \alpha = \tfrac{1}{2} P_r(r), \\ (x, y, w) = (\alpha^{-1}, r - \alpha, 5\alpha^2 - 3r\alpha - c\alpha^3), \qquad \alpha \neq 0. \end{aligned} \tag{418}$$

Here is the complete verification needed for this chart. At a point with $x \neq 0$, set $\alpha = 1/x$ and $r = y + 1/x$; solving $F_3 = c$ gives the third coordinate in (418). Substitution into the other original components gives

$$F_2 = 4r + 2\alpha - 3cr^2, \qquad F_1 = r^2 + r\alpha - cr^3.$$

The first equation equals b precisely when $2\alpha = P_r(r)$; after that substitution $2(F_1 - a) = P(r)$. This proves both directions and uniqueness of the root label. A multiple root gives no finite point in this chart. Directly, $F(0, y, w) = (w + 4y^2, y, 0)$, so $c = 0$ has the additional inverse point $(0, b, a - 4b^2)$ and $c \neq 0$ has none.

On $b = c = 0$ the root cover is $a = -r^2$, with $\alpha = -2r$. The arithmetic coordinate is the specific further map

$$a = 2s, \qquad s = -r^2/2. \tag{419}$$

Its unramified domain is $\mathbb{C}_r^* = \mathbb{C}_r \setminus \{0\}$ over $\mathbb{C}_s^*$. The point $r = 0$ is the ramification point of the completed cover and is excluded by $\alpha \neq 0$ from the original inverse chart. These are explicit restrictions of the original map.

The constant-coefficient heat evolution in the original root variable is a distinct, explicitly retained map of this very polynomial:

$$P(\tau, r) = P_{a_0 + 2\tau, b_0 + 6c\tau, c}(r), \qquad P_\tau = 6cr - 4 = P_{rr}.$$

Here τ is forward heat time. The explicit reversal $\widetilde{P}(t, r) = P(-t, r)$ satisfies $\widetilde{P}_t = -\widetilde{P}_{rr}$. This is the backward sign convention; the arithmetic Newman generator below acts in its specified s coordinate. The arithmetic pullback below transports differentiation in $s = a/2$, which must also carry the derivative of r with respect to s. Both operators will therefore be displayed with their exact coordinates.

For an open set $V \subset \mathbb{C}_s^*$, set $\widetilde{V} = \{r : -r^2/2 \in V\}$. Every holomorphic function on $\widetilde{V}$ has a unique decomposition

$$f(r) = q_0(s) + r q_1(s), \qquad q_0, q_1 \in \mathcal{O}(V). \tag{420}$$

Indeed its even part is $(f(r) + f(-r))/2$ and its odd coefficient is $(f(r) - f(-r))/(2r)$. Both are invariant under $r \mapsto -r$ and therefore define holomorphic functions of s on local inverse charts, agreeing on their overlaps. These formulas also prove uniqueness. They define an isomorphism $T_V : \mathcal{O}(V)^2 \to \mathcal{O}(\widetilde{V})$. The same decomposition holds over V containing zero: power series show that the odd numerator divided by r extends holomorphically and that each even power series is a power series in $s = -r^2/2$. The derivative operator below, however, is first defined only over $V \subset \mathbb{C}^*$.

27 Connection, square, and branch coupling

Write a section as the column $q = (q_0, q_1)^\mathsf{T}$ in the ordered basis $(1, r)$. On the unramified cover, $ds/dr = -r$, so differentiation in the original spectral coordinate is $\partial_s^{\mathrm{lift}} = -r^{-1}\partial_r$. In particular $\partial_s r = -1/r = r/(2s)$. Consequently

$$D = \partial_s + A(s), \quad A(s) = \begin{pmatrix} 0 & 0 \\ 0 & 1/(2s) \end{pmatrix}, \qquad T_V D = \partial_s^{\mathrm{lift}} T_V. \tag{421}$$

This is a connection: for $g \in \mathcal{O}(V)$, $D(gq) = g'q + gDq$. Its curvature in this one complex coordinate is zero; the nontrivial information is its ramification and monodromy.

Proposition 27.1 (The full second derivative and its commutators)**.** *On $\mathcal{O}(V)^2$ one has*

$$D^2 = \begin{pmatrix} \partial_s^2 & 0 \\ 0 & \partial_s^2 + s^{-1}\partial_s - 1/(4s^2) \end{pmatrix}, \tag{422}$$

$$R = \begin{pmatrix} 0 & -2s \\ 1 & 0 \end{pmatrix}, \quad R^2 = -2sI, \quad [D, R] = -R^{-1}, \tag{423}$$

$$[D^2, R] = -2R^{-1}D - R^{-3}. \tag{424}$$

Here R is multiplication by the original root r; commutators include differentiation of the matrix coefficients.

Proof. Expanding $(\partial_s + A)^2$ gives $\partial_s^2 + 2A\partial_s + A' + A^2$. Its lower diagonal constant is $-1/(2s^2) + 1/(4s^2) = -1/(4s^2)$, proving (422). Multiplication $r(q_0 + rq_1) = -2sq_1 + rq_0$ proves R and R^2. Matrix multiplication then gives

$$R' + AR - RA = \begin{pmatrix} 0 & -1 \\ 1/(2s) & 0 \end{pmatrix} = -R^{-1}.$$

Differentiating $RR^{-1} = I$ covariantly gives $[D, R^{-1}] = -R^{-1}[D, R]R^{-1} = R^{-3}$. Finally $[D^2, R] = D[D, R] + [D, R]D = -2R^{-1}D - R^{-3}$. This proves every term, including the zeroth-order branch term. □

The sheet involution is $J = \mathrm{diag}(1, -1)$; it commutes with D, D^2 and anticommutes with R. The trace is $(q_0, q_1) \mapsto 2q_0$, with kernel the odd summand, and its half has the section $q_0 \mapsto (q_0, 0)$. D preserves the two summands on the punctured base, whereas R exchanges them and couples the odd coefficient into $-2sq_1$. This describes exactly the information lost by tracing.

On a simply connected $V \subset \mathbb{C}^*$ choose a holomorphic root $r(s)$ satisfying $r(s)^2 = -2s$. The invertible matrix $G = \mathrm{diag}(1, r(s))$ obeys $D = G^{-1}\partial_s G$. Horizontal coefficients satisfy $q_0' = 0$ and $q_1' + q_1/(2s) = 0$, hence $(q_0, q_1) = (c_0, c_1/r(s))$. Continuation once around zero changes the sign of the latter coefficient. Thus the monodromy is exactly $\mathrm{diag}(1, -1)$; no single-valued global choice of $r(s)$ on $\mathbb{C}^*$ is implied by this local gauge formula.

28 The actual arithmetic heat function

We use the convention of [4, Section 1, equations (1)–(4)]:

$$\begin{aligned} \Phi(u) &= \sum_{n \ge 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \\ H_t(z) &= \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\, du, \\ Z_t(s) &= 8H_t(2s), \qquad Z_0(s) = \Xi(s) = \xi(1/2 + is), \\ \xi(\sigma) &= \tfrac{1}{2}\sigma(\sigma - 1)\pi^{-\sigma/2}\Gamma(\sigma/2)\zeta(\sigma). \end{aligned} \tag{425}$$

The Mellin and spectral coordinates are related by $\sigma = 1/2 + is$, with inverse $s = -i(\sigma - 1/2)$. They are not silently identified.

We rederive the integral identity and its analytic bounds. Set

$$h(x) = \frac{\pi}{2}x^2(2\pi x^2 - 3)e^{-\pi x^2}, \quad S(\lambda) = \sum_{n\geq 1} h(n\lambda), \quad k(v) = e^{v/2}S(e^v).$$

For the Fourier convention $\widehat{h}(q) = \int_{\mathbb{R}} h(x)e^{-2\pi i x q}dx$, the Gaussian transform and its second and fourth derivatives give

$$\widehat{x^2e^{-\pi x^2}} = (1/(2\pi) - q^2)e^{-\pi q^2},$$
$$\widehat{x^4e^{-\pi x^2}} = (q^4 - 3q^2/\pi + 3/(4\pi^2))e^{-\pi q^2}.$$

Multiplying by $-3\pi/2$ and π^2 respectively gives $\widehat{h} = h$. Since $h(0) = 0$, Poisson summation gives $S(\lambda) = \lambda^{-1}S(1/\lambda)$ and $k(v) = k(-v)$. For $\lambda \geq 1$, $|S(\lambda)| \leq C\lambda^4 e^{-\pi\lambda^2}$, because the remaining series is bounded by a constant times $\sum_{n\geq 1} n^4 e^{-\pi(n^2-1)} < \infty$. The symmetry gives the corresponding bound as $\lambda \downarrow 0$. Each fixed number of v derivatives introduces only finitely many additional powers of ne^v. It follows, with constants depending on the derivative order, that

$$|k^{(j)}(v)| \leq C_j e^{C_j|v|}e^{-\pi e^{2|v|}}. \tag{426}$$

Termwise differentiation on compact sets is justified by the same summable bounds. On the negative half-line differentiate the proved evenness identity. This proves (426) on both ends.

For $\Re\sigma > 1$, absolute integrability permits summation inside the Mellin integral. The substitution $q = \pi x^2$ gives

$$\begin{aligned}\int_0^\infty h(x)x^{\sigma-1}dx &= \pi^{-\sigma/2}\left[\tfrac{1}{2}\Gamma(\sigma/2+2) - \tfrac{3}{4}\Gamma(\sigma/2+1)\right] \\ &= \frac{\sigma(\sigma-1)}{8}\pi^{-\sigma/2}\Gamma(\sigma/2), \\ \int_0^\infty S(\lambda)\lambda^{\sigma-1}d\lambda &= \xi(\sigma)/4.\end{aligned}$$

The two-ended bounds make the last integral entire in σ, so the equality extends by analytic continuation. Substituting $\lambda = e^v$ and $\sigma = 1/2 + is$ proves

$$Z_t(s) = 4\int_{\mathbb{R}} e^{tv^2/4}k(v)e^{isv}\,dv. \tag{427}$$

Indeed direct expansion also gives $\Phi(u) = 2k(2u)$; the substitution $v = 2u$ and the evenness of k carry the stated H_t integral to (427), retaining the factors $2, 4, 8$. The bound (426) dominates every fixed derivative uniformly on compact complex (t, s) sets. Thus Z is jointly entire and differentiation under its integral proves exactly

$$\partial_t Z_t = -\tfrac{1}{4}\partial_s^2 Z_t. \tag{428}$$

This supplies the actual initial datum and the actual time orientation.

29 An explicit domain for the heat evolution

The differential expression alone does not specify a heat evolution on every entire-function space. We now give a precise invariant domain containing the actual kernel, without assigning it to the topology of another spectral construction.

Definition 29.1. Let $\mathscr{E}$ consist of $\phi \in C^\infty(\mathbb{R}, \mathbb{C})$ for which

$$p_{A,B,m,j}(\phi) = \int_{\mathbb{R}} |\phi^{(j)}(v)|(1+|v|)^m e^{Av^2+B|v|}\,dv < \infty$$

for every nonnegative integer A, B, m, j. Give it these seminorms. Define

$$T\phi(s) = 4\int_{\mathbb{R}} \phi(v)e^{isv}\,dv, \qquad \mathscr{A} = T\mathscr{E},$$

and equip $\mathscr{A}$ with the transported topology.

The map T is injective. To give the needed uniqueness argument, let $T\phi$ vanish on the real line and let $g_\epsilon(v) = (4\pi\epsilon)^{-1/2}e^{-v^2/(4\epsilon)}$. Its Gaussian integral representation is $g_\epsilon(v) = (2\pi)^{-1}\int_{\mathbb{R}} e^{-\epsilon s^2}e^{isv}ds$. Fubini is valid because $\phi \in L^1$ and the Gaussian is integrable; it expresses $\phi * g_\epsilon$ as a Gaussian-weighted integral of the vanishing Fourier transform. Thus $\phi * g_\epsilon = 0$. As $\epsilon \downarrow 0$, these convolutions converge to ϕ in L^1: translations are continuous in L^1, first for continuous compactly supported functions by uniform continuity, then for all L^1 functions by approximation; splitting the Gaussian integral into a small translation ball and its vanishing tail proves the claim. Consequently $\phi = 0$ almost everywhere and, being continuous, everywhere. This also justifies the transported topology without an unspecified choice of representative.

Proposition 29.2 (Heat evolution with its inverse)**.** *For every complex t, multiplication $E_t\phi(v) = e^{tv^2/4}\phi(v)$ is a continuous automorphism of $\mathscr{E}$. The operators $U_t = TE_tT^{-1}$ on $\mathscr{A}$ satisfy*

$$\begin{gathered} U_tU_u = U_{t+u}, \qquad U_t^{-1} = U_{-t}, \qquad \partial_tU_t = -\tfrac{1}{4}\partial_s^2U_t, \\ U_t\Xi = Z_t, \qquad U_t\partial_s = \partial_sU_t, \qquad U_tM_sU_{-t} = M_s - \tfrac{t}{2}\partial_s. \end{gathered} \tag{429}$$

Here M_s is multiplication by the unchanged spectral coordinate.

Proof. Every v derivative of $e^{tv^2/4}\phi$ is a finite sum of $e^{tv^2/4}$ times a polynomial in v times a derivative of ϕ. Each target seminorm is bounded by finitely many source seminorms with larger A, m; on compact t sets these bounds are uniform. The inverse multiplier is $e^{-tv^2/4}$. Differentiation in t inserts $v^2/4$, while two s derivatives in the transform insert $-v^2$, proving the generator and the group identities. The kernel bound proves $k \in \mathscr{E}$, so $U_t\Xi = Z_t$ follows from (427).

Integration by parts gives $M_sT\phi = T(i\phi')$. For completeness the boundary terms vanish: each derivative of $e^{isv}\phi$ is integrable on a horizontal real-v line and the function itself is integrable, so its limits at both ends exist and are zero. The same argument applies after exponential weights. Thus multiplication by s preserves $\mathscr{A}$; differentiation does also, since $\partial_sT\phi = T(iv\phi)$. Finally

$$e^{tv^2/4}i\partial_v(e^{-tv^2/4}\phi) = i\phi' - \tfrac{t}{2}iv\phi.$$

Applying T proves the last identity in (429). All displayed operators have the domains just specified. $\square$

30 Transport of the moving arithmetic relation

Put $B_t = M_s - (t/2)\partial_s$. For every polynomial p, the automorphism calculation proves

$$U_t(p(s)\Xi(s)) = p(B_t)Z_t(s). \tag{430}$$

The right side is the polynomial in the actual differential operator, with composition order retained. For example the first two nonconstant relations are

$$\begin{aligned} &B_tZ_t = sZ_t - \tfrac{t}{2}Z_t', \\ &B_t^2Z_t = (s^2 - t/2)Z_t - tsZ_t' + \tfrac{t^2}{4}Z_t''. \end{aligned}$$

These follow from $\partial_s(sq) = q + s\partial_s q$. In particular, an evolving polynomial multiple is not obtained by silently leaving multiplication by s fixed.

One can take a closure in the topology that has actually been defined: let $N_0 = \overline{\mathbb{C}[s]\Xi}^{\mathscr{A}}$ and $N_t = U_t N_0$. Continuity and invertibility give $N_t = \overline{\{p(B_t)Z_t : p \in \mathbb{C}[s]\}}^{\mathscr{A}}$. M_s preserves N_0, since it is continuous and preserves its dense generating subspace; hence B_t preserves N_t. The map $[f] \mapsto [U_t f]$ is a well-defined continuous isomorphism $\mathscr{A}/N_0 \to \mathscr{A}/N_t$ with inverse induced by U_{-t} and intertwines the induced M_s with the induced B_t. This proof applies to these explicit spaces. It makes no identification with the topology or completion of the Connes quotient [5, Section 3.6].

There is a second exact description, local at the moving divisor. Let $Z = Z_t(s)$ and $\mathcal{P} = \partial_t + \frac{1}{4}\partial_s^2$. For every holomorphic $g(t,s)$ the complete product calculation is

$$\mathcal{P}(Zg) = Z\mathcal{P}g + \tfrac{1}{2}Z_s g_s. \tag{431}$$

The term $\mathcal{P}Z$ is zero by (428); the mixed product term is retained. On the complement of $Z = 0$, set $b = Z_s/Z$. Direct differentiation of f/Z gives the exact conjugation

$$Z\mathcal{P}Z^{-1} = \partial_t + \tfrac{1}{4}\partial_s^2 - \tfrac{1}{2}b\partial_s + \tfrac{1}{2}b^2. \tag{432}$$

To check the potential term before canceling any expression, it is $\frac{1}{2}(Z_s/Z)^2 - \frac{1}{4}Z_{ss}/Z - Z_t/Z$; the last two terms cancel by the established heat equation. Thus the operator in (432) sends Zg exactly to $Z\mathcal{P}g$. Its coefficients are meromorphic at zeros of Z, but its restriction to the ideal of holomorphic multiples of Z has the displayed holomorphic value. Division by Z on this ideal is well-defined and unique by analytic continuation. There is no claim that division is holomorphic on arbitrary sections at a zero.

The identical calculation on the two-component module replaces ∂_s by D:

$$Z(\partial_t + \tfrac{1}{4}D^2)Z^{-1} = \partial_t + \tfrac{1}{4}D^2 - \tfrac{1}{2}bD + \tfrac{1}{2}b^2 I. \tag{433}$$

Indeed $D(Zq) = Z_s q + ZDq$, so the product expansion is exactly the scalar one, retaining $A(s)$ in D. This gives the full moving arithmetic relation on the retained cover without assuming a derivative acts on a frozen quotient.

31 The exact branch-sensitive heat defect

The actual pulled arithmetic solution is $q(t,s) = (Z_t(s), 0)$. Equation (422) proves $(\partial_t + \frac{1}{4}D^2)q = 0$. Its function on the cover is $f(t,r) = Z_t(-r^2/2)$. By differentiating $s = -r^2/2$ twice, the scalar generator is

$$L_r = -\frac{1}{4r^2}\partial_r^2 + \frac{1}{4r^3}\partial_r, \qquad \partial_t f = L_r f \quad (r \neq 0). \tag{434}$$

This retains the original spectral diffusion; it does not substitute the constant-coefficient r-heat equation of the fibre polynomial.

The branch observable rZ_t is the column $Rq = (0, Z_t)$. From the already proved commutator its complete defect is

$$\begin{aligned}(\partial_t + \tfrac{1}{4}D^2)(Rq) &= -\tfrac{1}{2}R^{-1}Dq - \tfrac{1}{4}R^{-3}q \\ &= \begin{pmatrix} 0 \\ Z_t'/(4s) - Z_t/(16s^2) \end{pmatrix}.\end{aligned} \tag{435}$$

Equivalently, as a function on the cover it is $-Z_t'/(2r) - Z_t/(4r^3)$. These terms are the exact dynamics of the retained odd observable, not a reason to discard that observable or declare it unrelated. For real t, $Z_t(0) > 0$ (proved from the integral in (446)), so its leading branch pole is nonzero. The original arithmetic solution itself remains entire at the branch point.

32 Keeping the other original fibre coefficients

The preceding branch at $s = 0$ belongs to the particular slice $b = c = 0$. We now retain the full source family, with b, c fixed and $a = 2s$. Its cover is defined by

$$cr^3 - 2r^2 + br - 4s = 0, \qquad s = \frac{cr^3 - 2r^2 + br}{4}. \tag{436}$$

On its original inverse chart $P_r = 3cr^2 - 4r + b = 2\alpha \neq 0$, implicit differentiation gives

$$\partial_s^{\text{lift}} = \frac{4}{P_r}\partial_r, \qquad L_{b,c} = -\frac{4}{P_r^2}\partial_r^2 + \frac{4P_{rr}}{P_r^3}\partial_r. \tag{437}$$

To check the second identity, apply the first operator twice: $16P_r^{-2}\partial_r^2 - 16P_{rr}P_r^{-3}\partial_r$, then multiply by $-1/4$. Thus the actual function $Z_t((cr^3 - 2r^2 + br)/4)$ satisfies this exact transported equation on $P_r \neq 0$. The branch equations, with no coefficient removed, are

$$b = 4r_0 - 3cr_0^2, \qquad s_0 = \frac{r_0^2 - cr_0^3}{2}. \tag{438}$$

They follow by solving $P_r = 0$ and substituting in (436).

For the two-sheet subfamily $c = 0$, retain arbitrary $b \in \mathbb{C}$ and put $\Delta = b^2 - 32s$. The algebraic relation is $r^2 - (b/2)r + 2s = 0$. In the unchanged ordered basis $(1, r)$,

$$R_b = \begin{pmatrix} 0 & -2s \\ 1 & b/2 \end{pmatrix}, \qquad D_b = \partial_s + A_b, \quad A_b = \begin{pmatrix} 0 & 4b/\Delta \\ 0 & -16/\Delta \end{pmatrix}. \tag{439}$$

These formulas hold on $\Delta \neq 0$. Indeed $(b - 4r)^2 = \Delta$ in the algebra, hence $\partial_s r = 4/(b - 4r) = 4(b - 4r)/\Delta$. This gives both coefficient entries of A_b. Multiplication by r gives R_b. Direct differentiation and multiplication then prove

$$\begin{aligned} R_b^2 - \tfrac{b}{2}R_b + 2sI &= 0, \\ [D_b, R_b] &= 4(bI - 4R_b)/\Delta, \\ D_b^2 &= \partial_s^2 + 2A_b\partial_s + (16/\Delta)A_b. \end{aligned} \tag{440}$$

For the last equality, $A_b' = (32/\Delta)A_b$ and $A_b^2 = (-16/\Delta)A_b$. At $b = 0$ these identities recover (421)–(422) exactly.

The other root is $b/2 - r$. Thus the sheet involution and trace in this original basis are respectively

$$J_b = \begin{pmatrix} 1 & b/2 \\ 0 & -1 \end{pmatrix}, \qquad \mathrm{Tr}_b(q_0, q_1) = 2q_0 + (b/2)q_1.$$

$J_b^2 = I$, and J_b commutes with D_b because b is fixed and $A_bJ_b = J_bA_b$. Also $\mathrm{Tr}_b\, D_b q = \partial_s \mathrm{Tr}_b\, q$, as the two added terms are $8bq_1/\Delta - 8bq_1/\Delta = 0$. The trace kernel is exactly the set of pairs $(-bq_1/4, q_1)$; the half-trace has the section $q \mapsto (q, 0)$. This explicitly exhibits the branch information and its coupling when the original linear coefficient is present.

The branch point is $r_0 = b/4$, $s_0 = b^2/32$. The exact coordinate change $\eta = r - b/4$, with inverse $r = \eta + b/4$, gives

$$s = s_0 - \eta^2/2, \qquad Z_t(s) = Z_t(s_0 - \eta^2/2). \tag{441}$$

The arithmetic function and its time origin have not changed. The displayed translation applies to the cover only. For any specified $s_0 \in \mathbb{C}$ there is a coefficient b with $b^2 = 32s_0$, so every base point occurs as a branch point of an original $c = 0$ fibre family. If $Z_t(s_0) = 0$ has multiplicity m, its factorization $Z_t(s) = (s - s_0)^m A(s)$, $A(s_0) \neq 0$, pulls back to $(-1/2)^m\eta^{2m}A(s_0 - \eta^2/2)$ and has multiplicity $2m$. This proves the exact contact with any actual arithmetic zero and keeps its analytic unit.

The original inverse point there has

$$\alpha = b/2 - 2r = -2\eta, \quad x = -1/(2\eta), \quad y = b/4 + 3\eta, \quad w = 26\eta^2 + (3b/2)\eta.$$

These follow directly from (418). At $\eta = 0$ its two chart branches escape, while the separate point $(0, b, 2s_0 - 4b^2)$ remains in the finite fibre. Thus the zero-free origin of the $b = 0$ arithmetic pullback does not remove the branch mechanism from the entire original coefficient family. Conversely, choosing a coefficient to place a branch at a specified base point does not prove that that point is an arithmetic zero, or determine whether a zero lies off the real s axis. The formulas state exactly what each operation retains.

33 The actual arithmetic zero divisor on the retained inverse cover

This section uses the arithmetic spectral coordinate s, the Newman time t, and the original inverse-root coordinate r. In particular, the letter s here is not the completed-zeta argument ρ. The complete coordinate dictionary is

$$a = 2s = -r^2, \qquad z = 2s = -r^2, \qquad \rho = \tfrac{1}{2} + is = \tfrac{1}{2} - \tfrac{i}{2}r^2. \tag{442}$$

The maps $s \mapsto a$ and $s \mapsto z$ have inverses $a \mapsto a/2$ and $z \mapsto z/2$; $s \mapsto \rho$ has inverse $\rho \mapsto -i(\rho - 1/2)$. The last root map is a two-sheet map, whose two local inverses on a simply connected open subset of $\mathbb{C}_s \setminus \{0\}$ are $r = \sqrt{-2s}$ and $r = -\sqrt{-2s}$. No single inverse on all of $\mathbb{C}_s \setminus \{0\}$ is asserted.

The primary analytic conventions below are those of Brad Rodgers and Terence Tao[4, Section 1, equations (1)–(4)] and D. H. J. Polymath[7, Section 3, Proposition 3.1]. The local calculations below prove their transport in the coordinates (442), including the analytic factors in zero motion.

33.1 The specified datum, its moments, and the branch point

Keep the exact kernel and constants

$$\begin{aligned}
\Phi(u) &= \sum_{n=1}^{\infty} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \\
H_t(z) &= \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\, du, \\
Z(t, s) &= 8H_t(2s), \qquad f(t, r) = Z(t, -r^2/2) = 8H_t(-r^2).
\end{aligned} \tag{443}$$

Thus $Z(0, s) = \Xi(s) = \xi(1/2 + is)$ by the cited Fourier identity, where $\xi(\rho) = \rho(\rho - 1)\pi^{-\rho/2}\Gamma(\rho/2)\zeta(\rho)/2$, with its entire continuation at the removable exceptional points. This identity fixes the initial datum; the inverse cubic below is never substituted for it.

For $u \geq 0$ each summand of Φ is positive, since its polynomial factor is

$$\pi n^2 e^{5u}(2\pi n^2 e^{4u} - 3) > 0.$$

For a completely explicit integrable majorant put $C_\Phi = \sum_{n \geq 1} n^4 e^{-\pi(n^2 - 1)} < \infty$. The finiteness follows by comparison with $\sum_{n \geq 1} n^4 e^{-\pi(n-1)}$. With $\lambda = \pi e^{4u} \geq \pi$, one has

$$0 < \Phi(u) \leq 2\pi^2 e^{9u} \sum_{n \geq 1} n^4 e^{-\lambda n^2} \leq 2\pi^2 C_\Phi e^{9u - \pi e^{4u}}.$$

On $|t| \leq T$, $|s| \leq S$, every derivative of the integrand in (443) is bounded by a constant times $u^N \exp(Tu^2 + (9 + 2S)u - \pi e^{4u})$, for an integer $N \geq 0$. This is integrable: $\pi e^{4u}/2$ eventually exceeds $Tu^2 + (9+2S)u + N\log(1+u)$, and $\exp(-\pi e^{4u}/2)$ is integrable. The same argument applies on compact r sets, with $2S$ replaced by a bound for $|r|^2$. Differentiation under the integral and locally uniform holomorphic integration therefore show that Z is entire on $\mathbb{C}_t \times \mathbb{C}_s$, and f is entire on $\mathbb{C}_t \times \mathbb{C}_r$. They give the exact equations

$$Z_t = -\tfrac{1}{4} Z_{ss}, \qquad f_t = \mathcal{L} f, \qquad \mathcal{L} = -\frac{1}{4r^2}\partial_r^2 + \frac{1}{4r^3}\partial_r \quad (r \neq 0). \tag{444}$$

Indeed $f_r = -rZ_s$ and $f_{rr} = -Z_s + r^2 Z_{ss}$, so the two terms in $\mathcal{L}f$ are $Z_s/(4r^2) - Z_{ss}/4$ and $-Z_s/(4r^2)$, respectively. Their sum is $-Z_{ss}/4$.

Define the actual moments, with their original measure,

$$M_k(t) = \int_0^\infty u^{2k} e^{tu^2} \Phi(u)\, du \quad (k = 0, 1, 2, \ldots).$$

The majorant just proved gives $M_k'(t) = M_{k+1}(t)$ and

$$\begin{aligned} \partial_s^{2k} Z(t,0) &= 8(-1)^k 2^{2k} M_k(t), \qquad\qquad \partial_s^{2k+1} Z(t,0) = 0, \\ f(t,r) &= 8\sum_{k=0}^{\infty} \frac{(-1)^k M_k(t)}{(2k)!} r^{4k}. \end{aligned} \tag{445}$$

The series converges locally uniformly in both variables, by expansion of the cosine and the same compact-set majorant. In particular

$$\begin{aligned} \Xi(0) &= 8M_0(0) = -\frac{\Gamma(1/4)\zeta(1/2)}{8\pi^{1/4}} > 0, \\ \Xi'(0) &= 0, \qquad \Xi''(0) = -32 M_1(0) < 0, \\ f(t,r) &= 8M_0(t) - 4M_1(t) r^4 + \tfrac{1}{3} M_2(t) r^8 + O(r^{12}). \end{aligned}$$

The equality with $\Gamma(1/4)\zeta(1/2)$ follows by evaluating the specified completed-zeta identity, while both strict signs follow directly from the positive integrals. Neither sign requires a numerical evaluation of $\zeta(1/2)$.

For every real t and v there is the stronger exact assertion, also recorded in the first paragraph of Polymath's Section 3[7],

$$Z(t, iv) = 8\int_0^\infty e^{tu^2} \Phi(u) \cosh(2vu)\, du > 0. \tag{446}$$

In particular the actual arithmetic zero divisor avoids $s = 0$ for every real Newman time. The completed root cover has a branch point there, but no arithmetic zero reaches that branch point at any real time. Evenness of the cosine also proves, for all complex t, s, r,

$$Z(t,-s) = Z(t,s), \qquad f(t,-r) = f(t,r), \qquad f(t,ir) = f(t,r). \tag{447}$$

For real t complex conjugation gives $\overline{Z(t,s)} = Z(t,\bar{s})$ and $\overline{f(t,r)} = f(t,\bar{r})$, because the kernel is real.

33.2 Exact function maps and the removable and nonremovable terms

Write $\mathcal{O}(\mathbb{C})$ for entire functions with the topology of uniform convergence on compact sets. The pullback

$$\pi^* : \mathcal{O}(\mathbb{C}_s) \longrightarrow \mathcal{O}(\mathbb{C}_r), \qquad q \longmapsto q(-r^2/2)$$

is injective, since every $s \in \mathbb{C}$ has a root of $r^2 = -2s$. Its image is exactly the even entire functions. To prove the latter statement without a choice of square root, expand an even entire function as $g(r) = \sum_{n\geq 0} a_{2n} r^{2n}$ and define $q(s) = \sum_{n\geq 0} (-2)^n a_{2n} s^n$. This series converges for every s: choose

$R > \sqrt{2|s|}$ and apply the Cauchy bound $|a_{2n}| \le \max_{|r|=R} |g(r)| R^{-2n}$. It gives $q(-r^2/2) = g(r)$, and proves uniqueness by comparing coefficients. It also proves continuity of the inverse on compact sets. Every entire cover function has the unique decomposition

$$g(r) = q_0(-r^2/2) + r q_1(-r^2/2).$$

Here q_0 is recovered from $(g(r) + g(-r))/2$ and q_1 from $(g(r) - g(-r))/(2r)$, with the latter expression extended at $r = 0$ by its power series. The sheet involution $\iota g(r) = g(-r)$ has trace map $g \mapsto g + \iota g$ with kernel exactly the odd entire functions. These are maps of actual holomorphic functions, before any arithmetic quotient is formed.

For every integer $n \ge 0$ the operator in (444) gives

$$\mathcal{L}(r^n) = \frac{n(2-n)}{4} r^{n-4}. \tag{448}$$

Consequently, for $g(r) = \sum_{n \ge 0} a_n r^n$, its Laurent expansion is

$$\mathcal{L}g = \frac{a_1}{4r^3} - \frac{3a_3}{4r} + \sum_{n \ge 4} \frac{n(2-n)}{4} a_n r^{n-4}.$$

The terms $n = 0, 2$ are zero. Thus $\mathcal{L}g$ extends to an entire function exactly when $a_1 = a_3 = 0$. Its value at zero in that case is $-2a_4 = -g^{(4)}(0)/12$. In particular $\mathcal{L}f(t, 0) = 8M_1(t) = f_t(t, 0)$: the singular coordinate coefficients cancel on the actual arithmetic datum.

The largest subspace of $\mathcal{O}(\mathbb{C}_r)$ on which every iterate $\mathcal{L}^j$ remains entire is the even subspace. For even monomials successive applications of (448) reach r^0 or r^2, which are annihilated, without producing a pole. Equivalently, $\mathcal{L}^j \pi^* q = (-1/4)^j \pi^* q^{(2j)}$. Conversely, let a_n be the first nonzero odd coefficient of g. When $n = 4j + 1$, applying $\mathcal{L}^{j+1}$ produces a nonzero multiple of r^{-3} from this term; when $n = 4j + 3$ it produces a nonzero multiple of r^{-1}. Every factor $(n - 4\ell)(2 - n + 4\ell)/4$ is nonzero for these odd exponents. Distinct initial powers retain distinct exponents after the same number of applications, so no cancellation with a higher power is possible. Even powers cannot contribute a pole. This proves the assertion and maximality. It asserts stability of all differential iterates, not existence of a heat evolution for every even entire datum.

There is an equivalent spacetime statement. A function g holomorphic on a product of a time disk with a full disk about $r = 0$, satisfying $g_t = \mathcal{L}g$ for $r \ne 0$, is even in r. Indeed successive time differentiation gives $\mathcal{L}^j g = \partial_t^j g$ on the punctured disk, and the right side is holomorphic at zero for every j. The preceding first-odd-coefficient argument, which is local in r, applies at every time. The equation therefore descends there through the proved even-function inverse to $q_t = -q_{ss}/4$. On a connected punctured plane the kernel of $\mathcal{L}$ consists exactly of $A + Br^2$: multiplying the equation by $4r^3$ gives $r g_{rr} = g_r$, hence $(g_r/r)' = 0$ and then $g = A + Br^2$.

33.3 The original cubic, all finite fibres, and the zero divisor

Keep the original map in spatial coordinates (x, y, w):

$$\begin{aligned}
F_1 &= (1 + xy)^3 w + y^2(1 + xy)(4 + 3xy),\\
F_2 &= y + 3x(1 + xy)^2 w + 3xy^2(4 + 3xy),\\
F_3 &= 2x - 3x^2 y - x^3 w.
\end{aligned}$$

Its retained polynomial is

$$P_{a,b,c}(r) = cr^3 - 2r^2 + br - 2a, \qquad \alpha = \tfrac{1}{2} P'_{a,b,c}(r).$$

For $x \ne 0$ the inverse chart is

$$x = \alpha^{-1}, \qquad y = r - \alpha, \qquad w = 5\alpha^2 - 3r\alpha - c\alpha^3, \qquad P_{a,b,c}(r) = 0, \quad P'_{a,b,c}(r) \ne 0. \tag{449}$$

Here is the needed exact verification. Set $r = y + 1/x$ and $\alpha = 1/x$. Solving $F_3 = c$ gives the displayed expression for w. Substitution in the other components gives $F_2 = 4r + 2\alpha - 3cr^2$ and $F_1 = r^2 + r\alpha - cr^3$. The equality $F_2 = b$ is $2\alpha = P'(r)$, and after its substitution $2(F_1 - a) = P(r)$. These identities prove both directions and recover r from the inverse image. In particular a multiple root $P' = 0$ has no finite point in this chart. On $x = 0$, direct substitution gives $F(0, y, w) = (w + 4y^2, y, 0)$; there is the additional point $(0, b, a - 4b^2)$ precisely when $c = 0$.

The arithmetic slice $a = 2s$, $b = c = 0$ now gives, with no omitted coefficient,

$$P_{2s,0,0}(r) = -2r^2 - 4s, \qquad \alpha = -2r.$$

Its root relation is exactly $s = -r^2/2$. For $r \neq 0$ the spatial inverse is the explicit map

$$\Gamma(r) = \left(-\frac{1}{2r},\ 3r,\ 26r^2\right), \qquad F(\Gamma(r)) = (-r^2, 0, 0). \tag{450}$$

The inverse root is $y + 1/x = 3r - 2r = r$. For $s \neq 0$, its two roots give two distinct finite points $\Gamma(r)$ and $\Gamma(-r)$, and the complete original fibre also has the point $(0, 0, 2s)$. For $s = 0$ the polynomial has its double root at zero, both inverse chart points are excluded by $\alpha = 0$, and the sole finite fibre point is $(0, 0, 0)$. The completed root point $r = 0$ is not silently assigned the undefined spatial coordinate $x = -1/(2r)$.

For real t, every zero s_* of the specified $Z(t, \cdot)$ is nonzero by (446). It therefore has exactly two zero lifts r_* and $-r_*$ on the retained cover, and these have the finite spatial representatives in (450). The actual scalar $Z(t, F_1(x, y, w)/2)$ vanishes on all three points of the original fibre above $(2s_*, 0, 0)$. Thus the odd sheet observable is retained by the cover, and the extra $x = 0$ fibre point is separately recorded.

Multiplicity is also transported exactly. At any fixed complex time, write $Z(t, s) = (s - s_*)^m A(s)$ at a zero, where $A(s_*) \neq 0$. For $r_* \neq 0$ with $r_*^2 = -2s_*$,

$$f(t, r) = \left[-\tfrac{1}{2}(r - r_*)(r + r_*)\right]^m A(-r^2/2).$$

The factor $[-(r + r_*)/2]^m A(-r^2/2)$ is a unit near r_*, so the multiplicity is m there and also at $-r_*$. The local coordinate $s - s_* = -(r - r_*)(r + r_*)/2$ has derivative $-r_* \neq 0$; its holomorphic inverse is the corresponding branch of $r = \sqrt{-2s}$. It gives isomorphisms of the local function rings and of their quotients by the respective zero ideals. At $s_* = 0$ the exact ideal is instead $(s^m)\mathcal{O}_{\mathbb{C}_r,0} = (r^{2m})$; the factor $(-1/2)^m A(-r^2/2)$ is a unit. Thus the multiplicity doubles there. The quotient at that point is the rank-two module

$$\mathbb{C}\{r\}/(r^{2m}) = \mathbb{C}\{s\}/(s^m) \ \oplus\ r\mathbb{C}\{s\}/(s^m), \qquad s = -r^2/2.$$

Separating even and odd coefficients proves existence and uniqueness of this decomposition; the trace has precisely its odd summand as kernel. These local divisor rings do not assert that differentiation acts on a frozen global arithmetic spectral quotient.

For $r = u + iv$ with $u, v \in \mathbb{R}$, the original arithmetic point is

$$s = \frac{v^2 - u^2}{2} - iuv, \qquad \rho = \frac{1}{2} + uv + i\frac{v^2 - u^2}{2}. \tag{451}$$

It lies on $\operatorname{Re}\rho = 1/2$ exactly when $uv = 0$, which is exactly the union of the real and imaginary r axes. For real t there are no cover zeros on $u = v$ or $u = -v$, since these lines map into the imaginary s axis of (446). In particular an imaginary root coordinate alone does not give an off-critical arithmetic zero; the exact arithmetic test in this coordinate is $uv \neq 0$ at a zero of $f(0, r)$.

33.4 Simple zero motion with every analytic-unit term

Consider a point of the actual zero locus with $Z_s(t_0, s_0) \neq 0$. The holomorphic implicit function theorem applies because Z is jointly entire and this derivative is nonzero. It gives a unique holomorphic zero $s = \sigma(t)$ in a product of small disks. Differentiating $Z(t, \sigma(t)) = 0$ and using (444) gives the exact velocity

$$\sigma'(t) = \frac{Z_{ss}(t, \sigma(t))}{4Z_s(t, \sigma(t))}. \tag{452}$$

More generally write a local factorization $Z = AQ$, where A is a nonvanishing holomorphic function of (t, s) and Q is the full local zero factor under consideration. The product rule gives

$$Q_t = -\tfrac{1}{4}Q_{ss} - \frac{A_s}{2A}Q_s - \left(\frac{A_t}{A} + \frac{A_{ss}}{4A}\right) Q. \tag{453}$$

All functions in this equality are evaluated on that product of disks; dividing by A is valid there. At a simple root it gives

$$\sigma' = \frac{1}{4}\frac{Q_{ss}}{Q_s} + \frac{1}{2}\frac{A_s}{A}. \tag{454}$$

If $Q = \prod_{j=1}^{m}(s - \sigma_j(t))$ has distinct roots at the time in question, factor off $s - \sigma_j$ and differentiate the remaining product. This proves

$$\frac{Q_{ss}(\sigma_j)}{Q_s(\sigma_j)} = 2\sum_{k\neq j} \frac{1}{\sigma_j - \sigma_k}, \qquad \sigma_j' = \frac{1}{2}\sum_{k\neq j} \frac{1}{\sigma_j - \sigma_k} + \frac{1}{2}\frac{A_s(t, \sigma_j)}{A(t, \sigma_j)}.$$

For a single local root $Q = s - \sigma$, its entire velocity is $A_s/(2A)$. Removing the unit would remove this actual velocity.

Away from $r = 0$, a lifted simple root obeys $r^2 = -2\sigma$, so differentiation and the computed spatial derivatives give

$$r' = -\frac{\sigma'}{r} = -\frac{1}{4r}\frac{Z_{ss}}{Z_s} = \frac{1}{4r^2}\frac{f_{rr}}{f_r} - \frac{1}{4r^3}. \tag{455}$$

All ratios are evaluated at the moving zero. For a factorization $f = BV$ on a cover disk, the complete product equation is

$$\begin{aligned} V_t = & -\frac{1}{4r^2}V_{rr} + \left(\frac{1}{4r^3} - \frac{B_r}{2r^2B}\right) V_r \\ & + \left(-\frac{B_{rr}}{4r^2B} + \frac{B_r}{4r^3B} - \frac{B_t}{B}\right) V, \end{aligned} \tag{456}$$

and at a simple root

$$r' = \frac{1}{4r^2}\frac{V_{rr}}{V_r} + \frac{B_r}{2r^2B} - \frac{1}{4r^3}.$$

For the two complete sheet lifts of a single base zero, take $V = r^2 + 2\sigma(t)$ and $B(t, r) = -A(t, -r^2/2)/2$, where $Z = A(s - \sigma)$. At a root $V_{rr}/V_r = 1/r$. Its term $1/(4r^3)$ cancels the displayed $-1/(4r^3)$ exactly, while $B_r/B = -rA_s/A$. The result is $r' = -A_s/(2rA) = -\sigma'/r$. This computation retains both sheet roots and the analytic unit; it identifies the cancellation rather than deleting either term. The original spatial velocity is also explicit:

$$\frac{d}{dt}\Gamma(r(t)) = \left(\frac{r'}{2r^2},\ 3r',\ 52rr'\right), \qquad \frac{d}{dt}F(\Gamma(r(t))) = (2\sigma', 0, 0).$$

33.5 A double zero: exact pair equations and the unit drift

At an actual double zero (t_0, s_0), put $h = t - t_0$ and write

$$Z(t_0, s_0 + \delta) = a_2\delta^2 + a_3\delta^3 + O(\delta^4), \qquad a_2 \neq 0.$$

The local zero factor can be expressed as $Q = (s - M)^2 - D^2$, where M and D^2 are holomorphic in t, $M(t_0) = s_0$, and $D^2(t_0) = 0$. A construction not depending on the naming of the two roots is to take a small circle containing only the double zero and no boundary zero. The integrals

$$p_k(t) = \frac{1}{2\pi i}\int_{\text{circle}} s^k \frac{Z_s(t,s)}{Z(t,s)}\, ds \quad (k = 1, 2)$$

are holomorphic and equal the first two root power sums with multiplicity, by the residue computation for a local zero factor. Thus $M = p_1/2$ and $D^2 = p_2/2 - M^2$ are holomorphic. The two zeros, counted with multiplicity, are $M + D$ and $M - D$. The holomorphic unit can itself be constructed on a smaller concentric disk by the contour formula

$$A(t, s) = \frac{1}{2\pi i}\int_{\text{circle}} \frac{Z(t,\zeta)}{Q(t,\zeta)(\zeta - s)}\, d\zeta.$$

The denominator is nonzero on the contour for all the retained times, so the integral is jointly holomorphic. At each fixed time Z/Q extends holomorphically across its two zeros, counted with their multiplicities, by their one-variable Taylor factorizations. Cauchy's formula therefore identifies the displayed integral with Z/Q, proving $Z = AQ$. At $t = t_0$ it is $Z(t_0, s)/(s - s_0)^2$, nonzero at s_0, so A is nonvanishing after shrinking the disks.

Put $b(t, s) = A_s(t, s)/A(t, s)$. At every nearby time with $D \neq 0$ the proved simple-zero equation gives the exact pair dynamics

$$\begin{aligned}
(M + D)' &= \frac{1}{4D} + \tfrac{1}{2}b(t, M + D),\\
(M - D)' &= -\frac{1}{4D} + \tfrac{1}{2}b(t, M - D),\\
M' &= \tfrac{1}{4}\big[b(t, M + D) + b(t, M - D)\big],\\
(D^2)' &= \tfrac{1}{2} + \tfrac{D}{2}\big[b(t, M + D) - b(t, M - D)\big].
\end{aligned} \tag{457}$$

The sum and D times the difference are even holomorphic functions of D, hence holomorphic functions of D^2. These equations extend to $h = 0$, and there give

$$(D^2)'(t_0) = \tfrac{1}{2}, \qquad M'(t_0) = \frac{a_3}{2a_2} = \frac{Z_{sss}(t_0, s_0)}{6Z_{ss}(t_0, s_0)}.$$

Indeed $A(t_0, s_0) = a_2$ and $A_s(t_0, s_0) = a_3$. The full equation (453) still carries A_t and A_{ss}; their factors multiply Q and vanish only when evaluated at a zero. They have not been assumed to vanish on the neighborhood.

The following explicit expansion proves the pair's first unit term and its error order. The heat equation gives the terms of weighted degree at most three, with δ of weight one and h of weight two:

$$Z(t_0 + h, s_0 + \delta) = a_2(\delta^2 - h/2) + a_3(\delta^3 - 3h\delta/2) + O((|\delta| + |h|^{1/2})^4).$$

Set $h = \varepsilon^2$. Dividing by ε^2 after $\delta = \varepsilon y$ gives a holomorphic function of (ε, y) whose value at $\varepsilon = 0$ is $a_2(y^2 - 1/2)$. Its two simple roots are $y = \pm 1/\sqrt{2}$, so the holomorphic implicit function theorem gives two holomorphic y branches. Their next coefficient is found by substituting $\delta = \lambda\varepsilon + \mu\varepsilon^2$: $\lambda^2 = 1/2$ and $2a_2\lambda\mu + a_3(\lambda^3 - 3\lambda/2) = 0$. Hence $\mu = a_3/(2a_2)$ and

$$\sigma_\pm(t) = s_0 \pm \sqrt{h/2} + \frac{a_3}{2a_2}h + O(h^{3/2}). \tag{458}$$

This means convergent holomorphic expansions in ε, with a chosen $\varepsilon^2 = h$; reversing ε exchanges the root labels. It follows that, on either punctured time branch,

$$\sigma'_{\pm} = \pm\frac{1}{2\sqrt{2h}} + \frac{a_3}{2a_2} + O(h^{1/2}).$$

For real t_0, s_0 the coefficients are real. For small positive h both branches are real, by uniqueness and conjugation of their real ε power series. For small negative real h their imaginary parts are $\pm\sqrt{|h|/2} + O(|h|^{3/2})$, so they are a nonreal conjugate pair. This describes an actual double-zero event wherever it occurs; it does not insert such an event at an unspecified real arithmetic time.

At real time $s_0 \neq 0$ by (446). Choose either $r_0^2 = -2s_0$. The inverse branch has $dr/ds = -1/r_0$ and $d^2r/ds^2 = -1/r_0^3$. Taylor substitution in (458) yields the two roots near that particular sheet point,

$$r_{\pm}(t) = r_0 \mp \frac{1}{r_0}\sqrt{h/2} - \left(\frac{a_3}{2a_2r_0} + \frac{1}{4r_0^3}\right)h + O(h^{3/2}). \tag{459}$$

The other sheet consists exactly of their negatives. Both the arithmetic unit drift and the curvature of the original root map occur in its displayed coefficient of h.

33.6 Every multiplicity and its interaction with ramification

At an actual zero of finite multiplicity $m \geq 1$ write $Z(t_0, s_0 + \delta) = \sum_{k\geq m} a_k\delta^k$, $a_m \neq 0$. There is finite multiplicity because $Z(t_0, \cdot)$ is not the zero function. For real t_0 positivity at zero already proves this. For complex t_0 put $K(u) = e^{t_0u^2}\Phi(|u|)$ on $\mathbb{R}$. The proved decay shows that K is continuous, bounded, and integrable, and $K(0) = \Phi(0) > 0$. Its Fourier transform satisfies $\widehat{K}(\xi) = 2H_{t_0}(\xi)$ for real ξ, by evenness. If $Z(t_0, \cdot)$ were identically zero, this transform would vanish. For $\epsilon > 0$, the Gaussian integral gives

$$G_\epsilon(x) = \frac{e^{-x^2/(4\epsilon)}}{\sqrt{4\pi\epsilon}} = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-\epsilon\xi^2}e^{ix\xi}\,d\xi.$$

Fubini is justified by $K \in L^1$ and the integrable Gaussian in ξ; it gives $(K * G_\epsilon)(0) = (2\pi)^{-1}\int \widehat{K}(\xi)e^{-\epsilon\xi^2}\,d\xi = 0$. On the other hand $(K * G_\epsilon)(0) \to K(0) > 0$ as $\epsilon \downarrow 0$: on $|u| < \delta$ continuity bounds $|K(u) - K(0)|$ arbitrarily small, while on its complement boundedness of K and the vanishing Gaussian tail bound the remaining integral. This contradiction proves the required nonidentity with the zero function for every complex time.

Repeated differentiation of the heat equation gives $\partial_t^\ell Z = (-1/4)^\ell\partial_s^{2\ell}Z$. The bivariate Taylor coefficient of $h^\ell\delta^j$ is therefore

$$\frac{(-1/4)^\ell}{\ell!}\frac{(j+2\ell)!}{j!}a_{j+2\ell}.$$

Set $h = \varepsilon^2$, $\delta = \varepsilon y/\sqrt{2}$, with the inverse substitutions retained whenever $\varepsilon \neq 0$. The holomorphic Taylor series is divisible by ε^m; grouping the terms of degree m and $m+1$ gives

$$\frac{Z(t_0 + \varepsilon^2, s_0 + \varepsilon y/\sqrt{2})}{a_m(\varepsilon/\sqrt{2})^m} = \mathrm{He}_m(y) + \frac{a_{m+1}}{a_m}\frac{\varepsilon}{\sqrt{2}}\mathrm{He}_{m+1}(y) + O(\varepsilon^2),$$

$$\mathrm{He}_m(y) = \sum_{\ell=0}^{\lfloor m/2\rfloor}\frac{m!(-1)^\ell}{2^\ell\ell!(m-2\ell)!}y^{m-2\ell}. \tag{460}$$

The quotient extends holomorphically at $\varepsilon = 0$; its remainder is uniform on each bounded y set. This follows directly from the convergent bivariate Taylor series after substitution, so the error does not assume a finite polynomial initial datum.

For completeness the polynomial has m distinct real roots. Differentiating its displayed finite sum proves $\mathrm{He}'_m = m\,\mathrm{He}_{m-1}$ and $\mathrm{He}_{m+1} = y\,\mathrm{He}_m - m\,\mathrm{He}_{m-1}$. Starting with $\mathrm{He}_0 = 1$ and $\mathrm{He}_1 = y$, induction shows that the roots of two successive polynomials strictly interlace: at consecutive roots of He_m, the values $-m\,\mathrm{He}_{m-1}$ alternate sign, providing one root of He_{m+1} between each consecutive pair. At the largest root, $\mathrm{He}_{m-1} > 0$, so $\mathrm{He}_{m+1} < 0$ there and is positive sufficiently far to the right. At the smallest root, the sign is $(-1)^m$ and the far-left sign of its monic degree $m+1$ polynomial is $(-1)^{m+1}$, providing a further root to the left. There are thus $m+1$ distinct roots. The induction begins with the single root of He_1 and gives the claimed reality, simplicity, and interlacing. Let these roots be λ_j. At a root of He_m the recurrences also give

$$\frac{\mathrm{He}_{m+1}(\lambda_j)}{\mathrm{He}'_m(\lambda_j)} = -1.$$

Apply the holomorphic implicit function theorem to (460) at each $(0, \lambda_j)$. The resulting roots are

$$\sigma_j(t) = s_0 + \lambda_j\sqrt{h/2} + \frac{a_{m+1}}{2a_m}h + O(h^{3/2}). \tag{461}$$

They are all the nearby roots: a small fixed circle about s_0 has no boundary zero at t_0, and continuity followed by Rouché's theorem keeps its zero count at m; the m disjoint roots just constructed account for that count. Writing $Z(t_0, s) = (s - s_0)^m A_0(s)$ gives $a_{m+1}/(2a_m) = A_0'(s_0)/(2A_0(s_0))$. Thus the first common drift at every multiplicity is exactly the initial analytic-unit derivative. Higher coefficients remain determined by the full Taylor series and the exact unit equation (453).

For $s_0 \neq 0$, on either chosen sheet $r_0^2 = -2s_0$ the same inverse coordinate Taylor expansion proves

$$r_j(t) = r_0 - \frac{\lambda_j}{r_0}\sqrt{h/2} - \left(\frac{a_{m+1}}{2a_m r_0} + \frac{\lambda_j^2}{4r_0^3}\right)h + O(h^{3/2}).$$

At $s_0 = 0$, an event of the actual family is possible only at nonreal complex time, by (446). On this locus the exact evenness $Z(t, -s) = Z(t, s)$ forces $m = 2k$ and $a_{m+1} = 0$. Its Hermite roots are nonzero, since $\mathrm{He}_{2k}(0) = (-1)^k(2k)!/(2^k k!)$. Set $h = \eta^4$ and $\varepsilon = \eta^2$. Equation (461) now gives

$$\sigma_j(t_0 + \eta^4) = \frac{\lambda_j}{\sqrt{2}}\eta^2 + O(\eta^6), \qquad r_{j,\pm}(\eta) = \pm\eta\sqrt{-\sqrt{2}\lambda_j} + O(\eta^5).$$

The square root of each nonzero displayed constant has two choices. To justify the expansion, $-2\sigma_j(t_0+\eta^4)/\eta^2$ extends holomorphically and has nonzero value $-\sqrt{2}\lambda_j$ at zero; either holomorphic square root of that unit gives exactly $r_{j,\pm} = \pm\eta\sqrt{-2\sigma_j/\eta^2}$. The $2m$ lifted roots account for the cover multiplicity $2m$ proved above. This is the precise fourth-root time ramification over an arithmetic multiple zero at the cover branch locus. It does not occur there at real Newman time, and it has no finite spatial inverse chart at its central point because $P' = 0$.

Verification status. The accompanying exact rational-polynomial checker verifies the chain rule coefficients, monomial pole coefficients, original spatial lift, analytic-unit product identities, and finite coefficients of the collision expansions. The convergence, positivity, local holomorphic factorizations, root counts, and complete multiplicity statements are proved in the text; the checker is not a substitute for those proofs. No off-critical zero of the distinguished arithmetic initial function is asserted by these calculations.

34 The actual arithmetic observable under spatial diffusion

Retain the original real spatial coordinates $q = (x, y, w)$ and the polynomial F in (417). The complex formulas below are their holomorphic extensions; a sum of derivative squares is a Euclidean squared norm only on the stated real locus. Define the exact observable

$$\sigma(q) = F_1(q)/2, \qquad \Psi(t, q) = Z_t(\sigma(q)).$$

The Newman time t and the spatial flow time τ remain distinct variables. The already proved entire function Z makes Ψ entire in $(t, q) \in \mathbb{C} \times \mathbb{C}^3$.

34.1 Original vector fields and the rank-one heat operator

Here is a direct determinant proof needed to define the vector fields. Use the already proved inverse-chart map

$$H(\alpha, r, c) = (\alpha^{-1}, r - \alpha, 5\alpha^2 - 3r\alpha - c\alpha^3), \quad \alpha \neq 0.$$

Its inverse on $x \neq 0$ is $(1/x, y + 1/x, F_3(q))$. The two derivative matrices, in the displayed coordinate order, are

$$DH = \begin{pmatrix} -\alpha^{-2} & 0 & 0 \\ -1 & 1 & 0 \\ -3c\alpha^2 + 10\alpha - 3r & -3\alpha & -\alpha^3 \end{pmatrix}, \quad D(F \circ H) = \begin{pmatrix} r & \alpha - 3cr^2 + 2r & -r^3 \\ 2 & 4 - 6cr & -3r^2 \\ 0 & 0 & 1 \end{pmatrix}.$$

The first determinant is α. Expansion along the last row of the second gives $r(4-6cr)-2(\alpha-3cr^2+2r) = -2\alpha$. The chain rule therefore proves $\det DF = -2$ on $x \neq 0$. Both sides are polynomials in (x, y, w), so their equality extends to $x = 0$ by continuity. Put $J = DF$, $W_i = J^{-1}e_i$, and

$$U = 2W_1 + 6F_3W_2.$$

These are globally defined polynomial vector fields, since $J^{-1} = -(1/2)\operatorname{adj} J$. With $\epsilon_{123} = 1$ and cyclic indices, the adjugate formula is $W_i = -\frac{1}{2}\nabla F_{i+1} \times \nabla F_{i+2}$. The divergence of each cross product is zero: its differentiated components contract a symmetric Hessian with an antisymmetric ϵ tensor. Also $W_iF_j = \delta_{ij}$, by $JW_i = e_i$. The product rule now gives

$$\operatorname{div} U = 2\operatorname{div} W_1 + 6F_3 \operatorname{div} W_2 + 6W_2F_3 = 0, \qquad U\sigma = 1.$$

For every twice differentiable scalar $h(s)$, direct differentiation proves $U(h \circ \sigma) = h'(\sigma)$ and $U^2(h \circ \sigma) = h''(\sigma)$. Consequently the actual observable satisfies the exact operator identity

$$\partial_t\Psi + \tfrac{1}{4}U^2\Psi = 0. \tag{462}$$

Here the full spatial operator is

$$U^2h = U^iU^j\partial_i\partial_jh + U^i(\partial_iU^j)\partial_jh = \partial_i(U^iU^j\partial_jh),$$

where repeated indices run over the three original coordinates. The last equality uses the proved divergence identity. Its real principal tensor is $UU^{\mathsf{T}}/4$, of rank one because $U\sigma = 1$ implies $U \neq 0$. On $C_c^\infty(\mathbb{R}^3)$, one integration by parts gives $-\int \bar{h}U^2h = \int |Uh|^2$; polynomial coefficients are bounded on the support and there are no boundary terms. The explicit reversal $\Phi(\tau, q) = \Psi(-\tau, q)$ obeys $\partial_\tau\Phi = U^2\Phi/4$. This proves an exact forward spatial heat realization on the specified observable, with every first-order coefficient retained.

34.2 Every Euclidean diffusion coefficient

Let $Q = 1+xy$, an abbreviation for the original polynomial factor. Differentiation before restriction to a fibre gives

$$\begin{aligned} g_1 &:= \partial_x\sigma = \tfrac{3}{2}yQ^2w + \tfrac{1}{2}y^3(7 + 6xy), \\ g_2 &:= \partial_y\sigma = \tfrac{3}{2}xQ^2w + \tfrac{1}{2}(8y + 21xy^2 + 12x^2y^3), \\ g_3 &:= \partial_w\sigma = \tfrac{1}{2}Q^3. \end{aligned}$$

Define $N = g_1^2 + g_2^2 + g_3^2$ and

$$L = \Delta_q\sigma = 3wQ(x^2 + y^2) + 3y^4 + 18x^2y^2 + 21xy + 4.$$

For example the second derivatives in the Laplacian are $\sigma_{xx} = 3wy^2Q + 3y^4$, $\sigma_{yy} = 3wx^2Q + 18x^2y^2 + 21xy + 4$, and $\sigma_{ww} = 0$, proving every term in L. The chain rule in each coordinate, followed by summation, proves

$$\Delta_q\Psi = N(q)Z_{ss}(t, \sigma(q)) + L(q)Z_s(t, \sigma(q)). \tag{463}$$

No term is suppressed by the one-variable presentation of Z. For real q, $N = |\nabla\sigma|^2 > 0$: an invertible J cannot have its first row zero.

On the original finite inverse lift $\Gamma(r) = (-1/(2r), 3r, 26r^2)$, $r \neq 0$, direct substitution gives

$$\nabla\sigma(\Gamma(r)) = (9r^3/4, 3r/8, -1/16)^{\mathsf{T}}, \qquad D^2\sigma(\Gamma(r)) = \begin{pmatrix} -108r^4 & 3r^2/4 & 9r/8 \\ 3r^2/4 & 13/4 & -3/(16r) \\ 9r/8 & -3/(16r) & 0 \end{pmatrix}.$$

Thus, retaining $\sigma(\Gamma(r)) = -r^2/2$,

$$\begin{aligned} N(\Gamma(r)) &= \frac{1296r^6 + 36r^2 + 1}{256}, \\ L(\Gamma(r)) &= \frac{13}{4} - 108r^4, \\ (\Delta_q\Psi)(t, \Gamma(r)) &= \frac{1296r^6 + 36r^2 + 1}{256} Z_{ss}(t, -r^2/2) + \left(\frac{13}{4} - 108r^4\right) Z_s(t, -r^2/2). \end{aligned} \tag{464}$$

These are identities of rational holomorphic functions for complex $r \neq 0$. Their Euclidean-norm interpretation uses real $r \neq 0$.

For a real differentiable time map $t = \theta(\tau)$ and the original viscosity $\nu > 0$, the complete advection–diffusion residual is

$$\begin{aligned} &(\partial_\tau + U \cdot \nabla_q - \nu\Delta_q)\Psi(\theta(\tau), q) \\ &\qquad = \left(-\frac{\theta'(\tau)}{4} - \nu N(q)\right) Z_{ss}(\theta(\tau), \sigma(q)) + (1 - \nu L(q))Z_s(\theta(\tau), \sigma(q)). \end{aligned} \tag{465}$$

The three preceding exact identities prove this formula term by term. It is the direct map between the actual arithmetic heat solution and the scalar spatial advection–diffusion operator of this polynomial flow. It does not assert that its displayed residual vanishes.

34.3 The particular arithmetic covector along the escaping orbit

Local existence and uniqueness of the smooth polynomial ODE $\partial_\tau X_\tau = U(X_\tau)$ follow on a closed finite ball from the contraction map $Y \mapsto q + \int_0^\tau U(Y(v))\,dv$ for sufficiently short time; bounded DU supplies its Lipschitz constant. Continuing these solutions defines the maximal local flow on its actual open domains. Smooth dependence follows by differentiating the integral equation; uniqueness supplies the inverse local flow. The identities $UF_1 = 2$, $UF_2 = 6F_3$, and $UF_3 = 0$ integrate to

$$F(X_\tau(q)) = (F_1(q) + 2\tau, F_2(q) + 6\tau F_3(q), F_3(q)).$$

In particular $\sigma(X_\tau(q)) = \sigma(q) + \tau$. Write $A_\tau = D_qX_\tau$ and $K_\tau = A_\tau^{-1}A_\tau^{-\mathsf{T}}$. Differentiating this last scalar identity gives the full covector map

$$A_\tau^{\mathsf{T}}\nabla\sigma(X_\tau(q)) = \nabla\sigma(q), \qquad \nu\nabla\sigma(q)^{\mathsf{T}}K_\tau\nabla\sigma(q) = \nu N(X_\tau(q)).$$

Every matrix is evaluated at the same initial point q and time τ; no arbitrary covector has replaced this actual one.

At $q_0 = (1, -3/2, 13/2) = \Gamma(-1/2)$ put $z = \sqrt{1 - 8\tau} > 0$, with inverse $\tau = (1 - z^2)/8$. For $-\infty < \tau < 1/8$ the explicit trajectory is $X_\tau(q_0) = \Gamma(-z/2) = (z^{-1}, -3z/2, 13z^2/2)$. To verify the

ODE, its F image is $(-1/4+2\tau,0,0)$; differentiation gives $J\gamma' = (2,0,0) = JU(\gamma)$, so invertibility of J gives $\gamma' = U(\gamma)$. Its value at $\tau = 0$ is q_0, and its escape as $\tau \uparrow 1/8$ precludes a finite continuous extension of the ODE solution. For the specific initial covector

$$\ell_\sigma = \nabla\sigma(q_0) = (-9/32, -3/16, -1/16)^{\mathsf{T}}$$

we have the entire transported vector and cost

$$\begin{aligned} A_\tau^{-\mathsf{T}}\ell_\sigma &= (-9z^3/32, -3z/16, -1/16)^{\mathsf{T}}, \\ \nu\ell_\sigma^{\mathsf{T}} K_\tau \ell_\sigma &= \nu\frac{81z^6+36z^2+4}{1024} \longrightarrow \frac{\nu}{256} \quad (z \downarrow 0). \end{aligned}$$

This computes the precise diffusive cost of the covector selected by the arithmetic observable, even at the spatial escape endpoint. The zero condition itself remains $Z_t(\sigma(q)) = 0$ with its full analytic divisor as proved above; the transported cost is not an assertion of such a zero or of a Euclidean Navier–Stokes counterexample.

35 The source fourth coefficient and its exact Newman heat packet

The result supplied in the linked task uses the coupling symbol ξ. Keep its original finite open cubic box, $L \geq 2$, with $N = 3(2L)(2L+1)^2$ links and $M = 3(2L)^2(2L+1)$ elementary faces. On product Haar probability measure on $SU(2)^N$, the actual Hamiltonian and scales are

$$H = \kappa\sum_e E_e + b\sum_p (2 - W_p), \qquad \kappa = \frac{2g^2}{a}, \quad b = \frac{1}{2g^2 a}, \quad \xi = \frac{b}{\kappa} = \frac{1}{4g^4}, \qquad a, g > 0.$$

Here a is the source lattice spacing; it is not the polynomial target coordinate $a = F_1$ in earlier sections. Each use below retains this distinction. The electric operator is $E_e = -\sum_{a'=1}^{3} X_{e,a'}^2$ for $T_{a'} = -i\sigma_{a'}/2$; the prime distinguishes the generator index from both meanings of a. The potential trace W_p and the original operator and form domains are specified and proved in Appendix A. Let ψ_ξ be the positive unit vacuum, $\mathcal{E}$ its energy,

$$A_\xi = (H - \mathcal{E})/\kappa, \qquad v_e = (E_e - \langle\psi_\xi, E_e\psi_\xi\rangle)\psi_\xi.$$

For two distinct links e, f sharing no elementary face, retain the original ordered incidence count

$$a_{ef} = \#\{(p,q) : e \in \partial p,\ f \in \partial q, |\partial p \cap \partial q| = 1\}.$$

This is a count, not a continuum limit or a change of physical scale. Appendix A proves, in the full original box,

$$\begin{aligned} \langle v_e, v_f\rangle &= \frac{127a_{ef}}{219024}\xi^4 + O_L(\xi^6), \\ \langle v_e, (H-\mathcal{E})v_f\rangle &= 0. \end{aligned} \tag{466}$$

It retains both shared-link spin channels, their exact norms, the disconnected subtraction, and the full vacuum and spectral-band Gram factors. The complete source proof is included, not replaced by these two consequences.

For the energy variable λ define the distribution

$$\mathfrak{d} = c_0\delta_3 + \beta\delta_3' + c_1\delta_{9/2} + c_2\delta_{13/2}, \qquad \begin{aligned} c_0 &= -\frac{59}{275184}, \quad \beta = \frac{1}{336}, \\ c_1 &= \frac{1}{1296}, \quad c_2 = \frac{3}{132496}. \end{aligned} \tag{467}$$

The convention is $\delta_3'(h) = -h'(3)$, so every sign is fixed. The proved source functional is, for bounded continuous h differentiable at 3,

$$\lim_{\xi \to 0} \xi^{-4} \langle v_e, h(A_\xi) v_f \rangle = a_{ef} \mathfrak{d}(h). \tag{468}$$

The variable λ is the dimensionless energy ω/κ; the physical measure is its pushforward under $\lambda \mapsto \kappa\lambda$. The source proves that the original matrix-valued spectral measure is positive semidefinite on every Borel set. An off-diagonal entry retains its signs, and its fourth Taylor coefficient retains δ_3'.

35.1 A four-dimensional spectral distribution and its heat map

Let $\mathcal{V}$ be the ordered span $(\delta_3, \delta_3', \delta_{9/2}, \delta_{13/2})$. Multiplication of a distribution by the unchanged energy coordinate acts in this basis by

$$Q = \begin{pmatrix} 3 & -1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 9/2 & 0 \\ 0 & 0 & 0 & 13/2 \end{pmatrix}. \tag{469}$$

Indeed $(\lambda\delta_3')(h) = -3h'(3) - h(3)$, while the other three basis vectors are multiplied by their displayed energies. This proves the entire matrix, including its off-diagonal entry. The exact exponential of $tQ^2/4$ has its first block

$$e^{9t/4} \begin{pmatrix} 1 & -3t/2 \\ 0 & 1 \end{pmatrix}$$

and remaining entries $e^{81t/16}, e^{169t/16}$. To verify it, Q^2 has first block $9I + \left(\begin{smallmatrix} 0 & -6 \\ 0 & 0 \end{smallmatrix}\right)$; the second matrix has square zero, so its exponential series stops after its linear term. The two other entries are scalar exponentials.

Define the exact Fourier morphism

$$\mathcal{F}_{\mathcal{V}}(d)(s) = d_\lambda(e^{is\lambda}).$$

Its basis images are $e^{3is}, -ise^{3is}, e^{9is/2}, e^{13is/2}$. It is injective. If a linear combination vanishes, divide it by e^{3is} and put $s = iy$, $y \to +\infty$. The remaining expression is a polynomial of degree at most one in y, plus constant multiples of $e^{-3y/2}$ and $e^{-7y/2}$. Its polynomial coefficients must vanish; multiplying the remaining identity by $e^{3y/2}$ then taking the same limit makes the third coefficient zero, and then the fourth is zero. Differentiation of the four displayed functions gives

$$\partial_s \mathcal{F}_{\mathcal{V}}(d) = i\mathcal{F}_{\mathcal{V}}(Qd), \qquad -\tfrac{1}{4}\partial_s^2 \mathcal{F}_{\mathcal{V}}(d) = \mathcal{F}_{\mathcal{V}}(Q^2 d/4).$$

Thus the source spectral coefficient has an exact heat intertwiner, not merely a shared dimension or notation.

Applying this map to $a_{ef} e^{tQ^2/4} \mathfrak{d}$ yields

$$\begin{aligned} J_t(s) = a_{ef} \Big[& \left(-\frac{59}{275184} - \frac{t}{224} - \frac{is}{336} \right) e^{9t/4 + 3is} \\ & + \frac{1}{1296} e^{81t/16 + 9is/2} + \frac{3}{132496} e^{169t/16 + 13is/2} \Big]. \end{aligned} \tag{470}$$

It is entire in both variables and satisfies $\partial_t J = -\partial_s^2 J/4$. Equivalently this follows directly from $J_t = a_{ef} \mathfrak{d}(e^{t\lambda^2/4 + is\lambda})$. For the actual original spectral measure, this test is bounded on the real energy axis whenever $\operatorname{Re} t < 0$, for every complex s, and also when $t = 0, s \in \mathbb{R}$. Completing the real square in $\operatorname{Re}(t)\lambda^2/4 - \operatorname{Im}(s)\lambda$ proves that bound. Consequently (468) gives

$$\lim_{\xi \to 0} \xi^{-4} \langle v_e, e^{tA_\xi^2/4 + isA_\xi} v_f \rangle = J_t(s)$$

on precisely these guaranteed domains. The finite coefficient (470) has the stated entire continuation; this does not assume Gaussian exponential moments of the full unbounded interacting spectral measure at positive real t.

Keep physical correlation time $\theta \geq 0$ separate from Newman time t. The unchanged dimensionful correlation coefficient is

$$a_{ef}\mathfrak{d}(e^{-\kappa\theta\lambda}) = a_{ef}\left[\left(c_0 + \frac{\kappa\theta}{336}\right)e^{-3\kappa\theta} + c_1 e^{-9\kappa\theta/2} + c_2 e^{-13\kappa\theta/2}\right].$$

It equals $J_0(i\kappa\theta)$ by substitution, giving the exact time dictionary rather than identifying the two evolutions. Its value at zero is $127a_{ef}/219024$, and its first physical-time derivative is zero because $3c_0 - \beta + (9/2)c_1 + (13/2)c_2 = 0$. These are exactly (466), with the physical factor κ retained in the derivative.

36 The original fourth coupling coefficient as an arithmetic heat automorphism

This section uses the complete finite-box Wilson calculation retained in the complete source proof in Appendix A. Its parameter is $\xi = b/\kappa = 1/(4g^4)$, whereas the completed Riemann function is $\xi(\rho)$ and the unchanged arithmetic heat function is $Z_t(s)$. These notations retain their original meanings. We construct the actual operator-valued arithmetic observable, prove its fourth coupling coefficient, and give the inverse of the resulting arithmetic map.

36.1 Original source operators and the complete spectral functional

Fix the original open box with vertices $\{-L, \dots, L\}^3$, integer $L \geq 2$, its $N = 3(2L)(2L+1)^2$ positively oriented physical edges, and its $M = 3(2L)^2(2L+1)$ contained elementary faces. On the gauge-invariant subspace of product Haar probability $L^2(SU(2)^N)$, retain

$$\begin{gathered} T_a = -i\sigma_a/2, \qquad E_e = -\sum_{a=1}^{3} X_{e,a}^2, \qquad H_0 = \sum_e E_e, \qquad S = \sum_p W_p, \\ H = \kappa H_0 + b\sum_p (2 - W_p) = \kappa(H_0 - \xi S) + 2bMI, \\ \kappa = \frac{2g^2}{a}, \quad b = \frac{1}{2g^2 a}, \quad \xi = \frac{1}{4g^4}, \quad a, g > 0. \end{gathered} \tag{471}$$

Here $X_{e,a}$ is the original left derivative generated by T_a, and W_p is the original oriented fundamental Wilson trace. The metric identity is $-\Delta_e^c = 4E_e$, so the electric coefficient in that metric is $\kappa/4$. Neither this factor nor the scalar $2bM$ is changed. Let ψ_ξ be the positive unit ground vector and write

$$\begin{gathered} \mathcal{E}(\xi) = 2bM + \kappa e(\xi), \qquad A_\xi = \frac{H - \mathcal{E}(\xi)}{\kappa} = H_0 - \xi S - e(\xi)I, \\ \gamma_e(\xi) = \langle \psi_\xi, E_e\psi_\xi \rangle, \qquad v_e(\xi) = (E_e - \gamma_e(\xi))\psi_\xi. \end{gathered} \tag{472}$$

The inner product is conjugate-linear in its first entry. For real ξ near zero, A_ξ is self-adjoint on the physical H^2 domain, nonnegative, and has the original smooth ground vector in its kernel. The formula for A_ξ is the exact physical excitation energy divided by its original κ; the scalar in H cancels its identical scalar in $\mathcal{E}$ in this displayed difference.

For distinct edges e, f sharing no face, let a_{ef} count the ordered-by-containment pairs of adjacent faces containing e and f, respectively. The complete spectral coefficient proved in the source calculation is, for bounded continuous complex $h : \mathbb{R} \to \mathbb{C}$ differentiable at 3,

$$\lim_{\xi \to 0} \xi^{-4} \langle v_e(\xi), h(A_\xi) v_f(\xi) \rangle = a_{ef} \mathcal{D}(h), \qquad \mathcal{D}(h) = c_0 h(3) - \frac{h'(3)}{336} + c_1 h(9/2) + c_2 h(13/2), \tag{473}$$

where every coefficient is retained:

$$c_0 = -\frac{59}{275184}, \qquad c_1 = \frac{1}{1296}, \qquad c_2 = \frac{3}{132496}. \tag{474}$$

Equivalently, $\mathcal{D} = c_0 \delta_3 + \delta_3'/336 + c_1 \delta_{9/2} + c_2 \delta_{13/2}$, with $\delta_3'(h) = -h'(3)$. Thus the derivative term is part of the original spectral coefficient. The source proof obtains it from the actual first interacting spectral band and its full Gram isometry, together with both exterior spin channels; it is not replaced here by an atomic measure.

36.2 Bounded arithmetic functional calculus at every complex heat time

Lemma 36.1 (The actual arithmetic test and its operator integral)**.** *For every complex t, s, the function $h_{t,s}(q) = Z_t(s + q)$ is bounded and continuously differentiable on the whole real q-axis. On real ξ near zero, its bounded spectral calculus is exactly*

$$Z_t(s + A_\xi) = 4 \int_{\mathbb{R}} k(v) e^{tv^2/4 + isv} e^{ivA_\xi} \, dv. \tag{475}$$

The integral is the strong integral of bounded operators, defined on each Hilbert-space vector; its operator norm is bounded by

$$4 \int_{\mathbb{R}} |k(v)| e^{(\operatorname{Re} t)v^2/4 + |\operatorname{Im} s||v|} \, dv. \tag{476}$$

It is jointly entire in t, s, in the operator norm, and obeys $\partial_t Z_t(s + A_\xi) = -\frac{1}{4} \partial_s^2 Z_t(s + A_\xi)$.

Proof. Use the unchanged arithmetic identity (427):

$$h_{t,s}(q) = 4 \int_{\mathbb{R}} k(v) e^{tv^2/4 + isv} e^{iqv} \, dv.$$

For real q, the last factor has modulus one. The previously proved two-ended superexponential bound on k dominates the absolute integral in (476). Its product with $|v|^j$ is also integrable for every nonnegative integer j. This proves boundedness of every real q-derivative, uniformly in q, by differentiating the integral. On compact complex (t, s) sets the same dominator works after every specified t, s, q derivative.

For any Hilbert-space vector u, the function $v \mapsto e^{ivA_\xi} u$ is strongly continuous and has constant norm. Thus the right side of (475) is a Bochner integral on that vector and satisfies the stated norm bound. For any two vectors, the corresponding spectral measure has finite total variation, bounded by the product of their norms. Fubini therefore interchanges this measure integral with the absolute v integral and gives its matrix element as that of $h_{t,s}(A_\xi)$. This proves the operator identity, rather than postulating an unbounded entire functional calculus.

Expand the scalar exponential about any fixed (t, s). On every smaller compact polydisc, its absolutely summed Taylor series is dominated by the same integral with larger fixed quadratic and linear weights. The integral norm inequality proves convergence of the resulting operator Taylor series in norm. This establishes joint entire dependence. Differentiation in t inserts $v^2/4$, whereas two s derivatives insert $-v^2$, which proves the heat identity with the original sign and factor. □

Theorem 36.2 (The actual arithmetic fourth coefficient)**.** *Define the entire arithmetic function*

$$\mathcal{K}_t(s) = c_0 Z_t(s + 3) - \frac{Z_t'(s + 3)}{336} + c_1 Z_t(s + 9/2) + c_2 Z_t(s + 13/2). \tag{477}$$

For every complex t, s, the actual original observable

$$G_\xi(t,s) = \langle v_e(\xi), Z_t(s + A_\xi)v_f(\xi)\rangle \tag{478}$$

satisfies

$$\lim_{\xi\to 0} \xi^{-4}G_\xi(t,s) = a_{ef}\mathcal{K}_t(s). \tag{479}$$

The limit retains the actual Riemann initial datum:

$$\mathcal{K}_0(s) = c_0\Xi(s+3) - \frac{\Xi'(s+3)}{336} + c_1\Xi(s+9/2) + c_2\Xi(s+13/2), \tag{480}$$

$$\partial_t\mathcal{K}_t = -\tfrac{1}{4}\partial_s^2\mathcal{K}_t, \qquad \partial_t G_\xi = -\tfrac{1}{4}\partial_s^2 G_\xi. \tag{481}$$

Proof. Lemma 36.1 verifies every test-function hypothesis of (473) for the actual $h(q) = Z_t(s+q)$. Its values and derivative at the four specified source locations give (477) and (479) literally. Every translated term and its retained derivative is jointly entire. Translation by each of the original constants and ∂_s commute with ∂_s^2, so (428) proves the first identity in (481); the operator identity from Lemma 36.1 proves the second one. Finally $Z_0 = \Xi$ gives (480) with no change of arithmetic initial function. □

36.3 Actual coupling analyticity and the sixth-order remainder

Proposition 36.3 (Uniform remainder on compact arithmetic parameter sets)**.** *For each original finite box and each compact $\mathcal{Q} \subset \mathbb{C}^2_{t,s}$, there are numbers $r > 0$ and $C_{\mathcal{Q},L,e,f} < \infty$ such that, for real $|\xi| \le r/2$,*

$$\sup_{(t,s)\in\mathcal{Q}} |G_\xi(t,s) - a_{ef}\xi^4\mathcal{K}_t(s)| \le C_{\mathcal{Q},L,e,f}|\xi|^6. \tag{482}$$

The construction gives a jointly holomorphic extension in a complex coupling disc and all complex t, s. The radius and constant retain their dependence on the original finite box.

Proof. We first specify the analytic continuation of the vectors and bras. The full free ground eigenvalue is simple and the next full-space eigenvalue is $3/4$; the original perturbation has $\|S\| = 2M$. The resolvent on $|z| = 3/8$ has norm at most $8/3$, so the source Neumann construction supplies the analytic rank-one vacuum projection in a coupling disc contained in $|\xi| < 3/(16M)$. Write $\chi(\xi) = P(\xi)1$. Complex conjugation C on functions satisfies $CP(\xi)C = P(\bar{\xi})$, since both H_0 and S have real coefficients. Thus $C\chi(\xi) = \chi(\bar{\xi})$.

The scalar $\mathfrak{n}(\xi) = \langle \chi(\bar{\xi}), \chi(\xi)\rangle$ is holomorphic and equals one at zero. On a smaller symmetric disc, it is nonzero and has the holomorphic square root with value one at zero. Set $\psi(\xi) = \chi(\xi)/\sqrt{\mathfrak{n}(\xi)}$. For small real ξ this is the original positive unit vacuum: it is the real normalized eigenvector with positive overlap with the constant vector. For complex ξ, this formula fixes its analytic extension; no assertion that its Hilbert norm remains one off the real axis is needed. Its eigenvalue $e(\xi)$ is holomorphic and real on the real axis.

Analyticity of ψ holds in every fixed Sobolev norm. Indeed the exact equation

$$\psi(\xi) = (I + H_0)^{-1}[(1 + e(\xi) + \xi S)\psi(\xi)]$$

increases the Sobolev index by two, since multiplication by the original smooth S preserves that index and the elliptic inverse increases it by two. Starting with the Hilbert-space analytic projection proves the assertion inductively. Consequently

$$\gamma_e(\xi) = \langle \psi(\bar{\xi}), E_e\psi(\xi)\rangle, \qquad v_e(\xi) = (E_e - \gamma_e(\xi))\psi(\xi)$$

are holomorphic in every required Sobolev space. Their restrictions to real ξ are exactly (472). The conjugate-linear-first functional $u \mapsto \langle v_e(\bar{\xi}), u\rangle$ is holomorphic in ξ; it is the precise continuation of the original bra.

On the same physical domain put $A_\xi = H_0 - \xi S - e(\xi)I$, now for complex ξ. The bounded perturbation of the free unitary group gives e^{ivA_ξ} for every real v. Its Dyson series is obtained by inserting $-\xi S$ between the unchanged free groups. The term with n insertions has norm at most $(2M|\xi||v|)^n/n!$; the ordered integrals are strong integrals on each vector. The resulting bounded-operator series converges in norm, uniformly on compact coupling sets. The scalar shift commutes with every term and contributes $e^{-ive(\xi)}$. Hence

$$\|e^{ivA_\xi}\| \le \exp\{|v|(2M|\xi| + |\operatorname{Im} e(\xi)|)\}. \tag{483}$$

This proves norm-holomorphic dependence on ξ for fixed real v, with the entire original bounded potential retained.

Define the continuation of (478) by

$$\widetilde{G}(\xi, t, s) = 4\int_{\mathbb{R}} k(v)e^{tv^2/4+isv}\langle v_e(\bar{\xi}), e^{ivA_\xi}v_f(\xi)\rangle\, dv. \tag{484}$$

Choose a closed coupling disc $|\xi| \le r$ strictly inside the disc just constructed. Put

$$B_r = \sup_{|\xi|\le r} \|v_e(\bar{\xi})\|\,\|v_f(\xi)\|, \qquad E_r = \sup_{|\xi|\le r} |\operatorname{Im} e(\xi)|.$$

Both numbers are finite by continuity on that closed disc. On a compact arithmetic parameter set $\mathcal{Q}$, set $T_{\mathcal{Q}} = \max_{\mathcal{Q}} \operatorname{Re} t$ and $S_{\mathcal{Q}} = \max_{\mathcal{Q}} |\operatorname{Im} s|$. The absolute integral in (484) is uniformly bounded by the explicit finite number

$$M_{r,\mathcal{Q}} = 4B_r \int_{\mathbb{R}} |k(v)|e^{T_{\mathcal{Q}}v^2/4+(S_{\mathcal{Q}}+2Mr+E_r)|v|}\, dv. \tag{485}$$

The superexponential arithmetic kernel absorbs all these retained weights. On larger compact arithmetic sets it also absorbs the weights introduced by each parameter derivative. The locally uniformly convergent Dyson and scalar exponential series, or their Cauchy integral formulas dominated by this same bound, therefore prove joint holomorphy of $\widetilde{G}$. For real coupling it equals the original observable by Lemma 36.1.

The source central-link unitary is

$$(\mathcal{Z}F)(U) = F\big(((-1)^{\sum_{j<i} n_j} U_{(n,i)})_{(n,i)}\big).$$

It preserves Haar measure, commutes with every original E_e, and changes the sign of every Wilson trace. Consequently $\mathcal{Z}(H_0 - \xi S)\mathcal{Z} = H_0 + \xi S$. Uniqueness and positivity of the real vacuum imply $\psi_{-\xi} = \mathcal{Z}\psi_\xi$, $e(-\xi) = e(\xi)$, $v_e(-\xi) = \mathcal{Z}v_e(\xi)$, and $A_{-\xi} = \mathcal{Z}A_\xi\mathcal{Z}$. Thus $G_{-\xi}(t,s) = G_\xi(t,s)$ for real coupling. The identity theorem extends this equality to the complex disc. No time or arithmetic coordinate changes in this symmetry.

Write the coupling Taylor series as $\widetilde{G} = \sum_{n\ge 0} g_n(t,s)\xi^n$. The actual limit (479) on the real axis forces $g_0 = g_1 = g_2 = g_3 = 0$ and $g_4 = a_{ef}\mathcal{K}_t(s)$: otherwise division by ξ^4 could not have the proved finite limit. Evenness makes every odd coefficient vanish. The Cauchy bound on $|\xi| = r$ gives $\sup_{\mathcal{Q}} |g_n| \le M_{r,\mathcal{Q}}r^{-n}$. For $|\xi| \le r/2$, summing the remaining series yields

$$\sup_{\mathcal{Q}} |\widetilde{G} - a_{ef}\xi^4\mathcal{K}| \le 2M_{r,\mathcal{Q}}r^{-6}|\xi|^6.$$

This proves (482), for example with the specified constant $2M_{r,\mathcal{Q}}r^{-6}$. All box dependence in M, N, r, B_r, E_r remains present. The estimate does not exchange a volume limit with a coupling derivative. $\square$

36.4 An invertible arithmetic map on the full retained Fourier domain

Retain the exact spaces $\mathscr{E}, \mathscr{A}$ and transform T already defined above, including every seminorm

$$p_{A,B,m,j}(\phi) = \int_{\mathbb{R}} |\phi^{(j)}(v)|(1+|v|)^m e^{Av^2+B|v|}\,dv, \qquad T\phi(s) = 4\int_{\mathbb{R}} \phi(v)e^{isv}\,dv.$$

Define the source multiplier and arithmetic operator by

$$m(v) = c_0 e^{3iv} - \frac{iv}{336}e^{3iv} + c_1 e^{9iv/2} + c_2 e^{13iv/2}, \tag{486}$$

$$(\mathcal{T}_{\mathcal{D}} f)(s) = c_0 f(s+3) - \frac{f'(s+3)}{336} + c_1 f(s+9/2) + c_2 f(s+13/2). \tag{487}$$

Lemma 36.4 (A uniform exact lower bound for the original multiplier)**.** *For every real v,*

$$|m(v)| \geq \delta, \qquad \delta = \frac{575}{1752192} > 0. \tag{488}$$

In particular its real-axis reciprocal exists everywhere, with no branch choice or removed spectral point.

Proof. Factoring the nonzero phase gives the exact identity

$$e^{-3iv}m(v) = c_0 - \frac{iv}{336} + c_1 e^{3iv/2} + c_2 e^{7iv/2}.$$

For $|v| \leq 1/2$, the elementary inequality $\cos x \geq 1 - x^2/2$ yields

$$\begin{aligned} \operatorname{Re}(e^{-3iv}m(v)) &= c_0 + c_1\cos(3v/2) + c_2\cos(7v/2) \\ &\geq c_0 + \frac{23}{32}c_1 - \frac{17}{32}c_2 = \frac{575}{1752192}. \end{aligned}$$

For completeness, this cosine bound follows directly by integration: for $x \geq 0$, $1 - \cos x = \int_0^x \sin u\,du \leq \int_0^x u\,du$, using $\sin u \leq u$; the latter follows from $u - \sin u = \int_0^u (1-\cos w)\,dw \geq 0$. Evenness gives the negative case. Since the real part is positive, its displayed lower bound also bounds the modulus.

For $|v| \geq 1/2$, the triangle inequality with the original linear term gives

$$\begin{aligned} |m(v)| &\geq \frac{|v|}{336} - (|c_0| + c_1 + c_2) \\ &\geq \frac{1}{672} - \frac{10825}{10732176} = \frac{10291}{21464352} > \frac{575}{1752192}. \end{aligned}$$

All coefficients in these estimates are the original rational coefficients. The two intervals cover the whole real axis and prove (488). ☐

Theorem 36.5 (Exact arithmetic automorphism and heat conjugacy)**.** *Multiplication by m is a continuous automorphism of $\mathscr{E}$. Its inverse is literal pointwise multiplication by $1/m$ on the unchanged real Fourier axis. The operator $\mathcal{T}_{\mathcal{D}}$ is a continuous automorphism of $\mathscr{A}$, with exact inverse*

$$\mathcal{T}_{\mathcal{D}}^{-1} f = T\left(\frac{T^{-1}f}{m}\right). \tag{489}$$

For every complex t,

$$\mathcal{T}_{\mathcal{D}} = TM_mT^{-1}, \qquad \mathcal{T}_{\mathcal{D}}U_t = U_t\mathcal{T}_{\mathcal{D}}, \qquad \mathcal{T}_{\mathcal{D}}^{-1}U_t = U_t\mathcal{T}_{\mathcal{D}}^{-1}, \tag{490}$$

$$\mathcal{K}_t = \mathcal{T}_{\mathcal{D}}Z_t, \qquad Z_t = \mathcal{T}_{\mathcal{D}}^{-1}\mathcal{K}_t. \tag{491}$$

Thus the actual arithmetic function is exactly recoverable from the propagated source coefficient, on this fully specified domain.

Proof. Every derivative of m has at most linear growth on the real axis. More explicitly, for $j \ge 0$ it satisfies $|m^{(j)}(v)| \le C_j(1+|v|)$, where one may take

$$C_j = |c_0|3^j + c_1(9/2)^j + c_2(13/2)^j + \frac{3^j + \mathbf{1}_{j\ge 1} j 3^{j-1}}{336}.$$

The product rule gives the seminorm estimate

$$p_{A,B,m,j}(m\phi) \le \sum_{\ell=0}^{j} \binom{j}{\ell} C_\ell p_{A,B,m+1,j-\ell}(\phi).$$

This proves continuity of M_m on the original $\mathscr{E}$.

Let $r(v) = 1/m(v)$, which is smooth by Lemma 36.4. The exact product identity $mr = 1$ and its derivatives give

$$r^{(j)} = -\frac{1}{m}\sum_{\ell=1}^{j} \binom{j}{\ell} m^{(\ell)} r^{(j-\ell)} \quad (j \ge 1), \qquad |r| \le \delta^{-1}.$$

Set $Q_0 = \delta^{-1}$ and recursively define the finite positive constants

$$Q_j = \delta^{-1} \sum_{\ell=1}^{j} \binom{j}{\ell} C_\ell Q_{j-\ell}.$$

Induction proves $|r^{(j)}(v)| \le Q_j(1+|v|)^j$: the term with index $\ell \ge 1$ has exponent at most $1 + j - \ell \le j$, and the denominator is bounded by δ^{-1}. Therefore

$$p_{A,B,m,j}(r\phi) \le \sum_{\ell=0}^{j} \binom{j}{\ell} Q_\ell p_{A,B,m+\ell,j-\ell}(\phi).$$

This proves that reciprocal multiplication is continuous on exactly the same $\mathscr{E}$. The two pointwise compositions are the identity, proving the Fourier automorphism and its inverse.

Applying the retained transform to each term of $m\phi$ gives the three translations and the derivative in (487); the factor iv is exactly the Fourier multiplier of ∂_s. The injectivity of T already proved above makes its inverse on $\mathscr{A}$ unambiguous. This proves the first identity in (490) and (489). Multiplication by m and by $1/m$ each commutes pointwise with multiplication by $e^{tv^2/4}$. All three maps preserve $\mathscr{E}$, so applying T proves the two full-domain heat commutation identities for every complex t. Finally (477) is exactly $\mathcal{T}_{\mathcal{D}} Z_t$, and applying the proved inverse gives (491). $\square$

The source factor a_{ef} is separate from this automorphism. When $a_{ef} > 0$, its division also recovers the unscaled $\mathcal{K}$ from the coefficient in (479). The original boxes contain such pairs. In particular, for $e = ((0,0,0),1)$ and $f = ((0,2,0),1)$, there is exactly one common edge-adjacency neighbor, $((0,1,0),1)$. To verify uniqueness directly, a direction-1 neighbor of e is its parallel translate by one step in direction 2 or 3; the corresponding lists for e and f meet only at the displayed midpoint edge. A direction-2 neighbor has its direction-2 base coordinate in $\{-1,0\}$ for e, and in $\{1,2\}$ for f, so these lists are disjoint. A direction-3 neighbor has direction-2 coordinate zero for e and two for f, also disjoint. All faces in the unique midpoint channel are contained in every original box with $L \ge 2$. The exact source channel bijection gives $a_{ef} = 1$, including these open boundaries.

Equations (484) and (491) provide the complete source-to-arithmetic heat correspondence. They assert an invertible map of the specified arithmetic function space, not preservation of the pointwise zero divisor by a translated derivative operator. The original Z_t, its original coordinates, and its zeros remain recoverable by the explicit inverse (489).

37 Propagation through the original inverse fibres and spatial flow

Use the exact per-channel arithmetic heat function $\mathcal{K}_t(s)$ defined by (477), with all source constants in (467). Retain the incidence count by setting $G_t(s) = a_{ef}\mathcal{K}_t(s)$. Define

$$g_t(r) = G_t(-r^2/2), \qquad \Psi^{(4)}(t,q) = G_t(\sigma(q)), \quad \sigma(q) = F_1(q)/2.$$

These are entire compositions of the actual transformed arithmetic function; they retain the incidence factor a_{ef}. Since the new transform commutes with the actual Newman heat evolution, the already proved coordinate chain rules apply to it verbatim and give

$$\begin{aligned} \partial_t g_t &= -\frac{1}{4r^2}\partial_r^2 g_t + \frac{1}{4r^3}\partial_r g_t \qquad\qquad (r \neq 0),\\ \partial_t \Psi^{(4)} + \tfrac{1}{4}U^2\Psi^{(4)} &= 0,\\ \Delta_q \Psi^{(4)} &= N(q)G_{ss}(t,\sigma(q)) + L(q)G_s(t,\sigma(q)). \end{aligned} \tag{492}$$

Every coefficient N, L, U is the original one proved in Section 34; no Euclidean diffusion coefficient has been replaced by the root-coordinate generator. For any retained time map $t = \theta(\tau)$, the full residual is explicitly

$$(\partial_\tau + U \cdot \nabla_q - \nu\Delta_q)G_{\theta(\tau)}(\sigma(q)) = \left(-\frac{\theta'}{4} - \nu N\right)G_{ss} + (1-\nu L)G_s,$$

with the derivatives on the right evaluated at $(\theta(\tau), \sigma(q))$. The proof is the time chain rule and the three identities in (492), so the force-sized remainder is calculated rather than assumed absent.

The three spectral translations also have exact maps in the retained original root coordinates. For $\lambda \in \{3, 9/2, 13/2\}$ define the entire algebraic correspondence

$$\mathcal{C}_\lambda = \{(r,u) \in \mathbb{C}^2 : u^2 = r^2 - 2\lambda\}.$$

It satisfies $-u^2/2 = -r^2/2 + \lambda$. Consequently

$$Z_t(-r^2/2 + \lambda) = f_t(u), \qquad Z_s(t, -r^2/2 + 3) = -\frac{\partial_u f_t(u)}{u} \quad ((r,u) \in \mathcal{C}_3,\ u \neq 0),$$

where $f_t(u) = Z_t(-u^2/2)$ is the original function, already proved even and entire. The second formula follows from $\partial_u f_t = -uZ_s(t, -u^2/2)$; thus its quotient has a unique entire continuation at $u = 0$ equal to $-Z_s(t, 0)$. Neither formula depends on the choice of sign of u.

In these actual correspondences the full propagated function is

$$g_t(r) = a_{ef}\left[c_0 f_t(u_3) + \beta\frac{\partial_u f_t(u_3)}{u_3} + c_1 f_t(u_{9/2}) + c_2 f_t(u_{13/2})\right], \qquad u_\lambda^2 = r^2 - 2\lambda.$$

Each potentially branched expression descends to the displayed entire function of r, by the proven evenness and removable quotient. The finite inverse point for a nonzero u_λ is exactly $\Gamma(u_\lambda) = (-1/(2u_\lambda), 3u_\lambda, 26u_\lambda^2)$; its target is

$$F(\Gamma(u_\lambda)) = (-r^2 + 2\lambda, 0, 0).$$

At $u_\lambda = 0$ the completed root point still has no finite inverse in the $x \neq 0$ chart, as its unchanged first coordinate shows. The additional finite point with $x = 0$ is retained by the earlier complete fibre calculation. The three new branch values are precisely $r^2 = 6, 9, 13$; they arise from the source's three specified energies.

All of this also retains the full original cubic coefficient family. For fixed b, c and $s(r) = (cr^3 - 2r^2 + br)/4$, set $g_{t;b,c}(r) = G_t(s(r))$. The exact same chain rule gives

$$\partial_t g_{t;b,c} = -\frac{4}{P_r^2}\partial_r^2 g_{t;b,c} + \frac{4P_{rr}}{P_r^3}\partial_r g_{t;b,c} \quad (P_r = 3cr^2 - 4r + b \neq 0).$$

Thus the source fourth coefficient is propagated into the coefficient family as well as its $b = c = 0$ slice. This is a completed map of the source observable and its actual heat dynamics. For the per-channel function, recovery of the original Ξ uses the full-domain inverse automorphism proved in Section 36. For any actual pair with $a_{ef} > 0$, the recovered input is $Z_t = \mathcal{T}_{\mathcal{D}}^{-1}(G_t/a_{ef})$; the explicit separated pair in the source proof has $a_{ef} \geq 1$. At $a_{ef} = 0$ this fourth coefficient is exactly zero and carries no pair observation. A zero of a finite linear combination is not silently substituted for a zero of its recovered arithmetic input.

38 The second Euclidean diffusion of the actual arithmetic observable

The first diffusion coefficient in the preceding calculation has a finite limit along the escaping inverse-fibre trajectory. We compute the next Euclidean diffusion without dropping any derivative of that coefficient. All spatial derivatives in this section refer to the original Cartesian coordinates (x, y, w), with their Euclidean metric. Newman time t remains distinct from the fluid trajectory time τ.

Keep

$$Q = 1 + xy, \qquad \sigma(x, y, w) = \tfrac{1}{2}\left[Q^3 w + y^2 Q(4 + 3xy)\right], \qquad \Psi_t(x, y, w) = Z(t, \sigma(x, y, w)).$$

Here Z is the actual entire function defined by (443); its initial datum is the specified completed zeta function. Its analyticity and all differentiated integral bounds were proved above. In particular, for each real t,

$$Z_s(t, 0) = Z_{sss}(t, 0) = 0, \qquad Z_{ss}(t, 0) = -32M_1(t) < 0.$$

The strict inequality retains the positive original moment $M_1(t) = \int_0^\infty u^2 e^{tu^2}\Phi(u)\, du$.

38.1 The full fourth-order chain rule

Write $g = \nabla\sigma$, $H = D^2\sigma$, $N = g \cdot g$, and $L = \Delta\sigma$, where $\Delta = \partial_x^2 + \partial_y^2 + \partial_w^2$. For any entire function h of one variable, ordinary differentiation gives

$$\Delta(h \circ \sigma) = Nh''(\sigma) + Lh'(\sigma), \tag{493}$$

$$\begin{aligned}\Delta^2(h \circ \sigma) = {} & N^2 h^{(4)}(\sigma) + (4g^\mathsf{T} Hg + 2NL)h^{(3)}(\sigma) \\ & + (2\|H\|_\mathrm{F}^2 + 4g \cdot \nabla L + L^2)h''(\sigma) + (\Delta L)h'(\sigma).\end{aligned} \tag{494}$$

Here $\|H\|_\mathrm{F}^2 = \sum_{i,j=1}^3 H_{ij}^2$ on the real locus. To prove every coefficient, differentiate the first formula by the product rule:

$$\Delta(Nh'') = (\Delta N)h'' + 2(g \cdot \nabla N)h^{(3)} + NLh^{(3)} + N^2 h^{(4)},$$
$$\Delta(Lh') = (\Delta L)h' + 2(g \cdot \nabla L)h'' + L^2 h'' + LNh^{(3)}.$$

In these lines every derivative of h is evaluated at σ. For the remaining two identities, write $N = \sum_i (\partial_i\sigma)^2$ and differentiate each summand:

$$\partial_j N = 2\sum_i (\partial_i\sigma)(\partial_j\partial_i\sigma), \qquad \Delta N = 2\sum_{i,j}(\partial_j\partial_i\sigma)^2 + 2\sum_i (\partial_i\sigma)\partial_i\Delta\sigma.$$

Thus $g \cdot \nabla N = 2g^\mathsf{T} Hg$ and $\Delta N = 2\|H\|_\mathrm{F}^2 + 2g \cdot \nabla L$. Substitution proves (494), including its second- and first-derivative terms.

38.2 All coefficients on the unchanged inverse trajectory

For $z > 0$ set

$$\gamma(z) = (z^{-1}, -3z/2, 13z^2/2), \qquad \tau = (1 - z^2)/8.$$

The inverse is $z = \sqrt{1 - 8\tau}$ on $\tau < 1/8$. This is precisely the original trajectory, parametrized by a positive coordinate; it has not been rescaled. Direct substitution gives $Q(\gamma) = -1/2$ and $\sigma(\gamma) = -z^2/8$. For clarity the original first derivatives and Laplacian before substitution are

$$\begin{aligned} g_1 &= \tfrac{3}{2} y Q^2 w + \tfrac{1}{2} y^3 (7 + 6xy), \\ g_2 &= \tfrac{3}{2} x Q^2 w + \tfrac{1}{2}(8y + 21xy^2 + 12x^2y^3), \\ g_3 &= \tfrac{1}{2} Q^3, \\ L &= 3wQ(x^2 + y^2) + 3y^4 + 18x^2y^2 + 21xy + 4. \end{aligned}$$

Differentiating these polynomials, followed by the stated substitution, gives

$$g(\gamma) = \begin{pmatrix} -9z^3/32 \\ -3z/16 \\ -1/16 \end{pmatrix}, \qquad H(\gamma) = \begin{pmatrix} -27z^4/4 & 3z^2/16 & -9z/16 \\ 3z^2/16 & 13/4 & 3/(8z) \\ -9z/16 & 3/(8z) & 0 \end{pmatrix}.$$

The full contractions needed in (494) are

$$\begin{aligned} N(\gamma) &= \frac{81z^6 + 36z^2 + 4}{1024}, & L(\gamma) &= \frac{13 - 27z^4}{4}, \\ \|H(\gamma)\|_{\mathrm{F}}^2 &= \frac{729z^8}{16} + \frac{9z^4}{128} + \frac{81z^2}{128} + \frac{169}{16} + \frac{9}{32z^2}, \\ (g \cdot \nabla L)(\gamma) &= \frac{9477z^8}{512} - \frac{405z^4}{64} + \frac{27z^2}{128} + \frac{81}{32} + \frac{3}{32z^2}, \\ (g^{\mathsf{T}} H g)(\gamma) &= -\frac{2187z^{10}}{4096} + \frac{81z^6}{4096} - \frac{81z^4}{4096} + \frac{117z^2}{1024} + \frac{9}{1024}, \\ (\Delta L)(\gamma) &= -111z^2 + \frac{36}{z^2}. \end{aligned}$$

These are polynomial differentiation and finite matrix products, with both occurrences of each off-diagonal Hessian entry retained in its squared Frobenius norm. They also give a direct way to check each term in the following fully expanded result:

$$(\Delta^2 \Psi_t)(\gamma(z)) = \sum_{j=1}^{4} C_j(z)\, \partial_s^j Z(t, -z^2/8), \tag{495}$$

where

$$\begin{aligned} C_4(z) &= \left(\frac{81z^6 + 36z^2 + 4}{1024} \right)^2, \\ C_3(z) &= -\frac{6561z^{10}}{2048} + \frac{243z^6}{2048} - \frac{135z^4}{1024} + \frac{351z^2}{512} + \frac{31}{512}, \\ C_2(z) &= \frac{26973z^8}{128} - \frac{4419z^4}{64} + \frac{135z^2}{64} + \frac{669}{16} + \frac{15}{16z^2}, \\ C_1(z) &= -111z^2 + \frac{36}{z^2}. \end{aligned}$$

For example, the coefficient of z^{-2} in C_2 is $2(9/32) + 4(3/32) = 15/16$; the L^2 term has no negative power. Thus this coefficient includes the differentiated first-order coefficient as well as the Hessian square.

Proposition 38.1 (Two successive Euclidean diffusion limits)**.** *For every fixed real Newman time t,*

$$\lim_{z\downarrow 0}(\Delta\Psi_t)(\gamma(z)) = \frac{Z_{ss}(t,0)}{256} = -\frac{M_1(t)}{8}, \tag{496}$$

$$\lim_{z\downarrow 0} z^2(\Delta^2\Psi_t)(\gamma(z)) = \frac{15}{16}Z_{ss}(t,0) = -30M_1(t) < 0. \tag{497}$$

In particular $\Delta^2\Psi_t$ is unbounded on $\mathbb{R}^3$, although Ψ_t is real analytic and smooth at every finite spatial point.

Proof. At this fixed t, the entire even Taylor expansion of $Z(t,s)$ and its termwise derivatives give, with $d_j = \partial_s^j Z(t,0)$,

$$\begin{aligned} Z_s(t,-z^2/8) &= -d_2 z^2/8 + O(z^6), \\ Z_{ss}(t,-z^2/8) &= d_2 + O(z^4), \\ Z_{sss}(t,-z^2/8) &= -d_4 z^2/8 + O(z^6), \\ Z_{ssss}(t,-z^2/8) &= d_4 + O(z^4). \end{aligned}$$

The remainders follow from convergence on a fixed closed disk around zero; they are bounded there by the convergent Taylor tails. Equation (493), $N(\gamma) \to 1/256$, and $L(\gamma) \to 13/4$ prove (496). In (495), C_4 and C_3 are bounded as $z \downarrow 0$, $z^2C_2 \to 15/16$, and $z^2C_1 \to 36$. Multiplying that equation by z^2 and using the four Taylor estimates proves (497). The last term tends to zero because $Z_s(t,-z^2/8) \to 0$; its unmultiplied finite contribution is $-36d_2/8 + O(z^4)$ and has not been discarded from the exact formula. The moment identity and positivity give the displayed strict sign. Consequently, for sufficiently small $z > 0$ the bilaplacian is at most $-15M_1(t)z^{-2}$, proving unboundedness. Entire composition proves smoothness at each finite point. Finally $|\gamma(z)| \geq z^{-1} \to \infty$, so this sequence is explicitly at spatial infinity. ☐

38.3 The exact relation to the material diffusion operator

Retain the original divergence-free polynomial flow X_τ and its Jacobian A_τ from the material-generator construction. On each of its open flow domains, X_τ is a smooth diffeomorphism, $\det A_\tau = 1$, and $K_\tau = A_\tau^{-1}A_\tau^{-\mathsf{T}}$. The scalar operator proved there is $\mathcal{L}_\tau = \operatorname{div}(K_\tau\nabla)$. For every smooth physical scalar v on the image domain the full chain and cofactor identity give

$$\mathcal{L}_\tau(v \circ X_\tau) = (\Delta v) \circ X_\tau.$$

Apply this same identity a second time, now to the smooth scalar Δv. At this fixed τ it proves the exact intertwining

$$\mathcal{L}_\tau^2(v \circ X_\tau) = (\Delta^2 v) \circ X_\tau.$$

The square is operator composition; it differentiates every spatial coefficient of K_τ. It is not the square of a coefficient evaluated at one point.

The already established original flow identity $F_1(X_\tau(q)) = F_1(q) + 2\tau$ gives

$$\Psi_t(X_\tau(q)) = Z(t,\sigma(q)+\tau).$$

At $q_0 = (1,-3/2,13/2)$ and $\tau < 1/8$ one has $X_\tau(q_0) = \gamma(\sqrt{1-8\tau})$. Therefore (496) and (497) give, without interchanging the two time variables,

$$\lim_{\tau\uparrow 1/8} \mathcal{L}_\tau[Z(t,\sigma+\tau)](q_0) = -M_1(t)/8,$$

$$\lim_{\tau\uparrow 1/8} (1-8\tau)\mathcal{L}_\tau^2[Z(t,\sigma+\tau)](q_0) = -30M_1(t).$$

For the physical operator $\nu\Delta$ with the unchanged $\nu > 0$, the respective limits acquire the factors ν and ν^2. The scalar in brackets itself has an entire extension in τ. Its material coefficients nevertheless have the displayed second diffusion behavior at the end of this incomplete chart. These identities are an exact relation between the arithmetic observable and Euclidean viscosity. They do not give a singular velocity at a finite physical point, admissible initial velocity or forcing, or a counterexample to the stated Navier–Stokes alternatives.

39 The exact Euler–Navier–Stokes force map and its terminal obstruction

All spatial variables in this section remain the original Cartesian variables $x = (x_1, x_2, x_3) \in \mathbb{R}^3$. Time remains $t \in [0, T)$, where $0 < T < \infty$. The viscosity is an arbitrary fixed real number $\nu > 0$; it is not set equal to one. Vector norms are Euclidean, matrix norms are Frobenius, and all L^p norms below are over $\mathbb{R}^3$. We use $(\operatorname{curl} v)_i = \sum_{j,k=1}^{3} \varepsilon_{ijk}\partial_j v_k$, $\operatorname{div} v = \sum_j \partial_j v_j$, and $\Delta v = \sum_j \partial_j^2 v$, component by component. In particular, $\varepsilon_{123} = 1$ fixes the orientation.

Source and the exact scope of the application. The released manuscript *Blowup for the Euler equations with smooth forcing*, Theorem 1.1, states the existence of a smooth Euler solution with compactly supported smooth initial velocity, a force smooth through a finite terminal time and supported in a fixed solid torus, finite energy and bounded spatial velocity gradients on each shorter time interval, and $\|\operatorname{curl} u(t)\|_\infty \longrightarrow \infty$ as $t \uparrow T$. Its Section 13.2 and Proposition 13.3 additionally state that the constructed velocity has fixed compact support and is uniformly bounded before T. The released `euler-blowup/README.md` states the finite-energy Lipschitz class and the same vorticity divergence. These are the specific source facts used for the application below. This section proves the force map and the obstruction to transferring that *same velocity* with a regular force. It does not repeat or independently certify the manuscript's complete Euler existence construction or its claimed Lean verification. No Navier–Stokes existence or singularity assertion is assumed in the proofs below.

39.1 The map, including every pressure adjustment

Let (u, p_E, f_E) solve

$$\partial_t u + (u \cdot \nabla)u + \nabla p_E = f_E, \qquad \operatorname{div} u = 0. \tag{498}$$

For any scalar distribution q on $\mathbb{R}^3 \times (0, T)$ define

$$p_{NS} = p_E + q, \qquad f_{NS} = f_E - \nu\Delta u + \nabla q. \tag{499}$$

Then exactly the same velocity u satisfies

$$\partial_t u + (u \cdot \nabla)u + \nabla p_{NS} = \nu\Delta u + f_{NS}, \qquad \operatorname{div} u = 0. \tag{500}$$

Indeed the left side of (500) is $f_E + \nabla q$ by (498), and its right side is $\nu\Delta u + f_E - \nu\Delta u + \nabla q = f_E + \nabla q$. Conversely, subtracting (498) from (500) proves (499) with $q = p_{NS} - p_E$. Thus (499) describes *all* forces and pressures for which the original velocity solves the viscous equation; there is no restriction on the pressure adjustment beyond the meaning of the displayed distributions.

Put $\omega = \operatorname{curl} u$. Commutation of distributional partial derivatives gives

$$\operatorname{curl} f_{NS} = \operatorname{curl} f_E - \nu\Delta\omega. \tag{501}$$

To verify the pressure cancellation componentwise, write its ith component as

$$(\operatorname{curl} \nabla q)_i = \sum_{j,k} \varepsilon_{ijk}\partial_j\partial_k q.$$

the terms indexed by (j,k) and (k,j) cancel because the derivatives commute and $\varepsilon_{ikj} = -\varepsilon_{ijk}$. The identity (501) is therefore independent of q, including pressure adjustments whose gradients have bad behavior at spatial infinity. An arbitrary distribution q need not possess time sections. Nevertheless, since u and f_E are smooth before T, the right side of (501) is smooth and is the canonical representative of the distribution $\operatorname{curl} f_{NS}$. Its time sections are therefore defined for every $t < T$, independently of whether q or f_{NS} has time sections.

39.2 A terminal energy bound from the stated preterminal class

Here is a complete justification that the source's preterminal energy hypothesis supplies a bound independent of the shorter interval. Suppose u is smooth on $\mathbb{R}^3 \times [0,T)$ and solves (498), with a distributional pressure, and for every $\tau < T$ satisfies

$$u \in L^\infty([0,\tau]; L^2), \qquad \nabla u \in L^\infty(\mathbb{R}^3 \times [0,\tau]). \tag{502}$$

Suppose f_E is continuous into L^2 on every such shorter interval and $\int_0^T \|f_E(s)\|_2\, ds < \infty$. The compactly supported smooth force in the source has these properties: a common compact containing set K gives $\|f_E(s)\|_2 \le |K|^{1/2} \sup_{K\times[0,T]} |f_E|$.

First, a spatially Lipschitz field $v \in L^2$ with Lipschitz constant $L = \|\nabla v\|_\infty > 0$ is bounded. If $A = |v(x)| > 0$, then $|v(y)| \ge A/2$ on the ball of radius $A/(2L)$ about x. Hence

$$\|v\|_2^2 \ge \frac{A^2}{4}\frac{4\pi}{3}\left(\frac{A}{2L}\right)^3 = \frac{\pi A^5}{24L^3}, \qquad \|v\|_\infty \le \left(\frac{24}{\pi}\right)^{1/5} \|v\|_2^{2/5} L^{3/5}.$$

If $L = 0$, v is constant and its finite L^2 norm forces $v = 0$. Smoothness and Fatou's lemma extend the almost-every-time bounds in (502) to every time section. The inequality therefore bounds u uniformly on each $[0,\tau]$.

We justify the elimination of pressure without a growth assumption on p_E. Use the unitary Fourier convention $\widehat{v}(\xi) = (2\pi)^{-3/2} \int e^{-ix\cdot\xi} v(x)\, dx$, extended by density to L^2. Let P be the orthogonal L^2 projection with multiplier $I - \xi \otimes \xi/|\xi|^2$ for $\xi \ne 0$; its value at 0 does not affect an L^2 function. This real symmetric idempotent multiplier has operator norm at most one. It projects onto the fields with distributional divergence zero and preserves curl, as is verified by $\xi \cdot Pv = 0$ and $\xi \times Pv = \xi \times v$.

Set $g = f_E - (u\cdot\nabla)u$. On $[0,\tau]$ this is an $L_t^\infty L_x^2$ field, since $\|(u\cdot\nabla)u\|_2 \le \|\nabla u\|_\infty \|u\|_2$. Taking curl of Euler and using $\operatorname{div} u = 0$ shows that both curl and divergence of the space-time distribution $\partial_t u - Pg$ vanish. For a compactly supported smooth time test η in $(0,\tau)$, its pairing with that test is the L^2 field

$$V_\eta = -\int_0^\tau u(t)\eta'(t)\, dt - \int_0^\tau Pg(t)\eta(t)\, dt.$$

It has zero curl and divergence. Fourier transformation gives $\xi \times \widehat{V}_\eta = 0$ and $\xi \cdot \widehat{V}_\eta = 0$ almost everywhere. For $\xi \ne 0$ the first relation makes $\widehat{V}_\eta$ parallel to ξ and the second makes it perpendicular, so it is zero. The remaining point $\xi = 0$ has measure zero. Thus $V_\eta = 0$, proving $\partial_t u = Pg$ as an L^2-valued distribution in time.

For completeness, the distributional conclusion supplies an absolutely continuous L^2 representative: subtract the Bochner primitive $\int_0^t Pg(s)\, ds$ from u; pairing the remainder against each L^2 test vector gives a scalar distribution with derivative zero and hence a constant. The constants define one L^2 vector, using the uniform L^2 bounds. This gives the claimed representative. Its strong L^2 continuity and the original local smoothness identify it with the given $u(t)$ at every t, including $t = 0$: choose times of almost-everywhere agreement converging to the chosen time and pair against compactly supported spatial tests. Consequently $u(t) = u(0) + \int_0^t Pg(s)\, ds$ in L^2.

The Hilbert-space norm identity now gives, for almost every $t < \tau$,

$$\frac{1}{2}\frac{d}{dt}\|u(t)\|_2^2 = \langle u, Pg\rangle_{L^2} = \langle u, f_E\rangle_{L^2} - \int u\cdot((u\cdot\nabla)u)\, dx.$$

The norm identity follows by expanding the square of $u(t+h) = u(t) + \int_t^{t+h} Pg(s)\,ds$ and using the Lebesgue differentiation theorem for the L^2-valued integrand. Orthogonality of P and $Pu = u$ justify its removal from the inner product.

Choose a smooth cutoff $\chi_R(x) = \chi(x/R)$, equal to one on $|x| \le R$, zero for $|x| \ge 2R$, with $0 \le \chi \le 1$ and $|\nabla\chi_R| \le C_\chi/R$. Local integration by parts and $\operatorname{div} u = 0$ give

$$\int \chi_R u \cdot ((u\cdot\nabla)u)\,dx = -\frac{1}{2}\int |u|^2 u \cdot \nabla\chi_R\,dx.$$

The right side has absolute value at most $C_\chi \|u\|_\infty \|u\|_2^2/(2R)$ and tends to zero. The left integrand is dominated by $\|\nabla u\|_\infty |u|^2$, so dominated convergence applies. We have proved the exact Euler energy identity

$$\frac{1}{2}\|u(t)\|_2^2 - \frac{1}{2}\|u(0)\|_2^2 = \int_0^t \langle u(s), f_E(s)\rangle_{L^2}\,ds. \tag{503}$$

For $\epsilon > 0$, differentiation of $(\|u(t)\|_2^2 + \epsilon^2)^{1/2}$, the identity just proved, and Cauchy–Schwarz bound its derivative above by $\|f_E(t)\|_2$. Integrating and then taking $\epsilon \downarrow 0$ proves

$$\|u(t)\|_2 \le \|u(0)\|_2 + \int_0^t \|f_E(s)\|_2\,ds \le E := \|u(0)\|_2 + \int_0^T \|f_E(s)\|_2\,ds < \infty. \tag{504}$$

The proof works on every $[0,\tau]$, and its final constant does not depend on τ. It therefore proves (504) for every $t < T$. For the actual constructed field, the fixed support and uniform velocity bound in Proposition 13.3 give a second direct verification of this conclusion.

39.3 A heat identity with no hidden assumption at spatial infinity

Let $v \in L^2(\mathbb{R}^3;\mathbb{R}^3)$ and put $\zeta = \operatorname{curl} v$ as a tempered distribution. Suppose the distribution $\Delta\zeta$ is represented by a bounded vector field h. Define, for $a > 0$,

$$G_a(x) = (4\pi a)^{-3/2}\exp\left(-\frac{|x|^2}{4a}\right), \qquad e^{a\Delta}w = G_a * w.$$

The following identity is an equality of tempered distributions, with a bounded continuous representative on its right:

$$\zeta = \operatorname{curl}(G_a * v) - \int_0^a G_r * h\,dr. \tag{505}$$

In particular,

$$\|\operatorname{curl} v\|_\infty \le C_0 a^{-5/4}\|v\|_2 + a\|\Delta\operatorname{curl} v\|_\infty, \qquad C_0 = \frac{1}{\sqrt{2}}(8\pi)^{-3/4}, \quad a > 0. \tag{506}$$

For a smooth v the bound concerns its ordinary pointwise curl.

Here are all convergence and constant computations. A function in L^2 defines a tempered distribution by Cauchy–Schwarz on Schwartz test functions; its distributional derivatives do so as well. For a Schwartz test function ϕ, the map $r \mapsto G_r * \phi$, with value ϕ at $r = 0$, is continuously differentiable in Schwartz space on every bounded interval of nonnegative r, and its derivative is $G_r * \Delta\phi$. This follows directly by Fourier transformation: the multipliers are $e^{-r|\xi|^2}$ and $-|\xi|^2 e^{-r|\xi|^2}$; differentiating each Schwartz seminorm is justified by the rapid decay of $\widehat{\phi}$ and polynomial bounds on all derivatives of these multipliers, uniform for bounded r. The fundamental theorem of calculus, paired with ζ, gives

$$\langle G_a * \zeta - \zeta, \phi\rangle = \int_0^a \langle \Delta\zeta, G_r * \phi\rangle\,dr = \int_0^a \langle G_r * h, \phi\rangle\,dr.$$

Fubini is valid because $|\langle h, G_r * \phi\rangle| \le \|h\|_\infty\|\phi\|_1$ for every $r \ge 0$; the Gaussian has integral one and is nonnegative. Differentiating the convolution with v in distributions gives $G_a * \zeta = \operatorname{curl}(G_a * v)$, proving (505). The function $\int_0^a G_r * h\,dr$ is bounded by $a\|h\|_\infty$. It is continuous: integrals over $[\delta, a]$ are continuous, and the integral over $[0,\delta]$ has uniform norm at most $\delta\|h\|_\infty$, which tends to zero. This proves the asserted representative without a pointwise or growth assumption on ζ.

For a real unit vector b and every x, the scalar triple product and Cauchy–Schwarz give

$$|b\cdot \operatorname{curl}(G_a * v)(x)| = \left|\int v(y)\cdot (b\times \nabla G_a(x-y))\,dy\right| \le \|v\|_2 \left(\int |b\times \nabla G_a(z)|^2\,dz\right)^{1/2}.$$

The sign in this formula follows from $\operatorname{curl}(G_a * v) = \int \nabla G_a(x-y)\times v(y)\,dy$. The Gaussian integrals are

$$\|G_a\|_2^2 = (4\pi a)^{-3}(2\pi a)^{3/2} = (8\pi a)^{-3/2}, \qquad \int (\partial_j G_a)(\partial_k G_a) = \frac{\delta_{jk}}{4a}\|G_a\|_2^2.$$

Indeed $\partial_j G_a = -x_j G_a/(2a)$, odd integrands have zero integral, and integration by parts in one dimension gives $\int x_j^2 G_a^2 = a\int G_a^2$. Thus

$$\int |b\times \nabla G_a|^2 = \int (|\nabla G_a|^2 - |b\cdot\nabla G_a|^2) = \frac{1}{2a}(8\pi a)^{-3/2}.$$

Taking the supremum over $|b|=1$ yields $\|\operatorname{curl}(G_a * v)\|_\infty \le C_0 a^{-5/4}\|v\|_2$ with precisely the constant displayed in (506). Together with (505) this proves the bound.

There is no surviving spatial harmonic term in this argument: the distributional identity started with $\zeta = \operatorname{curl} v$ for $v\in L^2$, kept its exact heat evolution, and integrated its actual Laplacian. No boundedness of ζ was used to prove its boundedness.

39.4 The pressure-independent obstruction and its rate

Apply (506) to each original time section $u(t)$. Write

$$M(t) = \|\omega(t)\|_\infty, \quad F_E(t) = \|\operatorname{curl} f_E(t)\|_\infty, \quad F_{NS}(t) = \|(\operatorname{curl} f_E - \nu\Delta\omega)(t)\|_\infty.$$

Thus F_{NS} always uses the canonical curl representative just proved to exist, even for an arbitrary distributional pressure adjustment. Whenever $F_{NS}(t) < \infty$, identity (501) shows that $\Delta\omega(t)$ is bounded and that

$$M(t) \le C_0 a^{-5/4} E + \frac{a}{\nu}(F_E(t) + F_{NS}(t)) \qquad \text{for every } a > 0. \tag{507}$$

In particular, a finite uniform bound on the two force curls gives a finite uniform bound on vorticity, contradicting the source's $M(t) \longrightarrow \infty$.

The conclusion already excludes uniformly bounded first spatial derivatives of f_{NS}. Explicitly, for any vector field f,

$$|\operatorname{curl} f|^2 = (\partial_2 f_3 - \partial_3 f_2)^2 + (\partial_3 f_1 - \partial_1 f_3)^2 + (\partial_1 f_2 - \partial_2 f_1)^2 \le 2|\nabla f|_F^2,$$

using $(a-b)^2 \le 2(a^2+b^2)$ on each displayed summand. The source's fixed support and smoothness through T give $F_E^* := \sup_{t<T} F_E(t) < \infty$. If f_{NS} is a smooth spacetime force and $\sup_{t<T}\|\nabla f_{NS}(t)\|_\infty = B < \infty$, then (507) gives

$$M(t) \le C_0 a^{-5/4} E + \frac{a}{\nu}(F_E^* + \sqrt{2}B) \quad (0 \le t < T)$$

for each fixed $a > 0$, an explicit contradiction. More generally, suppose the spacetime distributions $\partial_j (f_{NS})_k$ have bounded measurable representatives, whose matrix has essential supremum Frobenius norm B on $\mathbb{R}^3\times(0,T)$. The same componentwise inequality gives a spacetime essential bound $\sqrt{2}B$ on the curl. The canonical curl representative is smooth, so this bound holds everywhere: a strict violation at one point would persist on an open set of positive measure. The displayed contradiction therefore also rules out uniformly bounded spacetime weak first derivatives for arbitrary distributional forces, without assuming time sections of their gradients.

One can also quantify the necessary loss. The source has $E > 0$: otherwise (504) gives $u = 0$ and $M = 0$. Put $A = C_0 E > 0$. At a time with $M(t) > 0$, rearrange (506), using $\|u(t)\|_2 \le E$, to get

$$\|\Delta\omega(t)\|_\infty \ge \frac{M(t)}{a} - \frac{A}{a^{9/4}}.$$

Take the admissible positive value $a = (9A/(4M(t)))^{4/5}$. This is the maximizer: differentiating the right side gives $-M(t)a^{-2} + (9A/4)a^{-13/4}$, with a unique zero there, positive derivative before that zero and negative derivative after it. Substitution gives

$$\|\Delta\omega(t)\|_\infty \ge \frac{5}{9}\left(\frac{4}{9}\right)^{4/5}(C_0 E)^{-4/5} M(t)^{9/5}. \tag{508}$$

The reverse triangle inequality in (501) therefore proves, for every pressure adjustment,

$$F_{NS}(t) \ge \nu \frac{5}{9}\left(\frac{4}{9}\right)^{4/5}(C_0 E)^{-4/5} M(t)^{9/5} - F_E(t). \tag{509}$$

When a force curl is unbounded at the chosen time the inequality holds in the extended sense. If it is bounded, all preceding steps apply. For $M(t) = 0$ the same displayed lower bound is simply $F_{NS}(t) \ge -F_E(t)$ and also holds. Finally, whenever the force has smooth spacetime representatives, as every candidate in the target problem must have,

$$\|\nabla f_{NS}(t)\|_\infty \ge \frac{1}{\sqrt{2}}\left[\nu \frac{5}{9}\left(\frac{4}{9}\right)^{4/5}(C_0 E)^{-4/5} M(t)^{9/5} - F_E(t)\right]. \tag{510}$$

Consequently $M(t) \to \infty$ and $F_E^* < \infty$ imply $F_{NS}(t) \to \infty$ for every distributional pressure adjustment in (499). For smooth spacetime forces they also imply $\|\nabla f_{NS}(t)\|_\infty \to \infty$. This is stronger than excluding only the choice $p_{NS} = p_E$.

The precise result for the released construction. The original Euler velocity has a well-defined exact viscous realization with force $f_E - \nu\Delta u + \nabla q$ for every $\nu > 0$ and every pressure adjustment q. Every such force has a canonical smooth curl whose spatial supremum diverges as the original terminal time is approached, and none has uniformly bounded spacetime weak first derivatives. For smooth spacetime forces, their spatial first-derivative supremum also diverges at that time. Thus this realization cannot have a force smooth through the terminal time with a common compact support, nor can it satisfy a requirement of globally bounded first spatial derivatives through that time. On $\mathbb{R}^3$, smoothness alone on a noncompact spatial domain would not imply a global derivative bound; the fixed-support or stated decay requirement is material and has been kept explicit.

This obstruction concerns the original velocity and the exact map (499). It proves no nonexistence statement about a different Navier–Stokes velocity, a different construction, a viscosity-dependent change of the evolving fields, or a different mathematical correspondence. It also gives no Navier–Stokes counterexample with the initial data and forcing regularity required by a Millennium problem formulation. The map and the obstruction, rather than a mere comparison of equation names, identify exactly what a successful further transfer must change.

40 An exact audit of the 2024 rational cylinder construction

40.1 Scope and primary-source comparison

The primary source is T. Tao, *A possible approach to finite time blowup for the Euler equations*, 8 September 2024, `https://terrytao.wordpress.com/2024/09/08/a-possible-approach-t` o-finite-time-blowup-for-the-euler-equations/. The retrieved display contains the axial factor $(1+x^2)^{-1}$, the transverse factor $(1+x^2)^{-2}$, and a middle coefficient $-2\varphi v$. The calculations below retain these original rational functions and prove that the coefficient they generate is $-4\varphi v$. The source's equation (7) also has its two Gram-matrix rows interchanged. These are statements about the retrieved formulas, not claims about a singularity theorem. All further formulas in this section are derived here. The incoming report's equations (16.4)–(16.9), on physical PDF pages 30–31 (printed pages 25–26), have the correct derivative, coefficient, and Gram order. Its local metric argument is extended below to precise global and volume statements.

40.2 The full product equations and the pressure map

Let N be a smooth compact manifold without boundary with a smooth Riemannian metric h. A positive volume density suffices when N is nonorientable; when a volume form is used, fix an orientation. On $M = \mathbb{R}_x \times N$ retain the product metric $g = dx^2 + h$ and write $u = a\partial_x + U$ and $f = b\partial_x + F$. Here a, p, b are real-valued smooth functions of (t, x, y) and U, F are smooth sections of the pullback of TN. The sign convention is

$$\partial_t u + \nabla^g_u u = -\operatorname{grad}_g p + f, \qquad \operatorname{div}_g u = 0. \tag{511}$$

In a coordinate chart y^i on N, the only nonzero Christoffel symbols of g are $\Gamma^i_{jk}(h)$: this follows directly from $g_{xx} = 1$, $g_{xi} = 0$, $\partial_x h_{ij} = 0$ in $\Gamma^A_{BC} = \frac{1}{2} g^{AD}(\partial_B g_{CD} + \partial_C g_{BD} - \partial_D g_{BC})$. In particular

$$\partial_t a + a\partial_x a + U(a) = -\partial_x p + b, \tag{512}$$

$$\partial_t U + a\partial_x U + \nabla^h_U U = -\operatorname{grad}_h p + F, \tag{513}$$

$$\partial_x a + \operatorname{div}_h U = 0. \tag{514}$$

The last identity also follows from $\operatorname{div}_g u = (\sqrt{\det h})^{-1}[\partial_x(\sqrt{\det h}\, a) + \partial_{y^i}(\sqrt{\det h}\, U^i)]$.

There is an exact residual construction on this cylinder. For every smooth pair (a, U) obeying (514), set

$$A(t, x, y) = \partial_t a + a\partial_x a + U(a), \tag{515}$$

$$R(t, x, y) = \partial_t U + a\partial_x U + \nabla^h_U U, \tag{516}$$

$$p(t, x, y) = q(t, y) - \int_0^x A(t, s, y)\, ds, \tag{517}$$

$$b = 0, \qquad F = \operatorname{grad}_h p + R, \tag{518}$$

where q is any smooth real function on the time interval times N. The integral is over a finite interval for every x, so differentiation under it proves smoothness on the open spacetime domain. Its x derivative is $-A$ by the fundamental theorem of calculus. Substitution in (512) and (513) proves every component of (511). Conversely, a solution with $b = 0$ has p of (517), with $q = p(t, 0, y)$, and has the force (518). Thus the pressure map describes exactly, and not merely formally, all such force/pressure pairs for a specified divergence-free velocity. Changing q generally changes the transverse force; it is not the usual pressure gauge with the force held fixed.

The cancelled-pressure subclass is $R = 0$ and $F = \operatorname{grad}_h p$. Its closed velocity system is therefore

$$\partial_t U + a\partial_x U + \nabla^h_U U = 0, \qquad \partial_x a + \operatorname{div}_h U = 0. \tag{519}$$

On an axial circle, the same construction requires $\int_0^L A(t, s, y)\, ds = 0$ for pressure periodicity. The integral formula on $\mathbb{R}$ supplies no automatic periodic realization.

40.3 Energy and the exact constant-parameter scaling

Set $e = |U|_h^2/2$. Metric compatibility gives $(\partial_t + a\partial_x + U)e = \langle R, U\rangle_h$. Since (514) holds,

$$\partial_t e + \partial_x(ae) + \operatorname{div}_h(eU) = \langle R, U\rangle_h. \tag{520}$$

Indeed, the two product-rule terms added to the transport derivative are $e(\partial_x a + \operatorname{div}_h U) = 0$. Integration on $[-L, L] \times N$ yields the exact identity

$$\frac{d}{dt}\int_{-L}^{L}\int_N e\, dV_h\, dx = -\int_N [ae]_{x=-L}^{x=L}\, dV_h + \int_{-L}^{L}\int_N \langle R, U\rangle_h\, dV_h\, dx.$$

Here the integral of the N divergence vanishes: in an oriented chart $\operatorname{div}_h(eU)dV_h = d\iota_{eU}dV_h$, and Stokes' theorem on the compact boundaryless manifold gives zero; a partition of unity gives the identical density statement. Consequently the transverse energy is conserved for $R = 0$ when the energy is finite, the axial boundary flux tends to zero, and differentiation and the $L \to \infty$ limit are justified, for example by uniform integrable majorants for e and $\partial_t e$ on each compact time interval. Finite energy alone is not substituted for this flux/limit requirement.

For positive constants λ, μ, let

$$\widetilde{a}(t, x, y) = \frac{\lambda}{\mu} a(t/\mu, x/\lambda, y), \qquad \widetilde{U}(t, x, y) = \frac{1}{\mu} U(t/\mu, x/\lambda, y).$$

All three terms of R acquire the factor μ^{-2}; the divergence acquires the factor μ^{-1}. This follows from the ordinary chain rule and the x independence of h. The axial acceleration acquires λ/μ^2. Hence in the cancelled-pressure subclass $\widetilde{p} = (\lambda^2/\mu^2)p(t/\mu, x/\lambda, y)$ and $\widetilde{F} = \operatorname{grad}_h \widetilde{p}$ give the exact full forced equations as well. This is a symmetry of that subclass with its force transformed, not an assertion about an arbitrary fixed force in (511). The energy transformation is exactly

$$\widetilde{E}_N(t) = \frac{\lambda}{\mu^2} E_N(t/\mu), \qquad \widetilde{E}_x(t) = \frac{\lambda^3}{\mu^2} E_x(t/\mu), \tag{521}$$

by $x = \lambda\xi$. Equal nonzero transverse energies give $\mu^2 = \lambda$; comparable energies give corresponding two-sided bounds on λ/μ^2. No original dimensional constant or metric has been changed by these statements.

40.4 All rational coefficient equations, with the discrepancy retained

For smooth $\varphi\colon N \to \mathbb{R}$ and smooth vector fields $v, w \in \Gamma(TN)$, write

$$D = 1 + x^2, \qquad a = \frac{\varphi}{D}, \qquad U = \frac{v + xw}{D^2}. \tag{522}$$

Differentiation, without altering either denominator, gives

$$\partial_x a = -\frac{2x\varphi}{D^2}, \tag{523}$$

$$\partial_x U = \frac{wD - 4x(v + xw)}{D^3} = \frac{w - 4xv - 3x^2 w}{D^3}, \tag{524}$$

$$\nabla^h_U U = \frac{\nabla_v v + x(\nabla_v w + \nabla_w v) + x^2\nabla_w w}{D^4}. \tag{525}$$

The last formula uses $v(D) = w(D) = 0$, including all connection terms in each ∇. Define the three actual vector fields

$$C_0 = \nabla_v v + \varphi w, \quad C_1 = \nabla_v w + \nabla_w v - 4\varphi v, \quad C_2 = \nabla_w w - 3\varphi w. \tag{526}$$

The full residual and divergence, for arbitrary φ, v, w, are

$$R = \frac{C_0 + xC_1 + x^2 C_2}{D^4}, \qquad \operatorname{div}_g u = \frac{\operatorname{div}_h v + x(\operatorname{div}_h w - 2\varphi)}{D^2}. \tag{527}$$

Since $D > 0$ for every real x, a vanishing vector polynomial has all coefficients zero in any chart. Thus the stationary closed system is equivalent in both directions to the five equations

$$C_0 = C_1 = C_2 = 0, \qquad \operatorname{div}_h v = 0, \qquad \operatorname{div}_h w = 2\varphi. \tag{528}$$

There are no omitted derivatives of φ in its transverse equation: the advective derivative there differentiates U, not a. Those derivatives occur in the axial pressure equation below.

For the primary display's middle expression set instead $B_1 = \nabla_v w + \nabla_w v - 2\varphi v$. The exact relation is $C_1 = B_1 - 2\varphi v$. In particular, retaining all four other equations and $B_1 = 0$ gives

$$R = -\frac{2x\varphi v}{(1+x^2)^4}. \tag{529}$$

The discrepancy disappears exactly when the vector field φv vanishes identically. The residual construction (517)–(518) maps any such printed-system solution to full forced Euler with $F = \operatorname{grad}_h p - 2x\varphi v/(1+x^2)^4$. This proves the precise bridge after the coefficient failure; it makes no assertion that a nondegenerate solution of either system has been found. More generally (527) gives that bridge for every triple satisfying only its two displayed divergence equations.

40.5 Explicit pressure, force tails, and energy of the original ansatz

For (522), the axial acceleration is

$$a\partial_x a + U(a) = \frac{-2x\varphi^2 + v(\varphi) + xw(\varphi)}{D^3}.$$

Define

$$A_0 = 2\varphi^2 - w(\varphi), \qquad B_0 = v(\varphi), \tag{530}$$

$$J(x) = \frac{x}{4D^2} + \frac{3x}{8D} + \frac{3}{8}\arctan x, \tag{531}$$

$$p(x,y) = -\frac{A_0(y)}{4D^2} - B_0(y)J(x). \tag{532}$$

The derivative of the three summands in J is respectively $1/(4D^2) - x^2/D^3$, $3/(8D) - 3x^2/(4D^2)$, and $3/(8D)$; putting these over D^3 gives $J' = D^{-3}$. Also $(D^{-2})' = -4xD^{-3}$. Consequently $p_x = (xA_0 - B_0)/D^3$, the negative of the displayed acceleration. Thus p and

$$F = -\frac{\operatorname{grad}_h A_0}{4D^2} - J(x)\operatorname{grad}_h B_0 + \frac{C_0 + xC_1 + x^2C_2}{D^4}, \qquad b = 0 \tag{533}$$

are a completely explicit, smooth stationary force/pressure pair for every divergence-compatible triple. Every spatial covariant derivative of this pair is bounded on $\mathbb{R}\times N$: derivatives in y of its coefficients are bounded by compactness, J is bounded, and all positive-order x derivatives of J and of the displayed rational factors are bounded. These are stationary fields, so positive time derivatives vanish. This proves smooth bounded forcing for this stationary map; it does not imply bounds for a singularly modulated map.

Its force tails have an exact further restriction. Since $J(+\infty) = 3\pi/16$ and $J(-\infty) = -3\pi/16$,

$$\lim_{x\to\pm\infty} F(x,\cdot) = \mp\frac{3\pi}{16}\operatorname{grad}_h B_0. \tag{534}$$

The other summands tend to zero uniformly on N. If $\operatorname{grad}_h B_0$ is nonzero at a point, it is bounded away from zero on a nonempty open set; the uniform limit gives a positive lower bound for $|F|$ on that set for all sufficiently large $|x|$. Then $F \notin L^2(M)$. If $\operatorname{grad}_h B_0 = 0$, the remaining terms are $O(|x|^{-4})$ or faster, and $F \in L^2(M)$. For the divergence-compatible triples under consideration, $\int_{N_j} B_0\, dV_h = \int_{N_j} \operatorname{div}_h(\varphi v)\, dV_h = 0$ on every connected component N_j. A function with zero gradient is constant on each component, so here the exact L^2 criterion is $B_0 = v(\varphi) = 0$. Adding a different $q(y)$ to the pressure adds $\operatorname{grad}_h q$ to both limits in (534); both can vanish only when $\operatorname{grad}_h B_0 = 0$ and $\operatorname{grad}_h q = 0$. Thus this tail issue cannot be removed by that pressure choice.

The velocity always has finite total energy. The needed integrals are

$$\int_{\mathbb{R}} \frac{dx}{D^2} = \frac{\pi}{2}, \quad \int_{\mathbb{R}} \frac{dx}{D^4} = \frac{5\pi}{16}, \quad \int_{\mathbb{R}} \frac{x\,dx}{D^4} = 0, \quad \int_{\mathbb{R}} \frac{x^2\,dx}{D^4} = \frac{\pi}{16}. \tag{535}$$

For completeness put $x = \tan s$, $-\pi/2 < s < \pi/2$. The first, second, and fourth integrands become respectively $\cos^2 s$, $\cos^6 s$, and $\sin^2 s \cos^4 s$. Integration by parts of $\int_{-\pi/2}^{\pi/2} \cos^{2m} s\,ds$ gives $I_m = (2m-1)I_{m-1}/(2m)$ with $I_0 = \pi$; hence $I_1 = \pi/2$, $I_2 = 3\pi/8$, $I_3 = 5\pi/16$ and the fourth integral is $I_2 - I_3 = \pi/16$. Oddness gives the third. Expanding $|v + xw|^2 = |v|^2 + 2x\langle v, w\rangle + x^2|w|^2$ therefore proves

$$E_N = \frac{\pi}{32} \int_N (5|v|_h^2 + |w|_h^2)\,dV_h, \qquad E_x = \frac{\pi}{4} \int_N \varphi^2\,dV_h. \tag{536}$$

Algebraic decay, compact transverse geometry, and smooth bounded force are the properties proved here; rapid decay in Euclidean three-space is not a consequence of these formulas.

40.6 Exact one-form and Lie-derivative equations

Let $\theta = h(v,\cdot)$ and $\eta = h(w,\cdot)$; the source calls the second one-form λ. The notation η here prevents collision with the positive axial scaling parameter. For any vector field X, metric compatibility and vanishing torsion give

$$\begin{aligned}(\iota_v d\theta)(X) &= v[\theta(X)] - X[\theta(v)] - \theta([v,X]) \\ &= h(\nabla_v v, X) - h(\nabla_X v, v), \\ \tfrac{1}{2}d[\theta(v)](X) &= h(\nabla_X v, v).\end{aligned}$$

Adding proves

$$(\nabla_v v)^\flat = \iota_v d\theta + \tfrac{1}{2}d[\theta(v)]. \tag{537}$$

Apply this identity to $v + w$, expand every bilinear term, and subtract the individual v and w identities. The resulting mixed identity is

$$(\nabla_v w + \nabla_w v)^\flat = \iota_v d\eta + \iota_w d\theta + \tfrac{1}{2}d[\eta(v) + \theta(w)]. \tag{538}$$

Since h is positive definite, lowering indices is a vector-bundle isomorphism. The three equations $C_i = 0$ are consequently equivalent to

$$\varphi\eta + \iota_v d\theta + \tfrac{1}{2}d[\theta(v)] = 0, \tag{539}$$
$$-4\varphi\theta + \iota_v d\eta + \iota_w d\theta + \tfrac{1}{2}d[\eta(v) + \theta(w)] = 0, \tag{540}$$
$$-3\varphi\eta + \iota_w d\eta + \tfrac{1}{2}d[\eta(w)] = 0. \tag{541}$$

No antisymmetric covariant term has been dropped: the mixed expression is the symmetric sum of the two accelerations, exactly as in C_1.

Define the Lie derivative on one-forms by $(\mathcal{L}_v\alpha)(X) = v[\alpha(X)] - \alpha([v,X])$. Expanding $d\alpha(v,X)$ shows directly that $\mathcal{L}_v\alpha = \iota_v d\alpha + d[\alpha(v)]$. Thus the identical system in Lie-derivative notation is

$$\mathcal{L}_v\theta - \tfrac{1}{2}d[\theta(v)] + \varphi\eta = 0, \tag{542}$$
$$\mathcal{L}_v\eta + \mathcal{L}_w\theta - \tfrac{1}{2}d[\eta(v) + \theta(w)] - 4\varphi\theta = 0, \tag{543}$$
$$\mathcal{L}_w\eta - \tfrac{1}{2}d[\eta(w)] - 3\varphi\eta = 0. \tag{544}$$

With a specified smooth positive density ω, define divergence by $\mathcal{L}_v\omega = (\operatorname{div}_\omega v)\omega$. The divergence constraints eliminate φ exactly by $\varphi = \frac{1}{2}\operatorname{div}_\omega w$ and $\operatorname{div}_\omega v = 0$. In the middle one-form equation the resulting coefficient is therefore $-2(\operatorname{div}_\omega w)\theta$, rather than $-(\operatorname{div}_\omega w)\theta$. An arbitrary choice of ω requires the precise metric/volume realization proved in the following section; it is not implicitly equal to dV_h.

For completeness the full time-dependent transverse equation also has an exact metric-free acceleration identity. Put $\zeta = h(U,\cdot)$, a time- and x-dependent one-form on N. The same calculation, with h fixed in t, x, gives

$$R^\flat = \partial_t \zeta + a\partial_x \zeta + \mathcal{L}_U \zeta - \tfrac{1}{2} d_N[\zeta(U)].$$

This equation includes all dynamics of the closed cylinder system; there is no derivative of a in this transverse one-form identity.

40.7 Time-dependent concentration gives an explicit residual

Let (a_*, U_*) be any fixed smooth stationary divergence-free pair on this same cylinder and put $R_* = a_*\partial_\xi U_* + \nabla_{U_*} U_*$. For arbitrary smooth positive functions $L(t), M(t)$, define

$$\xi = x/L(t), \qquad a_{L,M} = \frac{L(t)}{M(t)} a_*(\xi, y), \qquad U_{L,M} = \frac{1}{M(t)} U_*(\xi, y).$$

Its divergence is $M^{-1}(\partial_\xi a_* + \operatorname{div}_h U_*) = 0$. Direct differentiation proves the complete transverse residual

$$R_{L,M} = \frac{1}{M^2} R_*(\xi, y) - \frac{M'}{M^2} U_*(\xi, y) - \frac{L'}{ML} \xi \partial_\xi U_*(\xi, y). \tag{545}$$

Indeed $\xi_t = -L'\xi/L$, the two transport terms have factor M^{-2}, and differentiating $M^{-1}U_*(\xi, y)$ gives exactly the last two terms. Its axial acceleration, with none of the modulation terms discarded, is

$$\begin{aligned} A_{L,M} = {} & \left(\frac{L'}{M} - \frac{LM'}{M^2} \right) a_* - \frac{L'}{M} \xi \partial_\xi a_* \\ & + \frac{L}{M^2} (a_* \partial_\xi a_* + U_*(a_*)). \end{aligned} \tag{546}$$

Formula (517) with this $A_{L,M}$ and (518) with (545) supply an exact forced Euler solution on every open time interval where $L, M > 0$ are smooth. No assertion of force regularity at an endpoint where these functions vanish follows from this construction.

If $R_* = 0$ and the transverse-energy scaling factor is held fixed by $M = \kappa\sqrt{L}$ with a fixed $\kappa > 0$, then

$$R_{L,M} = -\frac{L'}{ML} \left(\tfrac{1}{2} U_* + \xi \partial_\xi U_* \right). \tag{547}$$

For a nonzero smooth U_* on the entire cylinder, the expression in parentheses cannot vanish identically. To prove this, on each half-line the equation is $\partial_\xi(|\xi|^{1/2} U_*) = 0$ in each local bundle component. Thus $U_* = c_+(y)\xi^{-1/2}$ for $\xi > 0$ and $U_* = c_-(y)(-\xi)^{-1/2}$ for $\xi < 0$. Continuity at $\xi = 0$ forces both coefficient fields to vanish; the equation at zero also gives $U_*(0, y) = 0$. Hence U_* would be zero everywhere. Therefore nonconstant L in this pure scaling construction produces a nonzero residual at every time with $L' \neq 0$. This proves only the defect of pure modulation of a fixed smooth profile. The full residual and pressure formulas above remain an exact force bridge and do not exclude more general profile dynamics.

There is a concrete nonzero stationary member, including its degeneracy. On a flat torus $N = \mathbb{R}^m/\Lambda$ with its specified constant positive metric and lattice Λ, choose a nonzero constant vector V and take $\varphi = 0$, $w = 0$, $v = V$. All five equations (528) hold because constant vector fields have zero covariant derivatives and divergence. Thus $a = 0$, $U = V/(1 + x^2)^2$, $p = 0$, $f = 0$ is a smooth unforced stationary Euler solution on this product cylinder. Its energy is $5\pi \operatorname{Vol}_h(N)|V|_h^2/32$ by (536). The pair (v, w) has rank one, so this example does not realize the independent-pair Gram condition or a singularity. It verifies directly that the rational construction has actual solutions while identifying the exact rank of this example.

41 Exact metric realization and the circle volume lift

This section proves the metric and volume maps needed by the rational cylinder reduction. It does not assert existence of a nonzero solution to the nonlinear reduced equations. All constructions below are explicit maps on the stated data, and all residual identities hold for arbitrary smooth data in their domains, including data with nonzero residual.

41.1 Domains, conventions, and the source ordering

Let N be a smooth, Hausdorff, second-countable manifold of dimension $n \geq 2$. Let $v, w \in \Gamma(TN)$ be smooth and pointwise linearly independent, and let $\theta, \eta \in \Omega^1(N)$. The independence requirement is pointwise on all of N; the same results apply on any open independent locus without extending a formula across its rank-changing boundary. Define

$$H = \begin{pmatrix} a & b \\ c & d \end{pmatrix} := \begin{pmatrix} \theta(v) & \theta(w) \\ \eta(v) & \eta(w) \end{pmatrix}, \qquad A(X) := \begin{pmatrix} \theta(X) \\ \eta(X) \end{pmatrix}. \tag{548}$$

No rescaling of v, w, their order, or the given one-forms is made. A Riemannian metric has positive definiteness as part of its definition. When a volume form is prescribed, N is additionally oriented and $\mu \in \Omega^n(N)$ is smooth and everywhere positive for that orientation. The metric volume form is positive for the same orientation. Divergence is defined by $\mathcal{L}_X \mu = (\operatorname{div}_\mu X)\mu$. For a two-form ω, the contraction convention is $(\iota_X \omega)(Y) = \omega(X, Y)$.

The primary 2024 post defines $\theta = h(v, \cdot)$ and writes λ for $h(w, \cdot)$, corresponding to η here. Its displayed equation (7) orders the rows as (λ, θ) while ordering the columns as (v, w). Thus its displayed matrix is $\Pi_{\text{swap}} H$, with

$$\Pi_{\text{swap}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \det(\Pi_{\text{swap}} H) = -\det H. \tag{549}$$

For an independent pair realized by a positive metric, $\det H > 0$, so the displayed $\Pi_{\text{swap}} H$ is not a positive definite Gram matrix. Simultaneously reordering both arguments would give $\Pi_{\text{swap}} H \Pi_{\text{swap}}^T$, which is a Gram matrix in the order (w, v). Equation (549) identifies the exact permutation; the correct duality map remains the one proved below. The incoming report uses the correctly ordered H. See https://terrytao.wordpress.com/2024/09/08/a-possible-approach-to-finite-time-blowup-for-the-euler-equations/.

41.2 All realizing metrics, globally

Proposition 41.1 (Complete realization on the independent locus)**.** *There exists a smooth Riemannian metric h on N satisfying*

$$h(v, X) = \theta(X), \qquad h(w, X) = \eta(X) \quad (X \in TN) \tag{550}$$

exactly when H is symmetric positive definite at every point. In this case put $D = \operatorname{span}(v, w)$, $E = \ker \theta \cap \ker \eta$, and define

$$P(X) = (v \;\; w) H^{-1} A(X), \qquad Q(X) = X - P(X). \tag{551}$$

Then P and Q are smooth complementary projections with images D and E. The full set of metrics satisfying (550) is

$$h_k(X, Y) = A(X)^T H^{-1} A(Y) + k(QX, QY), \tag{552}$$

where k ranges over the smooth positive definite metrics on E. For $n = 2$, E is the zero bundle and the first term is the unique metric.

Proof. If (550) holds, its restriction to (v, w) gives the Gram matrix of the independent pair. Symmetry follows from symmetry of h; for $(s, t) \neq (0, 0)$ its quadratic form equals $h(sv + tw, sv + tw) > 0$. This proves necessity, including $b = c$.

Conversely suppose $H = H^T > 0$. The map $A : TN \to N \times \mathbb{R}^2$ has rank two because its restriction to D is the invertible matrix H. In local coordinates its kernel is a smooth rank-$(n-2)$ subbundle: an invertible two-column minor allows solving two components smoothly in terms of the remaining components. Thus E is a smooth vector bundle. For $X \in TN$, $A(PX) = HH^{-1}A(X) = A(X)$, so $QX \in E$; also $Pv = v$, $Pw = w$, and $P|_E = 0$. These equalities prove $P^2 = P$, $Q^2 = Q$, $PQ = QP = 0$, and $TN = D \oplus E$. The entries of H^{-1} are smooth because $\det H$ never vanishes, so both projections are smooth globally.

For completeness, a global positive metric on E can be constructed as follows. Choose a locally finite trivializing cover and a smooth partition of unity subordinate to it, and give each local trivialization its coordinate Euclidean quadratic form. Their partition-weighted sum is smooth. At a point, at least one nonzero partition weight multiplies a strictly positive quadratic form on any nonzero vector in that fibre; the other summands are nonnegative. This produces a smooth positive k. This construction uses only the stated smooth-manifold hypotheses, which give a locally finite refinement and smooth partitions of unity.

Formula (552) is smooth and symmetric. If $X \neq 0$, then either $A(X) \neq 0$, making its first quadratic term positive, or $A(X) = 0$ and $QX = X \neq 0$, making the second term positive. Moreover $A(v) = He_1$, $A(w) = He_2$, $Qv = Qw = 0$. Hence $h_k(v, X) = e_1^T A(X) = \theta(X)$ and $h_k(w, X) = e_2^T A(X) = \eta(X)$, proving sufficiency.

To prove exhaustion, any realizing h has $h(D, E) = 0$ by its two prescribed dualities. Its restriction to D is H in the given frame (v, w). Writing $X = PX + QX$, $Y = PY + QY$ therefore gives (552) with $k = h|_E$. Conversely every such k was already shown to work. If $n = 2$, E has rank zero, leaving no free term. □

41.3 Exact compatibility with a prescribed volume

Proposition 41.2 (Dimension two and dimensions at least three). *Retain $H = H^T > 0$ and the specified orientation and positive μ. If $n = 2$, a metric satisfying (550) and $d\,\mathrm{vol}_h = \mu$ exists exactly when*

$$\mu(v, w)^2 = \det H = \theta(v)\eta(w) - \theta(w)\eta(v). \tag{553}$$

If $n \geq 3$, such a metric always exists. Explicitly choose any $k_0 > 0$ on E, form $h_0 = h_{k_0}$, define the positive smooth ratio q by

$$\mu = q\, d\,\mathrm{vol}_{h_0}, \qquad m = n - 2, \tag{554}$$

and set $k = q^{2/m} k_0$ in (552).

Proof. For $n = 2$, the ordered frame (v, w) is either positive or negative for the specified orientation on each connected component. Let $\varepsilon = \operatorname{sgn} \mu(v, w) \in \{1, -1\}$, which is locally constant because $\mu(v, w)$ never vanishes. The metric volume of this frame is exactly $\varepsilon\sqrt{\det H}$. Thus its equality to $\mu(v, w)$ is equivalent to (553), because both have the sign ε. Equality on this frame is equality of the two top-degree forms. The metric was unique in Proposition 41.1, proving both directions without changing the orientation or setting any determinant to a normalized value.

For $n \geq 3$, choose any local frame $e_1, \ldots, e_m$ of E. In the unchanged frame $(v, w, e_1, \ldots, e_m)$, the metrics h_0 and h_k have matrices $\operatorname{diag}(H, K_0)$ and $\operatorname{diag}(H, q^{2/m}K_0)$, respectively. Consequently

$$\frac{\det \operatorname{diag}(H, q^{2/m}K_0)}{\det \operatorname{diag}(H, K_0)} = \frac{(\det H) q^{2m/m} \det K_0}{(\det H) \det K_0} = q^2.$$

Both metric volume forms have the specified orientation and their positive ratio is therefore q. It follows from (554) that $d\,\mathrm{vol}_{h_k} = \mu$. The function $q^{2/m}$ is globally smooth and positive, so the metric is global. No trivialization of E was required globally. □

The assertion in the incoming report that the local construction does not freely prescribe volume requires this dimensional refinement: with an everywhere independent pair the complete same-manifold obstruction is (553) in dimension two, whereas there is no prescribed-positive-volume obstruction in dimensions at least three. Changing k can change acceleration as a vector. The exact acceleration relation, including the inverse metric, is proved next.

41.4 Dualization of each reduced acceleration

For a realizing metric h, let ∇ denote its Levi-Civita connection and let $\flat_h : TN \to T^*N$ and $\sharp_h : T^*N \to TN$ be its mutually inverse metric dualities. Equations (537) and (538) have already proved the full acceleration and polarization identities for this metric, with the same contraction convention. We now keep their inverse metric maps explicit while varying the complement metric.

For a prescribed scalar $\varphi \in C^\infty(N)$ define, with all coefficients retained,

$$B_0 = \varphi\eta + \iota_v d\theta + \tfrac{1}{2} d[\theta(v)], \tag{555}$$

$$B_1 = -4\varphi\theta + \iota_v d\eta + \iota_w d\theta + \tfrac{1}{2} d[\theta(w) + \eta(v)], \tag{556}$$

$$B_2 = -3\varphi\eta + \iota_w d\eta + \tfrac{1}{2} d[\eta(w)]. \tag{557}$$

The exact vector residuals are

$$\begin{aligned} \varphi w + \nabla_v v &= \sharp_h B_0, \\ -4\varphi v + \nabla_v w + \nabla_w v &= \sharp_h B_1, \\ -3\varphi w + \nabla_w w &= \sharp_h B_2. \end{aligned} \tag{558}$$

These are identities for arbitrary smooth data in the domain of Proposition 41.1. In particular vanishing of the three one-form residuals is equivalent to vanishing of the three vector residuals, for each realizing h, since $\sharp_h$ is an isomorphism. The residuals B_j are independent of k; their vector representatives are related for any two realizing metrics by the explicit fibrewise map $\sharp_{h_{k_2}} \flat_{h_{k_1}}$. Together with the volume construction this proves the exact realization relation for the reduced system, including its two divergence constraints $\operatorname{div}_\mu v = 0$ and $\operatorname{div}_\mu w = 2\varphi$. It does not assume these constraints or $B_j = 0$ have been solved.

41.5 The circle lift as a map of metric and volume data

Here the base metric h is fixed; it may be any smooth Riemannian metric, including the metric already realizing the given pair. The construction in this subsection needs no pair of independent fields and works in every base dimension. Keep the given μ and define $f > 0$ by

$$\mu = f \, d\operatorname{vol}_h . \tag{559}$$

Fix a circle period $L > 0$ without normalizing it. Put

$$\widehat{N} = N \times (\mathbb{R}/L\mathbb{Z}), \qquad \pi : \widehat{N} \to N, \quad \pi(y,s) = y, \qquad \widehat{h} = \pi^* h + (f \circ \pi)^2 ds \otimes ds. \tag{560}$$

The form ds is the global invariant one-form induced from $\mathbb{R}$; the coordinate s itself is only local modulo L. The product orientation is specified by a positive base form followed by ds. Define the injective linear lift

$$\mathcal{J} : \Gamma(TN) \longrightarrow \Gamma(T\widehat{N}), \qquad (\mathcal{J}X)_{(y,s)} = (X_y, 0). \tag{561}$$

Its image consists precisely of the horizontal vector fields invariant under all circle translations. In coordinates the coefficients of a translation-invariant horizontal field depend only on y, yielding the inverse of $\mathcal{J}$. For functions and differential forms the map is the ordinary pullback π^*, which is also injective: restriction to the slice $y \mapsto (y, s_0)$ is a left inverse.

Proposition 41.3 (Volume and first-order operators of the lift). *The data* (560) *satisfy*

$$d\,\mathrm{vol}_{\widehat{h}} = \pi^*\mu \wedge ds, \tag{562}$$

$$(\mathcal{J}X)^{\flat_{\widehat{h}}} = \pi^*(X^{\flat_h}), \qquad \mathrm{grad}_{\widehat{h}}\,\pi^*F = \mathcal{J}(\mathrm{grad}_h F), \tag{563}$$

$$\mathrm{div}_{\widehat{h}}\,\mathcal{J}X = \pi^*(\mathrm{div}_\mu X), \tag{564}$$

$$\widehat{\nabla}_{\mathcal{J}X}\mathcal{J}Y = \mathcal{J}(\nabla_X Y), \tag{565}$$

$$[\mathcal{J}X, \mathcal{J}Y] = \mathcal{J}[X,Y], \qquad d\pi^*\alpha = \pi^*d\alpha, \qquad \iota_{\mathcal{J}X}\pi^*\alpha = \pi^*(\iota_X\alpha). \tag{566}$$

For a time-dependent field $X(t)$ the lift commutes with ∂_t.

Proof. Take positively oriented local coordinates $y^1, \ldots, y^n$ on N. Write $h = (h_{ij})$, $h^{-1} = (h^{ij})$, and $D_h = \det(h_{ij}) > 0$. The matrix of $\widehat{h}$ in the unchanged coordinates (y, s) is $\mathrm{diag}(h_{ij}, f^2)$, with determinant $D_h f^2$. Its positive volume is therefore $f\sqrt{D_h}\,dy^1 \wedge \cdots \wedge dy^n \wedge ds$, proving (562) with its orientation and coefficient. Pairing $(X, 0)$ with $(Y, a\partial_s)$ gives $h(X,Y)$, proving the duality identity. Pairing $(\mathrm{grad}_h F, 0)$ with the same arbitrary vector gives $dF(Y)$, proving the gradient identity.

If $\mu = \rho\,dy^1 \wedge \cdots \wedge dy^n$, expand its Lie derivative using $\mathcal{L}_X dy^j = dX^j$ and the product rule. Only the coefficient of $dy^1 \wedge \cdots \wedge dy^n$ survives, giving

$$\mathrm{div}_\mu X = \rho^{-1}\partial_i(\rho X^i) = \mathrm{div}_h X + X(\log f), \qquad \rho = f\sqrt{D_h}. \tag{567}$$

The lifted field has zero s component, so the identical divergence calculation with density ρ in (y,s) proves (564).

For clarity, all potentially nonzero lifted Christoffel coefficients are computed from $\widehat{\Gamma}^A_{BC} = \frac{1}{2}\widehat{h}^{AD}(\partial_B\widehat{h}_{CD} + \partial_C\widehat{h}_{BD} - \partial_D\widehat{h}_{BC})$:

$$\begin{aligned} &\widehat{\Gamma}^k_{ij} = \Gamma^k_{ij}, && \widehat{\Gamma}^s_{ij} = 0, && \widehat{\Gamma}^k_{is} = \widehat{\Gamma}^k_{si} = 0, \\ &\widehat{\Gamma}^s_{is} = \widehat{\Gamma}^s_{si} = f^{-1}\partial_i f, && \widehat{\Gamma}^k_{ss} = -fh^{kj}\partial_j f, && \widehat{\Gamma}^s_{ss} = 0. \end{aligned} \tag{568}$$

For instance the middle coefficient in the second line equals $\frac{1}{2}h^{kj}(-\partial_j f^2) = -fh^{kj}\partial_j f$; the first equals $\frac{1}{2}f^{-2}\partial_i f^2$. The other equalities follow because there are no mixed metric entries, $\partial_s h_{ij} = \partial_s f = 0$, and the horizontal inverse block is h^{-1}. Since $\mathcal{J}X$ and $\mathcal{J}Y$ have zero vertical components, (568) proves (565), including its zero vertical component.

The coordinate formula for a bracket contains only derivatives of the horizontal coefficients, proving the bracket identity. Pulling back a form replaces its coefficient functions by functions of y and retains the differentials dy^i; taking d differentiates those coefficients in the same y directions, proving the exterior-derivative identity. Evaluation of the alternating forms on arbitrary product tangent vectors proves the contraction identity. The map $\mathcal{J}$ is fixed in time, and its coordinate coefficients are those of X, proving commutation with ∂_t. □

The warping in (560) equals the determinant construction in Section 3 of the primary 2017 paper: defining $D_\mu(h)$ by $d\,\mathrm{vol}_h = \sqrt{D_\mu(h)}\mu$ gives $D_\mu(h) = f^{-2}$ and thus $D_\mu(h)^{-1} = f^2$. Our circle has retained period L, while that source takes $L = 1$. Its Section 4 proves a metric realization by another complement; (552) provides the full family in the intrinsic kernel complement. See Sections 3–4 of `https://arxiv.org/html/1707.07807v2`.

In particular set $(\widehat{v}, \widehat{w}, \widehat{\theta}, \widehat{\eta}, \widehat{\varphi}) = (\mathcal{J}v, \mathcal{J}w, \pi^*\theta, \pi^*\eta, \pi^*\varphi)$. Equations (563) and (566) give the exact identities $\widehat{B}_j = \pi^*B_j$ for all three residuals (555)–(557); (565) gives the corresponding lift of each vector residual, with coefficients $1, -4, -3$ unchanged. The divergence residuals lift as

$$\mathrm{div}_{\widehat{h}}\,\widehat{v} = \pi^*(\mathrm{div}_\mu v), \qquad \mathrm{div}_{\widehat{h}}\,\widehat{w} - 2\widehat{\varphi} = \pi^*(\mathrm{div}_\mu w - 2\varphi).$$

Injectivity of pullback and of $\mathcal{J}$ proves both directions of each zero-residual equivalence. Thus the volume obstruction in dimension two has an explicit exact bridge to dimension three. This statement constructs the bridge for every input tuple; it is not an existence claim for a tuple with the nonlinear residuals zero.

41.6 Time-dependent dynamics and energy

Fix a time interval $I \subset \mathbb{R}$ and smooth $U : I \times N \to TN$, $p : I \times N \to \mathbb{R}$, and $F : I \times N \to TN$, with the vector fields based at $y \in N$. Define the momentum residual, without discarding pressure or force, by

$$\mathcal{E}_h(U, p, F) = \partial_t U + \nabla_U U + \operatorname{grad}_h p - F. \tag{569}$$

The lifted fields are $\widehat{U} = \mathcal{J}U$, $\widehat{p} = \pi^* p$, $\widehat{F} = \mathcal{J}F$. Proposition 41.3 proves

$$\mathcal{E}_{\widehat{h}}(\widehat{U}, \widehat{p}, \widehat{F}) = \mathcal{J}\mathcal{E}_h(U, p, F), \qquad \operatorname{div}_{\widehat{h}} \widehat{U} = \pi^* \operatorname{div}_\mu U. \tag{570}$$

Hence $\mathcal{J}$ is a bijection between the base smooth solution pairs of the weighted-volume Euler system $\mathcal{E}_h = 0$, $\operatorname{div}_\mu U = 0$ and the lifted smooth Euler solution pairs whose velocity is horizontal and whose velocity, pressure, and specified force are circle-invariant with the given lifts. This is a statement about solution sets, proved by the two residual identities and the explicit inverses on their images. It neither assumes existence for arbitrary initial data nor changes the base divergence constraint from div_μ to div_h. Indeed those two divergences differ by the explicit term $U(\log f)$.

If N is compact without boundary, the energy and its balance are also exact. For any smooth X, Y,

$$\int_{\widehat{N}} \widehat{h}(\mathcal{J}X, \mathcal{J}Y) d\operatorname{vol}_{\widehat{h}} = L \int_N h(X, Y)\, \mu. \tag{571}$$

This follows by (562), the unchanged horizontal inner product, and $\int_{\mathbb{R}/L\mathbb{Z}} ds = L$. Let $E_\mu(U) = \frac{1}{2}\int_N h(U, U)\mu$. For any smooth solution of the base system in (570), metric compatibility and time independence of h, μ give

$$\frac{dE_\mu}{dt} = -\int_N U\left(\tfrac{1}{2}h(U, U) + p\right)\mu + \int_N h(F, U)\mu = \int_N h(F, U)\mu.$$

For the final equality, let $G = \frac{1}{2}h(U, U) + p$ and use $\operatorname{div}_\mu(GU) = U(G) + G \operatorname{div}_\mu U = U(G)$. Its integral is zero: $\mathcal{L}_{GU}\mu = d(\iota_{GU}\mu)$ because $d\mu = 0$ in top degree, and Stokes' theorem has no boundary term. The lifted kinetic energy is LE_μ and its work balance is multiplied by the same L. No energy coefficient has been normalized away.

41.7 Second-order operators: the exact additional terms

The lift preserves the horizontal connection in (565), but tracing over all lifted directions adds terms. Define $Z = \operatorname{grad}_h \log f$. The scalar Laplacian convention here is $\Delta_h F = \operatorname{div}_h \operatorname{grad}_h F$, whose Euclidean expression is $\sum_i \partial_i^2 F$. By the gradient and divergence identities already proved,

$$\Delta_{\widehat{h}} \pi^* F = \pi^*(\Delta_h F + h(Z, \operatorname{grad}_h F)). \tag{572}$$

Indeed $\operatorname{div}_\mu \operatorname{grad}_h F = \operatorname{div}_h \operatorname{grad}_h F + (\operatorname{grad}_h F)(\log f)$ by (567); the last term is the displayed inner product.

For vector fields use the positive-trace rough operator

$$\mathscr{L}_h X = \sum_{a=1}^{n} \left(\nabla_{e_a}\nabla_{e_a} X - \nabla_{\nabla_{e_a} e_a} X\right), \tag{573}$$

in any local h-orthonormal frame. This is the metric trace of the second covariant derivative, so its definition is independent of frame. Set $e_0 = f^{-1}\partial_s$ on the lift and lift each e_a horizontally. The Christoffel formulas prove

$$\widehat{\nabla}_{\mathcal{J}X} e_0 = 0, \qquad \widehat{\nabla}_{e_0} \mathcal{J}X = h(Z, X)e_0, \qquad \widehat{\nabla}_{e_0} e_0 = -\mathcal{J}Z. \tag{574}$$

For example the first equality follows from differentiating f^{-1}: $-f^{-2}X(f)\partial_s + f^{-1}X(\log f)\partial_s = 0$. The second is $f^{-1}X(\log f)\partial_s$. The third is $f^{-2}(-f\operatorname{grad}_h f) = -Z$, since $\partial_s f = 0$. The horizontal trace in (573) is exactly $\mathcal{J}\mathscr{L}_h X$. The remaining term is

$$\widehat{\nabla}_{e_0}(h(Z,X)e_0) - \widehat{\nabla}_{-\mathcal{J}Z}\mathcal{J}X = -h(Z,X)\mathcal{J}Z + \mathcal{J}(\nabla_Z X),$$

because $e_0[h(Z,X)] = 0$. Thus the full vector formula is

$$\mathscr{L}_{\widehat{h}}\mathcal{J}X = \mathcal{J}\left(\mathscr{L}_h X + \nabla_Z X - h(Z,X)Z\right). \tag{575}$$

The form version can be calculated without an unspecified viscosity convention. Define the codifferential on differential forms by $\delta_h\alpha = -\sum_a \iota_{e_a}\nabla_{e_a}\alpha$ and the nonnegative Hodge Laplacian by $\Delta_h^{\mathrm{H}} = d\delta_h + \delta_h d$. For every k-form α,

$$\delta_{\widehat{h}}\pi^*\alpha = \pi^*(\delta_h\alpha - \iota_Z\alpha), \qquad \Delta_{\widehat{h}}^{\mathrm{H}}\pi^*\alpha = \pi^*(\Delta_h^{\mathrm{H}}\alpha - \mathcal{L}_Z\alpha). \tag{576}$$

Here contractions of zero-forms are zero. To prove the first identity, the horizontal frame terms agree with the base terms by (565) and $\widehat{\nabla}_{\mathcal{J}e_a}e_0 = 0$. For $k \geq 1$ and horizontal $Y_1, \ldots, Y_{k-1}$, the vertical contribution before its minus sign is

$$(\widehat{\nabla}_{e_0}\pi^*\alpha)(e_0, \mathcal{J}Y_1, \ldots, \mathcal{J}Y_{k-1}) = \alpha(Z, Y_1, \ldots, Y_{k-1}).$$

The derivative of the initially zero evaluation is zero, the derivative of its first slot is $-Z$, and derivatives of the other slots are vertical and still give zero by alternation. Evaluations with an additional vertical slot are zero. This proves the first formula on all tangent arguments. Applying d, commuting it with pullback, and also applying the first identity to $d\alpha$ gives $\Delta_{\widehat{h}}^{\mathrm{H}}\pi^*\alpha = \pi^*(\Delta_h^{\mathrm{H}}\alpha - d\iota_Z\alpha - \iota_Z d\alpha)$. The equality $d\iota_Z + \iota_Z d = \mathcal{L}_Z$ follows in coordinates by differentiating the coefficients of α and of Z: its coefficient for indices $i_1, \ldots, i_k$ is $Z^j\partial_j\alpha_{i_1\cdots i_k} + \sum_{r=1}^k(\partial_{i_r}Z^j)\alpha_{i_1\cdots i_{r-1}ji_{r+1}\cdots i_k}$, which is the coordinate definition of the Lie derivative. This proves the second formula, including its sign. For functions it is consistent with (572) since $\Delta_h^{\mathrm{H}}F = -\Delta_h F$ under the specified conventions.

These identities supply an exact viscous bridge with all added terms. For example retain a dimensional viscosity $\nu > 0$ and define

$$\mathcal{V}_h(U,p,F) = \partial_t U + \nabla_U U + \operatorname{grad}_h p - \nu\mathscr{L}_h U - F.$$

Equation (575) yields, for every input triple,

$$\mathcal{V}_{\widehat{h}}(\mathcal{J}U, \pi^*p, \mathcal{J}F) = \mathcal{J}\left(\mathcal{V}_h(U,p,F) - \nu[\nabla_Z U - h(Z,U)Z]\right). \tag{577}$$

Accordingly the explicitly corrected force

$$\widehat{F}_{\mathrm{corr}} = \mathcal{J}\left(F - \nu[\nabla_Z U - h(Z,U)Z]\right) \tag{578}$$

gives the exact identity $\mathcal{V}_{\widehat{h}}(\mathcal{J}U, \pi^*p, \widehat{F}_{\mathrm{corr}}) = \mathcal{J}\mathcal{V}_h(U,p,F)$, with divergence still given by (564). The signs follow by substituting (578) into (577); the two drift contributions cancel. Equivalently, without changing force, the base diffusion operator that lifts is $\mathscr{L}_h + \nabla_Z - (X \mapsto h(Z,X)Z)$. If a base field becomes singular, this formula alone gives no bound on the corrected force; its exact extra term has been retained. Nor is a metric rough operator asserted to equal the Euclidean three-dimensional Navier–Stokes operator on a different manifold. The proven morphism is (577), together with its precise force correction, between the stated domains and operators.

42 IPM: exact pressure, Fourier, and affine-wave correspondences

This section audits equations (4.1)–(4.6) of the incoming research report, physical PDF pages 10–11, and proves the additional identities used here. Its starting equation is retained with the report's coefficients and signs:

$$\partial_t\rho + u\cdot\nabla\rho = F, \qquad u = -\nabla p - \rho e_2, \qquad \nabla\cdot u = 0, \qquad e_2 = (0,1). \tag{579}$$

The spatial coordinates are $x = (x_1, x_2)$; neither coordinate, time, nor a coefficient in this equation is rescaled without an explicit map below.

42.1 The primary-source coefficient and the Fourier zero mode

The primary manuscript of Alpöge, Buckmaster, and Coiculescu, *Extending the Córdoba–Martínez-Zoroa IPM Blow-up to Uniformly Space-Time Smooth Forcing*, specifies on physical page 3, equations (1)–(2), the torus $\mathbb{T}^2_{2\pi} = (\mathbb{R}/2\pi\mathbb{Z})^2$ and operator

$$\widehat{u_{\mathbb{T}}(\sigma)}(k) = 2\pi m(k)\widehat{\sigma}(k) \quad (k \in \mathbb{Z}^2 \setminus \{0\}), \qquad \widehat{u_{\mathbb{T}}(\sigma)}(0) = 0, \qquad \partial_t \sigma + u_{\mathbb{T}}(\sigma) \cdot \nabla \sigma = G, \tag{580}$$

where

$$\widehat{f}(k) = \frac{1}{(2\pi)^2} \int_{[0,2\pi]^2} f(x) e^{-ik\cdot x}\, dx, \qquad m(k) = \left(\frac{k_1 k_2}{k_1^2 + k_2^2}, -\frac{k_1^2}{k_1^2 + k_2^2} \right). \tag{581}$$

The factor 2π is explicit in the source and absent in report equation (4.2). The following results give the exact relation. The source's Theorem 2.1 states smooth forcing on $[0, 1] \times \mathbb{T}^2_{2\pi}$, odd mean-zero density and force, smoothness before time 1, and divergence of density and velocity gradients. This statement
The retrieved file, its hash, and the inspection locations are recorded in the accompanying source inventory; the primary URL is https://cims.nyu.edu/~tristanb/ipm.pdf.

Proposition 42.1 (Pressure elimination with the zero mode retained)**.** *For any $\rho \in C^\infty(\mathbb{T}^2_{2\pi})$, set $\overline{\rho} = \widehat{\rho}(0)$ and define the mean-zero periodic pressure $\mathcal{P}\rho$ and the periodic vector field $\mathcal{M}\rho$ by*

$$\widehat{\mathcal{P}\rho}(k) = \frac{ik_2}{k_1^2 + k_2^2}\widehat{\rho}(k) \quad (k \neq 0), \qquad \widehat{\mathcal{P}\rho}(0) = 0, \tag{582}$$

$$\widehat{\mathcal{M}\rho}(k) = m(k)\widehat{\rho}(k) \quad (k \neq 0), \qquad \widehat{\mathcal{M}\rho}(0) = -\overline{\rho}e_2. \tag{583}$$

Then $\mathcal{M}\rho = -\nabla\mathcal{P}\rho - \rho e_2$ and $\nabla \cdot \mathcal{M}\rho = 0$. Every smooth periodic solution of the last two equations in (579) has this velocity and has pressure $\mathcal{P}\rho + c$ for a spatial constant c. If $\mathcal{M}_0$ has the same nonzero symbol and zero symbol at $k = 0$, then

$$\mathcal{M}\rho = \mathcal{M}_0\rho - \overline{\rho}e_2, \qquad \mathcal{M}_0\rho = -\nabla\mathcal{P}\rho - (\rho - \overline{\rho})e_2. \tag{584}$$

On the universal cover $\mathbb{R}^2$, the same field $\mathcal{M}_0\rho$ satisfies the unprojected Darcy law with pressure $\mathcal{P}\rho - \overline{\rho}x_2$. This pressure is periodic exactly when $\overline{\rho} = 0$.

Proof. Taking divergence of $u = -\nabla p - \rho e_2$ gives $\Delta p = -\partial_2 \rho$. For $k \neq 0$ its Fourier coefficients obey $-(k_1^2 + k_2^2)\widehat{p}(k) = -ik_2\widehat{\rho}(k)$, giving (582). Hence

$$\widehat{u}(k) = -ik\frac{ik_2}{k_1^2 + k_2^2}\widehat{\rho}(k) - e_2\widehat{\rho}(k) = \left(\frac{k_1 k_2}{k_1^2 + k_2^2}, \frac{k_2^2}{k_1^2 + k_2^2} - 1 \right) \widehat{\rho}(k) = m(k)\widehat{\rho}(k).$$

The zero Fourier coefficient of a periodic gradient is zero, so $\widehat{u}(0) = -\overline{\rho}e_2$, with no freedom to set it to zero unless $\overline{\rho} = 0$. Direct multiplication gives $k \cdot m(k) = (k_1^2 k_2 - k_1^2 k_2)/(k_1^2 + k_2^2) = 0$. Smoothness follows because smooth Fourier coefficients decay faster than every polynomial, whereas all displayed multipliers grow at most polynomially; this also justifies every termwise derivative. The difference of two periodic pressures with the same density has every nonzero Fourier coefficient zero, and is therefore constant. Subtracting the zero-mode formulas proves (584). Finally $\nabla(-\overline{\rho}x_2) = -\overline{\rho}e_2$ proves the cover-space formula. Translation by $2\pi e_2$ changes that pressure by $-2\pi\overline{\rho}$, proving the stated periodicity criterion. $\square$

Proposition 42.2 (Exact coefficient map without changing space or time)**.** *On the same torus and time interval, define the source and report residuals*

$$\mathcal{N}_{2\pi}(\sigma) = \partial_t \sigma + 2\pi\mathcal{M}_0\sigma \cdot \nabla\sigma, \qquad \mathcal{N}_1(\rho) = \partial_t \rho + \mathcal{M}\rho \cdot \nabla\rho.$$

For every smooth spatially mean-zero σ the following is an exact operator identity:

$$\mathcal{N}_1(2\pi\sigma) = 2\pi\mathcal{N}_{2\pi}(\sigma). \tag{585}$$

Adjoin to the source scalar equation its periodic potential $q = \mathcal{P}\sigma + c_s(t)$, where c_s is an arbitrary smooth function. It gives the exact velocity relation $u_{\mathbb{T}}(\sigma) = 2\pi(-\nabla q - \sigma e_2)$. Consequently the invertible map of these full forced systems is

$$\rho = 2\pi\sigma, \quad F = 2\pi G, \quad u = u_{\mathbb{T}}(\sigma) = 2\pi\mathcal{M}_0\sigma, \quad p = 2\pi q = 2\pi\mathcal{P}\sigma + 2\pi c_s(t). \tag{586}$$

It sends source initial density σ^{in} to $\rho^{\mathrm{in}} = 2\pi\sigma^{\mathrm{in}}$; its inverse divides density, force, and pressure by 2π (recovering q) and leaves velocity, space, and time fixed. It multiplies every density or force derivative and every homogeneous norm of such a derivative by 2π. Velocity and its derivatives are unchanged.

Proof. Linearity and the mean-zero condition give $\mathcal{M}(2\pi\sigma) = 2\pi\mathcal{M}_0\sigma$ and $\mathcal{P}(2\pi\sigma) = 2\pi\mathcal{P}\sigma$. Therefore

$$\partial_t(2\pi\sigma) + (2\pi\mathcal{M}_0\sigma)\cdot\nabla(2\pi\sigma) = 2\pi[\partial_t\sigma + 2\pi\mathcal{M}_0\sigma\cdot\nabla\sigma].$$

Proposition 42.1 proves both Darcy's law and incompressibility for the pressure in (586); the gradient of $2\pi c_s(t)$ is zero. Thus every spatially constant pressure function is retained by the map and its inverse. All inverse formulas follow by dividing by the nonzero constant 2π. Differentiation and absolute homogeneity of norms prove the last assertion. Integrating the source transport equation on the torus gives $\partial_t\overline{\sigma} = \overline{G}$, since the advective integral is $\int \nabla\cdot(\sigma u_{\mathbb{T}}(\sigma)) = 0$; thus the source's mean-zero forcing and initial density preserve the required subspace. □

There is also an exact treatment of the mean mode. Let $b(t)$ be any smooth real function, let $B(t) = \int_{t_0}^t b(s)\,ds$, and write $y = x + B(t)e_2$. For a smooth mean-zero source density σ and an arbitrary smooth real function $c(t)$ define

$$\begin{aligned} \rho(x,t) &= 2\pi\sigma(y,t) + b(t), & u(x,t) &= u_{\mathbb{T}}(\sigma)(y,t) - b(t)e_2, \\ p(x,t) &= 2\pi\mathcal{P}\sigma(y,t) + c(t), & F(x,t) &= 2\pi G(y,t) + b'(t). \end{aligned} \tag{587}$$

These are periodic in x and solve (579) whenever the source residual is G. Indeed Darcy's law is $-2\pi\nabla\mathcal{P}\sigma - (2\pi\sigma + b)e_2 = u_{\mathbb{T}}(\sigma) - be_2$ at y. Incompressibility is preserved by translation. The transport residual is exactly

$$2\pi\sigma_t + 2\pi B'\partial_2\sigma + b' + (u_{\mathbb{T}}(\sigma) - be_2)\cdot 2\pi\nabla\sigma = 2\pi G + b',$$

because $B' = b$. Conversely, any smooth periodic report solution is put in this form by $b = \overline{\rho}$, $c = \overline{p}$, $B(t) = \int_{t_0}^t b$, and $\sigma(y,t) = [\rho(y - B(t)e_2, t) - b(t)]/(2\pi)$; the preceding equalities read backwards prove its source equation. This retains the mean density, the mean velocity, and their exact dynamical relation.

42.2 Every mean-zero periodic phase profile

Proposition 42.3 (Wave, pressure, cancellation, and mean correction)**.** *Let $k = (k_1, k_2) \in \mathbb{Z}^2\setminus\{0\}$, let $h \in C^\infty(\mathbb{R}/2\pi\mathbb{Z};\mathbb{R})$, and suppose $\int_0^{2\pi} h(s)\,ds = 0$. Put $\kappa = k_1^2 + k_2^2$ and let H be the unique mean-zero 2π-periodic function with $H' = h$. For every real a,*

$$\vartheta(x) = ah(k\cdot x), \qquad v(x) = am(k)h(k\cdot x), \qquad \pi(x) = -\frac{ak_2}{\kappa}H(k\cdot x) \tag{588}$$

are smooth periodic fields satisfying $v = \mathcal{M}\vartheta = \mathcal{M}_0\vartheta = -\nabla\pi - \vartheta e_2$, $\nabla\cdot v = 0$, and $v\cdot\nabla\vartheta = 0$. For the primary source, velocity and pressure are respectively $2\pi v$ and $2\pi\pi$ at the unchanged source density ϑ. If the mean-zero hypothesis on h is dropped and $\overline{h} = (2\pi)^{-1}\int_0^{2\pi} h$, the report operator instead gives

$$\mathcal{M}[ah(k\cdot x)] = am(k)(h(k\cdot x) - \overline{h}) - a\overline{h}e_2, \qquad \mathcal{M}[ah]\cdot\nabla(ah) = -a^2\overline{h}k_2h'(k\cdot x). \tag{589}$$

Proof. The function $H_*(s) = \int_0^s h(r)\,dr$ is periodic because the integral over a period is zero. Subtracting its mean gives H, and the difference of two such primitives is a constant of mean zero. Write $h(s) = \sum_{n\in\mathbb{Z}\setminus\{0\}} h_n e^{ins}$. For each integer $N \geq 0$, integration by parts N times bounds $|h_n| \leq C_N|n|^{-N}$, so the series and all derivatives converge absolutely. The frequency nk is nonzero, and direct substitution gives

$$m(nk) = \left(\frac{n^2k_1k_2}{n^2(k_1^2+k_2^2)}, -\frac{n^2k_1^2}{n^2(k_1^2+k_2^2)}\right) = m(k).$$

Distinct n give distinct frequencies nk. Thus the spatial mean is zero and the multiplier acts on every harmonic by the same vector $m(k)$, including negative n. Applying the pressure formula harmonic by harmonic, or differentiating π, gives

$$-\nabla\pi - \vartheta e_2 = a\left(\frac{k_1k_2}{\kappa}, \frac{k_2^2}{\kappa} - 1\right) h(k\cdot x) = am(k)h(k\cdot x).$$

Finally $\nabla\cdot v = a(m(k)\cdot k)h' = 0$ and $v\cdot\nabla\vartheta = a^2(m(k)\cdot k)hh' = 0$. The source assertion follows by multiplying velocity and pressure by the explicit source coefficient. For nonzero mean, apply the proved formulas to $h - \overline{h}$, retain the Darcy zero-mode term $-a\overline{h}e_2$, and take its dot product with akh'. □

The same direct pressure proof of (588) holds on $\mathbb{R}^2$ for every nonzero real k, without imposing an integer lattice. On a rectangular torus with periods $L_1, L_2 > 0$ it holds whenever $k_jL_j \in 2\pi\mathbb{Z}$ for $j = 1, 2$. These are explicit descent conditions, not changes of coordinates. A profile with a smaller fundamental period may also descend for additional k; no necessity claim is made for an arbitrary non-faithful profile.

42.3 The frozen growth rate is realized by a full affine solution

Theorem 42.4 (Exact unforced affine IPM solution and flow)**.** *Fix $A, a_0, t_0 \in \mathbb{R}$, $k \in \mathbb{R}^2\setminus\{0\}$, and a real smooth 2π-periodic mean-zero profile h with the primitive H above. Retain $\kappa = k_1^2 + k_2^2$ and define*

$$\lambda = \frac{Ak_1^2}{k_1^2+k_2^2}, \qquad a(t) = a_0\exp\left(\frac{Ak_1^2}{k_1^2+k_2^2}(t-t_0)\right). \tag{590}$$

On the entire domain $\mathbb{R}^2\times\mathbb{R}$ set

$$\begin{aligned}
\rho(x,t) &= Ax_2 + a(t)h(k_1x_1+k_2x_2), \\
u(x,t) &= a(t)\left(\frac{k_1k_2}{k_1^2+k_2^2}, -\frac{k_1^2}{k_1^2+k_2^2}\right)h(k_1x_1+k_2x_2), \\
p(x,t) &= -\frac{A}{2}x_2^2 - \frac{a(t)k_2}{k_1^2+k_2^2}H(k_1x_1+k_2x_2).
\end{aligned} \tag{591}$$

These fields solve the full system (579) with $F = 0$, and are smooth at every finite time. The exact Lagrangian map and its inverse are

$$\Phi_t(y) = y + m(k)h(k\cdot y)I(t), \qquad \Phi_t^{-1}(x) = x - m(k)h(k\cdot x)I(t), \qquad I(t) = \int_{t_0}^t a(s)\,ds. \tag{592}$$

They preserve area and satisfy $\rho(\Phi_t(y),t) = Ay_2 + a_0h(k\cdot y)$.

Proof. No pressure multiplier is applied to the affine function Ax_2 at frequency zero: its pressure is the explicitly displayed polynomial. Direct differentiation gives

$$-\nabla\left(-\frac{A}{2}x_2^2\right) - (Ax_2)e_2 = 0, \qquad -\nabla\left(-\frac{a(t)k_2}{\kappa}H(k\cdot x)\right) - a(t)h(k\cdot x)e_2 = a(t)m(k)h(k\cdot x).$$

Adding these identities proves Darcy's law. Write $s = k \cdot x$. All terms of the divergence and transport equations are

$$\begin{aligned}\nabla \cdot u &= a(t)h'(s)(k_1 m_1(k) + k_2 m_2(k)) = 0,\\ \partial_t \rho &= a'(t)h(s),\\ \nabla \rho &= Ae_2 + a(t)kh'(s),\\ u \cdot \nabla \rho &= Aa(t)m_2(k)h(s) + a(t)^2(m(k)\cdot k)h(s)h'(s)\\ &= -\frac{Ak_1^2}{k_1^2+k_2^2}a(t)h(s) + 0.\end{aligned}$$

Since $a' = \lambda a$, their sum is identically zero. For a spatial multi-index $\alpha = (\alpha_1, \alpha_2)$ and any integer $j \geq 0$, the wave part has derivative

$$\partial_t^j \partial_x^\alpha [a(t)h(k\cdot x)] = a_0 \lambda^j e^{\lambda(t-t_0)} k_1^{\alpha_1} k_2^{\alpha_2} h^{(|\alpha|)}(k \cdot x).$$

Together with the derivatives of the retained polynomial Ax_2 and $-Ax_2^2/2$, this proves smoothness and boundedness of every derivative on every compact subset of $\mathbb{R}^2 \times \mathbb{R}$; wave derivatives are also bounded on all space for every compact time interval.

Because $k \cdot m(k) = 0$, $k \cdot \Phi_t(y) = k \cdot y$. Differentiating in time yields $\partial_t \Phi_t(y) = a(t)m(k)h(k\cdot y) = u(\Phi_t(y), t)$. The same phase identity verifies both compositions of the displayed inverse. The derivative matrix is $D_y\Phi_t = \mathrm{Id} + I(t)h'(k\cdot y)m(k) \otimes k$. Its two-dimensional determinant, expanded directly, is

$$(1 + Ih'm_1k_1)(1 + Ih'm_2k_2) - (Ih'm_1k_2)(Ih'm_2k_1) = 1 + Ih'(m_1k_1 + m_2k_2) = 1.$$

Finally

$$\rho(\Phi_t(y), t) = Ay_2 + [Am_2(k)I(t) + a(t)]h(k\cdot y).$$

The derivative of the bracket is $Am_2(k)a(t) + a'(t) = 0$ and its value at $t = t_0$ is a_0, proving the transported-density identity. This argument does not divide by A, a_0, or k_1, so the stationary degenerate cases are included exactly. $\square$

For $A > 0$ and $k_1 \neq 0$ the report's rate $-m(k)\cdot(Ae_2) = Ak_1^2/(k_1^2 + k_2^2)$ is therefore the exact growth rate of this nonlinear solution. Its arbitrary-amplitude wave has identically zero self-advection. The statement concerns an exponential in time, so it establishes no finite-time singularity.

For completeness, retaining the primary coefficient in the cover-space Darcy law $u = -\nabla p - 2\pi\sigma e_2$ gives the exact fields

$$\begin{aligned}\sigma &= A_\mathrm{s}x_2 + a_\mathrm{s}(t)h(k\cdot x), & u &= 2\pi a_\mathrm{s}(t)m(k)h(k\cdot x),\\ p &= -\pi A_\mathrm{s}x_2^2 - \frac{2\pi a_\mathrm{s}(t)k_2}{\kappa}H(k\cdot x), & a_\mathrm{s}' &= \frac{2\pi A_\mathrm{s}k_1^2}{\kappa}a_\mathrm{s}.\end{aligned} \tag{593}$$

Indeed the substitutions $A = 2\pi A_\mathrm{s}$ and $a = 2\pi a_\mathrm{s}$ in (591) give exactly these velocities and pressures, and divide the scalar equation by 2π. Thus the apparent rates have the same value under the proved density map; no factor has been discarded.

42.4 The exact domain and energy limitations

If $A \neq 0$, ρ in (591) cannot be a periodic function on a full two-dimensional lattice. To prove this, suppose a lattice period ℓ has $\ell_2 \neq 0$. Periodicity under every integer multiple $n\ell$ would give

$$0 = An\ell_2 + a(t)[h(k\cdot x + nk\cdot \ell) - h(k\cdot x)].$$

The second term has absolute value at most $2|a(t)|\|h\|_\infty$ whereas $|An\ell_2|\to\infty$, a contradiction. A full two-dimensional lattice contains such an ℓ. When $A=0$ and $k\in\mathbb{Z}^2\setminus\{0\}$, the formulas do descend to $\mathbb{T}^2_{2\pi}$ and reduce to the exact stationary wave of Proposition 42.3.

If $a_0\neq 0$, $h\not\equiv 0$, and $k_1\neq 0$, the velocity has infinite kinetic energy on $\mathbb{R}^2$ at every finite time. Here is the exact integral. Set $J(k_1,k_2)=(-k_2,k_1)$ and use the invertible coordinates $x=(sk+rJk)/\kappa$, so $k\cdot x=s$ and $dx=\kappa^{-1}ds\,dr$. Moreover

$$|m(k)|^2=\frac{k_1^2k_2^2+k_1^4}{\kappa^2}=\frac{k_1^2}{\kappa}.$$

For any $R>0$ the integral over $0\le s\le 2\pi$, $-R\le r\le R$ is

$$\frac{2R}{\kappa}\,|a(t)|^2\frac{k_1^2}{\kappa}\int_0^{2\pi}|h(s)|^2\,ds.$$

All factors except R are strictly positive, so letting $R\to\infty$ proves the assertion. If $k_1=0$, $m(k)=0$ and the velocity is zero; that degenerate case is retained rather than included in the infinite-energy claim. The affine scalar background also does not lie in L^2 when $A\neq 0$: if $k_1\neq 0$ it grows linearly in r in these same coordinates; if $k_1=0$ it is a nonzero function of x_2 independent of x_1. For the latter assertion, a nonzero affine function cannot be canceled globally by the bounded periodic wave. The exact localized residual below is the next map toward a finite-energy field; a failure of direct periodic descent is only that descent obstruction.

43 Exact IPM smoothing and localization with every residual retained

The pressure and coefficient correspondences above are retained here. The following notation admits both source and report coefficients and either torus zero-mode convention; it also states the complete operator needed for the smoothing estimates. Fix $L_1,L_2>0$, put

$$\mathbb{T}^2=\mathbb{R}^2/(L_1\mathbb{Z}\times L_2\mathbb{Z}),\qquad V=L_1L_2,\qquad \kappa(n)=\left(\frac{2\pi n_1}{L_1},\frac{2\pi n_2}{L_2}\right),\qquad n\in\mathbb{Z}^2.$$

Coordinates are $x=(x_1,x_2)$, $e_2=(0,1)$, Lebesgue volume is $\mathrm{d}x$, and

$$\widehat{f}(n)=\frac{1}{V}\int_{\mathbb{T}^2}f(x)e^{-i\kappa(n)\cdot x}\mathrm{d}x,\qquad \overline{f}=\widehat{f}(0),\qquad f(x)=\sum_{n\in\mathbb{Z}^2}\widehat{f}(n)e^{i\kappa(n)\cdot x}.$$

The formulas below are initially identities for smooth periodic functions. Their L^2 extensions are stated separately. Let $\sigma\in\mathbb{R}$ be fixed and let $a\in\mathbb{R}^2$ be fixed. Define the scalar-to-vector linear operator $M_{\sigma,a}:C^\infty(\mathbb{T}^2;\mathbb{R})\to C^\infty(\mathbb{T}^2;\mathbb{R}^2)$ by the full symbol

$$b_{\sigma,a}(n)=\begin{cases}\sigma\left(\dfrac{\kappa_1(n)\kappa_2(n)}{|\kappa(n)|^2},-\dfrac{\kappa_1(n)^2}{|\kappa(n)|^2}\right), & n\neq 0,\\ a, & n=0,\end{cases}\qquad \widehat{M_{\sigma,a}f}(n)=b_{\sigma,a}(n)\widehat{f}(n).\tag{594}$$

The symbol is real and even, so it preserves the reality of f. It is bounded, so smooth Fourier series and their differentiated series converge absolutely after multiplication by this symbol. In particular the codomain asserted above is correct. Throughout, M denotes this fixed $M_{\sigma,a}$, $u=M\rho$, and

$$\mathcal{N}_{\sigma,a}(\rho)=\partial_t\rho+\operatorname{div}(\rho M\rho)=\partial_t\rho+(M\rho)\cdot\nabla\rho.\tag{595}$$

For completeness, these operator bounds require only the following calculation. At nonzero n, $\kappa(n)\cdot b_{\sigma,a}(n)=0$ and $|b_{\sigma,a}(n)|^2=\sigma^2\kappa_1(n)^2/|\kappa(n)|^2\le\sigma^2$; at $n=0$ the divergence multiplier is zero and the velocity symbol is a. Thus $\operatorname{div}Mf=0$ and $\overline{Mf}=a\overline{f}$. Parseval gives $\|Mf\|_2\le B\|f\|_2$ with $B=\max\{|\sigma|,|a|\}$, and density of Fourier polynomials extends M uniquely to L^2. Derivatives eliminate the zero mode, so $\|\nabla Mf\|_2\le|\sigma|\|\nabla f\|_2$ for $f\in H^1$. The report with periodic pressure is $(\sigma,a)=(1,-e_2)$; the primary manuscript's operator is $(\sigma,a)=(2\pi,0)$ on $L_1=L_2=2\pi$, as established above.

44 The exact fixed-scale defect, including weak solutions

Choose $\eta \in C_c^\infty(\mathbb{R}^2)$ with $\eta \geq 0$ and $\int_{\mathbb{R}^2} \eta(z)\mathrm{d}z = 1$. No radiality or vanishing moment is assumed. For $\varepsilon > 0$, define

$$S_\varepsilon f(x) = \int_{\mathbb{R}^2} \eta(z) f(x - \varepsilon z)\mathrm{d}z = \int_{\mathbb{T}^2} K_\varepsilon(y) f(x-y)\mathrm{d}y, \tag{596}$$

where $K_\varepsilon(y) = \sum_{j\in\mathbb{Z}^2} \varepsilon^{-2}\eta((y+(L_1 j_1, L_2 j_2))/\varepsilon)$. The sum is locally finite; K_ε is a smooth periodic kernel with integral 1. The first equality follows by partitioning $\mathbb{R}^2$ into translates of a fundamental cell and changing variables. Its Fourier multiplier is $s_\varepsilon(n) = \int_{\mathbb{R}^2} \eta(z) e^{-i\varepsilon\kappa(n)\cdot z}\mathrm{d}z$. In particular $s_\varepsilon(0) = 1$. Scalar multiplication of each Fourier coefficient proves, without omitting its zero mode,

$$MS_\varepsilon = S_\varepsilon M, \qquad \partial_j S_\varepsilon = S_\varepsilon \partial_j, \qquad \partial_t S_\varepsilon = S_\varepsilon \partial_t \quad \text{when } \varepsilon \text{ is fixed.} \tag{597}$$

The last identity also follows by testing the time derivative against a compactly supported test function and interchanging finite kernel integrals.

Proposition 44.1 (Flux defect and its sign)**.** *Let $\rho \in L^2_{\mathrm{loc}}(I \times \mathbb{T}^2)$ and let F be a distribution such that $\partial_t\rho + \operatorname{div}(\rho u) = F$, where $u = M\rho$. Put*

$$\rho_\varepsilon = S_\varepsilon\rho, \quad u_\varepsilon = S_\varepsilon u = M\rho_\varepsilon, \quad R_\varepsilon = S_\varepsilon(\rho u) - \rho_\varepsilon u_\varepsilon.$$

All products in this definition are locally integrable. In distributions,

$$\mathcal{N}_{\sigma,a}(\rho_\varepsilon) = S_\varepsilon F - \operatorname{div} R_\varepsilon. \tag{598}$$

Define $C_\varepsilon = -\operatorname{div} R_\varepsilon$ for this weak class. For smooth ρ this states exactly

$$C_\varepsilon(\rho) := u_\varepsilon \cdot \nabla\rho_\varepsilon - S_\varepsilon(u\cdot\nabla\rho) = -\operatorname{div} R_\varepsilon. \tag{599}$$

Both R_ε and C_ε are independent of a.

Proof. The L^2 bound from the preceding operator estimate, integrated over any compact time interval, puts u in L^2. Cauchy–Schwarz puts ρu in L^1. Kernel convolution is bounded on L^2 and L^1 by Jensen's inequality and translation invariance of Lebesgue volume, so the smoothed products are also in L^1. Applying S_ε to the weak equation and using (597) gives

$$\partial_t\rho_\varepsilon + \operatorname{div} S_\varepsilon(\rho u) = S_\varepsilon F.$$

Substitution of $S_\varepsilon(\rho u) = \rho_\varepsilon u_\varepsilon + R_\varepsilon$ proves (598). Since both u and u_ε are divergence free, the smooth conservative form equals (599), proving its sign. To compare zero modes, write $u = M_{\sigma,0}\rho + a\bar{\rho}$. This last term is spatially constant. Its contribution to R_ε is $a\bar{\rho}S_\varepsilon\rho - a\bar{\rho}\rho_\varepsilon = 0$, because smoothing preserves the mean. This proves both independence claims. □

Proposition 44.2 (Exact increment formula and convergence)**.** *For $\delta_{\varepsilon z} f(x) = f(x-\varepsilon z) - f(x)$, the vector flux is*

$$\begin{aligned} R_\varepsilon(x) = &\int_{\mathbb{R}^2} \eta(z)\delta_{\varepsilon z}\rho(x)\delta_{\varepsilon z}u(x)\mathrm{d}z \\ &- \left(\int_{\mathbb{R}^2} \eta(z)\delta_{\varepsilon z}\rho(x)\mathrm{d}z\right)\left(\int_{\mathbb{R}^2} \eta(z)\delta_{\varepsilon z}u(x)\mathrm{d}z\right). \end{aligned} \tag{600}$$

For the L^2_{loc} class of Proposition 44.1, $R_\varepsilon \to 0$ in $L^1_{\mathrm{loc}}(I\times\mathbb{T}^2)$ as $\varepsilon \downarrow 0$, and $C_\varepsilon \to 0$ in distributions. If at a fixed time $\rho, u \in H^1(\mathbb{T}^2)$, then, with $A_j = \int_{\mathbb{R}^2} \eta(z)|z|^j\mathrm{d}z$,

$$\|R_\varepsilon\|_{L^1(\mathbb{T}^2)} \leq \varepsilon^2(A_2 + A_1^2)\|\nabla\rho\|_2\|\nabla u\|_2 \leq |\sigma|\varepsilon^2(A_2+A_1^2)\|\nabla\rho\|_2^2. \tag{601}$$

For spatially Hölder continuous $\rho \in C^{0,\alpha}$ and $u \in C^{0,\beta}$, $0 < \alpha, \beta \leq 1$, measured using the flat torus distance, there is the explicit pointwise bound

$$\|R_\varepsilon\|_\infty \leq \varepsilon^{\alpha+\beta}(A_{\alpha+\beta} + A_\alpha A_\beta)[\rho]_\alpha[u]_\beta. \tag{602}$$

Proof. Expand the first integral in (600) as $S_\varepsilon(\rho u) - \rho u_\varepsilon - u\rho_\varepsilon + \rho u$. The product in its second line is $\rho_\varepsilon u_\varepsilon - \rho u_\varepsilon - u\rho_\varepsilon + \rho u$. Their difference is precisely R_ε. This expansion is legitimate almost everywhere by the integrability established above and Fubini's theorem.

Spatial translations are continuous in $L^2(J \times \mathbb{T}^2)$ for compact $J \subset I$: first prove it for finite spatial Fourier polynomials with $L^2(J)$ coefficients, whose factors $e^{-i\kappa \cdot h}$ converge to 1, and then approximate an L^2 function by these polynomials using Parseval and the norm-preserving property of translation. They are likewise continuous in L^1 by approximation by continuous functions and the isometry of translation. Integrating these convergences against η with dominated convergence proves $S_\varepsilon \rho \to \rho$ and $S_\varepsilon u \to u$ in L^2, and $S_\varepsilon(\rho u) \to \rho u$ in L^1. Moreover

$$\|\rho_\varepsilon u_\varepsilon - \rho u\|_1 \le \|\rho_\varepsilon - \rho\|_2 \|u_\varepsilon\|_2 + \|\rho\|_2 \|u_\varepsilon - u\|_2 \longrightarrow 0.$$

This proves the asserted flux convergence. For every compactly supported smooth scalar test function ϕ, $\langle C_\varepsilon, \phi\rangle = \int R_\varepsilon \cdot \nabla \phi \, \mathrm{d}t\mathrm{d}x \to 0$, proving distributional convergence with the correct sign.

For $f \in H^1$, the line integral formula, first for smooth approximations and then by H^1 convergence, gives $\|f(\cdot - h) - f\|_2 \le |h| \|\nabla f\|_2$. Apply Cauchy–Schwarz in x to the first line of (600), and Minkowski followed by Cauchy–Schwarz to the second line. The resulting constants are respectively $\varepsilon^2 A_2$ and $\varepsilon^2 A_1^2$. Derivatives eliminate the zero mode; Parseval with the nonzero symbol bound gives $\|\nabla u\|_2 \le |\sigma| \|\nabla \rho\|_2$, proving (601). Finally the torus distance between $x - \varepsilon z$ and x is at most $\varepsilon |z|$. Thus $|\delta_{\varepsilon z} \rho| \le [\rho]_\alpha \varepsilon^\alpha |z|^\alpha$ and the corresponding inequality holds for u. Substitution into the two terms of (600) proves (602). □

The convergence just proved is a distributional residual statement. It does not assert a uniform bound for all mixed derivatives of a construction's force. The exact fixed-scale mixed derivative formula can also be retained: if $\mathcal{R}_\varepsilon(f, g) = S_\varepsilon(fg) - (S_\varepsilon f)(S_\varepsilon g)$, then for a time order $q \ge 0$ and spatial multi-index γ,

$$\partial_t^q \partial_x^\gamma \mathcal{R}_\varepsilon(f, g) = \sum_{r=0}^{q} \sum_{\delta \le \gamma} \binom{q}{r} \binom{\gamma}{\delta} \mathcal{R}_\varepsilon(\partial_t^r \partial_x^\delta f, \partial_t^{q-r} \partial_x^{\gamma - \delta} g). \tag{603}$$

Indeed the first derivative of the bilinear expression is the sum with the derivative on each of its two arguments, because fixed S_ε commutes with that derivative. Repeating this rule proves the binomial formula by induction, including its coefficients. Applied componentwise with $g = u$, it is an exact formula for every mixed derivative of R_ε.

45 Moving scales and Fourier cutoffs

Proposition 45.1 (Time-dependent smoothing)**.** *Let ρ, F be smooth, let $\mathcal{N}_{\sigma,a}(\rho) = F$, and let $\varepsilon \in C^1(I; (0, \infty))$. The precise identity is*

$$\mathcal{N}_{\sigma,a}(S_{\varepsilon(t)} \rho) = S_{\varepsilon(t)} F + C_{\varepsilon(t)}(\rho) + \varepsilon'(t)(\partial_\varepsilon S_\varepsilon)_{\varepsilon = \varepsilon(t)} \rho, \tag{604}$$

where

$$\partial_\varepsilon S_\varepsilon \rho(x) = -\int_{\mathbb{R}^2} \eta(z) z \cdot \nabla \rho(x - \varepsilon z) \mathrm{d}z. \tag{605}$$

More generally, a smooth scalar Fourier multiplier $S(t)$, including a smooth Fourier cutoff, satisfies

$$\mathcal{N}_{\sigma,a}(S(t)\rho) = S(t)F + \left((MS(t)\rho) \cdot \nabla S(t)\rho - S(t)((M\rho) \cdot \nabla \rho)\right) + (\partial_t S(t))\rho. \tag{606}$$

This last statement requires that $S(t)$ and its time derivative act on the smooth functions at issue; finite smooth spectral cutoffs satisfy it directly.

Proof. Differentiate the defining integral (596). Compact support of η and smoothness of ρ justify differentiation under the integral, giving (605) and $\partial_t(S_{\varepsilon(t)}\rho) = S_{\varepsilon(t)}\partial_t\rho + \varepsilon'(t)(\partial_\varepsilon S_\varepsilon)\rho$. Add the nonlinear term and use Proposition 44.1 for its frozen value of ε. This proves (604). For the general statement, differentiate the scalar Fourier multiplier at each frequency and apply the same product rule. Its commutation with M follows from multiplication of the corresponding scalar and vector symbols, including the zero mode, proving (606). □

For example, a cutoff with symbol $s(n/\mu(t))$, $s \in C_c^\infty(\mathbb{R}^2)$, $\mu(t) > 0$, has the explicitly retained extra symbol

$$\partial_t s(n/\mu(t)) = -\frac{\mu'(t)}{\mu(t)^2} n \cdot (\nabla s)(n/\mu(t)).$$

This is the chain rule, and is not part of the quadratic commutator.

46 Exact spatial localization and background–child residuals

For a smooth real cutoff $\chi(t, x)$, define the vector commutator

$$K_\chi f = [M, \chi]f = M(\chi f) - \chi M f. \tag{607}$$

There is no support assertion implicit in this definition: M is a Fourier operator and $K_\chi f$ can persist in a region where $\chi = 1$. In the present conventions its complete Fourier formula is

$$\widehat{K_\chi f}(n) = \sum_{\ell \in \mathbb{Z}^2} (b_{\sigma,a}(n) - b_{\sigma,a}(\ell))\widehat{\chi}(n - \ell)\widehat{f}(\ell). \tag{608}$$

Indeed Fourier multiplication gives the first term with symbol $b_{\sigma,a}(n)$ and the second with symbol $b_{\sigma,a}(\ell)$; subtracting gives exactly the displayed sum. Smoothness makes it absolutely convergent. In particular the terms $n = 0$ and $\ell = 0$ are included, and

$$\operatorname{div} K_\chi f = -(Mf) \cdot \nabla\chi, \qquad K_\chi^{\sigma,a} f = K_\chi^{\sigma,0} f + a(\overline{\chi f} - \chi\overline{f}). \tag{609}$$

The first equality follows by taking divergence of (607) and using $\operatorname{div} Mf = \operatorname{div} M(\chi f) = 0$; the second follows by inserting $M_{\sigma,a} f = M_{\sigma,0} f + a\overline{f}$ into that same definition.

Proposition 46.1 (Cutoff of a complete density)**.** *Let ρ, F, χ be smooth and let $\mathcal{N}_{\sigma,a}(\rho) = F$. Set $u = M\rho$, $q = \chi\rho$, and $v = Mq = \chi u + K_\chi\rho$. Then the complete residual of q is*

$$\begin{aligned}\mathcal{N}_{\sigma,a}(q) = \chi F + \rho\partial_t\chi + \chi(\chi - 1)u \cdot \nabla\rho + \chi(K_\chi\rho) \cdot \nabla\rho \\ + \chi\rho u \cdot \nabla\chi + \rho(K_\chi\rho) \cdot \nabla\chi.\end{aligned} \tag{610}$$

Proof. Using $\operatorname{div} v = 0$, apply the product rule without deleting any term:

$$\begin{aligned}\partial_t q + v \cdot \nabla q &= \chi\partial_t\rho + \rho\partial_t\chi + (\chi u + K_\chi\rho) \cdot (\chi\nabla\rho + \rho\nabla\chi) \\ &= \chi(F - u \cdot \nabla\rho) + \rho\partial_t\chi + \chi^2 u \cdot \nabla\rho + \chi(K_\chi\rho) \cdot \nabla\rho \\ &\quad + \chi\rho u \cdot \nabla\chi + \rho(K_\chi\rho) \cdot \nabla\chi.\end{aligned}$$

Combining the two explicitly displayed coefficients of $u \cdot \nabla\rho$ gives (610); all other original factors remain present. □

Proposition 46.2 (Adding and then localizing a child)**.** *For any smooth background r and child θ, put $U = Mr$, $w = M\theta$. Then, without requiring that either field solve an auxiliary equation,*

$$\mathcal{N}_{\sigma,a}(r + \theta) - \mathcal{N}_{\sigma,a}(r) = \partial_t\theta + U \cdot \nabla\theta + w \cdot \nabla r + w \cdot \nabla\theta. \tag{611}$$

Define the computed linearized residual $G = \partial_t\theta + U\cdot\nabla\theta + w\cdot\nabla r$. For $q = \chi\theta$, the corresponding full localized residual is

$$\begin{aligned}\mathcal{N}_{\sigma,a}(r+q) - \mathcal{N}_{\sigma,a}(r) = \chi G &+ \theta(\partial_t\chi + U\cdot\nabla\chi) + (K_\chi\theta)\cdot\nabla r \\ &+ \chi^2 w\cdot\nabla\theta + \chi(K_\chi\theta)\cdot\nabla\theta \\ &+ \chi\theta w\cdot\nabla\chi + \theta(K_\chi\theta)\cdot\nabla\chi.\end{aligned} \tag{612}$$

Proof. Linearity gives $M(r+\theta) = U + w$. Expanding $(U+w)\cdot\nabla(r+\theta)$ and subtracting $U\cdot\nabla r$ leaves the three advective terms in (611); the time derivative leaves $\partial_t\theta$. Apply this identity with q in place of θ. The three linear terms become

$$\begin{aligned}\partial_t q + U\cdot\nabla q + (Mq)\cdot\nabla r = \chi&(\partial_t\theta + U\cdot\nabla\theta + w\cdot\nabla r) \\ &+ \theta(\partial_t\chi + U\cdot\nabla\chi) + (K_\chi\theta)\cdot\nabla r.\end{aligned}$$

The remaining quadratic term is

$$(Mq)\cdot\nabla q = (\chi w + K_\chi\theta)\cdot(\chi\nabla\theta + \theta\nabla\chi),$$

whose four products are exactly the final four terms in (612). This proves the formula without treating G as an assumed vanishing quantity. $\square$

Taking $r = S_\varepsilon\rho$ in Proposition 46.2 and inserting (598) proves precisely the report's background–child expansion for fixed ε. For moving ε, the additional linear scale derivative from (604) must also be retained. For a nonzero lattice vector n, a smooth 2π-periodic mean-zero profile h, and $\theta(x) = h(\kappa(n)\cdot x)$, its only frequencies are jn, $j\in\mathbb{Z}\setminus\{0\}$. The symbol obeys $b_{\sigma,a}(jn) = b_{\sigma,a}(n)$ at all these frequencies. Hence $w = b_{\sigma,a}(n)h(\kappa(n)\cdot x)$ and

$$w\cdot\nabla\theta = h(\kappa\cdot x)h'(\kappa\cdot x)b_{\sigma,a}(n)\cdot\kappa(n) = 0.$$

This proves the exact transverse cancellation with its mean hypothesis. After localization, only the $\chi^2 w\cdot\nabla\theta$ term in (612) vanishes by this identity; the other displayed terms retain their exact values. They give an exact forced relation for the localized field, rather than a claim that no relation exists.

47 An explicit nonzero smoothing defect

In this section only, take $L_1 = L_2 = 2\pi$, set $U_\sigma = M_{\sigma,0}$, and let $S_s = e^{s\Delta}$ for fixed $s \ge 0$. Its scalar multiplier at $n\in\mathbb{Z}^2$ is $e^{-s|n|^2}$. Here $m(n) = (n_1n_2/|n|^2, -n_1^2/|n|^2)$ for $n\ne 0$ and $m(0)=0$; P_σ denotes the pressure multiplier $i\sigma n_2/|n|^2$ at $n\ne 0$ and 0 at $n=0$, with the Darcy identity proved above. All fields below have zero mean, so the arbitrary torus mean-velocity coefficient has no effect on this example.

Proposition 47.1 (The exact defect, with a complete forced solution)**.** *Let $\rho(t,x) = \cos x_1 + \cos(x_1+x_2)$ for every $t\in\mathbb{R}$ and $x\in\mathbb{T}^2$. For every $s\ge 0$ and $\sigma\in\mathbb{R}$, define*

$$N_\sigma = U_\sigma\rho\cdot\nabla\rho, \qquad C_{s,\sigma} = U_\sigma[S_s\rho]\cdot\nabla(S_s\rho) - S_sN_\sigma.$$

Then

$$N_\sigma(x) = \frac{\sigma}{4}\sin(2x_1+x_2) + \frac{3\sigma}{4}\sin x_2, \tag{613}$$

$$C_{s,\sigma}(x) = \frac{\sigma}{4}(e^{-3s} - e^{-5s})\sin(2x_1+x_2) + \frac{3\sigma}{4}(e^{-3s} - e^{-s})\sin x_2. \tag{614}$$

For every $s > 0$ and $\sigma \neq 0$, this defect is not the zero function. The stationary fields

$$\begin{aligned}
\rho &= \cos x_1 + \cos(x_1 + x_2), \\
u_\sigma &= \left(\frac{\sigma}{2} \cos(x_1 + x_2), -\sigma \cos x_1 - \frac{\sigma}{2} \cos(x_1 + x_2) \right), \\
p_\sigma &= -\frac{\sigma}{2} \sin(x_1 + x_2)
\end{aligned}$$

with forcing $F_\sigma(t, x) = N_\sigma(x)$ solve, at every point,

$$\partial_t \rho + u_\sigma \cdot \nabla \rho = F_\sigma, \qquad u_\sigma = -\nabla p_\sigma - \sigma \rho e_2, \qquad \nabla \cdot u_\sigma = 0.$$

Their heat transforms solve these same equations with fields $(S_s \rho, S_s u_\sigma, S_s p_\sigma)$ and the exact forcing

$$\widetilde{F}_{s,\sigma} = S_s F_\sigma + C_{s,\sigma} = \frac{\sigma e^{-3s}}{4} \sin(2x_1 + x_2) + \frac{3\sigma e^{-3s}}{4} \sin x_2.$$

Proof. The functions are real and smooth on the original torus and have zero mean. The four nonzero coefficients of ρ are $1/2$ at $(1, 0), (-1, 0), (1, 1), (-1, -1)$. Since

$$m(1, 0) = m(-1, 0) = (0, -1), \qquad m(1, 1) = m(-1, -1) = (1/2, -1/2),$$

the displayed formula for u_σ follows without any frequency rescaling. Directly,

$$-\nabla p_\sigma - \sigma \rho e_2 = \left(\frac{\sigma}{2} \cos(x_1 + x_2), \frac{\sigma}{2} \cos(x_1 + x_2) - \sigma \cos x_1 - \sigma \cos(x_1 + x_2) \right) = u_\sigma,$$

and the two divergence terms are $-\sigma \sin(x_1 + x_2)/2$ and $+\sigma \sin(x_1 + x_2)/2$, whose sum is zero. The original gradient is

$$\nabla \rho = (-\sin x_1 - \sin(x_1 + x_2), -\sin(x_1 + x_2)).$$

Multiplying both components, retaining all four terms, gives

$$\begin{aligned}
N_\sigma = &\; -\frac{\sigma}{2} \cos(x_1 + x_2) \sin x_1 - \frac{\sigma}{2} \cos(x_1 + x_2) \sin(x_1 + x_2) \\
&+ \sigma \cos x_1 \sin(x_1 + x_2) + \frac{\sigma}{2} \cos(x_1 + x_2) \sin(x_1 + x_2) \\
= &\; \sigma \cos x_1 \sin(x_1 + x_2) - \frac{\sigma}{2} \sin x_1 \cos(x_1 + x_2).
\end{aligned}$$

The exact trigonometric identities

$$\begin{aligned}
\cos x_1 \sin(x_1 + x_2) &= \frac{1}{2} \sin(2x_1 + x_2) + \frac{1}{2} \sin x_2, \\
\sin x_1 \cos(x_1 + x_2) &= \frac{1}{2} \sin(2x_1 + x_2) - \frac{1}{2} \sin x_2
\end{aligned}$$

yield (613) with both original interaction contributions retained.

The input squared wave lengths are exactly 1 and 2. Therefore

$$\begin{aligned}
S_s \rho &= e^{-s} \cos x_1 + e^{-2s} \cos(x_1 + x_2), \\
U_\sigma[S_s \rho] &= \left(\frac{\sigma}{2} e^{-2s} \cos(x_1 + x_2), -\sigma e^{-s} \cos x_1 - \frac{\sigma}{2} e^{-2s} \cos(x_1 + x_2) \right), \\
\nabla(S_s \rho) &= \left(-e^{-s} \sin x_1 - e^{-2s} \sin(x_1 + x_2), -e^{-2s} \sin(x_1 + x_2) \right).
\end{aligned}$$

Consequently the four product terms are

$$\begin{aligned}U_\sigma[S_s\rho]\cdot\nabla(S_s\rho) = {} & -\frac{\sigma}{2}e^{-3s}\cos(x_1+x_2)\sin x_1 - \frac{\sigma}{2}e^{-4s}\cos(x_1+x_2)\sin(x_1+x_2)\\ & +\sigma e^{-3s}\cos x_1\sin(x_1+x_2) + \frac{\sigma}{2}e^{-4s}\cos(x_1+x_2)\sin(x_1+x_2)\\ = {} & \frac{\sigma}{4}e^{-3s}\sin(2x_1+x_2) + \frac{3\sigma}{4}e^{-3s}\sin x_2.\end{aligned}$$

The output frequencies $(2,1)$ and $(0,1)$ have squared lengths 5 and 1, respectively. Thus

$$S_sN_\sigma = \frac{\sigma}{4}e^{-5s}\sin(2x_1+x_2) + \frac{3\sigma}{4}e^{-s}\sin x_2.$$

Subtracting this last expression in the stipulated order proves (614). At the explicit point $x=(\pi/4,0)$,

$$C_{s,\sigma}(\pi/4,0) = \frac{\sigma}{4}(e^{-3s}-e^{-5s}) = \frac{\sigma}{4}e^{-5s}(e^{2s}-1)\neq 0 \quad (s>0,\ \sigma\neq 0).$$

This proves nonvanishing for each positive s, not merely at one selected heat time. For $s=0$ or $\sigma=0$, formula (614) instead gives zero exactly.

All displayed original fields are time independent, so $\partial_t\rho = 0$. With the stated nonzero forcing for $\sigma\neq 0$, the transport equation follows from (613). The direct pressure and divergence computations already prove its two remaining equations. Heat multiplication commutes mode by mode with U_σ, P_σ, and the spatial derivatives. Therefore the heat-transformed fields satisfy Darcy's law and zero divergence. Their time derivative remains zero, and their transport term is the computed $\sigma e^{-3s}N_1$, exactly the stated $S_sF_\sigma + C_{s,\sigma}$. This proves all the transformed equations. □

48 A complete finite-energy forced field on the plane

This section uses the report's coefficient 1, its original Cartesian coordinates, and its Darcy law $v=-\nabla p - qe_2$ on $\mathbb{R}^2$. The affine field is retained explicitly; no multiplier is applied to its nonintegrable polynomial part.

Fix $A, a_0, t_0\in\mathbb{R}$, $k=(k_1,k_2)\in\mathbb{R}^2\setminus\{0\}$, and a smooth real 2π-periodic mean-zero function h. A smooth periodic primitive H exists: start with $H_0(s)=\int_0^s h(z)\mathrm{d}z$, whose increment under $s\mapsto s+2\pi$ is the zero integral of h, and subtract its period mean if a zero-mean pressure profile is desired. Set

$$\kappa = k_1^2+k_2^2,\quad m(k)=\left(\frac{k_1k_2}{\kappa}, -\frac{k_1^2}{\kappa}\right),\quad \lambda=\frac{Ak_1^2}{\kappa},\quad \alpha(t)=a_0e^{\lambda(t-t_0)},\quad s=k\cdot x,$$

$$\begin{aligned} r(t,x) &= Ax_2+\alpha(t)h(s), \qquad U(t,x)=\alpha(t)m(k)h(s),\\ P(t,x) &= -\frac{A}{2}x_2^2 - \frac{\alpha(t)k_2}{\kappa}H(s).\end{aligned} \tag{615}$$

Theorem 42.4 proves, by the complete Darcy and transport calculation for these unchanged fields, $U=-\nabla P - re_2$, $\operatorname{div} U = 0$, and $r_t + U\cdot\nabla r = 0$, including $A=0$, $a_0=0$, and $k_1=0$.

Let $I\subset\mathbb{R}$ be an open interval. Choose a fixed compact set $K\subset\mathbb{R}^2$ and a real smooth function $\chi\in C^\infty(I\times\mathbb{R}^2)$ such that $\operatorname{supp}\chi(t,\cdot)\subset K$ for every $t\in I$. The cutoff is permitted to depend on time; its time derivative is retained below. Define the complete density $q=\chi r$. On $\mathbb{R}^2$ our Fourier convention is

$$\widehat{f}(\xi)=\int_{\mathbb{R}^2} f(x)e^{-i\xi\cdot x}\mathrm{d}x, \qquad f(x)=\frac{1}{(2\pi)^2}\int_{\mathbb{R}^2}\widehat{f}(\xi)e^{i\xi\cdot x}\mathrm{d}\xi.$$

Define $M_{\mathbb{R}^2}$ and $\mathcal{P}_{\mathbb{R}^2}$ on $C_c^\infty(\mathbb{R}^2)$ by

$$\widehat{M_{\mathbb{R}^2} f}(\xi) = \left(\frac{\xi_1\xi_2}{|\xi|^2}, -\frac{\xi_1^2}{|\xi|^2}\right)\widehat{f}(\xi), \qquad \widehat{\mathcal{P}_{\mathbb{R}^2} f}(\xi) = \frac{i\xi_2}{|\xi|^2}\widehat{f}(\xi) \quad (\xi \neq 0). \tag{616}$$

Values assigned at the singleton $\xi = 0$ do not change these Lebesgue Fourier integrals. This is distinct from a discrete torus zero mode.

Proposition 48.1 (Exact localized finite-energy solution)**.** *The fields*

$$q = \chi r, \qquad v = M_{\mathbb{R}^2} q, \qquad p_c = \mathcal{P}_{\mathbb{R}^2} q, \qquad E = v - \chi U \tag{617}$$

are smooth on $I \times \mathbb{R}^2$ and obey

$$q_t + v \cdot \nabla q = F_c, \qquad v = -\nabla p_c - q e_2, \qquad \operatorname{div} v = 0, \tag{618}$$

with the following explicit force:

$$\begin{aligned} F_c &= \chi_t r + \chi(\chi - 1)U \cdot \nabla r + \chi r U \cdot \nabla\chi + rE \cdot \nabla\chi + \chi E \cdot \nabla r \\ &= \chi_t(Ax_2 + \alpha h) - \chi(\chi - 1)\frac{Ak_1^2}{\kappa}\alpha h \\ &\quad + \chi\alpha(Ax_2 + \alpha h)h\, m(k) \cdot \nabla\chi + (Ax_2 + \alpha h)E \cdot \nabla\chi \\ &\quad + \chi E \cdot (Ae_2 + \alpha k h'). \end{aligned} \tag{619}$$

The arguments of h, h' in this formula are the unchanged $s = k \cdot x$. Both q and F_c have spatial support in K. For every compact $J \subset I$, every time order $j \geq 0$, and every spatial multi-index γ, their mixed derivatives and those of v, p_c are bounded on $J \times \mathbb{R}^2$. The velocity and density have finite energy:

$$\|v(t)\|_2^2 \leq \|q(t)\|_2^2 = \int_{\mathbb{R}^2} \chi(t,x)^2[Ax_2 + \alpha(t)h(k \cdot x)]^2 \mathrm{d}x < \infty. \tag{620}$$

Thus (617)–(619) are an actual smooth forced IPM solution for every such cutoff, including each fixed compact cutoff. No smallness of F_c or E is asserted.

Proof. All derivatives of q are smooth and supported in K, hence are bounded on $J \times \mathbb{R}^2$. For any nonnegative integer N, integration by parts gives

$$\widehat{\partial_t^j q}(t,\xi) = (1 + |\xi|^2)^{-N}\widehat{(1 - \Delta_x)^N \partial_t^j q}(t,\xi).$$

The Fourier transform on the right has absolute value at most the L^1 norm of its argument, uniformly bounded for $t \in J$. Therefore every time derivative of $\widehat{q}$ decays faster than any inverse power of $|\xi|$, uniformly on J. The symbol of $M_{\mathbb{R}^2}$ has norm at most 1. The pressure symbol has absolute value at most $|\xi|^{-1}$, which is locally integrable in two dimensions since $\int_{|\xi|<1} |\xi|^{-1}\mathrm{d}\xi = 2\pi$. Consequently all inverse Fourier integrals for v, p_c, including every time derivative and every spatial derivative, converge absolutely and uniformly on $J \times \mathbb{R}^2$. Differentiating under those integrals proves their claimed smoothness and boundedness. The symbols are conjugate symmetric under $\xi \mapsto -\xi$ as required for real outputs from real q.

For $\xi \neq 0$ the divergence multiplier vanishes, and the Darcy multiplier is

$$-i\xi\frac{i\xi_2}{|\xi|^2} - e_2 = \left(\frac{\xi_1\xi_2}{|\xi|^2}, -\frac{\xi_1^2}{|\xi|^2}\right).$$

The exceptional singleton has measure zero, so these equalities give the pointwise smooth Darcy and divergence equations after Fourier inversion. Parseval with the retained factor $(2\pi)^{-2}$ gives

$$\|v\|_2^2 = \frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} \frac{\xi_1^2}{|\xi|^2}|\widehat{q}(\xi)|^2 \mathrm{d}\xi \leq \frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} |\widehat{q}(\xi)|^2 \mathrm{d}\xi = \|q\|_2^2,$$

which proves (620) because q has compact support.

The correction $E = v - \chi U$ is defined by actual smooth fields, so it is smooth and belongs to L^2 at each time. Taking divergence gives $\operatorname{div} E = -U \cdot \nabla\chi$, since $\operatorname{div} v = \operatorname{div} U = 0$. For the transport equation, the exact product expansion is

$$\begin{aligned} q_t + v \cdot \nabla q &= \chi_t r + \chi r_t + (\chi U + E) \cdot (r\nabla\chi + \chi\nabla r) \\ &= \chi_t r - \chi U \cdot \nabla r + \chi r U \cdot \nabla\chi + \chi^2 U \cdot \nabla r + rE \cdot \nabla\chi + \chi E \cdot \nabla r. \end{aligned}$$

The already proved unforced affine identity supplies $r_t = -U \cdot \nabla r$. Combining the two displayed coefficients gives the first line of (619). Substituting the complete fields and $U \cdot \nabla r = -Ak_1^2\alpha h/\kappa$ gives the remaining lines. Each term has a factor χ, $\nabla\chi$, or χ_t, all supported in K. Hence $\operatorname{supp} F_c \subset K$. Smoothness then implies all its mixed derivatives are uniformly bounded on $J \times K$, and they vanish outside K, proving the full uniform statement. □

In a spacetime open set on which $\chi = 1$, the exact map retains $q = r$, $v = U + E$, and $F_c = E \cdot \nabla r$. Thus agreement of the localized density with the affine density does not require discarding the actual nonlocal velocity or its force. The result constructs a finite-energy forced field and its complete pressure; it makes no finite-time blow-up claim.

49 Checked additions from the second half of the incoming report

This section records independently derived identities prompted by file pages 21–40 (printed pages 16–35) of the incoming reconstruction. It does not certify the incoming report or the cited source papers. The axisymmetric coordinate and force reconstruction is already proved in the cumulative notebook; here its exact variable-coefficient wave residual and its relation to phase diffusion are made explicit. Every formula below is local on a fixed coordinate patch $D \Subset \mathbb{R} \times (0, \infty)$ unless a whole-space domain is specified.

49.1 Exact material-wave and cutoff identities

Retain the coordinates $y_1 = z$, $y_2 = r^2/2$, the orientation $J(a, b) = (-b, a)$, and the operators

$$\begin{gathered} A = \operatorname{diag}((2y_2)^{-1}, 1), \quad C = \operatorname{diag}(1, 2y_2), \quad W = (2y_2)^{-2}, \\ L = \operatorname{div}(A\nabla), \quad Q = C : D^2, \quad M = Q + 4\partial_2. \end{gathered}$$

In particular $Q = (2y_2)L$ with multiplication after application of L. Let Ψ_0, Γ_0 be smooth on $D \times I$, where I is a compact interval, and write $v_0 = J\nabla\Psi_0$, $\xi_0 = L\Psi_0$, $D_0 = \partial_t + v_0 \cdot \nabla$. Let ϕ solve $D_0\phi = 0$. For a material-coordinate description, take a flow $X(a, t)$ of v_0, initially the identity, on the region under consideration, and put $\phi(X(a, t), t) = p \cdot a$ for a retained nonzero covector p. Differentiating this equality gives

$$\zeta := \nabla_y\phi = (D_aX)^{-T}p, \quad D_0\zeta = -(D_yv_0)^T\zeta, \quad \det D_aX = 1.$$

For the last identity, differentiation of a determinant gives $\partial_t \det D_aX = (\operatorname{div} v_0)(X, t) \det D_aX = 0$; the initial value is one. The gradient identity follows by differentiating $D_0\phi = 0$ in each coordinate. Thus these are exact maps, rather than a replacement of X by an affine approximation.

Fix $N > 0$, smooth amplitudes a, b, and smooth scalar functions F, P with $P' = F$. Set $s = N\phi$, $\gamma = aF(s)$, $\psi = bP(s)$ and

$$w = J\nabla\psi, \quad \eta = L\psi, \quad \kappa = \zeta^TA\zeta, \quad q = \zeta^TC\zeta = 2y_2\kappa, \quad h = 2(A\nabla b) \cdot \zeta + bL\phi, \quad e = Lb.$$

There is no approximation in

$$w = NbJ\zeta F(s) + J\nabla b\, P(s), \tag{621}$$

$$\eta = N^2\kappa bF'(s) + NhF(s) + eP(s), \tag{622}$$

$$Q\gamma = N^2qaF''(s) + N\{2(C\nabla a) \cdot \zeta + aQ\phi\}F'(s) + (Qa)F(s). \tag{623}$$

Indeed, for any scalar H and amplitude b,

$$\partial_i(bH(s)) = b_iH + Nb\phi_iH',$$

$$\partial_{ij}(bH(s)) = b_{ij}H + N(b_i\phi_j + b_j\phi_i + b\phi_{ij})H' + N^2b\phi_i\phi_jH''.$$

Contracting these explicit entries with A or C proves all the displayed formulas. Here $\partial_iA_{ij} = 0$ for each j, so $L = A : D^2$; no coefficient derivative has been dropped. The identity $C = (2y_2)A$ proves the equality of the two quadratic forms.

Define the reduced residuals, retaining $\nu \geq 0$, by

$$E_\Gamma(\Gamma, \Psi) = (\partial_t + J\nabla\Psi \cdot \nabla)\Gamma - \nu Q\Gamma,$$

$$E_\xi(\Gamma, \Psi) = (\partial_t + J\nabla\Psi \cdot \nabla)L\Psi - W\partial_1(\Gamma^2) - \nu ML\Psi.$$

Their exact differences are

$$\delta E_\Gamma = (D_0a)F + w \cdot \nabla\Gamma_0 + w \cdot \nabla\gamma - \nu Q\gamma, \tag{624}$$

$$\delta E_\xi = D_0\eta + w \cdot \nabla\xi_0 + w \cdot \nabla\eta - W\partial_1(2\Gamma_0\gamma + \gamma^2) - \nu M\eta. \tag{625}$$

In these expressions

$$D_0\eta = D_0(N^2\kappa b)F' + N(D_0h)F + (D_0e)P,$$

$$w \cdot \nabla\gamma = Nb(J\zeta \cdot \nabla a)F^2 + aN(J\nabla b \cdot \zeta)PF' + (J\nabla b \cdot \nabla a)PF,$$

because the remaining apparent N^2 term is proportional to $J\zeta \cdot \zeta = 0$. The swirl source is explicitly

$$\begin{aligned}\partial_1(2\Gamma_0\gamma + \gamma^2) &= 2\partial_1(\Gamma_0a)F + 2N\Gamma_0a\zeta_1F' \\ &\quad + \partial_1(a^2)F^2 + 2Na^2\zeta_1FF'.\end{aligned}$$

Expanding the two residual definitions gives (624)–(625) term by term, and the preceding differentiation formulas expand every quantity in them. This includes the old-vorticity advection and the quadratic swirl source. With $d = J\zeta \cdot \nabla\Gamma_0$, $c = 2W\Gamma_0$, and $\Omega = N^2\kappa b$, the indicated inviscid principal coefficients are therefore $D_0a + d\Omega/(N\kappa)$ and $D_0\Omega - Nc\zeta_1a$. Cancelling those two coefficients gives the report's material amplitude equations; the displayed residuals prove exactly what remains after that cancellation.

Localization can be done without modifying divergence: replacing b by χb replaces w by $\chi w + bP(s)J\nabla\chi$, which is still $J\nabla(\chi\psi)$. For any of $\mathcal{D} = L, Q, M$ write $\mathcal{D} = \sum_{i,j} B_{ij}\partial_{ij} + \sum_i d_i\partial_i$ with respectively $(B, d) = (A, 0), (C, 0), (C, (0, 4))$. Direct application of the product rule proves the exact commutator

$$[\mathcal{D}, \chi]u = 2\sum_{i,j} B_{ij}\chi_iu_j + \left(\sum_{i,j} B_{ij}\chi_{ij} + \sum_i d_i\chi_i\right)u. \tag{626}$$

This formula includes the $4\chi_2u$ term for M. It also supplies an all-order quantitative bound without an implicit derivative loss. For a mixed multi-index α in (y_1, y_2, t), each term in $\partial^\alpha(B_{ij}\chi_iu_j)$ is exactly

$$\sum_{\beta+\gamma+\delta=\alpha} \frac{\alpha!}{\beta!\gamma!\delta!}(\partial^\beta B_{ij})(\partial^\gamma\chi_i)(\partial^\delta u_j).$$

The other two products in (626) have the identical multinomial expansion with (χ_i, u_j) replaced by (χ_{ij}, u) or (χ_i, u) and B_{ij} by d_i in the latter. Taking absolute values and suprema gives the bound obtained by summing these explicitly listed nonnegative products, with the factor two on the first sum. All coefficient seminorms are finite on $D \times I$: derivatives of $(2y_2)^{-1}$ have size $m!/(2y_2^{m+1})$, and $2y_2$ has only its zeroth and first nonzero derivatives. Thus the formula is an actual bound at every mixed order, not a claimed summability estimate for an unspecified cascade.

49.2 The viscous principal operator retains the full phase profile

The leading viscous term in (623) is $\nu N^2 q\partial_s^2(aF)$ on the right side of the evolution equation. For the normalized vorticity profile $\eta = \partial_s\mathcal{O}$, the same principal operator acts on $\mathcal{O}$. With frozen coefficients, the principal profile system is

$$\partial_t\begin{pmatrix}\mathcal{A}\\ \mathcal{O}\end{pmatrix} = \nu N^2 q\partial_s^2\begin{pmatrix}\mathcal{A}\\ \mathcal{O}\end{pmatrix} + \begin{pmatrix}0 & -d/(N\kappa)\\ Nc\zeta_1 & 0\end{pmatrix}\begin{pmatrix}\mathcal{A}\\ \mathcal{O}\end{pmatrix}. \tag{627}$$

Here ∂_s integration of the vorticity equation concerns nonzero phase modes; its zero mode is a separate mean equation. For a nonzero integer m, inserting e^{ims}, retaining $\widehat{\mathcal{O}} = \sigma\mathcal{O}/N$ with $\sigma \neq 0$, gives

$$B_m = \begin{pmatrix}-\nu N^2 qm^2 & -d/(\sigma\kappa)\\ \sigma c\zeta_1 & -\nu N^2 qm^2\end{pmatrix}, \quad \det(\lambda I - B_m) = (\lambda + \nu N^2 qm^2)^2 + \frac{cd\zeta_1}{\kappa}. \tag{628}$$

Expanding the determinant proves the stated formula. The eigenvalues are

$$-\nu N^2 qm^2 \pm \sqrt{-cd\zeta_1/\kappa}.$$

Formula (12.9) of the report is the case $|m| = 1$. For a general periodic F with a linear germ, it must be read as a harmonic diagnostic. A nonzero such F cannot satisfy $F'' = -F$: on its linear germ $F(s) = s$, the left side is zero and the right side is $-s$. Consequently a single scalar damping coefficient does not evolve that entire germ-preserving profile. The exact relation is the phase operator (627) and its Fourier family (628), with the lower terms retained by (624)–(625).

49.3 The weighted propagator bound and all label derivatives

The finite-dimensional estimate in report Lemma 11.1 is correct. Here are its exact domain and differentiated version. Let matrices $B_0(a,t), E(a,t)$ be continuous in $t \in I = [t_0, t_1]$ and smooth in labels a in a compact set. The propagators exist by the ordered integral series: its term of degree n is bounded by $K^n|t-s|^n/n!$ when the coefficient norm is at most K. Uniform convergence permits differentiation of its integral equation, which proves both existence and the evolution equation. Iterating the integral equation for a difference proves uniqueness, since the same factorial bound tends to zero as $n \to \infty$.

Let P_0 denote the propagator for B_0 and P that for $B = B_0 + E$. For any positive weight $\rho(a,t)$ and constant $C \geq 1$ for which $\|P_0(a;t,s)\| \leq C\rho(a,t)/\rho(a,s)$ at $s \leq t$, the exact identity and resulting bound are

$$P(t,s) = P_0(t,s) + \int_s^t P_0(t,r)E(r)P(r,s)\,dr, \tag{629}$$

$$\|P(t,s)\| \leq C\frac{\rho(a,t)}{\rho(a,s)}\exp\left(C\int_s^t \|E(a,r)\|\,dr\right). \tag{630}$$

To prove the identity, denote its right side by $H(t)$. Differentiation gives $H' = B_0 H + EP$ and $H(s) = I$, while $P' = B_0 P + EP$ and $P(s) = I$. Thus $(P - H)' = B_0(P - H)$ with zero initial value, and uniqueness gives $P = H$. For the bound set $h(t) = \rho(a,s)\|P(t,s)\|/\rho(a,t)$. The identity gives $h(t) \leq C + C\int_s^t \|E(r)\|h(r)\,dr$. The right side $H(t)$ is absolutely continuous and obeys $H'(t) \leq C\|E(t)\|H(t)$, $H(s) = C$. Multiplication by $\exp(-C\int_s^t \|E\|)$ and integration prove (630). Only positivity and the displayed ratio bound are needed for the weight; differentiability of the weight is not used.

For $z' = Bz + f$ define $z_\alpha = \partial_a^\alpha z$. Repeated Leibniz differentiation gives the complete triangular equation

$$z_\alpha' = Bz_\alpha + f_\alpha + \sum_{0<\beta\leq\alpha}\binom{\alpha}{\beta}(\partial_a^\beta B)z_{\alpha-\beta}.$$

Consequently its exact solution is $P(t,s)z_\alpha(s)$ plus the Duhamel integral of the displayed source. Inserting (630) bounds this by

$$\frac{|z_\alpha(t)|}{\rho(a,t)} \le Ce^{C\int_s^t \|E\|}\left\{\frac{|z_\alpha(s)|}{\rho(a,s)} + \int_s^t \frac{|f_\alpha(r)| + \sum_{0<\beta\le\alpha}\binom{\alpha}{\beta}\|\partial_a^\beta B(r)\|\,|z_{\alpha-\beta}(r)|}{\rho(a,r)}\,dr\right\}.$$

This inductively gives finite bounds at every fixed derivative order on the stated compact domain. It does not supply amplification or smallness for a not-yet-constructed fluid iteration.

49.4 The report's Kelvin wave is an exact nonlinear local family

Keep $a>0$, $\nu>0$, a signed $k_0\neq 0$, and a real b_0. Set $k(t)=k_0e^{at}$ and

$$b(t) = b_0\exp\left[at - \frac{\nu k_0^2}{2a}(e^{2at}-1)\right], \qquad V(t) = b(t)/k(t),$$

$$u = (-ax_1, ax_2, -V(t)\sin(k(t)x_1)), \qquad p = -\tfrac{1}{2}a^2(x_1^2+x_2^2).$$

These fields solve the unforced Cartesian Navier–Stokes equation exactly on $\mathbb{R}^3\times[0,\infty)$, and $\operatorname{curl} u = b(t)\cos(k(t)x_1)e_2$. Indeed $k' = ak$, $b' = (a-\nu k^2)b$, and hence $V' = -\nu k^2 V$. The phase has zero material derivative under $(-ax_1, ax_2, 0)$. The added velocity has only a third component, is independent of x_3, and therefore has zero self-advection and zero divergence. Its derivative acting on the base also vanishes. Its only remaining momentum residual is $(-V'-\nu k^2V)\sin(kx_1)e_3 = 0$. The base convection is $(a^2x_1, a^2x_2, 0)$, cancelled by ∇p. The stated curl follows by differentiating u_3 in x_1 with the right-handed orientation. The family has infinite spatial energy, as its first component alone has infinite squared integral.

Here is its explicit compact interior bridge. Keep any $0<R_0<R_1$ and $0<T_0<T_1$. Define $b_*(s)=e^{-1/s}$ for $s>0$ and zero otherwise, $H(s)=b_*(s)/(b_*(s)+b_*(1-s))$, and

$$\chi(x) = H\left(\frac{R_1^2-|x|^2}{R_1^2-R_0^2}\right), \qquad \eta_*(t) = H\left(\frac{T_1-t}{T_1-T_0}\right).$$

The denominator of H is positive since at least one of $s, 1-s$ is positive. Every derivative of $e^{-1/s}$ is that exponential times a polynomial in $1/s$ and tends to zero at $s=0+$; thus these are smooth cutoffs with the stated inner plateaux and outer zero regions. Set

$$\mathcal{A} = (0, (b/k^2)\cos(kx_1), -ax_1x_2), \qquad w_* = \nabla\chi\times\mathcal{A}, \qquad V_* = \chi u + w_* = \operatorname{curl}(\chi\mathcal{A}),$$

$$u_c = \eta_* V_*, \qquad p_c = \eta_*\chi p.$$

Indeed the three components of $\operatorname{curl}\mathcal{A}$ are $-ax_1$, ax_2, and $-(b/k)\sin(kx_1)$ respectively, proving the potential identity including its orientation. Prescribe the finite force

$$\begin{aligned} f_c = {}& \eta_*' V_* + \eta_*\partial_t w_* - \nu\eta_*\{2\nabla\chi\cdot\nabla u + (\Delta\chi)u + \Delta w_*\} \\ &+ \eta_*\chi(\eta_*\chi - 1)(u\cdot\nabla)u + \eta_* p\nabla\chi \\ &+ \eta_*^2\{\chi(u\cdot\nabla\chi)u + \chi(u\cdot\nabla)w_* + \chi(w_*\cdot\nabla)u \\ &\qquad + (w_*\cdot\nabla\chi)u + (w_*\cdot\nabla)w_*\}. \end{aligned}$$

Expanding $\Delta(\chi u + w_*)$ gives $\chi\Delta u + 2\nabla\chi\cdot\nabla u + (\Delta\chi)u + \Delta w_*$. Expanding $(\chi u + w_*)\cdot\nabla(\chi u + w_*)$ gives $\chi^2(u\cdot\nabla)u$ and exactly the five terms in braces in the last two lines. Finally $\nabla p_c = \eta_*\chi\nabla p + \eta_* p\nabla\chi$. Inserting the already proved equation $u_t - \nu\Delta u = -(u\cdot\nabla)u - \nabla p$ leaves precisely f_c. Hence $\partial_t u_c + (u_c\cdot\nabla)u_c + \nabla p_c = \nu\Delta u_c + f_c$, and $\operatorname{div} u_c = 0$ because it is a spatial curl times a time scalar. On $|x|\le R_0$, $0\le t\le T_0$, all cutoff derivatives vanish and $(u_c, p_c, f_c) = (u, p, 0)$ exactly.

All these fields are smooth for every finite $t\ge 0$, since $k(t)$ never vanishes. They vanish for $|x|\ge R_1$ or $t\ge T_1$. Every mixed derivative of f_c has a finite maximum on the compact set

$\{|x| \leq R_1, 0 \leq t \leq T_1\}$. Multiplying that maximum by $(1 + R_1 + T_1)^K$ proves each required weighted force bound of order K; the same argument proves rapid decay of the smooth initial datum. Also $\int |u_c|^2 \leq (4\pi R_1^3/3) \max_{|x|\leq R_1, 0\leq t\leq T_1} |u_c|^2$, uniformly in time. This is the exact compact forced bridge, including its admissibility. It does not supply a finite-time singularity. For a fractional multiplier on the wave the damping exponent is $-\nu|k_0|^\alpha(e^{\alpha at} - 1)/(\alpha a)$ for $\alpha > 0$. The absolute value is required when the retained covector is signed.

49.5 Self-similar scales, their full residual, and an exact force bound

The report's substitution is correct in its stated convention. The following calculation additionally retains physical reference scales. Fix $T, \tau_*, L_*, U_* > 0$, a centre $x_* \in \mathbb{R}^3$, and an exponent $\beta \in \mathbb{R}$. For $0 \leq t < T$ put $\tau = T - t$, $s = \tau/\tau_*$, $y = (x - x_*)/(L_* s^\beta)$, and set

$$u = U_* s^{\beta-1} U(y), \qquad p = U_*^2 s^{2\beta-2} P(y).$$

Here U is a fixed smooth divergence-free profile and P is a fixed smooth scalar profile. The exact derivatives are

$$\begin{aligned}
\partial_t u &= \frac{U_*}{\tau_*} s^{\beta-2}\{(1-\beta)U + \beta y \cdot \nabla U\}, \\
(u \cdot \nabla)u &= \frac{U_*^2}{L_*} s^{\beta-2} (U \cdot \nabla)U, \\
\nabla p &= \frac{U_*^2}{L_*} s^{\beta-2} \nabla P, \qquad \Delta u = \frac{U_*}{L_*^2} s^{-\beta-1} \Delta U, \\
\operatorname{div} u &= \frac{U_*}{L_*} s^{-1} \operatorname{div} U = 0.
\end{aligned}$$

Each identity follows from $\partial_t s = -1/\tau_*$, $\partial_t y = \beta y/(\tau_* s)$, and $\partial_{x_i} y_j = \delta_{ij}/(L_* s^\beta)$. Thus the prescribed physical residual is exactly

$$f = \frac{U_*}{\tau_*} s^{\beta-2} \left[(1-\beta)U + \beta y \cdot \nabla U + \frac{U_* \tau_*}{L_*}\{(U \cdot \nabla)U + \nabla P\} - \frac{\nu \tau_*}{L_*^2} s^{1-2\beta} \Delta U\right]. \tag{631}$$

This is the exact map from any specified profile pair to a forced Navier–Stokes solution before T. The incoming formula is its own unit-reference convention $x_* = 0$, $U_* = L_* = \tau_* = 1$; the constants in the present computation have not been discarded. In particular $\beta = 1/2$ makes the dimensionless diffusion coefficient independent of time.

For a nonzero Schwartz profile U, change of variables gives, exactly,

$$\|u(t)\|_2^2 = U_*^2 L_*^3 s^{5\beta-2} \|U\|_2^2, \tag{632}$$

$$\|\nabla u(t)\|_2^2 = U_*^2 L_* s^{3\beta-2} \|\nabla U\|_2^2. \tag{633}$$

If $\beta > 0$, the physical length $\ell = L_* s^\beta$ and the velocity scale $U_* s^{\beta-1}$ have the exponent relation $\gamma = (1-\beta)/\beta$. Hence $1 < \gamma < 3/2$ is exactly $2/5 < \beta < 1/2$. The dissipation time integral near T is finite exactly when $\beta > 1/3$, because integration of $s^{3\beta-2}$ is finite exactly when its exponent exceeds -1.

There is a concrete obstruction for a fixed global profile in that open window. The pressure class in this calculation is a smooth P with $|P(y)| + |\nabla P(y)| \leq C_P(1 + |y|)^{K_P}$ for some finite $C_P, K_P \geq 0$; in particular it includes every Schwartz P. For this class the Schwartz velocity profile makes each pressure integration below absolutely convergent and makes its boundary term zero. If the residual (631) is not in L^2, it already fails the admissible force condition. Otherwise the following calculation applies. Integration by parts proves

$$\int f \cdot u = \tfrac{1}{2} \frac{d}{dt} \|u\|_2^2 + \nu \|\nabla u\|_2^2.$$

The convection integral is $\frac{1}{2}\int \operatorname{div}(|u|^2 u) = 0$, the pressure integral is $-\int p \operatorname{div} u = 0$, and the viscous integration gives $-\int u \cdot \Delta u = \|\nabla u\|_2^2$, proving every term and sign. Substituting (633) and applying Cauchy–Schwarz yields

$$\|f(t)\|_2 \geq \frac{U_* L_*^{3/2}}{\tau_*} \|U\|_2 s^{(5\beta-4)/2} \left[\frac{5\beta - 2}{2} - \frac{\nu \tau_*}{L_*^2} \frac{\|\nabla U\|_2^2}{\|U\|_2^2} s^{1-2\beta} \right] \tag{634}$$

whenever the bracket is nonnegative. For $2/5 < \beta < 1/2$ its second term tends to zero. For all sufficiently small $s > 0$ it is at least $(5\beta - 2)/4 > 0$. Since $(5\beta - 4)/2 < 0$, the force norm diverges. Smooth forces with the target's uniform rapid spatial decay have bounded L^2 norm near T: squaring a bound with any spatial exponent $K > 3/2$ and integrating proves this directly. Thus this explicit fixed, globally self-similar finite-energy family cannot have an admissible terminal force in the stated open window.

The same argument at $\nu = 0$, $f = 0$ forces $\beta = 2/5$ for a nonzero finite-energy fixed global Euler profile. These conclusions concern the specified family only. The exact residual map (631) remains valid, and changing profiles, outer flows or spatially local similarity regions produces additional terms and energy fluxes not set to zero by this calculation. The dimensional packet window in the report therefore supplies a compatible scale calculation, not the dynamics or a global fixed-profile solution.

49.6 Precisely what the all-order force estimate establishes

The report's proposed bound on projected residuals is a research target; no new viscous iteration is constructed by writing it. The convergence calculation itself has the following complete content. For actual smooth increments f_q on a common compact spatial set K and on $[0, T]$, with integers $k_q \to \infty$ and $\|f_q\|_{C^k(K\times[0,T])} \leq \varepsilon_q$ for $k \leq k_q$, $\sum_q \varepsilon_q < \infty$, fix any mixed derivative α. All but finitely many increments obey its summable bound. The derivative series therefore converges uniformly. Its sum is the derivative of the function series: along a coordinate segment, the fundamental theorem of calculus for partial sums passes to the uniform limit, giving the derivative identity; induction proves it at every order. At $t = 0, T$ the same argument gives compatible one-sided derivatives. If all increments have flat jets at T, uniform convergence of each derivative proves that the summed jets are zero, so extension by zero past T is smooth. Without flat jets, extension by zero is not valid: the constant function one already demonstrates a discontinuity. A separate smooth extension must then be constructed. Spatial compactness and a compact time extension give the target decay by taking the derivative supremum on their common compact support and multiplying by the largest required polynomial weight there. None of these series facts proves that an unspecified sequence of physical force increments satisfies the assumed estimates.

Nor does a uniform global derivative supremum diverging by itself identify a singular point without localization. For example choose a smooth bump $\chi \in C_c^\infty(\mathbb{R}^3)$ that is one near zero and let

$$g(x,t) = \chi(x - e_1/(T-t)) \sin((x_2)/(T-t)), \quad t < T, \qquad g(x,T) = 0.$$

For every compact spatial set K, the support is disjoint from K for all t sufficiently close to T. Hence g is smooth through T near every finite spatial point (indeed it vanishes there), while $\|\partial_2 g(t)\|_\infty \geq 1/(T-t)$ at the bump centre. This corrects any use of the incoming abstract closure statement that omits a compact concentration region or a target norm with a proved continuation criterion. The actual off-axis construction has a fixed compact region, so this example does not refute that construction.

50 Primary-source corrections for the incoming report

This is a claim-specific source audit of the supplied report, with an independent operator calculation. It does not certify the complete proofs of the cited papers. The downloaded snapshots and their hashes are listed in `retrieval_receipt.json`; the inspection date is 8 September 2026.

50.1 The fractional theorem and the exact endpoint statement

The published Theorem 1 of Córdoba, Martínez-Zoroa and Zheng, ARMA 250 (2026), article 38, uses $|\nabla|^\alpha = (-\Delta)^{\alpha/2}$ and includes $0 \le \alpha < (22 - 8\sqrt{7})/9$. It asserts a solution on $\mathbb{R}^3 \times [0, 1]$ with positive viscosity, forcing in $L^1([0, 1]; C_x^{1,\epsilon}) \cap L^\infty([0, 1]; L_x^2)$ for some $\epsilon > 0$, velocity in $L^\infty_{t,x} \cap L^\infty_t L^2_x$, spatial smoothness for $t < 1$, and $\lim_{t\uparrow 1} \int_0^t \|\nabla u(\cdot, s)\|_\infty \, ds = \infty$. Section 1.1 says both solution and forcing are smooth before blow-up; Section 4.4 gives the corresponding preterminal construction. The theorem does not assert all-order mixed smoothness through $t = 1$. The incoming report's positive range is valid as a subrange, but omits the endpoint $\alpha = 0$ from the source theorem. Its distinction between $(-\Delta)^s$ and $|\nabla|^\alpha$ is correct: $s = \alpha/2$. The cumulative section on the dissipative vortex-layer programme already contains the parameter calculation and viscosity transfer, so it should be retained instead of duplicated here.

50.2 Tao's averaging: exact operator and dynamics

In arXiv:1402.0290v3, equations (1.11)–(1.13) include dilations, rotations, and real order-zero Fourier multipliers. Footnote 8 records the role of dilation averaging in the non-degeneracy argument. The text preceding (1.16) explicitly says that arbitrary such averaging does not automatically preserve energy cancellation. Theorem 1.5 chooses a symmetric operator with cancellation and Schwartz divergence-free data for which no global H^{10} mild solution exists. Thus the report's broad description is supported, but the dilation and cancellation qualifications should be supplied. The following calculation proves the precise relation with classical forced dynamics.

Retain the source's Fourier convention $\widehat{u}(\xi) = \int_{\mathbb{R}^3} u(x) e^{-2\pi i x\cdot\xi} \, dx$. Let $H = H^{10}_{\mathrm{df}}(\mathbb{R}^3)$ be the real divergence-free Sobolev space, with norm $\|u\|_H^2 = \int (1 + 4\pi^2 |\xi|^2)^{10} |\widehat{u}(\xi)|^2 \, d\xi$. Its L^2 dual is denoted H^*. The Leray projection P has symbol $I - \xi \otimes \xi/|\xi|^2$ for $\xi \ne 0$; its value at zero is immaterial. Define

$$B(u, v) = -\frac{1}{2} P\big((u \cdot \nabla)v + (v \cdot \nabla)u\big).$$

For a probability space (Ω, μ), retain $R_{j,\omega} \in SO(3)$, $C^{-1} \le \lambda_{j,\omega} \le C$ with $C \ge 1$, and real order-zero symbols $m_{j,\omega}$ satisfying $m_{j,\omega}(-\xi) = \overline{m_{j,\omega}(\xi)}$ and

$$\|m\|_k = \sup_{\xi\ne 0} |\xi|^k |\nabla^k m(\xi)| < \infty, \qquad \int_\Omega \prod_{j=1}^{3} \|m_{j,\omega}\|_{k_j} \, d\mu < \infty \quad (k_j \ge 0).$$

Write

$$\mathrm{Rot}_R \, u(x) = Ru(R^{-1}x), \qquad \mathrm{Dil}_\lambda \, u(x) = \lambda^{3/2} u(\lambda x), \qquad A_{j,\omega} = m_{j,\omega}(D) \, \mathrm{Rot}_{R_{j,\omega}} \mathrm{Dil}_{\lambda_{j,\omega}}.$$

Retain the measurability of the source's parameter families: the maps $\omega \mapsto R_{j,\omega}$ and $\omega \mapsto \lambda_{j,\omega}$ are measurable, and $(\omega, \xi) \mapsto m_{j,\omega}(\xi)$ is jointly measurable. In particular, for every fixed $u \in H$ the map $\omega \mapsto A_{j,\omega}u$ is strongly measurable in H. Here is a direct justification of this last assertion. Its Fourier representative is jointly measurable by the displayed formulas for rotation, dilation and multiplication. Its scalar coordinates against a countable orthonormal basis of the weighted Fourier L^2 space are measurable integrals. The finite orthogonal projections onto that basis are measurable finite-dimensional maps and converge in H at every ω, proving strong measurability. The same argument in unweighted L^2, using inverse rotations and dilations and conjugate symbols, proves that $\omega \mapsto A^*_{j,\omega} g$ is strongly measurable for every fixed $g \in L^2$. These are measurability hypotheses and their consequences; norm integrability alone is not being used to infer them. The averaging is the bilinear map $H \times H \to H^*$ specified by

$$\langle \widetilde{B}(u, v), w\rangle = \int_\Omega \langle B(A_{1,\omega}u, A_{2,\omega}v), A_{3,\omega}w\rangle \, d\mu(\omega). \tag{WA1}$$

All three operators preserve divergence. For rotations this follows by the chain rule and $R^{-1}R = I$; for dilations divergence is multiplied by $\lambda^{5/2}$; scalar Fourier multipliers commute with each derivative. They preserve real fields by the stated symbol symmetry. Changing variables $y = \lambda x$, and then $y = R^{-1}x$, proves that dilations and rotations are L^2 isometries with respective adjoints $\mathrm{Dil}_{\lambda^{-1}}$ and $\mathrm{Rot}_{R^{-1}}$. Plancherel gives $m(D)^* = \overline{m}(D)$. Consequently

$$\widetilde{B}(u,v) = \int_\Omega A_{3,\omega}^* B(A_{1,\omega}u, A_{2,\omega}v)\, d\mu(\omega), \quad A_3^* = \mathrm{Dil}_{\lambda_3^{-1}} \mathrm{Rot}_{R_3^{-1}} \overline{m}_3(D). \tag{WA2}$$

Here the integral actually defines an L^2 field, as follows without an unstated convergence step. Fourier Cauchy–Schwarz gives $\|u\|_\infty \le C_S \|u\|_H$, where

$$C_S = \left(\int_{\mathbb{R}^3} (1 + 4\pi^2 |\xi|^2)^{-10}\, d\xi \right)^{1/2} < \infty;$$

the radial integrand at infinity is bounded by a constant times r^{-18}. Also $\|\nabla u\|_2 \le \|u\|_H$ and $\|P\|_{2\to 2} = 1$. Therefore $\|B(u,v)\|_2 \le C_S \|u\|_H \|v\|_H$. The Fourier formula for dilation is $\widehat{\mathrm{Dil}_\lambda u}(\xi) = \lambda^{-3/2}\widehat{u}(\xi/\lambda)$, which gives $\|A_j u\|_H \le C^{10}\|m_j\|_0 \|u\|_H$. The integrand in (WA2) thus has L^2 norm at most $C_S C^{20} \prod_j \|m_j\|_0 \|u\|_H \|v\|_H$. It is also strongly measurable: continuity of $B : H \times H \to L^2$ makes $g_\omega = B(A_{1,\omega}u, A_{2,\omega}v)$ strongly measurable. Approximate g_ω pointwise by measurable simple L^2-valued functions. Applying $A_{3,\omega}^*$ to each simple function gives a strongly measurable function by the preceding fixed-vector result. The images converge pointwise in L^2, since each $A_{3,\omega}^*$ has finite operator norm. Thus the integrand is strongly measurable. Together with its integrable norm bound this proves existence of its Bochner integral, the asserted map, and the duality identity (WA1). No operator is replaced by an equal-looking different operator.

For smooth fields, the Laplacian obeys

$$\Delta A_j = \lambda_j^2 A_j \Delta, \tag{WA3}$$

because it commutes with scalar Fourier multipliers and rotations, and each of the two derivatives of $u(\lambda_j x)$ contributes a factor λ_j. Thus the actual dissipative operator remains explicitly present.

Fix the particular cancellation-preserving $\widetilde{B}$ from Tao's theorem. For any $u \in C(I;H) \cap C^1(I;L^2)$ on a time interval I, set

$$f_u = (\widetilde{B} - B)(u,u), \qquad p_u = -\Delta^{-1}\partial_i \partial_j (u_i u_j).$$

Both fields are well defined in L^2 at each time: $u_i u_j \in L^2$ by $\|u\|_\infty \|u\|_2 < \infty$, and the scalar pressure multiplier is bounded. Set $N = (u \cdot \nabla)u$. Divergence freedom gives $N_i = \partial_j(u_j u_i)$, hence $\nabla p_u = -(I - P)N$ in distributions and in L^2. Since $B(u,u) = -PN$, the following is an identity of L^2 residuals:

$$\partial_t u + N + \nabla p_u - \nu \Delta u - f_u = \partial_t u - \nu \Delta u - \widetilde{B}(u,u), \qquad \nu > 0. \tag{WA4}$$

Indeed, $N + \nabla p_u = PN = -B(u,u)$; substitution proves every sign in (WA4). In particular the source's coefficient $\nu = 1$ is retained when applying this identity to its solution. The map $u \mapsto (u, p_u, f_u)$ transfers its averaged equation exactly to the ordinary forced equation, with the same spatial coordinates and the same time variable. Conversely the forced equation with this explicitly prescribed f_u gives the averaged equation by the same identity. The force is divergence free because both bilinear operators take values in divergence-free fields.

There is also an exact energy relation. For $u \in H$, integration against a cutoff $\chi_R(x) = \chi(x/R)$, equal to one on $|x| \le R$ and supported on $|x| \le 2R$, gives

$$\int \chi_R N \cdot u = -\frac{1}{2} \int |u|^2 u \cdot \nabla \chi_R.$$

The right side is bounded in absolute value by $\|\nabla \chi\|_\infty \|u\|_\infty \|u\|_2^2/(2R)$ and tends to zero. Dominated convergence applies on the left because $N \in L^2$ and $u \in L^2$. As P is self-adjoint and $Pu = u$, this proves $\langle B(u,u),u\rangle = 0$. The chosen averaging has the same cancellation, so $\langle f_u,u\rangle = 0$. Along a trajectory with zero residual in (WA4), Plancherel gives $\langle \Delta u,u\rangle = -\|\nabla u\|_2^2$ and therefore

$$\frac{d}{dt}\frac{1}{2}\|u(t)\|_2^2 = -\nu\|\nabla u(t)\|_2^2. \tag{WA5}$$

This proves the force-defect bridge and its energy behavior. It supplies no all-order terminal estimates for f_u; those estimates are not a conclusion of (WA4) or (WA5). Consequently this exact transfer does not itself establish the requested smooth-through-terminal-time forced singularity.

50.3 The inspected numerical Euler manuscript's actual status

The author-hosted Ganeshram–Duruisseaux–Anandkumar manuscript has 107 pages in the downloaded snapshot. Its abstract describes an approximate profile and evidence for stability. Page 5 explicitly identifies unfinished certification of constants and margins, allowing refinement of estimates. Page 13, Section 4.6, additionally identifies local well-posedness and continuation in rescaled time as necessary for an unconditional blow-up result. Page 14, Section 4.9, leaves terms outside the signed matrices to be certified. Thus the report's candidate-status description is supported; the remaining work should not be described solely as evaluating constants. This audit checks these status statements, not their eventual feasibility or the correctness of all individual certificates. Page 7, equations (3)–(7), defines λ as the common radial and axial length exponent, and retains a moving axial centre with $\dot z_c = -C(T-t)^{\lambda-1}$. The report's fixed-centre illustrative ansatz therefore does not reproduce all the source's coordinates. The exact length dictionary is $\beta = \lambda$. Here is the complete coordinate and equation correspondence, including the axial translation and its velocity.

Let $I \subset \mathbb{R}$ be an interval, $\nu \geq 0$, and $d \in C^\infty(I;\mathbb{R}^3)$. Write $\mathcal{N}_\nu(u,p) = \partial_t u + (u\cdot\nabla)u + \nabla p - \nu\Delta u$ on $\mathbb{R}^3 \times I$, and let $\mathcal{S}_\nu(I)$ be the set of smooth triples (u,p,f) on this domain with u,f vector valued, p scalar valued, $\operatorname{div} u = 0$, and $\mathcal{N}_\nu(u,p) = f$. No spatial decay restriction is imposed in this definition. Define

$$\begin{aligned} &u_d(x,t) = u(x-d(t),t), \qquad\qquad p_d(x,t) = p(x-d(t),t),\\ &f_d(x,t) = f(x-d(t),t) - \sum_{j=1}^{3} d_j'(t)\partial_j u(x-d(t),t). \end{aligned} \tag{635}$$

These formulas define a bijection $\mathcal{S}_\nu(I) \to \mathcal{S}_\nu(I)$. Indeed $\partial_t u_d = (\partial_t u - d'\cdot\nabla u)(x-d,t)$, $\partial_{x_i} u_d = (\partial_i u)(x-d,t)$, and $\partial_{x_i}\partial_{x_j} u_d = (\partial_i\partial_j u)(x-d,t)$. Consequently divergence, convection, pressure gradient, and Laplacian are the corresponding original quantities evaluated at $(x-d,t)$, while the displayed time derivative gives precisely f_d. The inverse, including its force component, is

$$u(y,t) = u_d(y+d(t),t), \quad p(y,t) = p_d(y+d(t),t), \quad f(y,t) = f_d(y+d(t),t) + d'(t)\cdot\nabla u_d(y+d(t),t).$$

Substitution in both orders proves the identity map on every component. The spatial Jacobian is the identity matrix with determinant one, so $\|u_d(t)\|_2 = \|u(t)\|_2$ and $\|\nabla u_d(t)\|_2 = \|\nabla u(t)\|_2$ whenever these norms are finite. Every spatial derivative and every compact support is translated exactly. Time derivatives retain derivatives of d; no bound through an endpoint is inferred when these derivatives are unbounded there.

For clarity the velocity expressed in the moving frame is a second, also exact, map. For $(u,p,f) \in \mathcal{S}_\nu(I)$ put

$$v(y,t) = u(y+d(t),t) - d'(t), \quad q(y,t) = p(y+d(t),t), \quad g(y,t) = f(y+d(t),t) - d''(t).$$

Here $\partial_t v = (\partial_t u + d'\cdot\nabla u)(y+d,t) - d''$ and $(v\cdot\nabla)v = ((u-d')\cdot\nabla u)(y+d,t)$; the two d' terms cancel exactly, proving $\mathcal{N}_\nu(v,q) = g$ and $\operatorname{div} v = 0$. Its inverse is $u(x,t) = v(x-d,t) + d'$,

$p(x,t) = q(x-d,t)$, $f(x,t) = g(x-d,t)+d''$. Equivalently $q_a = q + d'' \cdot y$ gives $\mathcal{N}_\nu(v, q_a) = f(y+d,t)$, with inverse pressure $p(x,t) = q_a(x-d,t) - d'' \cdot (x-d)$. The affine pressure and constant velocity are retained, not discarded. If $u(t) \in L^2(\mathbb{R}^3)$ and $d'(t) \neq 0$, then $v(t) \notin L^2$: otherwise the difference of two L^2 fields would be the nonzero constant d', whose squared integral on balls is $|d'|^2(4\pi R^3/3) \to \infty$. Thus the first translation map preserves finite energy; the second has the explicitly proved different velocity and pressure behavior.

Now retain $T, \tau_*, L_*, U_* > 0$, $x_* \in \mathbb{R}^3$, and $\beta \in \mathbb{R}$. For $0 \le t < T$ put $s = (T-t)/\tau_*$ and $\ell = L_* s^\beta > 0$. For a fixed smooth profile pair (U, P) on $\mathbb{R}^3$ with $\operatorname{div}_Y U = 0$, set

$$Y = \frac{x - x_* - d(t)}{\ell}, \qquad u_d = U_* s^{\beta-1} U(Y), \qquad p_d = U_*^2 s^{2\beta-2} P(Y).$$

At each fixed t, $Y \mapsto x = x_* + d(t) + \ell Y$ is a diffeomorphism $\mathbb{R}^3 \to \mathbb{R}^3$ with Jacobian ℓI_3 and determinant ℓ^3. The field inverse on its image is $U(Y) = U_*^{-1} s^{1-\beta} u_d(x_* + d + \ell Y, t)$ and $P(Y) = U_*^{-2} s^{2-2\beta} p_d(x_* + d + \ell Y, t)$. Because $\partial_t Y = \beta Y/(\tau_* s) - d'/\ell$, $\partial_{x_i} Y_j = \delta_{ij}/\ell$, direct differentiation of each component gives the complete prescribed force

$$\begin{aligned} f_d = \frac{U_*}{\tau_*} s^{\beta-2} \Big[&(1-\beta)U + \beta (Y \cdot \nabla_Y) U + \frac{U_* \tau_*}{L_*} \{ (U \cdot \nabla_Y) U + \nabla_Y P \} - \frac{\nu \tau_*}{L_*^2} s^{1-2\beta} \Delta_Y U \Big] \\ &- \frac{U_*}{L_*} s^{-1} (d'(t) \cdot \nabla_Y) U. \end{aligned} \tag{636}$$

All profiles on the right are evaluated at Y. The first line is the fixed-centre residual evaluated in translated coordinates; the second is exactly the force correction in (635). Conversely the displayed field inverse and inverse force translation recover the fixed-centre forced equation with its original viscosity.

In the cited axial convention put $d(t) = z_c(t) e_3$ and retain the source's constant C and exponent λ, so $z_c' = -C(T-t)^{\lambda-1}$. For every reference $t_0 < T$ the full primitive is

$$z_c(t) = \begin{cases} z_c(t_0) + \dfrac{C}{\lambda} \left((T-t)^\lambda - (T-t_0)^\lambda \right), & \lambda \neq 0, \\ z_c(t_0) + C \log \dfrac{T-t}{T-t_0}, & \lambda = 0. \end{cases}$$

Differentiation proves the required negative sign in both cases. When $\beta = \lambda$, the last line of (636) is

$$\frac{C U_* \tau_*^{\lambda-1}}{L_*} s^{\beta-2} \partial_{Y_3} U = \frac{U_*}{\tau_*} s^{\beta-2} \frac{C \tau_*^\lambda}{L_*} \partial_{Y_3} U.$$

This is the exact retained drift in the moving-centre profile equation. The arbitrary initial centre is present in the inverse coordinate map. All formulas hold on the preterminal domain $s > 0$, including $C = 0$; they make no unproved assertion of an extension through $s = 0$. They relate the two coordinate presentations and their forced equations; they do not identify their profile solutions or discard the source's additional profile equations.

References

[1] L. Caffarelli, R. Kohn, and L. Nirenberg, *Partial regularity of suitable weak solutions of the Navier–Stokes equations*, Communications on Pure and Applied Mathematics 35 (1982), 771–831. https://doi.org/10.1002/cpa.3160350604. Theorem B, p.772; Theorem A', pp.772–773; Section 2B, pp.780–781; Appendix A.1, pp.822–823. Original article text inspected through the indexed scan transcription at https://www.scribd.com/document/683377073/Caffarelli-1982; original page images were not available to this audit.

[2] Q. Jiu and Z. Xin, *On Liouville Type of Theorems to the 3-D Incompressible Axisymmetric Navier–Stokes Equations*, arXiv:1501.02412v2 (17 March 2015), p.2. https://arxiv.org/pdf/1501.02412v2.

[3] D. Chamorro, P.-G. Lemarié-Rieusset, and K. Mayoufi, *The role of the pressure in the partial regularity theory for weak solutions of the Navier–Stokes equations*, arXiv:1602.06137, pp.2–3. `https://arxiv.org/pdf/1602.06137`. This directly read research paper corroborates the force threshold, the sufficient pressure class, and the spatial regularity statement.

[4] B. Rodgers and T. Tao, *The de Bruijn–Newman constant is non-negative*, Forum of Mathematics, Pi 8 (2020), e6. arXiv:1801.05914, version 5.

[5] A. Connes, C. Consani and H. Moscovici, *Zeta zeros and prolate wave operators: Semilocal adelic operators*, Annals of Functional Analysis (2024), doi:10.1007/s43034-024-00388-z.

[6] T. Tao, *A digestion of the Jacobian conjecture counterexample*, 21 July 2026, author's exposition.

[7] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann ξ function, and a new upper bound for the de Bruijn–Newman constant*, arXiv:1904.12438v2 (2019).

[8] C. L. Fefferman, *Existence and Smoothness of the Navier–Stokes Equation*, Clay Mathematics Institute official statement, including errata. `https://www.claymath.org/wp-content/uploads/2022/06/navierstokes.pdf`.

[9] T. Buckmaster, statement released 8 September 2026. `https://cims.nyu.edu/~tristanb/statement.pdf`.

[10] L. Alpöge and T. Buckmaster, *Blowup for the Euler Equations with Smooth Forcing*, released September 2026. `https://cims.nyu.edu/~tristanb/euler.pdf`.

[11] L. Alpöge and T. Buckmaster, *Blowup for the Boussinesq Equations with Smooth Forcing*, released September 2026. `https://cims.nyu.edu/~tristanb/boussinesq.pdf`.

[12] T. Buckmaster et al., public fluid formalization repository, snapshot cited in Section 2. `https://github.com/tristanbuckmaster/fluid_lean`.

A Complete source proof of the separated-link fourth coefficient

8 September 2026. Two links which share no elementary face nevertheless develop a nonzero connected electric covariance through two adjacent faces. This note calculates that coefficient in the actual full-box vacuum. The calculation keeps both intermediate spin channels and the disconnected expectation subtraction. It does not assume a volume-uniform remainder.

A.1 Original Hilbert space and the coefficient to be calculated

Fix $L \geq 2$, vertices $\{-L, \ldots, L\}^3$, all positive contained links e, and all contained elementary faces p. Write their complete counts as $N = 3(2L)(2L+1)^2$ and $M = 3(2L)^2(2L+1)$. Link variables are $U_e \in SU(2)$, with the inverse on reverse traversal. The face trace is

$$W_{(n;i,j)} = \operatorname{tr}\{U_i(n)U_j(n+\mathbf{e}_i)U_i(n+\mathbf{e}_j)^{-1}U_j(n)^{-1}\}, \qquad i<j.$$

On product Haar probability space use $T_a = -i\sigma_a/2$, left derivatives $X_{e,a}$, $E_e = -\sum_a X_{e,a}^2$, and

$$H = \kappa\sum_e E_e + b\sum_p (2 - W_p), \quad \kappa = 2g^2/a, \quad b = 1/(2g^2 a), \quad \xi = b/\kappa = 1/(4g^4). \tag{S1}$$

The fundamental Casimir is $3/4$. In the original metric $-\operatorname{tr}(AB)/2$, the Laplacian is four times the sum-of-squares Laplacian in these generators, so the electric coefficient in that metric is $\kappa/4$. Neither this factor nor the Wilson scalar $2bM$ is changed.

The compact connected configuration manifold and bounded smooth potential give a self-adjoint compact-resolvent operator of domain H^2, form domain H^1, and unique smooth positive unit ground vector ψ_ξ. For precision, the modulus inequality supplies a nonnegative form minimizer, elliptic regularity and the maximum principle make it positive, and the identity $q_{H-\mathcal{E}}(\psi f) = \kappa \sum_{e,a} \int \psi^2 |X_{e,a} f|^2$ implies uniqueness by division by ψ. Gauge transformations preserve this vector and every link Casimir. All following vectors are thus physical.

Set $r_e = \#\{p : e \in \partial p\}$, $\gamma_e = \langle \psi, E_e \psi \rangle$, $v_e = (E_e - \gamma_e)\psi$, and

$$C_{ef} = \langle v_e, v_f \rangle = \langle \psi, E_e E_f \psi \rangle - \gamma_e \gamma_f. \tag{S2}$$

The operators commute for different links. Smoothness makes (S2) legitimate as an operator identity on the vacuum, not only a formal product of forms.

For distinct e,f that share no face, define the integer

$$a_{ef} = \#\{(p,q) : e \in \partial p,\ f \in \partial q, |\partial p \cap \partial q| = 1\}. \tag{S3}$$

The pair in (S3) is ordered by which face contains e and which contains f; there is no double counting because no face contains both links. All faces are those of the original open box. The result is

$$\boxed{C_{ef} = \frac{127}{219024}\, a_{ef} \xi^4 + O_L(\xi^6) \quad (e, f \text{ share no face}).} \tag{S4}$$

In particular the coefficient is positive whenever a two-face channel exists, even though the energy matrix between these links is exactly zero. The latter assertion is proved in Appendix A.4.

A.2 The full second vacuum coefficient, including both shared-link channels

Put $H_0 = \sum E_e$ and $S = \sum W_p$, so $H = \kappa(H_0 - \xi S) + 2bM$. On $|z| = 3/8$ in the full Hilbert space the resolvent of H_0 has norm at most $8/3$, since its first nonzero eigenvalue is $3/4$. Also $\|S\| = 2M$. Therefore the bounded perturbation Neumann series converges for $|\xi| < 3/(16M)$, gives the analytic rank-one bottom projection, and gives an analytic positive unit eigenvector for real ξ near zero. Elliptic regularity applied repeatedly to the eigen-equation upgrades this to analyticity in each fixed Sobolev space. Its radius and estimates may depend on L. This suffices for the finite derivative calculations below.

Product Haar integration gives $\int W_p = 0$, $\langle W_p, W_q \rangle = \delta_{pq}$, and $H_0 W_p = 3W_p$. Indeed a link in only one of two distinct face boundaries has odd central parity; for the same face integrate its Haar holonomy and use the fundamental character squared norm one. Comparing coefficients in the actual equation $(H_0 - \xi S)\psi = e(\xi)\psi$, including its unit-vector condition, gives

$$\psi = 1 + \xi A_1 + \xi^2 A_2 + O_{H^k,L}(\xi^3), \quad A_1 = S/3, \quad e(\xi) = -M\xi^2/3 + O_L(\xi^3),$$

$$H_0 A_2 = (S^2 - M)/3, \qquad \int A_2 = -M/18. \tag{S5}$$

The inverse in (S5) can be resolved entirely. For adjacent distinct faces p,q sharing g put

$$Z_{pq,0} = \int W_p W_q \, dU_g, \qquad Z_{pq,1} = W_p W_q - Z_{pq,0}.$$

After reversing one real trace if needed, their words are $\mathrm{tr}(U_g B)$, $\mathrm{tr}(U_g^{-1} D)$, with six distinct outer links. The fundamental Haar identity gives $Z_{pq,0} = \mathrm{tr}(BD)/2$. The shared-edge spin tensor $1/2 \otimes 1/2$ consists of spins zero and one, while all outer links have spin $1/2$. Thus

$$H_0 Z_{pq,0} = \tfrac{9}{2} Z_{pq,0}, \quad H_0 Z_{pq,1} = \tfrac{13}{2} Z_{pq,1}, \quad \|Z_{pq,0}\|^2 = \tfrac{1}{4}, \quad \|Z_{pq,1}\|^2 = \tfrac{3}{4}. \tag{S6}$$

The norms use orthogonal Haar projection on g and total product norm one. A repeated face has $W_p^2 - 1 = \chi_1(U_p)$ with free energy 8 and norm one. An edge-disjoint pair has energy 6 and norm one, even when the faces meet at a vertex. Consequently the full coefficient is

$$\boxed{\begin{aligned} A_2 = & -\frac{M}{18} + \sum_p \frac{W_p^2 - 1}{24} \\ & + \sum_{\{p,q\}\ \text{edge-disjoint}} \frac{W_p W_q}{9} \\ & + \sum_{\{p,q\}\ \text{adjacent}} \left(\frac{4}{27} Z_{pq,0} + \frac{4}{39} Z_{pq,1} \right). \end{aligned}} \tag{S7}$$

All distinct-pair sums in (S7) are unordered; the factor two in S^2 has already been included. No finite-face subsystem is substituted for H.

Different unordered pairs in (S7) have different odd-link parity supports. If their symmetric boundaries agreed, their symmetric face difference would be a nonempty closed mod-two collection of at most four squares. A face in that collection needs another face on each of its four boundary edges; two distinct elementary squares share at most one edge, so at least four other squares would be needed, a contradiction. Haar integration on a link where parity differs makes the cross term zero. Repeated-face terms are even everywhere, and different repeated-face terms have different joint link spins. Thus all the displayed components, with their two spin channels, are mutually orthogonal. This remains true after application of any link Casimir, which preserves those parity and spin subspaces.

A.3 Exact connected fourth coefficient

Suppose e,f share no face. Then $E_e E_f 1 = E_e E_f A_1 = 0$: each term in A_1 is a single face and depends on at most one of these links. Self-adjointness moves $E_e E_f$ to either of these vectors, so every term of degree less than four in $\langle \psi, E_e E_f \psi \rangle$ vanishes, and its degree-four term is exactly

$$\langle A_2, E_e E_f A_2 \rangle. \tag{S8}$$

In particular no third or fourth unknown coefficient of the vacuum is being omitted from (S8): its possible pairing with 1 or A_1 is identically zero for the stated reason. Similarly

$$\gamma_e = \xi^2 \langle A_1, E_e A_1 \rangle + O_L(\xi^3) = \frac{r_e}{12} \xi^2 + O_L(\xi^3). \tag{S9}$$

Only a pair p,q with $e \in p$, $f \in q$ contributes to (S8). Repeated faces contribute zero. There are exactly $r_e r_f$ such distinct unordered pairs when they are labeled as in (S3). For an edge-disjoint pair, both e and f carry fundamental spin, so its contribution is $(3/4)^2/9^2 = 1/144$. For an adjacent pair, neither e nor f can be its shared edge: otherwise the face containing the other link would contain both e and f. They are therefore both outer fundamental edges in each of the two channels (S6), and the contribution is

$$\left(\frac{3}{4}\right)^2 \left[\left(\frac{4}{27}\right)^2 \frac{1}{4} + \left(\frac{4}{39}\right)^2 \frac{3}{4} \right] = \frac{1}{324} + \frac{3}{676}. \tag{S10}$$

Equation (S9) subtracts $r_e r_f/144$ at degree four in (S2). Hence every edge-disjoint pair cancels its disconnected contribution exactly; every adjacent pair leaves

$$\frac{1}{324} + \frac{3}{676} - \frac{1}{144} = \frac{127}{219024} > 0. \tag{S11}$$

This proves the coefficient in (S4).

For its stated even error, the center transformation $U_{(n,i)} \mapsto (-1)^{\sum_{j<i} n_j} U_{(n,i)}$ changes every face trace sign, commutes with every E_e and preserves Haar measure. It intertwines $H_0 - \xi S$ with $H_0 + \xi S$ and hence the positive unit vacua at ξ and$-\xi$. Both terms in (S2) are consequently even analytic functions of ξ near zero. Their next possible order after four is six. This proves the error in (S4); it does not supply a volume-uniform analytic radius.

A.4 Exact support map and relation to the energy matrix

Define the edge adjacency matrix B by $B_{eg} = 1$ when e,g are distinct physical links belonging to a common elementary face, and zero otherwise. Two distinct links belong to at most one elementary square: perpendicular incident links determine its two plane directions and base, while parallel links in a square must be opposite sides and also determine that square. Thus, for the e,f considered above,

$$\boxed{a_{ef} = (B^2)_{ef}.} \tag{S12}$$

Indeed a pair p,q counted in (S3) has a unique shared edge g, which is a common neighbor of e,f. Conversely a common neighbor g determines the unique p containing e,g and q containing g,f; these are distinct because e,f share no face, and their unique shared edge is g. These two maps are inverse. The covariance coefficient thus has support at graph distance two, including boundary incidences, and no support beyond distance two at this order. It makes no all-orders finite-range claim about C.

There are actual such channels in every $L \geq 2$. Take $e = ((0,0,0),1)$ and $f = ((0,2,0),1)$. They share no face, but the faces in directions $1,2$ based at $(0,0,0)$ and $(0,1,0)$ share the edge $((0,1,0),1)$, giving $a_{ef} \geq 1$. All of their vertices are in the original box, including at L=2.

Put $\mathcal{N}_{ef} = \langle v_e, (H - \mathcal{E}) v_f \rangle$. Commuting the link operators on the smooth real vacuum gives

$$\mathcal{N}_{ef} = \tfrac{1}{2} \langle \psi, [E_e, [H, E_f]] \psi \rangle. \tag{S13}$$

To check (S13), expand the double commutator, use $H\psi = \mathcal{E}\psi$, commute E_e,E_f, and use the real symmetry of their two matrix elements. Only potential faces containing f occur in $[H, E_f]$. If e shares no face with f, E_e commutes with every one of their multiplication and derivative terms. Thus

$$\boxed{\mathcal{N}_{ef} = 0 \text{ exactly, at every positive coupling, when e,f share no face.}} \tag{S14}$$

There is no contradiction between (S4) and (S14): a matrix-valued spectral measure can have nonzero zeroth moment and zero first moment off diagonal. Explicitly, for the spectral resolution of A=H−E on the physical space,

$$d\Sigma_{ef}(\omega) = \langle v_e, \mathbf{1}_{d\omega}(A) v_f \rangle, \quad C_{ef} = \int d\Sigma_{ef}, \quad \mathcal{N}_{ef} = \int \omega \, d\Sigma_{ef}. \tag{S15}$$

The measure matrix is positive semidefinite on every Borel set because its quadratic form is the squared norm of a projected linear combination of the v_e. An individual off-diagonal entry, however, need not be a nonnegative measure; (S15) retains its signs. Smoothness gives the finite moments used here. These identities are the exact relation between the nonlocal covariance contribution and the local energy matrix, not an assumption that one determines the other's support.

Finally $\xi^4 = 1/(256 g^{16})$. The physical coefficient in (S4) is therefore $127 a_{ef}/(56070144 g^{16})$; C has no extra energy factor, whereas $\mathcal{N}$ has one. The scalar energy shift remains in H and cancels only in H−E. The fixed-box coefficient is a concrete next input to the extensive denominator calculation; control of all higher orders and summability at fixed coupling is not inferred from its finite range at order four.

A.5 The complete fourth-order spectral functional and time correlation

The same calculation can resolve the energy dependence, rather than only its zeroth moment. Put $A_\xi = (H - \mathcal{E})/\kappa$, and write

$$F_{ef,\xi}(h) = \langle v_e, h(A_\xi) v_f \rangle.$$

For each fixed box, for distinct links sharing no face, and every bounded continuous real or complex function h which is differentiable at 3,

$$\boxed{\begin{aligned}\lim_{\xi\to 0} \xi^{-4} F_{ef,\xi}(h) = a_{ef}\Big[&-\frac{59}{275184} h(3) - \frac{1}{336} h'(3) \\ &+ \frac{1}{1296} h(9/2) + \frac{3}{132496} h(13/2)\Big].\end{aligned}} \tag{S16}$$

Here the limit is a functional on the specified test functions. In particular the derivative term must not be omitted or represented as an ordinary positive spectral atom. Each original matrix-valued spectral measure is still the exact positive-semidefinite measure (S15).

We prove (S16), including the relation to the actual first interacting spectral subspace. Let P project onto the free physical energy-three space $\mathcal{P} = \operatorname{span}\{W_p\}$, and let

$$R_3 = (I - P)((H_0 - 3)|_{\mathcal{P}^\perp})^{-1}(I - P).$$

The inverse is on the physical space: its constant channel is -1/3. To verify isolation, a nonzero physical spin network has no degree-one support vertex. Its support contains a cycle of length at least four in the bipartite cubic graph, costing at least 3. With fewer than six nontrivial edges it must be one elementary square: a fifth edge would have a new degree-one endpoint, or be a forbidden same-bipartition chord of the four-cycle. The degree-two invariant contractions require equal spins on that square. Only spin 1/2 has energy below 9/2. A six-edge rectangular loop realizes energy 9/2. Thus the first physical free band is exactly P, with the next energy 9/2.

For sufficiently small real ξ, the contour $|z - 3| = 3/4$ defines the actual spectral projection P_ξ, and the exact isometry

$$V_\xi = P_\xi P (P P_\xi P|_{\mathcal{P}})^{-1/2} : \mathcal{P} \longrightarrow \operatorname{ran} P_\xi \tag{S17}$$

uses its full Gram factor. Indeed the free contour resolvent has norm at most 4/3. For $|\xi| \le 3/(64M)$, its Neumann factor is at most 1/8, and integrating the resolvent difference gives $\|P_\xi - P\| \le 1/7$. The displayed inverse square root therefore exists by its binomial series; idempotence and orthogonality of the real projection show $V_\xi^* V_\xi = I$. Its range is smooth, and the finite-rank resolvent formula and elliptic regularity justify all operator products and fixed Sobolev derivatives below. It is orthogonal to the actual vacuum for a smaller interval, for example $|\xi| \le 3/(128M)$, by the free gaps and the bounded perturbation estimate $2M|\xi|$.

Let Q be the unchanged open-face signless adjacency matrix, with diagonal the actual face degrees and off-diagonal 1 for shared-edge faces. Direct virtual-channel calculation gives

$$V_\xi^* A_\xi V_\xi = 3I + \xi^2 F + O_L(\xi^4), \qquad F = \frac{7}{15} I - \frac{1}{21} Q. \tag{S18}$$

Here is the coefficient calculation rather than an assumption of (S18). Differentiating (S17) gives $V_0' = R_3 S P$. The second matrix coefficient before vacuum subtraction is $-PSR_3SP$. The vacuum channel gives +1/3 to every entry; a repeated-face channel gives -1/5 to its diagonal; an edge-disjoint pair gives -1/3 to its two diagonals and mutual entries; an adjacent pair gives $-(1/4)/(9/2 - 3) - (3/4)/(13/2 - 3) = -8/21$ to that same four-entry pattern. Thus the diagonal is $7/15 - M/3 - d_p/21$, and the off-diagonal is -1/21 exactly for adjacent faces. The actual vacuum coefficient -M/3 from (S5) gives (S18). The central cochain used in Appendix A.3 acts as -I on P; it makes the matrix in (S18) even and accounts for the fourth-order remainder.

The electric vector can be treated with arbitrary fixed real link weights w, retaining its linear dependence on all of them. Define $\Gamma_w = \sum w_e E_e$, $a_p(w) = \sum_{e\in p} w_e/4$, and $v(w) = (\Gamma_w - \langle \Gamma_w \rangle)\psi$. Put

$$y_\xi(w) = V_\xi^* v(w), \qquad \rho_\xi(w) = (I - P_\xi)v(w).$$

These are exact orthogonal components, both orthogonal to the actual vacuum. Coefficient comparison using (S7) and $V_0' = R_3SP$ gives

$$y_\xi(w) = \xi x(w) + \xi^3 z(w) + O_L(\xi^5), \qquad x(w) = \sum_p a_p(w) W_p,$$

$$\rho_\xi(w) = \xi^2 r(w) + O_{H^k,L}(\xi^3), \tag{S19}$$

where, for $s = a_p(w) + a_q(w)$ and g the shared edge,

$$r(w) = \sum_p \frac{2a_p(w)}{15}(W_p^2 - 1) + \sum_{\{p,q\}\ \text{adjacent}} \left[-\frac{2(s+w_g)}{9} Z_{pq,0} + \frac{2(3s+7w_g)}{273} Z_{pq,1}\right]. \tag{S20}$$

For completeness, Γ_w has eigenvalues $8a_p$ on the repeated face, $3s$ on an edge-disjoint pair, and $3s - 3w_g/2, 3s + w_g/2$ on the adjacent zero and one channels. The shared edge is counted twice as fundamental in 3s; replacing that by its actual spin accounts for precisely the last two corrections. In $\Gamma_w A_2$, the respective coefficients are $a_p/3, s/3, (4s - 2w_g)/9, (12s + 2w_g)/39$. Subtract the full dressing $R_3Sx(w)$, whose coefficients are $a_p/5, s/3, 2s/3, 2s/7$. The two constant coefficients are both $-\sum_p a_p/3$ and also cancel. This proves (S20) with no omitted channel. The centered projection identity $V_\xi^*\psi_\xi = 0$ is exact. The same central cochain makes $y_\xi(w)$ odd, proving the first expansion (S19).

For a single link e use the original weight $w_g^{(e)} = \delta_{eg}$, and denote the resulting x,z,r by $x_e, z_e, r_e^{\text{vec}}$. This superscript distinguishes the vector from the incidence count r_e. Then $x_e = \sum_{p\ni e} W_p/4$. If e,f share no face, $\langle x_e, x_f\rangle = 0$, and (S18) gives exactly

$$\langle x_e, F x_f\rangle = -a_{ef}/336. \tag{S21}$$

Each ordered adjacent pair contributes $-1/(4\cdot 21\cdot 4)$. In the cross spectral measure of $r_e^{\text{vec}}, r_f^{\text{vec}}$, only the pairs counted by (S3) occur. Repeated-face cross terms vanish. For every contributing pair, e and f are outer links of different faces, $s = 1/4$ for each weight, and $w_g = 0$. Thus their zero-channel coefficients in (S20) are both -1/18, and their one-channel coefficients are both 1/182. Orthogonality and (S6) give the complete free cross measure

$$\langle r_e^{\text{vec}}, h(H_0) r_f^{\text{vec}}\rangle = a_{ef}\left[\frac{1}{1296}h(9/2) + \frac{3}{132496}h(13/2)\right]. \tag{S22}$$

No edge-disjoint channel survives in (S20).

As $A_\xi - H_0 = -\xi S - e(\xi)I$ tends to zero in operator norm, its resolvents converge in norm. For bounded continuous h, this implies convergence of the cross expectations on ρ_ξ/ξ^2 to (S22): first use continuous functions vanishing at infinity and norm resolvent convergence, then approximate h on a compact interval and control the discarded tail by the uniformly bounded second spectral moments. Those moments follow from the H^2 convergence in (S19). Polarization gives the cross statement from the four corresponding diagonal statements. Hence (S22) is the entire fourth-order out-of-band contribution.

The fourth coefficient of the band mass is obtained without knowing the third vacuum coefficient: subtract (S22) at h=1 from (S4). It is

$$\langle x_e, z_f\rangle + \langle z_e, x_f\rangle = a_{ef}\left(\frac{127}{219024} - \frac{1}{1296} - \frac{3}{132496}\right) = -\frac{59}{275184}a_{ef}. \tag{S23}$$

Differentiability of h at 3 and spectral calculus of the finite matrix (S18) give $h(V_\xi^* A_\xi V_\xi) = h(3)I + \xi^2 h'(3)F + o_L(\xi^2)$. Combine this with (S19), (S21), (S23), and the exact orthogonal spectral decomposition into the band and its complement. This proves (S16).

In particular the complete dimensionful ground-state time correlation, with physical $\kappa > 0$ and physical time $t \geq 0$ fixed, is

$$\boxed{\begin{aligned}\langle v_e, e^{-t(H-\mathcal{E})} v_f \rangle = a_{ef}\xi^4 \Big[&-\frac{59}{275184} e^{-3\kappa t} + \frac{\kappa t}{336} e^{-3\kappa t} \\ &+ \frac{1}{1296} e^{-9\kappa t/2} + \frac{3}{132496} e^{-13\kappa t/2} \Big] + O_{L,t,\kappa}(\xi^6).\end{aligned}} \tag{S24}$$

Here $h(q) = e^{-\kappa t q}$ is used on the nonnegative excitation spectrum, with any bounded continuous extension below zero, so it belongs to the test class of (S16). The bounded potential perturbation gives $e^{-\kappa t K_\xi}$ its norm-convergent Duhamel series: each order is bounded by $(2M|\xi|\kappa t)^n/n!$. The excitation semigroup retains the additional analytic scalar $e^{\kappa t e(\xi)}$. The vacuum and vectors are analytic in fixed Sobolev norms, and the central cochain makes the whole expression even. These facts justify the even remainder in (S24), not only a pointwise little-o limit. For varying physical κ, the same proof uses the explicit parameter $\tau = \kappa t$ and keeps its dependence; no estimate uniform in τ or L is implied.

At t=0 the bracket is 127/219024. Its derivative at t=0 is exactly zero, because

$$3\left(-\frac{59}{275184}\right) - \frac{1}{336} + \frac{9}{2 \cdot 1296} + \frac{39}{2 \cdot 132496} = 0. \tag{S25}$$

This agrees with the exact identity (S14). It exhibits which separated-link spectral contributions cancel in the first moment: the actual band-mass correction, the actual band-energy derivative, and both higher spin channels. The term proportional to t is essential to that cancellation. Equations (S16)–(S25) describe the fixed-box fourth coefficient of the original interacting correlations; they do not exchange a volume limit with the coupling expansion or assert a continuum spectrum.

A.6 Primary context

The original Hamiltonian framework is J. Kogut and L. Susskind, *Hamiltonian formulation of Wilson's lattice gauge theories*, Physical Review D **11** (1975), 395–408, https://doi.org/10.1103/PhysRevD.11.395. Full conventions appear in (S1). For historical linked-cluster calculations in the same spatial dimension, see A. C. Irving, T. E. Preece and C. J. Hamer, *Cluster expansion approach to non-abelian lattice gauge theory in (3+1)D (I). SU(2)*, Nuclear Physics B **270** (1986), 536–552, https://doi.org/10.1016/0550-3213(86)90567-5. This note derives (S4) directly, with no historical novelty assertion or imported coefficient.